

Dielectric Dispersion of Colloidal Suspensions in the Presence of Stern Layer Conductance: Particle Size Effects

F. J. Arroyo,^{*,1} F. Carrique,[†] T. Bellini,[‡] and A. V. Delgado^{*}

^{*}Departamento de Física Aplicada, Facultad de Ciencias, Universidad de Granada, 18071 Granada, Spain; [†]Departamento de Física Aplicada I, Facultad de Ciencias, Universidad de Málaga, 29071 Málaga, Spain; and [‡]Dipartimento di Elettronica, Università di Pavia, 27100 Pavia, Italy

Received July 30, 1998; accepted October 8, 1998

In this work, we discuss the role that particle size plays in the manifestations of surface conduction on the dielectric response of colloidal dispersions. To that aim, experimental data on the dielectric constant of polystyrene suspensions of two different particle diameters (23 and 530 nm) are first compared to the predictions of a classical or standard model (E. H. B. DeLacey and L. R. White, *J. Chem. Soc. Faraday Trans. 2* **77**, 2007 (1983)), and it is found that, while the latter explains reasonably the dielectric behavior of the smallest particles, it considerably underestimates the phenomenon in the case of large particles. To explain these results in terms of contributions of ion motions in the inner region of the double layer of the particles, the approach followed by C. S. Mangelsdorf and L. R. White (*J. Chem. Soc. Faraday Trans. 86*, 2859 (1990)) is used to incorporate surface conductance in the theory of dielectric response of suspensions. In ac fields it is found that the model considerably improves the comparison between theory and experiment, whereas its use seems unnecessary for the smallest particles, where, whatever the combination used for the parameters of the theory, its predictions do not differ from the standard theory. Only in the case of the larger particles studied does the introduction of surface conductance play any role. A comparison between both types of theoretical results in a wide range of particle sizes demonstrates that Stern layer conductance always increases the magnitude of the low-frequency dielectric constant of suspensions, but its effect is less important the smaller the particle size and the larger the zeta potential for fixed ionic conditions in the dispersion medium. © 1999 Academic Press

Key Words: low-frequency dielectric dispersion; dynamic Stern layer; colloidal suspensions.

INTRODUCTION

When an ac electric field $E = E_0 \exp(i\omega t)$ is applied to a dispersion of colloidal particles, the polarization induced in the double layer of the particles is, in general, out of phase with the applied field. As a consequence, a characteristic frequency dependence is shown by the real (ϵ'_r) and imaginary (ϵ''_r) parts of the complex dielectric constant of the suspension (1, 2):

$$\epsilon_r^*(\omega) = \epsilon'_r(\omega) - i\epsilon''_r(\omega). \quad [1]$$

Furthermore, $\epsilon'_r(\omega = 0)$ can be much larger than that of the dispersion medium.

If the particle volume fraction ϕ is low (dilute suspension), a linear relationship exists between ϵ_r^* and ϕ ,

$$\epsilon'_r(\omega) = \epsilon'_{rs} + \phi \Delta\epsilon'_r(\omega) \quad [2]$$

$$\epsilon''_r(\omega) = \phi \Delta\epsilon''_r(\omega), \quad [3]$$

where $\Delta\epsilon'_r(\omega)$ and $\Delta\epsilon''_r(\omega)$ are, respectively, the real and imaginary parts of the complex dielectric increment, $\Delta\epsilon_r^*(\omega)$, of the suspension. The latter quantity measures the contribution of the particles and their double layers to the overall dielectric behavior of the suspension. Typically, $\Delta\epsilon_r^*(\omega)$ presents a relaxation frequency in the kHz to MHz region; since other relaxations appear at higher frequencies (2), we speak of low-frequency dielectric dispersion (LFDD) to describe the frequency dependence of $\Delta\epsilon'_r(\omega)$ or $\Delta\epsilon''_r(\omega)$ in the frequency region where double layer polarization phenomena manifest.

In fact, the frequency spectrum of $\Delta\epsilon_r^*$ is intimately related to the properties of the particles (size, shape, and dielectric constant), to the properties of the ionic dispersion medium (concentration, valence, and mobility of each ionic species present), and, most important, to the characteristics of the solid/liquid interface. This is one of the reasons explaining the growing interest in LFDD as a method of electrokinetic characterization.

Like in any other electrokinetic phenomenon, a theoretical model is needed to relate the above-mentioned properties to the measured dielectric increment. A number of theories have been proposed since the pioneering work of Dukhin and Shilov (2), differing from each other in the starting assumptions. For instance, some authors propose that only a diffuse double layer contributes to polarization, the inner layer ions being immobile (1–6), while others assume that Stern layer ions can undergo electric migration, and diffusion, thus playing a role in LFDD (7–15), with either free or restricted ionic exchange between diffuse and Stern regions. Theories contemplating only diffuse layer polarization are sometimes considered as part of the so-called *classical* or *standard* model of dielectric dispersion.

¹ To whom correspondence should be addressed.

Among them, we will be concerned here mainly with the treatment elaborated by DeLacey and White (4) that, although requiring numerical integration, is applicable whatever the values of κa (κ is the reciprocal Debye length, and a is the radius of the spherical particles of the suspension), zeta potential (ζ), and concentrations, valencies, and mobilities of the ions in solution.

Rosen *et al.* (13) generalized the DeLacey and White (DW hereafter) model by incorporating Stern layer conductance to the explanation of LFDD, whereas Mangelsdorf and White (16), using a slightly different approach, calculated the electrophoretic mobility and DC conductivity of suspensions when ions in the Stern layer are allowed to move under the action of the applied field.

In this paper, we use the original approach of Mangelsdorf and White together with the DW model to include a dynamic Stern layer (DSL). The results are compared to experimental $\Delta\epsilon'$ values obtained with suspensions of spherical particles of very different sizes to discuss under which conditions the general DSL theory must be used or under which conditions the simpler DW model is sufficient to explain the LFDD data.

EXPERIMENTAL

The colloidal dispersions studied were obtained from polystyrene spheres supplied by IDC Latex Corporation (USA). Two latexes were used, and their characteristics are given in Table 1. The original dispersions were first thoroughly cleaned by serum replacement in a stirred filtration cell, using Milli-Q water. Subsequently, fractions of these *clean* latexes were subjected to the same process with KCl solutions of the desired ionic strength, and, finally, the volume fraction (0.01 for sample L-23 and 0.02 for sample L-530) was adjusted by addition of suitable amounts of solution.

Electrophoretic mobilities (μ_e) of the two latexes in the presence of KCl solutions were measured in very dilute suspensions (volume fractions around 10^{-4}) using a Malvern Zetasizer 2c (England) at $25.0 \pm 0.5^\circ\text{C}$. At least 10 independent determinations were performed for each system, and their standard deviation will be used as an error parameter.

The complex conductivity and dielectric response of the polystyrene suspensions were obtained by measuring (in the 1 kHz–1 MHz frequency range) the complex admittance of a conductivity cell with variable interelectrode distance with a Hewlett–Packard HP-4284A LCR meter (USA). The details of

TABLE 1
General Characteristics of the Polystyrene Latexes
Used in This Work

Sample	Average diameter (nm)	Surface charge density ($\mu\text{C}/\text{cm}^2$)
L-23	23 ± 3	1.2
L-530	530 ± 30	3.9

TABLE 2
Electrophoretic Mobilities (μ_e) and Zeta Potentials (ζ) of the
Polystyrene Latexes for the Electrolyte Concentrations Indicated

Latex	[KCl] (mM)	μ_e ($\mu\text{m s}^{-1}/\text{V cm}^{-1}$)	ζ^a (mV)	κa^b
L-23	0.1	-5.1 ± 0.8	-119	0.38
	0.5	-5.3 ± 0.4	-201	0.85
L-530	0.1	-3.23 ± 0.07	-56.3	8.7
	1	-4.65 ± 0.06	-72.75	27.5

^a Zeta potential deduced from μ_e using O'Brien and White's theory (20).

^b κ^{-1} is the double layer thickness, and a the particle radius.

the method used and its ability to reduce the undesired effects of electrode polarization can be found in Refs. (17–19). The conductivity cell was thermostatted at $25.0 \pm 0.1^\circ\text{C}$ by means of a Haake F3-K (Germany) thermostatic bath.

RESULTS AND DISCUSSION

Electrophoretic Mobility

A preliminary step in the electrokinetic characterization of the particles was to measure their electrophoretic mobility as a function of electrolyte concentration (KCl). Table 2 shows the results, together with the ζ potentials calculated from μ_e data according to O'Brien and White's theory (20). The absolute values of both the electrophoretic mobility and the ζ potential display a maximum when the concentration of KCl approaches 1 mM (sample L-530) and 0.5 mM (L-23). This behavior of ζ is unjustified from the point of view of the standard electrokinetic model, according to which increasing the concentration of an indifferent electrolyte can only compress the diffuse layer, thus reducing $|\zeta|$. In any case, this anomaly has been observed in many polystyrene latexes (21–23) and among the reasons that were offered to explain it was anomalous (i.e., Stern layer) surface conductance. It is expected that our further investigation on the dielectric behavior of the two latex systems will give us additional clues on the dynamics of the electric double layer of these colloidal particles.

Dielectric Dispersion of the Suspensions

Figure 1 shows the real part of the dielectric increment as a function of the frequency of the applied field for both samples (L-530 and L-23, see Table 1) and the KCl concentrations used. The dielectric relaxation is clearly observable in latex L-530, whereas the phenomenon is only suggested by the data obtained with the small particles, due to the fact that the maximum frequency available in the HP-4284A is 1 MHz and the characteristic frequency for L-23 particles is much higher. Nevertheless, in both cases a quantitative estimation of the main parameters of the relaxation curve (i.e., the low-frequency value of $\Delta\epsilon'$, also named dielectric amplitude, and the relaxation frequency, ω_{rel}) can be obtained by fitting the ex-

perimental data to the following relaxation pattern, proposed by Vogel and Pauly (24) and that has been found to describe suitably the relaxation behavior of $\Delta\epsilon'_r(\omega)$ in suspensions,

$$\Delta\epsilon'_r(\omega) = \Delta\epsilon'_r(\infty) + \frac{\Delta\epsilon'_r(0) - \Delta\epsilon'_r(\infty)}{(1 + \omega\tau)(1 + \sqrt{\omega\tau})}, \quad [4]$$

where $\tau (=1/\omega_{\text{rel}})$ is the characteristic time of the system, $\Delta\epsilon'_r(\infty)$ is the high-frequency dielectric increment, and $\Delta\epsilon'_r(0)$ is the relaxation amplitude mentioned above.

The results of these fittings are shown in Table 3. As expected, smaller particles mean shorter times for ion to flow back and forth around the particle under the action of the alternating field. Furthermore, $\Delta\epsilon'_r(0)$ is more significant for large particles (in fact, for large κa values) because of the larger phase lags between diffusive ionic flows and the applied field giving rise to more intense displacement currents, ulti-

TABLE 3
Parameters of Eq. [4] (See Text) for the Experimental $\Delta\epsilon'_r$ Values Shown in Fig. 1

[KCl] (mM)	Latex L-23		Latex L-530	
	$\Delta\epsilon'_r(0)$	τ (10^{-8} s)	$\Delta\epsilon'_r(0)$	τ (10^{-8} s)
0.1	2400	1.96	8300	1210
0.5	2520	2.69	—	—
1	—	—	25000	2950

mately responsible for the high observed values of $\Delta\epsilon'_r(\omega \rightarrow 0)$ (2, 25).

Comparison between Experimental Data and Predictions of the Standard Model

We now explore to what extent the results just shown are justified in the light of the standard model in the DW approach (4). Qualitative arguments given above confirm that the general trends of the results (in particular, larger values of $\Delta\epsilon'_r(0)$ and smaller characteristic times for L-530 as compared to L-23) are in agreement with the standard approach. To carry out a more quantitative comparison, we have calculated the DW predictions for $\Delta\epsilon'_r(\omega)$ pertaining to experimental data in Fig. 1, using the ζ potentials of Table 2. The results are displayed in Fig. 2: as observed, the classical explanation of the polarization of electric double layers significantly underestimates the dielectric increments found experimentally in the case of L-530 particles. Unlike the large particles, the dielectric constant of L-23 suspensions is reasonably described by the DeLacey and White's theory, despite the fact that it can be expected that the surfaces of both types of lattices will be quite similar, except for the presence of surfactant on the surface of L-23, which is necessary to produce stable suspensions with such small size.

Before proceeding with plausible explanations for these different behaviors, it is worth considering the possibility that particle–particle interactions might account for deviations from the standard model, one of whose hypotheses is that the observed dielectric response is just the sum of the contributions of individual particles. This can be qualitatively verified by comparing the average particle–particle distance (r_p) with the characteristic diffusion length of counterions in the polarized double layer (L_D). According to different authors (2, 25), L_D is typically of the order of the particle radius, so the ratio r_p/L_D for our systems will be ~ 15 (sample L-23) and ~ 10 (sample L-530). This means that the suspensions considered are dilute enough for the classical models to be applicable.

We try to analyze how their low-frequency dielectric dispersion can be managed by a more complete theory based on the simultaneous contributions of both diffuse and Stern layers to the overall polarization of the particle's ionic atmosphere (13, 16).

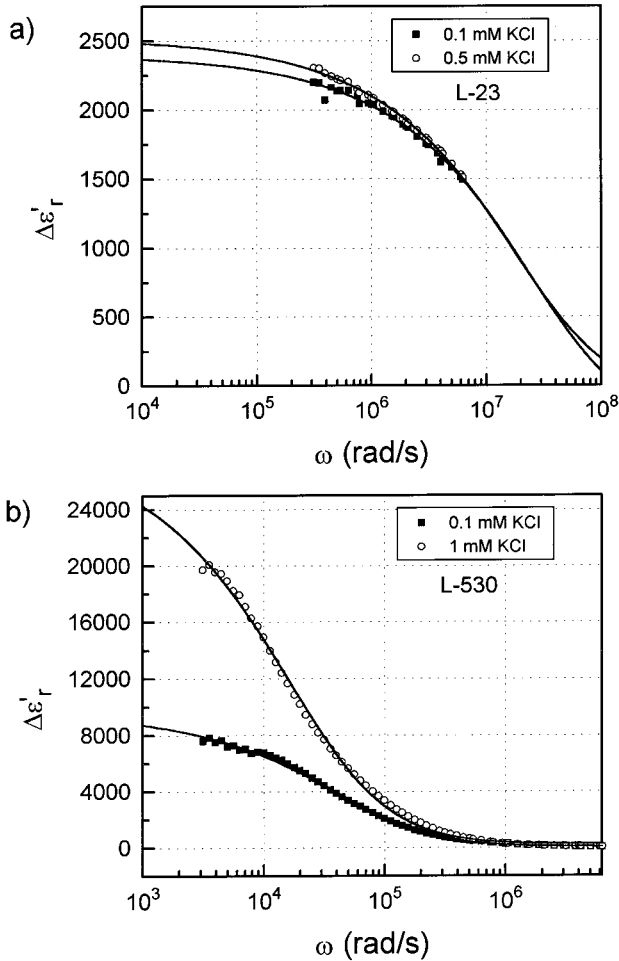


FIG. 1. Experimental values of the real part of the dielectric increment, $\Delta\epsilon'_r(\omega)$, as a function of frequency for latexes L-23 and L-530 and the concentrations of KCl indicated. Continuous lines: best fitting to Vogel and Pauly equation (cf. eq. [4]).

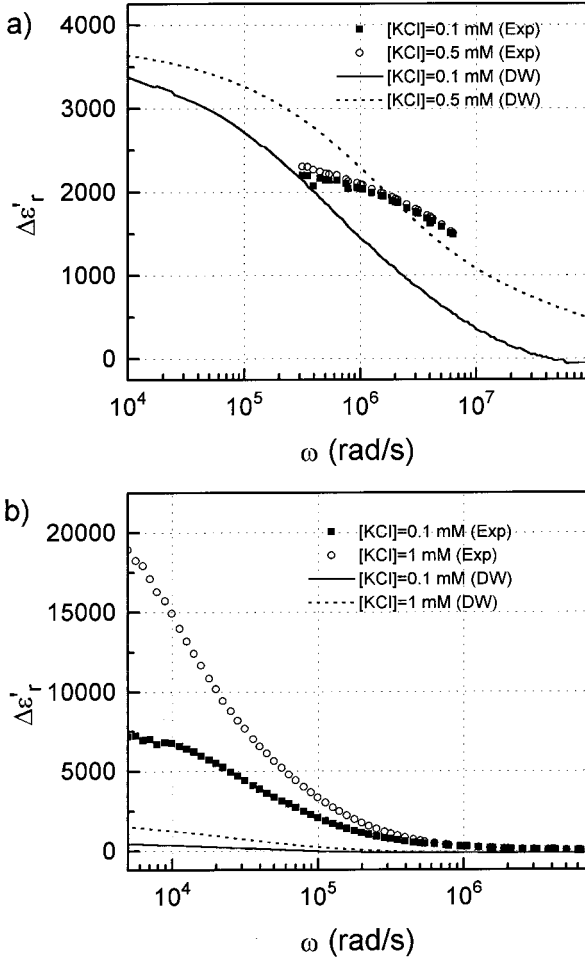


FIG. 2. Comparison between experimental (symbols) and predicted (lines) values of $\Delta\epsilon'_r$ for latexes L-23 (a) and L-530 (b) for the KCl concentrations indicated. Theoretical data were obtained with the DW model.

Comparison with DSL Model

According to Mangelsdorf and White (16), the lateral motion of Stern layer ions for a given external dc field depends on (i) the ζ potential (the shear plane is assumed to coincide with the beginning of the diffuse layer); (ii) the ratio between the drug coefficient of i th ionic species in the bulk solution (λ_i) and in the Stern layer (λ_i^s); (iii) the number N_i of sites available in the layer for adsorption of ions of type i ; (iv) the dissociation constant, K_i , of those sites; and (v) the capacity, C_2 , of the outer Stern layer. The effect of all these quantities is represented by a surface conductance parameter, δ_i , given by (16)

$$\delta_i = \frac{\frac{N_i}{K_i a} \frac{\lambda_i}{\lambda_i^s(\beta_1)} \exp\left(\frac{z_i e}{k_B T} \frac{\sigma_d}{C_2}\right)}{1 + \sum_{j=1}^N \frac{n_j^0(\infty)}{K_j} \exp\left[\frac{-z_j e}{k_B T} \left(\zeta - \frac{\sigma_d}{C_2}\right)\right]} \quad [5]$$

for each ionic species. In the case of ac fields, if the dissociation reactions in the Stern layer are considered fast enough, the reasonable hypothesis is made that both the overall surface charge σ_o and the Stern layer charge density σ_s are independent of time (26). The same can be said about the surface conductance parameters δ_i . In fact, Mangelsdorf and White (27, 28) have recently considered a general treatment in which both charge densities are allowed to change with time in the presence of an alternating field. These authors have demonstrated that the effect of time-varying surface charges on either the mobility or the dipole coefficient of the colloidal particle is negligible when compared to the DC surface conductance effect. Hence, in the present work, δ_i is considered constant, and stationary values are assumed for σ_o and σ_s , as in Refs. (16, 26).

Assuming that the supporting solution contains only two types of ions, and considering the predominant effect of counterions, we show only the results when δ_- changes (δ_- is the value of δ_i for counterions for a positively charged particle). Figure 3 demonstrates that, in agreement with results obtained by Rosen *et al.* (13), lateral motion of inner-layer ions is always favorable to increasing the dielectric constant of the suspensions. Furthermore, the effect can be extremely important, because the dielectric amplitude can increase by a factor of 10 for high (but physically reasonable) δ_- values.

To apply the model to our experimental data, the counterion parameters and ζ potential were changed until the differences between measured and predicted values of both $\Delta\epsilon'_r(\omega = 0)$ and μ_e were simultaneously minimized. Table 4 summarizes the

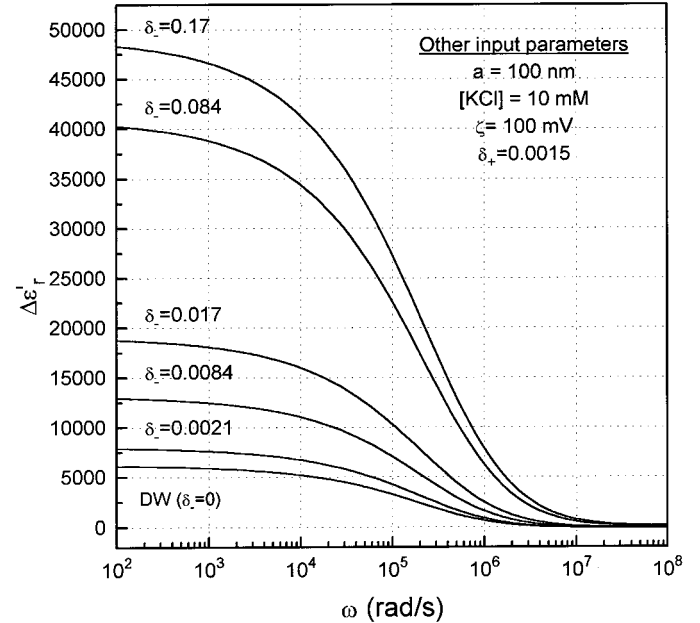


FIG. 3. Dielectric increment for suspensions of spherical particles, calculated with the Mangelsdorf-White DSL model for different values of δ_- , the surface conductance parameter of counterions. The DW predictions are included for comparison.

TABLE 4
Stern-Layer Conductance Parameters for Sample L-530
(Mangelsdorf–White Model, See Ref. (16))

	[KCl] = 0.1 mM	[KCl] = 1 mM
ζ (mV)	-130	-140
pK_+	2	2
eN_+ ($\mu\text{C}/\text{cm}^2$)	8	8
λ_+/λ_+^t	1	0.6
μ_e (predicted)		
μ_e (experimental)	1.046	0.933
$\Delta\epsilon_r'(0)$ (predicted)		
$\Delta\epsilon_r'(0)$ (experimental)	0.87	1.016

Note. Predicted values of μ_e and $\Delta\epsilon_r'(0)$ are also included. The following parameters were kept fixed: $eN = 4 \mu\text{C}/\text{cm}^2$; $\lambda_-/\lambda_-^t = 0.5$; $pK_- = 2$; Stern-layer capacitance = $130 \mu\text{F}/\text{cm}^2$.

results for samples L-530: note that ζ changes by less than 10% between $[\text{KCl}] = 0.1 \text{ mM}$ and $[\text{KCl}] = 1 \text{ mM}$ (compare with the change of $\sim 25\%$ if the standard model is used) and that the only parameter that has to be changed is the counterion drag ratio, λ_+/λ_+^t . According to these data, saturation effects cause the lateral ionic motion in the Stern layer to be somewhat hindered when the number of ions in the double layer is increased, which does not seem unreasonable. Figure 4 shows the full $\Delta\epsilon_r'(\omega)$ calculated curves together with experimental LFDD data; as observed, the agreement is considerably improved compared to classical DW predictions, except that the relaxation frequency is not well reproduced by DSL predictions.

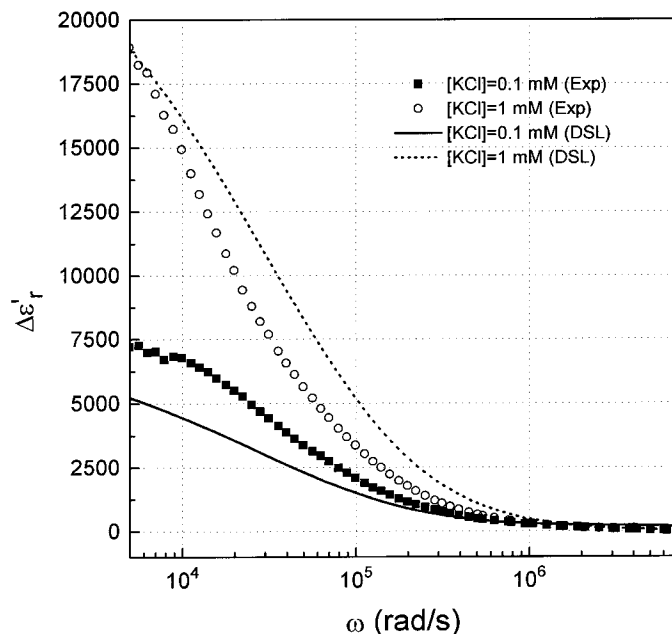


FIG. 4. Comparison between experimental (symbols) and predicted (lines) values of $\Delta\epsilon_r'$ for latex L-530 for the KCl concentrations indicated. Theoretical data were obtained with the DSL model, using the parameters in Table 4.

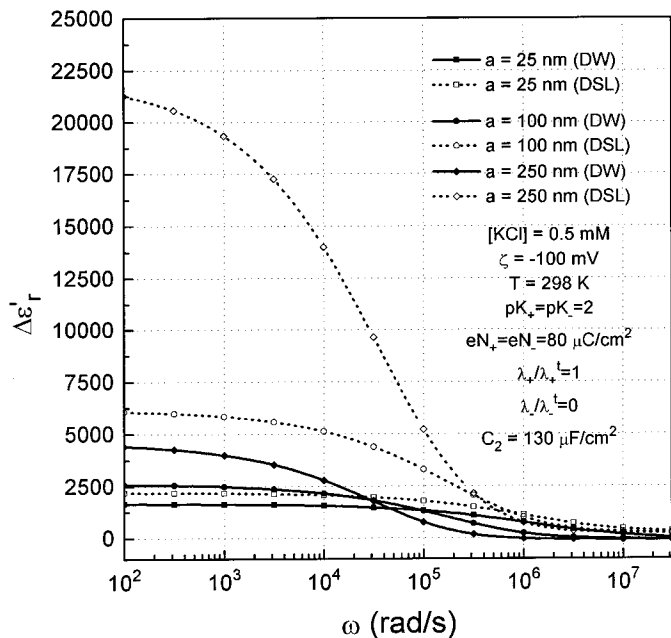


FIG. 5. Comparison between the frequency dependencies of $\Delta\epsilon_r'(\omega)$ calculated with the standard (DW) and DSL models, for different particle radii.

We repeated the same fitting process for L-23 particles. Whatever the changes introduced in the input parameters within reasonable physical ranges, we could never find significant differences between classical and DSL predictions. Only the ζ potential appeared to be the crucial quantity capable of modifying the model predictions on either the mobility or the dielectric increment. This suggests that DSL corrections are not needed if small particles (or, more correctly, small κa values) are used. Such a possibility has been further explored in Fig. 5, where the predictions of classical and DSL theories are compared for the frequency range of interest and three particle sizes. It is clearly seen that the low-frequency differences are negligible for small particles and become progressively more significant as the particle size is increased.

In Fig. 6 we show that, furthermore, if the ζ potential is reduced, the differences between DSL and DW approaches (that have been plotted as a function of particle size) increase. A qualitative explanation for the trends of variation of $(\Delta\epsilon_r'(0))_{\text{DSL}} - \Delta\epsilon_r'(0)_{\text{DW}}$ as a function of both ζ and particle radius a can be offered. Let us first consider the size effect: low particle radius (for any given ionic strength) means high double layer thickness (compared to a), that is, a very wide diffuse cloud. It is reasonable to assume that whatever the ionic conditions in the very thin Stern region (actually, it is considered two-dimensional in the DSL models), the role of the diffuse layer dipole will prevail over that of the inner layer. However, when the thickness of the former decreases (large κa), the influence of the latter will be enhanced, and the Stern layer effects may even (as shown in Figs. 5 and 6) largely exceed those generated in the diffuse layer. Concerning the ζ

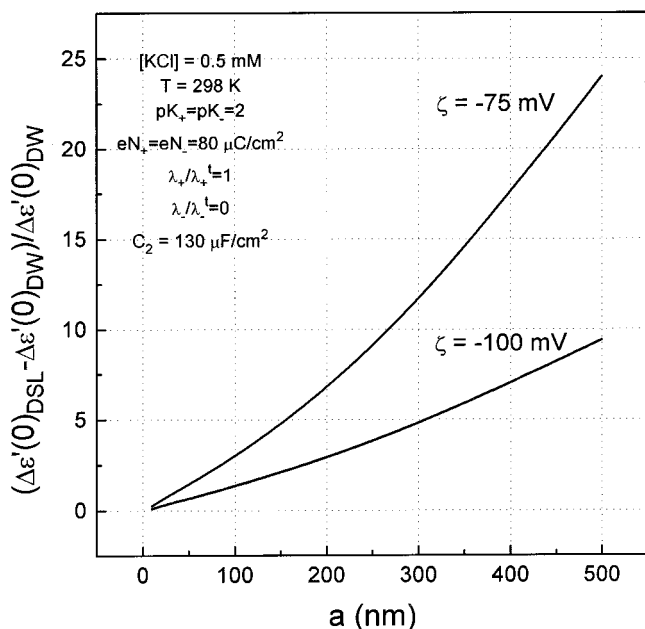


FIG. 6. Relative difference between the low-frequency dielectric increments predicted by DW and DSL theories, plotted as a function of particle radius, a , for two ζ potentials.

potential, a large ζ value will induce high counterion concentrations in the diffuse region of the double layer, thus hiding to some extent the polarization of Stern layer, less sensitive to ζ potential (N_i , λ_i^t , K_i are independent parameters, not directly linked to the value of ζ).

CONCLUSIONS

We have shown that, as previously reported for other electrokinetic phenomena, some discrepancies between theory and experiment can be explained by the existence of Stern layer surface conductance, that is, by using models (DSL) in which the inner layer ions can undergo tangential transport under the action of external fields. This is more likely for suspensions with large particles and/or high ionic strength ($\kappa a \gg 1$), since for low κa effects associated with polarization of the Stern layer are obscured by those of the thick diffuse layer. As a consequence, discrepancies found in the latter conditions are not likely to be justifiable under the assumption of Stern layer conductance alone.

ACKNOWLEDGMENTS

Thanks are due to DGICYT, Spain (Project PB94-0812-CO2), and INTAS (Project 95-IN/UA-0165) for financial support of this study.

REFERENCES

- Shilov, V. N., and Dukhin, S. S., *Kolloid Zh.* **32**, 293 (1970).
- Dukhin, S. S., and Shilov, V. N., "Dielectric Phenomena and the Double Layer in Disperse Systems and Polyelectrolytes." Wiley, New York, 1974.
- O'Brien, R. W., *Adv. Colloid Interface Sci.* **16**, 281 (1982).
- DeLacey, E. H. B., and White, L. R., *J. Chem. Soc. Faraday Trans. 2* **77**, 2007 (1981).
- Fixman, M., *J. Chem. Phys.* **78**, 1483 (1983).
- Mandel, M., and Odijk, T., *Am. Rev. Phys. Chem.* **35**, 75 (1984).
- Schwarz, G., *J. Phys. Chem.* **66**, 2636 (1962).
- Lyklema, J., Dukhin, S. S., and Shilov, V. N., *J. Electroanal. Chem.* **143**, 1 (1983).
- Razilov, I. A., and Dukhin, S. S., *Kolloid Zh.* **57**, 391 (1995); *Colloid J.* **57**, 364 (1995).
- Razilov, I. A., Pendze, G., and Dukhin, S. S., *Kolloid Zh.* **56**, 687 (1994); *Colloid J.* **56**, 612 (1994).
- Kijlstra, J., van Leeuwen, H. P., and Lyklema, J., *J. Chem. Soc. Faraday Trans.* **88**, 3441 (1992).
- Kijlstra, J., van Leeuwen, H. P., and Lyklema, J., *Langmuir* **9**, 1625 (1993).
- Rosen, L. A., Baygents, J. C., and Lyklema, J., *J. Chem. Phys.* **98**, 4183 (1993).
- Razilov, I. A., Zharkikh, N. I., and Dukhin, S. S., *Kolloid Zh.* **57**, 718 (1995); *Colloid J.* **57**, 678 (1995).
- Razilov, I. A., Zharkikh, N. I., González-Caballero, F., Delgado, A. V., and Dukhin, S. S., *Colloids Surf. A* **131**, 95 (1998).
- Mangelsdorf, C. S., and White, L. R., *J. Chem. Soc. Faraday Trans.* **86**, 2859 (1990).
- Springer, M. M., Ph.D. Thesis, Wageningen University, The Netherlands, 1979.
- Foster, K. R., and Schwan, H. P., in "Handbook of Biological Effects of Electromagnetic Fields" (C. Polk and E. Rostow, Eds.), p. 27. CRC Press, Boca Raton, FL, 1986.
- Carrique, F., Ph.D. Thesis, University of Granada, Spain, 1993.
- O'Brien, R. W., and White, L. R., *J. Chem. Soc. Faraday Trans.* **74**, 1607 (1978).
- Barchini, R., and Saville, D. A., *J. Colloid Interface Sci.* **173**, 86 (1995).
- Gittings, M. R., and Saville, D. A., *Langmuir* **11**, 798 (1995).
- Elimelech, M., and O'Melia, C. R., *Colloids Surf.* **44**, 165 (1990).
- Vogel, E., and Pauly, H., *J. Chem. Phys.* **89**, 3830 (1988).
- Lyklema, J., "Fundamentals of Interface and Colloid Science." Vol. II: "Solid-Liquid Interfaces." Academic Press, London, 1995.
- Ennis, J., and White, L. R., *J. Colloid Interface Sci.* **178**, 446 (1996).
- Mangelsdorf, C. S., and White, L. R., *J. Chem. Soc. Faraday Trans.* **94**, 2441 (1998).
- Mangelsdorf, C. S., and White, L. R., *J. Chem. Soc. Faraday Trans.* **94**, 2583 (1998).

Influence of pore fluid chemistry on the complex conductivity and induced polarization responses of Berea sandstone

David P. Lesmes and Kevin M. Frye¹

Department of Geology and Geophysics, Boston College, Chestnut Hill, Massachusetts

Abstract. The spectral induced-polarization (IP) response of rocks and soils is a complex function of pore solution chemistry, sample microgeometry, and surface chemical properties. We measure the complex conductivity and the time domain IP responses of Berea sandstone as a function of pore fluid ionic strength and pH. Complex conductivity is measured over the frequency range 10^{-3} to 10^6 Hz, and chargeability is computed using a time window of 0.16 to 1.74 s. The field IP parameters: phase, percent frequency effect, and chargeability are functions of both the surface and bulk electrical properties of the sample and are observed to decrease with increasing solution conductivity. Dividing these parameters by the sample resistivity yields normalized IP parameters (quadrature conductivity, metal factor, normalized chargeability) that are proportional to the imaginary component of the complex surface conductivity. Normalized IP parameters increase with ionic strength up to concentrations of 10^{-1} M NaCl and show a reduced response at pH 3, the point of zero charge for quartz-dominated systems. For concentrations $> 10^{-1}$ M NaCl, the normalized parameters decrease with increasing concentration. This decrease in surface polarization may indicate a decrease in the effective mobility of polarizing charges at high solution concentration. Our data indicate that normalized IP parameters are directly related to the physiochemical parameters that control the surface conductivity responses of rocks and soils. Normalization of IP measurements in environmental investigations should increase the effectiveness of IP surveys, especially in high-conductivity environments.

1. Introduction

Traditional “dc” resistivity surveys are primarily sensitive to the pore solution conductivity, the formation porosity, and the saturation. Resistivity surveys are used in environmental applications to map the distribution of contaminants and to characterize lithologic variations. However, resistivity methods often are not sensitive enough to detect small concentrations of toxic contaminants, and they are not very effective for predicting the hydraulic properties of the formation [Olhoeft, 1985; Ward, 1990].

Induced polarization (IP) methods are more sensitive to the surface chemical properties of rocks and soils. IP methods are often an effective means of mapping subsurface contamination in cases where dc resistivity methods are ineffective [Sadowski, 1988; Cahyna *et al.*, 1990; Börner *et al.*, 1993]. Furthermore, IP methods can also be used to predict the hydraulic properties of subsurface formations [Börner *et al.*, 1996; Sturrock, 1999; Sturrock *et al.*, 1999]. Further advances in the application of IP methods to the characterization and quantification of geochemical and hydrological properties will rely on a better understanding of the physiochemical processes which control the IP response of rocks and soils [e.g., Ward *et al.*, 1995].

The IP response of rocks and soils is controlled by electrochemical polarization mechanisms that are a function of the surface chemical and the microgeometrical properties of the sample. In earth materials containing water, ions from the saturating solution are attracted to the mineral surface, which usually has negative net surface charge. These ions form an electric double layer (EDL) that consists of a fixed layer of charge that is chemically adsorbed to the grain surface and a diffuse layer of charge that extends into the pore space. When an electric field is applied to the material, the ions in the EDL migrate tangentially to the grain surfaces and the grains become polarized. The polarization of the EDL leads to a frequency-dependent complex surface conductivity. This surface conductivity is a function of the surface charge density, the surface ionic mobility, and the grain size distribution [Lima and Sharma, 1992; Lesmes and Morgan, 2001]. The complex surface conductivity is usually assumed to add in parallel with the bulk electrical properties of the sample.

Commonly measured field IP parameters, such as phase, percent frequency effect, and chargeability, are dependent upon both the surface and bulk electrical properties of the material [Marshall and Madden, 1959; Collett and Katsube, 1973; Van Voorhis *et al.*, 1973; Vinegar and Waxman, 1984]. Dividing the field IP parameters by the formation resistivity, or multiplication by the formation conductivity, yields the following normalized IP parameters: quadrature conductivity, metal factor, and a normalized chargeability called “specific capacity” [Marshall and Madden, 1959; Keller, 1959]. These normalized IP parameters are more sensitive to the surface chemical properties of the sample and therefore are more

¹Now at Department of Earth, Atmospheric, and Planetary Sciences, Massachusetts Institute of Technology, Cambridge.

effective for characterizing geochemical and lithological variability.

In this investigation we measure the time domain and broadband frequency domain IP responses of Berea sandstone as a function of the bulk solution ionic strength and pH. We verify the equivalence of the time domain and frequency domain data and argue that IP parameter normalization is necessary to isolate the surface conductivity effects, especially at high solution conductivity. We find that pH effects are secondary to ionic strength effects. Surface conduction increases with ionic strength to a maximum value at $10^{-1} M$ NaCl, then decreases at higher ionic strengths. We attribute this maximum in surface conductivity to a trade-off between increasing surface charge density and decreasing surface ionic mobility with increasing ionic strength.

2. Complex Conductivity

The conduction current density $\mathbf{J}_C$ and the electric displacement $\mathbf{D}$ are related to the applied electric field $\mathbf{E}$ by the constitutive equations

$$\mathbf{J}_C = \sigma \mathbf{E} \quad (1)$$

$$\mathbf{D} = \kappa \epsilon_0 \mathbf{E}, \quad (2)$$

where ϵ_0 is the permittivity of free space, κ is relative permittivity, or dielectric constant, and σ is electrical conductivity. For time harmonic fields the displacement current density $\mathbf{J}_D$ may be expressed as

$$\mathbf{J}_D = \frac{\partial \mathbf{D}}{\partial t} = i\omega \kappa' \epsilon_0 \mathbf{E}, \quad (3)$$

where κ' is the real component of complex relative permittivity, ω is the angular frequency of the applied electric field, and i is $\sqrt{-1}$. The total current density $\mathbf{J}^* = \mathbf{J}_C + \mathbf{J}_D$ can be expressed as

$$\mathbf{J}^* = (\sigma' + i\omega \kappa' \epsilon_0) \mathbf{E} = (\sigma' + i\sigma'') \mathbf{E}, \quad (4)$$

where σ^* is the complex conductivity and σ' and σ'' denote the real and imaginary (or in-phase and quadrature) components of conductivity, respectively. The complex conductivity can also be expressed in polar form as a magnitude ($|\sigma^*|$) and a phase (θ):

$$\sigma^* = |\sigma^*| \exp(i\theta). \quad (5)$$

Alternatively, the intrinsic electrical properties can be represented by a complex resistivity (ρ^*) or a complex permittivity (ϵ^*):

$$\sigma^* = \frac{1}{\rho^*} = i\omega \epsilon^*. \quad (6)$$

Generally, the complex electrical properties of earth materials are frequency-dependent.

3. Complex Conductivity Models

Theoretically based effective medium theories can be used to model the complex electrical properties of porous media. These models are formulated in terms of the intrinsic electrical properties of the mixture and the shape of the embedded particles. Two-phase mixing models consisting of dry mineral grains and pore fluid (e.g., quartz and brine) can be used to predict the low-frequency conductivity and the high-frequency permittivity of rocks and soils [Sen et al.,

1981; Mendelson and Cohen, 1982]. However, the migration and polarization of ions within the EDL control the frequency dependence of the complex conductivity response. Therefore three-phase mixing models that account for the electrical properties of the surface phase are required to predict the large dispersions that are observed in the permittivity and conductivity responses of water saturated samples [Bussian, 1983; Knight and Endres, 1990; Samstag and Morgan, 1991].

A more simplistic alternative is to model the bulk conductivity and surface conductivity mechanisms as parallel conduction paths [Wyllie and Southwick, 1954; Marshall and Madden, 1959; Waxman and Smits, 1968; Clavier et al., 1984; Vinegar and Waxman, 1984]. Using the parallel equivalent circuit shown in Figure 1a, the complex conductivity response of the sample is given by

$$\sigma^* = (\sigma_{\text{bulk}} + i\omega \kappa_{\infty} \epsilon_0) + [\sigma'_{\text{surf}}(\omega) + i\sigma''_{\text{surf}}(\omega)]. \quad (7)$$

In this model, σ_{bulk} and κ_{∞} represent the low-frequency conductivity and the high-frequency dielectric constant of the bulk rock sample, and $\sigma'_{\text{surf}}(\omega)$ and $\sigma''_{\text{surf}}(\omega)$ are real and

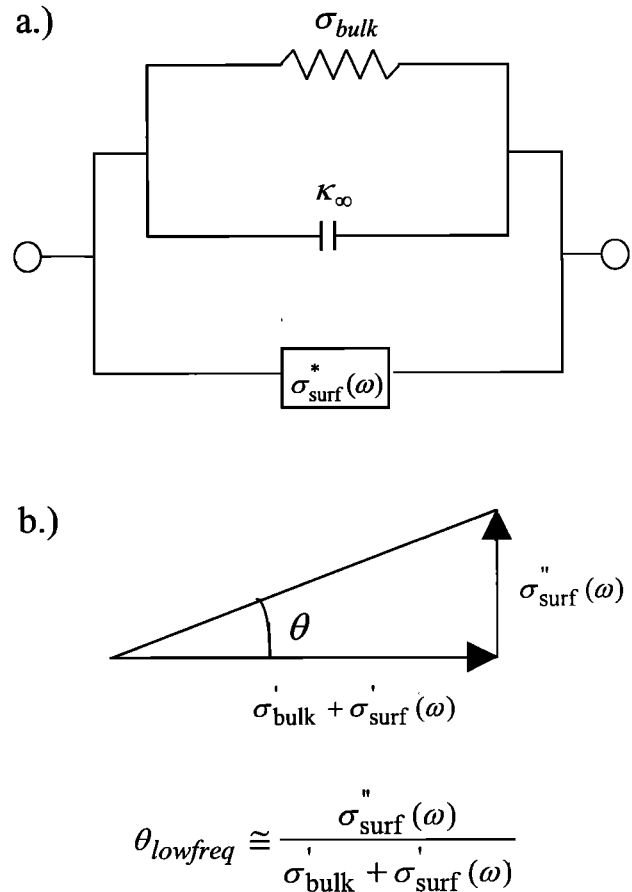


Figure 1. (a) Equivalent circuit model for the electrical response of rocks and soils. Electrolytic conduction in the bulk pore fluid is assumed to be purely resistive, the high-frequency dielectric constant κ_{∞} represents the capacitance of the system at high frequency, and the complex conductivity associated with " $\sigma'_{\text{surf}}(\omega)$ " accounts for low-frequency IP effects. (b) The relationship between the measured low-frequency phase (θ) and the real and imaginary components of the complex conductivity [after Vinegar and Waxman, 1984].

quadrature surface conductivity terms. Combining the real and the imaginary components in (7) gives

$$\sigma^* = [\sigma_{\text{bulk}} + \sigma'_{\text{surf}}(\omega)] + i[\omega\kappa_{\infty}\epsilon_0 + \sigma''_{\text{surf}}(\omega)]. \quad (8)$$

At low frequencies, where IP phenomena are typically measured in the field, $\omega\kappa_{\infty}\epsilon_0 \ll \sigma''_{\text{surf}}(\omega)$ and the low-frequency complex conductivity response of the sample is given by

$$\sigma^* = [\sigma_{\text{bulk}} + \sigma'_{\text{surf}}(\omega)] + i\sigma''_{\text{surf}}(\omega). \quad (9)$$

The quadrature conductivity is only a function of the surface conductivity, whereas the in-phase conductivity is a function of both the bulk and surface conductivity mechanisms. The vector diagram in Figure 1b shows how the sample phase is related to the real and imaginary components of the complex conductivity.

The bulk rock properties σ_{bulk} and κ_{∞} are independent of frequency. These parameters can be derived from the Hanai-Bruggeman (HB) effective medium theory [Sen et al., 1981]. The low-frequency conductivity response of the HB model is given by

$$\sigma_{\text{bulk}} = \sigma_w \phi^m, \quad (10)$$

where σ_w is the solution conductivity, ϕ is the porosity, and m is the cementation index which is a function of the effective grain shape. This relationship is called Archie's law as it was first derived empirically by Archie [1942]. The high-frequency dielectric response of the HB model is given by

$$\kappa_{\infty} = \kappa_w \phi^m \left(\frac{1 - \kappa_g / \kappa_w}{1 - \kappa_g / \kappa_{\infty}} \right)^m, \quad (11)$$

where κ_w and κ_g are the dielectric constants of the pore solution and the mineral grains, respectively.

The complex surface conductivity term $\sigma^*_{\text{surf}}(\omega)$ is frequency-dependent, and it arises from the electrochemical polarization of the water-wet rock matrix. At very low frequencies where the permittivity response approaches a constant value (κ_{static} in Figure 2) the imaginary component of surface conductivity goes to zero, and the real component of surface conductivity approaches a constant value $\sigma_{\text{surf}}(\text{dc})$. The dc component of surface conductivity is given by

$$\lim_{\omega \rightarrow 0} \sigma'_{\text{surf}}(\omega) = \sigma_{\text{surf}}(\text{dc}) = \frac{e\mu_s \Sigma_0 S_0}{f}, \quad (12)$$

where e is the electronic charge, μ_s is the effective surface ionic mobility, Σ_0 the mineral surface charge density, S_0 is a weighted surface-to-volume ratio, and f is a geometric factor that characterizes the "tortuosity" of the grain/pore interface [e.g., Johnson et al., 1986; Pride, 1994; Revil and Glover, 1998]. As frequency increases, the real and imaginary components of the surface conductivity also increase. The complex surface conductivity response can be represented by the product of the dc surface conductivity and a spectral response function $\Im[\omega, g(r), \mu_s]$:

$$\sigma^*_{\text{surf}}(\omega) = \frac{e\mu_s \Sigma_0 S_0}{f} \Im^*[\omega, g(r), \mu_s]. \quad (13)$$

$\Im[\omega, g(r), \mu_s]$ is obtained by convolving the response of the electrochemical polarization of the fixed and diffuse parts of the EDL for a single grain/pore of radius r with the distribution of grain/pore sizes $g(r)$ [Lesmes, 1993; Lesmes and Morgan, 2001; Lesmes et al., 2000]. The magnitude of the complex conductivity response is primarily determined by the product of S_0 , Σ_0 , and μ_s . The frequency dependence of

the complex conductivity response is primarily a function of the grain/pore size distribution [Sturrock, 1999; Sturrock et al., 1999; Lesmes et al., 2000].

3.1. Field IP Parameters

The IP phenomenon measured in the field is essentially a low-frequency dispersion in the complex conductivity response. In the time domain this dispersion manifests itself as a transient secondary voltage that exists in the material after the interruption of an externally applied dc current. The dispersion in the electrical response is controlled by physiochemical polarization mechanisms that result in a broad distribution of relaxation times. Full characterization of the electrical properties of a sample requires that the complex

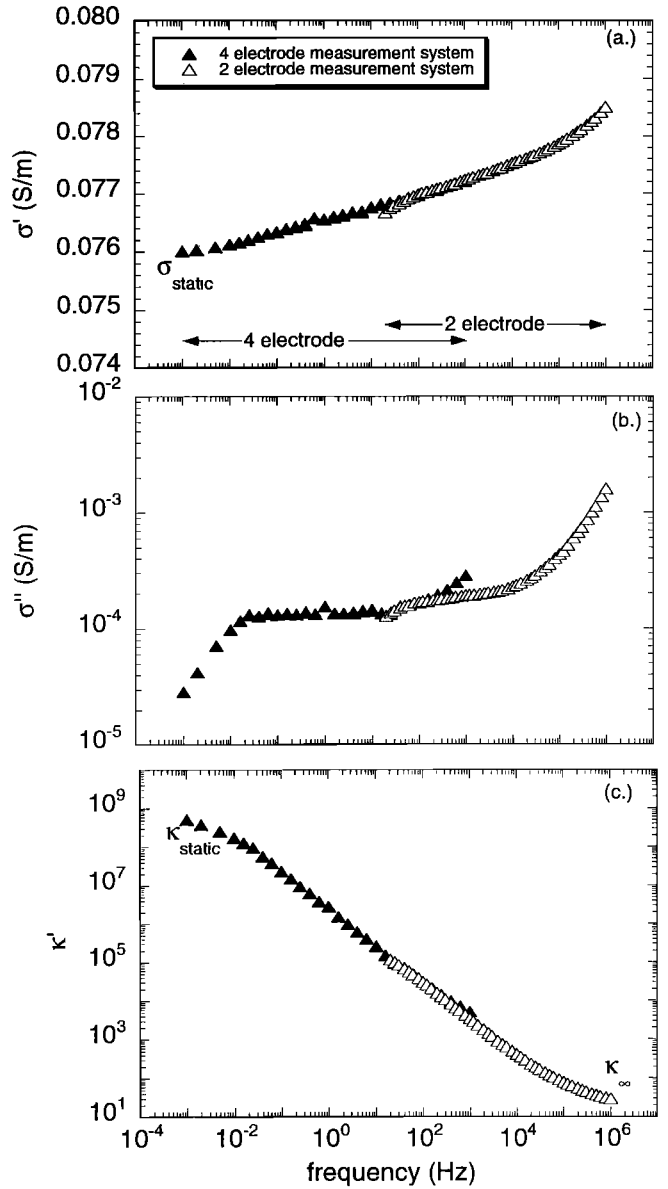


Figure 2. Real component of (a) formation conductivity, (b) quadrature conductivity, and (c) dielectric constant of Berea sandstone saturated with pH 8, 10^{-1} M NaCl solution. Low-frequency limits of dielectric constant and conductivity are labeled κ_{static} and σ_{static} . High-frequency limit of the dielectric constant is labeled κ_{∞} .

conductivity response be measured over a wide range of frequencies so that the entire distribution of relaxation times can be captured [Lesmes and Morgan, 2001]. The equivalent characterization in the time domain requires that the voltage decay be measured for a long duration with a very fast sampling rate. It is possible to make these broadband electrical impedance measurements in the laboratory [e.g., Olhoeft, 1986; Lesmes, 1993]; however, practical considerations and technical limitations such as electromagnetic (EM) coupling make it essentially impossible to capture the full relaxation time distribution in the field [Pelton et al., 1978].

Field IP surveys can be conducted using complex resistivity (CR), frequency domain IP, or time domain IP measurement systems. All of these systems are operationally similar to the dc resistivity method. However, in addition to measuring the conductive properties of the media, IP systems also measure, either directly or indirectly, the surface chemical properties of the media.

CR instruments measure the resistivity amplitude and phase. CR measurements made as a function of frequency are called spectral induced polarization (SIP). SIP surveys characterize the dispersions in both the conductive and capacitive properties of the subsurface. The CR phase for earth materials is typically small, and it is given by

$$\theta = \tan^{-1}(\sigma''/\sigma') \cong \sigma''/\sigma'. \quad (14)$$

The CR phase is essentially the ratio of the capacitive to conductive properties of a material.

Frequency domain IP methods measure the conductivity at two or more frequencies. A common frequency domain parameter is the percent frequency effect (PFE):

$$\text{PFE} = 100 \frac{\sigma(\omega_1) - \sigma(\omega_0)}{\sigma(\omega_0)}. \quad (15)$$

PFE is the relative dispersion in the conductivity response measured between a low frequency (ω_0) and a higher frequency (ω_1). The PFE values reported in this paper were calculated from conductivity measurements made at 1 and 10 Hz.

Time domain surveys measure the response of a sample to an applied direct current that is switched on and off. The apparent resistivity (ρ_a) is computed from the direct current measurements. Application of the current polarizes the material, and a decay voltage is observed when the current is removed. The voltage decay curve is used to quantify the IP response. One measure of the time domain IP response is the integral chargeability (M), which is defined as [e.g., Ward, 1990]:

$$M = \frac{1}{V_{\max}(t_1 - t_0)} \int_{t_0}^{t_1} V(t) dt. \quad (16)$$

$V(t)$ is the potential difference measured at a time t after the current is shut off, $V_{\max}$ is the maximum potential difference measured during current transmission, and t_0 and t_1 define the time window over which the voltage decay curve is integrated. We sampled the voltage decay curve from 0.16 to 1.74 s after the current was switched off. This time window represents a bandwidth similar to that of our frequency domain parameter, PFE, and is used as a default setting in the IRIS[®] ELREC T field IP system.

The proportionality between the CR phase, PFE, and M is both theoretically and experimentally well established [Seigel, 1959; Marshall and Madden, 1959; Madden and Cantwell,

1967; Collett and Katsube, 1973; Van Voorhis et al., 1973; Shuey and Johnson, 1973; Bertin and Loeb, 1976; Wait, 1984; Vinegar and Waxman, 1984]. These field IP parameters effectively measure the ratio of the capacitive to conductive properties of the material at low frequencies. The low-frequency capacitive component is primarily controlled by electrochemical polarization mechanisms, whereas the low-frequency conductive component is primarily controlled by electrolytic conduction in the bulk pore solution. Therefore the field IP parameters are sensitive to the ratio of surface conductivity to bulk conductivity effects.

3.2. Normalized IP Parameters

To isolate the surface conductivity effects, the field IP parameters can be weighted by the measured formation conductivity. It is clear from (14) that multiplication of the sample phase by the measured conductivity yields the quadrature conductivity (or the imaginary component of the complex conductivity). Marshall and Madden [1959] observed that PFE was negligible in materials saturated with high-conductivity pore fluids. They suggested a frequency domain IP parameter normalized for pore fluid resistivity, the metal factor (MF):

$$\text{MF} = 2\pi 10^3 \text{PFE} \sigma(\omega_0), \quad (17)$$

where the conductivity was to be measured in units of mhos per foot (S/foot). The MF is proportional to the conductivity dispersion measured between two frequencies:

$$\text{MF} = a[\sigma(\omega_1) - \sigma(\omega_0)], \quad (18)$$

where $a=2\pi 10^5$ if conductivity is measured in units of S/foot. However, if conductivity is measured in units of S/m, then $a=2(0.305)\pi 10^5$. To avoid this outdated and complicated nomenclature, we report our "metal factor" measurements as MF/a , which is equal to the conductivity dispersion measured in units of S/m. Keller [1959] proposed a similar normalization for the time domain chargeability parameter:

$$\text{MN} = M\sigma' \cong \frac{M}{\rho_a}. \quad (19)$$

Keller called this normalized chargeability the "specific capacity." We also report our normalized chargeability measurements in units of S/m. The normalized IP parameters help to isolate information about surface chemical processes, especially in samples with high pore fluid conductivity.

4. Experimental Procedure

Complex conductivity of Berea sandstone was measured from 10^{-3} to 10^6 Hz. NaCl brines were prepared with deionized, degassed water. The saturating solution was introduced to an evacuated chamber containing dried Berea sandstone in accordance with Society of Core Analysts (SCA) guidelines [Lerner et al., 1990; Maerefat et al., 1990; Worthington et al., 1990]. Low-frequency measurements (10^{-3} to 10^3 Hz) were performed in a four-electrode sample holder, and high-frequency measurements (20 Hz to 10^6 Hz) were performed in a two-electrode sample holder. The two measurement systems provide sufficient overlap to characterize the entire range of frequencies.

4.1. Description of Apparatus

Two electrode measurements were performed with platinum-black electrodes and a Hewlett-Packard 4284A LCR impedance meter. Open and short circuit calibrations were

made with the impedance meter to compensate for stray impedance effects associated with the sample cell and the connecting wires. The instrument makes a logarithmic sweep through the frequency range and records values for impedance and phase.

When the electrodes are in contact with the electrolytic solution, an additional impedance is placed in series with the sample, such that

$$z_{\text{total}}^* = z_s^* + z_e^*, \quad (20)$$

where z_{total}^* is the total measured complex impedance, z_s^* is the complex impedance of the sample under test, and z_e^* is the complex impedance due to electrode polarization. The polarization of electrodes exposed to brine is similar to the polarization mechanisms that control the IP response of earth materials containing metallic minerals. To fully account for this additional electrode impedance, at least three lengths of core are measured for one complex conductivity experiment [Cole and Curtis, 1937; Judy and Eberle, 1969; Lesmes, 1993]. The real and imaginary components of the complex impedance measurements made at a given frequency can then be plotted versus the core length divided by core cross-sectional area (Figure 3). The slopes of these plots are the real and imaginary components of the complex resistivity and the intercepts are the real and imaginary components of the electrode impedance:

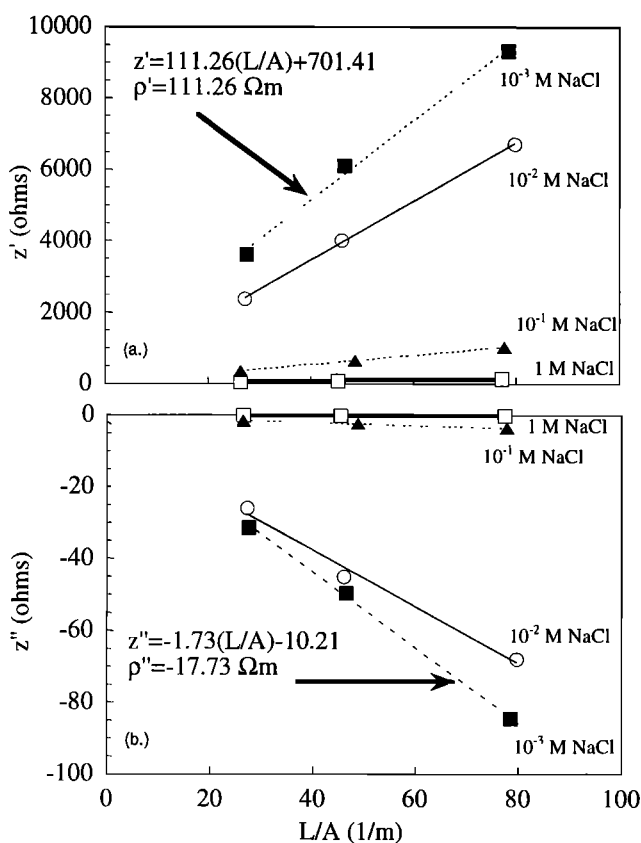


Figure 3. The length variation method for high-frequency electrical measurements yields more accurate estimates of the (a) real and (b) imaginary components of resistivity than measurement of a single length. The slope of the linear curve fit is taken as resistivity, while the intercept can be interpreted as chemistry-dependent electrode impedance. Data are from cores saturated with pH 8 solutions with NaCl concentrations at 10 kHz.

$$z_{\text{total}}^* = \rho^* \frac{L}{A} + z_e^*. \quad (21)$$

This length variation method was used to compute the complex resistivity response of the sample for every frequency measured. A linear least squares fit to the data points is performed, and the standard error of the slope provides an estimate of the imprecision in the measurement. At frequencies below 100 Hz the two-electrode measurements became inaccurate because the electrode impedance, which is in series with the sample, becomes very large.

The Boston College low-frequency impedance measurement system is similar to the four-electrode systems used by Vinegar and Waxman [1984], Olhoeft [1985], Morgan et al. [1986], and Lesmes [1993]. A waveform generator sends a sinusoidal current through the sample and a reference resistor, which are connected in series. Voltage waveforms are measured across the sample and the reference resistor. The number of cycles collected ranges from 1 at 10^{-3} Hz up to 900 cycles at 1 kHz. Sinusoids are fit to the voltage waveforms using a least squares method. The ratio of the amplitudes and the time delay between the waveforms are used to calculate the impedance and phase of the sample. The measurement system performs a logarithmic sweep through the frequency range 5 times. This system was built using a National Instruments data acquisition board and LabVIEW controlling software.

The separation of current and potential electrodes and the high input impedance (1 G Ω) of the analog-to-digital (A/D) converters minimize error due to the polarization of the potential electrodes, allowing the low-frequency measurement system to go to dc [Vinegar and Waxman, 1984]. However, above 10 Hz, capacitive coupling of the cables introduces phase error to the system [Olhoeft, 1985]. This cable capacitance is assumed to be in parallel with the reference resistor and the sample. By performing calibrations with reference resistors and standard solutions it is possible to determine the cable capacitance [Lesmes, 1993; Frye, 1997] and compensate for the resulting phase errors. However, at higher frequencies the phase correction for cable capacitance becomes much larger than the true phase, and the measurements are inaccurate. The high-frequency cutoff for four electrode systems typically occurs at ~ 100 to 10 kHz.

Time domain IP laboratory measurements are also performed with the four-electrode sample holder. A digital-to-analog converter in the data acquisition board produces a square wave in series with a reference resistor and the test material. After 2 s of current transmission, the voltage waveform goes to zero, the circuit is opened, and the polarization dissipates within the sample. The A/D converters sample the voltages across the core and the reference resistor. Decay curves are captured for both positive and negative pulses, and integral chargeability is calculated from 0.16 to 1.74 s after current cutoff. The values from the positive and negative decay curves are averaged to eliminate error due to any residual polarization from the first pulse. Apparent resistivity is calculated from data collected during dc current transmission.

4.2. Characteristics of Berea Sandstone

The Berea sandstone formation is an oil and gas reservoir within the Appalachian Basin. Berea sandstone has fairly uniform porosity and permeability, and it is easily accessible because it is a quarried building stone in South Amherst,

Table 1. Petrophysical Properties and Associated Uncertainties of Berea Sandstone Used in This Study^a

ϕ	k , mdarcy	F	m	Surface Area, m ² /g	S_{por} , cm ⁻¹
0.18	228	15.9	1.61	0.737	8.30×10^4
± 0.005	± 18	± 0.4	± 0.05	± 0.005	$\pm 10^2$

^aFrom Frye [1997] and Sturrock [1999].

Ohio. Therefore many rock physics experiments have been conducted using Berea sandstone and many of its physical and chemical properties are well characterized. The petrophysical properties of our Berea sandstone samples are listed in Table 1. The connected porosity ϕ is measured using Archimedes method, the permeability k is measured in a steady state flow test, and the specific surface area is measured using BET nitrogen adsorption [Sturrock, 1999]. Measurement of the “intrinsic” formation factor F and the cementation index m are described below. The mineralogical content of a Berea sandstone sample with similar petrophysical properties as our sample is listed in Table 2.

The formation factor (FF) is the ratio of the brine conductivity to the conductivity of the sample [Archie, 1942]:

$$FF = \frac{\sigma_w}{\sigma} \quad (22)$$

Surface conductivity effects, which are not accounted for in Archie's law (10), cause FF to increase with increasing solution conductivity. Surface conductivity effects are assumed to be negligible when FF approaches a relatively constant value at high salinity. In formations in which surface conductivity is negligible (i.e., high salinity and/or low surface area), Archie's law holds, and the formation factor is given by

$$F = \phi^{-m} \quad (23)$$

F is the intrinsic formation factor, which is characteristic of the grain shape and pore space topology (i.e., tortuosity). The cementation index m computed from F ranges from 1 to 3; for well-cemented sandstones, $m \approx 2$ [Jackson *et al.*, 1978; Schon, 1996]. For our Berea samples we obtained $m=1.61$, which was computed from F measured at 10 kHz when the sample is saturated with 1 M NaCl (pH 8) pore solution. The formation factor response FF plotted throughout the paper is simply the ratio of the brine conductivity to the sample conductivity as defined by (22).

4.3. Accuracy and Precision

The imprecision (95% confidence interval) of the low-frequency measurement system, based on resistor tests, is

Table 2. Mineralogical Composition of Berea Sandstone^a

Mineral	Weight Percent
Quartz	68
Chert	1
Feldspar	7
Carbonates	1
Clay (kaolinite)	3
Lithics	8
Unidentified	12

^aFrom Knight and Nur [1987]. Sample Berea 200 with $\phi=0.209$, $k=372$ mdarcy, and $S_{\text{por}}=8.40 \times 10^4$ cm⁻¹.

± 0.4 mrad phase and $\pm 0.1\%$ impedance amplitude. Although low-frequency electrical measurements are performed immediately after saturation of the cores, the time necessary to perform five logarithmic sweeps through the low-frequency range is ~ 150 min. Thus the largest source of error away from the high-frequency cutoff of the system is drift due to the quasi-equilibrated state of the rock/pore fluid system. Temperature variations (through Ohmic heating and change in room temperature) also influence the measured impedance and phase. These effects account for imprecision up to ± 1.5 mrad phase and $\pm 10\%$ impedance amplitude when measuring saturated cores. The resulting imprecision for the real component of conductivity is $< \pm 10\%$. However, the phase drift is coupled with the impedance drift (Figure 4), and the imaginary component of conductivity drifts $< \pm 6\%$. The accuracy of the system was tested by performing calibrating measurements of the electrical properties of water (Figure 5).

The precision of the high-frequency measurement system is improved by the length variation method. The real and imaginary components of conductivity have scatter of $< \pm 3\%$ and $\pm 5\%$, respectively. Mismatch between the high- and low-frequency measurement systems at 100 Hz is $< \pm 10\%$ and assumed to be due to the measurement drift occurring during low-frequency measurements.

5. Results

5.1. Complex Conductivity

Figure 2 is a plot of the complex conductivity response of Berea sandstone saturated with 0.1 M NaCl and pH 8. There is a small but steady increase in the in-phase conductivity (σ') with increasing frequency from the low-frequency conductivity limit (σ_{static}) at $\sim 10^{-3}$ Hz toward the high-frequency conductivity limit, which is not reached at 1 MHz. The dielectric response steadily decreases with increasing frequency from the low-frequency dielectric limit (κ_{static}) at $\sim 10^{-3}$ Hz toward κ_{∞} , which is approached but not fully reached at 1 MHz. The quadrature conductivity is equal to the dielectric response normalized by the measurement frequency. Between 10^{-2} and 10^4 Hz, there is a small but

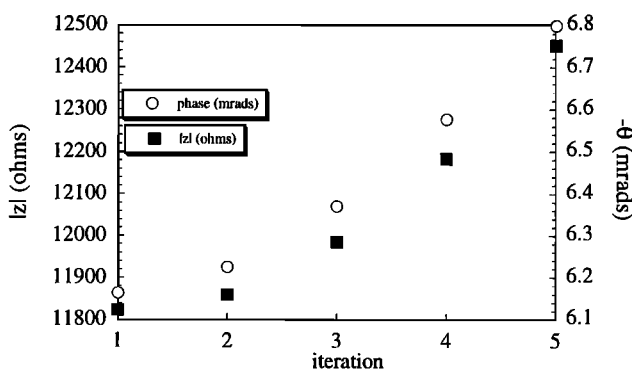


Figure 4. The modulus of the impedance ($|z|$) and phase angle ($-\theta$) at 10^{-2} Hz versus iteration demonstrates the drift which occurs when measuring electrical properties of Berea sandstone cores in the low-frequency measurement system. Note that the phase angle varies systematically with impedance. The data shown are for Berea sandstone saturated with 10^{-2} M NaCl, pH 8 solution.

steady increase in the quadrature conductivity with increasing frequency. For frequencies $<10^{-2}$ Hz the quadrature conductivity rapidly goes to zero. Presumably, this frequency is related to the relaxation time associated with the largest polarizing grains or aggregates of grains [Klein and Sill, 1982; Lesmes and Morgan, 2001]. For frequencies >10 kHz the quadrature conductivity rapidly increases as the displacement currents begin to be controlled by the high-frequency dielectric properties of the sample.

Figure 6 shows the effects of NaCl solution concentration on the broadband complex conductivity response of Berea sandstone. In these experiments the solution concentration was varied from 10^{-3} M to 1 M NaCl and the pH was held constant (pH 8). All of the formation factors decrease slightly with increasing frequency due to the dispersion in the measured conductivity. The formation factor ranges from ~ 3 to 16 and increases with increasing solution concentration, while the quadrature conductivity and dielectric constant increase with ionic strength to maximum values at 10^{-1} M NaCl and then decrease.

Figure 7 shows the influence of pH on the complex conductivity response of Berea sandstone. In this experiment the Berea sandstone cores were saturated with a 10^{-2} M NaCl solutions with pH values of ~ 3 , 6, 8, and 10. The formation factor varies from 8.6 to 14.6. The imaginary component of conductivity and the dielectric constant had similar

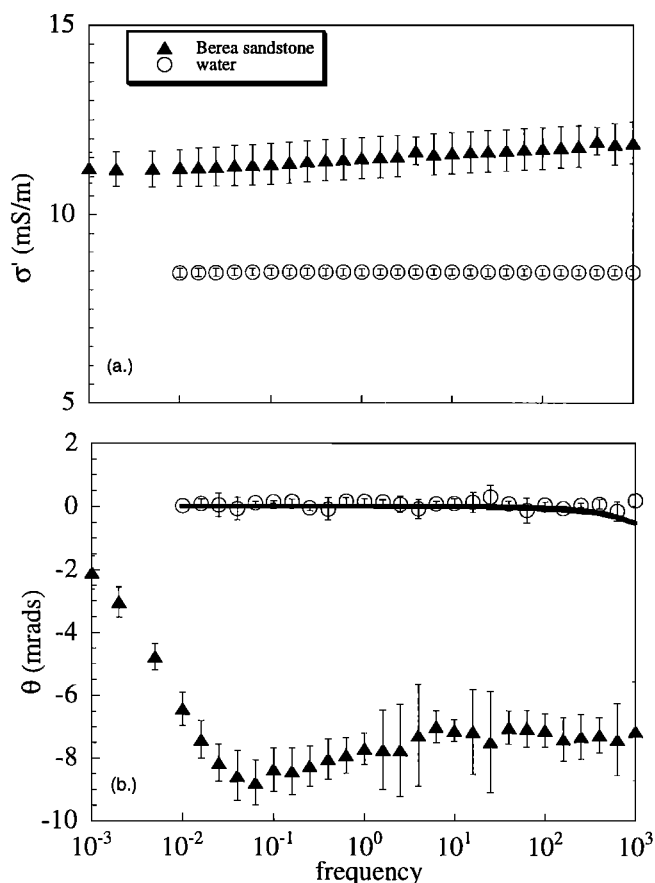


Figure 5. (a) Conductivity and (b) phase angle ($-\theta$) plotted versus frequency for water and Berea sandstone saturated with 10^{-2} M NaCl, pH 8 brine. Conductivity of water is independent of frequency, and the measured phase angle of water fits the expected response (solid line) well.

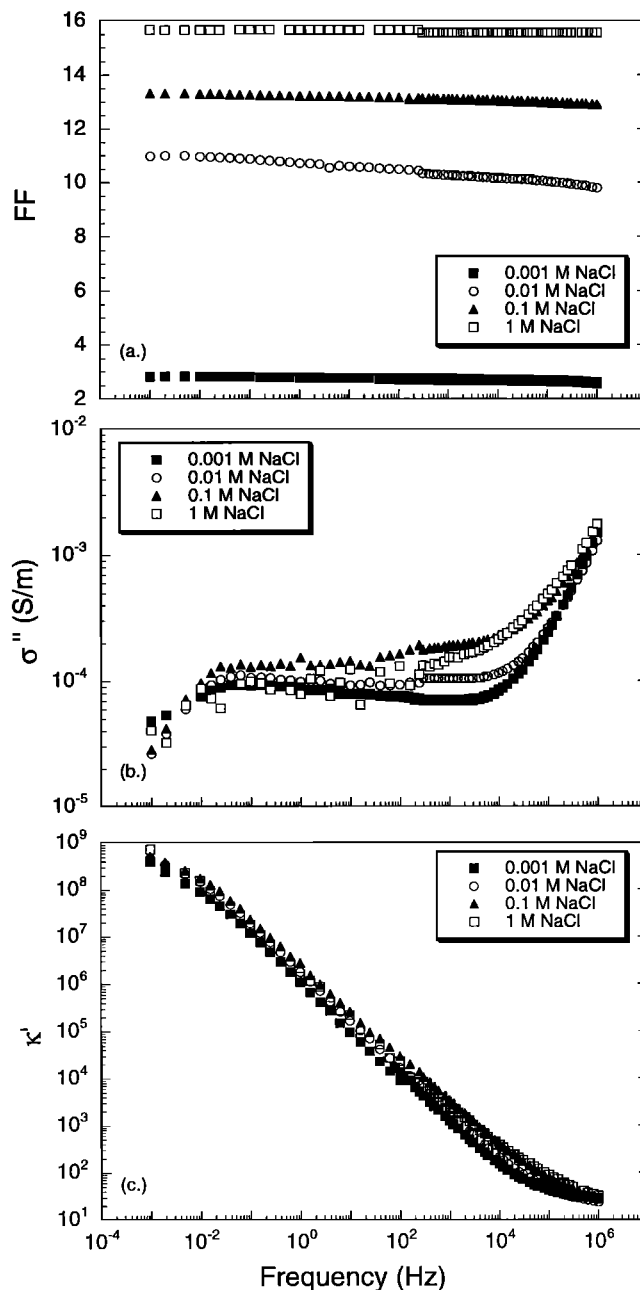


Figure 6. (a) Formation factor, (b) imaginary component of conductivity, and (c) dielectric constant plotted versus frequency for Berea sandstone saturated with pH 8 solutions of varying ionic strength (10^{-3} M, 10^{-2} M, 10^{-1} M, and 1 M NaCl).

dependencies on pH; however, the frequency variation in these measurements is large, and therefore the differences between the curves are less noticeable. The cores saturated with pH 3 solutions had the largest formation factors and the smallest imaginary conductivity and dielectric constants. These results indicate that surface conductivity and surface polarization effects are reduced at pH 3, which is the point of zero charge (PZC) for quartz-dominated systems [Li and DeBruyn, 1966].

Figure 8 shows the dependence of FF upon solution conductivity and pH. The FF data plotted in Figure 8 are the average responses measured over the frequency range of 1 to

10 Hz. Figure 8 demonstrates that FF increases with increasing solution conductivity and approaches a constant value at high solution concentrations (0.1 M to 1.0 M NaCl). The observed decrease in FF with decreasing solution conductivity is generally attributed to surface conductivity effects [Waxman and Smits, 1968]. At high salinity, surface conduction is less significant, and Archie's law becomes a more accurate model of rock conductivity. The FF versus solution conductivity curve for the pH 3 solution appears to be relatively insensitive to the solution conductivity. It appears that surface conductivity effects in the FF measurements are also reduced at pH 3, which is the PZC for quartz-dominated systems [Li and DeBruyn, 1966].

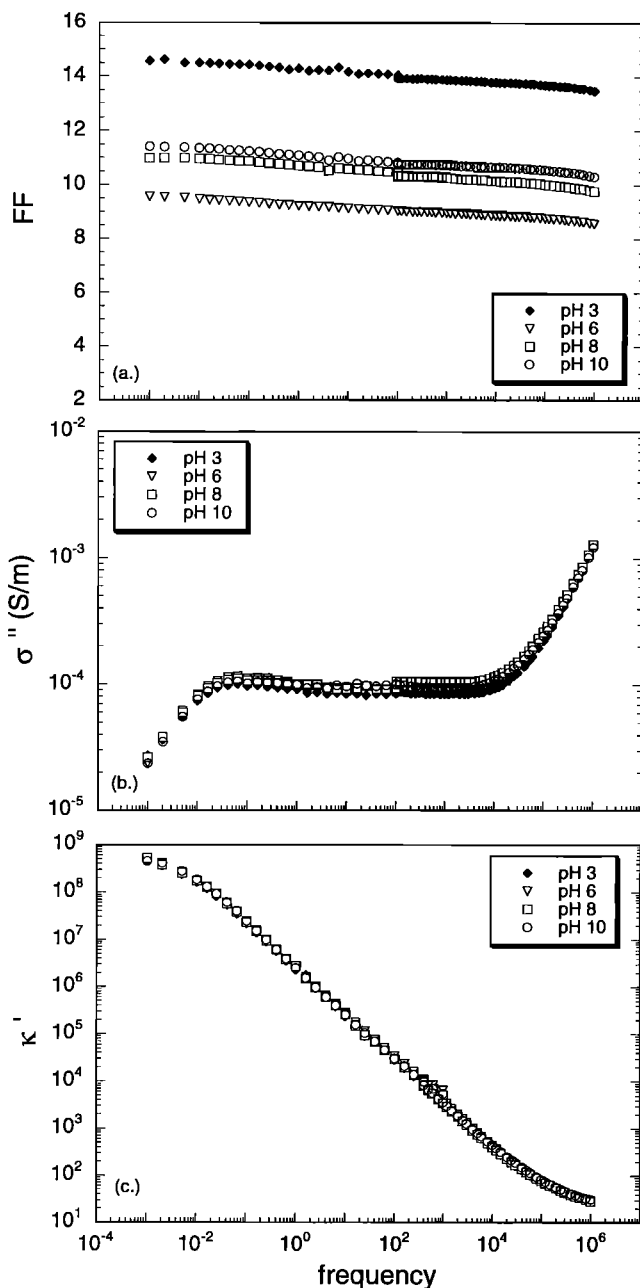


Figure 7. (a) Formation Factor, (b) imaginary component of conductivity, and (c) dielectric constant plotted versus frequency for Berea sandstone saturated with 10⁻² M NaCl brines with varying pH (3, 6, 8, and 10).

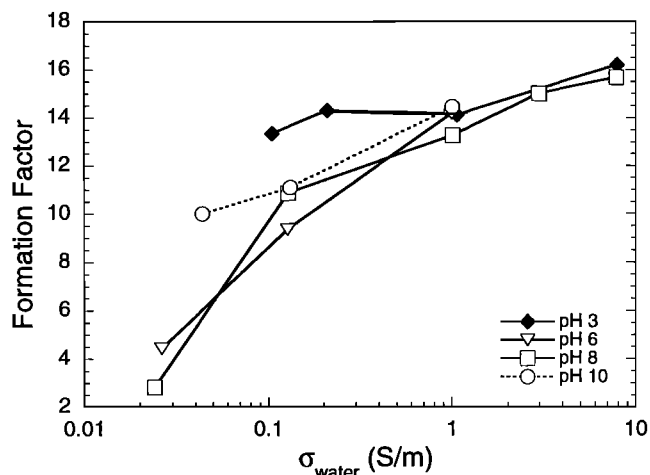


Figure 8. Average (1 to 10 Hz) formation factor of Berea sandstone saturated with NaCl solutions plotted versus brine conductivity as a function of pore fluid pH. Formation factor approaches a constant value (F in text) at high ionic strengths.

5.2. IP Field Parameters as a Function of Pore Fluid Chemistry

In Figure 9 the field IP parameters $-\theta$, PFE, and M are plotted versus brine conductivity for NaCl solutions of pH 3, 6, 8, and 10. The $-\theta$, PFE, and M have similar dependencies upon pore fluid chemistry. Increasing formation conductivity at higher ionic strengths masks the significant polarization effects observed at low pore fluid conductivity. The IP responses are smallest in the cores saturated with pH 3 solutions.

5.3. Normalized IP Parameters as a Function of Pore Fluid Chemistry

In saturated rocks and soils the dominant current path is usually through the bulk pore fluid, making surface conductivity a higher-order effect. Therefore, at high pore fluid conductivity the variability of traditional field IP parameters (θ , PFE, and M) are masked by the high overall formation conductivity. To better observe surface chemical effects in cores saturated with a high ionic strength fluid, one can separate the complex conductivity response into real and imaginary components. The imaginary component, or quadrature conductivity, is a function of only the surface conduction, whereas the real component is a function of both the surface conduction and bulk conductivity effects (9). The quadrature conductivity is obtained by essentially multiplying the phase by the conductivity of the sample (14).

The low-frequency quadrature conductivity response ($\sigma''_{\text{low ave}}$), averaged over the frequency range of 1 to 10 Hz, is shown plotted versus the solution conductivity in Figure 10a. Alternately, PFE and M can be weighted by the formation conductivity (or divided by resistivity). The resultant normalized IP parameters MF and MN demonstrate the same pore fluid chemistry dependence as the low-frequency quadrature conductivity (Figures 10b and 10c). The normalized IP parameters all increase with increasing solution conductivity up to solution concentrations of 10⁻¹ M NaCl. At higher solution concentrations the normalized IP parameters decrease. The pH dependence is secondary to ionic strength, but the normalized IP parameters are smallest at pH 3 for all of the NaCl concentrations measured.

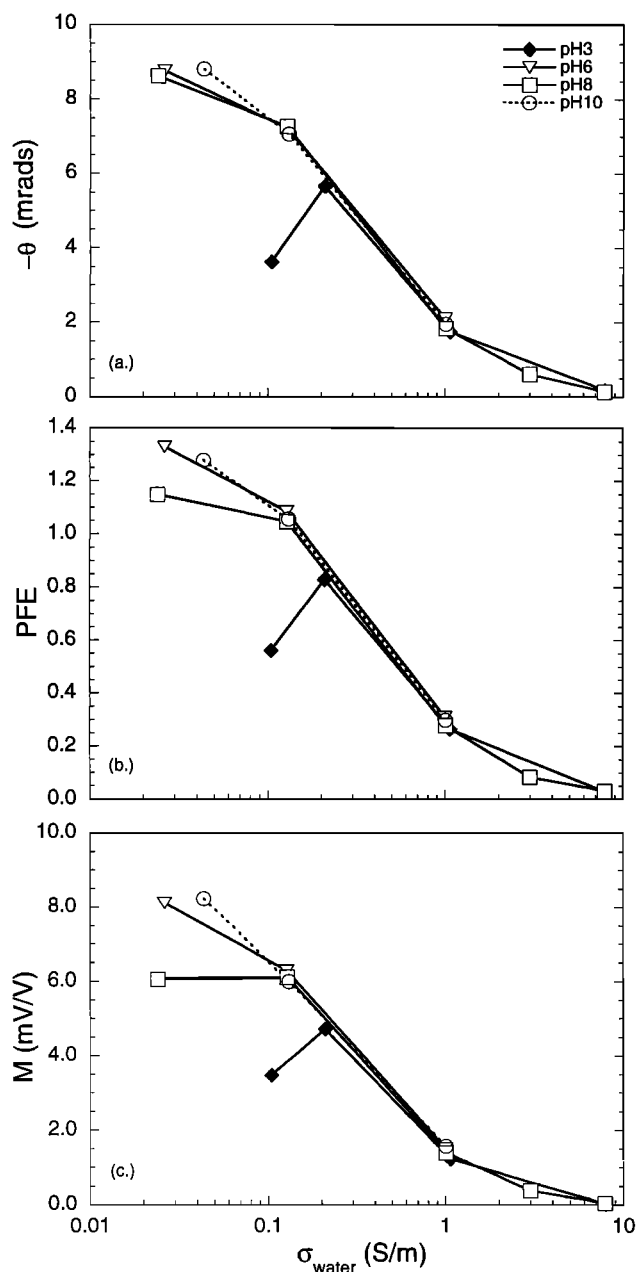


Figure 9. Phase angle (a) $-\theta$, (b) PFE, and (c) M versus brine conductivity as a function of pH. All three IP parameters go to zero at high brine conductivity.

6. Discussion

The complex conductivity spectra in Figures 2, 6, and 7 have similar shapes. At frequencies <0.02 Hz σ'' decreases rapidly as the permittivity response approaches κ_{static} , and the conductivity response approaches σ_{static} . At frequencies $>10^5$ Hz, σ'' increases rapidly as the permittivity response approaches κ_{∞} . The frequency dependence in the permittivity and conductivity responses is controlled by $\sigma_{\text{surr}}^*(\omega)$ in the equivalent circuit model of Figure 1a. Equations (12) and (13) express $\sigma_{\text{surr}}^*(\omega)$ as the product of the dc surface conductivity and a spectral response function, which is primarily controlled by the grain/pore size distribution. This simplified model predicts that changes in the pore solution chemistry will primarily affect the magnitude of the complex conductivity response and not the spectral shape. This

prediction is substantiated by the data in Figures 6 and 7, at least to within first order.

Effective media theories, which are based upon interfacial polarization mechanisms, predict that the dispersion in the permittivity and conductivity responses will shift to higher frequencies with increasing solution conductivity [Sen, 1981]. Kenyon [1984] measured the high-frequency (0.5 MHz to 1.3 GHz) complex conductivity response of carbonate rocks as a function of the solution conductivity (0.5 to 9 S/m). The permittivity measurements, when plotted as a function of the measurement frequency divided by the solution conductivity, all normalized to the same response. The permittivity response of Berea sandstone does not scale with the conductivity of the saturating solution. This result is to be expected from our simplified equivalent circuit model in which the frequency dependence in the complex conductivity

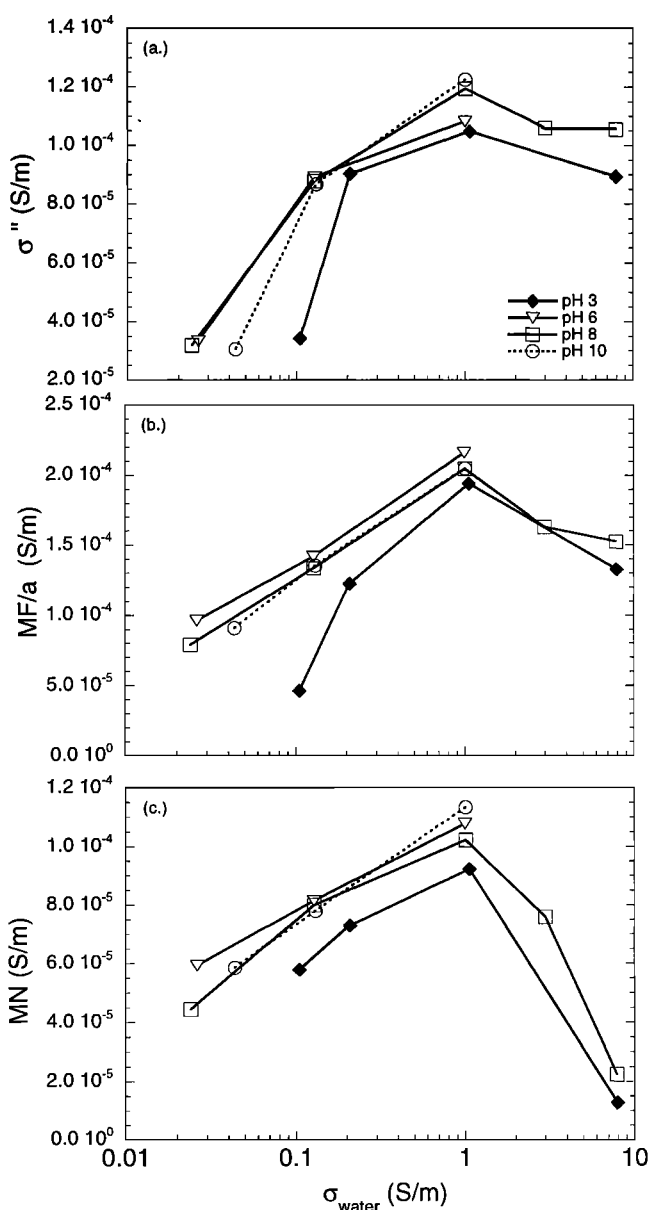


Figure 10. (a) The parameter $\sigma''_{\text{low ave}}$, (b) MF, and MN (c) versus brine conductivity as a function of pH. Normalized parameters are maximized at 10^{-1} M NaCl and pH dependence is observable over the entire conductivity range.

response arises from the electrochemical polarization of the EDL. Electrochemical polarization mechanisms in the carbonate rocks measured by Kenyon at high frequencies were much smaller effects compared to the interfacial polarization mechanisms.

It is important to note that broadband data, in which the full relaxation is captured, is required to uniquely interpret the complex conductivity response in terms of the underlying physical and chemical polarization mechanisms. In this paper we primarily focus on the effects of solution chemistry on the strength of the polarization. However, if we did not capture the full relaxation, then we would not know if the observed changes were caused by lateral shifts in the response caused by changes in the relaxation time distribution or vertical shifts in the response caused by changes in the strength of the polarization.

Common field IP parameters include the $-\theta$, PFE, and M . These IP field parameters are equivalent measures of the IP response of the sample [e.g., Hohman, 1990], and they depend upon the ratio of surface conductivity to bulk conductivity effects in the material. We observe that these field IP parameters decrease with increasing solution conductivity, which reflects the dominance of bulk conductivity over surface conductivity effects at high solution concentrations (Figure 9). The field IP parameters are less sensitive to pH than to concentration, but they are diminished at pH 3 (the PZC of quartz). Surface conductivity effects in the formation factor measurements are reduced at high solution conductivity and at pH 3.

Weighting $-\theta$ by the formation conductivity yields the quadrature component of the complex conductivity. The quadrature conductivity is only a function of the surface conductivity mechanisms, and it is independent of the bulk conductivity effects. Similarly, weighting the time domain M and the frequency domain PFE parameters by the formation conductivity (or normalization by resistivity) results in parameters equivalent to the imaginary component of the formation conductivity or the low-frequency dielectric constant. MF and MN display the same ionic strength pH dependence as the low-frequency imaginary conductivity (Figure 10). This includes a maximum value occurring at 0.1 M NaCl and a reduced response at pH 3. The quadrature conductivity and the normalized IP parameters isolate the surface conductivity effects and are directly related to the physiochemical parameters that control the polarization of the EDL.

Quadrature conductivity and the normalized IP parameters reflect the magnitude of the polarization of the complex conductivity response. They are primarily a function of the specific surface area, the surface charge density, and the effective surface ionic mobility of the ions in the EDL (13). Since our measurements were all made on Berea sandstone cores cut from the same block, the microstructure is the same for all of the samples. Therefore the observed dependence of quadrature conductivity and the normalized IP parameters on solution chemistry can be related to the surface charge density and the effective surface ionic mobility. These two geochemical parameters determine the structure and dynamics of ions within the EDL.

The surface conductivity and normalized IP parameters for Berea sandstone increase with increasing NaCl solution concentration up to concentrations of 0.1 M NaCl. This increase in surface conductivity results from an increase in the

surface charge density with increasing solution concentration. However, for NaCl concentrations >0.1 M the normalized IP parameters decrease with increasing solution concentration. This decrease in the quadrature conductivity can be explained by ion-ion interactions, which cause the effective surface ionic mobility to decrease at high solution concentrations. This effect is analogous to the decrease in solution activity at high ionic strength due to ion-ion interactions in the bulk pore solution [Wood *et al.*, 1984].

The effective mobility actually represents the migration of surface charge in both the fixed and diffuse parts of the EDL. The observed dependence of the effective mobility on solution concentration may also reflect the relative contributions of fixed layer and diffuse layer conductivity to the net surface conductivity response of the sample. It is generally assumed that the ions in the fixed layer will be less mobile than ions in the diffuse layer. Therefore, as the solution conductivity is increased, the diffuse layer is compressed, and there could be a transition from diffuse layer to fixed layer surface conduction and polarization. The reduction in the surface conductivity may also result from an increase in the tortuosity as the EDL is compressed. In actuality, many competing physiochemical effects are superimposed to yield the net surface ionic mobility and surface conductivity of the sample.

In this study, the mineralogy and the microgeometry remained constant as all of our measurements were made on Berea sandstone cores cut from the same block. In the field, these parameters will vary with the rock lithology and the soil texture. The effects of solution chemistry on the IP response will in general be a function of the mineralogy and microgeometry of the sample [e.g., Klien and Sill, 1982; Vinegar and Waxman, 1984]. Clay minerals, which have large specific surface area and high surface charge density, have a pronounced affect on the surface electrical properties of rocks and soils. The complex surface conductivity response of clay minerals results from both the electrochemical polarization of the EDL, which surrounds the grains, as well as membrane polarization mechanisms [e.g., Marshall and Madden, 1959; Vinegar and Waxman, 1984]. Membrane polarization is caused by the accumulation of anions at clay blockages that form at pore throats. Clay minerals, which are typically negatively charged, act as cation selective membranes, which preferentially repel anions and allow cations to pass through the blockage. Membrane polarization increases with increasing clay content and the efficiency of blockage, which is determined by the ratio of the cation-to-anion mobility through the clay membrane (i.e., the transference number) [Marshall and Madden, 1959].

Vinegar and Waxman [1984] measured the complex conductivity response of shaly sandstones with varying clay content (2 to 35% clay) and salinity (10^{-2} M to 2 M NaCl). They also observed a maximum in the quadrature conductivity response of their shaly sand samples measured as a function of salinity. They explained the peak in the quadrature conductivity response as a trade-off between increasing electrochemical polarization of the EDL with increasing salinity and decreasing membrane polarization at high-salinity. They argued that at high salinity the diffuse part of the EDL is compressed, making it easier for anions to migrate through the clay blockages thus decreasing the magnitude of the membrane polarization effect. We have explained our results just in terms of the electrochemical

polarization of the EDL. We argue that the decrease in polarization at high salinity is caused by a reduction in the effective mobility of the ions in the EDL. To test these two hypotheses, it may be useful to further analyze the surface conductivity contribution to the real part of the conductivity response. This surface conductivity term can be isolated using effective medium models [e.g., Bussian, 1983; Samstag and Morgan, 1991; Lesmes and Morgan, 2001]. If increased anion mobility through the clay blockages is responsible for the decrease in quadrature conductivity at high salinity, then the real part of the surface conductivity should increase at high salinity. Whereas if a reduction in surface ionic mobility in the EDL is responsible for the decrease in the quadrature conductivity at high salinity, then the real part of the surface conductivity should decrease at high salinity. Analysis of the broadband complex conductivity response of the samples and experiments on more uniform and well-characterized systems (e.g., silicate glasses) may also help to isolate and better understand the various surface conduction and polarization mechanisms.

7. Conclusions

SIP measurements are sensitive to surface chemical and microgeometrical properties of rocks and soils. Capturing the entire relaxation time distribution, which characterizes the SIP response, requires that the complex conductivity of the sample be measured over a wide range of frequencies (e.g., 10^{-3} to 10^9 Hz). This study investigates the effects of pore solution concentration and pH on the broadband complex conductivity response of Berea sandstone (10^{-3} to 10^6 Hz). The results indicate that the shape of the conductivity response is similar for all tested pore fluid concentrations and pH values. As the specific surface area is constant in this experiment, the surface charge density and the effective mobility of the ions in the fixed and diffuse parts of the EDL primarily determine the magnitude of the complex surface conductivity response (i.e., surface polarization).

IP field instruments typically measure the ρ_a and one of the following field IP parameters: $-\theta$, PFE, or M . These parameters are equivalent measures of the IP response and are proportional to the ratio of surface polarization to bulk conductivity within the sample. Weighting the field IP parameters by the sample conductivity yields the normalized IP parameters: quadrature conductivity, MF, and MN (or specific capacity). The normalized IP parameters are relatively independent of the bulk conductivity, and they are more directly related to the surface chemical properties of the porous media (i.e., specific surface area, surface charge density, and surface ionic mobility). Utilization of normalized IP parameters in field IP surveys should significantly improve the sensitivity of the IP measurements to the in situ geochemical and lithological properties of the formation. Therefore normalized IP parameters should improve the effectiveness of IP surveys in contaminant mapping and hydraulic parameter estimation.

Acknowledgments. The authors thank Dale Morgan for his encouragement and advice on this project. We also thank John Sturrock and John Ebel for their input, as well Associate Editor Kathy Whaler and two anonymous reviewers for their comments and criticisms. This work was funded by the Department of Energy (DOE grant DE-FG07-96ER14714) and Boston College.

References

- Archie, G. E., The electrical resistivity log as an aid in determining some reservoir characteristics, *Trans. Am. Inst. Min. Metall. Pet. Eng.*, **146**, 54-62, 1942.
- Bertin, J., and J. Loeb, *Experimental and Theoretical Aspects of Induced Polarization*, vols. I and II, Gebrüder Borntraeger, Stuttgart, Germany, 1976.
- Börner, F., M. Grubne, and J. Schön, Contamination indications derived from electrical properties in the low frequency range, *Geophys. Prospect.*, **41**, 83-98, 1993.
- Börner, F. D., J. R. Schopper, and A. Weller, Evaluation of transport and storage properties in the soil and groundwater zone from induced polarization measurements, *Geophys. Prospect.*, **44**, 583-601, 1996.
- Bussian, A. E., Electrical conductance in a porous medium, *Geophysics*, **48**, 1258-1268, 1983.
- Cahyna, F., O. Mazac, and D. Venhodova, Determination of the extent of cyanide contamination by surface geoelectrical methods, in *Geotechnical and Environmental Geophysics*, vol. II, *Environmental and Groundwater, Invest. Geophys.* vol. 5, edited by S.H. Ward, pp. 97-99, Soc. of Explor. Geophys., Tulsa, Okla., 1990.
- Clavier, C., G. Coates, and Dumanoir, Theoretical and experimental bases for the dual-water model for the interpretation of shaly sands, *Soc. Pet. Eng. J.*, **24**, 153-168, 1984.
- Cole, K. S., and R. H. Cole, Dispersion and adsorption in dielectrics, I, Alternating current characteristics, *J. Chem. Phys.*, **9**, 341-351, 1941.
- Cole, K. S., and H. J. Curtis, Wheatstone bridge and electrolytic resistor impedance measurements over a wide frequency range, *Rev. Sci. Instrum.*, **8**, 333, 1937.
- Collett, L. S., and T. J. Katsube, Electrical parameters of rocks in developing geophysical techniques, *Geophysics*, **38**, 76-91, 1973.
- Frye, K. M., The effect of pore fluid chemistry on the spectral induced polarization response of Berea sandstone, M. S. thesis, Boston College, Chestnut Hill, Mass., 1997.
- Glover P. W. J., P. G. Meredith, P. R. Sammonds and S. A. F. Murrell, Ionic surface electrical conductivity in sandstone, *J. Geophys. Res.*, **99**, 21635-21650, 1994.
- Hohman, G. W., Three-dimensional IP models, in *Induced Polarization: Applications and Case Histories, Invest. Geophys.* vol. 4, edited by J. B. Fink et al., pp. 150-178, Soc. of Explor. Geophys., Tulsa, Okla., 1990.
- Jackson, P. D., D. T. Smith, and P. N. Stanford, Resistivity-porosity-particle shape relationships for marine sands, *Geophysics*, **43**, 1250-1268, 1978.
- James, R. O., and G. A. Parks, Characterization of aqueous colloids by their electrical double layer and intrinsic surface chemical properties, *Surf. Colloid Sci.*, **12**, 199-216, 1982.
- Johnson, D. L., J. Koplik, and Schwartz, New pore-size parameter characterizing transport in porous media, *Phys. Rev. Lett.*, **57**, 2564-2567, 1986.
- Judy, M. M., and W. R. Eberle, A laboratory method for the measurement of the dielectric constant of rock and soil samples in the frequency range of 10^2 - 10^8 Hz, *U.S. Dep. of Inter. Geol. Surv. Tech. Rep.*, AFWL-TR-69-41, 1969.
- Keller, G. V., Analysis of some electrical transient measurements on igneous, sedimentary, and metamorphic rocks, in *Overvoltage Research and Geophysical Applications*, edited by J. R. Wait, pp. 92-111, Pergamon, Tarrytown, N.Y., 1959.
- Kenyon, W. E., Texture effects on megahertz dielectric properties of calcite rock samples, *J. Appl. Phys.*, **55**, 3153-3159, 1984.
- Klein, J. D., and W. R. Sill, Electrical properties of artificial clay-bearing sandstone, *Geophysics*, **47**, 1593-1605, 1982.
- Knight, R. J., and A. Endres, A new concept in modeling the dielectric response of sandstones: Defining a wetted rock and bulk water system, *Geophysics*, **55**, 586-594, 1990.
- Knight, R. J., and A. Nur, The dielectric constant of sandstones, 60 Hz to 4 MHz, *Geophysics*, **52**, 644-654, 1987.
- Lerner, D.B., J. M. Dacy, C. J. Raible, J. J. Rathmell, G. Swanson, and J. D. Walls, SCA guidelines for sample preparation and porosity measurement of electrical resistivity samples, part II, Sample preparation and porosity measurement, *Log Anal.*, **31**, 57-63, 1990.
- Lesmes, D. P., Electrical-impedance spectroscopy of sedimentary rocks, Ph.D. thesis, Texas A&M Univ., College Station, 1993.

- Lesmes, D. P., and F. D. Morgan, Dielectric spectroscopy of sedimentary rocks, *J. Geophys. Res.*, in press, 2001.
- Lesmes, D. P., J. Sturrock, and K. M. Frye, A physiochemical interpretation of the Cole-Cole dielectric model, paper presented at the Symposium on the Application of Geophysics to Environmental and Engineering Problems (SAGEEP), The Environmental and Engineering Geophysical Society, Arlington, Va., 2000.
- Li, H. C., and P. L. DeBruyn, Electrokinetic and adsorption studies on quartz, *Surf. Sci.*, 5, 203-220, 1966.
- Lima, O. A. L. de, and M. M. Sharma, A generalized Maxwell-Wagner theory for membrane polarization in shaly sands, *Geophysics*, 57, 431-440, 1992.
- Madden, T. R., and T. Cantwell, Induced polarization, a review, in *Mining Geophysics*, vol. II, pp. 373-400, Soc. of Explor. Geophys., Tulsa, Okla., 1967.
- Maerefat, N. L., B. A. Baldwin, A. A. Chaves, G. A. La Torraca, and B. F. Swanson, SCA guidelines for sample preparation and porosity measurement of electrical resistivity samples, part IV, Guidelines for saturating and desaturating core plugs during electrical resistivity measurements, *Log Anal.*, 31, 68-75, 1990.
- Marshall, D. J., and T. R. Madden, Induced polarization: A study of its causes, *Geophysics*, 24, 790-816, 1959.
- Mendelson, K. S., and M. H. Cohen, The effect of grain anisotropy on the electrical properties of sedimentary rocks, *Geophysics*, 47, 257-263, 1982.
- Morgan, F. D., T. R. Madden, and B. R. Bennett, An instrument system for low-frequency (10^{-3} - 10^3 Hz) impedance measurements, *IEEE Trans., Instrum. Meas.*, 35, 287-292, 1986.
- Morgan, F. D., E. R. Williams, and T. R. Madden, Streaming potential properties of Westerly granite with applications, *J. Geophys. Res.*, 94, 12,449-12,461, 1989.
- Olhoeft, G. R., Low-frequency electrical properties, *Geophysics*, 50, 2492-2503, 1985.
- Olhoeft, G. R., Electrical properties from 10^{-3} to 10^9 Hz- physics and chemistry, in *Physics and Chemistry of Porous Media, II*, edited by J. R. Banavar, J. Koplik, and K. W. Winkler, *AIP Conf. Proc.*, 154, pp. 775-786, 1986.
- Pelton, W. H., L. Rijo, and C. M. Swift Jr., Inversion of two-dimensional resistivity and induced-polarization data, *Geophysics*, 43, 788-803, 1978.
- Pride, S. R., Governing equations for the coupled electromagnetics and acoustic of porous media, *Phys. Rev. B*, 50, 15678-15696, 1994.
- Revil, A., and P. W. J. Glover, Nature of surface electrical conductivity in natural sands, sandstones, and clays, *Geophys. Res. Lett.*, 25, 691-694, 1998.
- Sadowski, R. M., Clay-organic interactions, M.Sc. thesis, Colo. Sch. of Mines, Golden, 1988.
- Samstag, F. J., and F. D. Morgan, Induced polarization of shaly sands: Salinity domain modeling by double embedding of the effective medium theory, *Geophysics*, 56, 1749-1756, 1991.
- Schon, J. H., *Physical Properties of Rocks: Fundamental and Principles of Petrophysics*, Elsevier Sci., New York, 1996.
- Seigel, H. O., Mathematical formulation and type curves for induced polarization, *Geophysics*, 24, 547-565, 1959.
- Sen, P. N., Relation of certain geometrical features to the dielectric anomaly of rocks, *Geophysics*, 46, 1714-1720, 1981.
- Sen, P. N., C. Scala, and M. H. Cohen, A self-similar model for sedimentary rocks with application to the dielectric constant of fused glass beads, *Geophysics*, 46, 781-795, 1981.
- Shuey, R. T., and M. Johnson, On the phenomenology of electrical relaxation in rocks: The electrical parameters of rocks, *Geophysics*, 38, 37-48, 1973.
- Sillars, R. W., The properties of a dielectric containing semi-conducting particles of various shapes, *J. Inst. Electr. Eng.*, 80, 378-394, 1937.
- Sturrock, J. T., Estimates of permeability from broadband (spectral) induced polarization measurements, M. S. thesis, Boston College, Chestnut Hill, Mass., 1999.
- Sturrock, J. T., D. P. Lesmes, and F. D. Morgan, The influence of micro-geometry on the hydraulic permeability and the induced polarization response of sandstones, paper presented at the Symposium on the Application of Geophysics to Environmental and Engineering Problems (SAGEEP), The Environmental and Engineering Geophysical Society, Chicago, Ill., 1998.
- Sturrock, J. T., D. P. Lesmes and F. D. Morgan, Permeability estimation using spectral induced polarization measurements, paper presented at the Symposium on the Application of Geophysics to Environmental and Engineering Problems (SAGEEP), The Environmental and Engineering Geophysical Society, Oakland, Calif., 1999.
- Tadros, T. F. and J. Lyklema, The electrical double layer on silica in the presence of bivalent counter-ions, *J. Electroanal. Chem.*, 22, 1-7, 1968.
- Van Voorhis, G. D., P. H. Nelson, and T. L. Drake, Complex resistivity spectra of porphyry copper mineralization, *Geophysics*, 38, 49-60, 1973.
- Vinegar, H. J., and M. H. Waxman, Induced polarization of shaly sands, *Geophysics*, 49, 1267-1287, 1984.
- Wait, J. R., Relaxation phenomena and induced polarization, *Geoexploration*, 22, 107-127, 1984.
- Ward, S. H., Resistivity and Induced Polarization Methods, in *Geotechnical and Environmental Geophysics*, vol. 1, *Review and Tutorial, Invest. Geophys.* vol. 5, edited by S.H. Ward, pp. 141-190, Soc. of Explor. Geophys., Tulsa, Okla., 1990.
- Ward, S. H., B. K. Sternberg, D. J. LaBrecque and M. M. Poulton, Recommendations for IP research, *Leading Edge*, 14, 243-247, 1995.
- Waxman, M. H., and L. J. M. Smits, Electrical conductivities in oil-bearing shaly sands, *Soc. Pet. Eng. J.*, 243, 107-122, 1968.
- Wood, S. A., D. A. Crerar, S. L. Brantley, and M. Borsik, Mean molal stoichiometric activity coefficients of alkali halides and related electrolytes, *Am. J. Sci.*, 284, 668-705, 1984.
- Worthington, A. E., J. H. Hedges, and N. Pallatt, SCA guidelines for sample preparation and porosity measurement of electrical resistivity samples, part I, Guidelines for preparation of brine and determination of brine resistivity for use in electrical resistivity measurements, *Log Analyst*, 31, 20-28, 1990.
- Wyllie, M. R. J., and P. F. Southwick, An experimental investigation of the SP and resistivity phenomena in dirty sands, *J. Pet. Technol.*, 6, 44-57, 1954.

K. M. Frye, Department of Earth, Atmospheric, and Planetary Sciences, Massachusetts Institute of Technology, Cambridge, MA 02139. (kfrye@mit.edu)

D. P. Lesmes, Department of Geology and Geophysics, Boston College, Chestnut Hill, MA 02167. (lesmes@bc.edu)

(Received November 30, 1999; revised October 18, 2000; accepted October 26, 2000.)

Thin double layer theory of the wide-frequency range dielectric dispersion of suspensions of non-conducting spherical particles including surface conductivity of the stagnant layer

V.N. Shilov ^{a,*}, A.V. Delgado ^b, F. Gonzalez-Caballero ^b, C. Grosse ^c

^a *Institute of Biocolloid Chemistry, National Academy of Sciences of Ukraine, Vernadskogo Prospect 42, 252042 Kiev, Ukraine*

^b *Department of Applied Physics, University of Granada, Granada, Spain*

^c *Department of Physics, University of Tucuman and CONICET, Tucuman, Argentina*

Abstract

The well-known results of the Dukhin–Shilov theory of the thin diffuse double layer polarization in low-frequency alternating fields is generalized in two directions: to include the surface conductivity of the stagnant layer and to describe a wide frequency range including both the low-frequency and the Maxwell–Wagner–O’Konski dispersions. The generalized expression obtained for the induced dipole moment is confirmed by comparison with numerical results of the DeLacey and White approach. Expressions describing the wide-range spectra of dielectric permittivity and conductivity of suspensions are also obtained. The results are presented in the terms of the ζ -potential and the parameter κa , taking also into account the surface conductivity behind the plane of shear. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: Double layer; Wide-frequency; Stagnant layer

1. Introduction

The investigation of the dielectric dispersion of suspensions involves determinations of its dielectric constant $\varepsilon(\omega)$ and conductivity $K(\omega)$ as functions of the angular frequency ω of the applied AC field. These quantities are related to the properties of the constituents of the suspension: the

bulk, surface, and geometrical characteristics of the disperse particles, their concentration (usually expressed in terms of volume fraction ϕ_p), and the characteristics of the dispersion medium (electrolyte solution viscosity, ion concentrations, valences and diffusion coefficients of all the ions).

The dielectric constant is the macroscopic manifestation of the electrical polarization of the suspension components. Different times are necessary for the development of the various mechanisms contributing to the system’s polarization. Therefore, the investigation of the frequency dependence of the dielectric permittivity and con-

* Corresponding author.

E-mail address: shilov@i.kiev.ua (V.N. Shilov).

ductivity makes it possible to obtain information about the characteristics of the disperse system that are responsible for the polarization of the particles.

For suspensions in electrolyte solutions there are two characteristic mechanisms of dielectric dispersion, which are both sensitive to the polarization of the electric double layer of the disperse particles. The first is the low-frequency dielectric dispersion, caused by the influence of field-induced electrolyte concentration variations on the local electric currents in the double layer (concentration polarization of the double layer, or volume diffusion mechanism of double layer polarization). The second is the well-known Maxwell-Wagner dielectric dispersion, caused by the formation of the field-induced free (ionic) charge distributions near the surfaces that separate the phases with different local conductivities and dielectric constants.

The theory of the low-frequency dielectric dispersion has been first elaborated in [1–3] for dilute suspensions of homogeneous spherical particles with no volume conductivity and surrounded by a thin double layer. It is based on the well-known system of electrokinetic equations [4] where it is considered that the liquid and ions are immobile in the dense part of the double layer while, in its diffuse part, the viscosity, diffusion coefficients, and dielectric constant, are considered to have the same values as in the bulk electrolyte solution (the so-called *standard model* of the electric double layer). The theory was later confirmed [5] and generalized in different directions: for rotating applied fields [6], for particles with volume conductivity [7–9], for non-spherical particles [10–13], for concentrated suspensions [14–16], for weak electrolytes [17,18].

In the original Maxwell–Wagner theory [19,20], the surface was considered only as a geometrical boundary between homogeneous phases without any specific properties. This was generalized by O’Konski [21], who completed the model by taking into account the surface conductivity K^s , without any specification of its physical nature. Assuming that the surface conductivity is due to excess ion concentration in the proximity of the charged particle surface, it opened the possibility

for the characterization of the double layer using measured parameters of the Maxwell–Wagner (or rather Maxwell–Wagner–O’Konski) dielectric dispersion.

All analytical theories of the dielectric dispersion of suspensions deal separately with the MW dispersion or the low-frequency one. No formula is available that considers the wide frequency interval including both mechanisms, except for those presented by O’Brien [22] and by Grosse [23], which are based on simplified treatments of the low-frequency dispersion. The present work is devoted to the deduction of such an analytical formula that includes, furthermore, the contribution of the surface conductivity behind the slipping or shear plane (the so-called *generalized standard model* [24]).

A very important progress in the theory of both the low-frequency and the MW dielectric dispersions is due to the numerical theory by DeLacey and White [25], which was developed in the frame of the standard system of electrokinetic equations for dilute suspensions of non-conducting spherical particles in the absence of surface conductivity behind the slipping plane (*standard model*), and without any restriction on the diffuse layer thickness or the ζ -potential values. These numerical results will be used in this work to check the range of applicability of the analytical theory.

2. Theory

2.1. The natural frequency scale for suspensions in electrolyte solutions

The frequency range in which the dielectric dispersion of suspensions in electrolyte solution is usually measured roughly extends between 1 kHz and several hundred MHz. In order to define in this frame the *low frequency* and *high-frequency* ranges, it is useful to introduce an important concept dealing with a point in the frequency scale. This point corresponds to the *reciprocal of the relaxation time of the electrolyte solution* τ_m :

$$\omega_m \equiv \frac{1}{\tau_m} \equiv \frac{K_m}{\epsilon_m} \quad (1)$$

where K_m and ε_m are the conductivity and absolute dielectric constant of the dispersion medium. The value of τ_m characterizes the time required for the screening of charge and electric field perturbations in the electrolyte solution. Its role in the time dimension is similar to that played by the Debye screening length

$$L_D \equiv 1/\kappa = \sqrt{\frac{(D^+ + D^-)\varepsilon_m}{2K_m}} \quad (2)$$

in the space dimension (here $D^\pm$ are the ion diffusion coefficients and a symmetric electrolyte is assumed). The order of magnitude of τ_m corresponds to the time of ion diffusion through a distance equal to the Debye length:

$$\tau_m \approx \frac{L_D^2}{2D} \equiv \frac{1}{2D\kappa^2} \quad (3)$$

where D is the average ion diffusion coefficient.

It is so advisable to define the low-frequency range by the inequality:

$$\omega < < \omega_m. \quad (4)$$

For these frequencies, the characteristic value of the *conduction current density* in the electrolyte solution exceeds the *displacement current density*, and the space distribution of the local electric fields in the disperse system is mainly determined by the distribution of ionic currents. In the high-frequency range determined by the inequality:

$$\omega > > \omega_m \quad (5)$$

the characteristic value of the displacement current density exceeds that of the conduction current, and the space distribution of the local electric fields is determined by the polarization of the molecular dipoles, rather than by the distribution of ionic currents.

In the forthcoming considerations it is convenient to represent the time dependence of the harmonic external field $E(t)$ with frequency ω , by means of the complex multiplier $e^{i\omega t}$

$$E(t) = E(\omega) \cdot e^{i\omega t} \quad (6)$$

The well-known advantage of this representation is that all the field-induced magnitudes $X(t)$ have also this same time dependence (in the linear approximation with respect to the external field):

$$X(t) = X^*(\omega) \cdot e^{i\omega t} \quad (7)$$

where the asterisk denotes a complex quantity. If any magnitude X had a relaxation at a frequency close to ω , it would be out of phase with respect to the external field. In the representation used, this corresponds to the appearance of an imaginary part of $X^*(\omega)$ and also to its frequency dependence.

Using the representation given in Eqs. (6) and (7), the full electric response of the disperse system can be characterized by its complex permittivity

$$\varepsilon^*(\omega) = \varepsilon(\omega) - \frac{iK(\omega)}{\omega} \quad (8)$$

or its complex conductivity $K^*(\omega)$

$$K^*(\omega) = K(\omega) + i\omega\varepsilon(\omega). \quad (9)$$

It should be emphasized that the suspension permittivity $\varepsilon(\omega)$ and conductivity $K(\omega)$ are real functions of the frequency.

The contributions $\delta\varepsilon^*(\omega)$ and $\delta K^*(\omega)$ of the dispersed particles to the complex permittivity and conductivity of a dilute suspension are defined as:

$$\delta\varepsilon^*(\omega) \equiv \delta\varepsilon(\omega) - \frac{i\delta K(\omega)}{\omega} = \varepsilon^*(\omega) - \varepsilon_m^*(\omega) \quad (10)$$

$$\delta K^*(\omega) \equiv \delta K(\omega) + i\omega\delta\varepsilon(\omega) = K^*(\omega) - K_m^*(\omega) \quad (11)$$

where the functions

$$\varepsilon_m^*(\omega) = \varepsilon_m - \frac{iK_m}{\omega} \quad (12)$$

$$K_m^*(\omega) = K_m + i\omega\varepsilon_m \quad (13)$$

represent the complex permittivity and complex conductivity of the dispersion medium (ε_m , K_m , $\delta\varepsilon(\omega)$, and $\delta K(\omega)$ are real).

The functions $\delta\varepsilon^*(\omega)$ and $\delta K^*(\omega)$ are determined by the superposition of the long-range electric fields of the polarized particles. This can be expressed through the dipole coefficient $\gamma_p^*(\omega)$ of a particle and the volume fraction ϕ_p :

$$\delta\varepsilon^*(\omega) = 3\phi_p\varepsilon^*(\omega)\gamma_p^*(\omega) \quad (14)$$

$$\delta K^*(\omega) = 3\phi_pK_m^*(\omega)\gamma_p^*(\omega) \quad (15)$$

The dipole coefficient $\gamma_p^*(\omega)$ appears in the expression for the electric potential distribution φ_m^* , valid in the electro-neutral electrolyte solution outside the double layer at large distances r from the center of the particle:

$$\varphi_m^* = -Er \cos \theta + \frac{\gamma_p^* a^3 E}{r^2} \cos \theta \quad (16)$$

where a is the particle radius. The first term in the righthand side of this expression is the potential of the homogeneous external field, and θ is the angle between the radius vector of a point and the direction of the field. The second term in Eq. (16) represents the long-range potential distortion produced by the polarized particle together with its double layer.

According to [1–3,5,25], the frequency dependencies of the suspension permittivity and conductivity $\varepsilon(\omega)$ and $K(\omega)$ (in the frequency range where the characteristics of the dispersion medium may be considered as frequency-independent) are caused by the frequency dependence of the dipole coefficient $\gamma_p^*(\omega)$. Note that in the low-frequency part of the natural scale where the strong inequality (4) holds, the absolute value of the imaginary part of the complex permittivity of the dispersion medium (K_m/ω), exceeds its real part by a large factor. This results in the well-known very large values of the low-frequency suspension's permittivity due to the contribution of $\frac{K_m}{\omega} \operatorname{Im} \gamma_p^*(\omega) = \varepsilon_m \frac{\omega_m}{\omega} \operatorname{Im} \gamma_p^*(\omega)$ in the real part of Eq. (14).

It is interesting to point out that a symmetrical phenomenon, consisting in a very large contribution of the particles to the suspension conductivity, $\delta K(\omega)$, may be expected in the high-frequency part of the natural scale ($\omega > \omega_m$). It is due to the contribution of the addend $\varepsilon_m \omega \operatorname{Im} \gamma_p^*(\omega)$ that can be very large in the case that there is some mechanism that causes a frequency dependence of $\gamma_p^*(\omega)$ in the high-frequency range.

2.2. Middle-frequency range:

Maxwell–Wagner–O’Konski relaxation

There are various mechanisms contributing to the particle polarization, and each of them is

always associated to some property that differs for the solid, the liquid, and their interface. As mentioned above, the most widely known mechanism of dielectric dispersion, the *Maxwell–Wagner dispersion*, occurs when the phases in contact have different volume conductivities and dielectric permittivities. Let K_p and ε_p denote the volume conductivity and permittivity of the particles. If the ratio between these two quantities differs for the particles and the medium, i.e. if

$$\frac{\varepsilon_m}{K_m} \neq \frac{\varepsilon_p}{K_p} \quad (17)$$

then the condition of continuity of the normal components of the current density and the displacement vector on both sides of the surface are inconsistent with one another, resulting in the formation of free ionic charge near the surface. The finite time needed for the formation of this free charge (of the order of τ_m) is in fact responsible for the Maxwell–Wagner dielectric dispersion approximately at $\omega_m = 1/\tau_m$.

The conductivity of the particle K_p must be modified, according to O’Konski [21], to include the contributions of both the volume and the surface conductivity K^σ in an effective particle conductivity K_{per} :

$$K_{\text{per}} = K_p + \frac{2K^\sigma}{a} \quad (18)$$

where K^σ includes both the surface conductivity within the stagnant layer and the surface conductivity of the diffuse layer beyond the plane of shear.

Furthermore, both conductivity and dielectric constant can be considered as parts of the complex dielectric permittivity of any of the system components, as shown by Eq. (12) for the dispersion medium, and by

$$\varepsilon_{\text{per}}^*(\omega) = \varepsilon_p - \frac{iK_{\text{per}}}{\omega} \quad (19)$$

for the particle. In terms of these quantities, the Maxwell–Wagner–O’Konski theory gives the following expression for the complex dipole coefficient of a disperse particle:

$$\gamma_p^*(\omega) = a^3 \frac{\epsilon_{\text{pef}}^* - \epsilon_m^*}{\epsilon_{\text{pef}}^* + 2\epsilon_m^*}. \quad (20)$$

In what follows, we shall consider the usual situation that the particle is non-conducting ($K_p = 0$), so that

$$K_{\text{pef}} = \frac{2K^\sigma}{a}. \quad (21)$$

Under this condition (non conducting particles with surface conductivity), the Maxwell–Wagner–O’Konski dipole coefficient tends to a frequency independent value in the low-frequency range characterized by inequality (4):

$$\gamma_p^*(\omega) = \frac{2Du - 1}{2Du + 2} \quad (22)$$

where Du is Dukhin’s number defined as:

$$Du = \frac{K^\sigma}{K_m a}. \quad (23)$$

2.3. Low-frequency dielectric dispersion caused by the concentration polarization of a thin double layer

Whereas the Maxwell–Wagner dielectric dispersion arises due to the different space dependencies of the local conductivity and permittivity, the difference in the space dependence of the relative contributions of cations and anions to the local conductivity (space dependence of the *transfer numbers* of cations and anions) gives raise to variations in the local electrolyte concentration. This in turn gives rise to the *volume diffusion mechanism of the low-frequency dielectric dispersion*. The contributions of cations $K^{\sigma+}$ and anions $K^{\sigma-}$ to the surface conductivity, are usually quite different (the contribution of counterions predominates). On the contrary, the contributions of cations K_m^+ and anions K_m^- to the bulk electrolyte solution conductivity K_m have comparable values. This means that almost always, the following inequality holds:

$$\frac{K^{\sigma+}}{K_m^+} \neq \frac{K^{\sigma-}}{K_m^-}. \quad (24)$$

Under this condition, the balance between the flows of the ions of both signs is only possible at

the expense of a field-induced variation of electrolyte concentration $\delta c(\vec{r})$ near the polarized particle. The ionic diffusion caused by $\delta c(r)$ gives raise to a low-frequency dependence of the particle’s dipole coefficient, $\gamma_p^*(\omega)$.

For a z -valent symmetrical electrolyte, the values of $K_m^\pm$ are determined by the expressions:

$$K_m^\pm = \frac{z^2 e^2 n_0}{kT} D^\pm \quad (25)$$

where n_0 is the anion or cation number concentration in the bulk electrolyte solution. Eq. (25) correspond to the well-known expression for the electrolyte solution conductivity:

$$K_m = K_m^+ + K_m^- = \frac{z^2 e^2 n_0}{kT} (D^+ + D^-). \quad (26)$$

As for the expressions for the contributions of the two types of ions to the surface conductivity $K^{\sigma\pm}$, they depend on the model used for the description of the kinetic properties of the double layer. In the framework of the *standard model*, they are given by:

$$K_d^{\sigma\pm} = K_m^\pm \left\{ 2 \frac{\exp\left(\mp \frac{z\tilde{\zeta}}{2}\right) - 1}{\kappa} + 3m^\pm \left[2 \frac{\exp\left(\mp \frac{z\tilde{\zeta}}{2}\right) - 1}{\kappa} \pm \frac{z\tilde{\zeta}}{\kappa} \right] \right\}. \quad (27)$$

$$\tilde{\zeta} = \frac{e}{kT} \zeta \quad (28)$$

$$m^\pm = \frac{2\epsilon_m}{3\eta D^\pm} \left(\frac{kT}{ze} \right)^2 \quad (29)$$

which correspond to Bikerman’s formula to the surface conductivity [27]:

$$K_d^\sigma = \frac{2K_m}{\kappa} \left\{ \frac{D^+}{D^+ + D^-} \left[\exp\left(-\frac{z\tilde{\zeta}}{2}\right) - 1 \right] (1 + 3m^+) + \frac{D^-}{D^+ + D^-} \left[\exp\left(+\frac{z\tilde{\zeta}}{2}\right) - 1 \right] (1 + 3m^-) \right\}. \quad (30)$$

In Eqs. (27) and (30) the lower index d denotes that the surface conductivity is solely due to ions in the *diffuse* part of the double layer. Nevertheless, in the framework of the so-called *generalized standard model*, the surface conductivity expression includes an additional term that accounts for

the existence of conduction behind the plane of shear.

The analytical theory of the low-frequency dipole coefficient in AC fields was elaborated in [1,3]. It is based, first of all, on the local equilibrium approximation among the field-induced variations of the parameters of the polarized double layer and the field-induced variations of the electrolyte solution concentration adjoining the double layer. This approximation is well justified when:

(i) Due to the small thickness of the double layer,

$$\kappa a \gg 1 \quad (31)$$

the diffuse layer near every small portion of the particle's surface may be considered as a small, and hence a quasi-equilibrium part of a large system, represented by the adjoining solution.

(ii) Due to the low frequency of the external field, i.e. when the inequality (4) is fulfilled, the period of the field greatly exceeds the time needed for the re-establishment of the screening charge of the diffuse layer.

The solution of the standard electrokinetic equation system obtained in [1–3] using the local equilibrium approximation, can be expressed in the following compact form for the low-frequency dipole coefficient [6,26]:

$$\begin{aligned} \gamma_p^*(\omega) &= \frac{2Du(\zeta) - 1}{2Du(\zeta) + 2} - \frac{3(R^+ - R^-)H}{2B} \\ &+ \frac{3(R^+ - R^-)H}{2B} \frac{i\omega\tau_\alpha}{1 + \sqrt{2/S}\sqrt{i\omega\tau_\alpha} + i\omega\tau_\alpha} \end{aligned} \quad (32)$$

where

$$Du(\zeta) = \frac{K_d^\sigma}{K_m a} \quad (33)$$

$$R^\pm = \frac{2K_d^{\sigma\pm}}{K_m^\pm a} \quad (34)$$

$$\tau_\alpha = \frac{a^2(D^+ + D^-)S}{4D^+ D^-} \quad (35)$$

$$A = 4Du(\zeta) + 4 \quad (36)$$

$$\begin{aligned} B &= (R^+ + 2)(R^- + 2) - U^+ - U^- \\ &- (U^+ R^- + U^- R^+)/2 \end{aligned} \quad (37)$$

$$S = \frac{A}{B} \quad (38)$$

$$\begin{aligned} H &= \\ &= \frac{(R^+ - R^-)(1 - z^2 \Delta^2) - U^+ + U^- + z\Delta(U^+ + U^-)}{A} \end{aligned} \quad (39)$$

$$U^\pm = \frac{48m^\pm}{\kappa a} \ln \left[\cosh \frac{z\zeta}{4} \right] \quad (40)$$

$$\Delta = \frac{D^- - D^+}{z(D^- + D^+)} \quad (41)$$

while η is the viscosity of the suspending medium, k is the Boltzmann constant, and T the absolute temperature.

The first addend in the right-hand side of Eq. (32) coincides with Eq. (22) and represents simultaneously the high-frequency limit of the low-frequency dispersion of the dipole coefficient, and the low-frequency limit of the Maxwell–Wagner–O’Konski dispersion. The second addend represents the amplitude of the variation of the dipole coefficient corresponding to the low-frequency dispersion. Hence, the first two addends taken together, represent the low-frequency limit of the dipole coefficient with respect to both the low-frequency and the Maxwell–Wagner–O’Konski dispersions. The third frequency-dependent addend, provides the frequency dependent transition of the dipole coefficient between its low- and high-frequency limits.

The quantity S weakly depends on the Dukhin number and the product κa , being always of the order of unity (except for very small κa):

$$S \cong 1. \quad (42)$$

Hence, a comparison of Eqs. (3) and (35) leads to the important conclusion that:

$$\frac{\tau_\alpha}{\tau_m} \approx (\kappa a)^2. \quad (43)$$

This means that for thin double layers ($\kappa a \gg 1$), the characteristic frequency for the low-frequency dispersion is $(\kappa a)^2$ times lower than that of the Maxwell–Wagner dispersion.

2.4. Inclusion of the surface conductivity of the stagnant layer

Eqs. (30), (32–40) represents the characteristics of the low-frequency dispersion of the dipole coefficient calculated in the frame of the standard electrokinetic model. According to the latter, the only parameter of the solid/liquid interface that is needed to account for LFDD is the ζ -potential. However, with the exception of rare cases (one example is presented in [26,28] for the suspension of AgI), the contribution K_d^σ of the diffuse layer (expressed by Bikerman's formula), is only a part, and in many cases even a small part, of the total value of the surface conductivity. This means that in the majority of real systems, the main part of the total surface conductivity is due to counterions in the inner part of double layer, which are responsible for the conductivity behind the slipping plane. The latter includes the contribution K_i^σ of mobile counterions, such as ions localized in the hydrodynamically immobile (due to surface roughness or polymer adsorption) part of the diffuse layer, and to mobile counterions in the Stern layer.

In order to deduce an expression for the low-frequency behavior of the dipole coefficient taking into account the surface conductivity of the stagnant layer, let us examine the physical meaning of the different terms intervening in Eq. (32). The most important are the parameters $R^\pm$, which are related to the surface and electrolyte solution conductivities of cations and anions [Eqs. (34) and (30)]. The parameters $U^\pm$ [Eq. (36)] which characterize the contribution of the convection induced by $\delta c(\vec{r})$ (capillary–osmotic convection) to the partial surface currents of cations and anions, have a very little influence on the overall values of these currents.

Neglecting the tangential flow of coions in the double layer as compared to the flow of counterions, as well as the capillary osmotic term for counterions, all the magnitudes entering in the expression of the dipolar coefficient can be expressed in terms of the surface conductivity and the electrolyte solution properties:

$$(R^+ - R^-)H \Rightarrow \frac{Du^2(\zeta)}{Du(\zeta) + 1} \frac{(1 - z^2\Delta^2)}{t_c^2} \quad (44)$$

$$A = 4Du(\zeta) + 4 \quad (45)$$

$$B \Rightarrow 4 \left(\frac{Du(\zeta)}{t_c} + 1 \right) \quad (46)$$

$$S \Rightarrow \frac{Du(\zeta) + 1}{\frac{Du(\zeta)}{t_c} + 1} \quad (47)$$

where

$$t_c = \frac{D_c}{D^+ + D^-} \quad (48)$$

and D_c is the diffusion coefficient of counterions.

The contribution of the surface conductivity due to ions situated behind the slipping plane, K_i^σ , can be characterized by the parameter:

$$\Theta = \frac{K_i^\sigma}{K_d^\sigma} \quad (49)$$

To emphasize the influence of K_i^σ on the Dukhin number, both arguments, ζ (for K_d^σ) and Θ (for K_i^σ) must be used:

$$Du(\zeta, \Theta) = \frac{K^\sigma}{K_{ma}} = \frac{K_d^\sigma}{K_{ma}} (1 + \Theta). \quad (50)$$

The problem of the generalization the theory of the volume diffusion mechanism by taking into account the surface conductivity due to ions situated behind the slipping plane has been dealt with in Refs. [29–31] for the case when $\kappa a \gg 1$. It can be shown that simplified formulae obtained by neglecting the coion contribution to the surface current in the double layer, and also by neglecting the contributions of the currents of capillary–osmotic origin, allow to write the low-frequency dispersion in terms of only one surface characteristic, namely $Du(\zeta, \Theta)$. The following substitutions must be carried out in Eqs. (32), (36)–(38):

$$(R^+ - R^-)H \Rightarrow \frac{Du^2(\zeta, \Theta)}{Du(\zeta, \Theta) + 1} \frac{(1 - z^2\Delta^2)}{t_c^2} \quad (51)$$

$$A = 4Du(\zeta, \Theta) + 4 \quad (52)$$

$$B \Rightarrow 4 \left(\frac{Du(\zeta, \Theta)}{t_c} + 1 \right) \quad (53)$$

$$S \Rightarrow \frac{Du(\zeta, \Theta) + 1}{\frac{Du(\zeta, \Theta)}{t_c} + 1}. \quad (54)$$

2.5. Wide-frequency range polarizability and dielectric dispersion

A general formula for the induced dipole moment will now be described that yields simultaneously the frequency dependence of both dispersion ranges, namely, the α -dispersion and the Maxwell–Wagner–O’Konski dispersion. The treatment was recently elaborated in [32], for conditions corresponding to the quasi-equilibrium polarization of the Stern layer in iso-electric point (in the absence of equilibrium diffuse layer), but it has not been applied yet in the frame of an analytical theory of thin double layer polarization. However, the coincidence [see Eq. (32)] of the high-frequency limit of the dipole coefficient obtained from the analytical theory of the low-frequency-dispersion, with the low-frequency limit of dipole coefficient obtained by using the theory of Maxwell–Wagner–O’Konski [22], creates the necessary prerequisites for a simple procedure of derivation of the general formula for the dipole coefficient γ_p^* .

In order to derive this formula, a superposition approximation will be used, whereby we simply replace the frequency-independent term $[2Du(\zeta) - 1]/[2Du(\zeta) + 2]$, representing the high-frequency limit of Eq. (32) by the Maxwell–Wagner–O’Konski dipole coefficient, $[\varepsilon_{\text{per}}^* - \varepsilon_m^*]/[\varepsilon_{\text{per}}^* + 2\varepsilon_m^*]$. After some algebra, this leads to:

$$\gamma_p^* = \frac{\frac{\varepsilon_p}{\varepsilon_m} - 1}{\frac{\varepsilon_p}{\varepsilon_m} + 2} - \frac{3}{2} \frac{\frac{\varepsilon_p}{\varepsilon_m} - 2Du}{\left(\frac{\varepsilon_p}{\varepsilon_m} + 2\right)(Du + 1)} \frac{1}{1 + i\omega\tau_{\text{MW}}} - \frac{3(R^+ - R^-)H}{2B} \frac{1 + \sqrt{2/S}\sqrt{i\omega\tau_\alpha}}{1 + \sqrt{2/S}\sqrt{i\omega\tau_\alpha} + i\omega\tau_\alpha} \quad (55)$$

where

$$\tau_{\text{MW}} = \frac{\varepsilon_m}{2K_m} \frac{\frac{\varepsilon_p}{\varepsilon_m} + 2}{Du + 1} \quad (56)$$

$R^\pm$, H , S , A , and Du are given either by the standard electrokinetic model expressions (23), (32)–(41), or by Eqs. (49)–(54) if surface conductivity behind the slipping plane is taken into account.

The asymptotic validity of the superposition approximation is determined by the strong inequality expressing the very small value of relaxation time τ_{MW} (corresponding to the Maxwell–Wagner–O’Konski dispersion) as compared to τ_α , corresponding to the low-frequency dispersion (related to the volume diffusion mechanism):

$$\tau_{\text{MW}} \ll \tau_\alpha. \quad (57)$$

The superposition approximation does not take into account the mutual influence of the volume diffusion mechanism (inherent to the low-frequency dispersion) and the displacement current (inherent to the Maxwell–Wagner–O’Konski dispersion) in the process of particle polarization. But if condition (57) holds, the frequencies at which the volume diffusion mechanism is essential for the particle’s polarization ($\omega \approx 1/\tau_\alpha$) happen to be very small ($\omega \ll 1/\tau_{\text{MW}}$), so that displacement currents are negligible with respect to conduction currents. On the other hand, the frequencies at which these two types of currents are comparable ($\omega \approx 1/\tau_{\text{MW}}$) are so large that the volume diffusion mechanism has a negligible influence on the fields and currents ($\omega \gg 1/\tau_\alpha$). The value of τ_{MW} is close to the value of electrolyte relaxation time (1) for the majority of typical systems:

$$\tau_{\text{MW}} \cong \frac{\varepsilon_m}{K_m} = \tau_m. \quad (58)$$

From Eqs. (31) and (58) it follows that:

$$\frac{\tau_\alpha}{\tau_{\text{MW}}} \cong (\kappa a)^2 \gg 1. \quad (59)$$

This means that the strong inequality (57) that allows to use the superposition approximation, is closely related to the inequality (31), describing the small thickness of the Debye atmosphere as compared to particle radius. Therefore, the superposition approximation that leads to the general formula (55) for the dipole coefficient, happens to

be applicable just in the case of thin double layers, for which both theories: the theory of volume diffusion mechanism of the low-frequency dispersion and the theory of the Maxwell–Wagner–O’Konski dispersion, are applicable.

In order to check the accuracy of the wide-frequency range formula (55), the frequency depen-

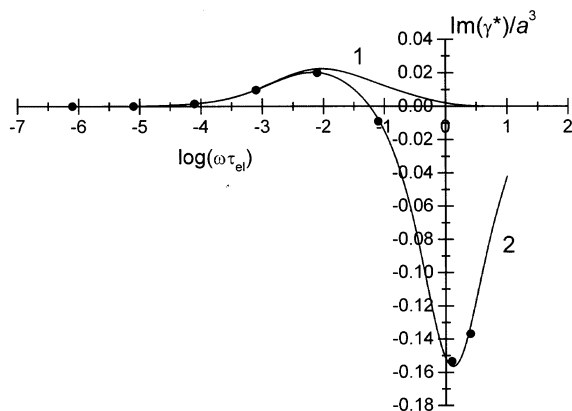


Fig. 1. Comparison between dipole coefficient data obtained by means of Eq. (55) (curve 2) with numerical calculations [25] (symbols), for a KCl solution with $\kappa a = 25.12$. Dipole coefficient data obtained with the LFDD theory [3,6] are also included (curve 1). Parameters used: $\epsilon_m = 80\epsilon_0$; $\epsilon_p = 2\epsilon_0$; $K_{\text{per}} = 0.875 \cdot 10^{-1} \text{ S m}^{-1}$; $D^+ = D^- = 2 \cdot 10^{-9} \text{ m}^2 \text{ s}^{-1}$; $\eta = 10^{-3} \text{ g (m s}^{-1})$; $\zeta = 100 \text{ mV}$.

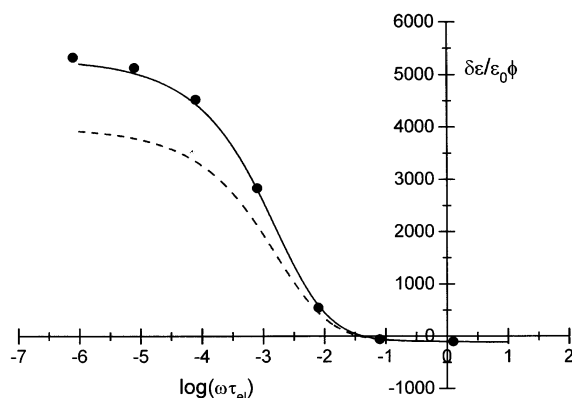


Fig. 2. Comparison of analytical (curves) and numerical (symbols) results of the wide-frequency range dependence of the suspension permittivity. Solid line: full calculation based on the standard model; dashed line: results obtained neglecting coion and capillary–osmotic contributions. Same parameters as in Fig. 1.

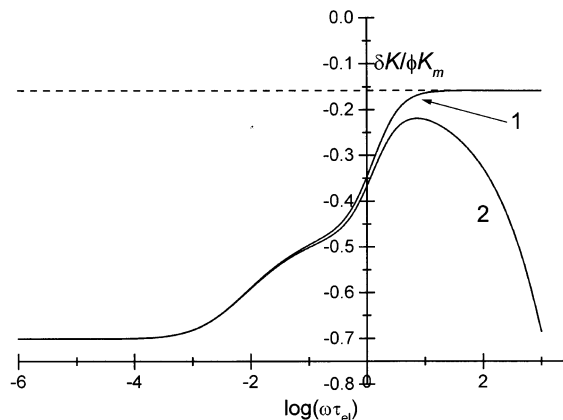


Fig. 3. Frequency dependence of the suspension conductivity. Curve 1: modified formula (61); curve 2: direct substitution of Eq. (55) into (11). The dashed horizontal line shows the high frequency limit of the Maxwell–Wagner–O’Konski dispersion of the conductivity, Eq. (62). Same parameters as in Fig. 1.

dence of the dipole coefficient, for the case of the standard model, has been compared with numerical calculations [25]. The results for the imaginary part of the dipole coefficient, $\text{Im} [\gamma_p^*(\omega)]$ are represented in Fig. 1 (curve 2). As can be seen, the analytical results are confirmed by numerical calculations, with a relative error lesser than 3% for $\kappa a > 25$, in the frequency range including both the volume diffusion and the Maxwell–Wagner dispersions. Curve 1 shows the dipole coefficient calculated using the low-frequency expression [Eq. (32)].

The frequency dependence of the suspension permittivity in the wide-frequency range that including both dispersions, can be obtained by direct substitution of Eq. (55) into Eq. (10). Fig. 2 demonstrates a good agreement of $\delta\epsilon(\omega)$ calculated in this way using the coefficients corresponding to the standard model, Eqs. (30), (32)–(41), with the numerical results of DeLacey and White [25]. The dashed curve represents the analytical results obtained using the coefficients (44)–(48) (in which the coion and the capillary–osmotic contributions to the surface ionic flows are neglected) with $\Theta = 0$.

As for the calculation of the frequency dependence of the suspension conductivity, a direct substitution of (55) into (11) leads to a wrong

behavior of $\delta K(\omega)$ for very high frequencies, exceeding ω_m (see Fig. 3, curve 2). The origin of this incorrect behavior is the high frequency asymptotic form of the last term in the right-hand side of Eq. (55), which is of $O(1/\sqrt{\omega\tau_\alpha})$ in its imaginary part. The product of this asymptotic dependence and the imaginary part of K_m^* [equal to $\omega\epsilon_m$, Eq. (11)] causes the divergence of $\delta K(\omega)$. This slowly decreasing high-frequency asymptotic term is a consequence of the approximation of local equilibrium in the theory of the volume diffusion mechanism. This approximation works well when the period of the alternating field greatly exceeds the relaxation time of the screening charge (diffuse layer) τ_m . However, when the frequency reaches values comparable to τ_m and higher, the diffuse layer can not follow the field-induced concentration variations. Furthermore, for these high frequencies, the concentration variations themselves occur inside the diffuse layer, and it is no longer possible to assume the existence of 'local equilibrium of the field-induced variations of the parameters of the polarized double layer with the field-induced variations of the electrolyte solution concentration adjoining the double layer'.

Due to the these two factors, the slowly decreasing high frequency asymptotic form of $\gamma_p^*(\omega)$, given in Eqs. (32) or (46), must fully vanish for high frequencies, i.e., above ω_m . The influence of the concentration polarization on the dipole coefficient, is reflected by the last addend in the right hand side of Eq. (55). Hence, a possible way to take into account these arguments in the expression for the wide frequency-range dipole coefficient $\gamma_p^*(\omega)$, is to modify Eq. (55) multiplying its last addend by some function $F(\omega\tau_m)$, which is equal to unity at low frequencies and rapidly decreases when the frequency increases. A suitable function is:

$$F(\omega\tau_{cl}) = e^{-\omega\tau_{cl}} = \exp\left(-\omega\frac{\epsilon_m}{K_m}\right). \quad (60)$$

This dependence guarantees the correct influence of the concentration polarization when condition (3) is fulfilled (i.e. when the double layer may be considered as locally in equilibrium), and simultaneously eliminates the parasitic, slowly de-

creasing high frequency behavior of $\gamma_p^*(\omega)$. The resulting modified expression for the wide-frequency range dipole coefficient is:

$$\begin{aligned} \frac{\gamma_p^*}{a^3} = & \frac{\epsilon_p}{\epsilon_m} - 1 - \frac{3}{2} \frac{\epsilon_p - 2Du}{\epsilon_m} \frac{1}{\left(\frac{\epsilon_p}{\epsilon_m} + 2\right)(Du + 1)} \frac{1}{1 + i\omega\tau_{MW}} \\ & - \frac{3(R^+ - R^-)H}{2AS} \frac{1 + \sqrt{2/S}\sqrt{i\omega\tau_\alpha}}{1 + \sqrt{2/S}\sqrt{i\omega\tau_\alpha} + i\omega\tau_\alpha} \\ & \times \exp\left(-\omega\frac{\epsilon_m}{K_m}\right). \end{aligned} \quad (61)$$

In the high frequency range, where the relative difference between Eqs. (55) and (61) is significant, the absolute value of $\text{Im} [\gamma_p^*(\omega)]$ is very small, and the difference cannot be appreciated in Fig. 1. The particle's contribution to the dielectric permeability, $\delta\epsilon(\omega)$, calculated combining Eq. (10) and either Eq. (55) or (61), are also very close and indistinguishable in Fig. 2.

3. Discussion

The frequency dependence of the particle's contribution to the conductivity $\delta K(\omega)$, calculated combining Eqs. (11) and (55) for the case of the standard model, is represented by curve 2 in Fig. 3. The dashed horizontal line shows the value of the well-known (see, for example [33]) high-frequency limit of the Maxwell–Wagner–O'Konski conductivity dispersion, δK_{MW}^h :

$$\frac{\delta K_{MW}^h}{\phi K_m} = \frac{3(\epsilon_p - \epsilon_m)}{\epsilon_p + 2\epsilon_m} + \frac{9\epsilon_m \left(\frac{K_{pef}}{K_m} \cdot \epsilon_m - \epsilon_i\right)}{(\epsilon_p + 2\epsilon_m)^2}. \quad (62)$$

For non-conducting particles with thin diffuse double layer, aDu must be substituted for the quotient K_{pef}/K_m appearing in this expression. It is clear from Fig. 3 that the modified formula (61) effectively eliminates the parasitic high-frequency behavior of $\gamma_p^*(\omega)$ and $\delta K(\omega)$, while simultaneously maintaining the correct low-frequency behavior. This is demonstrated by the fact that $\delta K(\omega)$ reaches exactly the high-frequency limit (62).

The low-frequency limit of $\delta K(\omega)$ is, in agreement with Eq. (11), proportional to the low frequency limit of $\gamma_p^*(\omega)$, which is represented by the sum of the first two terms in the right-hand side of Eq. (32). Expressing this limit in terms of Dukhin's number with the help of Eqs. (44)–(47), and using Eq. (41) (for Δ) and (48) (for t_c), leads to:

$$\begin{aligned} \delta K|_{\omega \rightarrow 0} &= 3\phi K_m \gamma_p^*(\omega)|_{\omega \rightarrow 0} \\ &= 3\phi_p \left[\frac{Du - 1}{4Du + 2} \pm \frac{3z\Delta Du^2}{(4Du + 2)(Du + t_c)} \right]. \end{aligned} \quad (63)$$

The upper (lower) sign in the right hand side of this expression, corresponds to a positively (negatively) charged particle. The first addend between the square brackets corresponds to the static (direct current) dipole coefficient. It should be emphasized that, according to (63), the low-frequency limits of $\delta K(\omega)$ and $\gamma_p^*(\omega)$ only coincide with their corresponding static values for the case $\Delta = 0$, i.e. when the diffusion coefficients of coions and counterions are the same.

The cause of this difference is the contribution to the dipole coefficient of the volume charge distribution that arises in the non-static clouds of electrolyte concentration variations δc , which are due to the different values of the anion and cation

diffusion flows. Formally, this volume charge manifests itself by the fact that the alternating field-induced potential distribution outside the double layer is described by the Poisson (not Laplace's) equation with its right-hand side proportional to $\sqrt{\omega\tau_\alpha}\delta c\Delta$ (see, for example, [1,3,6]). It is evident that since the ions with the highest mobility leave faster the regions with positive δc , the sign of the volume charge will coincide with the sign of the ion with lesser mobility (correspondingly the field-induced volume charge in the regions with negative δc will have the sign of the most mobile ion).

The field-induced concentration variation δc is positive near that particle pole where counterions move from the double layer into the bulk (and negative near the opposite pole where counterions move from the bulk towards the double layer. Therefore, the low-frequency limits of the dipole coefficient and of the suspension conductivity increment are larger than the corresponding static values if coions are more mobile than counterions, and vice versa. Consequently, in the case of more mobile coions, the low-frequency limit of the conductivity increment may substantially exceed the value $\frac{3}{4}\phi_p K_m$, which is the maximum value attainable by the static conductivity increment. The influence of the difference of diffusion coefficients on the low-frequency conductivity is shown in Fig. 4 for the case of a negatively charged particle.

Another interesting behavior corresponds to the high-frequency increment of the suspension conductivity, which may reach very high values exceeding by several decades the value $\frac{3}{4}\phi_p K_m$. This is to be expected for $\omega \gg \omega_{MW}$ if $Du(\zeta, \Theta)$ is so large that it gives rise to the inequality $\tau_{mW} \gg \tau_m$, [see Eq. (57)]. This phenomenon, shown in Fig. 5, is the result of the large real value of the product of the imaginary parts of the medium conductivity and the dipole coefficient.

As a final remark it is worth noting that while the theory presented in this work was developed for homogeneous non-conducting particles, some of the obtained results are applicable to more complex systems such as conducting particles surrounded by an insulating membrane (biological

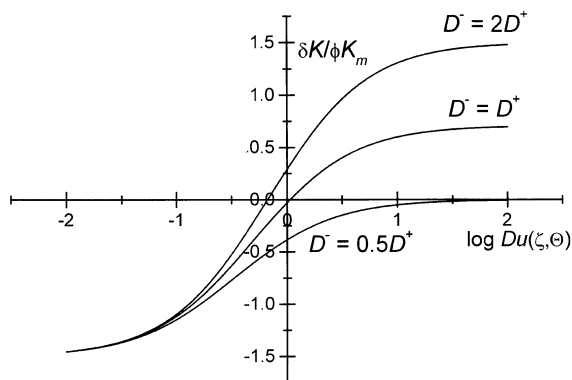


Fig. 4. Dependence of the low-frequency limit of the conductivity increment on Dukhin's number for the case of positive counterions. Same parameters as in Fig. 1 with the exception of coion diffusion coefficients, D^- , which are shown in the figure.

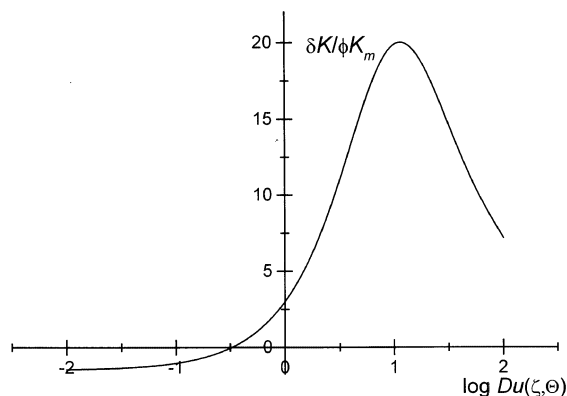


Fig. 5. Dependence of the high-frequency ($\omega = 10\omega_m$) increment of the suspension conductivity on Dukhin's number. Same parameters as in Fig. 1.

cells, for example). The limits of applicability are determined by the characteristic frequency associated to the charging of the cell membrane, which is of the order of [33,34]:

$$\omega_\beta = \frac{\frac{K_m h}{\varepsilon_{\text{mem}} a}}{\frac{K_m}{K_i} + \frac{1/2}{1 + Du(\zeta, \Theta)}} \quad (64)$$

where h is the thickness of the membrane, ε_{mem} its absolute permittivity, and K_i the conductivity of the internal medium. The results deduced for the low-frequency dispersion can be used for these systems as long $\omega < \omega_\beta$. On the contrary, the high-frequency results cannot be used in the form presented here, since at higher frequencies the capacitive current across the membrane becomes significant and the electric response of the system becomes dependent on the inner conductive properties that were not considered in this work. An approximate wide-frequency theory for these systems is presented in [34].

Acknowledgements

Financial support by CICYT, Spain (MAT98-0940, and sabbatical grant to V.N.S.) and by as CIUNT, Argentina, under Project 26/E120, are gratefully acknowledged.

References

- [1] V.N. Shilov, S.S. Dukhin, *Kolloidn. Zh.* 32 (1970) 117.
- [2] V.N. Shilov, S.S. Dukhin, *Kolloidn. Zh.* 32 (1970) 293.
- [3] S.S. Dukhin, V.N. Shilov, *Dielectric Phenomena and the Double Layer in Disperse Systems and Polyelectrolytes*, Wiley, New York, 1974.
- [4] J.Th.G. Overbeek, *Koll. Beihefte* 54 (1943) 287.
- [5] M. Fixman, *J. Chem. Phys.* 72 (1980) 5177.
- [6] C. Grosse, V.N. Shilov, *J. Phys. Chem.* 100 (1996) 1771.
- [7] I.N. Simonov, V.N. Shilov, *Kolloidnii Zh.* 39 (1977) 878.
- [8] I.N. Simonov, V.N. Shilov, *Kolloidnii Zh.* 39 (1977) 891.
- [9] C. Grosse, V.N. Shilov, *J. Colloid Interface Sci.* 178 (1996) 18.
- [10] Ju.Ja. Eremova, V.N. Shilov, *Kolloidnii Zh.* 37 (1975) 635.
- [11] Ju.Ja. Eremova, V.N. Shilov, *Kolloidnii Zh.* 57 (1995) 255.
- [12] C. Grosse, V.N. Shilov, *J. Colloid Interface Sci.* 193 (1997) 178.
- [13] C. Grosse, S. Pedrosa, V.N. Shilov, *J. Colloid Interface Sci.* 220 (1999) 31.
- [14] Y.U.B. Borkovskaja, V.N. Shilov, *Kolloidnii Zh.* 54 (1992) 173.
- [15] V.N. Shilov, Yu.B. Borkovskaya, *Kolloidnii Zh.* 56 (1994) 647.
- [16] A.V. Delgado, F.J. Arroyo, V.N. Shilov, Yu.B. Borkovskaja, *Colloids and Surfaces A* 140 (1998) 139.
- [17] C. Grosse, V.N. Shilov, *J. Colloid Interface Sci.* 211 (1999) 160.
- [18] C. Grosse, V.N. Shilov, *J. Colloid Interface Sci.* 225 (2000) 340.
- [19] J.C. Maxwell, *Electricity and Magnetism*, vol. 1, Dover, New York, 1954.
- [20] K.W. Wagner, *Arch. Electrotech* 2 (1914) 371.
- [21] C.T. O'Konski, *J. Phys. Chem.* 64 (1960) 605.
- [22] R.W. O'Brien, *J. Colloid Interface Sci.* 113 (1986) 81.
- [23] C. Grosse, *J. Chem. Phys.* 92 (1988) 3905.
- [24] S.S. Dukhin, V.N. Shilov. In: A.V. Delgado, (Ed.), *Interfacial Electrokinesics and Electrophoresis*, Marcel Dekker: New York, 2001, 55–85.
- [25] E.H.B. DeLacey, L.R. White, *J. Chem. Soc. Faraday. Trans. 2* (77) (1981) 2007.
- [26] J. Lyklema, *Fundamentals of Interface and Colloid Science*, Academic Press, New York, 1995 Chap. 4.
- [27] J.J. Bikerman, *Z. Physik.Chem.*, A163: (1933) 378; *Kolloid-Z.*, 72: (1935) 100
- [28] T.L. Chelidze, V.N. Shilov, *Kolloidnii Zh.* 35 (1973) 195.
- [29] J. Kilstra, H.P. van Leeuwen, J. Lyklema, *J. Chem.Soc. Faraday Trans.* 88 (1992) 3441.
- [30] J. Kilstra, H.P. van Leeuwen, J. Lyklema, *Langmuir* 9 (1993) 1625.

- [31] M. Minor, Ph.D. thesis, University of Wageningen, The Netherlands, 1998.
- [32] C. Grosse, M.C. Tirado, W. Pieper, R. Pottel, J. Colloid Interface Sci. 205 (1998) 26.
- [33] T. Hanai, Electric Properties of Emulsions, in: P. Sherman (Ed.), Emulsion Science, Academic Press, London, 1998.
- [34] C. Grosse, In: A.V. Delgado (Ed.), Interfacial Electrokinetics and Electrophoresis, Marcel Dekker: New York, 2000.

Theoretical and experimental study of time domain-induced polarization in water-saturated sands

K. Titov^{a,*}, V. Komarov^b, V. Tarasov^a, A. Levitski^c

^a*VIRG-Rudgofizika, 20 Fayansovaya Street, 193019 St. Petersburg, Russia*

^b*Geological Department, Saint-Petersburg State University, 6/7 Universitetskaya Emb., 199034 St. Petersburg, Russia*

^c*ExploTech Ltd., POB 3 Ariel, 44837 Israel*

Received 10 November 2000; accepted 17 May 2002

Abstract

A theoretical model of spectral-induced polarization (IP) of sand is presented. In the proposed model, contacts of sand grains and intergrain solution-filled space are considered as electrical current passages of varying thickness, which differ in values of ion transport number. Ion-selective narrow passages are considered as active zones, large passages as passive. The proposed model describes spectral IP characteristics for the medium where the length of passive zones is much greater than the length of active ones. The model is called short narrow pores (SNP) model. The SNP model predicts a growth of IP time constant with increase of length of ion-selective zone. Both the time domain and frequency domain parameters are described. The parameters of Cole–Cole model corresponding to the SNP model were also found. The behaviour of model parameters is compared with experimental data obtained on natural and sieved sands using time domain technique. The natural sand spectra correspond neither to the simple SNP model nor simple Cole–Cole model with single time constant because the lengths of ion-selective zones vary, reflecting the grain-size distribution. The spectra of sieved sand compared with the theoretical SNP spectra reveal close correspondence between experimental data and theoretical parameters. For four sieved sands, both the theoretical and experimental data show that the time constant of the IP is proportional to the square of the average grain size.

© 2002 Elsevier Science B.V. All rights reserved.

Keywords: Membrane polarization; Diffusion; Spectral-induced polarization; Ion transport; Number; Transient characteristic; Time constant

1. Introduction

The problem of distinguishing between sand units containing saline water and clay units on the basis of surface electrical methods is very important in environmental investigations at oil production fields. The DC or low frequency resistivity of these units is

frequently almost the same. Thus, the application of induced polarisation (IP) is recommended. We started this work with the purpose to use IP spectral characteristics to distinguish between different types of rocks and soils.

The phenomenon of induced polarization in rocks has been discovered by Schlumberger (1920). Pelton et al. (1978), Komarov (1980), Olhoeft (1985) and others studied the characteristics of spectral-induced polarization (SIP) in frequency domain (FD) or time domain (TD). Most of these studies were conducted in

* Corresponding author. Tel.: +7-812-567-98-95; fax: +7-812-567-87-41.

E-mail address: tito@kt4096.spb.edu (K. Titov).

the systems that included both electron and ion conductors (ore minerals and pore moisture). The most common approach is an approximation of frequency dispersion in rocks (both electron-ion and ion-conductive) using the Cole–Cole equation (Pelton et al., 1978):

$$\rho(\omega) = \rho_0 \left[1 - \eta_0 \left(1 - \frac{1}{1 + (i\omega\tau)^c} \right) \right], \quad (1)$$

where ρ is the complex resistivity (in $\Omega \cdot m$), ω is the angular frequency of the polarizing current (in s^{-1}), ρ_0 is the DC resistivity (in $\Omega \cdot m$), $i = (-1)^{1/2}$, $\eta_0 = U_p/U_i$ is the chargeability (in V/V) corresponding to polarization seen immediately after switch-off of an infinite long current step (U_p and U_i are the electrical potential drops in off-time and on-time, respectively), τ is the time constant (in s) and c is the dimensionless exponent.

The time domain analogue of the Cole–Cole equation (Pelton et al., 1978) describes the transient IP characteristic during switching-on of the polarizing current (on-time):

$$\eta(t) = \eta_0 \Psi_c(t) = \eta_0 \left[\sum_{k=1}^{\infty} (-1)^{k-1} \frac{(x)^{kc}}{\Gamma(1 + kc)} \right] \quad (2)$$

where $\Psi_c(t)$ is the normalised Cole–Cole transient characteristic that vary between 0 and 1, $\Gamma(z)$ is the gamma-function, $x = t/\tau$ is the dimensionless argument and t is on-time counted from the beginning of the current step.

The empirically defined constants (η_0 , τ and c) depend on the petrophysical properties of rocks and soils. However, it is not entirely clear which coefficients correspond to which properties of the rocks.

Marshall and Madden (1959) offered a theoretical model of IP (formulated in terms of frequency domain) based on the membrane polarization. In the model, sediments are liken to serial connection of active (ion-selective) and passive (nonselective) zones. The zones have different numbers of cation and anion transport, which produces local concentration gradients under applied external voltage. The local solute flows, appearing under these concentration gradients, result in additional electrical current in ion-selective zones, which yields additional electrical field and causes the frequency dependency of the

resistivity. The model predicts increase of the time constant with the growth of nonselective zones.

Both the Cole–Cole and Marshall–Madden models predict a maximum in the phase spectrum.

Fridriksberg and Sidorova (1961) developed similar approach (in terms of time domain) to predict chargeability of ion-conductive rocks. They expressed chargeability as follows:

$$\eta_0 = \frac{4(\Delta n)^2}{\left(\frac{l_1}{S_1} + \frac{l_2}{\alpha_2 S_2} \right) \left(\frac{S_1}{l_1} + \frac{S_2}{l_2} \right)} \quad (3)$$

where Δn is the difference of ion transport numbers in the active and passive zones; l_1 , S_1 , l_2 , S_2 are the lengths and sections of passive and active zones, respectively; α_2 is the coefficient of efficiency of active zone, which depends on the surface conductivity and represents a ratio of the average capillary conductivity and the bulk solution conductivity.

Based on this equation, Fridriksberg and Sidorova estimated the chargeability of water saturated soils to be 1...2%. The artificial model containing serial connection of selective membranes (cellophane or collodion) and passive zones (capron cloth) was experimentally studied. The predicted and experimental magnitudes of chargeability were found to be close, which supports accuracy of the model.

Kormiltsev (1963) offered the TD formulation of Marshall–Madden model and performed calculations showing influence of current waveform on the measured decay. The lengths of active and passive zones were assumed to be equal. The resulting formula was derived from basic solution of the equation of diffusion, and represented infinite series. The IP decay was found to be dependent on the difference in ion transport number in active and passive zones, the diffusion coefficient and the length of the zones.

Numerous empirical data on SIP in soils were published for the low frequency range ($10^{-2} \dots 10^4$ Hz) (Vinegar and Waxman, 1984; Börner et al., 1993, 1996; Vanhala, 1997, etc.). Three essential types of spectra emerge from these data.

(1) The phase spectrum is nearly constant over the observed frequency range. Therefore, the IP spectrum corresponds to neither Cole–Cole nor Marshall–Madden models. This type of spectra produce the Constant Phase Angle model (CPA) (Vinegar and Waxman, 1984; Börner et al., 1993, 1996).

(2) The phase spectrum has a maximum. Therefore, IP can be approximated by the Cole–Cole equation (Eq. (1)). The Marshall–Madden model can be also applied. Examples of such spectra are published in Vanhala, 1997.

(3) The phase spectrum increases with frequency increase (Vanhala, 1997). This type of spectra corresponds to neither the Cole–Cole nor the CPA models.

The goal of this paper is further development of spectral IP theory for water-saturated sand, based on

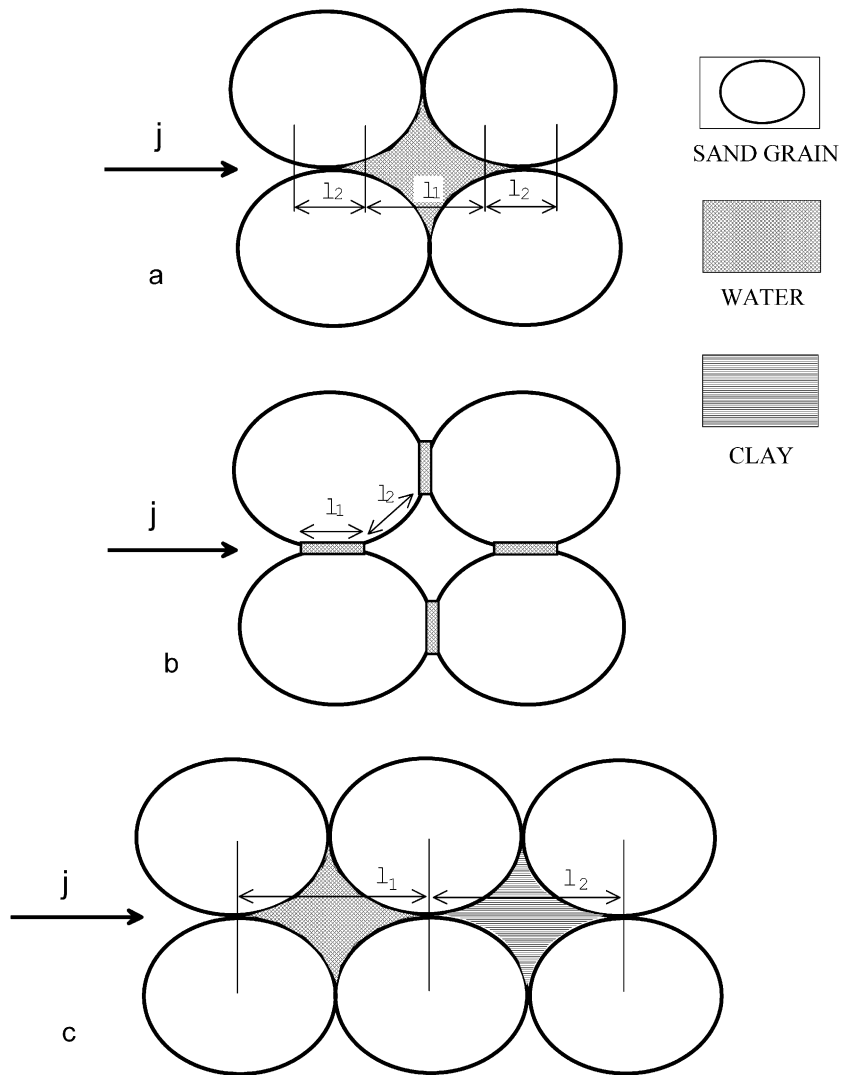


Fig. 1. Schematization of the electrical current passageway geometry in sandy-clay sediments. (a) Pure water-saturated sand. (b) Pure undersaturated sand. (c) Saturated sandy-clay sediments. In the saturated sand, water-filled intergrain space (where physico-chemical characteristics of the water are close to those of a free solution) can be considered as large pores of length l_1 . Areas of grain contact, where the water is in a closer bond on the grain surface, can be considered as narrow pores of length l_2 . In the undersaturated sand, the water film at the surface of moistened grains can be considered as a relatively narrow passage for electrical current (narrow capillaries of length l_2). Areas of grain contact, where water is more abundant, can be viewed as large capillaries of length l_1 . In sandy-clay water-saturated sediments, the large passages of the electrical current are water-filled pores between the sand grains (l_1), while the narrow pores are those filled with clay particles (l_2).

the fundamental Marshall–Madden model, as well as experimental verification of this theory. Because experiments were conducted in TD, theoretical assessments were also formulated in TD. The correspondence between TD and FD characteristics is described in Appendix C.

2. Theory

In ion-conductive rocks and soils, the electrical current passes through interconnected serial and/or parallel large and narrow pores. The scheme showing these large and narrow current passages in soils (Fig. 1) does not cover all possible types of pore space. However, it appears to be sufficient for our IP modelling. Let us assume that the pore moisture contains one type of cations and one type of anions, both of valence 1. The IP phenomenon is caused by

the difference between the ion transport numbers in large and narrow pores (Marshall and Madden, 1959; Fridrikhsberg and Sidorova, 1961; Vinegar and Waxman, 1984). This difference, in turn, is stipulated by different concentrations of cations and anions in the diffusion zone of the electrical double layer (EDL) (Fig. 2a) (Kruyt, 1952). Since the surface of the solid phase of soils is usually charged negatively, the cation concentration in the pore solution is higher than that of the anions. The thinner the pores, the greater the part of the pore volume occupied by EDL. Therefore, the greater the difference in average concentration between the cations and anions (Kruyt, 1952). It was shown that the electrical current flow through a sequence of large and narrow capillaries (Fig. 2b) results in deficiency of anions and surplus of cations at the boundaries of the thick and thin capillaries (Marshall and Madden, 1959; Fridrikhsberg and Sidorova, 1961). The cations surplus/deficiency q at

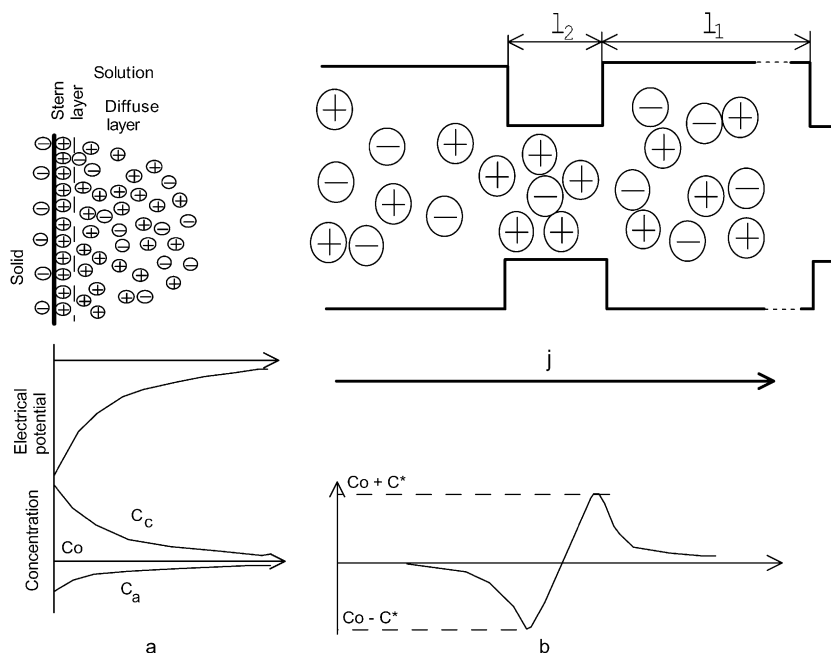


Fig. 2. IP scheme in ion-conductive rocks (according to Fridrikhsberg and Sidorova, 1961). (a) Electrical double layer (EDL) scheme, distribution of equilibrium potential and concentration in EDL. (b) Pore system and distribution of concentration under the influence of electrical current. For each pore, l_1 , S_1 , l_2 , S_2 are lengths and sections. Narrow pore is characterised also by the coefficient of efficiency α_2 (see also Section 1). C_0 —equilibrium concentration. C_c , C_a —cation and anion concentrations, respectively. C^* —additional concentration under the influence of electrical current.

the large/narrow capillary boundary can be described as follows:

$$q = \pm \frac{dm}{dt} = \frac{\Delta nj}{zF} \quad (\text{in mol s}^{-1}) \quad (4)$$

where j is the electrical current that flows through the capillary (in A), $z=1$ is the valence of the ions, F is Faraday's constant (in C/mol), $\Delta n = n_2^c - n_1^c = n_1^a - n_2^a$ is the difference of ion transport numbers in the large and narrow capillaries (n —ion transport number; the lower index 1—in large capillaries, the lower index 2—in narrow capillaries, the upper index c—cations, the upper index a—anions).

For small current density, we consider ion transport numbers $n(t)$ as invariable with time because the variation in the concentration is much smaller than the equilibrium concentration (the linear IP).

The total density of the pulse electrical current J^t includes a migration (Ω) component J^Ω and a diffusion (IP) component J^d , which can be expressed using the Ohm's and Fick's laws as:

$$\begin{aligned} J^t(\mathbf{r}, t) &= J^\Omega(\mathbf{r}, t) + J^d(\mathbf{r}, t) \\ &= \sigma E^\Omega(\mathbf{r}, t) - FRTu(\text{grad } C_c(\mathbf{r}, t) \\ &\quad - \text{grad } C_a(\mathbf{r}, t)) \end{aligned} \quad (5)$$

where σ is the conductivity of the solution (in $\Omega \cdot \text{m}$)⁻¹, E^Ω is the migration component of electrical field (in V/m), R is the universal gas constant (in J·(mol·K)⁻¹), T is the absolute temperature (in K), u is the ion mobility in (m·mol·N⁻¹s⁻¹), $C_c(\mathbf{r}, t)$ and $C_a(\mathbf{r}, t)$ are the cation and anion average concentrations in the pore solution (in mol/m³), $\mathbf{r}$ is the vector of space coordinates and t is the time.

Note that the diffusion coefficient and ion mobility are related as: $D = RTu$.

The ion average concentrations in pores C_c and C_a are linearly related to the equilibrium concentration in bulk solution C (that corresponds to concentration on the pore axis for the pore much larger than the Debye length):

$$C_c = \mu_c C \quad (6a)$$

$$C_a = \mu_a C \quad (6b)$$

$$\bar{C} = C_a + C_c \quad (6c)$$

where μ_a and μ_c are the dimensionless coefficients depending on the Debye length and pore shape; $\bar{C}$ is the total concentration averaged by pore section.

The Debye length is expressed as:

$$\lambda = \frac{1}{zF} \sqrt{\frac{\varepsilon_0 \varepsilon RT}{2C}} \quad (\text{in m}) \quad (7)$$

where ε_0 is the dielectric constant (in F/m) and ε is the dimensionless solution permittivity.

If cation and anion mobilities are equal (that is valid, e.g. for KCl), ion transport depends on average concentrations only:

$$n^c = \frac{C_c}{\bar{C}} = \frac{\mu_c}{\mu_c + \mu_a} \quad (8a)$$

$$n^a = \frac{C_a}{\bar{C}} = \frac{\mu_a}{\mu_c + \mu_a} \quad (8b)$$

The total electrical field E^t also consists of the migration component E^Ω and diffusion component E^{IP} . Taking into consideration the notation of ion transport number (Eqs. (8a) and (8b)), as well as relationship between the concentration and conductivity in the pore solution:

$$\sigma = (zF)^2 u \bar{C} \quad (\text{in } (\Omega \text{ m})^{-1}) \quad (9)$$

and using expression (5), E^{IP} can be described as follows:

$$E^{\text{IP}}(\mathbf{r}, t) = -\frac{RT}{F} (n^c - n^a) \frac{\text{grad } \bar{C}(\mathbf{r}, t)}{\bar{C}(\mathbf{r}, t)}, \quad (10)$$

Let us assume that the total ion concentration averaged by a pore section consists of the equilibrium concentration $\bar{C}_0$, and the additional concentration, due to the passing electrical current $C^*(\mathbf{r}, t)$: $\bar{C}(\mathbf{r}, t) = C^*(\mathbf{r}, t) + \bar{C}_0$. Since the process is considered linear (i.e. $C^*(\mathbf{r}, t) \ll \bar{C}_0$), Eq. (10) can be reduced to:

$$E^{\text{IP}}(\mathbf{r}, t) \approx -\frac{RT}{F} (n^c - n^a) \frac{\text{grad } C^*(\mathbf{r}, t)}{\bar{C}_0} \quad (11)$$

Expression (11) is integrated over the corresponding lengths l_1 and l_2 to obtain the electrical potential

drop at the system of one large and one narrow capillary (Fig. 2b):

$$\Delta U_1^{\text{IP}}(t) = -2 \frac{RT}{F} (n_1^c - n_1^a) C_{12}^*(t) / \bar{C}_0 \quad (12a)$$

$$\Delta U_2^{\text{IP}}(t) = 2 \frac{RT}{F} (n_2^c - n_2^a) C_{12}^*(t) / \bar{C}_0 \quad (12b)$$

where $C_{12}^*(t)$ is the positive additional concentration at the large/narrow pore boundary, resulting from the passing electrical current.

In Eqs. (12a) and (12b), we ignored the difference in average concentration in large and narrow pores, which appears to be not of importance for sand.

The IP voltage drop at the system of the large and narrow pores is the sum of the potential differences in Eqs. (12a) and (12b):

$$\Delta U^{\text{IP}} = 4 \frac{RT}{F} \Delta n \frac{C_{12}^*(t)}{\bar{C}_0} \quad (13)$$

The migration component of the voltage drop at the system of the large and narrow pores is expressed using Ohm's law:

$$\Delta U^{\Omega} = \frac{j}{\sigma_0} \left(\frac{l_1}{S_1} + \frac{l_2}{\alpha_2 S_2} \right) \quad (14)$$

where σ_0 (in $\Omega \cdot \text{m}^{-1}$) is the conductivity of equilibrium bulk solution.

Combining Eqs. (13) and (14), one can obtain an expression for polarizability where time dependence is represented by $C_{12}^*(t)$:

$$\eta = 4 \frac{RT \sigma_0 \Delta n}{jF \bar{C} \left(\frac{1}{L_1} + \frac{1}{\alpha_2 L_2} \right)} C_{12}^*(t) \approx 4 \times \frac{RT \Lambda \Delta n}{jF \left(\frac{1}{L_1} + \frac{1}{\alpha_2 L_2} \right)} C_{12}^*(t) \quad (15)$$

where Λ (in $\text{m}^2 \Omega^{-1} \text{mol}^{-1}$) is the molar conductivity of the solution, $\Lambda = (zF)^2 u$; $L_1 = S_1/l_1$, $L_2 = S_2/l_2$ (in m) are geometric coefficients of the pores.

The IP voltage is usually normalised by the total measured voltage (which includes both Ohm and IP components). However, in ion-conductive rocks and soils, the IP voltage does not exceed few percents of the

total voltage. Therefore, normalisation by the migration (Ω) voltage does not produce an important error.

Expression (15) shows that the TD characteristic of the IP is completely defined by the time variation in the ion concentration at the large/narrow capillaries boundary during the polarization cycle (on-time plus off-time). To evaluate the time variation in concentration $C_{12}^*(t)$, we have to solve the equation of diffusion for the conditions defined by the current waveform, as well as by the length and section of large and narrow capillaries.

According to definition of TD transient characteristic (Wait, 1982), let us consider the polarization of a medium caused by the infinite electrical current step. Based on the schematic structure of the current passages in the sand (Fig. 1a), let us also assume that the length of narrow capillaries is significantly less than the length of large ones:

$$l_2 \ll l_1 \quad (16)$$

Assumption (16) produces the short narrow pores (SNP) model. The assumption is not valid for shale sand (Fig. 1c), but appears to be correct for saturated sand (Fig. 1a). The above assumption enables us to simplify the solution of equation of diffusion. Let us further assume that the ion diffusion coefficients in the large and narrow capillaries have the same value.

As long as concentration C^* is averaged by pore section, the equation of diffusion to be solved contains one space and one time variables:

$$D \frac{\partial^2 C^*}{\partial x^2} - \frac{\partial C^*}{\partial t} = q(l_2/2) - q(-l_2/2) \quad (17)$$

where $q(l_2/2)$ and $-q(-l_2/2)$ are the source and discharge active during the polarization pulse, situated on the boundary of narrow/large pores with coordinates $(l_2/2)$ and $(-l_2/2)$, respectively. The capacity of the sources $q = \pm dm/dt$ (see expression (4)). The sought concentration C^* is obtained by solving Eq. (17):

$$C_{12}^* = \frac{ql_2}{D(S_1 + S_2)} \gamma(t, \tau) \quad (18)$$

$$\gamma(t, \tau) = \frac{1}{\sqrt{\pi}} \sqrt{\frac{t}{\tau}} \left(1 - \exp\left(-\frac{\tau}{t}\right) \right) + \text{erfc}\left(\sqrt{\frac{\tau}{t}}\right) \quad (19)$$

$$\tau = \frac{l_2^2}{4D} \quad (20)$$

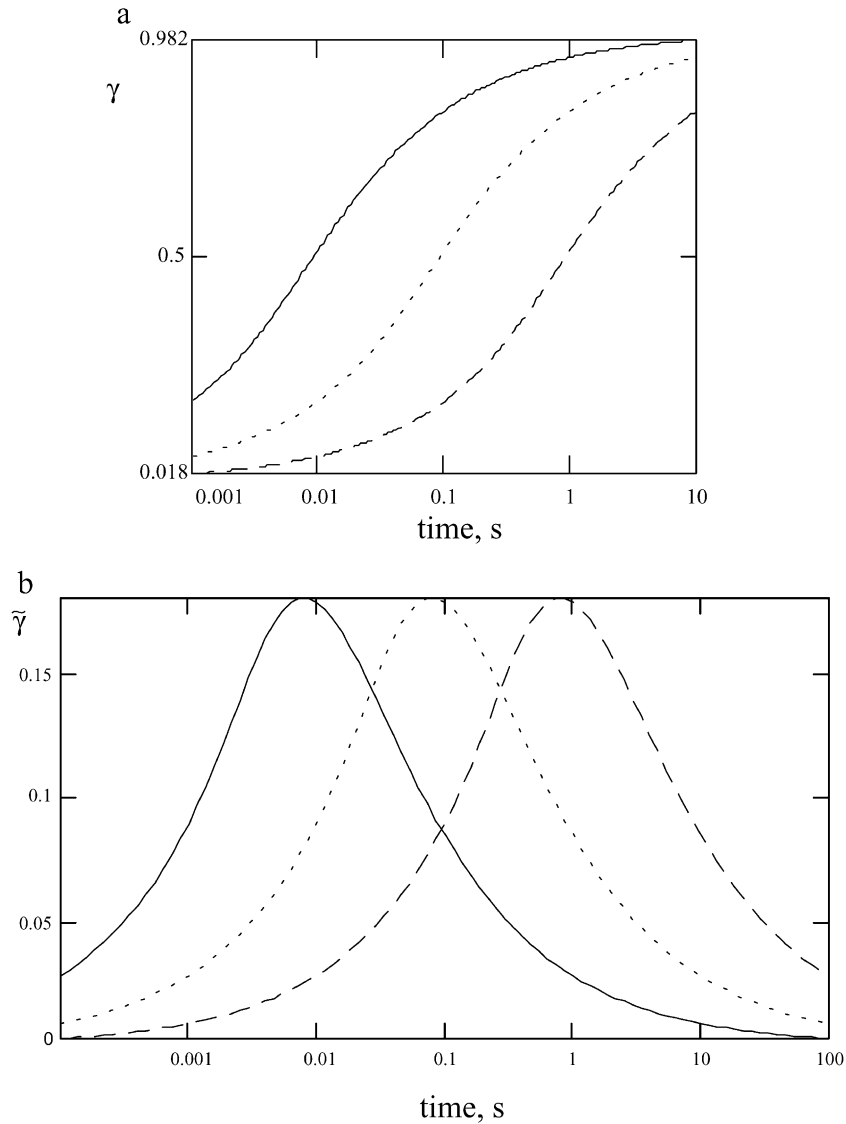


Fig. 3. Normalised transient characteristic of IP (a) and normalised differential polarizability (b) vs. time calculated using Eqs. (19) and (24), respectively. Solid line: $\tau = 0.01$ s, dashed line: $\tau = 0.1$ s, dash-dotted line: $\tau = 1.0$ s. Note that the maxima are slightly shifted to the left from the corresponding values of τ .

where $\gamma(t, \tau)$ is the dimensionless normalised transient characteristic that vary between 0 and 1; τ (in s) is the time constant.

For the solution details, see Appendix A.

Substituting Eq. (4) to Eq. (18) and Eq. (18) to Eq. (15), taking into consideration relationships between ion mobility and diffusion coefficient, and between

molar conductivity and ion mobility, one can obtain transient characteristic of polarization:

$$\eta = \frac{4l_2(\Delta n)^2}{(S_1 + S_2)\left(\frac{1}{L_1} + \frac{1}{\alpha_2 L_2}\right)} \gamma(t, \tau) = K(\Delta n)^2 \gamma(t, \tau) \quad (21)$$

where K is the dimensionless geometric factor. Since $\gamma(t, \tau)$ is the normalised transient characteristic, the product $K(\Delta n)^2$ corresponds to chargeability (one can compare with Eq. (2)):

$$\eta_0 = K(\Delta n)^2 \quad (22)$$

Fig. 3a depicts normalised transient characteristics of IP calculated for three values of τ . The differential polarizability and normalised differential polarizability formulae are obtained by differentiating Eqs. (21) and (19) by time logarithm (for details of these parameters, see Section 3):

$$\eta_d = K(\Delta n)^2 \tilde{\gamma} \quad (23)$$

$$\begin{aligned} \tilde{\gamma} &= \frac{\partial \gamma}{\partial (\ln t)} = \sqrt{\frac{t}{4\pi\tau}} \left(1 - \exp\left(-\frac{\tau}{t}\right) \right) \\ &= \sqrt{\frac{x}{4\pi}} \left(1 - \exp\left(-\frac{1}{x}\right) \right) \end{aligned} \quad (24)$$

where dimensionless $x = \{t/\tau\}$.

The Eq. (24) was used to calculate the normalised differential polarizability shown in Fig. 3b. The abscises of the maxima ($t_{\max}$) in the graphs are related to the values of τ as:

$$t_{\max} = 0.794\tau \quad (25)$$

We also show comparison between Cole–Cole and SNP models in TD (Appendix B) and FD (Appendix C).

Eqs. (23) and (24), taking into consideration the notation (20) and formula (25), can be used to interpret experimental data.

3. Experimental

3.1. Studied characteristics

Wait (1982) defined the TD transient characteristic as a time variation in the IP voltage in the course of polarization caused by an infinite unit current step. Komarov (1980) proposed to use the derivative of this transient characteristic by the

logarithm of time (differential polarizability, η_d) for theoretical and experimental evaluation of the IP decay. Note that the time derivative of the TD transient characteristic corresponds to IP decay from the pulse of infinitesimal duration. Therefore, η_d represents the product of the above IP decay and the time. In the Cole–Cole model, this differential polarizability versus time produces a maximum corresponding to the maximum in phase shift (ϕ) versus frequency. The behaviours of the $\eta_d(t)$ and $\phi(1/\omega)$ are similar.

Komarov (1980) also developed a calculation technique to obtain the $\eta_d(t)$. An electrical current pulse is considered as a superposition of two current steps of the opposite polarity, which correspond to the front and end of the pulse, respectively. Therefore, the signal measured in the off-time represents the finite difference of the two transient characteristics, mutually displaced along the time axis for the duration of the pulse. If the transient characteristic versus time logarithm is nearly linear, the differential polarizability can be obtained by dividing the measured signal by the corresponding coefficient:

$$P = \log(t_i + t) - \log(t) = \log(1 + t_i/t) \quad (26)$$

where t is the measurement time, counted from the end of the pulse, t_i is the duration of the pulse.

This differential polarizability presents the following advantages comparing to IP decay:

- the η_d slowly changes with time (for data corresponding to the constant phase angle model) or contains maximum (for data corresponding to the Cole–Cole and similar models), which makes it easier to analyse instead of IP decay;
- the data obtained using various duration of pulses can be rewritten in term of differential polarizability and then compared;
- the behaviour of $\eta_d(t)$ is similar to $\phi(1/\omega)$, which allows easy comparison of data obtained in TD and FD.

Examples of the use of differential polarizability are abundant in Russian publications. One can find examples presented in English in Mikhailov et al. (1986).

3.2. Experimental procedure

The measurement setup is shown in Fig. 4. STROB-M (Shereshevsky et al., 1999) is a multifunctional measuring device developed in the Laboratory of Electrical and Electromagnetic Methods of the Russian Institute of Exploration Geophysics (VIRG-Rudgeofizika). It was designed for field and laboratory measurements in the TD electromagnetic and IP methods. The dynamic range of the measured voltage values is 160 dB, time range varies from 5 μ s to 10 min. Measurements are controlled by a built-in micro-processor.

The laboratory transmitter used in the experiments was capable of switching the current off within several microseconds. The receiver and transmitter were synchronised using quartz timer. Rectangular current pulses of opposite polarity were transmitted through the sand samples. We used pulses and pauses of the same duration. The measurements were conducted during off-time, when the transmitter resistivity was 200 M Ω , input resistivity of the measuring device was 90 M Ω . In the above conditions, the influence of the sample discharge through the current and potential circuits was minimal. Voltage was measured in the pauses between the current pulses of the

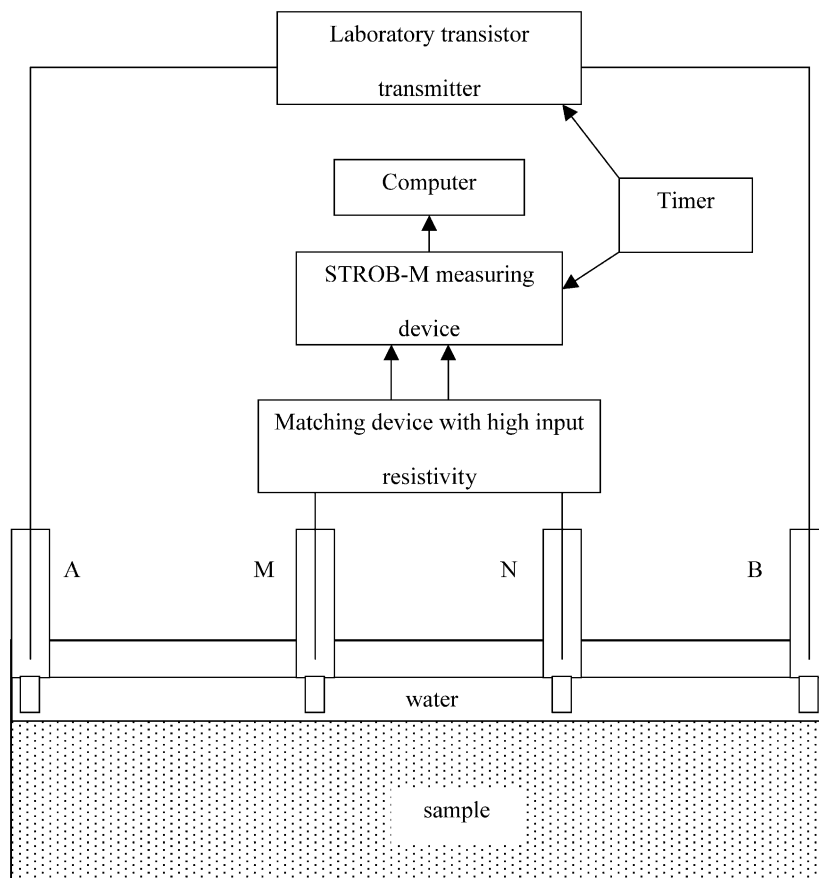


Fig. 4. Block-diagram of the laboratory TD spectral IP measurement system. Cu–CuSO₄ nonpolarizing electrodes (A, M, N and B) were positioned in the water layer (2–3 mm) that covered the sand that filled the vessel. A symmetric measurement array was used.

opposite polarity. As usual in the TD technique, the voltage decay was sampled through a sequence of windows. We sampled each IP decay 50–200 times. The value recorded for each window is expressed as:

$$\eta = \frac{100}{U_i \cdot (t_{k+1} - t_k)} \int_{t_k}^{t_{k+1}} U(t) dt \quad (\text{in } \%) \quad (27)$$

where t_k, t_{k+1} are the limits on the integration and U_i is the time average of the measured voltage during the current on-time.

To reduce electromagnetic noise, the pulse duration, time windows and number of stacked pulses were varied when early (0.001–0.1 s), medium (0.1–10 s) or late (10–300 s) polarization was studied. The pulses of 0.1–300 s were used depending on the time range studied. When the early time polarization was studied, the short pulses (0.1–1 s) and narrow time windows (up to 10 ms) were used; up to 100 pulses was stacked. When the late time polarization was studied, the long pulses (30–300 s) were used, time windows became longer (up to 160 ms); up to 10 pulses were stacked.

As long as the stack mode was used, the simple method, described in Section 3.1, was modified to obtain the differential polarizability. The measured voltage decay can be represented by the superposition of transient characteristics corresponding to each front and end of the pulses:

$$\frac{\Delta U_p(t)}{\Delta U_i} = \eta_0 \sum_{k=0}^N (-1)^k (\Psi(t + (2k+1)t_i) - \Psi(t + 2kt_i)) \quad (28)$$

where N is the number of stacked pulses, the product $\eta_0 \Psi(t)$ is an experimental transient characteristic, t is the time counted from the end of the last pulse. Unknown quantity $\eta_0 \Psi(t)$ is represented as a polynomial:

$$\eta_0 \Psi(t) = \sum_{m=1}^M a_m (\log(t))^m \quad (29)$$

of order $M=3$. Coefficients of polynomial a_m are obtained using the least-square method. The differential polarizability was calculated using logarithmic differentiation of the determined polynomial. We applied the above algorithm to the IP decay in a

floating time window to obtain the better correspondence between the measured and approximate voltages.

First measurements on the tap water (salinity 100 mg/l) without sand were conducted to control accuracy of the measurement setup. Fig. 5 shows the apparent differential polarizability obtained in the water. The values stipulated by the electrode polarization and other types of noise greatly diminished with time, and after the first 20 ms read less than 0.01%.

Then two samples of natural silicate sand with different grain-size distribution were studied. The coarse-grained white sand No. 1 was taken from the beach of Gulf of Finland. The sample No. 2, from eluvium, contained dust particles and had reddish tint, showing presence of colloidal iron-containing minerals. The unwashed samples were saturated with the same tap water.

The natural silicate sand of mixed composition was sieved to four grain sizes: 0.8–1.5, 0.5–0.8, 0.3–0.5 and 0.2–0.32 mm. Clay and dust particles were removed by repetitive washing of the sand in the same tap water. The IP decays of these sieved sands were measured.

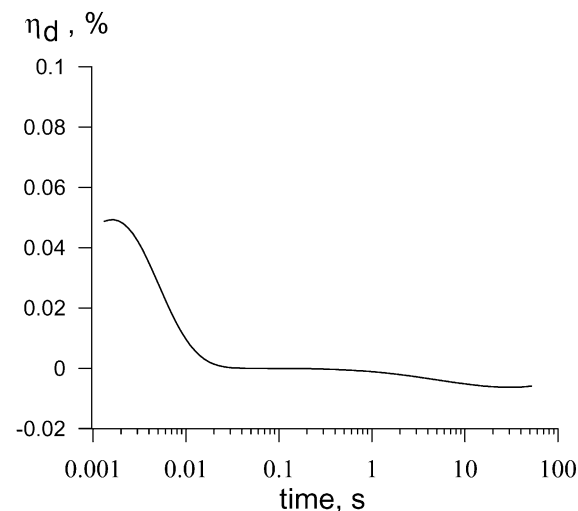


Fig. 5. Apparent differential polarizability obtained on the water. Note that when $t > 0.02$ s, the apparent differential polarizability becomes lower than 0.01%. Low values stipulated by the electrode polarization confirm accuracy of the measurement setup.

3.3. Results

3.3.1. Natural sands

The experimental data are presented in the Fig. 6 as the differential polarizability versus time. The differential polarizability values shown in the graphs are noticeably low (less than 1%), which is typical of sands. The η_d decreases with time. Sands studied in FD show the same behaviour in phase angle $\phi(1/\omega)$ (Vanhala, 1997). Therefore, neither the Cole–Cole nor constant phase angle model corresponds with the data in the entire time range.

The grain-size distributions show that for the first sample the dominant fraction is two steps greater than

that for the second sample. Small particles are less abundant in the first sample.

One can see the following features of differential polarizability in these natural samples:

- in the early time range (0.001–0.1 s), the first sample shows greater values than the second sample;
- in the medium time range (0.1–1 s), the behaviour of the both samples is similar;
- in the end of medium and the late time range (>1 s), the differential polarizability is small but well readable for the first sample, and practically disappears for the second sample.

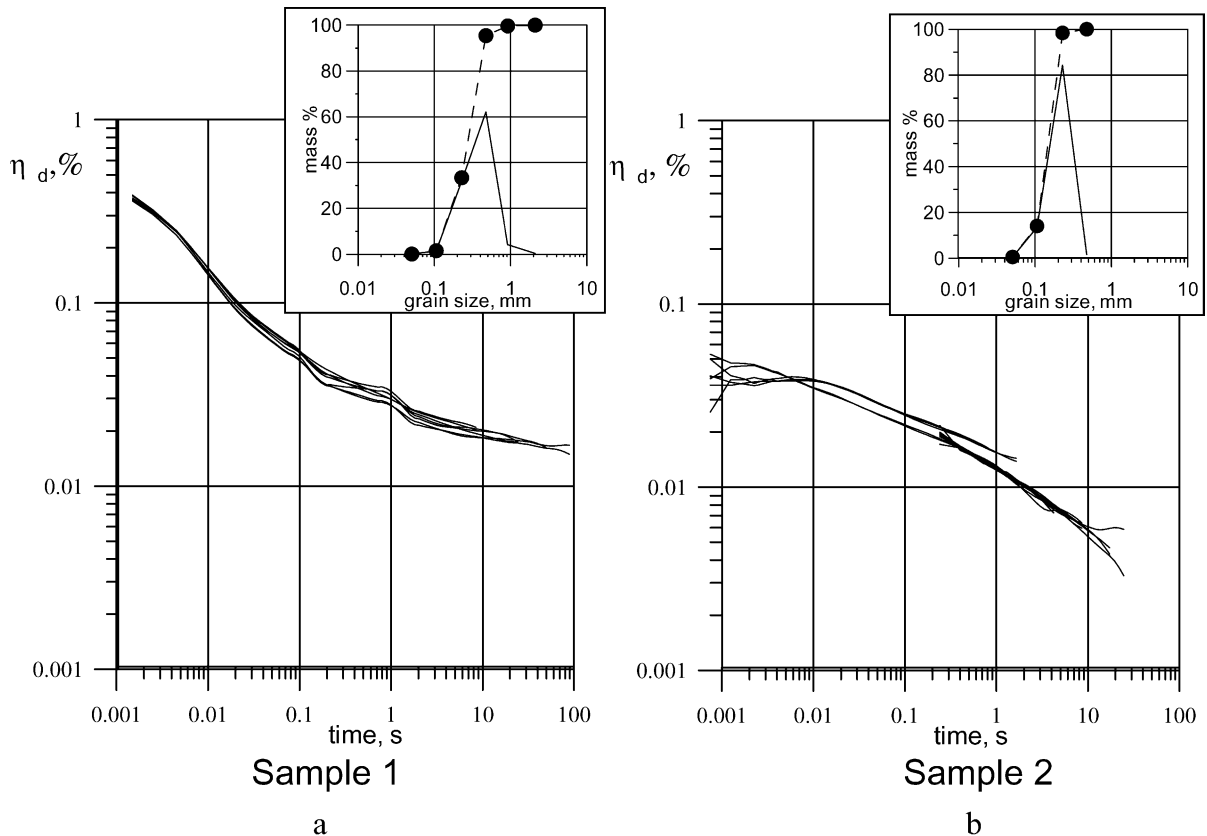


Fig. 6. Experimental data on differential polarizability in the water-saturated natural sand (a—sample 1, b—sample 2). Each continuous curve represents the differential polarizability obtained from current pulse of the constant duration t_i . Convergence of data obtained from pulses of various duration confirms accuracy of measurements and data treatment. Each measurement was repeated four to six times. To demonstrate good reproducibility, all the data are presented. Grain size distribution is given in solid lines and the corresponding cumulative in dashed lines with filled circles.

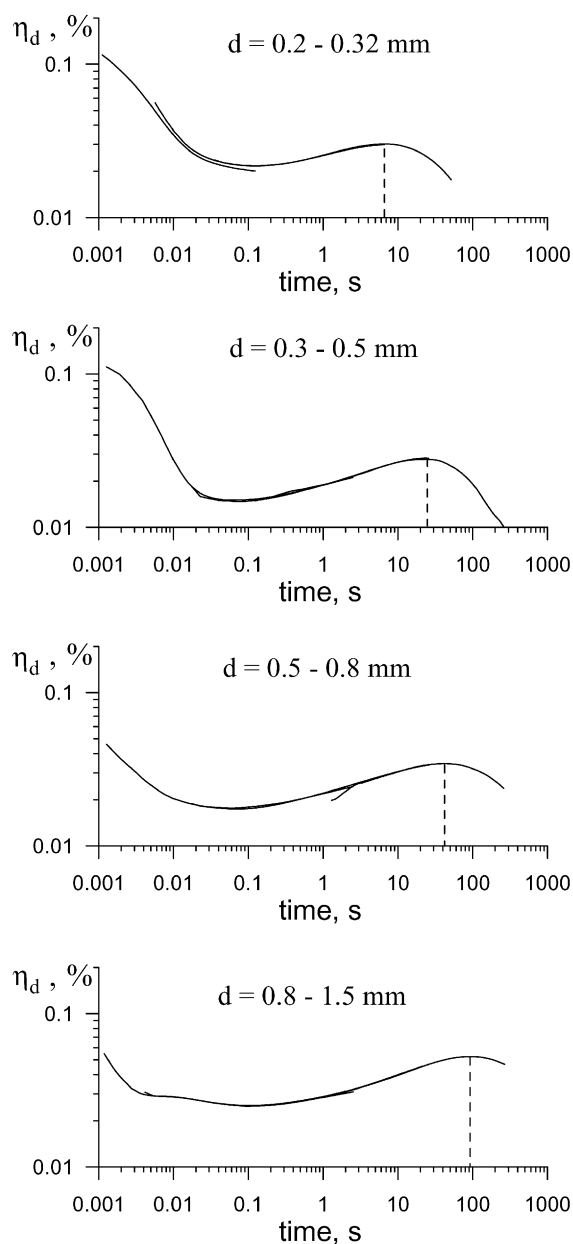


Fig. 7. Experimental data on the differential polarizability in the water-saturated sieved sand. Each continuous curve represents differential polarizability obtained from the current pulse of a constant duration t_i . Note very close convergence of the data obtained from pulses of various duration.

3.3.2. Sieved sands

The data are presented in Fig. 7 as differential polarizability versus time. The sharp decrease in the

η_d values can be observed at early times. A similar decrease was also observed in pure water measurements. It is probably due to the polarization processes that take place at the potential and current electrodes. The electrodynamic (Maxwell–Wagner) polarization may also occur in early times range for the sand (see also Fig. 6, first sample) (Dukhin, 1971; Olhoeft, 1985). In all four graphs for sieved sands, a maximum of the differential polarizability (0.025–0.05%) is observed in the 7–100 s range. One can see that the time of maxima increases when average grain size increases. The parts of spectra containing maxima (0.1–300 s) can be approximated by the Cole–Cole formula.

4. Discussion

Like the Cole–Cole and Marshall–Madden models, the proposed SNP model is valid for a single type of polarization sources in soil. Therefore, the model is comparable with the data obtained in sieved sands where the lengths of narrow and large current passages are nearly constant.

Based on the experimental data (Fig. 7), we obtain the values of τ for each grain size of the sand from times of maximum (see Eq. (25)) to verify the correctness of the SNP model. The effective lengths of the narrow pores (l_2 values) in Table 1 are calculated on the basis of Eq. (20) using the diffusion coefficient for potassium chloride in bulk solution ($\sim 10^{-9}$ m²/s). In a system of spherical grains, the ratio of the particle size and the effective length of the narrow pore (linear size of the grain

Table 1
Experimental characteristics of differential polarizability in the sands of different grain sizes

Sample no.	Grain size (mm)	Average particle diameter (d , mm)	$t_{\max}$ (s)	τ (s)	l_2 (mm)	d/l_2
1	0.8–1.5	1.15	90	110	0.66	1.7
2	0.5–0.8	0.65	40	50	0.45	1.4
3	0.3–0.5	0.40	22	28	0.33	1.2
4	0.2–0.32	0.26	7.0	8.8	0.19	1.4

$t_{\max}$ —the time of maximum of differential polarizability, τ —the time constant according to Eq. (25), l_2 —the length of narrow pore according to Eq. (20).

contact) is nearly constant ($d/l_2 \approx l_1/l_2 \sim \text{constant}$). Close values of the calculated ratio d/l_2 for four sieved sands confirm the accuracy of the proposed model (Table 1). The average value of the above ratio is 1.4. Note that the calculated ratio strongly depends on the diffusion coefficient value.

Based on the aforementioned average value of d/l_2 , the time constant τ and the sand grain size (d , mm) can be related as:

$$d \approx 0.09\sqrt{\tau} \quad (30)$$

where the empirical factor 0.09 is given in $\text{mm} \cdot \text{s}^{-1/2}$ (Fig. 8).

The obtained ratio d/l_2 allows estimation of chargeability η_0 of sands on the basis of the SNP model. Let us assume that $l_1^2 \sim S_1$, $l_2^2 \sim S_2$. In the above conditions, the geometric factor $K \sim 4 \cdot l_2^2/l_1^2$ (see Eq. (21)). $K \sim 4 \cdot l_2^2/d^2 \sim 4 \cdot 1.4^{-2} \approx 2$ because $l_1 \sim d$. Thus, $\eta_0 \sim 2(\Delta n)^2$, and the range of difference of ion transport number 0.1–0.05 produces $\eta_0 \sim 2 \dots 0.5\%$ that corresponds to the experimental data.

The data obtained on the natural sand (Fig. 6) do not contain maxima in the range 1–100 s, where

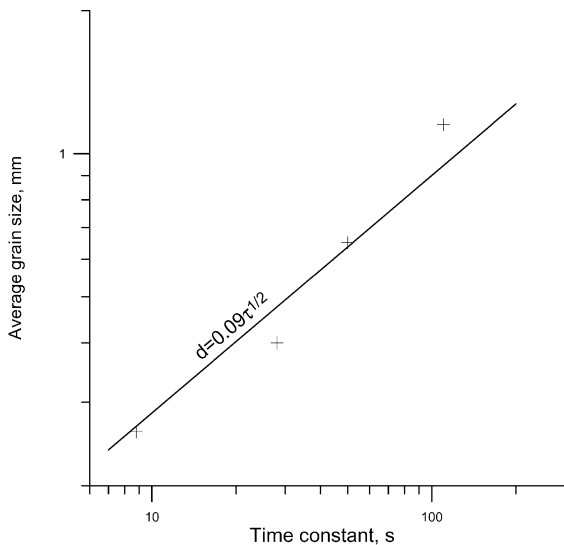


Fig. 8. Average grain size vs. time constant for sieved sands: crosses are experimental data, solid line is the approximation.

sieved sands show maximum. As long as the different lengths of narrow and large pores occur in natural sand, different values of τ are produced. If the soil presents a monotonous distribution of τ in the studied time range (which corresponds to the monotonous grain-size distribution), the differential polarizability versus time (or phase angle versus frequency) can be monotonous. In case of constant distribution of τ , the constant phase angle occurs. Thus, models containing single τ do not properly describe soils with mixed grain size.

We assume that the quantity of particles of a given size “ d ” in a volume unit determines the contribution of polarization with the time constant “ τ ” to the total polarization. Therefore, the inversion of IP spectra to grain-size distribution is theoretically possible.

The frequency dependency of IP contains three parameters in the Marshall–Madden’s model: selective zone length (l_2), ratio of selective/nonselective lengths (l_2/l_1) and number of cation transport (n^c). The proposed SNP model differs from the Marshall–Madden model because the l_1 is assumed to be noticeable greater than l_2 . Thus, the model contains only two variables (Δn) (which related to n^c) and τ (which related to l_2). Therefore, the SNP model represents a special case of the Marshall–Madden’s model proposed for sands. Since the SNP model contains only two variables (one of which represents the multiplier), this model appears to be useful for an inversion of IP spectra to obtain grain-size distribution of sands.

Comparison of the IP spectra of the 1st and 2nd natural sand samples shows that for the coarse-grained sand the differential polarizability in the late time range is noticeable higher than that for the fine-grained sand, which is in agreement with other data obtained for sands (Vanhala, 1997).

In the early time range, large values of η_d were obtained only for washed sieved sands and for unwashed natural sand No. 1. However, the first sample was naturally washed because it was taken from the sea beach. The second sample, which contained dust and colloidal particles, did not show large polarization in the early time range. Although early polarization was observed only in few samples, we can assume that colloidal and dust particles control the early polarisation.

5. Conclusions

- Based on the phenomenon of membrane polarization, theoretical characteristics of spectral IP in pure sands, called small narrow pores (SNP) model, are obtained. The proposed model represents a special case of the Marshall–Madden model. The SNP model is based on the assumption that the narrow pores are noticeably shorter than the large ones. This assumption allows us to use fewer model parameters than in the Marshall–Madden and Cole–Cole models. Thus, the validity of the proposed model is limited, yet the model seems to be useful for inversion of IP spectra for obtaining grain-size distribution of sands.

- According to the proposed model, the time constant τ in sieved sands increases with the growth of the grain size as: $\tau \sim d^2$.

- Two samples of natural sands with different grain-size distribution show IP spectra decreasing with time. In the early time range, these spectra are assumed to be related to the Maxwell–Wagner polarization. The spectra in the medium and late time ranges are produced by the membrane polarization, characterised by various relaxation times. This variety of the relaxation time is related to different grain sizes in natural sands, which result in different lengths of large and narrow current passages.

Acknowledgements

This study was supported by INTAS (contract No. 32046/97). V. Chernysh provided the authors with the laboratory measurement equipment. The research was significantly advanced by discussions with L. Byten-sky, A. Tarasov, Yu. Ilyin, A. Potapov, P. Konosavsky and I. Kharhordin. We also thank anonymous reviewers for their helpful suggestions.

Appendix A. Solution of the equation of diffusion for the SNP model

(1) An instantaneous source of Q (mol) is placed on boundary of two tubes with different sections S_1 and S_2 and the coefficients of diffusion D_1 and D_2 (Fig. 9).

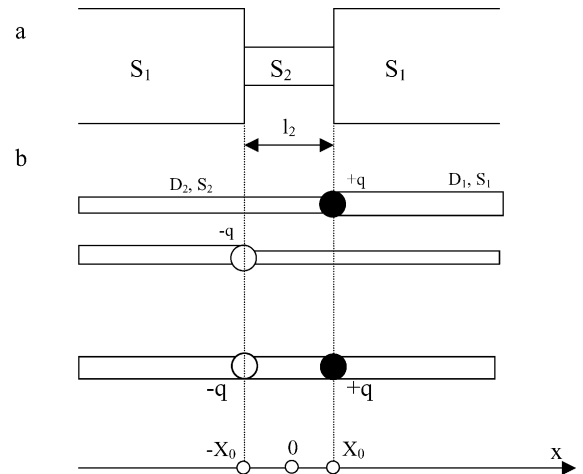


Fig. 9. Contacts of two large pores and a narrow pore (a) and equivalent scheme for solving the equation of diffusion (b). Source $+q$ and discharge $-q$ model concentration surplus and deficiency produced by the polarizing current.

The fundamental solution of equation of diffusion for each medium is expressed as (Carslow and Eger, 1964):

$$C_1 = \frac{a_1}{\sqrt{\pi D_1 t}} \exp\left(-\frac{(x-x_0)^2}{4D_1 t}\right) \quad (\text{A1.1})$$

$$C_2 = \frac{a_2}{\sqrt{\pi D_2 t}} \exp\left(-\frac{(x-x_0)^2}{4D_2 t}\right) \quad (\text{A1.2})$$

where a_1, a_2 are constants depending on the S_1, S_2, D_1, D_2 .

To determine a_1 and a_2 , two conditions are used:

$$C_1 = C_2(x = x_0) \quad (\text{A1.3})$$

$$S_1 \int_{-\infty}^{x_0} C_1 dx + S_2 \int_{x_0}^{\infty} C_2 dx = Q \quad (\text{A1.4})$$

After substitution of C_1, C_2 in Eqs. (A1.3) and (A1.4) and simplifications, the conditions are reformulated:

$$\frac{a_1}{\sqrt{D_1}} = \frac{a_2}{\sqrt{D_2}} \quad (\text{A1.5})$$

$$a_1 S_1 + a_2 S_2 = Q \quad (\text{A1.6})$$

The constants a_1 and a_2 are defined from Eqs. (A1.5) and (A1.6):

$$a_1 = \frac{Q\sqrt{D_1}}{S_1\sqrt{D_1} + S_2\sqrt{D_2}} \quad (\text{A1.7})$$

$$a_2 = \frac{Q\sqrt{D_2}}{S_1\sqrt{D_1} + S_2\sqrt{D_2}} \quad (\text{A1.8})$$

If $D_1 = D_2$ and $a_1 = a_2$, the concentration in the both media is expressed as:

$$C = \frac{Q}{(S_1 + S_2)\sqrt{\pi Dt}} \exp\left(-\frac{(x - x_0)^2}{4Dt}\right) \quad (\text{A1.9})$$

(2) The negative source introduced at the point $-x_0$ (Fig. 9) is taken into consideration on the basis of superposition. Thus, the concentration caused by positive and negative sources at the points $+x_0$ and $-x_0$ is expressed as:

$$C = \frac{Q}{(S_1 + S_2)\sqrt{\pi Dt}} \times \left[\exp\left(-\frac{(x - x_0)^2}{4Dt}\right) - \exp\left(-\frac{(x + x_0)^2}{4Dt}\right) \right] \quad (\text{A1.10})$$

where $2x_0 = l_2$. Note that $C|_{x=0} = 0$ at any time.

The concentration at the point x_0 is expressed as:

$$C = \frac{Q}{(S_1 + S_2)\sqrt{\pi Dt}} \left[1 - \exp\left(-\frac{l_2^2}{4Dt}\right) \right] \quad (\text{A1.11})$$

(3) During an infinite current step, the sources of capacity $\pm q$ (mol/s) produce concentration $C_{12}^*(t)$ obtained by integration of Eq. (A1.11):

$$C_{12}^*(t) = \frac{Q}{(S_1 + S_2)\sqrt{\pi D}} \int_0^t \left[1 - \exp\left(-\frac{l_2^2}{4D(t-t')}\right) \right] \frac{dt'}{\sqrt{t-t'}} \quad (\text{A1.12})$$

The result of integration of Eq. (A1.12) produces the solution of equation of diffusion representing time dependence of transient characteristic:

$$C_{12}^* = \frac{ql}{D(S_1 + S_2)} \times \left(\frac{1}{\sqrt{\pi}} \sqrt{\frac{t}{\tau}} \left(1 - \exp\left(-\frac{\tau}{t}\right) \right) + \operatorname{erfc}\left(\sqrt{\frac{\tau}{t}}\right) \right) \quad (\text{A1.13})$$

where q is the modulus of intensity of positive and negative sources in mol/s,

$$\tau = \frac{l_2^2}{4D} \quad (\text{A1.14})$$

Since τ is measured in time units, it can be considered as the “relaxation constant” of the polarization process.

Appendix B. Correspondence between Cole–Cole and SNP models in time domain

The logarithmic derivative Y of function Ψ_c (see Eq. (2)) is expressed as:

$$Y(x) = \frac{d\Psi_c(x)}{d\ln(x)} = - \sum_{k=1}^{\infty} (-1)^k \frac{ck(x)^{ck}}{\Gamma(1+ck)} \quad (\text{A2.1})$$

This normalised differential polarizability $Y(x)$ contains a maximum (Fig. 10). The maximum is shifted from the value of “1” at the x -axis. The value of $t_{\max}$ relates to τ as:

$$t_{\max} = N(c)\tau \quad (\text{A2.2})$$

where $N(c)$ is the dimensionless shift correction coefficient depending on the exponent c in Eqs. (1) and (2). The values of $N(c)$ and corresponding maxima of Y are shown in Table 2.

We found that the value of Cole–Cole exponent $c=0.62$ produces close maxima of $Y(x)$ and $\gamma(x)$, calculated on the basis of Eqs. (A2.1) and (24).

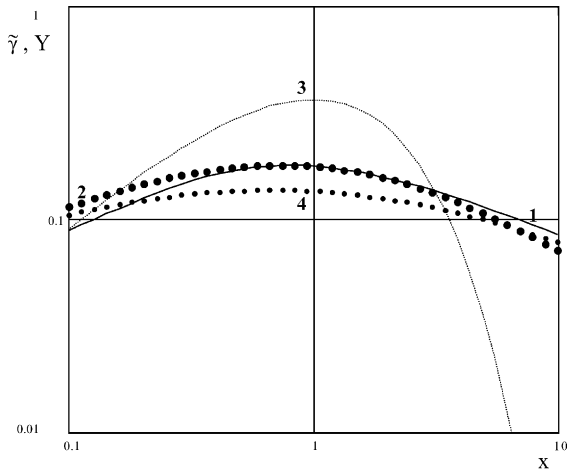


Fig. 10. Comparison between the Cole–Cole and SNP models in TD. Normalised differential polarizability for SNP ($\tilde{\gamma}(x)$) is calculated using Eq. (24) (graph 1). Normalised differential polarizability for the Cole–Cole model ($Y(x)$) is calculated using Eq. (A2.1) with various exponents [$c=0.62$ (graph 2), $c=1$ (graph 3), $c=0.5$ (graph 4)]. Dimensionless argument $x=t/\tau$. Note that the maximum of $Y(x)$ is not shifted from the value of “1” at x -axis if the Cole–Cole exponent $c=1$ only. The smaller is the c , the larger is the shift between $t_{\max}$ and τ .

One can compare the corresponding $N(c)$ equal to 0.735 (see Table 2) with the multiplier determining shift between maximum of normalised differential

Table 2
Parameters of the Cole–Cole model in the time domain

c	$N(c)$	$Y_{\max}$
0.1	0.566	0.0251
0.2	0.580	0.0508
0.3	0.602	0.0778
0.4	0.634	0.1067
0.45	0.653	0.1222
0.5	0.675	0.1384
0.55	0.698	0.1554
0.6	0.724	0.1734
0.62	0.735	0.1810
0.63	0.741	0.1848
0.65	0.752	0.1926
0.7	0.782	0.2129
0.75	0.814	0.2346
0.8	0.848	0.2577
0.85	0.883	0.2825
0.9	0.921	0.3090
0.95	0.960	0.3374
1.0	1.000	0.3679

c —the Cole–Cole exponent, $N(c)$ —the shift correction coefficient, $Y_{\max}$ —the maxima of the normalised differential polarizability.

polarizability and τ according the SNP model (see Eq. (25)). The difference between abscissas of maxima of functions Y and γ is 7.4%. Thus, these functions are in satisfactory correspondence, and in practice one can use any of them. For comparison, we also show graphs of $Y(x)$ corresponding to $c=1$ and 0.5 (Fig. 10).

Appendix C. Correspondence between the Cole–Cole and SNP models in frequency domain

Applying the Laplace–Carson transform to the SNP normalised transient characteristic (Eq. (19)), we can reformulate the SNP model in the FD as follows:

$$\rho(\omega) = \rho_0 \left(1 - \eta_0 \left(1 - \frac{1 - \exp(-2(i\omega\tau)^{1/2})}{2(i\omega\tau)^{1/2}} \right) \right) \quad (\text{A3.1})$$

where $\rho(\omega)$ is the frequency dependent resistivity (in $\Omega\cdot\text{m}$); ρ_0 is the DC resistivity (in $\Omega\cdot\text{m}$); η_0 is the chargeability (in V/V) given in Eq. (22).

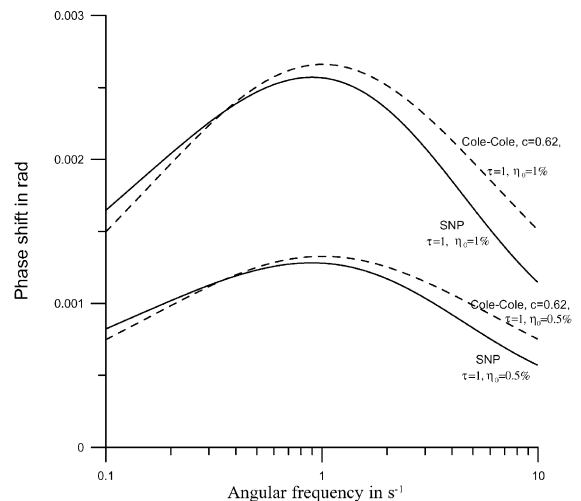


Fig. 11. Comparison between the Cole–Cole and SNP models in FD. Cole–Cole graphs are dashed line, SNP graphs are continuous lines. The graphs of phase shift are calculated using two values of chargeability typical of clean sands (1% and 0.5%). The value of Cole–Cole exponent c , which gives correspondence to the SNP model, was determined in the previous section ($c=0.62$).

One can compare the phase spectra produced by the SNP and Cole–Cole models. Based on Eq. (A3.1), we calculated the phase shift of $\rho(\omega)$ as:

$$\varphi = \operatorname{atan}\left(\frac{\operatorname{Im}(\rho(\omega))}{\operatorname{Re}(\rho(\omega))}\right) \quad (\text{A3.2})$$

The Cole–Cole phase shift was calculated on the basis of Eq. (1). Fig. 11 shows the Cole–Cole and SNP phase shifts calculated using values of chargeability typical of clean sands (1% and 0.5%), and the Cole–Cole exponent determined in Appendix B ($c=0.62$). One can note the agreement between Cole–Cole and SNP phase spectra.

References

- Börner, F., Gruhne, M., Schön, J., 1993. Contamination indications derived from electrical properties in the low frequency range. *Geophysical Prospecting* 41, 83–98.
- Börner, F.D., Schopper, J.R., Weller, A., 1996. Evaluation of transport and storage properties in the soil and groundwater zone from induced polarization measurements. *Geophysical Prospecting* 44, 583–601.
- Carslow, G., Eger, D., 1964. *Teploprovodnost tverdyh tel* (Thermal Conductivity of Solids) Nauka, Moscow (In Russian).
- Dukhin, S.S., 1971. Dielectric properties of disperse systems. In: Matievic, E. (Ed.), *Surface and Colloid Science*, vol. 3. Wiley, NY-London, pp. 83–166.
- Fridrikhsberg, D.A., Sidorova, M.P., 1961. Issledovanie svyazi yavleniya vyzvannoi polarizatsii s elektrokineticheskimi svoystvami kapillarnykh sistem (A study of relationship between the induced polarization phenomenon and the electro-kinetic properties of capillary systems). *Vestnik Leningradskogo Universiteta, Seria Khimii* 4, 222–226 (In Russian).
- Komarov, V.A., 1980. *Electrorazvedka metodom vyzvannoi polarizatsii* (Electrical Prospecting Using Induced Polarisation Method) Nedra, Leningrad (In Russian).
- Kormiltsev, V.V., 1963. O vzbuzhdenii i spade vyzvannoi polarizatsii v kapillarnoi srede (On excitation and decay of Induced Polarization in capillary medium). *Izvestia AN SSSR, Seria Geofizicheskaya* (Solid Earth Physics) 11, 1658–1666 (In Russian).
- Krulyt, H.R. (Ed.), 1952. *Colloid Science*, vol. 1. Elsevier, New York, pp. 115–193.
- Marshall, D.J., Madden, T.K., 1959. Induced polarization, a study of its causes. *Geophysics* 24 (1), 790–816.
- Mikhailov, H., Vargin, G., Shereshevsky, S., 1986. Peculiarities of applying the IP method when solving certain geological problems involved in ore prospecting in the Karelia–Kola region. In: Eskola, L., Fokin, A. (Eds.), *Electrical Prospecting for Ore Deposits in the Baltic Shield: Part 1. Galvanic Methods. Report of Investigation 73 - Geological Survey of Finland, Ep500*, vol. 73, pp. 85–90.
- Olhoeft, G.R., 1985. Low-frequency electrical properties. *Geophysics* 50 (12), 2492–2503.
- Pelton, W.H., Ward, S.H., Hallof, P.G., Sill, W.R., Nelson, P.H., 1978. Mineral discrimination and removal of inductive coupling with multifrequency IP. *Geophysics* 43 (3), 588–609.
- Schlumberger, C., 1920. *Etude sur la prospection électrique du sous-sol*. Gauthier-Villars et Cie, Chap. VIII, Gauthier-Villars et Cie, Paris, pp. 70–72.
- Shereshevsky, S.N., Chernysh, V.Yu., Golubev, A.M., 1999. Apparatura “STROB” (MPP, VP) i ee vozmozhnosti pri poiskakh mesorozhdenii urana (The instrument STROB (TDEM, IP) and its capabilities in exploration for uranium deposits). *Russkii Geofizicheskii Zhurnal* 15–16, 134–137 (In Russian).
- Vanhala, H., 1997. Mapping oil-contaminated sand and till with the spectral induced polarization (SIP) method. *Geophysical Prospecting* 45, 303–326.
- Vinegar, H.J., Waxman, M.H., 1984. Induced polarization of shaly sands. *Geophysics* 49 (8), 1267–1287.
- Wait, J.R., 1982. *Geoelectromagnetism*. Academic Press, NY.

On the relationship between induced polarization and surface area in metal-sand and clay-sand mixtures

Lee Slater¹, Dimitrios Ntarlagiannis¹, and DeBonne Wishart¹

ABSTRACT

Induced polarization (IP) measurements were conducted on saturated kaolinite-, iron-, and magnetite-sand mixtures as a function of varying percentage weight of a mineral constituent: 0%–100% for iron and magnetite and 0%–32% for kaolinite. We determined the specific surface area for each mineral using nitrogen gas adsorption, where the porosity of each mixture was calculated from weight loss after drying. We fit a Cole-Cole model (Cole and Cole, 1941) to the electrical data obtained for the magnetite and iron mixtures. In contrast, the kaolinite mixtures showed a power-law dependence of phase on frequency. The global polarization magnitude we obtained from the Cole-Cole modeling of the iron and magnetite mixtures displays a single, near-linear dependence on the ratio of surface area to pore volume (S_p) calculated for the mixtures. A similar relationship is found using a local measure of polarization (imaginary conductivity at 1 Hz) for the clay-sand mixtures. The S_p appears to be a critical parameter for determining IP in both metallic- and clay-containing soils. This result is not easily reconciled with traditional models of induced polarization.

INTRODUCTION

Interest is growing in the application of induced polarization (IP) for the investigation of environmental processes. The IP signal results from a diffusive and/or electrochemical polarization occurring within the electrical double layer (EDL) at the mineral-fluid interface. The physicochemical, interfacial properties controlling IP are important parameters for understanding environmental processes such as groundwa-

ter flow, solute retardation, chemical alteration (precipitation/dissolution), and microbial-mineral interactions.

The need to better understand environmental processes has stimulated research on the physicochemical controls on IP in soils. Lesmes and Frye (2001) studied the relationship between IP measurements recorded with geophysical instrumentation and electrical parameters defining polarization. Their research shows that imaginary conductivity (the effective dielectric permittivity scaled by the measurement frequency) is directly related to the interfacial properties, whereas measured IP parameters (phase angle and chargeability) reflect both polarization and conduction processes occurring in soils.

The purpose of our study is to improve understanding of the dependence of IP on interfacial properties of soils. Studies show that the mineral surface area in contact with fluid determines the IP response of soils and rocks: Power law relationships between imaginary conductivity and specific surface area normalized to pore volume (S_p) (Börner and Schön, 1991) (Börner et al., 1996) (Slater and Glaser, 2003) are reported. However, this observation appears inconsistent with traditional models of induced polarization that describe the IP responses of metallic mineral deposits and clay soils.

We examine the dependence of IP on the interfacial surface in contact with pore fluid for (1) a metal-sand mixture exhibiting electrode polarization and (2) a clay-sand mixture wherein membrane polarization might occur. The relationship between IP magnitude and S_p for both systems is compared with relationships observed for natural soils in recent literature. Our results suggest that S_p is the critical structural parameter determining IP — irrespective of mineralogy, i.e., metal or clay.

METHODS

Synthetic samples were created by mixing Ottawa sand (0.6–0.7 mm in diameter, density (ρ) = 2.64 g/cm³ with (1) iron filings (mean grain size 0.4 mm; ρ = 3.29 g/cm³, mean aspect

Manuscript received by the Editor August 7, 2005; revised manuscript received September 20, 2005; published online March 1, 2006.

¹Rutgers University, Department of Earth and Environmental Sciences, Newark, New Jersey 07102. E-mail: LSlater@andromeda.rutgers.edu.
© 2006 Society of Exploration Geophysicists. All rights reserved.

ratio of 5:1); (2) magnetite sand (mean grain size 0.43 mm; $\rho = 4.7 \text{ g/cm}^3$; trace amounts of quartz, ilmenite, zircon, garnet); and (3) kaolinite clay (mean grain size 0.002 mm, $\rho = 2.6 \text{ g/cm}^3$). We used a nitrogen gas-adsorption BET method (Brunauer et al., 1938) — repeated three times — to determine for a single sample the specific surface area (S_s) of the Ottawa sand ($0.0485 \pm 0.0015 \text{ m}^2/\text{g}$), iron ($2.84 \pm 0.09 \text{ m}^2/\text{g}$), magnetite ($1.08 \pm 0.03 \text{ m}^2/\text{g}$), and kaolinite ($10.97 \pm 0.33 \text{ m}^2/\text{g}$).

Mixtures were prepared by combining the metallic or clay constituent with Ottawa sand. Nine iron mixtures were prepared (0.50%, 1%, 5%, 10%, 20%, 35%, 50%, 75%, and 100% iron by weight). Nine magnetite mixtures were also prepared (0%, 5%, 15%, 30%, 50%, 70%, 85%, 95%, and 100% magnetite by weight). Saturation of samples with kaolinite concentration greater than 35% was hindered by the low permeabilities resulting from the dispersion of clay particles throughout the sand matrix. Therefore, twelve mixtures were prepared from 0% to 32% kaolinite by weight (0%, 1.3%, 1.6%, 2%, 2.5%, 3.2%, 4%, 5%, 6.3%, 8%, 10%, and 32%). Mixtures were dry-packed into cylindrical sample holders (16 cm $\times$ 3 cm ID). Samples were prepared as nearly identically as possible with respect to packing and composition. Mixtures were purged with CO_2 and evacuated (at -1.5 psi) prior to saturation with a 0.01 mol NaNO_3 solution.

Electrical measurements were made with a dynamic signal analyzer (DSA). The conductivity magnitude ($|\sigma|$) and phase angle (ϕ) were measured relative to a resistor at 40 frequency intervals between 10^{-1} and 10^3 Hz (10^{-1} and 10^4 Hz for the magnetite mixtures). Current was injected via stainless

steel electrodes placed at either end of the column. Potential measurements were recorded using silver-silver chloride electrodes. The real (σ') and imaginary (σ'') conductivity were calculated from the conductivity magnitude ($|\sigma|$) and phase angle (ϕ) measurements. Instrumentation errors were reduced by calibrations on water samples of known complex conductivity ($\sigma^*(\phi)$). Between 0.5 and 100 Hz, errors are generally less than $\pm 0.5 \text{ mrad}$ for ϕ and $\pm 0.5\%$ for $|\sigma|$ respectively. Capacitive coupling at the potential electrode connection increases the errors in ϕ above 100 Hz. For details of instrumentation and performance, see Slater and Lesmes (2002).

At the end of the experiment, saturation and porosity (ϕ) were determined from weight loss between saturated and dried samples. Saturation exceeded 0.96 for all samples. For each mixture S_p (μm^{-1}) was calculated using

$$S_p = [(S_{s_x} \times \rho_x \times W_x) + (S_{s_{OS}} \times \rho_{OS} [1 - W_x])] \times \frac{1 - \phi}{\phi} \quad (1)$$

where ρ is density (g/cm^3), S_s is m^2/g , and W is fractional weight; subscript OS refers to the Ottawa sand and subscript x refers to the mineral (iron, magnetite, or kaolinite) in the mixture.

A Cole-Cole (Cole and Cole, 1941) relaxation was used to fit data for the iron and magnetite sample. This was formulated in terms of complex conductivity (Jones, 2002) as

$$\sigma^*(\omega) = \sigma_0 \left[1 + m \left(\frac{(i\omega\tau)^c}{1 + (i\omega\tau)^c(1 - m)} \right) \right] \quad (2)$$

where σ_0 is the conductivity at DC frequency, τ is the mean relaxation time, c is a shape exponent that typically ranges from 0.1 to 0.6, and m is the chargeability (given by $m = 1 - \sigma_0/\sigma_\infty$, where σ_∞ is the conductivity at high frequency). The parameter $m_n = m \times \sigma_0$, represents a global measure of polarization accounting for the frequency dependence of σ^* (Lesmes and Frye, 2001). We used an algorithm employing a least-squares approach with Marquardt regularization (Kemna, 2000).

RESULTS

Figure 1 shows representative spectral data sets obtained for three mixtures: 10% iron, 15% magnetite, and 5% kaolinite. The iron and magnetite mixtures can be fit with a Cole-Cole model, as illustrated for the iron. Although the kaolinite data set could also be fit to a Cole-Cole model, it can be approximated by a power-law dependence of saturation and porosity (ϕ) on frequency. Since the power-law exponent does not vary greatly between the 12 kaolinite data sets (average = 0.24, standard deviation = 0.04), a single-frequency measurement of σ'' [e.g., at 1 Hz ($\sigma''_{1 \text{ Hz}}$)] represents the polarization magnitude of these mixtures. The linear correlation coefficient (R^2) is shown for the fit of the kaolinite data set.

The polarization magnitude as a function of S_p for all three mixtures is shown in Figure 2. The iron and magnetite mixtures are plotted in terms of m_n derived from Cole-Cole modeling, whereas the kaolinite mixtures are plotted in terms of $\sigma''_{1 \text{ Hz}}$. Power-law fits to the data sets show that the relationship between polarization magnitude and S_p is almost linear. A single power-law relationship is able to fit the magnetite and iron mixtures (exponent of 1.03). The exponent is 0.96 for the kaolinite mixtures.

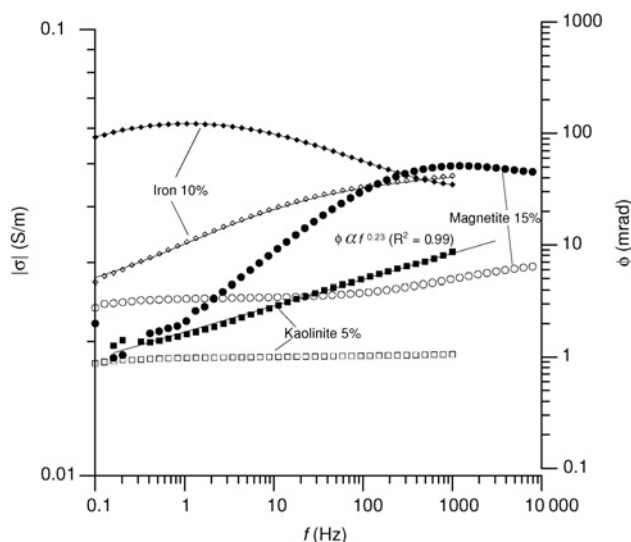


Figure 1. Spectral data sets obtained for three mixtures: 10% iron, 15% magnetite, 5% kaolinite. Solid symbols are the phase (ϕ) and open symbols are the conductivity magnitude ($|\sigma|$). The fit of the iron data set to a Cole-Cole model ($\sigma_0 = 0.0271 \text{ S/m}$; $m = 0.51$; $\tau = 0.33 \text{ s}$; $c = 0.424$) is shown. The fit of the kaolinite data set to a power-law expression is also shown. Note that the $|\sigma|$ axis covers only one order of magnitude, whereas the ϕ axis covers four orders of magnitude [frequency (f) dependence of $|\sigma|$ is thus greatly exaggerated]. Linear correlation coefficient (R^2) is shown for the kaolinite data set.

The dependence of the global measure of polarization (m_n) and imaginary conductivity ($\sigma''_{1\text{ Hz}}$) on pore volume (S_p) for the iron and magnetite mixtures is contrasted in Figure 3. Here m_n data for both mixtures are shown as a single data set since they follow the same near linear relation with S_p (Figure 2). Two differences arise in the $\sigma''_{1\text{ Hz}}$ dependence on S_p between the mixtures: (1) The magnitude of $\sigma''_{1\text{ Hz}}$ is greater for the iron mixtures, and (2) the relationship is approximately linear for the magnetite mixtures (exponent of 1.07) but not for the iron mixtures (exponent of 1.6).

DISCUSSION

Measures of polarization magnitude in these metal-sand and clay-sand mixtures ($R^2 > 0.93$ in every case) show a strong power-law dependence on S_p . The relationship is near linear except for $\sigma''_{1\text{ Hz}}$ versus S_p in the sand-iron mixtures where the power-law exponent is 1.6. The frequency of the dispersion peak in this data set is close to 1 Hz, and the position of this peak in frequency (hence Cole-Cole τ) varies with S_p (Slater et al., 2005, Figure 2). This effect increases the dependence of $\sigma''_{1\text{ Hz}}$ on S_p , increasing the power-law exponent. The dispersion peak in the magnetite data set occurs above 100 Hz; hence, shifts in this peak exert minimal control on $\sigma''_{1\text{ Hz}}$. In the magnetite mixture, the higher-frequency polarization peak also causes $\sigma''_{1\text{ Hz}}$ to be an order of magnitude less than the iron mixture at equivalent S_p (Figure 3). Using a global measure of the polarization (accounting for the frequency dependence) computed from Cole-Cole modeling, a

single, near-linear dependence on S_p is found for both data sets.

These results conflict with traditional concepts of IP in soils. Textbooks typically refer to electrode polarization associated with metals blocking electrolyte-filled pore passages (e.g., Reynolds, 1997, Figure 9.5). In our samples, iron filings dispersed within Ottawa sand do not block pore spaces, and the mixture porosity increases with the percentage iron (from 0.30 for 0% iron to 0.38 for 100% iron). Our results show that the polarization simply scales linearly in relation to the amount of the interface in contact with pore fluid (as quantified by S_p) and does not imply pore blockages.

We made IP measurements on iron particles that were suspended in a solution by attaching them to tape. Although $\phi < 5$ mrad, Cole-Cole dispersions were observed at IP frequencies (data not shown for brevity). Similar arguments pertain to the kaolinite mixtures. The traditional concept of membrane polarization invokes formation of cation-selective membranes impeding electrolytic conduction through pores. A common belief is that membrane polarization is strongest at relatively low concentrations of clays that are well dispersed throughout a soil. We find the polarization to scale near linearly with S_p up to 32% kaolinite concentration.

We observed a similar dependence of polarization magnitude on S_p for both metal and clay mixtures even though the polarization mechanisms at the interface are somewhat different. Note the very similar $\sigma''_{1\text{ Hz}}$ versus S_p relationships for the clay (Figure 3) and the magnetite (Figure 4) mixtures. In the clay mixtures, ionic diffusion occurs predominantly along the mineral surface. The metallic minerals in the iron and magnetite mixtures are electronic conductors, so ions migrate to/from the conductive interface (Wong, 1979). Furthermore, redox-active ions contribute an electrochemical polarization

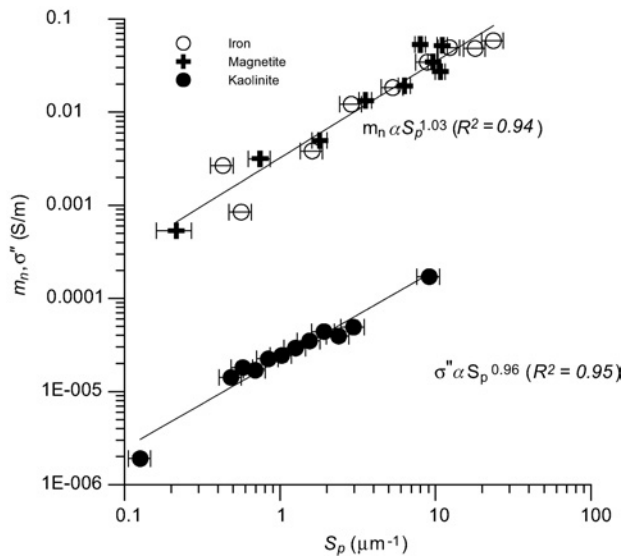


Figure 2. Polarization magnitude as a function of S_p for the sand-iron, sand-magnetite, and sand-clay mixtures. Polarization magnitude for the sand-iron and sand-clay mixtures is defined as m_n obtained from Cole-Cole modeling. A local measure of the polarization magnitude ($\sigma''_{1\text{ Hz}}$) is plotted for the sand-kaolinite mixtures. Power-law relations for 1) the combined sand-iron and sand-magnetite data set and 2) the sand-kaolinite data set are shown. Error bars are plotted when larger than symbol size. Linear correlation coefficients (R^2) are shown.

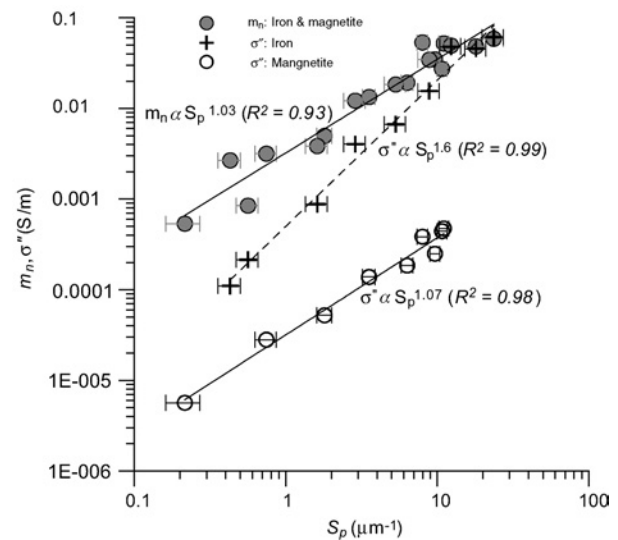


Figure 3. Local ($\sigma''_{1\text{ Hz}}$) and global (m_n) measures of the polarization magnitude for iron-sand and magnetite-sand mixtures as a function of S_p . The mixtures are shown as a single m_n data set, since they follow the same trend. Error bars are plotted when larger than symbol size. Linear correlation coefficients (R^2) are shown.

that facilitates conversion between electrolytic and electronic conduction at the interface. These differences are revealed in the high m_n associated with the metal mixtures (Figures 1 and 2) but not in the dependence on S_p .

Natural soils and rocks are more complex than the mixtures studied here because pore-scale properties other than S_p , i.e., mineral-fluid interface geometry, potentially affect the IP of natural soils. Previous studies adopted an interfacial-formation factor in empirical equations used to describe the IP response of soils and rocks (Vinegar and Waxman, 1984) (Börner et al., 1996). Scott and Barker (2005) show m_n as a function of S_p for sixteen sandstones (wide range of mineralogy, porosity, and S_s) from five formations within the United Kingdom. Their results are reproduced in Figure 4: The power-law relationship ($R^2 = 0.66$) between m_n and S_p is nonlinear (exponent of 0.5). Scott and Barker attribute the weak relationship between m_n and S_p to differences in the distribution of clay minerals throughout the sand matrix between samples. A second data set for natural soils found in Slater and Glaser (2003) is reproduced in Figure 4. These sands and silts from the Kansas City River floodplain exhibit limited variation in S_p (less than an order of magnitude) and a nonlinear power-law relationship ($R^2 = 0.6$) between σ'' and S_p (exponent of 1.2).

Although petrophysical properties other than S_p undoubtedly affect the IP of natural soils and rocks, we recognize an alternative explanation for the discrepancy between our results and others reported for natural soils. For the two constituents of each mixture, we measured S_s and then calculated the effective S_s of the sample using equation 1. Assuming the constituents were well mixed, S_s from the BET measurement should be representative of S_s at the scale of the IP measurement. In contrast, BET measurements on soils are typically

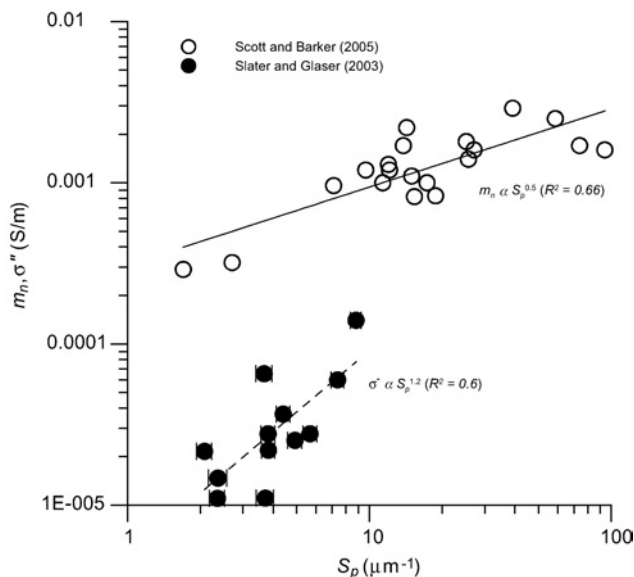


Figure 4. Power-law relationships between polarization magnitude and S_p observed in natural soils/rocks. In the sandstone data from Scott and Barker (2005), m_n is used to define the polarization magnitude. In the sandy alluvial sediments from Slater and Glaser (2003), $\sigma''_{1\text{ Hz}}$ is used to define the polarization magnitude. Linear correlation coefficients (R^2) are shown.

made on a small subsample of the soil used for electrical measurements. We must then question whether the BET-derived S_s adequately reflects the S_s at the scale of the IP measurement. For example, a small amount of clay or metal within a soil can profoundly influence the polarization magnitude. If this clay or metal is not present in identical concentrations in the subsample extracted for BET measurement, then the correct dependence of polarization magnitude on BET-derived S_p naturally cannot be measured.

In our study, measurements were made at the same electrolyte conductivity. The effect of electrolyte chemistry on IP remains uncertain; however, it appears to exert a relatively minor control in nonmetallic soils over a conductivity range typical of groundwater (Lesmes and Frye, 2001). The IP dependence on electrolyte chemistry is greater in metallic soils (Slater et al., 2005).

Although the IP response of soils remains an uncertain function of the physicochemical properties of soil and electrolyte, it is clear that S_p is a critical petrophysical property of the soil determining IP. The relationship between polarization magnitude and S_p is comparable for metallic and non-metallic soils. Similar near-linear relationships between measures of the interfacial polarization and S_p are observed at kHz to MHz frequencies (Knight and Nur, 1987), suggesting that common mechanisms may control electrical properties across a wide frequency range. Current efforts to develop theoretical models to describe IP in soils clearly should incorporate this dependence on S_p .

ACKNOWLEDGMENTS

Yuxin Wu of Rutgers University performed the measurements on the iron mixtures reported in this study. Andreas Kemna of the Institute of Chemistry and Dynamics of the Geosphere, Forschungszentrum Jülich, provided the Cole-Cole relaxation modeling algorithm that we used.

REFERENCES

- Börner, F. D., and J. H. Schön, 1991, A relation between the quadrature component of electrical conductivity and the specific surface area of sedimentary rocks: *Log Analyst*, **32**, 612–613.
- Börner, F. D., J. R. Schopper, and A. Weller, 1996, Evaluation of transport and storage properties in the soil and groundwater zone from induced polarization measurements: *Geophysical Prospecting*, **44**, 583–602.
- Brunauer, S., P. H. Emmett, and E. Teller, 1938, Adsorption of gases in multimolecular layers: *Journal of the American Chemical Society*, **60**, 309–319.
- Cole, K. S., and R. H. Cole, 1941, Dispersion and absorption in dielectrics. v. I, Alternating current field: *Journal of Chemical Physics*, **9**, 341–351.
- Jones, D. P., 2002, Investigation of clay-organic reactions using complex resistivity: M.S. thesis, Colorado School of Mines.
- Kemna, A., 2000, Tomographic inversion of complex resistivity—theory and application: *Der Andere Verlag*, Osnabrück, Germany.
- Knight, R., and A. Nur, 1987, The dielectric constant of sandstones, 60 kHz to 4 MHz: *Geophysics*, **52**, 644–654.
- Lesmes, D. P., and K. M. Frye, 2001, The influence of pore fluid chemistry on the complex conductivity and induced-polarization responses of Berea sandstone: *Journal of Geophysical Research*, **106**, 4079–4090.
- Reynolds, J. M., 1997, An introduction to applied and environmental geophysics: John Wiley & Sons.
- Scott, J. B. T., and R. D. Barker, 2005, Characterization of sandstone by electrical spectroscopy for stratigraphical and hydrogeological investigations: *Quarterly Journal of Engineering Geology and Hydrogeology*, **38**, 143–154.
- Slater, L., J. Choi, and Y. Wu, 2005, Electrical properties of

- iron-sand columns: implications for induced polarization investigation and performance monitoring of iron-wall barriers: *Geophysics*, **70**, G87–G94.
- Slater, L., and D. Glaser, 2003, Controls on induced polarization in sandy unconsolidated sediments and application to aquifer characterization: *Geophysics*, **68**, 1547–1558.
- Slater, L., and D. P. Lesmes, 2002, Electrical-hydraulic relationships observed for unconsolidated sediments: *Water Resources Research*, **38**, 10, 1213.
- Vinegar, H. J., and M. H. Waxman, 1984, Induced polarization of shaly sands: *Geophysics*, **49**, 1267–1287.
- Wong, J., 1979, An electrochemical model of the induced-polarization phenomenon in disseminated sulfide ores: *Geophysics*, **44**, 1245–1265.

A physical model of the low-frequency electrical polarization of clay rocks

Philippe Cosenza,¹ Ahmad Ghorbani,¹ André Revil,^{2,3} Maria Zamora,⁴ Myriam Schmutz,⁵ Damien Jougnot,^{2,3,6} and Nicolas Florsch¹

Received 3 December 2007; revised 7 May 2008; accepted 30 May 2008; published 5 August 2008.

[1] Low-frequency (0.18 Hz to 1.5 kHz) effective dielectric spectra have been measured on a set of near-saturated samples of argillite. The measured spectra of the real part of the complex apparent permittivity did not show significant correlation with cation exchange capacity (CEC) per unit mass of rock values. They satisfied a power law relationship with the frequency, at least for samples with CEC values lower than 10 cmol/kg. The Maxwell-Wagner-Hanai-Bruggeman formulation used for a two-phase mixture has been modified to account for mutual polarization between the pockets of water located in the micropores and those located in the macropores. The results of the modeling calculations illustrate (1) the ability of this new formulation to reproduce the power law relationships of the measured spectra of the real and imaginary components of the complex permittivity and (2) the strong impact of the pore electrical conductivity.

Citation: Cosenza, P., A. Ghorbani, A. Revil, M. Zamora, M. Schmutz, D. Jougnot, and N. Florsch (2008), A physical model of the low-frequency electrical polarization of clay rocks, *J. Geophys. Res.*, 113, B08204, doi:10.1029/2007JB005539.

1. Introduction

[2] Low-frequency measurements (mHz to kHz) of complex resistivity were initially used to detect small concentrations of disseminated mineralization in the exploration of ore bodies [e.g., Marshall and Madden, 1959; Van Voorhis *et al.*, 1973]. Recently, attention has been focused on LF dielectric properties of rocks and soils as a mean of indirectly determining their cation exchange capacity (CEC) and specific surface area [Börner and Schön, 1991; Slater *et al.*, 2005], their grain size distribution [Lesmes and Morgan, 2001], and their hydraulic conductivity [Börner *et al.*, 1996; de Lima and Niwas, 2000; Comas and Slater, 2004; Binley *et al.*, 2005] and to monitor microcracking during the excavation and the desiccation of clay rocks [Kruschwitz and Yaramanci, 2004; Cosenza *et al.*, 2007]. Experimental investigations have shown that the pore throat diameters and the pore geometry of water-saturated geomaterials contribute significantly to both in-phase and out-of-phase conduction at low frequencies [Scott and Barker, 2003]. Scott [2006] conducted low-frequency measurements of saturated sandstones, suggesting that low-frequency polarization originates in the polarization of the electrical

double layer (EDL) coating the grains. A similar result was obtained for packs of rough or smooth glass beads by Leroy *et al.* [2008].

[3] Although these investigations have identified key parameters (pore throat diameters, pore geometry, grain size distribution, roughness of the grains, and EDL), the physical mechanisms that control the low-frequency properties of clean sediments, and, a fortiori, of clayey materials, are still a subject of considerable debate. In their seminal book, Dukhin and Shilov [1974] explained that the main theories on the cause of the polarization of clay materials can be divided into two schools of thought.

[4] Those of the first school take into account explicitly the electrical surface phenomena (EDL and surface conduction along the surface of the clay minerals). Among these approaches, the theoretical works of de Lima and Sharma [1992] are likely the most popular for clay rocks: the authors propose to mix a Maxwell-Wagner formulation and microscopic theories related to the polarization of the EDL around a single grain on the basis of Dukhin and Shilov's [1974] theory.

[5] In the opposite school of thought, some authors neglect the polarization mechanisms related to the EDL and believe that the observed low-frequency polarization in clayey material is due to their particular texture and especially to the self-similar organization of their pore space [e.g., Korosak *et al.*, 2007]. Lunev *et al.* [2001] and Lounev *et al.* [2002] proposed a model describing the low-frequency electrical properties of porous material characterized by a fractal geometry of their pore space. They used this model to interpret their low-frequency dielectric data obtained from kaolinite clay samples: the electrical behavior of clay and water in the porous space is modeled by a self-similar chain of resistance-capacitance or “recap” (resistance plus

¹Laboratoire Sisyphe, UMR7619, Université Pierre et Marie Curie, CNRS, Paris, France.

²Department of Geophysics, Colorado School of Mines, Golden, Colorado, USA.

³LGIT, UMR5559, Université de Savoie, Equipe Volcan, CNRS, Le Bouget-du-Lac, France.

⁴Equipe de Géomatériaux et Environnement, Institut de Physique du Globe de Paris, Université Paris Diderot, CNRS, Paris, France.

⁵EGID, Université Michel de Montaigne, Pessac, France.

⁶ANDRA, Chatenay-Malabry, France.

Table 1. Mineralogical Composition of the Selected Samples

	TM90-1260	TM90-1280	Super-Sauze	EST05-687 ^a	EST12-161
Calcite carbonates as percent of total mass ^b	17.0	17.2	26.4	20	43.7
Composition of the clay fraction ^c					
Interstratified IS	-	-	-	15.4	-
Interstratified IS with >80% of illite	36.7	37.4	-	-	-
Interstratified IS with 30–50% of illite	-	-	-	-	83.4
Kaolinite	29.1	30.9	-	16.9	-
Illite	18.1	19.6	88.7	-	13.1
Chlorite	16.0	12.1	11.3	35.4	3.5
Smectite	-	-	-	30.8	-
Water content ^d (wt %)	3.33	3.42	10.97	-	6.0
Fraction of porosity ϕ	0.082	0.084	0.243	0.0787	0.142
CEC (cmol/kg)	9.62	9.53	6.36	18.3	15.4
Q_v^e (C/m ³)	270×10^6	295×10^6	56×10^6	608×10^6	264×10^6

^aFrom *Comparon* [2005].^bMixed layer IS is mixed layer illite-smectite.^cPercent by weight finer than 2 μm .^dWater content is not measured for EST05-687.^e Q_v is the total excess of charge per unit pore volume (Stern and diffuse layers). It is given by $Q_v = \rho_g \frac{1-\phi}{\phi} \text{CEC}$, where ρ_g is the grain density (taken to be equal to 2.6 g/cm³) and ϕ is the total porosity (taken to be equal to the average value measured in the site).

capacitance) electrical elements. In order to integrate the impact of the “bound water” filling the microporosity, these authors introduced a parallel circuit of two recap chains. However, it should be noted that *Lunev et al.* [2001] and *Lounev et al.* [2002] used a low CEC per unit mass of rock clay mineral (kaolinite) for which the influence of the EDL would be smaller in comparison with higher-CEC clay minerals like smectites. Consequently, the modeling approaches of both camps, which differ from the explicit introduction of EDL polarization, are likely not so contradictory.

[6] There is clearly a lack of experimental data for assessing the origin of the low-frequency polarization of clay rocks, especially on well-characterized rocks from the mineralogical point of view. In this paper, we provide (1) new measurements obtained with a variety of well-characterized clay rocks intensively studied for deep storage purposes and (2) a physical model explaining the data.

2. Phenomenological Aspects

2.1. Selected Clay Rocks and Experimental Investigations

[7] Five samples have been selected from three sites: the Tournemire site, the Super-Sauze site, and the Bure site, all located in France. The Tournemire site was chosen as a test site by the French Institute for Radioprotection and Nuclear Safety to study the confining properties of argillites for research purposes related to the long-term isolation of nuclear wastes. This site is located in southern France, at the western border of the Causses Basin, a Mesozoic sedimentary basin. Two samples were taken from well TM90, which was drilled in 2005 perpendicularly to the wall of a tunnel crossing a Toarcian argillite formation. Two cylindrical samples, labeled TM90-1260 and TM90-1280, each measuring 79.5 mm in diameter and 193–205 mm in length, were located 12 m from the gallery wall in a preserved zone outside the so-called excavation disturbed zone and outside the influence of the ambient air. Samples were immediately taken after the completion of drilling and

were put in Al-coated plastic bags under confined N₂ atmosphere.

[8] The mineralogical analyses of this Toarcian argillite have shown that clay minerals (kaolinite, illite, and interstratified illite-smectite) represent on average 40 wt % of the bulk rock composition [*De Windt et al.*, 1998]. The coarse fraction contains quartz (~20 wt %), K-feldspars, bioclasts, and pyrite (~2 wt %). Carbonates constitute about 15 wt % of the bulk rock composition. The porosity and the specific surface area of Tournemire argillites are in the ranges of 6–9% and 23–29 m²/g, respectively [*Bonin*, 1998]. The total CEC per unit pore volume (Q_v) and the mineral composition of the clay fraction obtained by X-ray diffractometry of the two samples, TM90-1260 and TM90-1280, are given in Table 1. The CEC was determined by the cobalthexamine method, which is a destructive analytical technique.

[9] The second site corresponds to Super-Sauze. This site is an earthflow occurring in the Callovo-Oxfordian black marls located in the Barcelonnette Basin (Alpes-de-Haute-Provence, southern France) [e.g., *Schmutz*, 2000]. Percussion drillings in dry conditions have been carried out from the surface in order to take a representative sample measuring about 600 mm in length and 50 mm in diameter, which was carefully closed in a tight tube sampler. From this sample, two smaller and unfractured samples (SS1 and SS2), measuring 80 and 160 mm in length, respectively, have been made by cutting slowly and directly the tight tube sampler immediately prior to performing the low-frequency electrical measurements.

[10] The Callovo-Oxfordian black marl is a compact shale formation. Mineralogical analyses indicate that clay minerals (mostly illite and chlorite) constitute on average 37 wt % of the bulk composition (Table 1) [*Schmutz*, 2000]. The coarse fraction contains quartz (~30 wt %) and carbonates (~30 wt %). The porosity of Super-Sauze black marls is in the range of 17–26%. The CEC and the mineral composition of the clay fraction of a small sample taken from the same sampler as SS1 and SS2 are given in Table 1.

[11] The third site, the Bure Underground Research Laboratory, is a test site of the Agence Nationale pour le

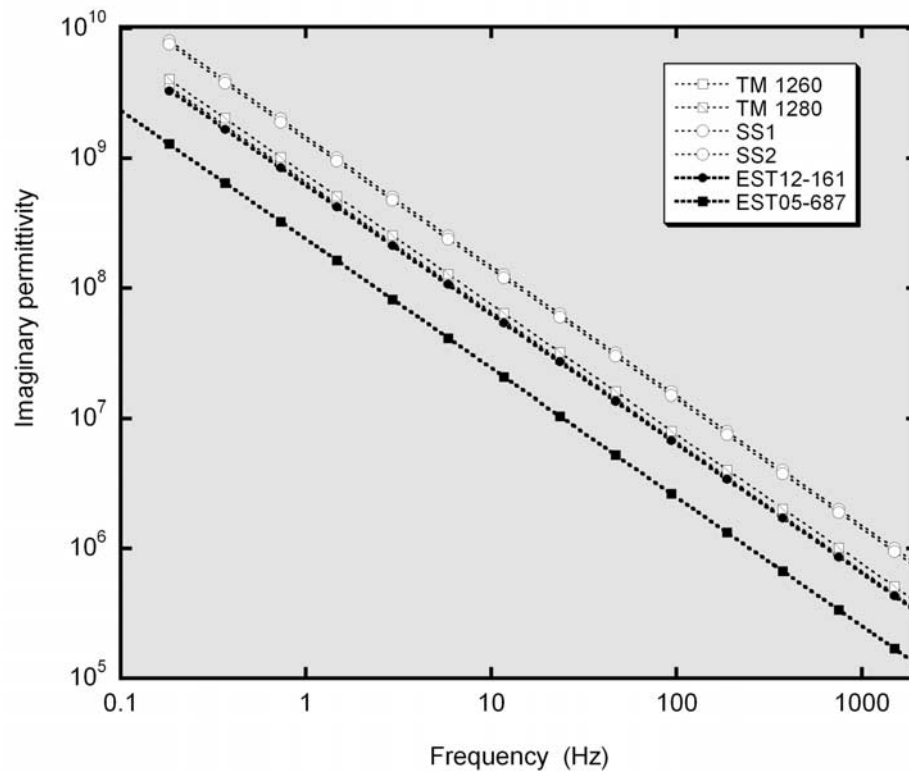


Figure 1. The low-frequency spectra of the imaginary part of complex permittivity ε'' . Open symbols, measurements on samples with CEC values lower than 10 cmol/kg. Solid symbols, measurements on samples with CEC values greater than 10 cmol/kg. Dotted lines, power law regressions for each sample.

Gestion des Déchets Radioactifs. This site has been created for the purpose of assessing the feasibility of a deep repository for isolation of radioactive wastes. This underground research laboratory is located in the eastern part of the Paris basin, at a depth of 490 m, in 130 m thick Callovo-Oxfordian argillites. Two samples were taken from two different boreholes, EST212 (sample 161 at a depth of 464.19 m) and EST205 (sample 687 at a depth of 489.51 m). The most abundant clay minerals include the R0-type interstratified illites-smectites (piles of unorderly flakes, 50–70% smectite) in the upper part of the formation (up to 490 m deep) [Yven *et al.*, 2007]. The samples EST12-161 and EST05-687 are 100–150 mm in length and 60–100 mm in diameter. In order to prevent damage and loss of water, after drilling, they were stored in confining and sealing cells named T1 cells. The CEC and the mineral composition of the clay fraction of samples are given in Table 1. The CEC and smectite content of this clay rock are significantly higher than those reported above.

[12] The low-frequency electrical measurements were obtained using the SIP FUCHS-II apparatus in the frequency range 0.18 Hz to 1.5 kHz. To minimize electrode polarization associated with electrochemical reactions occurring at the electrode sample contacts, measurements were performed with a four-electrode device using electrocardiogram Ag/AgCl electrodes as potential (or measuring) electrodes [Cosenza *et al.*, 2007]. The low-frequency measurements were performed immediately after removing the samples from their initial packaging (Al-coated plastic bags, tight tube samplers, and T1 cells), in a near-saturated state.

2.2. Results and Discussion

[13] The low-frequency spectra of the relative complex permittivity (imaginary part and real part) measured on the clay rock samples are given in Figures 1 and 2. The spectra of the effective complex permittivity have been found to deviate from the classical Debye relationship in two ways. First, the relative imaginary permittivity ε'' is inversely proportional to the frequency f (i.e., $\varepsilon'' \propto f^{-1}$). This clearly indicates that the dielectric losses in these clayey materials are purely ohmic (no influence of losses related to dipolar polarization phenomena). Indeed, the relative imaginary permittivity or the total dielectric losses ε'' is usually divided into two parts: a pure dielectric part ε''_d and an ohmic part $\sigma_{dc}/(2\pi f\varepsilon_0)$, where σ_{dc} is the direct current (DC) conductivity and ε_0 is the permittivity of vacuum [e.g., Guéguen and Palciauskas, 1994]. The second contribution, the ohmic contribution, is inversely proportional to the frequency.

[14] Second, the relative real permittivity ε' measurements given in Figure 2 show two remarkable features.

[15] 1. The relative real permittivity exhibits enormously high values (up to 3×10^{10}) in comparison with that of water (~ 80) and minerals (~ 5 – 10).

[16] 2. The spectra of ε' exhibit a strong nonlinear dependence with frequency. When clay rocks with low CEC values are considered (Figure 2), we observe that $\varepsilon' = Af^{-\alpha}$, where A is a constant in the range $(0.91$ – $2.90) \times 10^6$ and α is an exponent in the range 0.65–0.81 (regression coefficient R in the range 0.991–0.999). For clayey samples

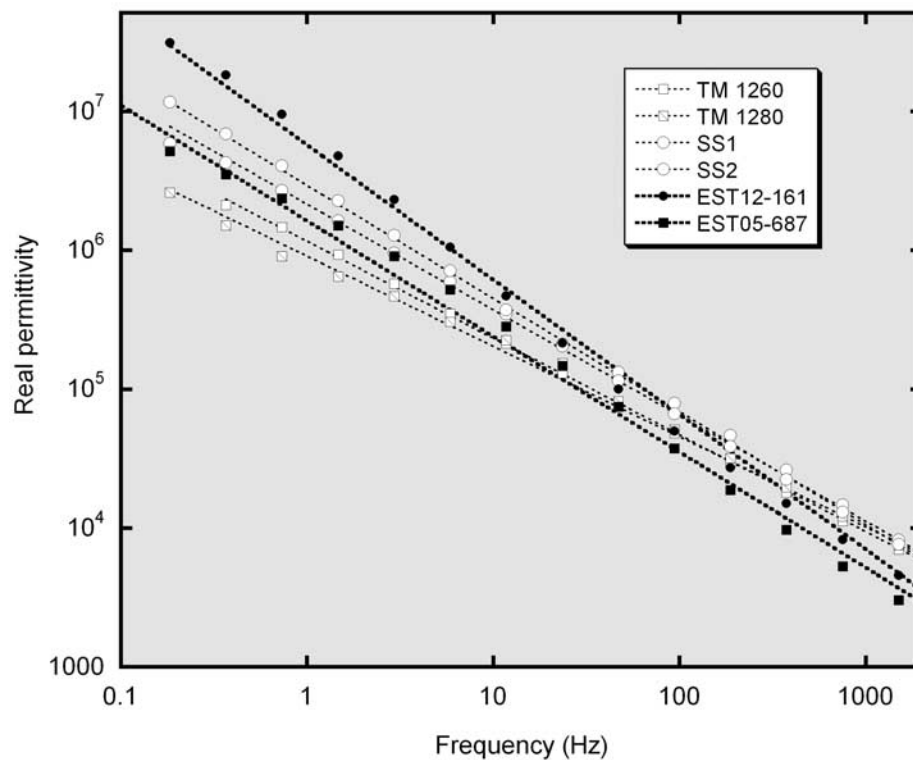


Figure 2. The low-frequency spectra of the real part of complex permittivity ϵ' . Open symbols, measurements on samples with CEC values lower than 10 cmol/kg. Solid symbols, measurements on samples with CEC values greater than 10 cmol/kg. Dotted lines, power law regressions for each sample.

with higher CEC values, the agreement with a power law is slightly poorer (R in the range 0.972–0.997).

[17] Considering the first point, the gigantic values of ϵ' have been extensively discussed in the literature, and two effects have often been proposed: (1) the Maxwell-Wagner effect due to the presence of thin platy particles perpendicular to the direction of the electric field [e.g., Sen, 1981] and (2) electrolyte-grain interface polarization associated with electrochemical phenomena [de Lima and Sharma, 1992].

[18] Considering the second point, the measured spectra shown in Figure 1 also exhibit an unexpected result: no clear relationship can be detected between the measured spectra of the real component $\epsilon'(f)$ and the CEC (or Q_v) values of each sample. The amplitude of ϵ' of the SS samples with the lowest CEC values and that of sample EST05-687 with the highest CEC value are very close. Moreover, the samples from the Bure site with similar petrophysical properties (CEC and water content) showed very different responses in the low-frequency range (0.18–10 Hz).

[19] Since most theoretical studies show that the dielectric responses associated with electrochemical phenomena [de Lima and Sharma, 1992] are strongly related to the surface conductivity that is proportional to CEC or Q_v , these comparisons suggest that the impact of the electrochemical phenomena is not, at least for the studied samples, of the first order. Consequently, these results confirm that the power law relationship that fits the measurements obtained with the samples with the lowest CEC values is related mainly to the texture of the clay rocks (particle size

distribution, pore size distribution, shape and location of clay aggregates, geometrical relationship between the clay aggregates and the nonclay minerals, microscale particle-to-particle contacts, etc.), as it is often mentioned in other studies with clay materials.

[20] Indeed, a power law has been evidenced by Thevanayagam [1997] and Lounev *et al.* [2002], who used unconsolidated and pure clay-water systems. Their results were interpreted as a pure textural effect associated with self-similar patterns at the microlevel structure. The first theoretical study [Thevanayagam, 1997] dealt with saturated kaolinite and montmorillonite in the kilohertz range to the high megahertz range. Lounev *et al.* [2002] studied dielectric spectra of artificial kaolinite-electrolyte systems in the range 10^{-2} – 10^6 Hz. Both approaches used electrical fractal networks assuming a self-similarity of the underlying physical processes and consequently invoked the concept of fractals (fractal solid skeleton and fractal pore surface) to explain the observed dielectric responses since experimental evidences for scale invariance in argillaceous soils and sediments exist [e.g., Krohn, 1988; Van Damme, 1995].

[21] However, at this stage, we feel that the application of an electrical fractal network model may be irrelevant in our context for two reasons.

[22] 1. Contrary to the materials used by Thevanayagam [1997] and Lounev *et al.* [2002], an argillaceous rock is not a pure clay material. It is composed of silica quartz and carbonates, often for the most part, with grains covering a large range of sizes and shapes. Consequently, it is difficult

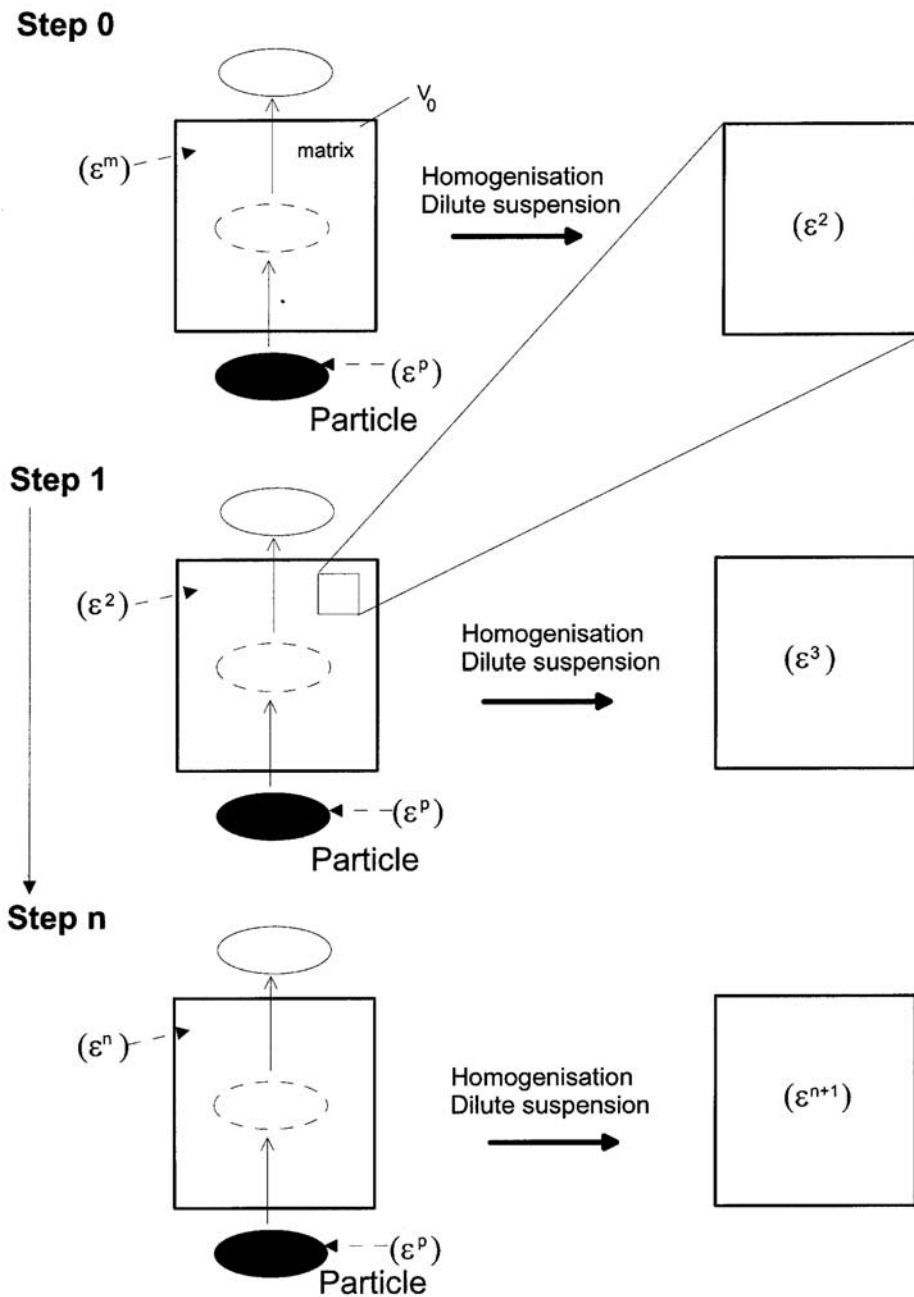


Figure 3. The iterative process of a DEM scheme for a two-phase mixture.

to appreciate how the fractal regime reflects the reality of the texture.

[23] 2. Electrical (fractal or not) network models suffer from at least two drawbacks. On the one hand, these models introduced empirical parameters for which the physical meaning is not clear. On the other hand, it is known that a measured spectrum can be represented by different electrical circuits with different physical meanings (uniqueness problem).

[24] In section 3, a new theoretical and simpler approach is proposed in order (1) to avoid the drawbacks of the electrical network models and (2) to model the low-frequency electrical responses of the clay rock samples with low CEC

values (i.e., the power law relationship), for which the impact of the EDL is assumed to be negligible.

3. Theoretical Aspects

[25] A simple and physically based approach for modeling the electrical properties of a mixture of different components is the Maxwell-Wagner-Hanai-Bruggeman (MWHB) equation, derived from the differential effective medium (DEM) theory [e.g., *Dukhin and Shilov*, 1974; *de Lima and Sharma*, 1992]. The DEM schemes offer four advantages: (1) all the parameters of this model have an explicit physical definition (i.e., no fitting parameters), (2) this model is easy to implement numerically, (3) the

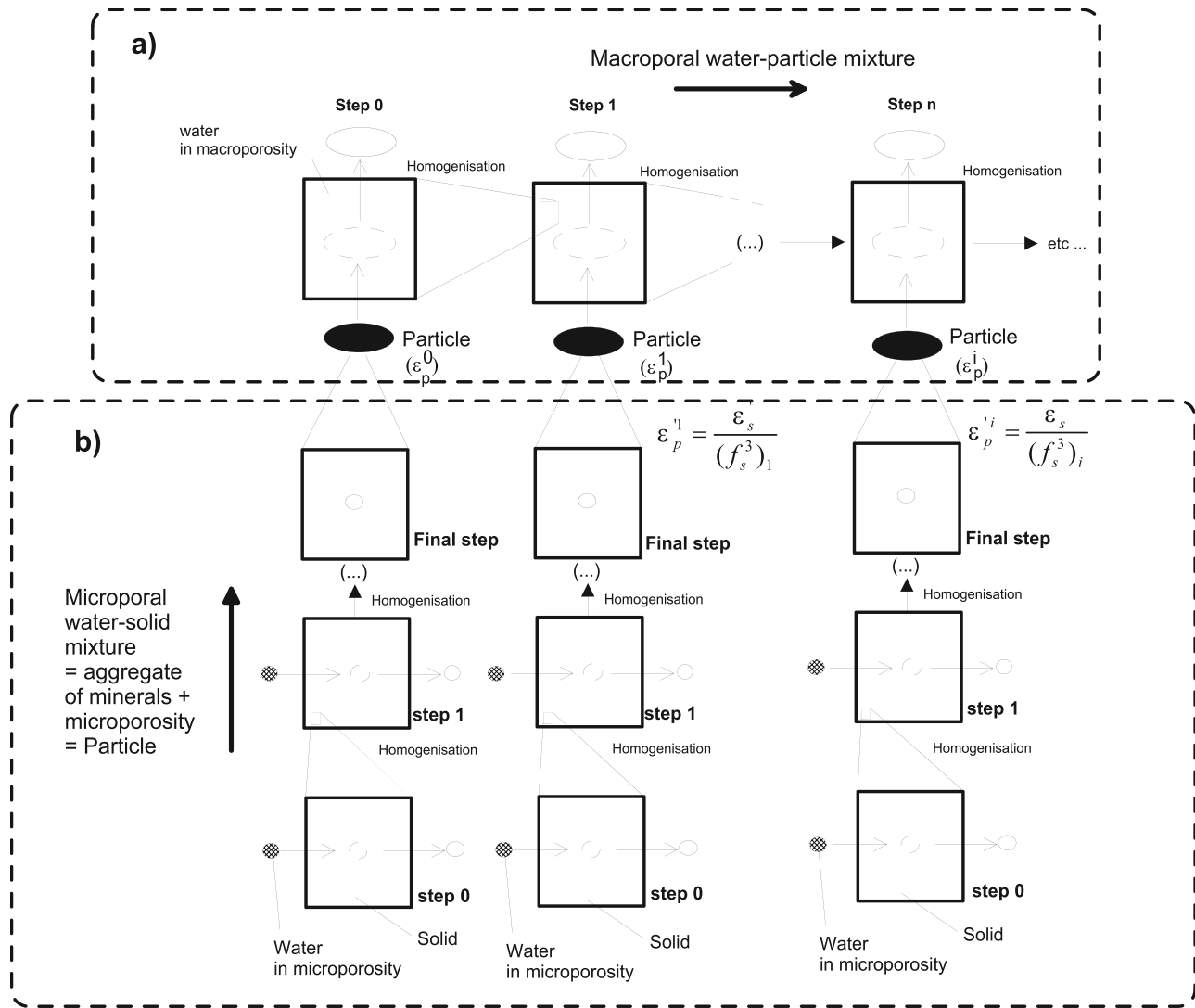


Figure 4. The imbricated DEM schemes. (a) Iterative process of the DEM scheme used for modeling the effective complex permittivity of macroporal water-particle mixture. A particle is an aggregate of clay minerals and nonclay minerals (quartz and carbonate grains) with water in the micropores. (b) Iterative process of the DEM scheme used for modeling the effective complex permittivity of particles (aggregates of clay and nonclay minerals with water in micropores). In the homogenization scheme (Figure 4b), water is in the micropores.

influence of the particle shape can be explicitly accounted for [e.g., Jones and Friedman, 2000], and (4) the model integrates the electromagnetic interactions between the different heterogeneities by an iterative process, leading to good agreements with dielectric data [e.g., Cosenza et al., 2003].

[26] Following an iterative process, the effective property of the mixture is explicitly calculated from an initial material through a series of incremental additions of elementary units. The mixture is built up by starting with a homogeneous medium and then replacing a small amount of the starting medium with another component with an ideal geometry (sphere or ellipsoid). For instance, consider a mixture of two components: spheroidal particles with an effective complex permittivity ϵ_p are embedded in a matrix characterized by an effective complex permittivity ϵ_m . The procedure begins with the initial material m corresponding to a relative permittivity ϵ_m in a volume V_0 . A small volume

of phase p , δV_p , is imbedded in phase m in such a way that the volume remains fixed at V_0 . The effective relative permittivity of the mixture is calculated considering a dilute suspension of particles p in a medium m . Now in the volume V_0 , the mixture has a homogenized effective complex permittivity ϵ_2 and constitutes the initial medium of the next step. A schematic of the iterative process is shown in Figure 3. The construction process continues such that (1) at each stage, the embedded particles are in dilute concentration and (2) the required volume fraction ϕ_p is satisfied. Finally, the effective complex permittivity of the mixture ϵ_{mix} is given by the MWHB equation [e.g., Chen and Or, 2006]:

$$\frac{\epsilon_p - \epsilon_{\text{mix}}}{\epsilon_p - \epsilon_m} \left(\frac{\epsilon_m}{\epsilon_{\text{mix}}} \right)^{1/m^*} = 1 - \phi_p, \quad (1)$$

where ϕ_p is the volume fraction of spheroidal inclusions and m^* is a particle shape factor related to the eccentricity of the

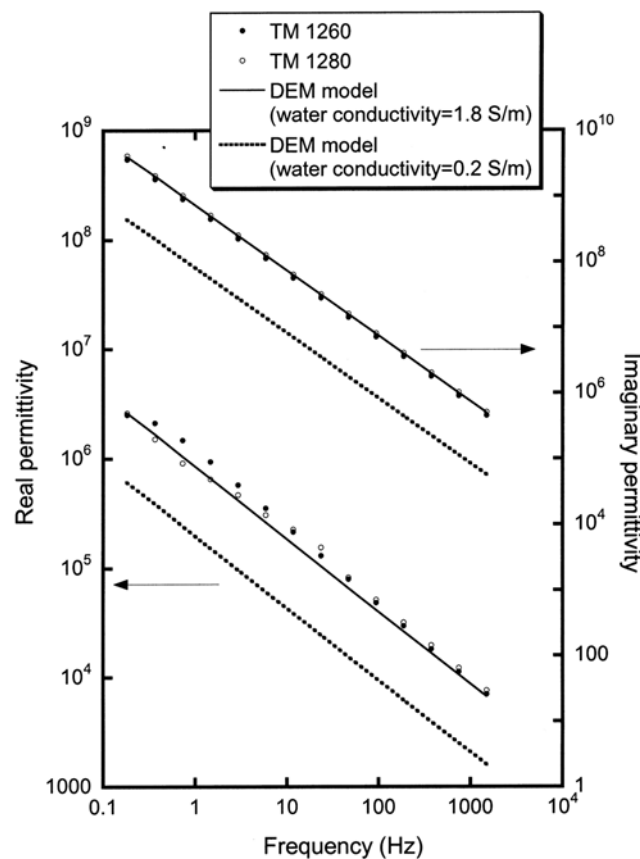


Figure 5. Simulated and experimental spectra of the complex permittivity of Tournemire samples. The dotted lines and the solid lines show the imbricated DEM model with a water conductivity equal to 0.2 and 1.8 S/m, respectively.

spheroidal particles. This exponent is related to the “cementation exponent” because this is the same exponent that enters into Archie’s first law. For spherical particles and in the DC limit, $m^* = 1.5$ [e.g., Sen, 1984].

[27] The iterative process associated with the DEM approach is not primarily used to model fractal or self-similar media. It is a better and intuitive way to account for the electromagnetic interactions (i.e., mutual polarizations) between heterogeneities in comparison with other effective medium approximations (i.e., the dilute approximation which neglects these interactions and the self-similar approximation which overestimates them) [Dukhin and Shilov, 1974].

[28] We propose to combine DEM schemes and the intuition of Lunev *et al.* [2001] and Lounev *et al.* [2002]. The low-frequency electrical properties of the low-CEC clay rocks result in the electromagnetic interaction between two states of water: the free water in the macroporosity and the water in the microporosity. Following the textural model proposed by Yven *et al.* [2007] for argillite, the macroporosity is mostly located at the interface between the clay matrix and the other minerals (carbonates and tectosilicates). It is characterized by micrometric to submicrometric pore sizes. The microporosity is the internal porosity of the

clay matrix (around clay aggregates and particles) characterized by inframetric to nanometric pore sizes.

[29] Classical DEM schemes (1) cannot reproduce power law spectra [Comparon, 2005] and (2) cannot account for heterogeneities with very different spatial scales. For this reason, in order to account for the mutual interaction between these states of water, two imbricated DEM schemes are considered: a first DEM scheme is introduced for modeling the mixture of “free” water and particles (aggregates of clayey units and nonclay minerals) in which a second DEM is imbricated to account for the microporal water in the particles (Figure 4).

[30] Regarding the latter scheme, the microporal water-solid mixture, two additional assumptions are considered.

[31] 1. The water electrical conductivity in the microporosity is much greater than the electrical conductivity of either the mixture or of the solid alone: the effects of the EDL and of surface conduction are supposed to be negligible for the lowest-CEC materials. As a matter of fact, microvolumes of water in the micropores are considered as conducting particles: the electrical field in these closed microvolumes tends to zero, and hence their permittivity tends to infinity. This assumption means that in agreement with experimental and thermodynamical arguments [Blackmore, 1978; Leroy *et al.*, 2007], the microporal water has to be very concentrated in comparison with that of macropores.

[32] 2. For the sake of simplicity, the microvolumes of water in the microporosity are spherical. Following these assumptions and the DEM scheme given in Figure 4b (conducting water microspheres in an insulating solid matrix), the real part of the complex dielectric constant of the microporal water-solid mixture (i.e., particle) ϵ'_p is given by [e.g., Dukhin and Shilov, 1974]

$$\epsilon'_p = \frac{\epsilon'_s}{(f_s)^3}, \quad (2)$$

where ϵ'_s is the real part of the complex dielectric constant of the solid and f_s is the volumetric fraction of the solid. It should be noted that equation (2) has been successfully used to model the permittivity of a mixture of water droplets in oil emulsions [Dukhin and Shilov, 1974].

[33] Moreover, since the electrical conductivity of the solid is considered negligible, the electrical conductivity of the microporal water-solid mixture is equal to zero, following the DEM scheme (Figure 4b). This assumption means that for these low-CEC argillites, the bulk rock electrical conductivity is mainly controlled by that of the macropores and hence obeys Archie’s first law. This assumption is usually checked by measuring rock conductivity as a function of water conductivity, i.e., with different brine salinities [e.g., Waxman and Smits, 1968]. However, considering argillites, this classical procedure is difficult to apply for at least two reasons. First, in order to control the brine conductivity in the clay rock sample, the sample has to be leached continuously: a chemical equilibrium should be reached when the brine conductivity extracted at the output of the sample is equal to that subjected at the input. From our own experience, in the case of a clay rock sample with a very low intrinsic permeability and diffusion coefficient

(typically equal to 10^{-20} m² and 10^{-11} m²/s, respectively [e.g., Boisson *et al.*, 2001]), this chemical equilibrium for a given brine salinity can take very long to reach, typically a few weeks for a decimetric sample. Second, when the macropore water chemistry is modified in a clay sample during a leaching experiment, this may also lead to a change in the micropore composition [Leroy *et al.*, 2007]. This may induce textural changes by shrinkage or swelling of the clay aggregates and changes in the surface conductivity, which are both difficult to control. To our knowledge, no study on the electrical properties of natural clay rock samples used this time-consuming and complex procedure. Moreover, experiments performed on artificial compacted clay samples wetted by means of a flow of a low-concentration (10^{-2} mol/L) NaCl solution [Mammari *et al.*, 2001] suggest that Archie's law can be used as a good first approximation, except for pure montmorillonite. This conclusion is also confirmed when Archie's law is expressed in terms of diffusion coefficients: a good fit with Archie's law is obtained using the transport through or effective porosity [Van Loon *et al.*, 2003; Rosanne *et al.*, 2003; Blum *et al.*, 2007].

[34] This approach was applied to Tournemire samples for which the electrical conductivity of interstitial water has been estimated to be equal to 0.2 S/m on average from total dissolved salts measurements obtained from the water of the site [Bonin, 1998]. We used $\epsilon'_w = 80$ (relative permittivity of water in macropores) and $\epsilon'_s = 5$ (typical value for silica). For the sake of simplicity, all particles (microporal water-solid mixtures) are assumed to be spherical (i.e., $m^* = 1.5$ or $1/m^* = 0.66$ in equation (1)). The total porosity was taken to be equal to 8% [Bonin, 1998]. The results of DEM calculations for different values of water conductivity (0.2 and 1.8 S/m) are shown in Figure 5. A satisfactory agreement with the experimental data can be observed with a water conductivity value much greater than the measured average from the water of the site. This discrepancy is not surprising since the water collected from the site is not necessarily in chemical equilibrium with that in the clay rock. However, despite the uncertainties of the used parameters, the imbricated DEM schemes can reproduce the observed order of magnitudes and the power law relationship of the measured spectra of both components of the complex permittivity. The theoretical exponent associated with the real permittivity (i.e., $1/m^* = 0.66$) is close to the observed power law exponents α obtained in both samples TM90-1260 and TM90-1280 (0.70 and 0.65, respectively).

4. Implications and Concluding Statements

[35] The experimental results of this study do not show a strong dependence on the measured CEC values and consequently on the physicochemical phenomena occurring at the surface of clay minerals. Consequently, in order to capture the main physical process involved in these measurements, we suggest that the existence of numerous conducting micropores is responsible for the observed power law relationships. This assumption is supported by a DEM approach accounting for the mutual electrical polarization between the water pockets of the micropores and those of the macropores. This new and general approach could be

extended to the low-frequency spectra of materials with dual porosity like zeolites, zeoponic, and pumice.

[36] **Acknowledgments.** This research was supported by the ANR-ECCO project (POLARIS) and the GDR FORPRO. The authors thank ANDRA and IRSN (J.-C. Gros, J. Cabrera, and K. Ben-Slimane) for providing the argillite samples from the Bure and Tournemire sites, respectively.

References

- Binley, A., L. D. Slater, M. Fukes, and G. Cassiani (2005), Relationship between spectral induced polarization and hydraulic properties of saturated and unsaturated sandstone, *Water Resour. Res.*, **41**, W12417, doi:10.1029/2005WR004202.
- Blackmore, A. V. (1978), Interpretation of electrical conductivity in a clay soil containing salts, *Aust. J. Soil Res.*, **16**, 311–318, doi:10.1071/SR9780311.
- Blum, P., M. Scheunemann, L. Van Loon, D. Coelho, N. Maes, and P. Grathwohl (2007), Evaluating the application of Archie's law for argillaceous rocks, paper presented at Clays in Natural and Engineered Barriers for Radioactive Waste Confinement, Agence Natl. pour la Gestion des Dechets Radioactifs, Lille, France.
- Boisson, J.-Y., L. Bertrand, J.-F. Heitz, and Y. Golvan (2001), In-situ and laboratory investigations of fluid flow through an argillaceous formation at different scales of space and time, Tournemire tunnel, southern France, *Hydrogeol. J.*, **9**, 108–123, doi:10.1007/s10040000119.
- Bonin, B. (1998), Deep geological disposal in argillaceous formations: Studies at the Tournemire test site, *J. Contam. Hydrol.*, **35**, 315–330, doi:10.1016/S0169-7722(98)00132-6.
- Börner, F. D., and J. H. Schön (1991), A relation between the quadrature component of electrical conductivity and the specific surface area of sedimentary rocks, *Log Anal.*, **32**, 612–613.
- Börner, F. D., W. Schopper, and A. Weller (1996), Evaluation of transport and storage properties in the soils and groundwater zone from induced polarization measurements, *Geophys. Prospect.*, **44**, 583–601, doi:10.1111/j.1365-2478.1996.tb00167.x.
- Chen, Y., and D. Or (2006), Effects of Maxwell-Wagner polarization on soil complex dielectric permittivity under variable temperature and electrical conductivity, *Water Resour. Res.*, **42**, W06424, doi:10.1029/2005WR004590.
- Comas, X., and L. Slater (2004), Low-frequency electrical properties of peat, *Water Resour. Res.*, **40**, W12414, doi:10.1029/2004WR003534.
- Comparon, L. (2005), Etude expérimentale des propriétés et diélectriques des matériaux argileux consolidés, Ph.D. thesis, Inst. de Phys. du Globe de Paris, Paris.
- Cosenza, P., C. Camerlynck, and A. Tabbagh (2003), Differential effective medium schemes for investigating the relationship between high-frequency relative dielectric permittivity and water content of soils, *Water Resour. Res.*, **39**(9), 1230, doi:10.1029/2002WR001774.
- Cosenza, P., A. Ghorbani, N. Florsch, and A. Revil (2007), Effects of drying on the low-frequency electrical properties of Tournemire argillites, *Pure Appl. Geophys.*, **164**, 2043–2066, doi:10.1007/s00024-007-0253-0.
- de Lima, O. A. L., and S. Niwas (2000), Estimation of hydraulic parameters of shaly sandstone aquifers from geoelectrical measurements, *J. Hydrol.*, **235**, 12–26, doi:10.1016/S0022-1694(00)00256-0.
- de Lima, O. A. L., and M. Sharma (1992), A generalized Maxwell-Wagner theory for membrane polarization in shaly sands, *Geophysics*, **57**(3), 431–440, doi:10.1190/1.1443257.
- De Windt, L., J. Cabrera, and J.-Y. Boisson (1998), Hydrochemistry in an indurated argillaceous formation (Tournemire tunnel site, France), in *Water-Rock Interaction*, edited by G. B. Aherhart and J. R. Hulston, pp. 145–148, A. A. Balkema, Rotterdam, Netherlands.
- Dukhin, S. S., and P. Shilov (1974), *Dielectric Phenomena and the Double Layer in Disperse Systems and Polyelectrolytes*, John Wiley, New York.
- Guéguen, Y., and V. Palciauskas (1994), *Introduction to the Physics of Rocks*, 299 pp., Princeton Univ. Press, Princeton, N. J.
- Jones, S. B., and S. P. Friedman (2000), Particle shape effects on the effective permittivity of anisotropic or isotropic media consisting of aligned or randomly oriented ellipsoidal particles, *Water Resour. Res.*, **36**(10), 2821–2833, doi:10.1029/2000WR00198.
- Korosak, D., B. Cvikel, J. Kramer, R. Jecl, and A. Praprotnik (2007), Fractional calculus applied to the analysis of spectral electrical conductivity of clay-water system, *J. Contam. Hydrol.*, **92**, 1–9, doi:10.1016/j.jconhyd.2006.11.005.
- Krohn, C. E. (1988), Fractal measurements of sandstones, shales, and carbonates, *J. Geophys. Res.*, **93**, 3297–3305, doi:10.1029/JB093iB04p03297.

- Kruschwitz, S., and U. Yaramanci (2004), Detection and characterization of the disturbed rock zone in claystone with complex resistivity method, *J. Appl. Geophys.*, **57**, 63–79, doi:10.1016/j.jappgeo.2004.09.003.
- Leroy, P., A. Revil, S. Altmann, and C. Tournassat (2007), Modeling the composition of the pore water in a clay-rock geological formation (Callovo-Oxfordian, France), *Geochim. Cosmochim. Acta*, **71**, 1087–1097, doi:10.1016/j.gca.2006.11.009.
- Leroy, P., A. Revil, A. Kemna, P. Cosenza, and A. Ghorbani (2008), Spectral induced polarization of water-saturated packs of glass beads, *J. Colloid Interface Sci.*, **321**, 103–117, doi:10.1016/j.jcis.2007.12.031.
- Lesmes, D. P., and F. D. Morgan (2001), Dielectric spectroscopy of sedimentary rocks, *J. Geophys. Res.*, **106**, 13,329–13,346, doi:10.1029/2000JB900402.
- Lounev, I., R. Nigmatullin, A. Zavidonov, Y. Gusev, I. Manurov, S. Karunov, and R. Muslimov (2002), Analysis of dielectric relaxation data in water-saturated sands and clays, *J. Non Cryst. Solids*, **305**, 255–260, doi:10.1016/S0022-3093(02)01100-6.
- Lunev, I., R. Nigmatullin, A. Zavidonov, Y. Gusev, and I. Manurov (2001), The effect of clay morphology on water relaxation, *Tech. Phys.*, **46**(11), 1473–1474, doi:10.1134/1.1418517.
- Mammar, N., M. Rosanne, B. Prunet-Foch, J.-F. Thovert, E. Tevissen, and P. M. Adler (2001), Transport properties of compact clays I. Conductivity and permeability, *J. Colloid Interface Sci.*, **240**, 498–508, doi:10.1006/jcis.2001.7697.
- Marshall, D. J., and T. R. Madden (1959), Induced polarization: A study of its causes, *Geophysics*, **24**(4), 790–816, doi:10.1190/1.1438659.
- Rosanne, M., N. Mammar, N. Koudina, B. Prunet-Foch, J.-F. Thovert, E. Tevissen, and P. M. Adler (2003), Transport properties of compact clays II. diffusion, *J. Colloid Interface Sci.*, **260**, 195–203, doi:10.1016/S0021-9797(02)00240-0.
- Schmutz, M. (2000), Apports des méthodes géophysiques à la connaissance des glissements-coulées développés dans les marnes noires: Application à Super-Sauze (Alpes de Haute Provence, France), Ph.D. thesis, Univ. Louis Pasteur Strasbourg, Strasbourg, France.
- Scott, J. B. T. (2006), The origin of the observed low-frequency electrical polarization in sandstones, *Geophysics*, **71**(5), G235–G238, doi:10.1190/1.2258092.
- Scott, J. B. T., and R. D. Barker (2003), Determining pore-throat size in Permo-Triassic sandstones from low-frequency electrical spectroscopy, *Geophys. Res. Lett.*, **30**(9), 1450, doi:10.1029/2003GL016951.
- Sen, P. N. (1981), Relation of certain geometrical features to the dielectric anomaly of rocks, *Geophysics*, **46**(12), 1714–1720, doi:10.1190/1.1441178.
- Sen, P. N. (1984), Grain shape effects on dielectric and electrical properties of rocks, *Geophysics*, **49**(5), 586–587, doi:10.1190/1.1441695.
- Slater, L., D. Ntarlagiannis, and D. Wishart (2005), On the relationship between induced polarization and surface area in metal-sand and clay-sand mixtures, *Geophysics*, **71**(2), A1–A5, doi:10.1190/1.2187707.
- Thevanayagam, S. (1997), Dielectric dispersion of porous media as a fractal phenomenon, *J. Appl. Phys.*, **82**(5), 2538–2547, doi:10.1063/1.366065.
- Van Damme, H. (1995), Scale invariance and hydric behaviour of soils and clays, *C. R. Acad. Sci., Ser. IIa Sci. Terre Planetes*, **320**, 665–681.
- Van Loon, L. R., J. M. Soler, A. Jakob, and M. H. Bradbury (2003), Effect of confining pressure on the diffusion of HTO, $^{36}\text{Cl}^-$ and $^{125}\text{I}^-$ in a layered argillaceous rock (Opalinus clay): Diffusion perpendicular to the fabric, *Appl. Geochem.*, **18**, 1653–1662, doi:10.1016/S0883-2927(03)00047-7.
- Van Voorhis, G. D., P. H. Nelson, and T. L. Drake (1973), Complex resistivity spectra of porphyry copper mineralization, *Geophysics*, **38**(1), 49–60, doi:10.1190/1.1440333.
- Waxman, M. H., and L. J. M. Smits (1968), Electrical conductivities in oil-bearing shaly sand, *Soc. Pet. Eng. J.*, **8**, 107–122, doi:10.2118/1863-A.
- Yven, B., S. Sammartino, Y. Géraud, F. Homand, and F. Villières (2007), Mineralogy, texture and porosity of Callovo-Oxfordian argillites of the Meuse/Haute Marne region (Eastern Paris basin), *Bull. Soc. Geol. Fr.*, **178**, 73–90.

P. Cosenza, N. Florsch, and A. Ghorbani, Laboratoire Sisyphe, UMR7619, Université Pierre et Marie Curie, CNRS, 4 Place Jussieu, F-75252 Paris, CEDEX 05, France. (philippe.cosenza@upmc.fr)

D. Jougnot and A. Revil, Department of Geophysics, Colorado School of Mines, Golden, CO 80401, USA.

M. Schmutz, EGID, Université Michel de Montaigne, F-33607 Pessac, France.

M. Zamora, Equipe de Géomatériaux et Environnement, Institut de Physique du Globe de Paris, Université Paris Diderot, CNRS, F-75252 Paris, France.



Complex conductivity of water-saturated packs of glass beads

P. Leroy^{a,1}, A. Revil^{b,c,*}, A. Kemna^{d,3}, P. Cosenza^e, A. Ghorbani^e

^a CNRS, Université Aix-Marseille III, Aix-en-Provence, France

^b Department of Geophysics, Colorado School of Mines, Golden, CO 80401, USA

^c LGIT (UMR 5559), CNRS, Université de Savoie, Equipe Volcan, Le Bourget-du-Lac, France

^d Agrosphere (ICG 4), Forschungszentrum Jülich, Germany

^e UMR 7619 Sisyphe, Université Pierre et Marie Curie, Paris 6, Paris, France

Received 21 November 2007; accepted 22 December 2007

Available online 12 February 2008

Abstract

The low-frequency conductivity response of water-saturated packs of glass beads reflects a combination of two processes. One process corresponds to the polarization of the mineral/water interface coating the surface of the grains. The other process corresponds to the Maxwell–Wagner polarization associated with accumulation of the electrical charges in the pore space of the composite medium. A model of low-frequency conductivity dispersion is proposed. This model is connected to a triple-layer model of electrochemical processes occurring at the surface of silica. This model accounts for the partition of the counterions between the Stern and the diffuse layers. The polarization of the mineral/water interface is modeled by the electrochemical polarization model of Schurr for a spherical grain. We take into account also the DC surface conductivity contribution of protons of the sorbed water and the contribution of the diffuse layer. At the scale of a macroscopic representative elementary volume of the porous material, the electrochemical polarization of a single grain is convoluted with the grain size distribution of the porous material. Finally, the Maxwell–Wagner polarization is modeled using the complex conductivity of a granular porous medium obtained from the differential effective medium theory. The predictions of this model agree well with experimental data of spectral induced polarization. Two peaks are observed at low frequencies in the spectrum of the phase. The first peak corresponds to the distribution of the size of the beads and the second peak is due to the roughness of the grains.

© 2007 Elsevier Inc. All rights reserved.

Keywords: Spectral impedance; Porous media; Induced polarization; Complex conductivity; Double layer; Particle size distribution

1. Introduction

Complex conductivity appears a very promising method to image nonintrusively the three-dimensional distribution of the permeability of porous materials [1–5] and its evolution over time, to detect the presence of various types of contaminants in the ground [6], and to study the properties of colloidal suspensions in the laboratory [7–11]. The complex conductivity response of a water-saturated porous material usually exhibits

relaxation phenomena in the frequency range extending from milliHertz to few kiloHertz [1,7–11]. Some researchers [12–14] have proposed phenomenological models, such as the Cole–Cole model, to fit the effective conductivity data in this frequency range. Whereas these models are very useful for practical applications, they do not relate the effective conductivity response to intrinsic physical and chemical properties of the porous medium.

Several studies have investigated the fundamental processes leading to the low-frequency dispersion of colloids. Two processes are known to be responsible for the observed data. One process is related to the polarization of the electrical double or triple layer coating the surface of the grains: the electrochemical (EC) contribution [15–17]. A fixed charge density is located at the surface of all minerals in contact with water. This fixed charge density is counterbalanced by the charge densities of the

* Corresponding author.

E-mail address: arevil@mines.edu (A. Revil).

¹ Present address: BRGM, Orléans, France.

² Also at CNRS-ANDRA/GDR FORPRO 0788, France.

³ Present address: Department of Geodynamics and Geophysics, University of Bonn, Germany.

Stern and diffuse layers. The theory of the low-frequency EC conductivity dispersion was first elaborated for dilute suspensions of homogeneous spherical insulating particles surrounded by a thin mineral/water interface [15–19]. In this case, the diffuse layer and/or the Stern layer and the electrolyte located in the vicinity of the interface become polarized. However, in a water-saturated granular porous material with a finite contiguity between the grains, the overlapping of the electrical diffuse layer at the grain-to-grain contacts probably impedes the polarization of the electrical diffuse layer (the electrical diffuse layer is above a percolation threshold of the material and therefore does not polarize). It follows that the electrochemical polarization may be due to the polarization of the Stern layer alone.

The theory of Schwarz [15] takes into account the diffusional relaxation of the counterion distribution along tangential concentration gradients in a conductive layer located at the surface of spherical insulating grains. This model is valid for a binary symmetric 1:1 electrolyte such as NaCl or KCl. Schurr [16] generalized the model of Schwarz by including the effects of DC surface conductivity. In this model, he partitioned the fixed charge layer into two parts: an inner layer, which was tightly bound to the particle surface, and an outer layer, which was loosely bound to the particle surface. Dukhin and Shilov [17] studied the flux of ions in thin double layers and the surrounding symmetric electrolyte. A quite similar analysis was provided later by Fixman [18], O'Brien [20], and Hinch et al. [21]. Lyklema et al. [19] modified the Schwarz theory to include the polarization of the diffuse layer but not the polarization of the surrounding electrolyte.

The other process needed to model the complex conductivity of a water-saturated porous material is the Maxwell–Wagner (MW) polarization [22,23]. In a porous composite, the different phases have different conductivities and dielectric constants. The MW polarization is caused by the formation of field-induced free charge distributions near the interface between the phases of the composite. In the case of granular media, models based on the effective medium approximation [24–26] have successfully modeled the Maxwell–Wagner polarization of a pack of spherical grains. A closed-form solution for the electrical conductivity response of a granular porous composite was introduced [27,28]. Note that because the diffuse and Stern layers modify the conductivity of micropores and grains, respectively, they also influence the MW polarization of saturated porous materials.

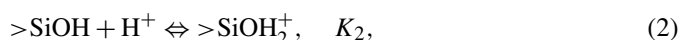
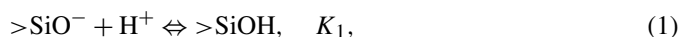
Recently, various researchers have combined the theories of the EC and MW polarizations to predict the complex dielectric responses of conductive porous media over a wide frequency range [29,30]. However, these models were not fully connected with an electrical triple-layer model (TLM) showing explicitly the role of the partition of the counterions between the Stern and the diffuse layers. Because the value of the partition coefficient describing the distribution of the counterions between the Stern and the diffuse layers depends on the ionic strength of the pore water, its composition, and its pH, this partition coefficient needs to be considered explicitly in polarization models. It is therefore crucial to combine the EC polarization models with an electrical double- or triple-layer model that accounts for the

complexation between the ions in the pore water solution and the active sites covering the surface of the mineral.

In this paper, we propose a model of low-frequency conductivity dispersion accounting for both electrochemical and Maxwell–Wagner polarizations, including a full connection to the electrical triple-layer model, and accounting for the grain size distribution of the porous material. This model was tested against experimental data.

2. Electrical triple-layer model

We consider silica grains in contact with a binary symmetric electrolyte such as NaCl for the simplicity of the presentation and the comparison with the experimental data. In the pH range 4–10, the surface mineral reactions at the silanol surface sites can be written as follows (see [31,32]):



and



The symbol $>$ refers to the mineral framework and K_1 , K_2 , and K_3 are the associated equilibrium constants for the different reactions reported above. Additional reactions for a multicomponent electrolyte can be easily incorporated by adding reactions similar to Eq. (3) or exchange reactions [33]. Therefore the present model is not limited to a binary salt. The protonation of surface siloxane groups $>\text{SiO}_2$ is extremely low and these groups can be considered inert [34]. We neglect here the adsorption of anion Cl^- at the surface of the $>\text{SiOH}_2^+$ sites which occurs at $\text{pH} < \text{pH}(\text{pzc}) \approx 2\text{--}3$ [35], where pzc is the point of zero charge of silica:

$$\text{pH}(\text{pzc}) = \frac{1}{2}(\log K_1 + \log K_2). \quad (4)$$

Consequently, K_2 is determined from the value of K_1 and the value of the pH of the zero point of charge. In addition, we have $\text{pH}(\text{pzc}) \approx 3$.

We develop an electrical TLM to determine the distribution of the coions and counterions at the mineral/water interface of silica minerals. The surface charge density Q_0 (in C m^{-2}) at the surface of the minerals (see Fig. 1) can be expressed as

$$Q_0 = e(\Gamma_{\text{SiOH}_2^+}^0 - \Gamma_{\text{SiO}^-}^0 - \Gamma_{\text{SiONa}}^0), \quad (5)$$

where Γ_i^0 is the surface site density of species i (in sites m^{-2}). The sites $>\text{SiONa}$ contribute to Q_0 in Eq. (5) because they are actually $>\text{SiO}^- \text{Na}^+$ sites with $>\text{SiO}^-$ located on the surface of the mineral and Na^+ located in the Stern layer (see Eq. (6) below). The surface charge density Q_β in the Stern layer is determined according to

$$Q_\beta = e\Gamma_{\text{SiONa}}^0. \quad (6)$$

The surface charge density in the diffuse layer is calculated using the classical Gouy–Chapman relationship in the case of a

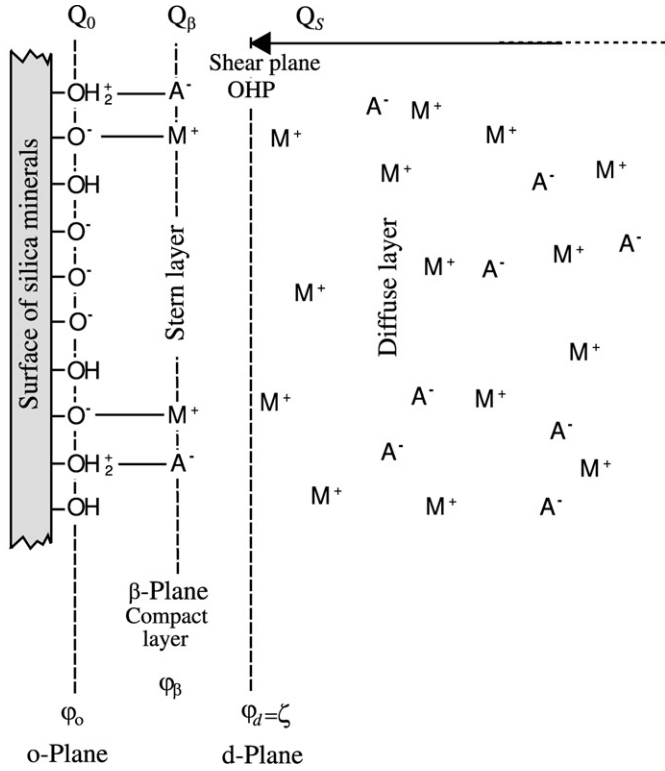


Fig. 1. Sketch of the electrical triple-layer model as suggested by Davis and Leckie [42]. M represents the metal cations (e.g., Na^+ or K^+) and A the anions (e.g., Cl^-). OHP represents the outer Helmholtz plane, which corresponds here to the shear plane where the zeta potential is defined. Q_0 is the surface charge density of the surface of the mineral, Q_β is the surface charge density of the Stern layer, and Q_S is the equivalent surface charge density of the diffuse layer.

symmetric monovalent electrolyte [36],

$$Q_S = -\sqrt{8\varepsilon_f k_b T C_f} \sinh(e\varphi_d/2k_b T), \quad (7)$$

where C_f is the salinity in the free electrolyte (in mol L^{-1}), T is the temperature (in K), ε_f is the permittivity of the pore water ($\varepsilon_f = 81\varepsilon_0$, $\varepsilon_0 \sim 8.85 \times 10^{-12} \text{ F m}^{-1}$), e represents the elementary charge (taken positive, $e = 1.6 \times 10^{-19} \text{ C}$), and k_b is the Boltzmann constant ($1.381 \times 10^{-23} \text{ J K}^{-1}$). The electrical potential φ_d (in V) is the electrical potential at the outer Helmholtz plane (OHP; see Fig. 1). We make the assumption that the electrical potential φ_d is equal to the zeta potential ζ placed at the shear plane. The shear plane is the hydrodynamic surface on which the relative velocity between the mineral grains and the pore water is null [37].

The continuity equation for the surface sites yields

$$\Gamma_1^0 = \Gamma_{\text{SiO}^-}^0 + \Gamma_{\text{SiOH}}^0 + \Gamma_{\text{SiOH}_2^+}^0 + \Gamma_{\text{SiONa}^+}^0, \quad (8)$$

where Γ_1^0 (in sites m^{-2}) is the total surface site density of the mineral. We use the equilibrium constants associated with the half-reactions to calculate the surface site densities Γ_i^0 . Solving Eq. (8) with the expressions of the equilibrium constants defined through Eqs. (1) to (3) yields

$$\Gamma_{\text{SiO}^-}^0 = \Gamma_1^0 / A, \quad (9)$$

$$\Gamma_{\text{SiOH}}^0 = \frac{\Gamma_1^0}{A} K_1 C_{\text{H}^+}^f \exp\left(-\frac{e\varphi_0}{k_b T}\right), \quad (10)$$

$$\Gamma_{\text{SiOH}_2^+}^0 = \frac{\Gamma_1^0}{A} K_1 K_2 C_{\text{H}^+}^{f^2} \exp\left(-\frac{2e\varphi_0}{k_b T}\right), \quad (11)$$

$$\Gamma_{\text{SiONa}^+}^0 = \frac{\Gamma_1^0}{A} K_3 C_{\text{Na}^+}^f \exp\left(-\frac{e\varphi_\beta}{k_b T}\right), \quad (12)$$

and

$$A = 1 + K_1 C_{\text{H}^+}^f \exp\left(-\frac{e\varphi_0}{k_b T}\right) + K_1 K_2 C_{\text{H}^+}^{f^2} \exp\left(-\frac{2e\varphi_0}{k_b T}\right) + K_3 C_{\text{Na}^+}^f \exp\left(-\frac{e\varphi_\beta}{k_b T}\right), \quad (13)$$

where φ_0 and φ_β are, respectively, the electrical potential at the o-plane corresponding to the mineral surface and the electrical potential at the β -plane corresponding to the plane of the Stern layer (Fig. 1). The symbols $C_{\text{H}^+}^f$ and $C_{\text{Na}^+}^f$ represent the concentrations of the protons and the sodium cations in the free electrolyte. The electrical potentials are related by

$$\varphi_0 - \varphi_\beta = Q_0 / C_1 \quad (14)$$

and

$$\varphi_\beta - \varphi_d = -Q_S / C_2 \quad (15)$$

[36], where C_1 and C_2 (in F m^{-2}) are the (constant) integral capacities of the inner and outer parts of the Stern layer, respectively. The global electroneutrality equation for the mineral/water interface is

$$Q_0 + Q_\beta + Q_S = 0. \quad (16)$$

We calculate the φ_d potential using Eqs. (5) to (16) and the procedure reported in [33,38]. We use $\Gamma_1^0 = 5 \text{ sites nm}^{-2}$ [39,40] and $C_2 = 0.2 \text{ F m}^{-2}$ [40–42]. We use the values of K_1 , K_3 , and C_1 given by Bolt [43] (see these values at Figs. 2a and 2b) to calculate the surface charge density Q_0 at the surface of silica mineral and the potential φ_d . Figs. 2 and 3 show good agreement between measured charge densities and zeta potentials and the model at various pHs and salinities.

As shown above, the counterions are located in both the Stern and the diffuse layers. The fraction of counterions located in the Stern layer is defined by

$$f_Q = \frac{\Gamma_{\text{SiONa}^+}^0}{\Gamma_{\text{SiONa}^+}^0 + \Gamma_{\text{Na}^+}^d}, \quad (17)$$

$$\begin{aligned} \Gamma_{\text{Na}^+}^d &\equiv \int_0^\infty [C_{\text{Na}^+}^d(r) - C_{\text{Na}^+}^f] dr \\ &= C_{\text{Na}^+}^f \int_0^\infty \left\{ \exp\left[-\frac{e\varphi(r)}{k_b T}\right] - 1 \right\} dr, \end{aligned} \quad (18)$$

and

$$\varphi(r) = \frac{4k_b T}{e} \tanh^{-1} \left[\tanh\left(\frac{e\varphi_d}{4k_b T}\right) \exp(-\kappa r) \right], \quad (19)$$

where r is the local distance defined perpendicularly from the interface between the pore water and the solid grain, φ is the

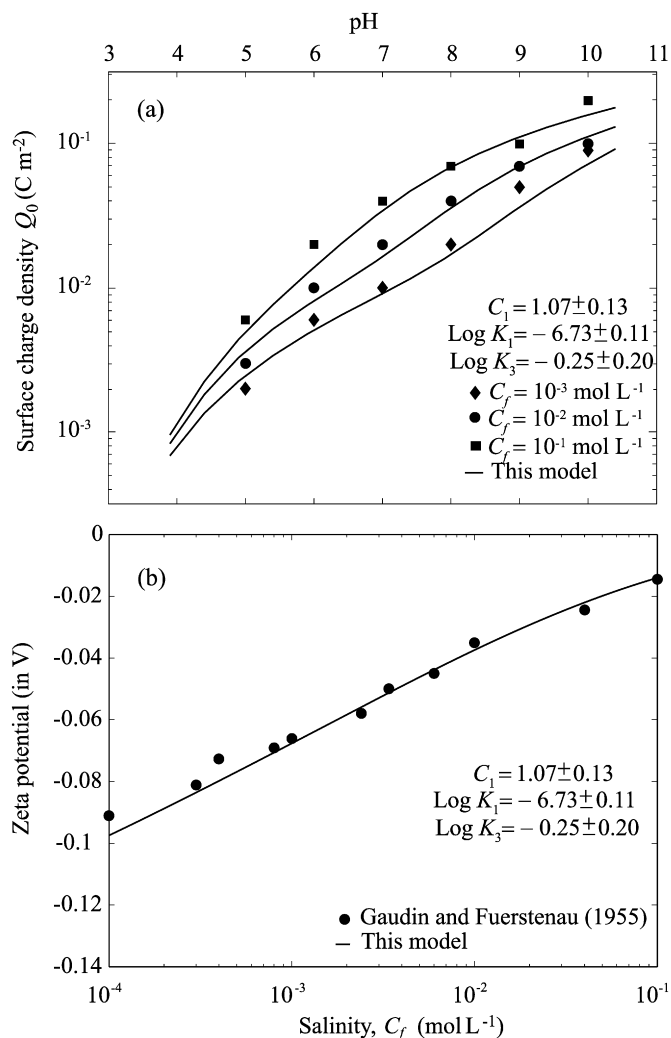


Fig. 2. Comparison between the predictions of the TLM model and the experimental data in the case of silica minerals. (a) Comparison between the prediction of the model and the surface charge density measurements obtained by potentiometric titrations at three different salinities (NaCl) and in the pH range 5–10 (data from [31]). (b) Comparison between the model prediction and the measurements of the zeta potential at different salinities and pH 6.5 (data from [61]). C_1 is expressed in F m⁻².

electrical potential in the diffuse layer, κ^{-1} is the Debye screening length (in m) ($2\kappa^{-1}$ corresponds approximately to the thickness of the diffuse layer; see [32]), and Γ_{Na}^d is the equivalent surface density of the counterions in the diffuse layer. Equation (19) can be found for example in Hunter [36]. The superscript d for the bulk concentration and surface density of sodium refers to the diffuse layer. In the next section, we will replace Γ_{SiONa}^0 by Γ_{Na}^S to denote the surface concentration of counterions in the Stern layer.

Fig. 4 shows that the fraction of counterions located in the Stern layer depends strongly on the salinity and the pH of the solution. For example, at pH 9 and at low salinities ($\leq 5 \times 10^{-4}$ mol L⁻¹) most of the counterions are located in the diffuse layer whereas at high salinity ($> 5 \times 10^{-4}$ mol L⁻¹) the counterions are located mainly in the Stern layer. At pH 9.5, most of the counterions are adsorbed in the Stern layer.

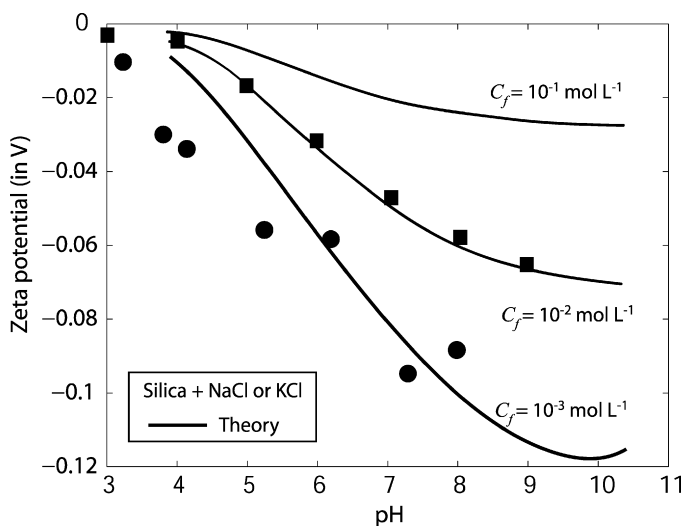


Fig. 3. Zeta potential versus pH at different background pore water salinities (NaCl). Data: filled squares (pure silica, 10⁻² M NaCl, from streaming potential measurements); filled circles (glass beads, 10⁻³ M NaCl) (data from [56], streaming potential measurements).

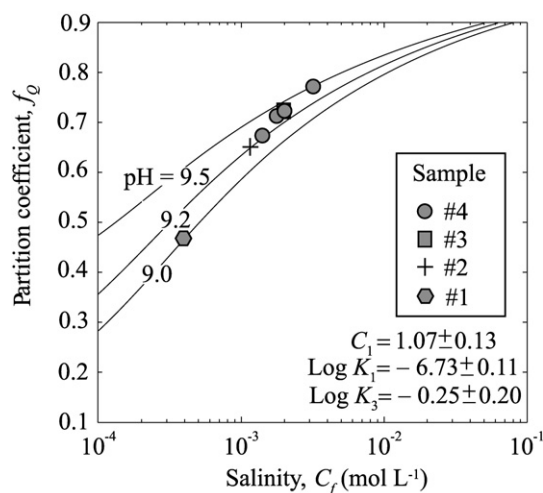


Fig. 4. Partition coefficient versus salinity of the free electrolyte with the TLM parameters indicated in Fig. 2 for NaCl (pH 9, 9.2, 9.5). The symbols correspond to the partition coefficient determined from the complex conductivity data for the seven experiments described in the main text. The data show an increase of the partition coefficient with the salinity and the pH in fair agreement with the model. C_1 is expressed in F m⁻².

3. Electrochemical polarization

As discussed in Section 2, the electrical double or triple layer comprises the Stern and the diffuse layers. The Stern layer is sometimes divided into two sublayers. These layers, coating a grain, are polarized when the grains are submitted to an alternating electric field. The polarization of the electrical double layer corresponds to the so-called EC polarization of the medium [19]. We first consider the case of a single spherical silica grain with a homogeneous charged surface and a thin diffuse layer satisfying the condition $a_0 \gg 2\kappa^{-1}$, where a_0 is the radius of the particle (expressed in meters and where κ^{-1} is the Debye screening length). We restrict ourselves to the polariza-

tion of the Stern layer and will not consider the polarization of the diffuse layer. Indeed, in a granular porous material with a small grain-to-grain contiguity, the overlapping of the electrical diffuse layer at the grain-to-grain contacts prevents its polarization. Therefore we believe that models based on the polarization of the diffuse layer do not represent correctly the EC polarization of a granular porous medium whereas these models are correct for a dilute suspension of grains with no contiguity between the grains. In Section 3.2, we will convolve the response obtained for a single grain size with the particle size distribution of the porous material.

3.1. Polarization of the Stern layer

A porous material made of spherical silica grains of radius a_0 is saturated by an electrolyte solution comprising M species. We note $q_i = \pm e z_i$, the charge of species i (expressed in C), where z_i is the valence of species i . We denote C_i (in mol L^{-1}) the concentration of species i (number of ions per unit pore volume). We consider an oscillating electrical field with magnitude $E_0 = |\mathbf{E}_0|$ and

$$\mathbf{E} = -\nabla\psi = \mathbf{E}_0 \exp(i\omega t), \quad (20)$$

where $i^2 = -1$ and ω is the angular frequency. Note that the electrical potential ψ due to the applied (external) electrical field is different from the electrical potential φ resulting in the electrical double layer. The superposition principle yields $\psi + \varphi$ for the total electrical potential in the pore space. The electrical potential ψ and the ionic concentrations C_i^f in the pore fluid are related by the Poisson equation, which combines Gauss's law and the Boltzmann distribution for the ionic concentrations:

$$\nabla \cdot (\varepsilon_f \nabla \psi) = - \sum_{i=1}^M q_i C_i^f. \quad (21)$$

In Eq. (21) the concentrations of the ionic species are described by Boltzmann distributions in the Coulombic field created by the charge density existing at the surface of the mineral. In the diffuse layer, the Poisson equation connects the electrical potential φ with the volumetric charge density $\bar{Q}_V$ (in C m^{-3}):

$$\nabla \cdot (\varepsilon_f \nabla \varphi) = -\bar{Q}_V. \quad (22)$$

The volumetric charge density per unit pore volume $\bar{Q}_V$ is related to the partition coefficient f_Q by [44],

$$\bar{Q}_V = (1 - f_Q) Q_V, \quad (23)$$

where Q_V represents the total bulk charge density corresponding to the excess of charge per unit pore volume of the porous material. For clays, this is this bulk charge density that can be obtained from cation exchange capacity measurements.

The complex effective conductivities of the particle and of the surrounding electrolyte are defined by

$$\sigma_s^* = \sigma_s + i\omega\varepsilon_s \quad (24)$$

and

$$\sigma_f^* = \sigma_f + i\omega\varepsilon_f. \quad (25)$$

The variables σ_s , σ_f , ε_s , and ε_f are the conductivities (in S m^{-1}) and the dielectric permittivities of the particle (s) and the fluid (f), respectively.

We consider now a species i adsorbed onto the mineral surface. The boundary condition at the surface of the grain is given by the integration of Gauss's law over a small pillbox enclosing the charged surface [16],

$$\varepsilon_s \frac{\partial \psi_s}{\partial r} \Big|_{a_0} - \varepsilon_f \frac{\partial \psi_f}{\partial r} \Big|_{a_0} = q_i \bar{F}_i^S + Q_{\text{DC}}, \quad (26)$$

where the subscripts s and f refer, respectively, to the electrical properties inside and outside the particle. The spherical coordinates r and θ are defined with $r = 0$ at the center of the grain and $\theta = 0$ in the direction of the electrical field. The expression containing $q_i \bar{F}_i^S$ (in C m^{-2}) is the deviation of the surface charge density of bound counterions in the Stern layer from its value in thermodynamic equilibrium. The parameter Q_{DC} denotes the surface charge density produced by divergent DC currents and due to the counterions in the diffuse layer and the protons of the water sorbed onto the mineral surface [32].

Schwarz [15] stated that the free charge transported to the surface of the sphere by normal currents may not respond to tangential electric fields. He considered only the diffusive flux of counterions in a spherical ionic atmosphere and no exchange of charges between the bound layer of counterions and the electrolyte. A previous theory by O'Konski [45] took into account the surface conductance, which was considered independent of frequency over the region of the low-frequency dispersion. O'Konski [45] introduced an integral surface conductivity operating at the solid/solution interface to represent its excess electrical conduction. He also considered a free exchange of charges between the surface layer and the electrolyte, but he neglected the ionic diffusion around the surface of the particle.

Schurr [16] combined the treatments of Schwarz and O'Konski and distinguished the bound charge current, which is defined as that which is completely canceled by the current due to chemical potential gradients in the steady state and the DC current which persists in accordance with Ohm's law in the steady state. In Schurr's theory, the ionic atmosphere around a charged sphere immersed in an aqueous electrolyte is divided into two parts: one represents a thin layer of counterions tightly bound to the particle surface as in the Schwarz model ("fixed layer") and another represents the diffuse layer. The counterions in the thin layer move tangentially to the surface. The diffuse layer charges exchange readily with the bulk electrolyte. There is no polarization of the diffuse layer and there is no exchange of ions between the Stern and the diffuse layer.

Far away from the grain (outside the electrical diffuse layer), the ionic concentrations approach their bulk values and the electrical field is equal to the applied electrical field:

$$C_i \rightarrow C_i^f \quad \text{as } r \rightarrow \infty \quad (27)$$

and

$$\nabla\psi = -\mathbf{E} \quad \text{as } r \rightarrow \infty. \quad (28)$$

The ionic flux density $\mathbf{j}_i$ (in $\text{mol m}^{-2} \text{s}^{-1}$) generated by the fluid velocity $\mathbf{v}_w$ (in m s^{-1}) and by the electrochemical potential gra-

dient $\nabla \tilde{\mu}_i$ is given by the Nernst–Planck equation

$$\mathbf{j}_i = C_i \mathbf{v}_w - \ell_{ii} \nabla \tilde{\mu}_i, \quad (29)$$

where ℓ_{ii} is a phenomenological coefficient (the terms ℓ_{ij} ($i \neq j$) represent interaction terms that are important only at high ionic strengths [46]). Equation (29) includes a convective term and an electrodiffusive term. We will consider the Coulombic interaction in the definition of the electrochemical potentials. This yields

$$\mathbf{j}_i = C_i \mathbf{v}_w - \frac{\beta_i C_i}{|q_i|} \nabla \tilde{\mu}_i, \quad (30)$$

where β_i is the mobility of species i (in $\text{m}^2 \text{s}^{-1} \text{V}^{-1}$) and $\mathbf{v}_w$ is the velocity of the fluid. The motion of the fluid results from electroosmotic coupling [63]. The gradient of the electrochemical potential of species i is defined by

$$\nabla \tilde{\mu}_i = k_b T \nabla \ln C_i + q_i \nabla \psi. \quad (31)$$

The counterions in the Stern layer are assumed to be able to move tangentially along the Stern layer. However, they are also assumed to be unable to enter or leave the Stern layer. This implies

$$\mathbf{j}_i \cdot \mathbf{n} = 0 \quad \text{at } r = a_0, \quad (32)$$

where $\mathbf{n}$ is the unit vector normal to the surface of the grain. In absence of chemical reactions with the mineral surface, the local continuity equation for the species i is

$$\nabla \cdot \mathbf{j}_i = -\frac{\partial C_i}{\partial t}. \quad (33)$$

Schwarz [15] assumed that, when an electrical field is applied to a spherical grain in solution, the charges in the fixed layer will migrate only in a direction tangential to the particle surface. From Eqs. (30) and (31), the flux of counterions is due to electromigration and diffusion. In spherical coordinates (with the center of the coordinate system at the center of a spherical grain), this flux is given by

$$j_i = -\frac{D_i^S}{a_0} \frac{\partial \Gamma_i^S}{\partial \theta} - \frac{\beta_i^S q_i \Gamma_i^S}{|q_i| a_0} \frac{\partial \psi_S}{\partial \theta}, \quad (34)$$

where the subscript or superscript S refers to the Stern layer, the diffusion coefficient of species i (in $\text{m}^2 \text{s}^{-1}$) is related to the mobility of this species by the Nernst–Einstein relationship $D_i^S = k_b T \beta_i^S / |q_i|$, and Γ_i^S is the concentration of species i in the Stern layer (in ions m^{-2}).

The variation of the surface concentration of bound counterions Γ_i^S can be due solely to electromigration within the electrical field and diffusion in a tangential direction. This implies according to Eqs. (33) and (34) and in spherical coordinates

$$\frac{1}{a_0^2 \sin \theta} \frac{\partial}{\partial \theta} \left(D_i^S \sin \theta \frac{\partial \Gamma_i^S}{\partial \theta} + \frac{\beta_i^S q_i \Gamma_i^S}{|q_i|} \sin \theta \frac{\partial \psi_S}{\partial \theta} \right) = \frac{\partial \Gamma_i^S}{\partial t}. \quad (35)$$

As stated by Schwarz [15], the applied electrical field does not deform the “fixed charge” at the surface of the particle. Consequently, the deviation of the surface concentration of bound

ions from its value in thermodynamic equilibrium is very small:

$$\bar{\Gamma}_i^S(\theta, t) = \Gamma_i^S(\theta, t) - \Gamma_i^{S0} \ll \Gamma_i^{S0}. \quad (36)$$

Equation (35) can be linearized to the form

$$\frac{D_i^S}{a_0^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \bar{\Gamma}_i^S}{\partial \theta} + \frac{q_i \Gamma_i^{S0}}{k_b T} \sin \theta \frac{\partial \psi_S}{\partial \theta} \right) = \frac{\partial \bar{\Gamma}_i^S}{\partial t}. \quad (37)$$

The DC current conduction is supposed to be independent of the bound charge polarization when the condition in Eq. (36) is satisfied. The DC surface conductivity Σ_S^0 resulting from electromigration in the diffuse layer and from the protons of the water adsorbed onto the mineral surface [32,45] is given by

$$\Sigma_S^0 = \int_{a_\zeta}^{2\kappa^{-1}} (\sigma_f(r) - \sigma_f) dr + \Sigma_S^0(\text{H}^+) \quad (38)$$

and

$$\Sigma_S^0 = e \sum_{i=1}^N \beta_i^f \Gamma_i^d + \Sigma_S^0(\text{H}^+), \quad (39)$$

where the subscript or superscript f refers to the electrolyte with conductivity σ_f (in S m^{-1}) and N is the number of species of counterions adsorbed onto the surface. In Eq. (38), $\sigma_f(r)$ is the local value of the conductivity of the pore water at distance r from the surface of the grains while σ_f is the electrical conductivity of the free pore water $\sigma_f = \sigma_f(r) \gg 2\kappa^{-1}$). We restrict ourselves now to the case where there are only counterions in the diffuse layer. In the following, we will suppose also that only cations exist at the mineral/water interface. The position a_ζ represents the position between the Stern and the diffuse layer around a spherical grain taken from the center of the grain. The density Γ_i^d is the surface site density of adsorbed counterions in the diffuse layer in the equilibrium state,

$$\Gamma_i^d \equiv \int_0^\infty [C_i^d(r) - C_i^f] dr \quad (40)$$

and

$$\Gamma_i^d = C_i^f \int_0^\infty \left\{ \exp \left[-\frac{q_i \varphi(r)}{k_b T} \right] - 1 \right\} dr, \quad (41)$$

where the electrical potential in the diffuse layer is given by Eq. (19).

The charge transport equation for the DC current is given by O’Konski [45]:

$$\frac{\Sigma_S^0}{a_0^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi_S}{\partial \theta} \right) + \sigma_f \frac{\partial \psi_f}{\partial r} \Big|_{a_0} - \sigma_s \frac{\partial \psi_s}{\partial r} \Big|_{a_0} = \frac{\partial Q_{\text{DC}}}{\partial t}. \quad (42)$$

The potential ψ_S must be derived from boundary conditions and the conservation equation for the electrical charge. The potential in the electrolyte ψ_f and the potential inside the sphere ψ_s satisfy the Laplace equation:

$$\nabla^2 \psi_f = 0 \quad \text{for } r > a_0 \quad (43)$$

and

$$\nabla^2 \psi_s = 0 \quad \text{for } r < a_0. \quad (44)$$

The boundary conditions are

$$\psi_s \quad \text{finite for } r \rightarrow 0, \quad (45)$$

$$\lim_{r \rightarrow a_0} \psi_f = \lim_{r \rightarrow a_0} \psi_s = \psi_s, \quad (46)$$

$$\psi_f \rightarrow -E_{0r} \cos \theta \exp(i\omega t) \quad \text{for } r \rightarrow \infty, \quad (47)$$

where E_{0r} is the radial component of $\mathbf{E}_0$, and

$$\varepsilon_s \frac{\partial \psi_s}{\partial r} \Big|_{a_0} - \varepsilon_f \frac{\partial \psi_f}{\partial r} \Big|_{a_0} = q_i \bar{\Gamma}_i^S + Q_{DC} \quad \text{for } r = a_0. \quad (48)$$

The deviation of the surface concentration of bound ions from its value in thermodynamic equilibrium and the electrical potential ψ_s are determined using Eqs. (35)–(48) and Legendre polynomials. We restrict ourselves now to the case of a binary monovalent electrolyte such as NaCl (or KCl). Equation (37) leads to the following relationship between $\bar{\Gamma}_{Na^+}^S$ and ψ_s (see [15]):

$$\bar{\Gamma}_{Na^+}^S = -\frac{e\Gamma_{Na^+}^{S0}}{(1 + i\omega\tau_{Na^+}^0)k_b T} \psi_s \quad (49)$$

and

$$\tau_{Na^+}^0 = \frac{a_0^2}{2D_{Na^+}^S}. \quad (50)$$

The relaxation time $\tau_{Na^+}^0$ (in s) is calculated using the known diffusion coefficients of the counterions in the Stern layer $D_{Na^+}^S$ (in $\text{m}^2 \text{s}^{-1}$).

The total current density $\mathbf{j}_S$ due to the movements of counterions in the Stern layer is defined by

$$\mathbf{j}_S = e\mathbf{j}_{Na^+}^S. \quad (51)$$

The flux of counterions is determined by the following expression in spherical coordinates:

$$j_{Na^+}^S = -\frac{\beta_{Na^+}^S k_b T}{ea_0} \left(\frac{\partial \Gamma_{Na^+}^S}{\partial \theta} + \frac{e\Gamma_{Na^+}^S}{k_b T} \frac{\partial \psi_s}{\partial \theta} \right). \quad (52)$$

According to Eqs. (49) and (52) and using the assumptions $\partial \Gamma_{Na^+}^S / \partial \theta \cong \partial \bar{\Gamma}_{Na^+}^S / \partial \theta$ and $\Gamma_{Na^+}^S \cong \Gamma_{Na^+}^{S0}$, we obtain

$$j_S = -\frac{1}{a_0} \beta_{Na^+}^S e \Gamma_{Na^+}^{S0} \left(\frac{i\omega\tau_{Na^+}^0}{1 + i\omega\tau_{Na^+}^0} \right) \frac{\partial \psi_s}{\partial \theta}, \quad (53)$$

$$\mathbf{j}_S = \beta_{Na^+}^S e \Gamma_{Na^+}^{S0} \left(\frac{i\omega\tau_{Na^+}^0}{1 + i\omega\tau_{Na^+}^0} \right) \mathbf{e}_S, \quad (54)$$

and

$$\mathbf{j}_S = \Sigma_S^* \mathbf{e}_S, \quad (55)$$

where the local electrical field $\mathbf{e}_S = -(1/a_0)(\partial \psi_s / \partial \theta) \mathbf{t}_S$ ($\mathbf{t}_S$ is one of the two unit vectors that are tangential to the surface of the grain), Σ_S^* is the complex surface conductivity (sometimes called the complex surface conductance because it is expressed

in Siemens) due to the fluxes of counterions in the Stern layer:

$$\Sigma_S^* = \beta_{Na^+}^S e \Gamma_{Na^+}^{S0} \left(\frac{i\omega\tau_{Na^+}^0}{1 + i\omega\tau_{Na^+}^0} \right). \quad (56)$$

In the case of insulating spherical particles coated by a conductive and fixed layer, the complex conductivity of the particle is determined using Eqs. (56) and (42)–(48). This yields

$$\sigma_s^* = i\omega\varepsilon_s + \sigma_s, \quad (57)$$

$$\sigma_s = \frac{2(\Sigma_S^* + \Sigma_S^0)}{a_0}, \quad (58)$$

$$\sigma_s = \frac{2}{a_0} \left[\beta_{Na^+}^S e \Gamma_{Na^+}^{S0} \left(\frac{i\omega\tau_{Na^+}^0}{1 + i\omega\tau_{Na^+}^0} \right) + \beta_{Na^+}^S e \Gamma_{Na^+}^d + \Sigma_S^0(\text{H}^+) \right], \quad (59)$$

and

$$\Sigma_S^0 = \beta_{Na^+}^S e \Gamma_{Na^+}^d + \Sigma_S^0(\text{H}^+), \quad (60)$$

where Σ_S^0 is the DC specific surface conductivity and $\Sigma_S^0(\text{H}^+)$ is a contribution from the protons of the water adsorbed onto the mineral surface (see Revil and Glover [32] who give $\Sigma_S^0(\text{H}^+) = 2.4 \times 10^{-9} \text{ S}$ at 25 °C).

If there is only one characteristic relaxation time (one specific adsorption mechanism), one can rewrite Eq. (59) as

$$\sigma_s = \frac{2}{a_0} (\Sigma_S^0 + \Sigma_S^\infty) - \frac{2}{a_0} \frac{\Sigma_S^\infty}{(1 + i\omega\tau_0)} \quad (61)$$

and

$$\Sigma_S^\infty = \beta_{Na^+}^S e \Gamma_{Na^+}^{S0}, \quad (62)$$

where Σ_S^∞ is the contribution of the Stern layer at high frequencies ($\omega \gg \tau_0^{-1}$). Equation (61) is similar to an equation used by [29].

3.2. Convolution with the particle size distribution

If the porous material is characterized by a particle size distribution (PSD), the effective surface conductivity is obtained by the convolution of the PSD with the conductivity response obtained for a Dirac distribution of grain size as a result of the superposition principle [47]

$$\sigma_s^* = f(a) \otimes \sigma_s^*(a, \omega) \quad (63)$$

and

$$\sigma_s^* = \sum_{i=1}^Q f(a_i) \sigma_s^*(a_i, \omega), \quad (64)$$

where $\otimes$ is the convolution product, a_i is the particle radius of the grain i , Q is the number of classes in the grain size distribution, and $f(a_i)$ represents the discretized version of the PSD $f(a)$. The total grain volume distribution is normalized to

unity:

$$\int_0^{\infty} f(a') da' = 1 \quad (65)$$

and

$$\sum_{i=1}^Q f(a_i) = 1. \quad (66)$$

This behavior can also be connected with the so-called Cole–Cole distribution widely used to represent the dispersive electrical properties of porous media [12,48]. In dealing with the dispersive complex surface conductivity function, we can write

$$\sigma_s^*(\omega) = \sigma_s^{\infty} + (\sigma_s^0 - \sigma_s^{\infty}) \int_0^{\infty} \frac{f(\tau)}{1 + i\omega\tau} d\tau, \quad (67)$$

where σ_s^{∞} is the high-frequency surface conductivity and σ_s^0 is the low-frequency surface conductivity. Both σ_s^{∞} and σ_s^0 are real parameters. The function $f(\tau)$ represents the distribution of the relaxation times. It is normalized such that

$$\int_0^{\infty} f(\tau) d\tau = 1. \quad (68)$$

In the case of a single relaxation time $f(\tau) = \delta(\tau - \tau_0)$, we recover the Debye distribution. The complex surface conductivity can be written as

$$\sigma_s^*(\omega) = \sigma_s'(\omega) + i\sigma_s''(\omega). \quad (69)$$

In the case of a Debye distribution, the real and imaginary parts of the surface conductivity are given by

$$\sigma_s'(\omega) = \sigma_s^{\infty} + (\sigma_s^0 - \sigma_s^{\infty}) \int_0^{\infty} \frac{f(\tau)}{1 + \omega^2\tau^2} d\tau, \quad (70)$$

$$\sigma_s'(\omega) = \sigma_s^{\infty} + \frac{(\sigma_s^0 - \sigma_s^{\infty})}{1 + \omega^2\tau_0^2}, \quad (71)$$

$$\sigma_s''(\omega) = (\sigma_s^{\infty} - \sigma_s^0) \omega \int_0^{\infty} \frac{\tau f(\tau)}{1 + \omega^2\tau^2} d\tau, \quad (72)$$

and

$$\sigma_s''(\omega) = \frac{(\sigma_s^{\infty} - \sigma_s^0)\omega\tau_0}{1 + \omega^2\tau_0^2}. \quad (73)$$

If we return to the mechanistic model developed in Section 3.1, we can identify the parameters

$$\sigma_s^0 = \frac{2}{a_0} \Sigma_s^0 \quad (74)$$

and

$$\sigma_s^{\infty} = \frac{2}{a_0} (\Sigma_s^0 + \Sigma_s^{\infty}). \quad (75)$$

A special form of relaxation time distribution is the Cole–Cole distribution

$$\sigma_s^* = \sigma_s^{\infty} + \frac{\sigma_s^0 - \sigma_s^{\infty}}{1 + (i\omega\tau_0)^{\alpha}}, \quad (76)$$

where α is an empirical exponent called the Cole–Cole exponent. In the limit $\alpha = 1$, we recover the Debye distribution. The probability distribution of the relaxation time for a Cole–Cole distribution is given by [49]

$$f(\tau) = \frac{1}{2\pi} \frac{\sin[\pi(\alpha - 1)]}{\cosh[\alpha \ln(\tau/\tau_0)] - \cos[\pi(\alpha - 1)]}. \quad (77)$$

This distribution is symmetric about $\tau = \tau_0$. For $0.5 \leq \alpha \leq 1$, the Cole–Cole distribution looks like a log normal distribution. The tail of the Cole–Cole distribution is becoming increasingly broad as α decreases [47]. Consequently, the Cole–Cole distribution can be used to represent the approximate dispersion associated with a log normal distribution of grain size.

4. Maxwell–Wagner polarization

Theories of effective dielectric properties of composite media are based on the Maxwell approach [22]. Wagner [23] generalized Maxwell's theory in the case of dilute suspensions of spheres. The so-called differential self-consistent model [24] allows the determination of the final concentration of inclusions by addition of infinitesimal portions of inclusions $\delta\Phi$. The increment $\delta\sigma_n$ of the effective conductivity of the mixture due to the addition of an infinitesimal portion of inclusions $\delta\Phi_n$ is given by

$$\delta\sigma_n = 3\sigma_n \frac{\sigma_s - \sigma_n}{2\sigma_n + \sigma_s} \delta\Phi_n, \quad (78)$$

where σ_s is the electrical conductivity of the particle. After integrating from $\sigma_n = \sigma_f$ and $\Phi_n = 0$ to $\sigma_n = \sigma$ and $\Phi_n = 1 - \varphi$ with φ the porosity of the medium, we obtain the conductivity of the mixture σ [24]:

$$\frac{\sigma - \sigma_s}{\sigma_f - \sigma_s} \left(\frac{\sigma_f}{\sigma} \right)^{1/3} = \varphi. \quad (79)$$

This equation was extended by Hanai [25] to model the complex electrical conductivity of a porous granular composite:

$$\frac{\sigma^* - \sigma_s^*}{\sigma_f^* - \sigma_s^*} \left(\frac{\sigma_f^*}{\sigma^*} \right)^{1/3} = \varphi, \quad (80)$$

$\sigma^* = \sigma + i\omega\varepsilon$ represents the total conductivity function proposed by Dias [50] and De Lima and Sharma [29] (Appendix A).

For nonspherical grains, Eq. (80) is generalized by [26,51]

$$\frac{\sigma^* - \sigma_s^*}{\sigma_f^* - \sigma_s^*} \left(\frac{\sigma_f^*}{\sigma^*} \right)^D = \varphi, \quad (81)$$

where D is the polarization exponent, which depends on the shape of the solid grains. The total conductivity response of a two-phase mixture consisting of rock grains and pore solution

is given by

$$\sigma^* = \frac{\sigma_f^*}{F} \left(\frac{1 - \sigma_s^*/\sigma_f^*}{1 - \sigma_s^*/\sigma^*} \right)^m \quad (82)$$

and

$$F = \varphi^{-m}, \quad (83)$$

where $m = 1/(1 - D)$ is called the cementation or Archie exponent. Equation (82) is consistent with the electrical conductivity σ^* of a pack of grains saturated by a solution of conductivity σ_f^* given by [52]:

$$\sigma^* = \frac{\sigma_f^*}{F}. \quad (84)$$

The exponent m depends on the shape of the grains. It is equal to 1.5 for spherical grains and it takes values > 1.5 for nonspherical grains [26,51–55]. However, for loosely packed granular materials, m can be as low as 1.3. In our model, Eq. (82) is solved numerically with a Matlab routine.

The complex conductivity can be also written as

$$\sigma^* = \sigma_{\text{eff}} + i\omega\epsilon_{\text{eff}}, \quad (85)$$

$$\sigma_{\text{eff}} = \text{Re}(\sigma^*), \quad (86)$$

and

$$\epsilon_{\text{eff}} = \text{Im}(\sigma^*/\omega). \quad (87)$$

The phase of the complex conductivity response is defined by

$$\Theta = \arctan(\omega\epsilon_{\text{eff}}/\sigma_{\text{eff}}). \quad (88)$$

In the following discussion, the absolute value of the phase of packs of glass beads will be given according to the definitions given in Appendix A.

A simulation of the phase response of a pack of glass beads is shown in Fig. 5 at different salinities. The first peak in the phase spectra corresponds to electrochemical polarization of the Stern layer around the grains and is directly sensitive to the grain size distribution. At higher frequencies, the response is dominated by the Maxwell–Wagner polarization. The comparison with the experimental data in the next section will also evidence electrochemical polarization associated with second-order heterogeneities in our opinion corresponding to the roughness of the surface of the grains. The effect of the roughness of the grains upon the phase spectra is situated in a frequency band that is usually intermediate between the polarization of the grains and the MW polarization.

5. Comparison with experimental data

We performed seven experiments with four samples of glass beads pack of different size and roughness and at different salinities. The experimental setups and results are displayed in Figs. 6–15 and summarized in Table 1. We first test the present model with experimental data obtained with one pack of glass beads of size 40–120 μm . Their chemical composition is $\text{SiO}_2 \sim 60.5\%$, $\text{Na}_2\text{O} \sim 12\text{--}18\%$, $\text{CaO} \sim 5\text{--}12\%$, $\text{MgO} \sim 4\%$,

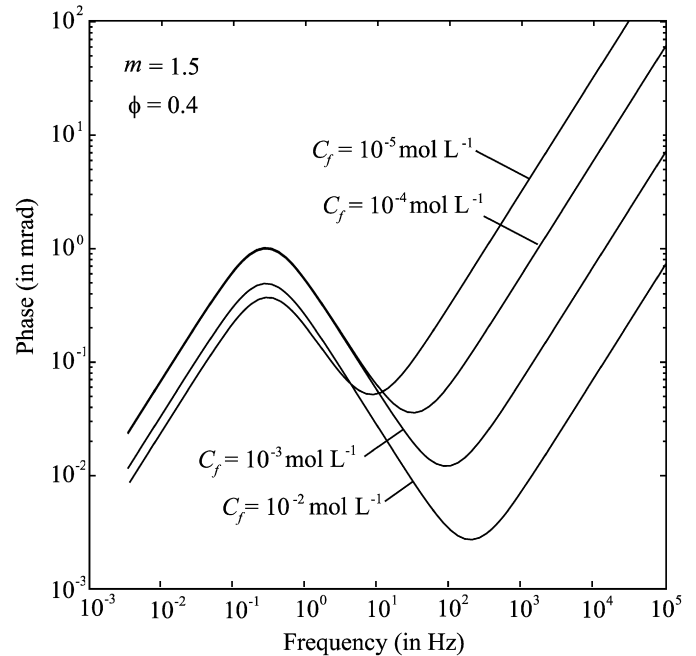


Fig. 5. Simulation of the response of a pack of glass beads (with no roughness of the grains) for four different salinities using the TLM model discussed in Section 2. The low-frequency peak corresponds to the polarization of the grains. The increase of the phase at higher frequencies corresponds to the Maxwell–Wagner polarization. We have assumed a log normal distribution for the PSD.

and $\text{Al}_2\text{O}_3 \sim 1\%$. They have electrochemical properties very similar to those of pure silica [56].

When the glass beads are in contact with water, the pH of the pore water is observed to increase. For example, we put 64 g of deionized water in contact with 255 g of these washed glass beads. The pH and the conductivity of the pore water reach values of 8.95 and $55 \times 10^{-4} \text{ S m}^{-1}$ after 45 min. Another experiment was performed by putting 111 g of deionized water in contact with 250 g of washed glass beads. The pH reaches a value of 9.24 and the pore water conductivity reaches a value of $1.03 \times 10^{-2} \text{ S m}^{-1}$ after 1 week. Results from the evolution of the composition of the pore water are reported in Table 2. The dominating cation resulting from the dissolution of the grains is sodium. The elevation of the electrical conductivity is probably related to the dissolution of the glass beads. This dissolution process brings cations constituting the glass beads to the solution. As a result, water dissociates to ensure electrical neutrality of the electrolyte and the electrical conductivity of the solution increases. This leads also to an increase of the concentration of hydroxide anions and to an elevation of the pH of the solution with time.

The experimental setup used to perform the first spectral-induced polarization experiment is shown in Fig. 6. Four non-polarizing electrodes are inserted in a box filled with the pack of glass beads and the geometrical factor is determined using the box filled with the electrolyte NaCl at a given electrical conductivity. The array is the Wenner- α array (the two electrodes used to measure the difference of electrical potential are the inner electrodes and the two other electrodes are used to inject

Table 1
Summary of the results of the seven experiments

Experiment	C_f (in mol L ⁻¹)	a_0 (μm)	Du (dimensionless)	Γ_i^S (10 ¹⁶ m ⁻²)	Γ_i^d (10 ¹⁶ m ⁻²)
1	4×10^{-4}	35	0.66	6.5	7.5
2	1.1×10^{-3}	187.5	0.24	5.5	3.0
3 and 4	2×10^{-3}	187.5	0.06	6.0	2.3
5	1.3×10^{-3}	187.5	0.11	6.0	2.9
6	1.8×10^{-3}	187.5	0.08	6.0	2.5
7	3.1×10^{-3}	187.5	0.05	6.0	1.8

Salinity, C_f ; peak of the grain size distribution, a_0 ; Dukhin number, Du (defined by $Du = \sigma'_s / \sigma_f$); density of the counterions in the Stern layer, Γ_i^S ; density of the counterions in the diffuse layer, Γ_i^d . In the text, the word “sample” is used to describe a couple “glass bead” and “electrolyte.” There are actually only three different glass beads used in our experiments: glass beads of size 1 (sample 1), glass beads of size 2 with rough surface (sample 2), and glass beads of size 2 with smooth surface (sample 3).

Table 2
Evolution of the composition of the pore water in contact with the glass beads (pack 1) at two different time lapses

Species	Concentration (in mol/L) (45 min)	Concentration (in mol/L) (1 week)
Ca ²⁺	1.8×10^{-4}	1.75×10^{-4}
K ⁺	7.7×10^{-6}	5.1×10^{-6}
Mg ²⁺	4.11×10^{-5}	4.11×10^{-5}
Na ⁺	4.48×10^{-4}	8.2×10^{-4}
Al ³⁺	3.1×10^{-6}	2.76×10^{-6}
pH	8.95	9.24
SiO ₂	5×10^{-4}	1.04×10^{-3}

In the first case, 255 g of glass beads were put in contact with 64 g of deionized water for 45 min. The mixture was in contact with the atmosphere. In the second case, 250 g of glass beads were put in contact with 111 g of deionized water (the gas is nitrogen). Chemistry is determined after 1 week.

the electrical current). This insures a high signal-to-noise ratio. In addition, we checked that the drift of the electrodes can be neglected. Spectral induced polarization (SIP) measurements were performed with the SIP-Fuchs system operating in the frequency range 10⁻¹–10⁴ Hz (see http://www.radic-research.de/Flyer_SIP-Fuchs_II_151104.pdf).

Given the chemical composition of the glass beads and the kinetics associated with the dissociation of Na₂O, the major cationic constituent of the solution is sodium. However, Ca²⁺ and Mg²⁺ may be present in nonnegligible quantities compared to the quantity of sodium. These cations have similar ionic mobilities in solution ($5.19 \times 10^{-8} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$ for Na⁺, $6.16 \times 10^{-8} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$ for Ca²⁺, and $5.48 \times 10^{-8} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$ for Mg²⁺ [27]). Therefore the peaks associated with the spectrum of the phase, at low frequencies, would be located at the same characteristic frequency (see Eq. (50)). As in Revil and Leroy [44], we assume that the value of the partition coefficient f_Q in the case of a binary monovalent electrolyte is similar to its value for a multiionic electrolyte. In the case of a smectite mineral immersed in a multiionic solution and at high ionic strength Leroy et al. [33] found a value of f_Q equal to 0.94. This value is not too far from the value calculated by Leroy and Revil [38] (0.98) for the same mineral but in the case of a binary monovalent electrolyte such as NaCl or KCl.

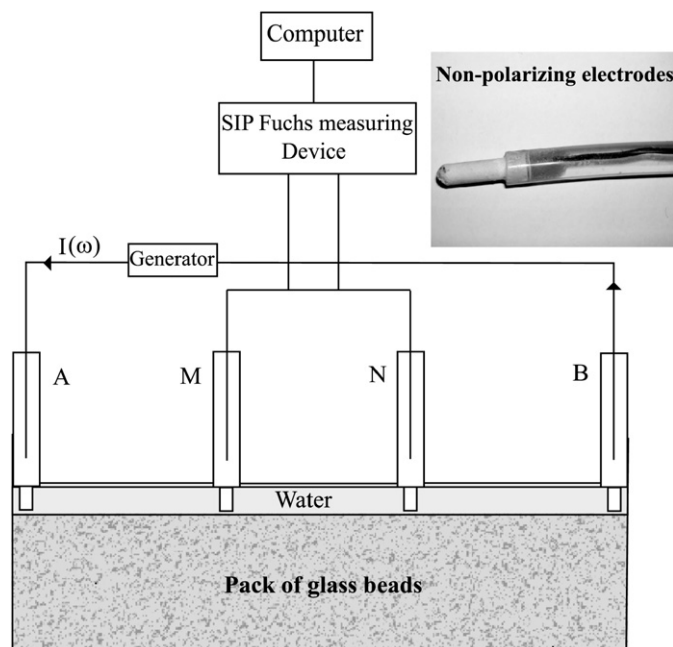


Fig. 6. Sandbox used for the spectral induced polarization (SIP) experiment. We use four Cu/CuSO₄ nonpolarizing electrodes with the Wenner- α configuration. The geometrical factor used to convert the conductance to conductivity is determined from a similar experiment using an electrolyte of known conductivity. The current at a given frequency is injected through the outer electrodes (A, B), and the inner electrodes (M, N) measure the electrical field associated with the polarization of the porous medium. The length of the sandbox is 30 cm and its width is 20 cm.

The measured pH of the solution was 9.0 and the electrical conductivity of the solution (at 25 °C) was equal to $5 \times 10^{-3} \text{ S m}^{-1}$.

The phase measurements and the model response as a function of the frequency are plotted in Fig. 8. Taking $D_S = 2.45 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$ and $a_0 = 35 \mu\text{m}$, the critical frequency for the electrochemical polarization is equal to 2 Hz (inverse of the relaxation time calculated by Eq. (50)), which corresponds to the peak of the first relaxation observed in Fig. 8. At frequencies higher than 1 kHz, the response of the medium is dominated by the MW polarization. However, above 1 kHz, the measured phase data are also likely to be contaminated by coupling effects associated with the wire layout of the experimental setup and overlapping with the Maxwell–Wagner polarization.

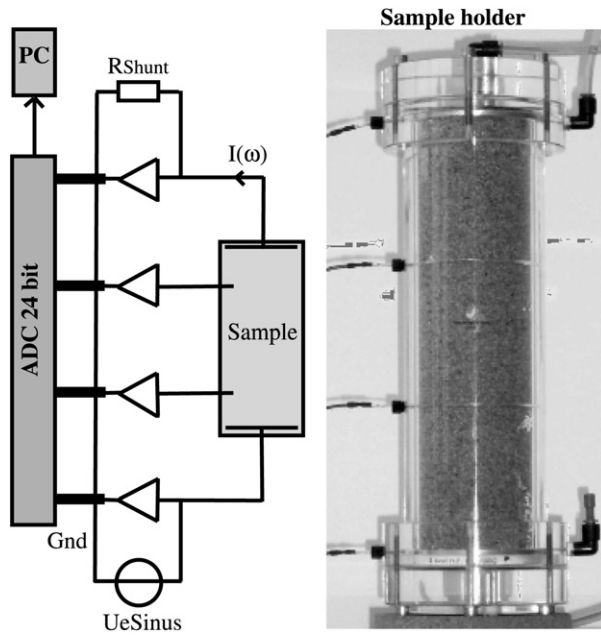


Fig. 7. Second experimental setup used for the measurements. The employed cylindrical sample holder has a length of 30 cm and an inner diameter of 6 cm. A four-electrode array consisting of two outer current electrodes (porous bronze plates with an effective pore diameter of 15 μm) and two inner potential electrodes (silver rings fixed in grooves 10 cm apart) is used.

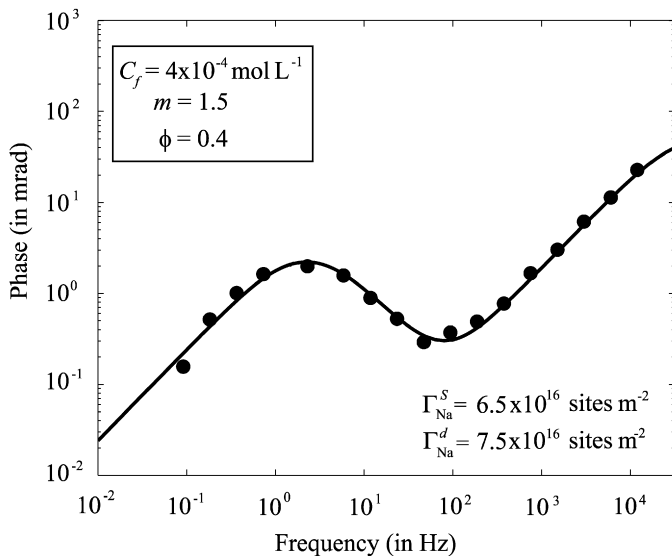


Fig. 8. Measured (filled circles) and calculated (solid line) angular frequency ω dependence of the phase of a pack of glass beads (sample 1). Measured porosity 0.40; pH (measured) 9.0. The cementation exponent ($m = 1.5$) is determined from electrical conductivity measurements [56]. The salinity of the pore solution used in the model is equal to $4 \times 10^{-4} \text{ mol L}^{-1}$ and the electrical conductivity is $5 \times 10^{-3} \text{ S m}^{-1}$ which corresponds to the measured electrical conductivity. Parameters determined from the TLM model: density of the counterions in the diffuse layer ($7.5 \times 10^{16} \text{ counterions m}^{-2}$) and density of the counterions in the Stern layer ($6.5 \times 10^{16} \text{ counterions m}^{-2}$). We use $D_S = 2.45 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$.

The mean particle diameter was determined with a laser granulometer (Laser Scattering Mastersizer MALVERN operating in the range 50 nm–2 mm; Fig. 9). The PSD shows the

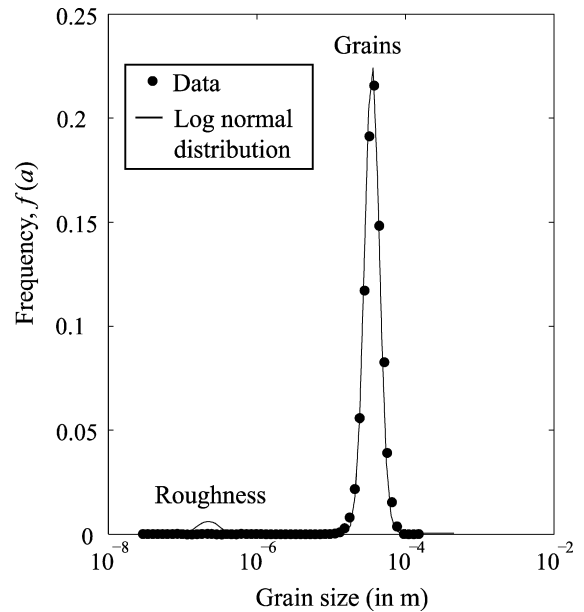


Fig. 9. Particle size distribution of the investigated glass beads pack (sample 1) determined using a laser granulometer. The solid line represents a fit of the data with two log normal distributions (mean grain radius $a_0 = 35$ and $0.2 \mu\text{m}$).

main distribution in the range 20–50 μm . The particle size distribution (shown in Fig. 9) is fitted with the following log normal distribution,

$$f(a) = \frac{1}{\sqrt{2\pi}\hat{\sigma}} \exp \left\{ - \left[\frac{\log(a/a_0)}{\sqrt{2}\hat{\sigma}} \right]^2 \right\}, \quad (89)$$

with $a_0 = 35 \mu\text{m}$ corresponding to the first peak of the PSD and $\hat{\sigma} = 0.1$ as the standard deviation. The roughness is described by a second log normal distribution and the parameters of this second distribution are fitted to the phase data. The second peak may be caused by the formation of silica gel layers at the surface of the mineral created at $\text{pH} > 8$ [62] and by other heterogeneities at the surface of the glass beads. The measured mass density of the grains is 2430 kg m^{-3} and the measured porosity of the various packs used in this study is $\phi = 0.40$ [56] irrespective of their grain sizes.

A comparison between our model and the experimental data is shown in Fig. 8. The porosity, the salinity, and the pH are measured, the value of the cementation exponent ($m = 1.5$) is assumed, and the values of the parameters describing the log normal distribution associated with the roughness of the grains are fitted to the data assuming that the roughness of the grains is described by a single log normal distribution. There is a good agreement between the model and the experimental data over the frequency range investigated. Note that this can be achieved only if the mobility of the counterions in the Stern layer is equal to the mobility of the counterions in the diffuse layer. This is in agreement with the results obtained by Lesmes and Morgan [47] using spectral induced polarization data and Tarasov and Titov [57] using time-domain induced polarization data.

Experiments 2 to 7 were performed on glass beads of size 250–500 μm using the experimental setup shown in Fig. 7 (see [5]). The used instrumentation, developed at the Central

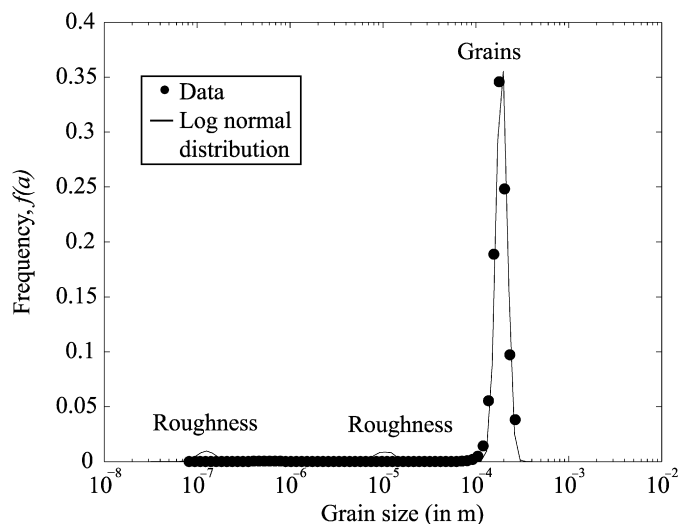


Fig. 10. Particle size distribution of the investigated glass beads pack (sample 2) determined using a laser granulometer. The solid line represents a fit of the data with a log normal distribution plus two small log normal distributions to include the effect of the roughness of the grains.

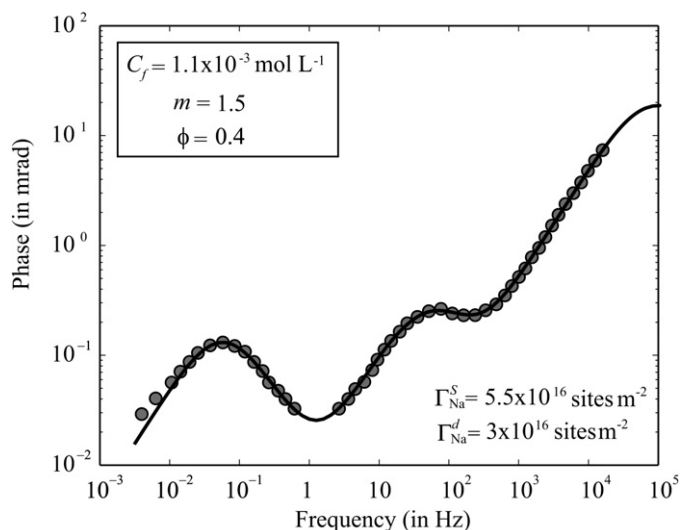


Fig. 11. Measured (filled circles) and calculated (solid line) angular frequency ω dependence of the phase of a pack of glass beads (sample 2). Measured porosity 0.40; pH (measured) 9.2; we assume $m = 1.5$. The salinity of the pore solution used in the model is equal to $1.1 \times 10^{-3} \text{ mol L}^{-1}$ and the electrical conductivity is $1.5 \times 10^{-2} \text{ S m}^{-1}$ which corresponds to the measured electrical conductivity. Parameters determined from the TLM model: density of the counterions in the diffuse layer ($3 \times 10^{16} \text{ counterions m}^{-2}$) and density of the counterions in the Stern layer ($5.5 \times 10^{16} \text{ counterions m}^{-2}$). We use $D_S = 2.45 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$.

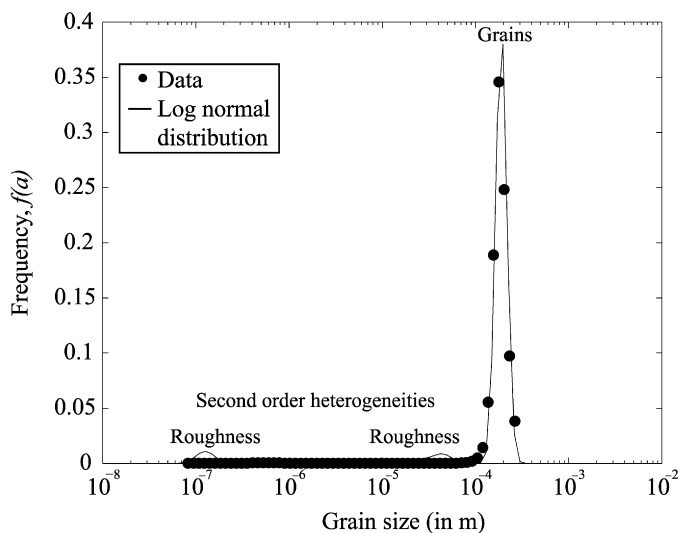


Fig. 12. Particle size distribution of the investigated glass beads pack determined using a laser granulometer (sample 3). The solid line represents a fit of the data with a log normal distribution plus two small log normal distributions to include the effect of the roughness of the grains.

to the size of the roughness at the surface of the grains (see Fig. 15).

In the third experiment (glass beads with smooth surface), the pH of the solution was measured at 9.3 and the electrical conductivity of the solution determined experimentally (at 25 °C) was equal to $2.6 \times 10^{-2} \text{ S m}^{-1}$ (the glass beads remained 3 weeks in a closed box). The grain size distribution of sample 3 is reported in Fig. 12. The surface of the beads is smooth, with a small roughness. The measured phase spectrum is shown in Fig. 13. For sample 3, taking $D_S = 2.45 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$ and

Laboratory for Electronics (ZEL) at the Forschungszentrum Jülich (Germany), offers a broader measurement spectrum (1 mHz–45 kHz) and a higher phase accuracy (below 1 kHz better than 0.1 mrad) [5] than the device used in experiment 1. The glass beads were put initially with deionized water and remained 1 to several weeks in a closed box. The samples of glass beads have the following chemical composition: 72% SiO_2 , 14% Na_2O , 9% CaO , 4% MgO , and 1% Al_2O_3 . The SIP measurements were done at the laboratory Agrosphere (ICG 4) at the Forschungszentrum Jülich (Germany). The employed cylindrical sample holder has a length of 30 cm and an inner diameter of 6 cm (Fig. 7). A four-electrode array consisting of two outer current electrodes (porous bronze plates with an effective pore diameter of 15 μm) and two inner potential electrodes (silver rings fixed in grooves 10 cm apart) is used. The measuring system was in particular tested on electrical networks [5] with phase responses in a range typical of sedimentary rocks.

Fig. 10 shows the particle size distribution of the second sample (glass beads with rough surface) measured by a laser granulometer. The phase angle distribution is shown in Fig. 11. The agreement between simulated and measured phase spectra (taking into account a parasitic capacitance of 0.5 pF) is very good, even for phase values much below 1 mrad. The measured pH of the solution was 9.2 and the measured electrical conductivity of the solution (at 25 °C) was equal to $1.5 \times 10^{-2} \text{ S m}^{-1}$ (the glass beads remained 2 weeks in a closed sample holder). For the second sample, the peak of the PSD is at $1.875 \times 10^{-4} \text{ m}$ (187.5 μm). Taking $D_S = 2.45 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$ and $a_0 = 187.5 \mu\text{m}$, the critical frequency for the electrochemical polarization is equal to 75 mHz, which corresponds to the peak of the first relaxation observed in Fig. 11 (first peak at ~ 70 –80 mHz). The peak of the second maximum is situated at a critical frequency ω_c approximately equal to 80 Hz. Using $a_0 = (2D_i^S/\omega_c)^{1/2}$, this would correspond to heterogeneities with a size of 6 μm . These heterogeneities correspond very well

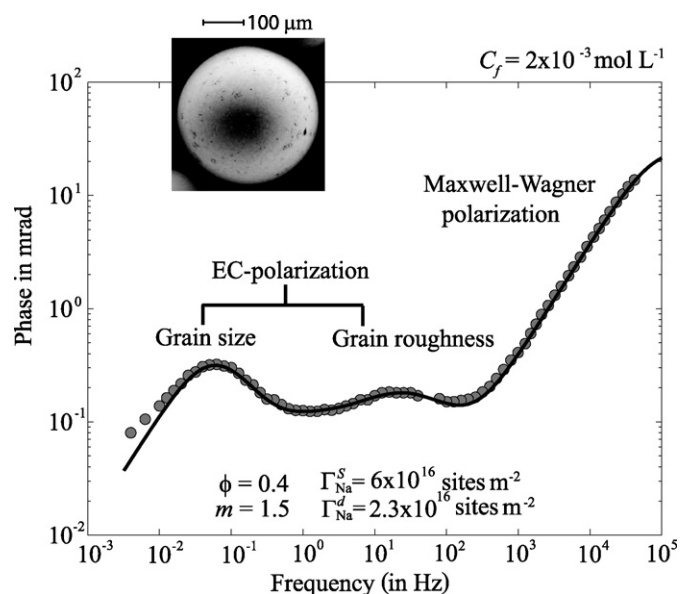


Fig. 13. Measured (filled circles) and calculated (solid line) angular frequency ω dependence of the phase of a pack of smooth glass beads (sample 3). Measured porosity 0.40; pH (measured) 9.3; we assume $m = 1.5$. The salinity of the pore solution used in the model is equal to $2 \times 10^{-3} \text{ mol L}^{-1}$ and the electrical conductivity is $2.6 \times 10^{-2} \text{ S m}^{-1}$ which corresponds to the measured electrical conductivity. Parameters determined from the TLM model: density of the counterions in the diffuse layer ($2.3 \times 10^{16} \text{ counterions m}^{-2}$) and density of the counterions in the Stern layer ($6 \times 10^{16} \text{ counterions m}^{-2}$). The picture shows the size of the glass beads and their roughness. The beads are relatively smooth.

$a_0 = 187.5 \mu\text{m}$, the critical frequency for the electrochemical polarization is equal to 75 mHz (as for sample 2). This value is close to the value of the frequency corresponding to the peak of the first relaxation observed in Fig. 13 (first peak at a critical frequency of ~ 70 – 80 mHz). The second peak is around 20 Hz and corresponds to heterogeneity of the size of $12 \mu\text{m}$. This is the size of the heterogeneities associated with the roughness of the grains (Fig. 13).

Fig. 14 presents a comparison of the phase spectra of packs of glass beads with smooth and rough surfaces (samples 3 and 4; sample 4 was treated with an acid to increase the roughness of the surface of the beads). In the case of the beads pack with a rough surface, the second peak of the phase spectrum is almost as high as the first peak. Fig. 14 shows very well the effect of the roughness upon the phase spectra and especially the increase of the amplitude of the second peak when roughness increases.

The grain size distribution and phase data corresponding to beads pack with rough surfaces (sample 4) are reported in Fig. 15. The measurements were realized at three different ionic strengths. The electrical conductivities of the solution were measured as $1.7 \times 10^{-2} \text{ S m}^{-1}$ (pH 9.2), $2.3 \times 10^{-2} \text{ S m}^{-1}$ (pH 9.3), and $4.1 \times 10^{-2} \text{ S m}^{-1}$ (pH 9.5). Our model is able to reproduce the experimental data quite well by accounting for the effect of the strong roughness of the glass beads. Note that the agreement between the data and the model decreases at very low frequency ($< 10^{-2} \text{ Hz}$). This is due probably to the multi-ionic character of the solution and/or electrode effects in the phase measurements. As shown in Fig. 4 the predictions of the

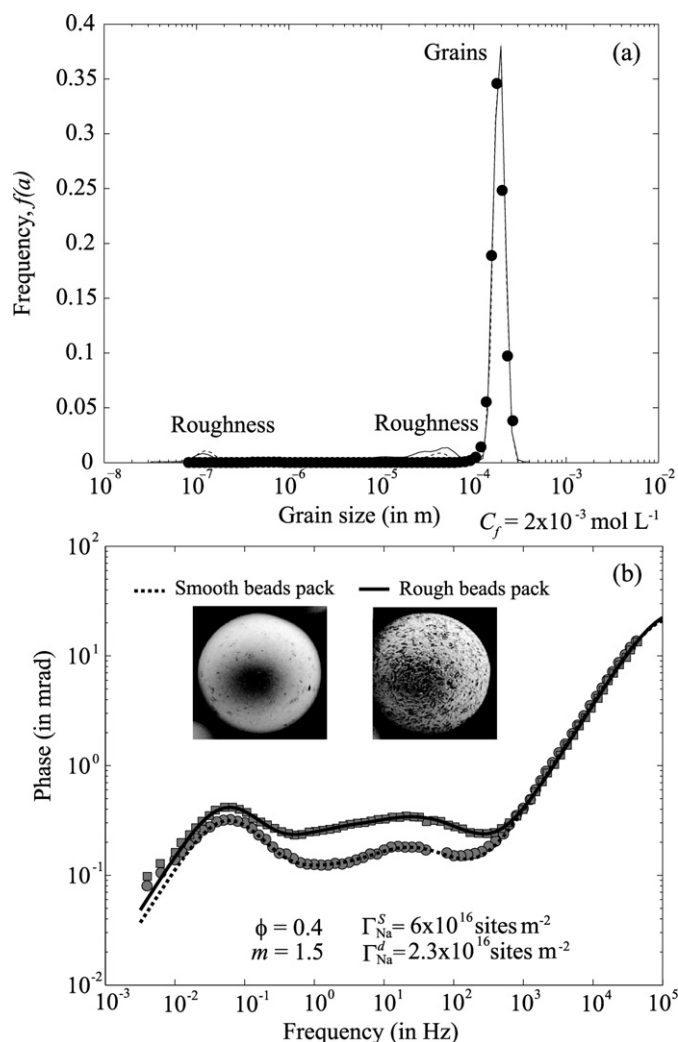


Fig. 14. Measured (filled circles) and calculated (solid and dashed lines) grain size (a) and phase (b) for smooth and rough glass beads (samples 3 and 4). The measured porosity is 0.40 and the measured pH is 9.3 for an electrical conductivity of $2.6 \times 10^{-2} \text{ S m}^{-1}$. We assume $m = 1.5$. Parameters determined from the TLM model: density of the counterions in the diffuse layer ($2.3 \times 10^{16} \text{ counterions m}^{-2}$) and density of the counterions in the Stern layer ($6 \times 10^{16} \text{ counterions m}^{-2}$).

TLM concerning the partition of the counterions between the Stern and the diffuse layers are in agreement with the values calculated by the SIP model. The partition coefficient increases with the ionic strength and with the pH. Spectral induced polarization seems to offer the possibility to monitor nonintrusively ongoing dissolution of a mineral surface and the concomitant variation of the roughness of this surface.

6. Concluding statements

A low-frequency polarization model of water-saturated granular media accounting for both the electrochemical polarization of the Stern layer and the Maxwell–Wagner polarization is developed. The electrochemical polarization is connected to the properties of the electrical triple layer coating the surface of the grains and especially to the partition coefficient of the counterions between the Stern and the diffuse layers. The elec-

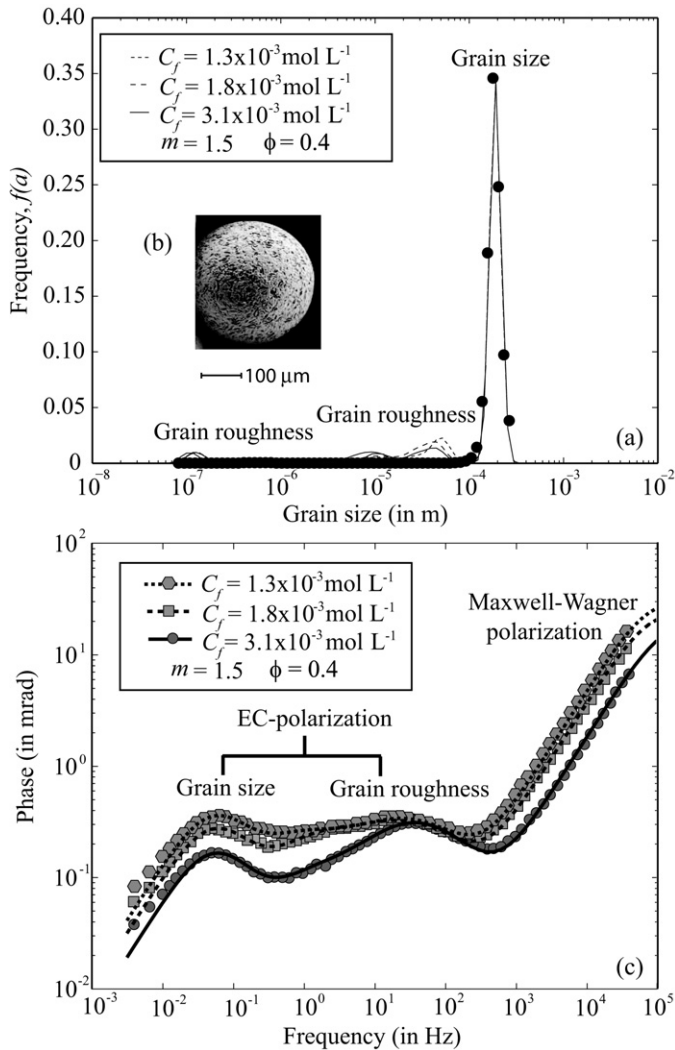


Fig. 15. Measured (filled circles) and calculated (solid and dashed lines) grain size (a) and phase (c) of a pack of rough glass beads (sample 4). The measured porosity is 0.40 and the measured pH is variable and depends on the ionic strength of the solution: 9.2 for an electrical conductivity of $1.7 \times 10^{-2} \text{ S m}^{-1}$, 9.3 for $\sigma_f = 2.3 \times 10^{-2} \text{ S m}^{-1}$, and 9.5 for $\sigma_f = 4.1 \times 10^{-2} \text{ S m}^{-1}$. We assumed $m = 1.5$. Parameters were determined from the TLM model (see the values of the densities of counterions in Table 1). The picture (b) shows the size of the glass beads and their strong roughness. The size of the heterogeneities at the surface of the grains is on the order of $10 \mu\text{m}$ and smaller.

trochemical polarization is controlled by the Stern layer. The polarization of the diffuse layer is neglected because of the connectivity of the diffuse layers at the macroscopic scale and therefore the diffuse layer has only a DC conductivity contribution. We consider also the DC conductivity contribution from the protons of the water adsorbed onto the mineral surface. It follows that surface conductivity of the grains is frequency dependent with a relaxation time that depends on the size of the grain. For a grain size distribution given by a log normal distribution, we use a Cole–Cole distribution to represent the influence of the grain size distribution upon the distribution of the relaxation times. In the general case, we can use the superposition principle and therefore a convolution product between the particle size distribution and the response of a single grain. The

Maxwell–Wagner polarization is accounted for by using the differential effective medium theory at the scale of a representative elementary volume of the porous material. Comparisons of the phase spectra of infiltrated glass beads are successful assuming that the mobility of the counterions is the same in the Stern layer and in the diffuse layer. The model implies that the partition coefficient of the counterions in the electrical triple layer increases with the salinity of the pore water in agreement with TLM calculations.

Acknowledgments

This work was supported by the NSF (Contract NSF-0710949), the GDR-FORPRO, the French National Agency for Radioactive Waste Management (ANDRA), and ANR-CNRS-INSU-ECCO. The post-doctoral grant of P. Leroy was supported by ANDRA. This is contribution FORPRO 2007/11 A. We thank T. Conte (BRGM) for the geochemical measurements and A. Lassin for fruitful discussions. Moreover, we are grateful to a number of colleagues at the Forschungszentrum Jülich, in particular Odilia Esser (ICG 4), who assisted with the spectral induced polarization measurements, Oliver Kranendonck (formerly ICG 5), who performed the laser granulometric measurements, Walter Schröder (ICG 3) and Hans-Martin Münch (formerly ICG 4), who provided the scanning electron microscope images, and Egon Zimmermann and Walter Glaas (both ZEL), who helped optimizing the SIP data acquisition.

Appendix A

Ampère's law is written as [22],

$$\nabla \times \mathbf{H} = \mathbf{j} + \frac{\partial \mathbf{D}}{\partial t}, \quad (\text{A.1})$$

where t is time (in s), $\mathbf{j}$ is the electrical current density (in A m^{-2}), $\mathbf{H}$ is the magnetic field intensity (in A m^{-1}), and $\mathbf{D} = \epsilon \mathbf{E}$ is the dielectric displacement (in C m^{-2}). We neglect the streaming current density in the expression of the current density and therefore the current density is given by Ohm's law ($\mathbf{j} = \sigma \mathbf{E}$) and we assume that the electrical field is given by

$$\mathbf{E} = \mathbf{E}_0 \exp(i\omega t) \quad (\text{A.2})$$

(engineer's convention). According to Eq. (A.1), the current density is the sum of a conduction current plus a displacement current. It follows that the total current density is given by

$$\mathbf{J}_t = (\sigma + i\omega\epsilon)\mathbf{E}, \quad (\text{A.3})$$

where σ and ϵ are complex scalars dependent upon frequency, $\sigma = \sigma' + i\sigma''$ and $\epsilon = \epsilon' + i\epsilon''$ (with the sign convention taken from Fuller and Ward [58] and Keller [59]) in an isotropic medium (note that Olhoeft [60] used a different sign convention, $\sigma = \sigma' + i\sigma''$ and $\epsilon = \epsilon' - i\epsilon''$, because with this convention σ' , σ'' , ϵ' , and ϵ'' have positive values). The total current density can be written as

$$\mathbf{J}_t = \sigma^* \mathbf{E}, \quad (\text{A.4})$$

where $\sigma^* = \sigma_{\text{eff}} + i\omega\epsilon_{\text{eff}}$ is the total current conductivity function and σ_{eff} and ϵ_{eff} are real scalars dependent upon frequency. These effective parameters are the parameters that are measured during an experiment. It follows that $\sigma_{\text{eff}} = \sigma' - \omega\epsilon''$ and $\epsilon_{\text{eff}} = \epsilon' + \sigma''/\omega$ (note that $-\omega\epsilon''$ is always positive). Therefore the effective conductivity includes contributions from charge conduction and polarization mechanisms. Similarly, the effective permittivity has contributions from both conduction and polarization.

References

- [1] G.R. Olhoeft, *Geophysics* 50 (1985) 2492–2503.
- [2] L. Slater, D. Lesmes, *Geophysics* 67 (2002) 77–88.
- [3] O.A.L. De Lima, *Geophysics* 58 (1992) 1689–1702.
- [4] F. Börner, J.R. Schopper, A. Weller, *Geophys. Prospect.* 44 (1996) 583–601.
- [5] A. Kemna, H.-M. Münch, K. Titov, E. Zimmermann, H. Vereecken, Extended Abstracts “Near Surface 2005”, 11th European Meeting of Environmental and Engineering Geophysics, Eur. Assn. Geosci. Eng., 2005, P054, 4 pp;
- A. Kemna, A. Binley, L. Slater, *Geophysics* 69 (2004) 97–107;
- E. Zimmermann, H.-M. Münch, J. Berwix, W. Glaas, H. Meier, A. Kemna, in: U. Brill, D.U. Sauer (Eds.), *Impedanzspektroskopie—Grundlagen und Anwendungen, Sonderausgabe Technische Mitteilungen*, vol. 99 (1/2), Organ des Haus der Technik e. V., Essen, 2006, pp. 244–249.
- [6] F. Börner, M. Gruhne, J. Schön, *Geophys. Prospect.* 41 (1993) 83–98.
- [7] H.P. Schwan, G. Schwarz, J. Maczuk, H. Pauly, *J. Phys. Chem.* 66 (1962) 2626–2635.
- [8] C. Ballarín, A. Bonincontro, C. Cametti, *J. Colloid Interface Sci.* 54 (1976) 415–423.
- [9] J.P. Poley, J.J. Nooteboom, P.J. deWaal, *Log Analyst* 19 (1978) 8–30.
- [10] N.C. Lockhart, *J. Colloid Interface Sci.* 74 (1980) 509–519.
- [11] V.N. Shilov, A.V. Delgado, F. Gonzalez-Caballero, C. Grosse, *Colloids Surf. A Physicochem. Eng. Aspects* 192 (2001) 253–265.
- [12] K.S. Cole, R.H. Cole, *J. Chem. Phys.* 9 (1941) 341–351.
- [13] D.W. Davidson, R.H. Cole, *J. Chem. Phys.* 19 (12) (1951) 1484–1490.
- [14] A.K. Jonscher, *Dielectric Relaxation in Solids*, Chelsea Dielectric, London, 1983.
- [15] G. Schwarz, *J. Phys. Chem.* 66 (1962) 2636–2642.
- [16] J.M. Schurr, *J. Phys. Chem.* 68 (1964) 2407–2413.
- [17] S.S. Dukhin, V.N. Shilov, *Dielectric Phenomena and the Double Layer in Disperse Systems and Polyelectrolytes*, John Wiley and Sons, New York, 1974.
- [18] M. Fixman, *J. Phys. Chem.* 72 (1980) 5177–5186.
- [19] J. Lyklema, S.S. Dukhin, V.N. Shilov, *J. Electroanal. Chem.* 143 (1983) 1–21.
- [20] R.W. O'Brien, *J. Colloid Interface Sci.* 92 (1) (1983) 204–216.
- [21] E.J. Hinch, J.D. Sherwood, W.C. Chen, P.N. Sen, *J. Chem. Soc. Faraday Trans. 2* (80) (1984) 535–551.
- [22] J.C. Maxwell, *A Treatise on Electricity and Magnetism*, third ed., Oxford University Press, London, 1892.
- [23] K.W. Wagner, *Arch. Elektrotechn.* 2 (1914) 371–387.
- [24] D.A.G. Bruggeman, *Annal. Phys.* 24 (1935) 636–679.
- [25] T. Hanai, in: P. Sherman (Ed.), *Emulsions Science*, Academic Press, New York, 1968, pp. 354–477.
- [26] P.N. Sen, C. Scala, M.H. Cohen, *Geophysics* 46 (1981) 781–795.
- [27] A. Revil, L.M. Cathles, S. Losh, J.A. Nunn, *J. Geophys. Res.* 103 (B10) (1998) 23925–23936;
- A. Revil, D. Hermitte, E. Spangenberg, J.J. Cochémé, *J. Geophys. Res.* 107 (B8) (2002) 2168.
- [28] A. Revil, *J. Geophys. Res.* 105 (B7) (2000) 16749–16768.
- [29] O.A.L. De Lima, M.M. Sharma, *Geophysics* 57 (3) (1992) 431–440.
- [30] A.L. Endres, R.J. Knight, *J. Colloid Interface Sci.* 157 (2) (1993) 418–425.
- [31] A. Kitamura, K. Fujiwara, T. Yamamoto, S. Nishikawa, H. Moriyama, *J. Nuclear Sci. Technol.* 36 (12) (1999) 1167–1175.
- [32] A. Revil, P.W.J. Glover, *Phys. Rev. B* 55 (3) (1997) 1757–1773;
- A. Revil, P.W.J. Glover, *Geophys. Res. Lett.* 25 (5) (1998) 691–694.
- [33] P. Leroy, A. Revil, S. Altmann, C. Tournassat, *Geochim. Cosmochim. Acta* 71 (5) (2007) 1087–1097.
- [34] T. Hiemstra, W.H. Van Riemsdijk, *J. Colloid Interface Sci.* 136 (1990) 132–150.
- [35] E. Papirer, *Adsorption on Silica Surfaces*, Marcel Dekker, New York, 2000.
- [36] R.J. Hunter, *Zeta Potential in Colloid Science: Principles and Applications*, Academic Press, New York, 1981.
- [37] A. Szymczyk, P. Fievet, A. Foissy, *J. Colloid Interface Sci.* 255 (2002) 323–331.
- [38] P. Leroy, A. Revil, *J. Colloid Interface Sci.* 270 (2) (2004) 371–380.
- [39] C.G. Armistead, A.J. Tyler, F.H. Hambleton, S.A. Mitchell, J.A. Hockey, *J. Phys. Chem.* 73 (11) (1969) 3947.
- [40] M. Aydin, T. Yano, S. Kilic, *Soil Sci. Soc. Am. J.* 68 (2004) 450–459.
- [41] J. Lyklema, J.T.G. Overbeek, *J. Colloid Interface Sci.* 16 (1961) 595–608.
- [42] J.A. Davis, J.O. Leckie, *J. Colloid Interface Sci.* 67 (1) (1978) 90–107.
- [43] G.H. Bolt, *J. Phys. Chem.* 61 (1957) 1166–1169.
- [44] A. Revil, P. Leroy, *J. Geophys. Res.* 109 (2004) B03208;
- A. Revil, P. Leroy, K. Titov, *J. Geophys. Res.* 110 (2005) B06202.
- [45] C.T. O'Konski, *J. Phys. Chem.* 64 (1960) 605–619.
- [46] D.G. Miller, *J. Phys. Chem.* 70 (8) (1966) 2639–2659.
- [47] D.P. Lesmes, F.D. Morgan, *J. Geophys. Res.* 106 (B7) (2001) 13329–13346.
- [48] J.R. Wait, *Geoexploration* 22 (1984) 107–127.
- [49] C.J.F. Bötcher, P. Bordewijk, *Theory of Electric Polarization*, vol. II, Dielectrics in Time-Dependent Fields, Elsevier Science, 1978.
- [50] C.A. Dias, *J. Geophys. Res.* 77 (1972) 4945–4956.
- [51] K.S. Mendelson, M.H. Cohen, *Geophysics* 47 (2) (1982) 257–263.
- [52] G.E. Archie, *Trans. AIME* 146 (1942) 54–61.
- [53] A. Revil, N. Linde, *J. Colloid Interface Sci.* 302 (2006) 682–694.
- [54] O.A.L. De Lima, M.M. Sharma, *Soc. Prof. Well Log Analysts, 32nd Ann. Logging Symp., Paper G*, 1991.
- [55] W.E. Kenyon, *Dielectric dispersion in laboratory measurements on cores*, Schlumberger Wireline Interpretation Sym., 1983.
- [56] A. Crespy, A. Bolève, A. Revil, *J. Colloid Interface Sci.* 305 (2007) 188–194.
- [57] A. Tarasov, K. Titov, *Geophys. J. Int.* 170 (1) (2007) 31–43.
- [58] B.D. Fuller, S.H. Ward, *IEEE Trans. Geosci. Electron.* GE-8 (1970) 7–18.
- [59] G.V. Keller, in: M.N. Nabighian (Ed.), *Electromagnetic Methods in Applied Geophysics*, in: *Soc. of Expl. Geophys.*, vol. 1, 1988, pp. 13–51.
- [60] G.R. Olhoeft, in: *Initial Report of the Petrophysics Laboratory*, U.S. Geol. Survey, Circular 789, 1979, pp. 1–25.
- [61] A.M. Gaudin, D.W. Fuerstenau, *Trans. AIME (Am. Inst. Min. Metall. Petrol. Eng.)* 202 (1955) 66–72.
- [62] A. Revil, P. Pezard, P.W.J. Glover, *J. Geophys. Res.* 104 (B9) (1999) 20021–20031.
- [63] C. Grosse, V.N. Shilov, *J. Phys. Chem.* 100 (1996) 1771–1778.

Spectral induced polarization of partially saturated clay-rocks: a mechanistic approach

D. Jougnot,^{1,2,3} A. Ghorbani,⁴ A. Revil,^{2,3} P. Leroy⁵ and P. Cosenza⁶

¹ANDRA, 1-7 rue Jean Monnet, 92298 Châtenay-Malabry, France. Email: arevil@mines.edu

²CNRS-UMR 5559-LGIT, Université de Savoie, Equipe volcan, 73376 Le-Bourget-du-Lac, France

³Colorado School of Mines, Department of Geophysics, Golden, 80401, CO, USA

⁴Department of Mining and Metallurgy, Yazd University, Yazd, Iran

⁵BRGM, 3 avenue C. Guillemin, BP 36009, 45060 Orléans, France

⁶FRE CNRS 3114 HydrASA, Université de Poitiers, France

Accepted 2009 October 20. Received 2009 October 19; in original form 2009 August 22

SUMMARY

We have developed a mechanistic model to interpret spectral induced polarization data of partially saturated clay-rocks. This model accounts for the polarization of the grains through an electrical double layer model with a polarization model of the inner part of the electrical double layer called the Stern layer. The polarization model accounts also for the Maxwell–Wagner polarization at frequencies higher than 100 Hz. The Maxwell–Wagner polarization is modelled by using a conductivity model modified to account for the presence of a non-wetting immiscible phase like air in the pore space. The resulting model is consistent with the first and second Archie's laws in the case where surface conductivity can be neglected. The volumetric charge density of the diffuse layer at saturation is divided by the saturation of the water phase to account for the partial water saturation of the porous material. The model comprises seven fundamental parameters: the formation factor, the second Archie's exponent, a critical water saturation level, the mean electrical potential of the pore space at saturation, the density of the counterions in the Stern layer, and at least two parameters describing the grain size distribution. Most of these parameters can be derived independently using alternative measurements and electrochemical models. Measurements were performed in the frequency range 10 mHz–45 kHz using five samples from the Callovo–Oxfordian formation in the eastern part of the Paris Basin, France. The model agrees fairly well with the experimental data at saturation and for partially saturated clay-rocks down to 1 Hz. Most of the seven physical parameters entering the model were independently evaluated.

Key words: Electrical properties; Electromagnetic theory; Hydrogeophysics; Permeability and porosity.

1 INTRODUCTION

Spectral induced polarization reports the modulus of the conductivity and the phase between the current and the voltage in the frequency domain (typically from few millihertz to few tens of kilohertz, sometimes few Megahertz). The modulus and the phase can also be recasted into an equivalent complex conductivity or complex resistivity. This method has been developed first to locate ore bodies (Marshall & Madden 1959; Van Voorhis *et al.* 1973) and more recently in environmental sciences (Kemna *et al.* 2000, 2004; Slater & Binley 2006; Hördt *et al.* 2007; Blaschek *et al.* 2008). In environmental sciences, spectral induced polarization has been used to map contaminants in the ground (Vanhala 1997b), to determine grain size distribution (Lesmes & Morgan 2001; Leroy *et al.* 2008) and permeability (Börner *et al.* 1996; de Lima & Niwas 2000; Binley *et al.* 2005; Hördt *et al.* 2007). In parallel to these new types

of investigations, the improvement of high accuracy acquisition systems able to measure the phase with a precision of 1–0.1 mrad (Zimmermann *et al.* 2008) has allowed to investigate rock samples and soils characterized by very low quadrature conductivity including glass beads (Leroy *et al.* 2008), clean (clay-free), sandstones (Binley *et al.* 2005) and clay materials (Ghorbani *et al.* 2009).

During the last few years, several works have been done in using spectral induced polarization to study the properties of clay-rocks (Kruschwitz & Yaramanci 2004; Cosenza *et al.* 2008; Tabbagh *et al.* 2009). These works were motivated by the use of the spectral induced polarization method to study nuclear waste disposals (Kruschwitz & Yaramanci 2004; Pusch 2006) and generally speaking to study the role of clay barriers for the confinement of different types of hazardous materials. Indeed, because of their high-specific surface area and very low permeabilities ($<10^{-19}$ m²) (Gaucher *et al.* 2004), clay-rocks are good candidates for host rocks (Homand

et al. 2004). Various national agencies (like ANDRA in France) have been commissioned to study the transport properties, speciation and thermohydro-mechanical behaviour of clay-rocks. These works will allow to assess the long-term safety of future potential radioactive waste disposals in these materials.

In order to study these clay-rocks formations, underground research laboratories have been built like Bure in France and Mol in Belgium or test sites have been selected, like Tournemire in France, and Mont Terry in Switzerland. Gallery excavations in these laboratories have generated perturbations of the rock: the Excavated Damaged Zone (EDZ) along the walls of the galleries (Tsang *et al.* 2005). To improve the knowledge of these zones, geophysical methods can provide very useful information especially spectral induced polarization as shown by Kruschwitz & Yaramanci (2004). Additional investigations were performed in the laboratory, which confirm this possibility (Cosenza *et al.* 2007; Ghorbani *et al.* 2009). However, all these authors used semi-empirical approaches based on the Cole–Cole model (Cole & Cole 1941; Ghorbani *et al.* 2007; Chen *et al.* 2008) or similar semi-empirical models like the Davidson–Cole model (Davidson & Cole 1950) and the generalized Cole–Cole model (Vanhala 1997a; Ghorbani *et al.* 2009). Leroy & Revil (2009) have proposed recently a mechanistic model to quantify spectral induced polarization in saturated clay-rich materials. However, a quantitative understanding of spectral induced polarization data in unsaturated clay-rocks is still missing.

We propose below a mechanistic model of spectral induced polarization of unsaturated clays-rocks. This model is based on an extension of the theory developed recently by Leroy & Revil (2009) for clay-rich materials. This model extends also some previous modelling efforts to understand cross-coupling properties of clay

materials (Revil & Leroy 2004; Revil & Linde 2006). We also compare the prediction of this new model to new spectral induced polarization measurements performed on five core samples from the Callovo–Oxfordian clay-rock formation. This formation is presently studied by ANDRA, the French Nuclear Waste Agency, as a potential host for nuclear wastes.

2 DESCRIPTION OF THE CALLOVO–OXFORDIAN CLAY-ROCK

ANDRA has developed an underground research facility, the test site of Bure, to study the possibility of long-term underground disposal of nuclear wastes. The site of Bure (Meuse/Haute-Marne, France) is located in the eastern part of the Paris Basin in a thick clay-stone formation of Callovian to Oxfordian age (Fig. 1). At Bure, the Callovo–Oxfordian formation has a thickness of approximately 130–160 m. The Callovo–Oxfordian formation can be subdivided in four main lithofacies units. At Borehole EST205 (see location in Fig. 1), we find first the C2d unit for depths comprised between approximately 417 and 437 m. This unit presents some smooth spatial variations in the mineralogy and petrophysical properties. The C2c and C2b units are located between approximately 437 and 456 m. These units are relatively homogeneous. The bottom unit C2b of the Callovo–Oxfordian formation unit is itself divided into two subunits: C2b2 (approximately 456–473 m) and C2b1 (approximately 473–508 m). The subunit C2b1 corresponds to the unit where the proportion of clay minerals is the highest through the Callovo–Oxfordian formation and is located at depths comprised between approximately 486 and 489 m. The underground laboratory of ANDRA at Bure is located just below this level. The bottom unit

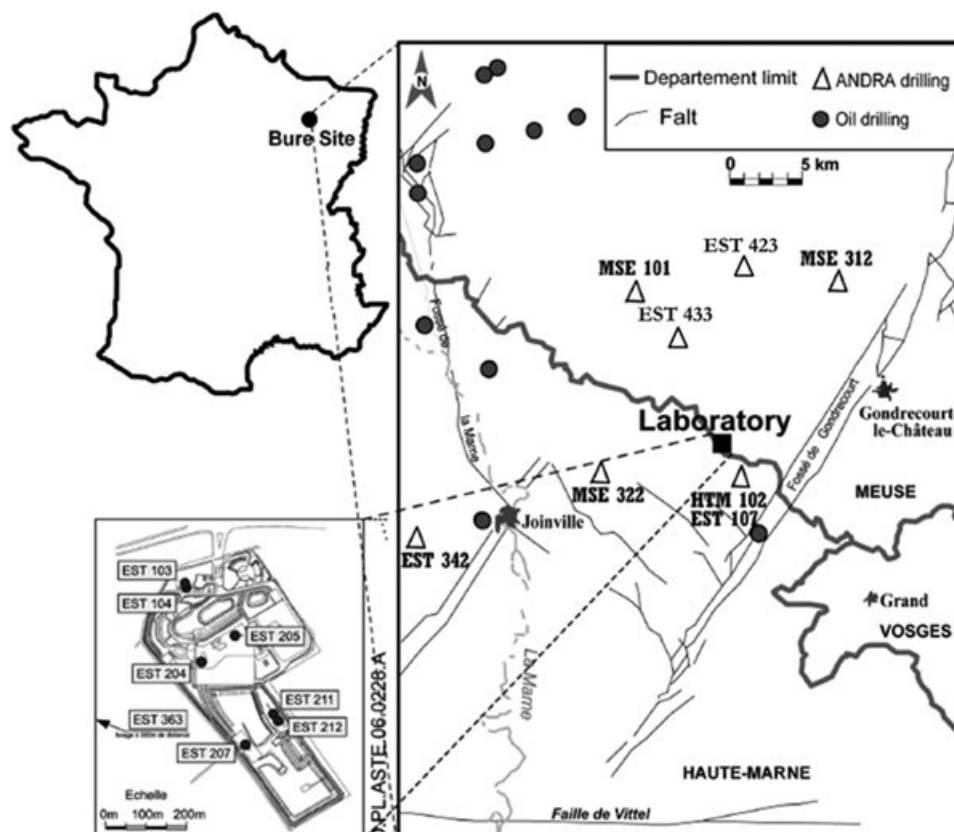
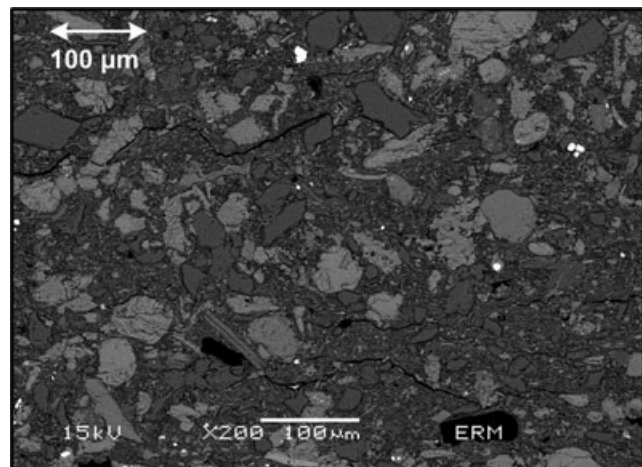


Figure 1. Location of the boreholes drilled by ANDRA where the core samples used in this study were extracted (modified from Trouiller 2006).

Table 1. Gravimetric water contents and porosities of the various Callovo–Oxfordian clay-rock units determined from the interpretation of borehole measurements (ANDRA 2008).

Unit	Gravimetric water content ^a		Porosity ^b	
	Mean (per cent)	Standard deviation	Mean (–)	Standard deviation
C2d	6.00	2.41	0.137	0.047
C2c	7.35	2.71	0.163	0.048
C2b2	8.04	1.03	0.178	0.019
C2b1	7.41	1.21	0.166	0.022
C2a	7.06	2.66	0.157	0.043
COx ^c	7.35	1.66	0.16	0.030

^aValues obtained from a 150 °C reference (dry).^bCalculated from the gravimetric water content with $\rho_w = 1000 \text{ kg m}^{-3}$ and $\rho_g = 2690 \text{ kg m}^{-3}$.^cMean values of the Callovo–Oxfordian formation**Figure 2.** SEM (Scanning Electron Micrograph) picture of the Callovo–Oxfordian clay-rock (credit: J.C. Robinet, ERM). The picture shows grains of silica and carbonates (typical diameter comprised between 10 and 30 μm) and a clay matrix (in dark grey).

of the Callovo–Oxfordian formation corresponds to the C2a unit (<508 m). The mean gravimetric water contents and porosities of the different units are summarized in Table 1 (ANDRA 2005).

The Callovo–Oxfordian clay-rock can be conceptualized as a clay matrix with some imbedded grains of silica and carbonates (calcite and a minor fraction of dolomite). (Fig. 2). The clay matrix represents 20–50 per cent of the rock volume in the total formation (up to 45–50 per cent in the subunit C2b1). This clay fraction is mainly composed of illite and interstratified illite-smectite clays. There is also a small amount of kaolinite in the lower part of the formation. Smectite based clay-minerals form aggregates called tactoids. The size of the tactoid is typically 10 μm (ANDRA 2005). The size of the silica and carbonate grains is usually larger than 4 μm with a mean around 10–20 μm . The volume of the silica and carbonate grains represent between 20 and 40 per cent of the rock assemblage. In addition, there are small amounts of potassic feldspar, plagioclase and pyrite. An extensive geochemical analysis of the Callovo–Oxfordian formation has been proposed recently by Gaucher *et al.* (2004).

3 THEORETICAL DEVELOPMENTS

3.1 Theory

We consider a representative elementary volume of the Callovo–Oxfordian clay-rock of porosity ϕ . The pore volume is

Table 2. Callovo–Oxfordian porewater composition.

THERMOAR model (Gaucher <i>et al.</i> 2006)		Simplified solute composition ^a (only charged species)	
Species	$C_i \text{ (mol L}^{-1}\text{)}$	Species	$C_i \text{ (mol L}^{-1}\text{)}$
Na	32×10^{-3}	Na^+	31.5×10^{-3}
K	7×10^{-3}	K^+	6.5×10^{-3}
Ca	15×10^{-3}	Ca^{2+}	9.5×10^{-3}
Mg	14×10^{-3}	Mg^{2+}	8.1×10^{-3}
Cl	30×10^{-3}	Cl^-	30×10^{-3}
S(+6)	34×10^{-3}	SO_4^{2-}	21×10^{-3}
pH	7.3	–	–
pCO ₂	–2.51	HCO_3^-	1.2×10^{-3}

^aObtained using PHREEQC2 (see Leroy *et al.* 2007). The software PHREEQC2 is described in Parkhurst & Appelo (1999).

filled by two continuous and immiscible fluid phases, a wetting fluid (subscript w) and a non-wetting fluid. In our analysis, the wetting fluid is water and the non-wetting fluid is air. We denote $s_w \in [0; 1]$ the saturation of the water phase. This saturation is related to the volumetric water content θ by $s_w = \theta/\phi$. In soil science, many authors define also a gravimetric water content w defined as the ratio between the mass of the water phase m_w relative to the mass of the solid phase m_s of the core sample. Following Revil *et al.* (2007) and Revil & Jougnot (2008), we define a percolation threshold s_w^c for the water phase. Below this threshold, the water phase is not continuous at the scale of the representative elementary volume (see also Hunt & Ewing 2003 for the use of percolation theory in the context of this problem).

The minerals of the Callovo–Oxfordian clay-rock, especially the clay minerals, have negatively charged surface at the pH of the pore water (pH = 7.2, see Table 2; Leroy *et al.* 2007). To counterbalance this negative charge on the mineral surface, the pore water has an excess of cations (see Leroy *et al.* 2007 for a geochemical modelling of the pore water composition of the Callovo–Oxfordian clay-rock). These counterions are partitioned between the Stern layer, which is attached to the mineral surface, and the diffuse layer located inside the pore water phase (Fig. 3). In our modelling approach and in unsaturated condition, the Callovo–Oxfordian clay-rock is composed by three phases: (1) the solid phase, including the insulating minerals coated by the conductive Stern layer, (2) the water phase including the diffuse layer and locally some free electrolyte pockets in some macropores and (3) the non-wetting phase corresponding to air. Neglecting the surface charge of the air–water interface (S_{aw}) (see Linde *et al.* 2007; Revil *et al.* 2007), the charge balance

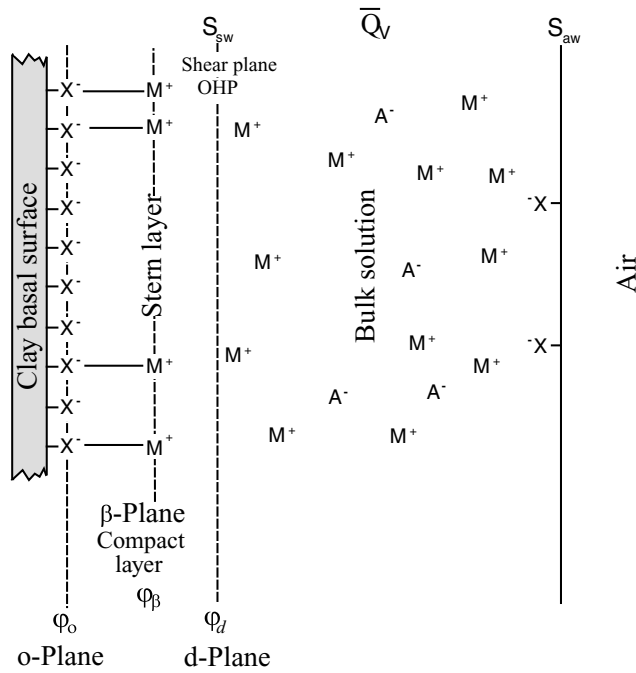


Figure 3. Sketch of the distribution of the ionic species in the pore space of a charged unsaturated porous medium. We note S_{sw} the shear plane between the mineral surface and the pore water and S_{wa} the interface between the air phase and the pore water (these two interfaces have complex shapes). The pore water is characterized by a volumetric charge density $\bar{Q}_V$. φ_o , φ_β and φ_d are the electrical potentials of the surface of the mineral surface, the mid-plane of the β -Stern layer, and the shear plane, respectively. When the mean distance between the air/water interface and the mineral/water interface decreases, the ions of the pore water phase are confined in a smaller volume and therefore the conductivity of the pore water phase increases.

condition is

$$\frac{\bar{Q}_V}{s_w} + \frac{1}{V_w} S_{sw} Q_S = 0, \quad (1)$$

where Q_S is the total surface charge density (in Cm^{-2}) at the solid-solution interface of area S_{sw} (including the true charge density on the mineral surface plus the surface charge density of the Stern layer), V_w (in m^3) denotes the volume of the water phase and $\bar{Q}_V/s_w$ represents the excess of charge per unit of volume of water in the unsaturated porous material ($\bar{Q}_V$ is the volumetric charge density at saturation of the water phase). Considering a solution with Q ionic species i , the effective charge density of the pore water is written as

$$\frac{\bar{Q}_V}{s_w} = \sum_{i=1}^Q q_i \bar{C}_i, \quad (2)$$

where q_i is the charge ($\pm z_i e$, where z_i is the valency of the ion and e the charge of the electron, that is, $\sim 1.6 \times 10^{-19}$ C) and $\bar{C}_i$ the ionic concentration of the ion i in the medium bulk solution (in m^{-3}). The bulk pore water solution of the clay matrix is mainly composed by the diffuse layer due to the small size of the pores (Leroy *et al.* 2007; Jougnot *et al.* 2009). We can use the Donnan approach to determine the ionic concentrations in the pore space of the clay matrix $\bar{C}_i$ (Revil & Linde 2006; Leroy *et al.* 2007; Revil 2007). The ionic species that are present in the pore space of the clay-rock are assumed to be in local thermodynamic equilibrium with an infinite reservoir of the same Q ionic species at concentrations C_i .

Equilibrium implies (Revil & Linde 2006)

$$\bar{C}_i = C_i \exp\left(-\frac{q_i \varphi_m}{k_B T}\right), \quad (3)$$

where φ_m represents the mean electrical potential in the pore water phase (in V), k_B represents the Boltzmann constant ($k_B = 1.3806 \times 10^{-23} \text{ JK}^{-1}$) and T is the absolute temperature (in K).

A very important parameter in the study of the electrical double layer is the partition coefficient f_Q (Leroy & Revil 2004; Leroy *et al.* 2007). This partition coefficient represents the fraction of the countercharge located in the Stern layer to compensate the true surface charge density of the mineral surface. We note Γ_{Xi}^0 the surface site density of species i in the Stern layer and Γ_i^d the surface site density of species i in the diffuse layer (both in m^{-2}). The partition coefficient f_Q is defined as (Leroy *et al.* 2007).

$$f_Q = \frac{\sum_{i=1}^Q z_i \Gamma_{Xi}^0}{\sum_{i=1}^Q z_i \Gamma_{Xi}^0 + \sum_{i=1}^Q z_i \Gamma_i^d}, \quad (4)$$

where z_i is the charge number of species i . For unsaturated conditions, the total charge density of counterions per unit of pore volume, Q_V , can be related to the excess of charge per unit of volume of solution in the medium, $\bar{Q}_V/s_w$ and to the partition coefficient defined above by

$$\frac{\bar{Q}_V}{s_w} = (1 - f_Q) Q_V, \quad (5)$$

and Q_V can be related in turn to the cation exchange capacity (CEC, in C kg^{-1}) of the rock:

$$Q_V = \rho_s \left(\frac{1 - \phi}{\phi} \right) \text{CEC}, \quad (6)$$

where ρ_s is the bulk density of the solid phase of the material. The CEC characterizes the density of charge on the mineral surface because it can be measured directly using chemical titration methods. The electrical double layer theory developed by Leroy *et al.* (2007) includes a specific speciation model of the different crystalline planes of a mineral. This model can be used to infer the dependence of the partition coefficient f_Q with the ionic strength of the pore water,

$$I \equiv \frac{1}{2} \sum_{i=1}^Q z_i^2 \bar{C}_i. \quad (7)$$

Leroy & Revil (2009) proposed recently a model to study the spectral induced polarization of clay-rich porous materials saturated by water. This model is based on two different polarization mechanisms: (1) a polarization of the Stern layer associated with the discontinuity of the solid phase and (2) a Maxwell–Wagner polarization process. This model has been built for the case of a binary symmetric electrolyte (like NaCl or KCl). This is a major assumption of this study as discussed further below. Considering the composition of the pore water proposed by Gaucher *et al.* (2006) and Leroy *et al.* (2007) (see Table 2), we see that the sodium Na^+ is the dominating counterion of the pore water. Therefore using a model with a single type of counterions is a reasonable assumption. However other counterions may be present and may play some role in widening the spectrum of the relaxation times. The sodium cation is located on the Outer Helmholtz Plane of the Stern layer, which is also called the d-plane (see Fig. 3), because sodium keeps its hydration shell on the mineral surface (Tournassat *et al.* 2009). Therefore it keeps some mobility along the mineral surface. If sodium would form an inner sphere complex with the mineral surface, it would have no tangential mobility along the mineral surface.

To extend the spectral induced polarization model developed by Leroy *et al.* (2007) to unsaturated conditions, and following Revil *et al.* (2007), we use the following change of variables to introduce the effect of the saturation of the water phase: $1/F \rightarrow (s_w - s_w^c)^n/F$ and $\bar{Q}_V \rightarrow \bar{Q}_V/s_w$ (see also Revil 1999), where F is the formation factor and n is the saturation exponent also called the second Archie's exponent (Archie 1942).

The polarization of the Stern layer is obtained by using the approach developed by Schwarz (1962). This approach is based on the assumption that there are no exchange of ions between the Stern and the diffuse layers. This assumption seems valid because the kinetics of ion sorption/desorption in the Stern layer is slow (typically several hours). We consider therefore that the movement of the ions in the Stern layer follows tangentially the pore water/mineral interface and that the characteristic length scale associated with this polarization mechanism is the size of the grains. In the model developed by Schwarz (1962), the relaxation time for the electromigration of an ion of species i , τ_i (in s), is therefore related to the particle radius a (in m) and to the diffusion coefficient of the ion i in the Stern layer D_S^i by,

$$\tau_i = \frac{a^2}{2D_S^i}. \quad (8)$$

This time constant corresponds to the relaxation time of the diffusion process of the counterions in the Stern layer with a characteristic distance a . In our model, the diffuse layer is homogenized with the pore water phase using the Donnan equilibrium approach (Revil & Linde 2006). We do not consider polarization for the diffuse layer because it forms a continuous phase through the porous material. In contrast, the Stern layer polarizes because it is discontinuous.

Because the clay fraction of the Callovo–Oxfordian clay-rock is mainly composed of illite and interstratified illite–smectite clays, we will consider only the response of illite and smectite to define the spectral induced polarization of the clay-rock. These minerals are coated by a Stern layer. We assume that all counterions are Na^+ . If all grains had the same grain radius a_0 , the Stern layer complex conductivity $\bar{\sigma}_S$ could be computed by (Leroy & Revil 2009).

$$\bar{\sigma}_S(a_0, \omega) = \left(\frac{2\Sigma_S^\infty}{a_0} - \frac{2}{a_0} \frac{\Sigma_S^\infty}{(1 - i\omega\tau_{\text{Na}}^0)} \right), \quad (9)$$

with ω the angular frequency (in rad s^{-1} , $\omega = 2\pi f$ is the frequency in Hertz), a_0 is a grain size and τ_{Na}^0 is the associated relaxation time associated with the backdiffusion of the counterions in the Stern layer. At very low frequencies $\omega \ll 1/\tau_{\text{Na}}^0$, there is no contribution from the Stern layer to surface conductivity because of the diffusion of the counterions. At high frequencies, diffusion does not have the time to occur and therefore conductivity increases because of the contribution of the Stern layer conductivity. The parameter Σ_S^∞ corresponds therefore to the contribution of the Stern layer at high frequencies. This contribution is defined by (Leroy & Revil 2009).

$$\Sigma_S^\infty = e\beta_{\text{Na}}^S \Gamma_{\text{XNa}}^0, \quad (10)$$

(see also Revil *et al.* 1999; Revil & Leroy 2001) where e is the elementary charge, β_{Na}^S is the ionic mobility of the sodium in the Stern layer (in $\text{m}^2 \text{s}^{-1} \text{V}^{-1}$). The mobility β_{Na}^S and the diffusion coefficient D_S^i are related to each other by the Nernst–Einstein relationship. The complex conductivity of the Stern layer σ_S^* (in Sm^{-1}) is determined as (Leroy & Revil 2009).

$$\sigma_S^* = \bar{\sigma}_S + i\omega\varepsilon_s, \quad (11)$$

$$\varepsilon_s = 4.5\varepsilon_0, \quad (12)$$

where ε_s represents the dielectric permittivity of the clay minerals ($\varepsilon_0 = 8.85 \times 10^{-12} \text{ Fm}^{-1}$, the dielectric permittivity of vacuum). The effect of the grain size distribution is obtained by convoluting the response obtained for a single grain with the particle size distribution (PSD) (see Leroy & Revil 2009). The PSD is related to the distribution of the relaxation times through eq. (8).

The conductivity of the pore water needs to be computed accounting for the presence of Q ionic species and the effect of the diffuse layer. Generalizing the result obtained by Revil & Linde (2006) for the saturated case, the conductivity of the water phase (in Sm^{-1}) in unsaturated condition is defined by,

$$\bar{\sigma}_f = \sum_{i=1}^Q q_i \beta_i \frac{\bar{C}_i}{s_w}, \quad (13)$$

where β_i is the ionic mobility of the species i in the solution (in $\text{m}^2 \text{s}^{-1} \text{V}^{-1}$). We see therefore that the conductivity of the water phase depends strongly on the saturation of the water phase and increases with the desaturation. A complex pore water conductivity σ_f^* can be defined as,

$$\sigma_f^* = \bar{\sigma}_f + i\omega\varepsilon_f, \quad (14)$$

with $\varepsilon_f = 81\varepsilon_0$ the dielectric permittivity of water (at standard atmospheric pressure and at 25°C).

We need now an expression for the conductivity of the representative elementary volume, σ^* , in unsaturated conditions. This is done by extending the result obtained by de Lima & Sharma (1992) and Revil *et al.* (2007) using the change of variable $1/F \rightarrow (s_w - s_w^c)^n/F$ and $\bar{Q}_V \rightarrow \bar{Q}_V/s_w$. This yields,

$$\sigma^* = \frac{1}{F} [(s_w - s_w^c)^n \sigma_f^* + (F - 1) \sigma_s^*] \quad (15)$$

This model accounts for the Maxwell–Wagner polarization. This electrical conductivity model is part of the generalized framework proposed by Revil & Linde (2006) and Revil *et al.* (2007) to describe transport properties of saturated and unsaturated microporous media. This framework presents the important advantage to propose the determination of the macroscopic transport properties in a porous medium from a couple of intrinsic material properties that can be transposed from one macroscopic property to the other. From these fundamental properties, it becomes therefore possible to predict a wide variety of transport properties of the porous material (e.g. Jougnot *et al.* 2009 used the formation factor determined from electrical conductivity measurements to determine the diffusion coefficient of ionic tracers in core samples of the Callovo–Oxfordian clay-rock).

3.2 Sensitivity analysis

The previous model of induced polarization is based on seven independent parameters: (1) the formation factor, F , (2) the saturation index, n , (3) the percolation threshold, s_w^c , (4) the Donnan equilibrium potential, φ_m , (5) the surface site density of the counterions in the Stern layer, Γ_{Na}^0 and (6) and (7) at least two parameters describing the grain size distribution (a mean and a standard deviation for a log normal probability distribution for instance).

The formation factor of the Callovo–Oxfordian clay-rock, in unit C2b1 and C2b2, is comprised between 15 and 140 (Jougnot *et al.* 2009). The saturation exponent n has been studied by Jougnot & Revil (2008) using a model for the thermal conductivity in unsaturated conditions that was applied to the Callovo–Oxfordian

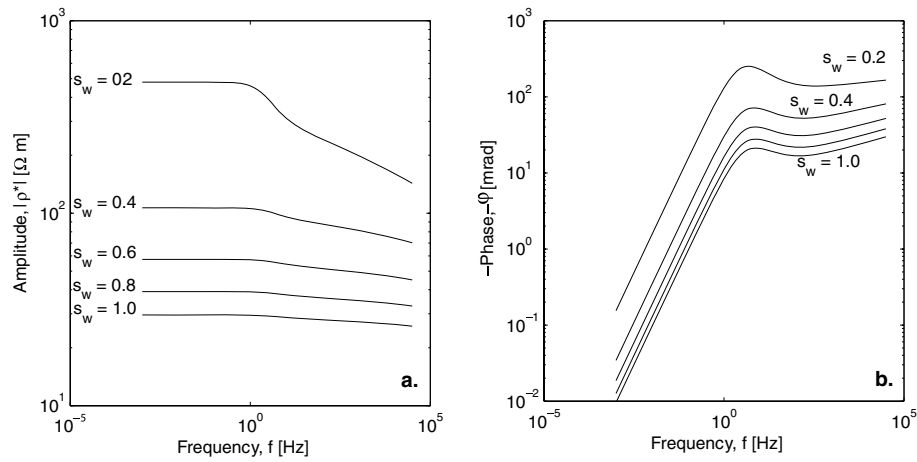


Figure 4. Sensitivity of the spectral induced polarization model with respect to the saturation of the water phase s_w . (a). Amplitude of the resistivity response versus frequency. (b). Phase versus frequency.

clay-rock. They obtained $n = 2.0$. No values exist for s_w^c for the Callovo–Oxfordian clay-rock. Revil & Jougnot (2008) previously considered $s_w^c = 0.1$ as a reasonable value. We will see below that a correct estimate seems to be higher than this value. Considering the pore water composition determined by Gaucher *et al.* (2006) (the THERMOAR model reported in Table 2), the Donnan equilibrium potential, φ_m , is comprised between -54 and -24 mV (see Jougnot *et al.* 2009). From Leroy & Revil (2009), the surface site density, Γ_{Na}^0 , is comprised between 0.632 nm^{-2} for pure smectite to 1.12 nm^{-2} for pure illite. The particle size distribution in consolidated clay-stones is difficult to determine experimentally because of the existence of the tactoids. As discussed above in Section 3.1, the distribution of the relaxation times, τ , is related to the grain size distribution. We consider below a Cole–Cole distribution to characterize the distribution of the relaxation times (Cole & Cole 1941; Bötcher & Bordewijk 1978):

$$f(\tau) = \frac{1}{2\pi} \frac{\sin[\pi(c-1)]}{\cosh[c \ln(\tau/\tau_0)] - \cos[\pi(c-1)]}, \quad (16)$$

$$\int_0^\infty f(\tau) d\tau = 1, \quad (17)$$

where τ_0 , is the main relaxation time (in s) and c is the Cole–Cole exponent. This distribution is symmetric for $\tau = \tau_0$. Leroy & Revil (2009) obtained $c = 0.76 \pm 0.05$ for the Mancos clay-rock and $c \in [0.60; 0.84]$ for illites and smectites. This distribution is convoluted with the expression obtained for the Stern layer surface conductivity for a single grain, see eq. (9). In our model, the grain size distribution follows a Cole–Cole distribution and to this distribution is associated a distribution of relaxation time that follows also a Cole–Cole distribution. This is very different than taking an empirical Cole–Cole model to describe the modulus and the phase of the resistivity.

To perform a sensitivity analysis, the values of the seven key-parameters of our spectral induced polarization model can be chosen as follow: $F = 80$, $n = 2.0$, $s_w^c = 0.10$, $\varphi_m = -40$ mV, $\Gamma_{Na}^0 = 1 \text{ nm}^{-2}$, $c = 0.85$ (smectite-A in Leroy & Revil 2009) and $\tau_0 = 3.80 \times 10^{-2}$ s (this corresponds to a grain diameter equal to 10^{-5} m and an ionic mobility in the Stern layer of $\beta_{Na}^S = 5.19 \times 10^{-8} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$, according to eq. (8)). We present the spectral induced polarization data of our sensitivity test by showing the mod-

ulus of the complex resistivity $|\rho^*|$ (the inverse of the complex conductivity in $\Omega \cdot m$) and the phase φ ($\varphi = -\arctan[\text{Im}(\sigma^*)/\text{Re}(\sigma^*)]$, in rad where $\text{Im}(\sigma^*)$ represents the quadrature conductivity and $\text{Re}(\sigma^*)$ represents the in phase conductivity). By convention, we will always plot $(-\varphi)$ expressed in mrad.

The first parameter we investigate is the saturation s_w . Fig. 4 shows the influence of the saturation with $s_w \in \{0.2; 1\}$. As expected, the amplitude of the complex resistivity increases with the desaturation of the porous material because of the insulating non-wetting phase (Fig. 4a). The phase also increases when the saturation decreases (Fig. 4b). This is because the presence of a non-wetting and insulating phase creates a new interface having an effect on the Maxwell–Wagner polarization. However, because the density of the counterions has also changed in the pore water, the presence of air has also an effect on the polarization of the Stern layer.

Fig. 5 shows the influence of F , n and s_w^c upon the spectral induced polarization response of a clay-rock. All these parameters influence the response in the same way: when they increase, both the amplitude of the complex resistivity, $|\rho^*|$ and the phase, $-\varphi$, increase (see Fig. 5). The sensitivity of our model to the two geochemical parameters (φ_m and Γ_{Na}^0) is shown in Fig. 6. The Donnan equilibrium potential influences the signal in the same way as F , n and s_w^c . The surface site density influences the model in a completely different way. When Γ_{Na}^0 increases, the amplitude increases at high frequencies (>1 Hz) but not at low frequency (<1 Hz) where the phase decreases (Fig. 6). This implies that the effect of the surface site densities is different on the Maxwell–Wagner polarization, which dominates at high frequencies. This influence is made through the effect of surface conductivity, which changes the apparent conductivity of the grains. At low frequencies (below 10 Hz in the present case), the response is dominated by the polarization of the Stern layer.

We investigate now the influence of the distribution of the relaxation times by analysing the sensitivity of the model to the two parameters τ_0 and c (see Fig. 7). We note that the mean relaxation time and the Cole–Cole exponent influence the high frequency part of the amplitude (>10 Hz) (Figs 7a and d). c also modifies the phase mainly at high frequency (Fig. 8b), while τ_0 completely shifts the peak of the phase. A lower value of τ_0 (smaller mean grain size or higher ionic mobility in the Stern Layer) corresponds to a shift of the critical frequency f_c to the high frequency [$f_c = 1/(2\pi\tau_0)$]. A

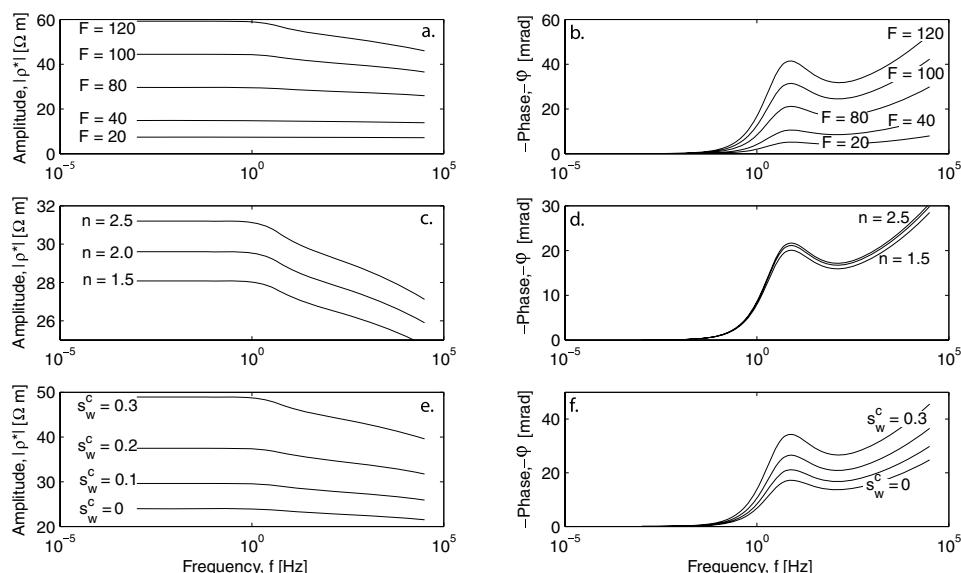


Figure 5. Sensitivity analysis of the spectral induced polarization model with respect to the formation factor, F , the saturation index, n and the value of the critical water saturation, s_w^c . The frequency range investigated is comprised between 1 mHz and 10 kHz and the calculations are performed for a water-saturated material. (a, c, e) Amplitude of the resistivity versus frequency. (b, d, f) Phase versus frequency.

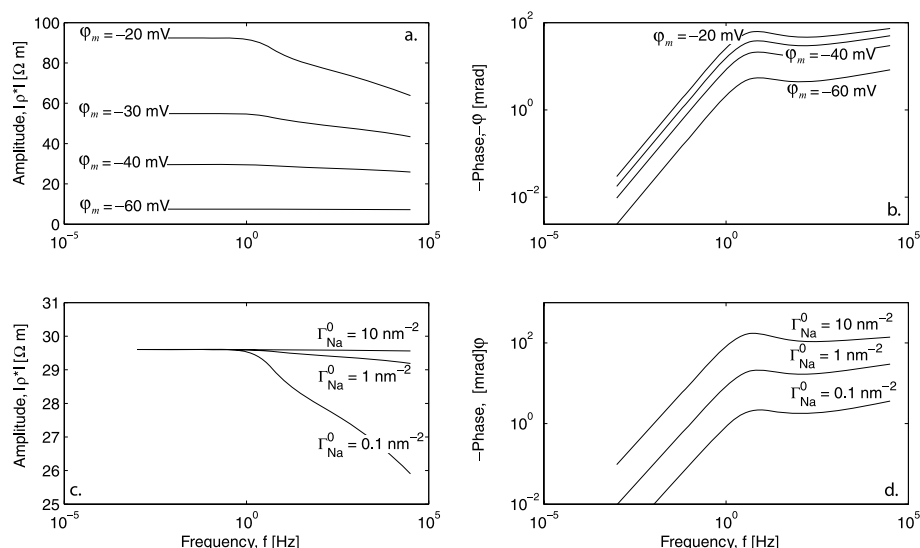


Figure 6. Influence of the two geochemical key parameters of the model (namely the Donnan equilibrium potential, ϕ_m (a and b), and the surface site density in the Stern layer, Γ_{Na}^0) on the spectral induced polarization response for the frequency range 1 mHz–10 kHz at saturation. (a and c) Amplitude of the resistivity versus the frequency. (b and d) Phase versus the frequency.

lower value of c corresponds to a broader distribution of grain size. Consequently, this leads to a broader phase distribution, because the phase is strongly linked to the Stern layer polarization [the Stern layer contribution to the surface conductivity depends on the grain size, see eq. (9)].

4 EXPERIMENTAL MEASUREMENTS

4.1 Description of the rock samples

In the present contribution, we only study the spectral induced polarization response of Callovo–Oxfordian clay-rock sam-

ples. Due to the formation depth and the diagenetic phenomena, the Callovo–Oxfordian clay-stone is a very consolidated rock (ANDRA 2005) and samples do not fall apart after extraction. The samples used for the experiments have been extracted from the Callovo–Oxfordian layer at Bure (Meuse/Haute-Marne, France) by drilling with a NaCl saturated mud. The core samples have been extracted from the C2b1 and C2b2 units. The clay-rock has been sampled as a series of ~ 20 cm long cylinders at different depths. The EST433 samples have been resampled to avoid the NaCl contamination and then became parallelepiped-like. All the samples have been conditioned in double hermetic aluminium bags and confined in polystyrene during the transport to avoid the perturbations. However a fraction of the samples have been broken during the

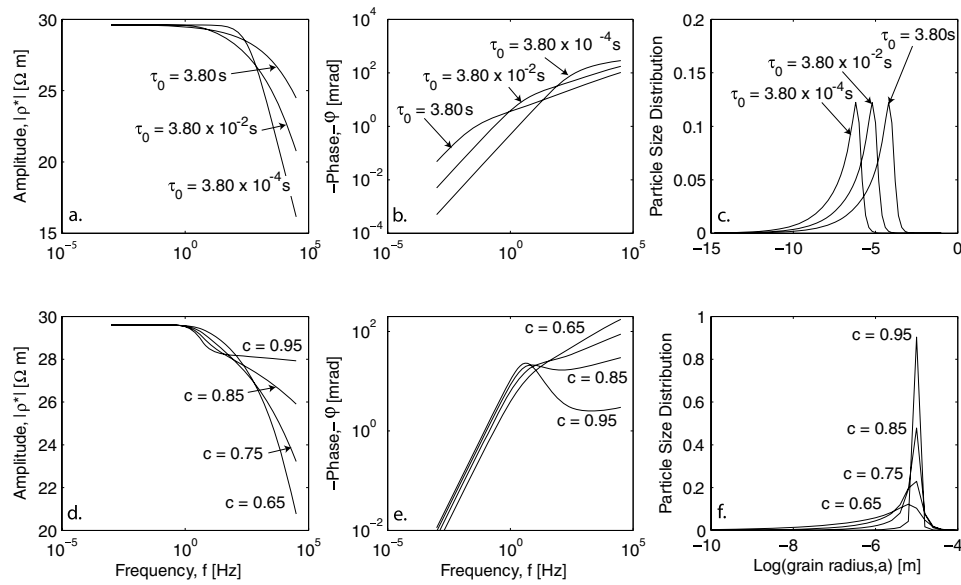


Figure 7. Influence of the two parameters controlling the particle size distribution (namely the mean relaxation time, τ_0 and the Cole-Cole exponent, c) on the amplitude of the resistivity versus frequency and the phase versus frequency distributions. The sensitivity analysis is performed for the frequency range 1 mHz–10 kHz and for the saturated case ($s_w = 1$). (a and d) Amplitude of the resistivity. (b and e) Phase. (c and f) Particle size distribution.

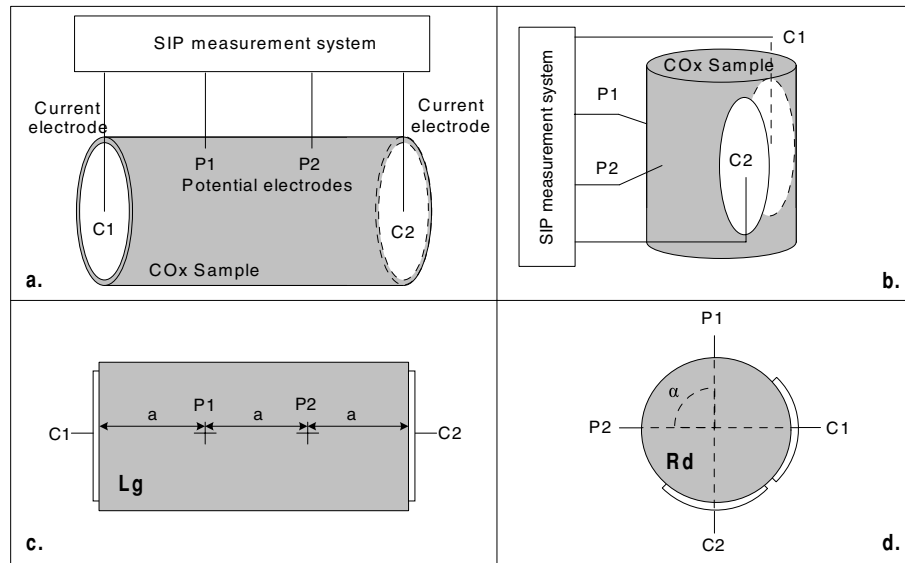


Figure 8. Experimental setups used for the spectral induced polarization measurements. (a, c) Sketch of the Lg-setup. Views from the side and the top, respectively. (b and d) Sketch of the Rg-setup. Views from the side and the top, respectively.

transportation. While opening, the samples were already partially unsaturated ($s_w^i \in [0.75; 0.81]$). In order to limit the perturbation in the clay-stone pore space, we have decided not to resaturate the samples prior to our experiments. For the experiments, we used five samples from two boreholes at different depths (from units C2b1

and C2b2) and with different sizes (Table 3). We resampled some of the broken samples to make them usable for our experiments. Note that the resampling has been done with dry air to avoid sample pollution and/or osmotic perturbation (which could yield to the sample disintegration).

Table 3. Description of the core samples (the depth corresponds to the middle of the sample).

Sample	Borehole	Depth (m) ^a	Lithofacies	Length (m)	Diameter (m)
EST27906	EST423	518.16	C2b2	0.194	0.100
EST28144	EST423	582.16	C2b1	0.059	0.098
EST29296	EST433	617.52	C2b1	0.108	0.063
EST29300	EST433	624.38	C2b1	0.167	0.055
EST30749	EST433	608.07	C2b2	0.078	0.063

4.2 Experimental setup

The Spectral Induced Polarization method is based on the measurements of the complex resistivity ρ^* (in Ωm) over several decades of frequency (usually from 1 mHz to 10 kHz). The complex resistivity is the inverse of the complex conductivity: $\rho^* = 1/\sigma^*$. What is actually measured is the complex impedance Z^* (in Ω) between the end-faces of a cylindrical sample for instance (Pelton *et al.* 1983):

$$Z^*(\omega) = \frac{U}{I} = |Z^*(\omega)| e^{i\varphi(\omega)}, \quad (18)$$

where U is the voltage between the end-face boundaries of the sample, I the magnitude of the current which goes through the sample, and $|Z^*(\omega)|$ and $\varphi(\omega)$ are the amplitude and the phase of the complex impedance, respectively. The complex resistivity ρ^* is related to Z^* by a geometrical factor K (in m) by: $\rho^*(\omega) = K Z^*(\omega)$. This geometrical factor takes into account not only the position of the electrodes on the sample, but also the size and the shape of the samples. The complex resistivity ρ^* can be recasted into a modulus $|\rho^*(\omega)|$ and a phase $\varphi(\omega)$. In the present contribution, amplitude will always stand for the resistivity amplitude. The phase is the same for the resistivity and the impedance.

Considering the geometry of samples #EST27906, #EST29296 and #EST29300, we choose the same electrode array than proposed by Cosenza *et al.* (2007). It is a Wenner- α type array (see Figs 8a and c for details). This setup is called the Lg-setup below. The lengths of samples #EST28144 and #EST30749 were too short to use the Lg-setup. Therefore, we tried a different approach by putting the electrodes on the side of the core samples (name the Rd-setup below). Instead of the length of the sample, we used a constant angle $\alpha = 45^\circ$ between each electrode (see Figs 8b and d for details). Due to the orientation of the boreholes in the Callovo–Oxfordian formation, the Lg-setup corresponds to electrical measurements perpendicular to the stratification while the Rd-setup provides measurements parallel to the stratification.

Both setups are composed by two current electrodes (C1 and C2) and two potential electrodes (P1 and P2). All these electrodes were developed in the medical world. We found that they allow the best contact, and thus the lowest contact resistance, among various tested electrodes. The current (or source) electrodes are used to inject an alternative current at different frequencies. We choose the round 50 mm diameter electrodes to fit the different sample sizes. They have thin carbon films to improve the contact (Dura-Stick II). The potential electrodes are electrocardiogram (ECG) electrodes: their silver–silver chloride (Ag/AgCl) composition made them non-polarizable (3M Red Dot). Each of these electrodes correspond to a round 10 mm diameter metal disk, galvanized with silver and covered with a soft sponge imbibed of a AgCl gel. For the computation of the geometrical constant K , it is important to note that the current electrodes have a diameter of 50 mm while the potential electrodes are nearly punctual.

The non-polarizability of the potential electrodes has been validated using porous and electrically inert samples. For example, a clean (clay-free) limestone sample saturated with a saline NaCl brine (0.78 S m^{-1} at 25°C) was used to perform a test as prescribed by Vinegar & Waxman (1984). As predicted, we measured no significant polarization in this case (see Fig. 9).

Spectral induced polarization measurements were performed using the apparatus developed at the Central Laboratory for Electronics, ZEL, at the Forschungszentrum Jülich, in Germany (Zimmermann *et al.* 2008). It is based on a four-electrode system operating in the frequency range 1 mHz–45 kHz. To our knowledge,

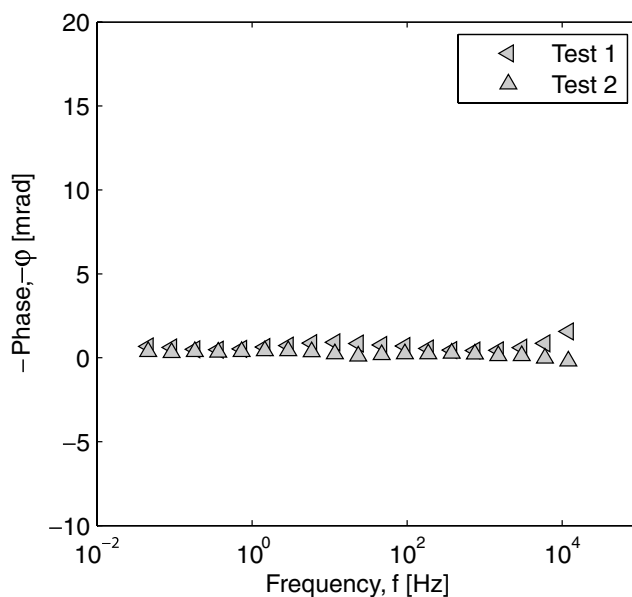


Figure 9. Test of the spectral induced polarization equipment and the electrodes (3M Red Dot) used to measure the potential response. This experiment is performed with a clean (clay-free) limestone sample saturated with a conductive brine (at 0.78 S m^{-1} at 25°C). In this condition, we expect the phase to be close to zero in the investigated frequency range in agreement with the experimental data.

its accuracy is unmatched (0.1 mrad between 1 mHz and 1 kHz). We have tested the linearity of the system to check the influence of the input voltage on the measured currents (Fig. 10). This test shows that the spectral induced polarization mechanisms involved in these experiments are linear for applied voltage differences higher than 0.1 V. In all the experiment, we used $U = 5 \text{ V}$ for the applied voltage.

For each electrode configuration set-up, we need to convert the measured resistance into a resistivity by using a geometrical coefficient, which depends on the position of the electrode, their extension, and the geometry of the sample. Considering the sample and the measurement setup (Lg or Rd) discussed above, we determined the values of K by solving the Laplace equation with source current terms using the finite element code COMSOL Multiphysics 3.4 accounting for the geometry of the core sample and the geometry of the electrodes discussed above. The values of these coefficients are reported in Table 4.

4.3 Drying procedure

We dried the core samples in two phases: (1) We call the first phase the desaturation phase while (2) the second phase is termed the heating phase (see Cosenza *et al.* 2007 and Ghorbani *et al.* 2009 for more details about this procedure). The studied samples were already partially unsaturated at the beginning of the experiments. We choose not to resaturate them to avoid possible perturbations like the swelling associated with changes in pore water composition. The saturations of the samples have been monitored through their weight with a precise balance ($\pm 0.1 \text{ g}$). The sample water saturations have been calculated from:

$$s_w = \frac{w}{\phi} \frac{\rho_d}{\rho_w}, \quad (19)$$

where $w = (m_{\text{meas}} - m_d)/m_d$ is the gravimetric water content with m_{meas} , the measured mass, and m_d , the dry mass of the sample. ϕ

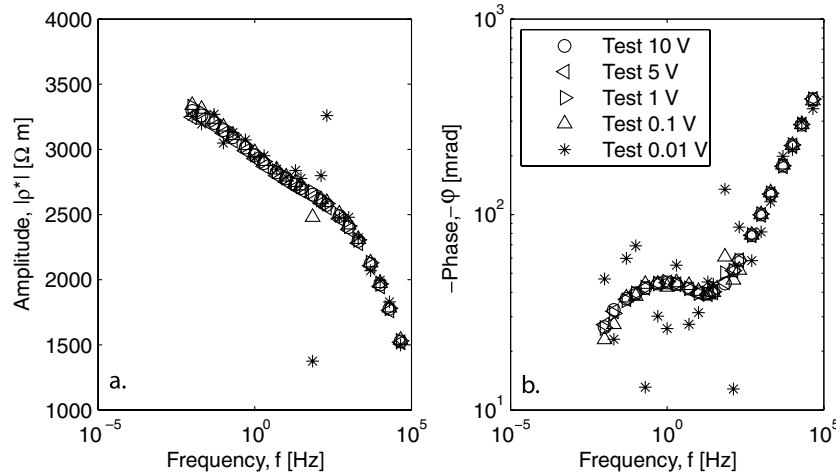


Figure 10. Linearity test of the equipment used to perform the spectral induced polarization measurements in the frequency range 10 mHz–45 kHz. We used a partially saturated Callovo–Oxfordian core sample EST29300. (a) Amplitude of the resistivity versus the frequency. (b) Phase versus the frequency. We used the following input voltages: 10, 5, 1, 0.1 and 0.01 V. Note that the response is independent on the applied voltage (at least for voltages higher than 1 V). The responses conform to the linear Ohm’s law.

Table 4. Properties of the samples and experimental setups.

Sample	Setup	Geometrical factor, K (m)	Porosity, ϕ (–) ^a	Initial saturation, s_w^i (–) ^b
EST27906	Lg	0.1219	0.157	0.805
EST28144	Rd	0.1756	0.176	0.797
EST29296	Lg	0.0913	0.163	0.770
EST29300	Lg	0.0703	0.146	0.748
EST30749	Rd	0.3835	0.181	0.808

^aValues from TCMR measurement in the borehole at the sample depth (ANDRA).

^bValues calculated from the weight measurements with the TCMR porosity.

is the ‘Total Combinable Magnetic Resonance’ (TCMR) porosity, $\rho_w = 1000 \text{ kg m}^{-3}$, and $\rho_d = (1 - \phi)\rho_g$ are the water and the dry density, respectively (with $\rho_g = 2690 \text{ kg m}^{-3}$ the grain density, see Gaucher *et al.* 2004 for details). We have decided to use the TCMR porosity (from borehole measurements) as a reliable estimate of the connected porosity because a study by ANDRA (2009) demonstrated that this was indeed the best estimate of the real porosity. The dry mass of the sample have been measured at the end of each experiment with the core samples being 24 hr at 105 °C in an oven. Table 4 presents the values of the porosity and the initial saturation, s_w^i , for each core sample.

We describe now quickly the desaturation phase. In this phase, the saturation decreases naturally by evaporation at $T = 22 \pm 2^\circ\text{C}$ and the relative humidity $h_r = 22 \pm 4$ per cent. Spectral induced polarization measurements have been done at different times after we removed the protective bags of the core samples, thus at different levels of the water saturations. The second drying phase is called the ‘heating phase’ because the samples have been put into an oven for 24 hr at three different temperatures (70, 90 and 105 °C). The higher temperature allowed the measurement of the dry mass of the sample, m_d , for the saturation determination (see above). After each heating step, we have waited a couple of hours (typically 2–4 hr) before the spectral induced polarization measurements were performed. This is the time required for the sample to cool down at room temperature ($T = 22 \pm 2^\circ\text{C}$). Fig. 11 shows the monitored saturation of the five samples during these two phases.

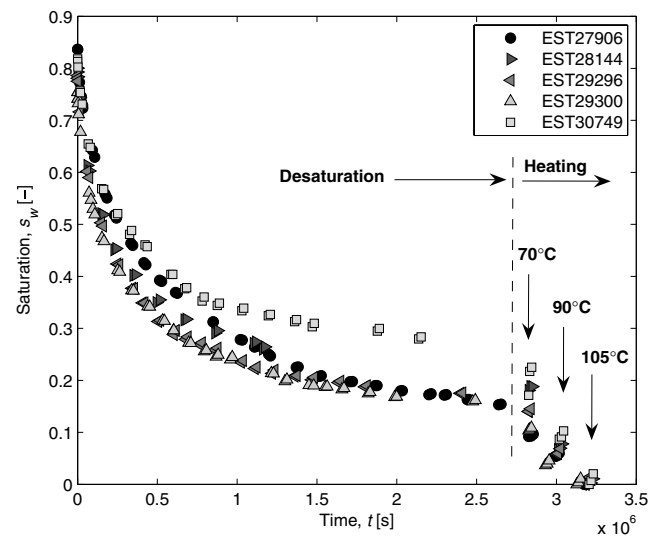


Figure 11. Complete desaturation curves for the five investigated samples as a function of the time and the three heating steps (70, 90 and 105 °C). Note the two phases for the desaturation: the desaturation phase and the heating phase.

5 INTERPRETATION

The spectral induced polarization signal can be decomposed into an amplitude $|\rho^*|$ and a phase ϕ . The behaviour of the amplitude can be compared to the DC resistivity except for the Stern contribution to surface conductivity (there is no contribution of the Stern layer at zero frequency according to the model described by Leroy *et al.* 2008 and Leroy & Revil 2009). We first analysed below the amplitudes (at a given low frequency) and then later the amplitude and the phase versus the frequency.

5.1 Interpretation of the amplitudes

We first focus on the amplitude of the complex resistivity at 10 Hz. We use the conductivity model developed by Revil *et al.* (2007) (eq. 15 above), which accounts for the saturation of the water phase. The conductivity of the pore water is calculated from the

THERMOAR geochemical model (see Table 2) and the Donnan equilibrium model discussed by Revil & Linde (2006). In this approach, the contribution of the diffuse layer is accounted for inside the conductivity of the pore water phase. Using a ‘non-DC’ input current, we have to consider the Stern contribution to the surface electrical conductivity σ_S . Therefore the conductivity model has five parameters: F , n , s_w^c , φ_m and σ_S , which is itself determined from a grain size distribution. We determined the values of these five parameters for each core sample using the Simplex algorithm to solve a non-linear optimization procedure. This algorithm is able to optimize parameters value by minimizing a cost function G defined by the sum of the square residual between the N measured and calculated conductivities, σ_{meas}^k and σ_{calc}^k , respectively

$$\text{Min}G = \sum_{k=1}^N (\sigma_{\text{meas}}^k - \sigma_{\text{calc}}^k)^2. \quad (20)$$

This problem being well posed, we do not need any regularization term. Minimization is performed for both the real and the imaginary parts of the complex conductivity.

Fig. 12 shows that the Revil *et al.* (2007) model fit pretty well the experimental data with the optimized parameters given in Table 5. The optimized values of the different parameters agree with previous data obtained on core samples from the C2b1 and C2b2 units of the Callovo–Oxfordian formation. For example, $F \in [38; 81]$ and

$\varphi_m \in [-49.5; -37.1]\text{mV}$ are consistent with the results obtained by Jougnot *et al.* (2009) and based on the diffusion of ionic tracers and *in situ* osmotic pressures in the Callovo–Oxfordian formation. The range of saturation exponents $n \in [1.44; 2.04]$ is consistent with the values reported by Jougnot & Revil (2008) who developed a thermal conductivity model for unsaturated Callovo–Oxfordian clay-rocks. This model is an extension of the model developed by Revil (2000) to represent, in a unified framework, the electrical and thermal conductivities of granular media at saturation. The range of values of the saturation threshold $s_w^c \in [0.10; 0.24]$ obtained from the conductivity measurements provide the first estimate of this parameter for the Callovo–Oxfordian clay-rock. The distribution of the surface conductivity associated with the Stern layer $\sigma_S \in [0.96; 3.25] \times 10^{-5} \text{ S m}^{-1}$ is quite narrow. It is controlled by the density of counterions (sodium in the present case) in the Stern layer. As discussed further, this density is consistent with electrical triple layer calculations (see Leroy *et al.* 2007 and Leroy & Revil 2009).

The two experimental setups (Lg and Rd) have allowed measurements perpendicular and parallel to the bedding planes (Table 5). From these considerations, we see that anisotropy has an influence upon n and s_w^c . In fact, the value of n normal to the bedding is higher than the value of n parallel to the bedding plane. The opposite occurs for s_w^c . However these observations should be considered with caution. Indeed in the Rd-setup, the current electrodes are very close to each other. This geometry implies that the current lines are not

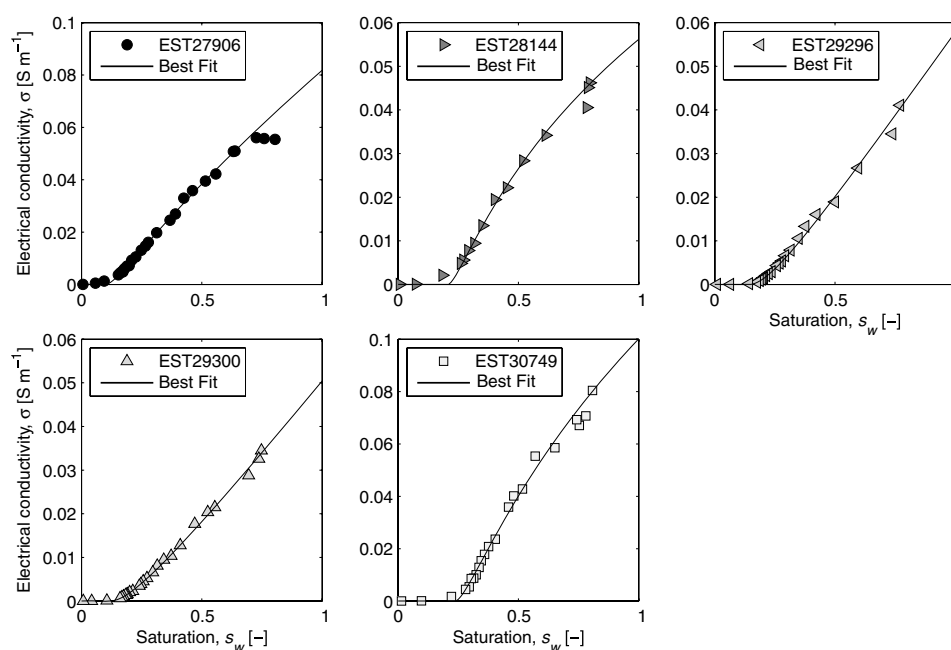


Figure 12. Comparison between the experimental data for the amplitude of the conductivity as a function of the saturation (measured at a frequency at 10 Hz). The plain line corresponds to the best fit obtained with the model proposed by Revil *et al.* (2007) (see main text). The uncertainties are smaller than the size of the symbols.

Table 5. Optimized values for the parameters entering the electrical conductivity model.

Sample	F (–)	n (–)	s_w^c (–)	φ_m (mV)	σ_S (S m^{-1})	Stratification
EST27906	44.6	1.79	0.10	–44.1	1.01×10^{-5}	Normal
EST28144	80.7	1.44	0.21	–49.5	1.33×10^{-5}	Parallel
EST29296	80.0	1.99	0.14	–49.4	0.99×10^{-5}	Normal
EST29300	42.6	2.04	0.12	–37.1	0.96×10^{-5}	Normal
EST30749	38.0	1.50	0.24	–48.0	3.25×10^{-5}	Parallel

homogeneously distributed through each core sample and then the measurements may not be representative of the complete volume of the core samples. The only way to check these observations would be to make electrical measurements with the Lg-setup on core sampled parallel to the bedding planes. This has been done by Cosenza *et al.* (2007) using the Tournemire clay-rock but not in this work using the Callovo–Oxfordian clay-rock.

5.2 Amplitude and phase versus frequency

We consider now the whole set of complex resistivity data and compare it to our mechanistic model for unsaturated clay-rocks. Fig. 13 presents the measured data for the spectral induced polarization measurements on the sample #EST29300 at seven different saturations. We can note the similarity in this trend with our model prediction as shown in Fig. 4. Indeed, as expected, the amplitude and the phase increase when the saturation of the water phase decreases. We also observed that after the heating phases, the experimental data are noisy and characterized by high uncertainties (especially after the heating step at 105 °C), which makes them difficult to interpret (Fig. 13). This could be explained by the increase of the contact resistance between the electrodes and the surface of the core samples. In order to avoid misinterpretation of our results, we focus below only with the experiments performed with the Lg-setup excluding the data obtained in the heating phase (samples #EST27906, #EST29296 and #EST29300). In addition, in the heating phase, the formation of microcracks was observed and this creates a strong response upon the spectral induced polarization data (Cosenza *et al.* 2007; Ghorbani *et al.* 2009) and the presence of these microcracks has not yet been accounted for in our model.

In order to compare the experimental data and the proposed model for spectral induced polarization in unsaturated clay-rocks, we need to determine the unknown parameters of the model. In the previous section, we have determined four of the seven key parameters for the

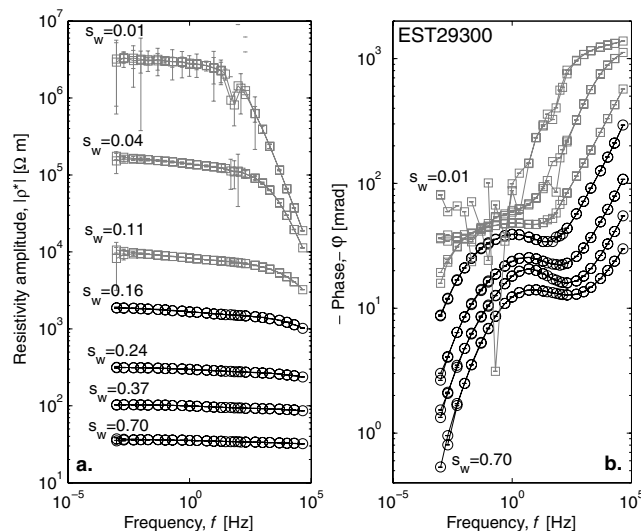


Figure 13. Experimental spectral induced polarization data (sample EST29300) at seven different saturations of the water phase and for the frequency range 1 mHz–45 kHz. (a) Amplitude of the resistivity versus the frequency. (b) Phase versus frequency. The black lines and open circles represent the data obtained during the desaturation phase (values of the water saturations: $s_w = 0.70, 0.37, 0.24$ and 0.16). The grey lines and the grey open squares correspond to the heating phase ($s_w = 0.11, 0.04$ and 0.01).

five core samples: F , n , s_w^c and φ_m (the values of these parameters are reported in Table 5). We used Leroy & Revil (2009) measurements in saturated clay-rich porous media for the three other parameters (Γ_{Na}^0 , τ_0 and c). These sets of values are considered to be prior values in the following. In order to find the best fit for each measurement, thus for the different saturations, we used a simple trial and error method from the prior parameters. Note that we optimize the whole complex signal: amplitude and phase at the same time.

The measurements on the sample #EST27906 (at $s_w = 0.82$) have been fitted by the proposed model with the following parameters: $F = 54.5$, $n = 1.79$, $s_w^c = 0.10$, $\varphi_m = -44.1$ mV, $\Gamma_{Na}^0 = 0.891$ nm⁻², $\tau_0 = 3.46 \times 10^{-2}$ s and $c = 0.855$. Fig. 14 shows that our model is able to fit pretty well the measured complex resistivities (total rms = 4.18 per cent with equal weight given to the in phase and quadrature conductivities). Dividing the frequency range in two parts, we see that the model perfectly fits the amplitude and the phase for frequencies higher than 1 Hz (Leroy *et al.* 2008; Leroy & Revil 2009). At lower frequencies, the predicted amplitudes are still pretty close to the experimental data, but the modelled phase underpredicts the data.

Fig. 15 shows the comparison between the measurements, during the desaturation procedure, and the prediction of the model. At low saturations, the measured Maxwell–Wagner polarization is higher than predicted by the model (from 1 to 15 kHz). The measured polarization is very well fitted between 1 Hz and 1 kHz.

Three explanations could explain the observed mismatch at low frequencies. (1) Another mechanism (like membrane polarization, see discussion in Leroy & Revil 2009) may occur at low frequencies and has not been accounted for in our model. (2) A second possibility is that we have only considered sodium as the only counterion responsible for the polarization of the Stern layer. Maybe this assumption is too restrictive. Other cations are sorbed and may participate to the polarization of the Stern layer broadening the distribution of the relaxation times of the polarization of this layer in the frequency domain. (3) We have considered an unimodal distribution of the relaxation time distribution. However, the distribution of the relaxation times may be multimodal (e.g. Ghorbani *et al.* 2009 used a double Cole–Cole model to fit their data). The mean relaxation time, τ_0 , corresponds to a grain radius of 9.5 μ m consistent both with the size of the tactoids and the size of the silica and carbonates grains.

The estimates of the model parameters determined by the best fits between the model and the data for core samples #EST27906, #EST29296 and #EST29300, are displayed in Fig. 16 as a function of the saturation. Note that the values of n , s_w^c and φ_m optimized using the amplitude data at 10 Hz gave the best fit (Table 5). The optimized values of F and c are independent on the saturation (see Figs 16a and b, respectively). This is consistent with the fact that the formation factor characterizes the topology of the pore space (Revil & Cathles 1999) and the exponent c characterizes the broadness of the grain size distribution (Leroy *et al.* 2008). The mean values of the formation factor are $\bar{F} = 46.1 \pm 1.8$, 85.5 ± 5.1 and 44.1 ± 2.2 for the core samples #EST27906, #EST29296 and #EST29300, respectively. These values are consistent with those reported in Table 5. The mean values of the Cole–Cole exponents are $\bar{c} = 0.852 \pm 0.005$, 0.854 ± 0.009 and 0.827 ± 0.006 , for the core samples #EST27906, #EST29296 and #EST29300, respectively.

We found that τ_0 and Γ_{Na}^0 vary slightly with the saturation. Fig. 16(c) shows that τ_0 increases when the saturation decreases. This observation is consistent with one the two τ of the fit by a double Cole–Cole model in the work of Ghorbani *et al.* (2009).

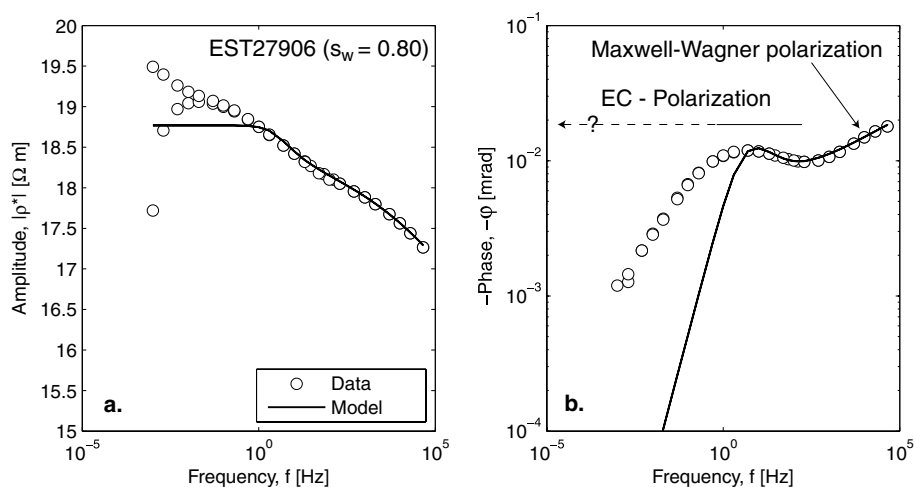


Figure 14. Comparison between the EST27906 measurement and the proposed model (for $s_w = 0.80$): (a) Amplitude versus frequency. (b) Phase versus frequency (from 1 mHz to 45 kHz) (rms = 4.18 per cent). The Maxwell–Wagner polarization dominates at frequencies higher than 1 kHz. Below 1 kHz, the induced polarization response of the core sample is dominated by the polarization of the Stern layer. EC stands for the electrochemical polarization of the Stern layer.

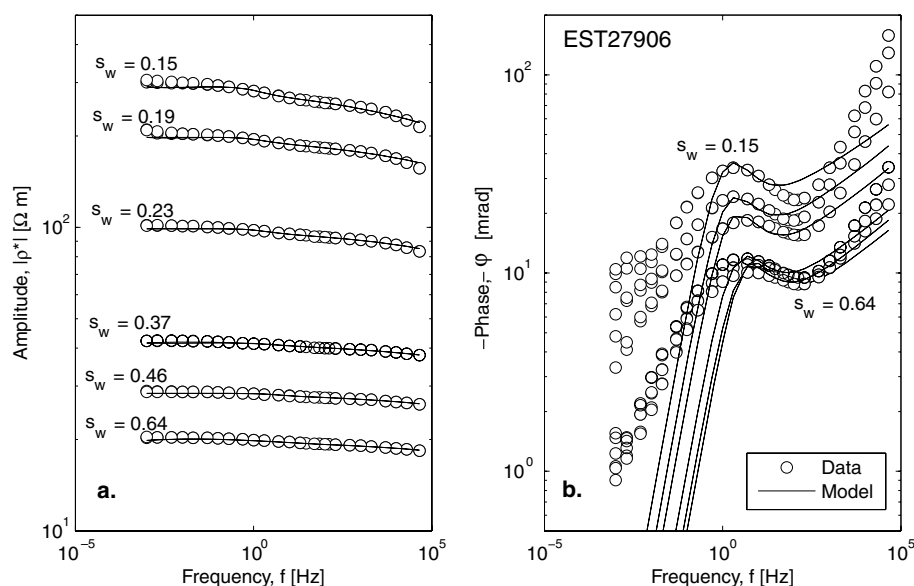


Figure 15. Comparison between the spectral induced polarization measurements from 1 mHz to 45 kHz (sample EST27906) and the proposed model (at six different saturations of the water phase). (a) Amplitude of the complex resistivity versus frequency. (b) Phase versus frequency. The plain lines correspond to the model predictions.

Ghorbani *et al.* (2009) explain this variation by textural changes. In fact, the clay mineral deformation (shrinkage) might break Stern layer connection inside the tactoids, and then increase the Stern layer diffusion path, thus the relaxation time. These results could confirm that the polarization of the Stern layer occurs in the tactoids and not around the coarse grains. This implies that the relaxation time distribution in such complex medium does not correspond directly to the particle size distribution, but more to a Stern diffusion path distribution inside the clay aggregates. In that case, τ would not be related to the grain radii but to the characteristic length of the tactoids. This explanation seems to be more realistic considering the complex geometry of clay assembly in consolidated clay-rocks (see Montes *et al.* 2004 for Scanning Electron Microscope pictures of the Callovo–Oxfordian at 10 and 50 μm scale). We observe also that the surface site density of sodium counterions in the Stern layer diminishes when the water saturation decreases.

This could be due to the disconnection of the Stern layers inside the tactoid.

6 CONCLUSIONS

We proposed a mechanistic model of spectral induced polarization to unsaturated clay-rocks. Sensitivity tests were conducted to determine the influence of saturation and seven key parameters on the modulus and phase of the complex resistivity. The model was successfully tested against experimental data on partially saturated core samples from the Callovo–Oxfordian formation. The model agreed with the amplitude of the conductivity (at 10 Hz) as a function of saturation. Then, we showed that the model was able to reproduce the complex conductivity behaviour of the medium at high frequencies (> 1 Hz) for water saturations comprised between 0.2 and 0.8. The polarization of the Stern layer that seems to take

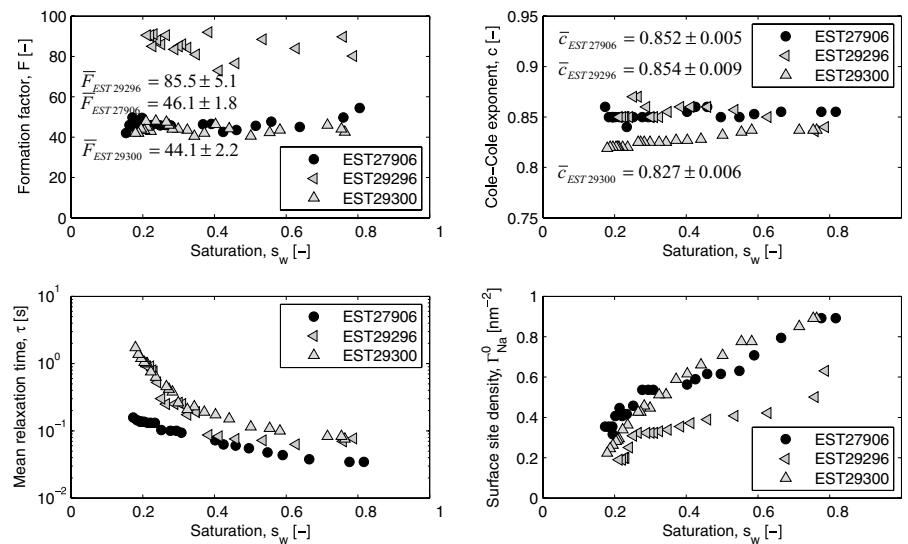


Figure 16. Evolution of the optimized model parameters as a function of the saturation: (a) the formation factor, F , (b) the Cole–Cole exponent, c , (c) the mean relaxation time, τ and (d) the surface site density, Γ_{Na}^0 . Note that the optimized values of n , s_w^c and φ_m are constant for all the saturations. The dependence of the relaxation time with the saturation is opposite to that observed by Binley *et al.* (2005).

place at the scale of clay aggregates (tactoids) is related to a characteristic length that depends on the saturation of the water phase. In a future work, we will take into consideration the polarization of the Stern by several counterions with different valences and mobilities. In our opinion that would allow us to improve the fit at low frequencies (< 1 Hz). We will also take into account the membrane polarization mechanism to produce a more complete theory.

ACKNOWLEDGMENTS

We thank the French National Agency for Radioactive Waste Management (ANDRA) for its support. The Ph.D. thesis of Damien Jougnot is supported by ANDRA. D. Jougnot strongly thanks S. Altmann, D. Coelho and J.C. Robinet for fruitful discussions. A. Revil and D. Jougnot thank T. Young for his support at CSM and M. Batzle for his advices, and G. Okay for her help. We thank Andreas Weller, an anonymous referee and the Associate Editor Jörg Renner for the quality of the constructive comments and the speed of the review process.

REFERENCES

- ANDRA, 2005. Dossier 2005 argile–Référentiel du site de Meuse/Haute Marne, Internal Report ANDRA n°C.RP.ADS.04.0022.
- ANDRA, 2008. Porosité totale des argilites du Callovo-Oxfordien, Internal Report ANDRA n°KMFAEAP080032.
- ANDRA, 2009. Dossier 2009 Argile, Référentiel du site de Meuse Haute-Marne, Internal report ANDRA.
- Archie, G.E., 1942. The electrical resistivity log as an aid in determining some reservoir characteristics, *Trans. AIME*, **146**, 54–62.
- Binley, A., Slater, L.D., Fukes, M. & Cassiani, G., 2005. Relationship between spectral induced polarization and hydraulic properties of saturated and unsaturated sandstones, *Water Resour. Res.*, **41**, W12417, doi:10.1029/2005WR004202.
- Blaschek, R., Hördt, A. & Kemna, A., 2008. A new sensitivity-controlled focusing regularization scheme for the inversion of induced polarization data based on the minimum gradient support, *Geophysics*, **73**, F45–54.
- Börner, F.D., Schopper, J.R. & Weller, A., 1996. Evaluation of transport and storage properties in the soils and groundwater zone from

- induced polarization measurements, *Geophys. Prospect.*, **44**, 583–601, doi:10.1111/j.1365-2478.1996.tb00167.x.
- Böttcher, C.J.F. & Bordewijk, P., 1978. *Theory of Electric Polarization*, Vol. II, Dielectrics in Time-Dependent Fields, Elsevier Science, Amsterdam.
- Chen, J., Kemna, A. & Hubbard, S.S., 2008. A comparison between Gauss-Newton and Markov-chain Monte Carlo-based methods for inverting spectral induced-polarization data for Cole-Cole parameters, *Geophysics*, **73**, 247–258.
- Cole, K.S. & Cole, R.H., 1941. Dispersion and adsorption in dielectrics (I), *J. Chem. Phys.*, **9**, 341–351.
- Cosenza, P., Ghorbani, A., Florsch, N. & Revil, A., 2007. Effects of drying on the low-frequency electrical properties of Tournemire argillites, *Pure appl. Geophys.*, **164**, 2043–2066, doi:10.1007/s00024-007-0253-0.
- Cosenza, P., Ghorbani, A., Revil, A., Zamora, M., Schmutz, M., Jougnot, D. & Florsch, N., 2008. A physical model of the low-frequency electrical polarization of clay-rocks, *J. geophys. Res.*, **113**, B08204, doi:10.1029/2007JB005539.
- Davidson, D.W. & Cole, R.H., 1950. Dielectric relaxation in glycerine, *J. Chem. Phys.*, **18**, 1417.
- de Lima, O.A.L. & Niwas, S., 2000. Estimation of hydraulic parameters of shaly sandstone aquifers from geoelectrical measurements, *J. Hydrol.*, **235**, 12–26, doi:10.1016/S0022-1694(00)00256-0.
- de Lima, O.A.L. & Sharma, M.M., 1990. A grain conductivity approach to shaly sandstones, *Geophysics*, **55**, 1347–1356.
- de Lima, O.A.L. & Sharma, M.M., 1992. A generalized Maxwell-Wagner theory for membrane polarization in shaly sands, *Geophysics*, **57**(3), 431–440, doi:10.1190/1.1443257.
- Gaucher, E. *et al.*, 2004. ANDRA underground research laboratory: interpretation of the mineralogical and geochemical data acquired in the Callovo-Oxfordian Formation by investigative drilling, *Phys. Chem. Earth*, **29**, 55–77, doi:10.1016/j.pce.2003.11.006.
- Gaucher, E. *et al.*, 2006. Modeling the porewater chemistry of the Callovo-Oxfordian formation at a regional scale, *C. R. Geosci.*, **338**(12–13), 917–930.
- Ghorbani, A., Camerlynck, C., Florsch, N., Cosenza, P.H., Tabbagh, A. & Revil, A., 2007. Bayesian inference of the Cole-Cole parameters from time and frequency-domain induced polarization, *Geophys. Prospect.*, **55**(4), 589–605, doi:10.1111/j.1365-2478.2007.00627.x.
- Ghorbani, A., Cosenza, P.H., Revil, A., Zamora, M., Schmutz, M., Florsch, N. & Jougnot, D., 2009. Non-invasive monitoring of water content and textural changes in clay-rocks using spectral induced

- polarization: a laboratory investigation, *Appl. Clay Sci.*, **43**, 493–502, doi:10.1016/j.clay.2008.12.007.
- Homand, F., Giraud, A., Escoffier, S., Koriche, A. & Hoxha, D., 2004. Permeability determination of a deep argillite in saturated and partially saturated conditions, *Int. J. Heat Mass Transfer*, **48**, 3517–3531.
- Hördt, A., Blaschek, R., Kemna, A. & Zisser, N., 2007. Hydraulic conductivity estimation from induced polarisation data at the field scale—the Krauthausen case history, *J. appl. Geophys.*, **62**, 33–46.
- Hunt, A.G. & Ewing, R.P., 2003. On the vanishing of solute diffusion in porous media at a threshold moisture content, *Soil Sci. Soc. Am. J.*, **67**, 1701–1702.
- Jougnot, D. & Revil, A., 2008. Thermal conductivity of unsaturated clay-rocks, *Hydrol. Earth Syst. Sci. Discuss.*, **5**, 2409–2423.
- Jougnot, D., Revil, A. & Leroy, P., 2009. Diffusion of ionic tracers in the Callovo-Oxfordian clay-rock using the Donnan equilibrium model and the formation factor, *Geochim. Cosmochim. Acta*, **73**, 2712–2726, doi:10.1016/j.gca.2009.01.035.
- Kemna, A., Binley, A., Ramirez, A. & Daily, W., 2000. Complex resistivity tomography for environmental applications, *Chem. Eng. J.*, **77**, 11–18.
- Kemna, A., Binley, A. & Slater, L., 2004. Crosshole IP imaging for engineering and environmental applications, *Geophysics*, **69**, 97–107.
- Kruschwitz, S. & Yaramanci, U., 2004. Detection and characterization of the disturbed rock zone in claystone with complex resistivity method, *J. appl. Geophys.*, **57**, 63–79, doi:10.1016/j.jappgeo.2004.09.003.
- Leroy, P. & Revil, A., 2004. A triple layer model of the surface electrochemical properties of clay minerals, *J. Colloid Interface Sci.*, **270**(2), 371–380.
- Leroy, P. & Revil, A., 2009. A mechanistic model for the spectral induced polarization of clay material, *J. geophys. Res.*, **114**, B10202, doi:10.1029/2008JB006114.
- Leroy, P., Revil, A., Altmann, S. & Tournassat, C., 2007. Modeling the composition of the pore water in a clay-rock geological formation (Callovo-Oxfordian, France), *Geochim. Cosmochim. Acta*, **71**(5), 1087–1097, doi:10.1016/j.gca.2006.11.009.
- Leroy, P., Revil, A., Kemna, A., Cosenza, P. & Ghorbani, A., 2008. Spectral induced polarization of water-saturated packs of glass beads, *J. Colloid Interface Sci.*, **321**, 103–117, doi:10.1016/j.jcis.2007.12.031.
- Lesmes, D.P. & Morgan, F.D., 2001. Dielectric spectroscopy of sedimentary rocks, *J. geophys. Res.*, **106**, 13 329–13 346, doi:10.1029/2000JB900402.
- Linde, N., Jougnot, D., Revil, A., Matthäi, S., Arora, T., Renard, D. & Doussan, C., 2007. Streaming current generation in two phase flow conditions, *Geophys. Res. Lett.*, **34**, L03306, doi:10.1029/2006GL028878.
- Marshall, D.J. & Madden, T.R., 1959. Induced polarization: a study of its causes, *Geophysics*, **24**(4), 790–816, doi:10.1190/1.1438659.
- Montes, H.G., Duplay, J., Martinez, L., Escoffier, S. & Rousset, D., 2004. Structural modifications of Callovo-Oxfordian argillite under hydration/dehydration conditions, *Appl. Clay Sci.*, **25**, 187–194.
- Parkhurst, D.L. & Appelo, C.A.J., 1999. User's guide to PHREEQC (Version 2)—a computer program for speciation, batch-reaction, one-dimensional transport, and inverse geochemical calculations: U.S. Geological Survey Water-Resources Investigations Report 99–4259, 312 p.
- Pelton, W.H., Sill, W.R. & Smith, B.D., 1983. Interpretation of complex resistivity and dielectric data, Part I, *Geophys. Trans.*, **29**(4), 297–330.
- Pusch, R., 2006. Clays and nuclear waste management, in *Handbook of Clay Science*, pp. 703–716, eds Bergaya, F., Theng, B.K.G. & Lagaly, G., Elsevier, Amsterdam.
- Revil, A., 1999. Ionic diffusivity, electrical conductivity, membrane and thermoelectric potentials in colloids and granular porous media: a unified model, *J. Colloid Interface Sci.*, **212**, 503–522.
- Revil, A., 2000. Thermal conductivity of unconsolidated sediments with geophysical applications, *J. geophys. Res.*, **105**, 16,749–16,768.
- Revil, A., 2007. Thermodynamics of transport of ions and water in charged and deformable porous media, *J. Colloid Interface Sci.*, **307**(1), 254–264.
- Revil, A. & Cathles, L.M., 1999. Permeability of shaly sands, *Water Resour. Res.*, **35**(3), 651–662.
- Revil, A. & Jougnot, D., 2008. Diffusion of ions in unsaturated porous materials, *J. Colloid Interface Sci.*, **319**, 226–235, doi:10.1016/j.jcis.2007.10.041.
- Revil, A. & Leroy, P., 2001. Hydroelectric coupling in a clayey material, *Geophys. Res. Lett.*, **28**(8), 1643–1646.
- Revil, A. & Leroy, P., 2004. Governing equations for ionic transport in porous shales, *J. geophys. Res.*, **109**, B03208, doi:10.1029/2003JB002755.
- Revil, A. & Linde, N., 2006. Chemico-electromechanical coupling in microporous media, *J. Colloid Interface Sci.*, **302**, 682–694, doi:10.1016/j.jcis.2006.06.051.
- Revil, A., Pezard, P.A. & Glover, P.W.J., 1999. Streaming potential in porous media. 1. Theory of the zeta-potential, *J. geophys. Res.*, **104**(B9), 20,021–20,031.
- Revil, A., Linde, N., Cerepi, A., Jougnot, D., Matthäi, S. & Finsterle, S., 2007. Electrokinetic coupling in unsaturated porous media, *J. Colloid Interface Sci.*, **313**, 315–327, doi:10.1016/j.jcis.2007.03.037.
- Schwarz, G., 1962. A theory of the low-frequency dielectric dispersion of colloidal particles in electrolyte solution, *J. Phys. Chem.*, **66**, 2636–2642.
- Slater, L. & Binley, A., 2006. Synthetic and field-based electrical imaging of a zero valent iron barrier: implications for monitoring long-term barrier performance, *Geophysics*, **71**, B129–137.
- Tabbagh, A., Cosenza, P.H., Guérin, R., Ghorbani, A. & Florsch, N., 2009. Modelling of Maxwell-Wagner induced polarisation amplitude, *J. appl. Geophys.*, **67**(2), 109–113.
- Tournassat, C., Chapron, Y., Leroy, P., Bizi, M. & Boulahya, F., 2009. Comparison of molecular dynamics simulations with Triple Layer and modified Gouy-Chapman models in a 0.1 M NaCl – montmorillonite system, *J. Colloid Interface Sci.*, **339**, 533–541, doi:10.1016/j.jcis.2009.06.051.
- Trouiller, A., 2006. The Callovo-Oxfordian of the Paris Basin: from its geological context to the modelling of its properties, *C. R. Geosci.*, **338**, 815–823.
- Tsang, C.-F., Bernier, F. & Davies, C., 2005. Geohydromechanical processes in the Excavation Damaged Zone in crystalline rock, rock salt, and indurated and plastic clays in the context of radioactive waste disposal, *Int. J. Rock Mech. Min. Sci.*, **42**, 109–125, doi:10.1016/j.ijrmms.2004.08.003.
- Van Voorhis, G.D., Nelson, P.H. & Drake, T.L., 1973. Complex resistivity spectra of porphyry copper mineralization, *Geophysics*, **38**(1), 49–60, doi:10.1190/1.1440333.
- Vanhala, H., 1997a. Laboratory and field studies of environmental and exploration applications of the Spectral Induced Polarization (SIP) method, *PhD thesis*. University of Technology, Helsinki.
- Vanhala, H., 1997b. Mapping oil-contaminated sand and till with the spectral induced polarization (SIP) method, *Geophys. Prospect.*, **45**, 303–326.
- Vinegar, H.J. & Waxman, M.H., 1984. Induced polarization of shaly sands, *Geophysics*, **49**(9), 1267–1287.
- Zimmermann, E., Kemna, A., Berwix, J., Glaas, W., Münch, H.M. & Huisman, J.A., 2008. A high-accuracy impedance spectrometer for measuring sediments with low polarizability, *Meas. Sci. Technol.*, **19**, doi:10.1088/0957-0233/19/10/105603.

Spectral induced polarization of shaly sands: Influence of the electrical double layer

A. Revil^{1,2}

Received 9 August 2011; revised 4 January 2012; accepted 5 January 2012; published 16 February 2012.

[1] I developed a new model named POLARIS describing the complex conductivity of (pyrite-free) shaly poorly sorted sands. This model is based on the solution given by the effective medium theory for grains coated by an electrical double layer and immersed in a background electrolyte. The electrical double layer comprises the Stern layer and the diffuse layer. Both layers play very distinct roles in the in-phase and quadrature conductivities. The polarization of the shaly sands is mainly controlled by the polarization of the Stern layer (except at very high salinities) with a very small mobility of the counterions contained in this layer. The in-phase component of the conductivity is controlled by the conductivity of the pore water with a contribution associated with the diffuse layer (the contribution of the Stern layer seems negligible). The fraction of counterions in the Stern layer is computed from a simple sorption isotherm and is used to infer the quadrature conductivity. The quadrature conductivity is assumed to be frequency independent, which is a reasonable approximation in clayey sands and sandstones, in agreement with observations. The polarization model is also based on the assumption that the Stern layer is discontinuous between grains, an assumption that is consistent with recent models of ionic transport in clayey sands. POLARIS explains the dependence of the quadrature conductivity on the salinity, cation exchange capacity, specific surface area (or specific surface per unit pore volume), and temperature. It can be used to predict the saturation and the permeability (inside 1 order of magnitude).

Citation: Revil, A. (2012), Spectral induced polarization of shaly sands: Influence of the electrical double layer, *Water Resour. Res.*, 48, W02517, doi:10.1029/2011WR011260.

1. Introduction

[2] Induced Polarization is a geophysical method that is sensitive to the ability of porous materials to store reversibly electrical charges. In environmental geophysics, induced polarization offers an extension to the classical direct current (DC) conductivity/resistivity method to image soils and shallow aquifers [Vanhal, 1997; Morgan *et al.*, 1999; Binley *et al.*, 2005]. In frequency domain induced polarization, in addition to a conductivity magnitude, a phase lag is measured between the current and the electrical field. This phase lag represents a polarization over a conduction term (like the chargeability in time domain induced polarization). The conductivity magnitude and the phase lag can be used to define a complex conductivity in which both an in-phase conductivity and a quadrature conductivity contribution can be defined [e.g., Olhoeft, 1986]. The complex conductivity can be investigated over a broad range of frequencies (typically from 1 mHz to 10 kHz).

[3] For clayey sands, the conductivity and the phase lag are observed to be quite frequency independent over a

broad range of frequencies (e.g., in the range 10 mHz to 100 Hz in the work by Vinegar and Waxman [1984] (their Figures 4 and 5 for clayey sands and sandstones) and Slater and Glaser [2003] (their Figure 3)). A constant phase lag behavior has been usually connected to the fractal nature of the material (see Hyslip and Vallejo [1997] and Drolon *et al.* [2003] for the fractal characterization of sand grains and Le Méhauté and Crépy [1983] and Wong [1987] for fractal-based low-frequency polarization models). A broad range of polarization length scales covering several decades can be responsible for this behavior. In some cases, however, shaly sands do show distinct frequency dependence with a distribution of relaxations around some central frequency. The associated relaxation times may be determined by the pore throat size of the material [Scott and Barker, 2003; Binley *et al.*, 2005; Zisser *et al.*, 2010] or possibly by the grain size distribution [e.g., Revil and Florsch, 2010].

[4] In the present paper, I will focus on the first class of materials exhibiting mainly a frequency-independent polarization (over the frequency range 10 mHz to 100 Hz). The case of materials exhibiting dispersion peak(s) will be examined in a future work. In addition, I am dealing with sandstones free of non-oxidized pyrite [see Pelton *et al.*, 1978]. The reason is that pyrite is responsible for a very strong polarization depending on its concentration and grain size. Because some authors do not report the pyrite content of their samples, it is possible that some observations made

¹Department of Geophysics, Colorado School of Mines, Golden, Colorado, USA.

²ISTerre, UMR CNRS 5275, Université de Savoie, Le Bourget du Lac, France.

in the literature of well-characterized phase peak could be due to the presence of pyrite.

[5] For a long time, the development of the induced polarization method has been slowed down by the lack of physical understanding of the measured response. In geophysics, the induced polarization response is usually modeled with empirical responses (like the Cole-Cole or the constant phase angle models). The use of these empirical models is limited because they have no predictive powers. In other words, they are merely a parametric way to represent the complex conductivity spectra using only few parameters.

[6] Several mechanistic models have been proposed in the literature but none of them have been able to explain, in a consistent way, all the observations available to date. These mechanistic models are usually divided into two groups: (1) polarization models of the electrical double layer [Schwarz, 1962; de Lima and Sharma, 1992; Lesmes and Morgan, 2001; Leroy et al., 2008] and (2) the membrane polarization model [Marshall and Madden, 1959; Dukhin and Shilov, 2002; Titov et al., 2002]. This separation is in reality very arbitrary as all the double layer polarization models developed in colloidal chemistry include a membrane polarization component, which involves the formation of neutral clouds of ions on both the side of a particle or the side of a pore throat in presence of a harmonic electrical field.

[7] In the recent literature, Revil and coworkers [Leroy et al., 2008; Revil and Florsch, 2010; Jougnot et al., 2010; Schmutz et al., 2010; Vaudelet et al., 2011a, 2011b; Revil and Skold, 2011] have developed a Stern polarization model, which has been used successfully to explain a number of observations made in the laboratory, especially for silica sands. However, as explained by Revil and Florsch [2010], there are some strong differences between the electrical double layer associated with silica grains and clay minerals. In the case of silica grains, the fraction of counterions populating the Stern layer is very salinity-dependent and usually smaller than 60%. Therefore, there is a lot of room for the counterions to move along the mineral surface in this layer. It was found that the mobility of the counterions in the Stern layer of silica grains is quite similar (if not equal) to the mobility of the same ions in the free water [Tarasov and Titov, 2007; Revil and Florsch, 2010].

[8] In clay minerals, the fraction of counterions in the Stern layer is very high (between 80% and 98% [Leroy and Revil, 2009, and references therein]). I expect therefore that the mobility of the counterions in the Stern layer of clays may be much smaller than in the bulk pore water because of the high density of counterions populating the Stern layer. What is the impact of this low mobility on the overall complex conductivity of shaly sands? One may think that this question could be answered independently by molecular dynamics (see, for instance, Bourg and Sposito [2011], who found that the mobility of the counterions in the Stern layer is roughly a 10th of the mobility in the pore water). That said, computing the mobility of the counterions in the Stern layer from molecular dynamics is subject to many assumptions and therefore this approach does not yield, yet, compelling values, as explained by Bourg and Sposito [2011].

[9] I developed below a new model named POLARIS on the basis of the effective medium theory. This type of upscaling method has been very successful to model the

conductivity of shaly sands [Bussian, 1983; de Lima and Sharma, 1990; Revil et al., 1998]. I will explain why the surface conductivity of the clay particles is dominated by the diffuse layer and the polarization of the clay particles by the Stern layer. POLARIS seems to explain an unprecedented number of observations with a minimum of input parameters. The end goal of this model is to interface between reactive transport modeling and geophysical time domain or frequency domain induced polarization. Also my goal in developing POLARIS is to keep this model simple enough so it can be used with a minimum of assumptions.

2. Complex Conductivity

[10] In section 2 I will define first the conventions used below to describe the electrical conductivity. Then, I will describe the result of the differential effective medium theory (DEM) for the in-phase conductivity (section 2.1) and its implication in terms of quadrature conductivity (section 2.2). I will show that the in-phase conductivity has a surface conductivity term dominated by the diffuse layer while the quadrature conductivity is dominated by the Stern layer. Therefore the determination of the partition coefficient of the counterions between the diffuse and the Stern layer is of paramount importance in POLARIS. I will develop in section 2.3 a simple sorption isotherm to compute this partition coefficient as a function of the pH (for near neutral pH values) and salinity.

[11] I start by defining the convention I use below. Ampère's law is written as

$$\nabla \cdot \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}, \quad (1)$$

where t is time (in s), $\mathbf{J}$ is the conduction (electromigration) current density, $\mathbf{J}_d = \partial \mathbf{D} / \partial t$ denotes the displacement current density (both in A m^{-2} , the subscript d is for displacement), $\mathbf{H}$ is the magnetic field (expressed in A m^{-1}), $\mathbf{D} = \epsilon^* \mathbf{E}$ is the dielectric displacement (in C m^{-2}), and ϵ^* is the permittivity (in F m^{-1}) of the material (ϵ^* / ϵ_0 denotes the relative complex permittivity and ϵ_0 the permittivity of vacuum). In linear models, the current density is given by a linear Ohm's law, $\mathbf{J} = \sigma^* \mathbf{E}$ with the harmonic electrical field $\mathbf{E} = \mathbf{E}_0 \exp(i\omega t)$, $i = \sqrt{-1}$ the pure imaginary number, and ω is the angular frequency with $\mathbf{E}_0$ a constant electrical field magnitude and direction. As a side note, the linear Ohm's law (i.e., a linear constitutive relationship between the current density and the electrical field) does not hold in presence of a membrane polarization mechanism except for small electrical fields [see, e.g., Urtenov et al., 2007; Sistat et al., 2008, and reference therein]. The total (subscript t) current density is the sum of the conduction current and displacement current densities. This yields $\mathbf{J}_t = (\sigma^* + i\omega\epsilon^*)\mathbf{E}$ [Vinegar and Waxman, 1984] where σ^* and ϵ^* are complex scalars dependent upon frequency, $\sigma^* = \sigma' + i\sigma''$ and $\epsilon^* = \epsilon' + i\epsilon''$. The total current density can be written as

$$\mathbf{J}_t = \sigma_{\text{eff}}^* \mathbf{E}, \quad (2)$$

where $\sigma_{\text{eff}}^* = \sigma_{\text{eff}} + i\omega\epsilon_{\text{eff}}$ is the effective or apparent complex conductivity and σ_{eff} and ϵ_{eff} are real scalars dependent upon frequency. These effective parameters are the

parameters that are measured during an experiment in the laboratory or in the field but these terms contain both electromigration and true dielectric polarization effects. They are given by

$$\sigma_{\text{eff}} = \sigma' - \omega \varepsilon'' \quad (3)$$

$$\varepsilon_{\text{eff}} = \varepsilon' + \sigma'' / \omega \quad (4)$$

(note that $-\omega \varepsilon''$ is always positive). Therefore, the effective conductivity and permittivity include contributions from charge conduction and polarization mechanisms. In this paper, I am only interested in the complex conductivity,

$$\sigma^* = \sigma' + i\sigma'', \quad (5)$$

which dominates the measured response at low frequencies (below 100 Hz). With the sign conventions taken above, the quadrature conductivity is negative as well as the phase lag (other conventions exist). In the following, I will consider that the in-phase conductivity and the quadrature conductivity are frequency independent. Actually, their frequency dependence is very small and can be modeled with the constant phase lag model, as shown by Börner [1992].

[12] I denote $\omega = 2\pi f$ the angular frequency, $f = \omega / (2\pi)$ the frequency (in Hertz). The magnitude of the conductivity $|\sigma^*|$ and the phase lag φ are related to the real (in-phase) and imaginary (out-of-phase or quadrature) components of the complex conductivity σ^* , σ' , and σ'' (expressed in S m^{-1}) by

$$\sigma^* = \sigma' + i\sigma'' = |\sigma| \exp(i\varphi), \quad (6)$$

with $|\sigma^*| = (\sigma'^2 + \sigma''^2)^{1/2}$ and $\tan \varphi = \sigma'' / \sigma'$. Note that some researchers use an alternative convention for the complex conductivity $\sigma^* = \sigma' - i\sigma''$, which yields a positively-defined quadrature conductivity [Olhoeft, 1979; Börner, 1992; Kemna, 2000].

[13] I introduce now the physical concepts used to describe the induced polarization model developed below. The first one is that the clay particles are coated by an electrical double layer model. These two layers include the Stern layer of sorbed counterions and the diffuse layer (Figure 1). This double layer model of clay particles is rather well established [see Leroy and Revil, 2004, and references therein] and the existence of the Stern layer is in agreement with molecular dynamic simulations showing a condensation of the counterions directly on the mineral surface [Wang et al., 2007; Tournassat et al., 2009]. We consider two types of granular porous materials: Either the porous material itself is made of grains of different sizes with the surface area dominated by the presence of clay particles (type I) or the material is made of silica grains, no clays, and the silica grains are assumed to be relatively well sorted (type II) (Figure 2a). I consider in both cases that the porous material is isotropic and homogeneous at the scale of a representative elementary volume (REV) and fully saturated by a pore water solution with hydrated ions. The mineral grains have some contiguity, which prevents a strong polarization of the diffuse layer. That said, the Stern layer remains discontinuous and therefore the distribution of grain size can represent a distribution of polarization length scales. In addition, for such dense grain assemblage,

the grain-to-grain interactions cannot be limited to a single dipole term. As a result, the distribution of relaxation times is expected to be very broad. The importance of the Stern layer in the polarization processes of porous media has been underlined by many authors [see Revil and Florsch, 2010; Grosse, 2011, and references therein]. The polarization of this layer (called also the stagnant layer in electrochemistry) is central to my model. The broad distribution of the polarization length scales implies weakly frequency-dependent phase lag and quadrature conductivity for type I materials (Figure 2b). At the opposite, for type II materials, a clear phase peak can be observed (Figure 2c). In the following, I am only interested in phase I materials.

[14] The second point is related to the upscaling approach used to obtain the macroscopic electrical conductivity at the scale of the REV. For the type of porous material I am dealing with (granular with a broad distribution of grain sizes; see Figure 2a), the solution of the DEM is effective for describing the macroscopic conductivity [e.g., Sen et al., 1981; Bussian, 1983; de Lima and Sharma, 1990, 1992; Revil et al., 1998]. Using the DEM, the complex conductivity can be described as a function of the conductivity of the pore water σ_f and the (complex) surface conductivity σ_s^* by [e.g., Sen et al., 1981]

$$\sigma^* = \frac{\sigma_f}{F} \left(\frac{1 - \sigma_s^* / \sigma_f}{1 - \sigma_s^* / \sigma^*} \right)^m, \quad (7)$$

$$F = \phi^{-m}, \quad (8)$$

where m is called the cementation or Archie exponent (dimensionless) [Archie, 1942], F is the electrical formation factor (dimensionless), ϕ is the connected porosity (dimensionless), σ_f (in S m^{-1}) denotes the conductivity of the pore water (real number), and $\sigma_s^*(\omega)$ denotes the (complex-valued) surface conductivity. Equation (7) is also known as the Bruggeman-Hanai equation [Bruggeman, 1935; Hanai, 1968]. Equation (8) is generally called Archie's law in the literature [Archie, 1942]. This deserves here three short comments: (1) First, there is no prefactor a in Archie's law as defined by Archie himself in 1942. (2) Archie's law *stricto sensu* is dealing with the linear correlation in a log-log plot between the formation factors and porosities for a collection of samples extracted from the same formation; this is not the case in equation (8) in which m is defined for a single material. (3) Equation (8) connects the inverse of the formation factor (which is exactly an effective porosity [see Revil and Cathles, 1999]) to the porosity considered to be a geometrical parameter. Therefore, m is perceived physically as a decoupling parameter between the total connected porosity and the fraction of this porosity dynamically connected by the electrical field lines.

[15] For shaly sands, the value of the cementation exponent can be predicted from the clay mineralogy and the clay content as discussed, for instance, by Revil et al. [1998, Figure 5]. The choice of the DEM is justified by the statistical analysis performed by Niwas et al. [2007] showing this model is much superior to the "generalized" Archie's law developed by Glover et al. [2000]. Also, as discussed by many authors [e.g., Sen and Goode, 1992, and references therein], the Waxman and Smits [1968] equation has an

Electrical double layer of clay minerals

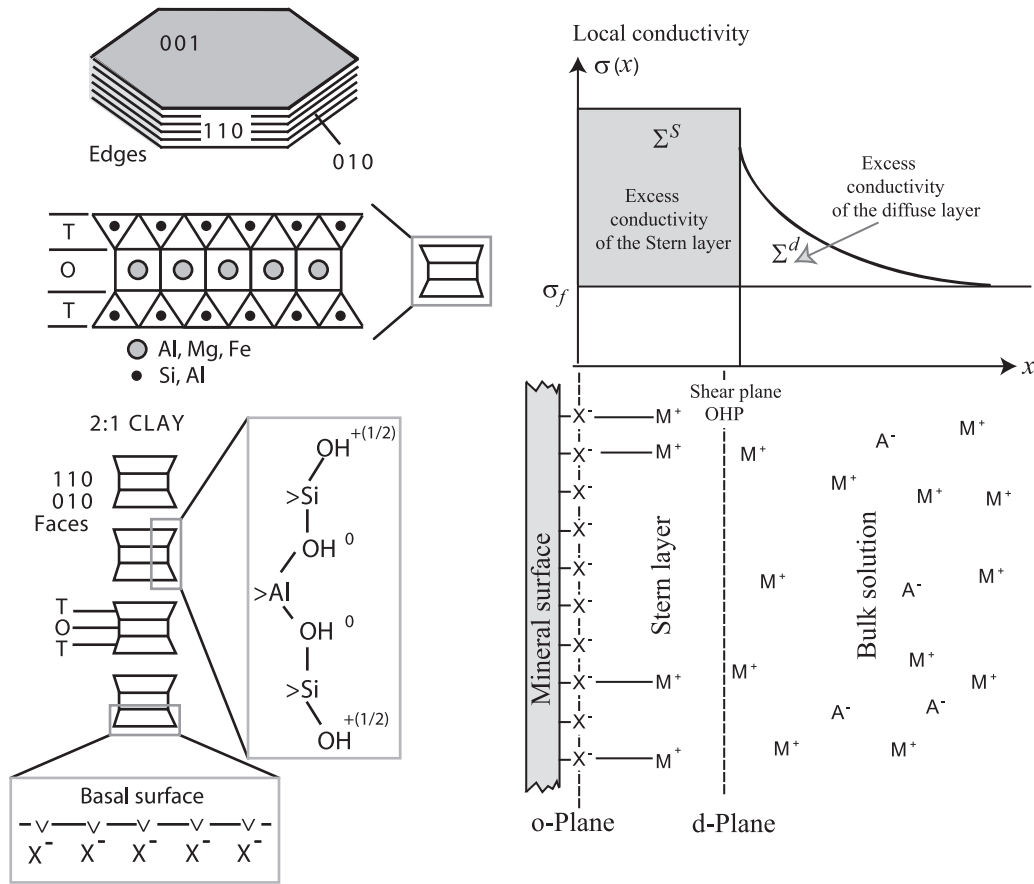


Figure 1. Electrical double layer at the surface of clays. The surface charge of the clays is heterogeneous and results from amphoteric sites located on the 110 and 010 crystalline planes, while the basal surfaces 001 carry negative sites associated with isomorphous substitutions in the crystalline framework (especially for illite and smectite). The electrical double layer is formed by the Stern layer of sorbed counterions (on the 110 and 010 planes) and the diffuse layer. The o plane defines the mineral surface, while the d plane defines the interface between the Stern layer and the diffuse layer on the 110 and 010 crystalline planes. The specific surface conductivities Σ_S and Σ_d define the excess surface conductivity associated with the Stern and diffuse layers with respect to the conductivity of the pore water.

unphysical dependence of the apparent mobility of the counterions with the salinity. *Revil et al.* [1998] performed a statistical analysis of the DEM versus the Waxman and Smits model and found that the DEM is also a better model to fit laboratory data avoiding such unphysical dependence for the mobility of the counterions.

2.1. In-Phase Conductivity

[16] *Revil et al.* [1998] used the DEM approach to find the in-phase component of the complex conductivity using an approximate analytical solution of the DEM at the scale of a REV. *Revil et al.* [2002] modified slightly this equation to account for the diffuse layer nature of surface conductivity at very low salinities. Their model yields the following expressions for the in-phase conductivity:

$$\sigma' = \sigma_f Du, Du \geq 1, \quad (9)$$

$$\sigma' = \frac{\sigma_f}{F} \left[F Du + \frac{1}{2} (1 - Du) \left(1 - Du + \sqrt{(1 - Du)^2 + 4 F Du} \right) \right], \quad Du \leq 1, \quad (10)$$

where Du is a dimensionless number called the Dukhin number [e.g., *Bolève et al.*, 2007]. This Dukhin number represents the ratio between the in-phase surface conductivity of the grains to the electrical conductivity of the pore water [e.g., *Dukhin and Shilov*, 2002]:

$$Du \equiv \frac{\sigma'_S}{\sigma_f}. \quad (11)$$

The approach of *Revil et al.* [1998] is slightly more complicated than equations (9) and (10) above because it accounts

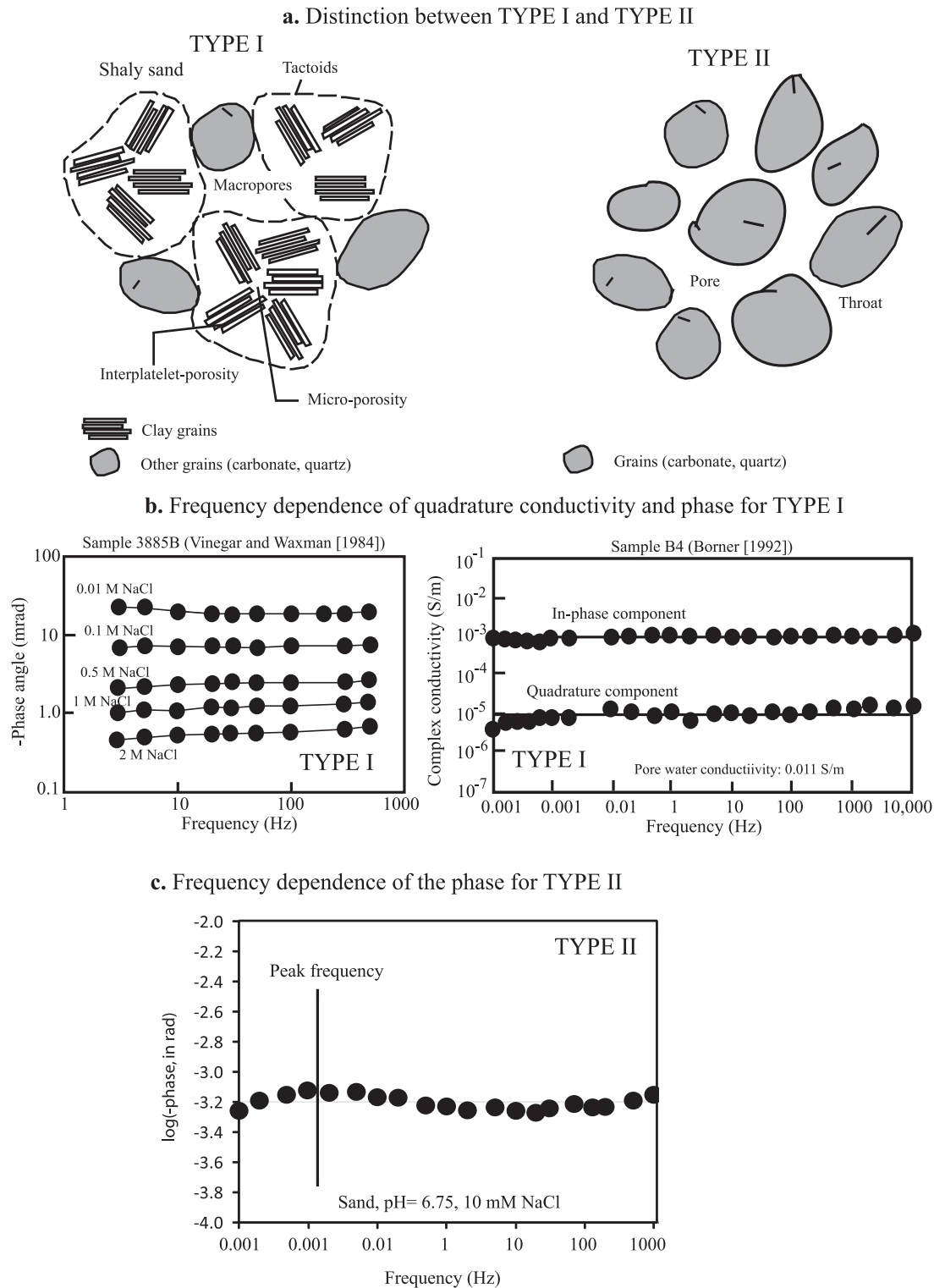


Figure 2. Shaly sand. (a) Microstructure with the distinction between media characterized by a broad range of polarization length scales like clayey sands (type I) and clean sand (type II) for which the pore volume is characterized by a succession of pores and throats. (b) Example of phase spectra for a sample of the database of *Vinegar and Waxman* [1984] (left) and the in-phase and quadrature conductivity spectra for a sample of the database of *Börner* [1992] (right). In both cases, the weak dependence of the phase or quadrature conductivity is the result of the broad distribution of polarization length scales. (c) The phase for type II materials (clean sand) shows a peak frequency.

for the different tortuosities felt by the anions and the cations during their electromigration in the pore space and along the mineral surfaces. However the difference between the *Revil et al.* [1998] model and equations (9) and (10) is rather small and will be ignored below.

[17] At low and high salinities, the asymptotic behaviors of equations (9) and (10) are

$$\sigma' = \sigma'_S, \text{ Du} \gg 1, \quad (12)$$

$$\sigma' = \frac{1}{F} [\sigma_f + m(F-1)\sigma'_S], \text{ Du} \ll 1, \quad (13)$$

respectively. Equation (13) is obtained from equation (7) by using binomial expansion [see *de Lima and Sharma*, 1992] and if equation (10) is used instead of equation (7), a similar expression is found with a value of cementation exponent $m = 2$ in the surface conductivity term [see *Niwas et al.*, 2007].

[18] I need now to express the surface conductivity term in equations (12) and (13) as a function of the mobility and density of the counterions. The counterions are defined as being the ions of opposite sign to the sign of the charge of the mineral surface. They are populating the electrical double layer to shield the charge of the clay particles. The counterions are located both in the Stern layer and in the diffuse layer (Figure 1). Triple-layer models (TLMs) have been developed to describe the charging properties of clays. The TLM of *Leroy and Revil* [2009] consider that the Stern layer is essentially associated with the edge of the clay particles. I will follow this idea below. I use f ($0 \leq f \leq 1$) to denote the relative fraction of counterions in the Stern layer ($f = 1$ if all the counterions are located in the Stern layer and $f = 0$ if all the counterions are located in the diffuse layer). This parameter (also called the partition coefficient hereinafter) will be of crucial importance to model the complex conductivity of shaly sands. The surface conductivity can be related to the excess charge density Q_V . This excess charge density represents the excess of electrical charges per unit pore volume of the porous material (in C m^{-3}). The relationship between the in-phase component of the surface conductivity and the excess charge density is [Kan and Sen, 1987; *Revil et al.*, 1998; *Brovelli and Cassiani*, 2011]

$$\sigma'_S = \frac{2}{3} \left(\frac{\phi}{1-\phi} \right) \beta_S Q_V, \quad (14)$$

where β_S represents an effective (or apparent) surface mobility for the counterions of the electrical double layer. As noted by *Vinegar and Waxman* [1984], the excess of charge per unit pore volume Q_V can be determined (1) intrusively through cation exchange capacity (CEC) measurements obtained by potentiometric titration methods [e.g., *Meier and Kahr*, 1999] or (2) nonintrusively through membrane potential measurements [Thomas, 1976]. The relationship between the excess of charge per unit pore volume and the CEC is given by

$$Q_V = \rho_g \left(\frac{1-\phi}{\phi} \right) \text{CEC}, \quad (15)$$

where ρ_g is the grain density (in kg m^{-3}) and the CEC denotes the cation exchange capacity (in meq g^{-1} or in C kg^{-1} in the international system of units) of the clay minerals (an arithmetic average can be used for a mixture of clay minerals or mixed layer clays [see *Meier and Kahr*, 1999, and references therein]). The CEC represents the amount of cations exchange sites on the mineral surface per unit mass of the minerals. This parameters is of crucial importance in many properties of porous media like defining sorption isotherms for clay-rich materials [Woodruff and *Revil*, 2011].

[19] Equation (14) does not account for the different contributions from the Stern and diffuse layers. To account for these contributions, the surface conductivity needs to be written explicitly as a function of the mobility of the counterions in the Stern and diffuse layers as well as the relative fractions f and $(1-f)$. This yields

$$\sigma'_S = \frac{2}{3} \left(\frac{\phi}{1-\phi} \right) (\beta_S^{(+)} f Q_V + \beta_{(+)} (1-f) Q_V), \quad (16)$$

where $\beta_{(+)}^S$ denotes the mobility of the counterions in the Stern layer and $\beta_{(+)}$ represents the mobility of the cation in the pore water. I will consider below that the pore water solution is an electrolyte of NaCl. Comparing equations (14) and (16), the effective (or apparent) surface mobility β_S is related to the mobility of the cations in the Stern and diffuse layers by

$$\beta_S = \beta_{(+)}^S f + \beta_{(+)} (1-f). \quad (17)$$

Revil et al. [1998] found that for NaCl solutions (the counterions are Na^+ for pH values in the range 5 to 7) $\beta_S = \beta_{(+)}^S / 10$. As the partition coefficient is $f \approx 0.9$ (as discussed in section 2.3 in more detail [see also *Leroy and Revil*, 2009; *Revil and Florsch*, 2010]), this means that

$$\beta_{(+)}^S \ll \beta_{(+)} \Rightarrow \begin{cases} \beta_S \approx \beta_{(+)} (1-f) \\ \sigma'_S \approx \frac{2}{3} \left(\frac{\phi}{1-\phi} \right) \beta_S Q_V \end{cases} \quad (18)$$

Therefore the mobility of the counterions in the Stern layer is expected to be much smaller than the mobility of the counterions in the diffuse layer. I investigate below how much smaller. This idea is consistent with the high density of counterions in the Stern layer of clay minerals. Indeed, this high density should lead to a smaller mobility of the counterions by comparison with its value in the bulk pore water. I will show in sections 2.2 and 2.3 that independent observations confirm this point of view and can be, at the same time, reconciled with previous modeling efforts.

[20] From equations (13) and (16), at high salinities, the in-phase component of the surface conductivity is written as

$$\sigma' = \frac{1}{F} [\sigma_f + A(\phi, m) \beta_S Q_V], \text{ Du} \ll 1, \quad (19)$$

$$A(\phi, m) = m(F-1) \frac{2}{3} \left(\frac{\phi}{1-\phi} \right), \quad (20)$$

where m is the same cementation exponent arising in Archie's law. Equation (19) is consistent with the high-salinity behavior described by *Sen and Goode* [1992]. Note at high porosities and for spherical grains ($m = 1.5$), the asymptotic behavior of equation (20) yields $A(\phi, m) \rightarrow m$ providing a rational for the *Waxman and Smits* [1968] equation, an approximation valid only at high salinities and high porosities.

[21] In section 3 I will use mainly the databases of *Vinegar and Waxman* [1984] and *Börner* [1992] to test my model. As discussed below, these databases are showing that the quadrature conductivity is independent of the frequency for a broad ranges of frequencies. As the conductivity data of *Vinegar and Waxman* [1984] are reported as a function of the salinity and temperature, I need to convert these data into pore water conductivities. For a NaCl solution, the conductivity of the pore water is given very accurately as a function of the salinity and temperature by the empirical formula obtained by *Sen and Goode* [1992]:

$$\sigma_f = (5.6 + 0.27T - 1.5 \cdot 10^{-4}T^2)C_f - \left(\frac{2.36 + 0.099T}{1.0 + 0.214C_f} \right) (C_f)^{\frac{3}{2}}, \quad (21)$$

where C_f is the salinity (expressed in Mol L⁻¹) and T the temperature (in °C).

2.2. Quadrature Conductivity

[22] As explained by *Vinegar and Waxman* [1984], the quadrature conductivity is controlled by two processes working in series, the weakest one dominates therefore the response (see Figure 2b). These two processes are namely the Stern layer polarization [see *Leroy et al.*, 2008] and the membrane polarization [Marshall and Madden, 1959; *Titov et al.*, 2002]. The rational for these two processes working in series is explained by *Vinegar and Waxman* [1984] and will be investigated in more detail in a future publication.

[23] The quadrature conductivity is dominated by the Stern layer contribution, which is generally much smaller than the membrane polarization except at very high salinities for which the membrane polarization is very small and therefore may control the overall response of the material [Vinegar and Waxman, 1984]. The second point is that the quadrature conductivity is frequency dependent but this frequency dependence is weak because of the very broad distribution of polarization length scales (see discussion by *Vinegar and Waxman* [1984] and their Figures 4 and 5). This implies that I can neglect the frequency dependence of the quadrature conductivity as a first-order approximation. This is not the case for well-sorted sands for which a clear relaxation peak is usually observed in the phase lag and in the quadrature conductivity. A complete theory with membrane polarization will be kept for a future work and I will focus below on the Stern layer polarization model. This limits however the application of my model to clayey sands and sandstones.

[24] The Stern polarization model is based on the assumption that the Stern layer is discontinuous between the grains. This assumption is consistent with transport models for ions through clayey material [Revil et al., 2011, and

references therein] and with compaction models by pressure solution for sandstones [Revil, 2001].

[25] For silica sands, neglecting the frequency dependence of the quadrature conductivity, the quadrature conductivity is related to the quadrature component of the surface conductivity by [Revil and Florsch, 2010]

$$\sigma'' = -\frac{F-1}{F} \sigma_s'' \approx -\sigma_s''. \quad (22)$$

The last term of equation (22) is obtained with $(F-1)/F \approx 1$ because $F \gg 1$ (typically $F > 10$ for shaly and clayey sandstones; see Tables 1 and 2). This assumption is however not crucial to the model and we can carry along the term $(F-1)/F$ further into the modeling if needed. *Revil and Florsch* [2010] developed the following formula for the quadrature conductivity of a clean sand with a single grain diameter d_0 (again neglecting the frequency-dependent term):

$$\sigma'' = -\frac{F-1}{F} \left(\frac{4\Sigma_s}{d_0} \right) \approx -\left(\frac{4\Sigma_s}{d_0} \right), \quad (23)$$

where Σ_s is the specific surface conductivity associated with the Stern layer (in S). If the grains have different grain diameters, the particle size distribution (PSD) is described by a (normalized) probability distribution $f(D)$ (D is the grain diameter). Equation (23) can be generalized to [Revil and Florsch, 2010]

$$\sigma'' = -4E_h \Sigma_s, \quad (24)$$

$$E_h = \int_0^{+\infty} f(D) d \ln D. \quad (25)$$

Therefore, the mean grain size needs to be replaced by the expectation of the moment of order (-1) of the PSD. I know from the conductivity theories developed by *Revil et al.* [1998] and *Revil and Florsch* [2010] that the following equivalence applies for the surface (interfacial) electrical conductivity [e.g., *Schmutz et al.*, 2010]:

$$\frac{2}{3} \left(\frac{\phi}{1-\phi} \right) \beta_{(+)}^s f Q_V \Longleftrightarrow 4E_h \Sigma_s. \quad (26)$$

There is therefore a bridge between expressing the surface conductivity as a function of the specific surface conductivity Σ_s (and the moment of order (-1) of the PSD) and as a function of the excess charge density per unit pore volume Q_V (or alternatively the CEC) (see *Brovelli and Cassiani* [2011] on the same topic).

[26] From equations (24) and (26), the quadrature conductivity model of *Revil and Florsch* [2010] can be written as

$$\sigma'' \approx -\frac{2}{3} \left(\frac{\phi}{1-\phi} \right) \beta_{(+)}^s f Q_V. \quad (27)$$

Like I have done above for the in-phase surface conductivity model with β_s , I introduce now an apparent

Table 1. Properties of the Sandstones From the *Vinegar and Waxman* [1984] Database

Sample	Mineralogy ^a	F^b	$\sigma'_s (\times 10^{-4} \text{ S m}^{-1})$	ϕ^d	$Q_V^e (\text{meq mL}^{-1})$	f_M^f	CEC ^g (C kg ⁻¹)	k^h (mD)
3477	K, C	15.1 ± 0.2	18.0 ± 2.0	0.201	0.026	0.92	237.74	833
3336A	K	23.1 ± 0.5	13.7 ± 2.2	0.205	0.042	0.96	393.65	71.7
3478	K, C	19.0 ± 0.6	12.6 ± 3.2	0.181	0.052	0.97	417.70	78.2
101	K, I	16.5 ± 0.3	26.2 ± 2.7	0.210	0.055	0.95	531.40	-
102	K, I	16.2 ± 0.4	7.40 ± 2.4	0.205	0.064	0.99	599.84	-
CZ10	C	18.0 ± 0.3	47.0 ± 3.0	0.233	0.070	0.93	772.91	116
3833A	C	13.9 ± 0.6	38.0 ± 8.0	0.224	0.110	0.96	1154.1	118
3126B	I, S	11.8 ± 0.3	64.0 ± 6.0	0.249	0.120	0.95	1446.1	468
3847A	I, C	48.4 ± 1.3	18.5 ± 1.9	0.126	0.144	0.97	754.56	0.66
3283A	K, C, I	19.5 ± 1.0	27.2 ± 7.0	0.186	0.150	0.98	1245.8	40
3885B	I, C	29.0 ± 1.5	32.0 ± 7.0	0.204	0.180	0.98	1676.7	4.95
3972E	S, M, I	21.9 ± 0.4	27.6 ± 2.4	0.187	0.185	0.98	1546.7	407
3258A	I	51.8 ± 3.0	62.0 ± 9.0	0.128	0.379	0.97	2022.1	0.61
3891A	C, I	42.1 ± 1.3	173 ± 9	0.192	0.385	0.94	3325.2	0.26
3308A	S, C	13.7 ± 0.7	320 ± 35	0.270	0.409	0.94	5498.4	5
3323F	S, C	15.1 ± 0.4	749 ± 48	0.288	0.690	0.92	10145	3.1
3324A	S, C	32.7 ± 1.1	670 ± 34	0.220	0.646	0.89	6622.6	1.48
3323E	S, C	15.8 ± 0.6	798 ± 62	0.281	0.690	0.91	9801.6	1.26
3324B	S, C	32.0 ± 1.0	650 ± 30	0.204	0.842	0.91	7843.3	1.14
3306F	S, C	50.0 ± 5.0	538 ± 95	0.171	0.950	0.92	7122.6	0.70

^aFrom *Vinegar and Waxman* [1984, Table 1]. K, kaolinite; C, chlorite; I, illite; S, smectite.^bFrom the fit of the in-phase conductivity data using the differential effective medium theory (DEM).^cFrom the fit of the in-phase conductivity data using the DEM.^dFrom *Vinegar and Waxman* [1984].^eFrom *Vinegar and Waxman* [1984].^fFrom the surface conductivity, porosity, and Q_V .^gCation exchange capacity. From Q_V and porosity assuming a mass density of 2650 kg m⁻³.^hMeasured brine permeability from *Vinegar and Waxman* [1984].

mobility, λ (in m² s⁻¹ V⁻¹, using the same notation as *Vinegar and Waxman* [1984]), for the surface quadrature conductivity as

$$\sigma'' \approx -\frac{2}{3} \left(\frac{\phi}{1-\phi} \right) \lambda Q_V \quad (28)$$

$$\lambda \equiv \beta_{(+)}^S f. \quad (29)$$

I need now to find an expression for the partition coefficient f , which is entering in both the surface conductivity and the

quadrature conductivity contributions. This is done in section 2.3.

2.3. A Model for the Partition Coefficient

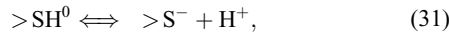
[27] As mentioned above, the last component of the model to investigate is the salinity dependence of the partition coefficient f . My goal is to develop a simplified, first-order solution of the TLM sorption problem of *Leroy and Revil* [2009]. I consider that the sorption of sodium in the Stern layer of kaolinite (illite and smectite are discussed below) and the dissociation of protons can be represented by the following reactions [e.g., *Leroy et al.*, 2007]

Table 2. Properties of the Sandstones From the *Börner* [1992] Database

Sample	Type ^a	F	$\sigma'_s (\times 10^{-4} \text{ S m}^{-1})$	ϕ	$S_{por} (\mu\text{m}^{-1})$	$S_{Sp} (10^3 \text{ m}^2 \text{ kg}^{-1})$	CEC ^c (C kg ⁻¹)	k (mD)
E6	Saxony	21.7 ^d	1.29e-2	0.1980	31.1	2.96	1125	-
E7	Saxony	11.5 ^d	1.70e-2	0.2762	13.2	1.95	741	157
E10	Saxony	17.0	1.85e-2	0.2238	41.6	4.64	1763	5.4
E12	Saxony	11.0	8.61e-3	0.2989	5.5	0.748	284	1490
E14	Saxony	11.1	9.58e-3	0.2403	10.7	1.31	498	1990
E17	Saxony	25.8 ^d	3.74e-3	0.1806	1.44	0.122	46	680
B2	Bentheim	15.6	1.57e-2	0.1943	2.15	0.198	74	1880
B4	Bentheim	11.7 ^b	7.63e-3	0.2315	2.22	0.261	99	-
R1	Altmark	150 ^e	3.96e-2	0.0463	43.5	0.809	307	-
R3	Altmark	45.1	1.07e-2	0.0970	24.7	1.01	384	0.52
F1	Fontainebleau	12.6	7.62e-3	0.2143	0.146	0.0153	6	-
F3	Fontainebleau	118	7.19e-4	0.0876	0.126	0.0036	1.4	138

^aThe Saxony sandstone contains quartz, feldspar, and kaolinite. The Bentheim sandstone contains silica (95%) and some kaolinite and dickite. The Altmark sandstone contains silica with a coating of smectite and illite and kaolinite and illite as cements. The Fontainebleau sandstone is nearly pure silica (99.8%).

^bUsing the porosity and $m = 1.68$ (from sample B2).^cUsing CEC = $Q_S S_{Sp}$, with $Q_S = 0.32 \text{ C m}^{-2}$.^dUsing the porosity and $m = 1.90$ (from the other Saxony samples).^eUsing the porosity and $m = 1.63$ (from sample R3).



where $>S$ refers the surface sites attached to the crystalline framework, H^0 are protons (which are assumed to be immobile) while weakly sorbed Na^+ are considered to be mobile in the Stern layer, eventually with a very small mobility. For illite and smectite, only equation (30) should be considered. I will consider the protons as immobile for clays but they seems to be mobile for silica as shown recently by *Skold and Revil* [2011]. Another component to consider is the charge associated with isomorphous substitutions in the crystalline framework. This involves mainly the substitution of Al^{3+} for Si^{4+} in the tetrahedral sheet of the clay particles and Mg^{2+} or Fe^{2+} for Al^{3+} in the octahedral sheet of the clay particles. These substitutions are important for illite and smectite but not for kaolinite. In agreement with the work of *Leroy and Revil* [2009], I will consider that the sites associated with the isomorphous substitutions (essentially the basal planes) are not associated with the formation of a Stern layer.

[28] To find a simplified sorption isotherm, I use the following equilibrium (constitutive) relationships corresponding to equations (30) and (31):

$$K_{Na} = \frac{\Gamma_{SNa}^0 [H^+]^0}{\Gamma_{SH}^0 [Na^+]^0}, \quad (32)$$

$$K_H = \frac{\Gamma_{S^-}^0 [H^+]^0}{\Gamma_{SH}^0}, \quad (33)$$

and a conservation equation for the surface species,

$$\Gamma_S^0 = \Gamma_{SNa}^0 + \Gamma_{SH}^0 + \Gamma_{S^-}^0 + \Gamma_X^0, \quad (34)$$

where Γ_S^0 denotes the (known) total surface site density (including the charge associated with isomorphous substitutions in the crystalline framework), Γ_{SNa}^0 , $\Gamma_{S^-}^0$, and Γ_{SH}^0 represent the surface charge density of the sites $>S^-Na^+$, $>S^-$, and $>SH^0$, respectively, and Γ_X^0 represents the number of equivalent sites corresponding to isomorphous substitutions (all expressed in sites m^{-2}). The value of the equilibrium constant is approximately $K_{Na} = 7.7 \times 10^{-2}$ [*Leroy et al.*, 2007], and the value of the value of K_H is between 10^{-10} for aluminol surface sites and 8×10^{-6} for silanol sites [*Leroy and Revil*, 2009]. The continuity equation (34) implies that the sites corresponding to the isomorphous substitutions are not attached with sorption in a Stern layer because they do not correspond to true surface sites. I simplify the notations using $pH = -\log_{10} [H^+]$ and $[Na^+] = C_f$, the salinity. The resolution of equations (32) to (34) is straightforward and yields the following sorption isotherm:

$$\Gamma_{SNa}^0 = (\Gamma_S^0 - \Gamma_X^0) \frac{K_{Na} C_f}{10^{-pH} + K_{Na} C_f + K_H}. \quad (35)$$

The partition coefficient is defined by

$$f \equiv \frac{\Gamma_{SNa}^0}{\Gamma_{S^-}^0 + \Gamma_X^0 + \Gamma_{SNa}^0}. \quad (36)$$

This means that the partition coefficient is defined as the surface concentration of counterions in the Stern layer divided by the total surface concentration of counterions in both the diffuse and the Stern layer (the negative charges $\Gamma_{S^-}^0 + \Gamma_X^0$ are compensated in the diffuse layer). This yields

$$f = f_M \left(\frac{K_{Na} C_f}{10^{-pH} (1 - f_M) + K_{Na} C_f + K_H} \right), \quad (37)$$

where the maximum value of the partition coefficient f_M (reached at high salinity) is defined from equations (35) and (36) as

$$f_M = 1 - \frac{\Gamma_X^0}{\Gamma_S^0}. \quad (38)$$

For illite and smectite, the activity of the silanol and aluminol surface groups can be neglected and therefore K_H can be neglected in equation (37). The maximum partition coefficients are shown in Figure 3 from the work of *Leroy and Revil* [2009]: $f_M(\text{kaolinite}) = 0.98$ (and therefore $\Gamma_X^0 \approx 0.02$ from equation (38), a very small value as expected from crystallographic considerations), $f_M(\text{illite}) = 0.90$ (and therefore $\Gamma_X^0 = 0.1 \Gamma_S^0$), and $f_M(\text{smectite}) = 0.85$ (and therefore $\Gamma_X^0 = 0.15 \Gamma_S^0$, which is consistent with the values reported by *Tertre et al.* [2011]).

[29] From equation (29) and equations (37) and (38), the apparent mobility entering into the expression of the quadrature conductivity can be expressed as follows:

$$\lambda = \lambda_M \left(\frac{K_{Na} C_f}{10^{-pH} (1 - f_M) + K_{Na} C_f + K_H} \right), \quad (39)$$

$$\lambda_M = \beta_{(+)}^S f_M, \quad (40)$$

where λ_M denotes the maximum value of the apparent mobility reached at high salinity. Again for smectite and illite, K_H can be neglected in equation (39). Equation (39) provides an explicit dependence of the apparent mobility of the counterions with the salinity. At high salinity, the model predicts that the apparent mobility λ is independent of the salinity (except if I account for membrane polarization effects) and at low salinity, the apparent mobility increases with the salinity. Therefore the Stern polarization model implies that there is an increase of the polarization with the salinity because the density of counterions in the Stern layer increases with the salinity. I will show in section 3 that this prediction is consistent with the available experimental data.

3. Validation of the Model

[30] I test ϕ below several predictions of the model with respect to the dependence of the quadrature conductivity with the variables entering into the model. These variables

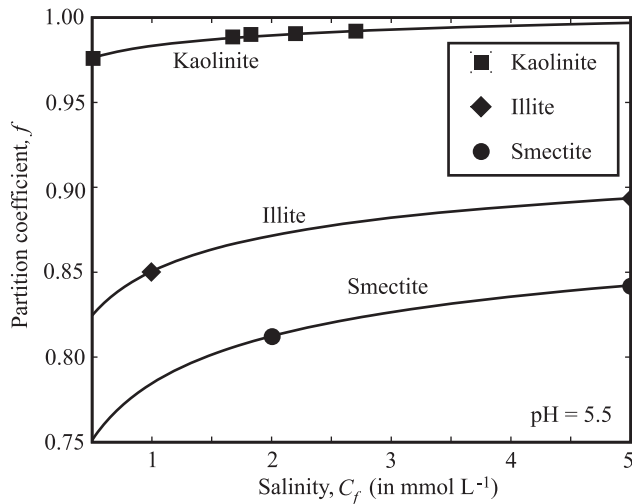


Figure 3. Partition coefficients for the three main types of clay minerals (chlorite has properties very similar to kaolinite). The lines correspond to the prediction of the triple-layer model (TLM) calculations [from Leroy and Revil, 2009] and the symbols correspond to estimates using the spectral induced polarization model of Leroy and Revil [2009] and complex conductivity measurements.

include textural variables (porosity, surface area), electrochemical variables (CEC or charge per unit surface area), and environmental variables (salinity and temperature). I use essentially two databases to test the model: one from Vinegar and Waxman [1984] (measured at 30 Hz and 25°C) and the second from Börner [1992] (measured at 1 Hz and room temperature) because of the availability of all the data. These databases corresponds to clayey sands and sandstones.

3.1. Quadrature Conductivity Versus Excess Charge Density

[31] The salinity dependence of the in-phase conductivity is analyzed in Figure 4. As all the data are above the iso-conductivity point (corresponding to $Du = 1$), I first use equation (10) and I minimize the data misfit (RMS) error between the measured (in-phase) conductivity and the in-phase conductivity predicted of the model. This first step yields the value of the formation factor F and the value of the in-phase surface conductivity σ'_s reported in Figure 4a for sample 3847A from Vinegar and Waxman [1984] (see Table 1) and in Figure 4b for sample B2 from Börner [1992] (see Table 2).

[32] The second step of the analysis is to compute the partition coefficient f_M using equation (18) and the value of the in-phase surface conductivity. For sample 3847A, using the value of the connected porosity ($\phi = 0.126$) and the value of the excess of charge per unit pore volume ($Q_V = 0.144 \text{ meq mL}^{-1} = 13.9 \times 10^6 \text{ C m}^{-3}$), and using $\beta_{(+)}(\text{Na}^+, 25^\circ\text{C}) = 5.2 \times 10^{-8} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$ and the value of the in-phase surface conductivity from the first step, I obtain the following value for the partition coefficient $f_M = 0.97$ using the value of the surface conductivity. For sample B2, the same analysis yields $f_M = 0.986$.

[33] The third step is to predict the value of the quadrature conductivity, neglecting at first its salinity dependence, which will be investigated only later on in this paper (see Figure 4b). So I use equations (28) and (29) with $f = f_M$. For sample 3847A, using $f_M = 0.97$ and the value of the mobility of the counterions in the Stern layer $\beta_{(+)}^S(\text{Na}^+, 25^\circ\text{C}) = 1.5 \times 10^{-10} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$, I obtain $\sigma'' = 1.94 \times 10^{-4} \text{ S m}^{-1}$, which is in good agreement with the measured values (see Figure 4a). For sample B2, using $f_M = 0.986$, $\beta_{(+)}^S(\text{Na}^+, 25^\circ\text{C}) = 1.5 \times 10^{-10} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$ at 25°C , I obtain $\sigma'' = 1.93 \times 10^{-5} \text{ S m}^{-1}$, also in good agreement with the measured values. The salinity dependence of the quadrature conductivity will be investigated in section 3.8.

[34] I use equation (10) to fit all the in-phase conductivity data reported by Vinegar and Waxman [1984] in order to determine the value of the (intrinsic) formation factor F and the value of the mean surface conductivity σ_s for each sample. These values are reported in Table 1 for all the samples of this database. Figure 5 shows the formation factors determined from the DEM model versus the formation factors given by Vinegar and Waxman [1984] for their database using the Waxman and Smits [1968] (WS) linear model. Both estimates are in fairly good agreement with each other, the formation factor estimated from the DEM being slightly smaller than the estimates provided by the WS model.

[35] The surface conductivities reported in Table 1 are shown as a function of the excess of charge density Q_V in Figure 6. According to equations (16) and (18), the slope of the trend is given by $s = (2/3)\beta_{(+)}(1 - f) = (2.8 \pm 0.2) \times 10^{-9} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$. Using the mobility of sodium in the pore water, this yields $f_M = 0.92$ as an average value for the partition coefficient for the complete database. Using a broader database, Revil *et al.* [1998] obtained $s = (5.1 \pm 0.2) \times 10^{-9} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$. This yields $f_M = 0.85$ as an average value for the partition coefficient. Usually these estimates are biased by the samples with the larger CEC (usually with smectite) and they are therefore more representative of smectite than of kaolinite. These estimates are therefore consistent with those reported in Figure 3 for the partition coefficient.

[36] I can also compute the partition coefficient f_M for each sample using

$$f_M = 1 - \frac{3}{2} \left(\frac{1 - \phi}{\phi} \right) \left(\frac{\sigma'_s}{\beta_{(+)} Q_V} \right), \quad (41)$$

where σ'_s denotes the surface conductivity determined from the in-phase conductivity data and ϕ and Q_V are independently measured in the database of Vinegar and Waxman [1984]. The computed partition coefficient are reported as a function of the CEC in Figure 7. I observe that small values of the CEC are associated with large values of $f_M > 0.95$, typical of kaolinite) while low values of f_M (in the range 0.87–0.95) are associated with high values of the CEC, typical of illite and especially smectite. This is again in agreement with the trends shown in Figure 3. These data can be fitted by $f_M = f_M(K) - 9.0 \times 10^{-6} \text{ CEC}$ where the CEC is expressed in C kg^{-1} and $f_M(K) = 0.98$ represents the value of the partition coefficient for kaolinite.

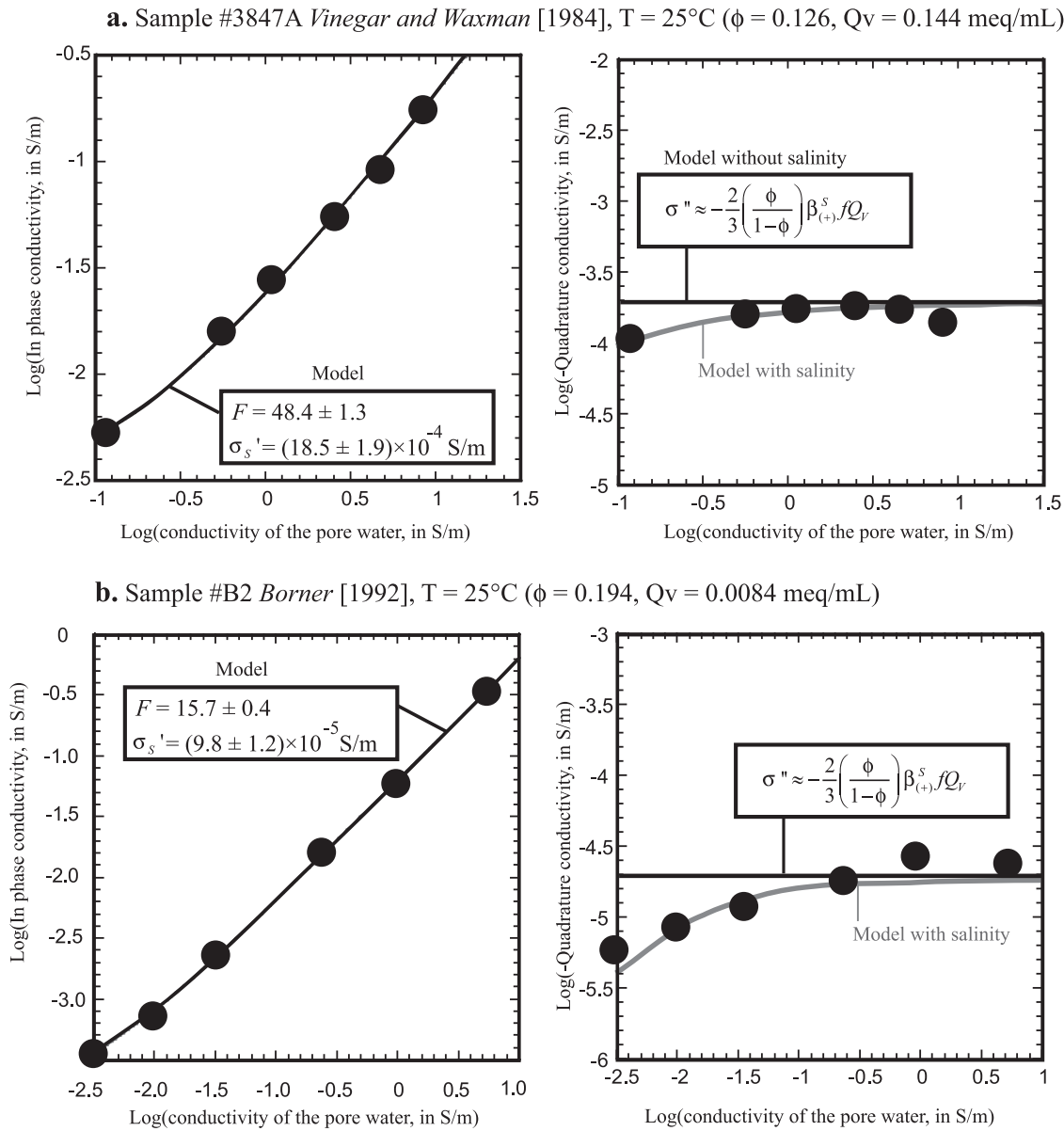


Figure 4. (a) Fit of the experimental data for a sample from the *Vinegar and Waxman* [1984] database. Note that the model captures well the nonlinear curvature of the in-phase conductivity data at low salinities. Note that all the data are above the isoconductivity point for which the conductivity of the sample is equal to the conductivity of the pore water and $D_u = 1$. In the right plot, the information gathered by fitting the in-phase conductivity data are used to predict the quadrature conductivity (at 30 Hz, 25°) without using, at this stage, the salinity dependence of the partition coefficient. There is no flush factor in the model. The errors bars are smaller than the size of the symbols. (b) The same for one core sample from the database of *Börner* [1992].

[37] In Figure 8, I use these values of the partition coefficient (reported in Table 1) to analyze the relationship between the surface conductivity and the quadrature conductivity. Such an analysis is important to see if I can predict the surface conductivity from the quadrature conductivity. When the surface conductivity is normalized with respect to the value of the partition coefficient, I find a high correlation between the quadrature conductivity and this normalized surface conductivity (Figure 8). Note that a good correlation is also obtained by taking an average

value of the partition coefficient (not shown here). That said, if one wants to apply this relationship to field data, a rough idea of the partition coefficient is required in the form $f_M/(1 - f_M)$ and the relationship determined above, $f_M = f_M(K) - 9.0 \times 10^{-6}$ CEC, can be used for that purpose.

[38] In Figure 9a, I plot the normalized quadrature conductivities $[(1 - \phi)/\phi]\sigma''$ as a function of the excess charge density Q_v at four distinct salinities (I did not choose the very high salinity 2 Mol L^{-1} because I suspect that

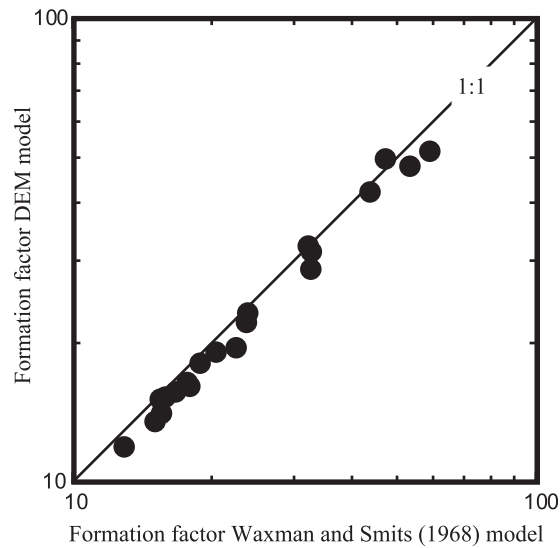


Figure 5. Comparison between the formation factor obtained from the fit of the data of *Vinegar and Waxman* [1984] using the differential effective medium (DEM) model and using the *Waxman and Smits* [1968] (WS) model. Note that the formation factors obtained using the DEM model are usually slightly smaller than those obtained with the WS model. The uncertainties are inside the size of the symbols.

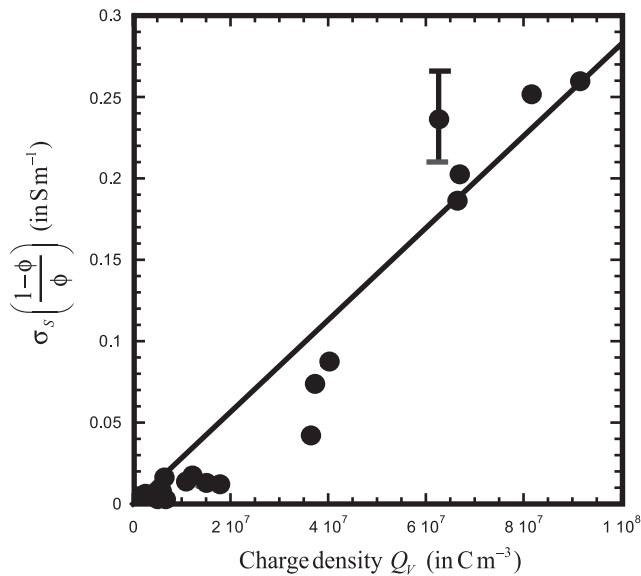


Figure 6. Comparison between the normalized (in-phase) surface conductivity (in S m⁻¹) and the excess charge density Q_v . The slope of the linear trend yields $s = (2/3)\beta_{(+)}(1-f) = (2.82 \pm 0.15) \times 10^{-9} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$. The uncertainty given on one data point is representative of the data set and is determined from the uncertainty obtained from the RMS fit of the conductivity data using the DEM model. The scatter is coming from the mineralogical composition, which influences the values of the partition coefficient f . The error bar provides an idea of the order of magnitude of the uncertainties in the surface conductivity.

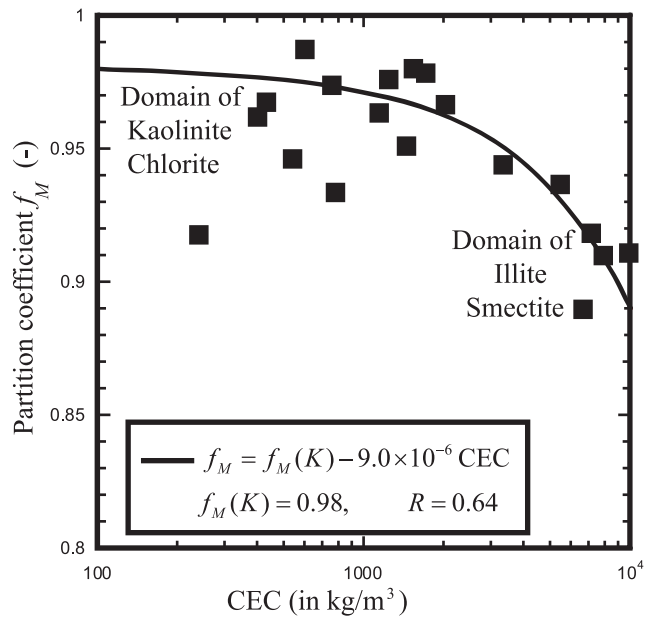


Figure 7. Value of the partition coefficient determined for each sample (from the surface conductivity) versus the cation exchange capacity (CEC) of the sample for the database of *Vinegar and Waxman* [1984]. At low CEC values, the partition coefficient is representative of kaolinite, while at high CEC the value of the partition coefficient is representative of illite and smectite. K stands for kaolinite. The main clay minerals of each sample are reported in Table 1.

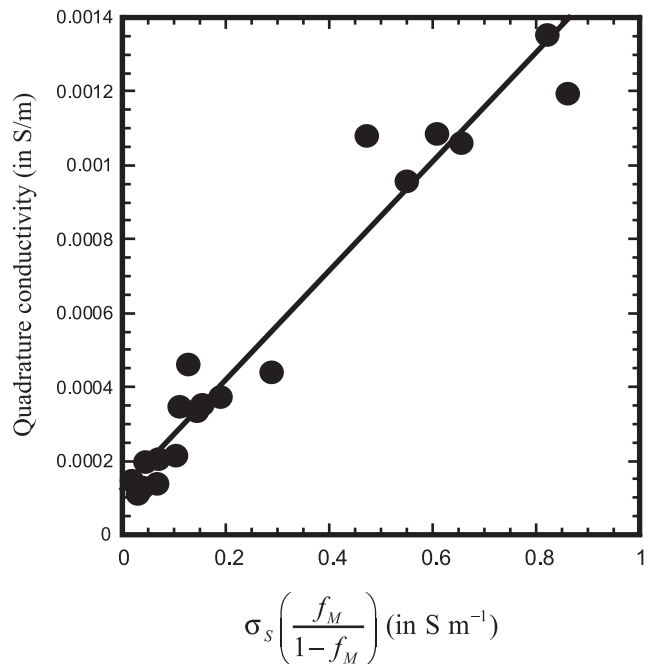


Figure 8. Quadrature conductivity versus scaled surface conductivity using the values of the partition coefficient reported in Table 1 for each sample. The slope of the linear trend ($R = 0.974$) is given by $s = \beta_{(+)}^S / \beta_{(+)} = 0.00148$ (dimensionless). This yields a counterion mobility in the Stern layer of $\beta_{(+)}^S = 0.8 \times 10^{-10} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$ at 25°C.

membrane polarization starts to be nonnegligible at this salinity). I see that the normalized quadrature conductivities and Q_V are highly correlated as predicted by the theory (see equation (27)) with a slope that is salinity dependent. At each salinity, I compute, for each sample, the apparent mobility λ according to

$$\lambda = -\frac{3}{2} \left(\frac{1-\phi}{\phi} \right) \frac{\sigma''}{Q_V}, \quad (42)$$

(from equation (28)). Then I compute the mean of λ for all the data, at each salinity, and I report these mean values of λ as a function of the salinity. The result is shown in Figure 9b. The result is fitted with equation (39). This fit provides the value of $\lambda_M = (1.32 \pm 0.03) \times 10^{-10} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$ and the value of the sorption coefficient $K_{\text{Na}} = (3.0 \pm 0.4) \times 10^{-4}$ ($\log_{10} K_{\text{Na}} = -3.5$). The value of the sorption coefficient is lower than the value proposed by Leroy *et al.* [2007] for smectite ($K_{\text{Na}} = 7.7 \times 10^{-2}$). This difference could be due to the mineralogical composition of the samples investigated by Vinegar and Waxman [1984], as shown in Table 1. The value of λ_M can be used to infer the value of the intrinsic mobility of the counterions in the Stern layer. Taking $f_M = 0.90$ yields $\beta_{(+)}^S (\text{Na}^+) = 1.5 \times 10^{-10} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$, so a counterion

mobility approximately 350 times smaller than in the bulk pore water for Na^+ .

3.2. Influence of the CEC

[39] Combining the expressions of the quadrature conductivity, equation (27), and the relationship between the excess charge density and the CEC, equation (15), I obtain a direct relationship between the quadrature conductivity and the CEC:

$$\sigma'' \approx -\frac{2}{3} \beta_{(+)}^S f \rho_g \text{CEC}. \quad (43)$$

In other words, the quadrature conductivity is a direct measure of the cation exchange capacity of the porous material, which has huge implications in agriculture as discussed in section 4. Taking $\beta_{(+)}^S (\text{Na}^+) = 1.5 \times 10^{-10} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$, $f = 0.9$, and $\rho_g = 2650 \text{ kg m}^{-3}$, I obtain $\sigma'' \approx -a \text{CEC}$ with $a \approx (2/3) \beta_{(+)}^S f \rho_g = 2.4 \times 10^{-7} \text{ S kg C}^{-1} \text{ m}^{-1}$. As shown in Figure 10a, the Stern layer polarization model is therefore able to explain the dependence of the quadrature conductivity with the cation exchange capacity of the material. In Figure 10b, I scale the values of the quadrature conductivity with the value of the partition coefficient f determined from

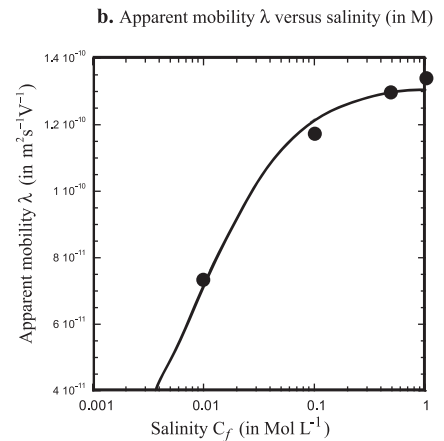
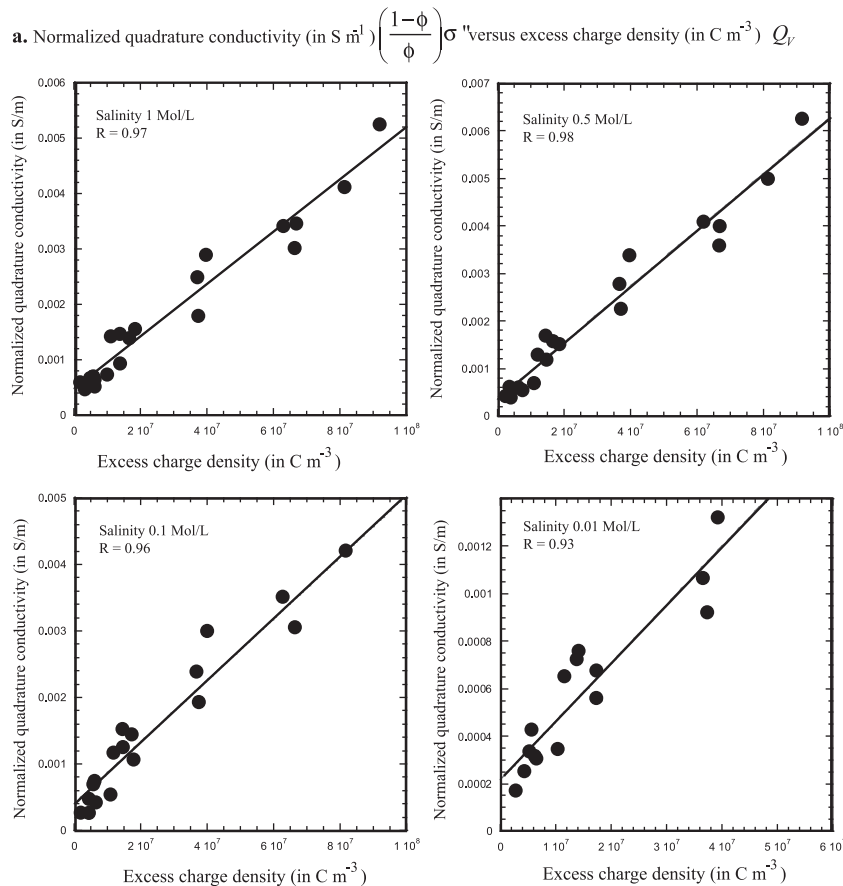


Figure 9. Quadrature conductivity versus the excess charge density and implication for the apparent mobility in the Stern layer. (a) Normalized quadrature conductivity versus excess charge density at four salinities. Note the excellent correlation for the linear trends. (b) Mean value of the apparent mobility obtained at each salinity versus the salinity of the pore water NaCl solutions. These data are fitted with the sorption isotherm described in section 2.3.

Quadrature conductivity versus CEC

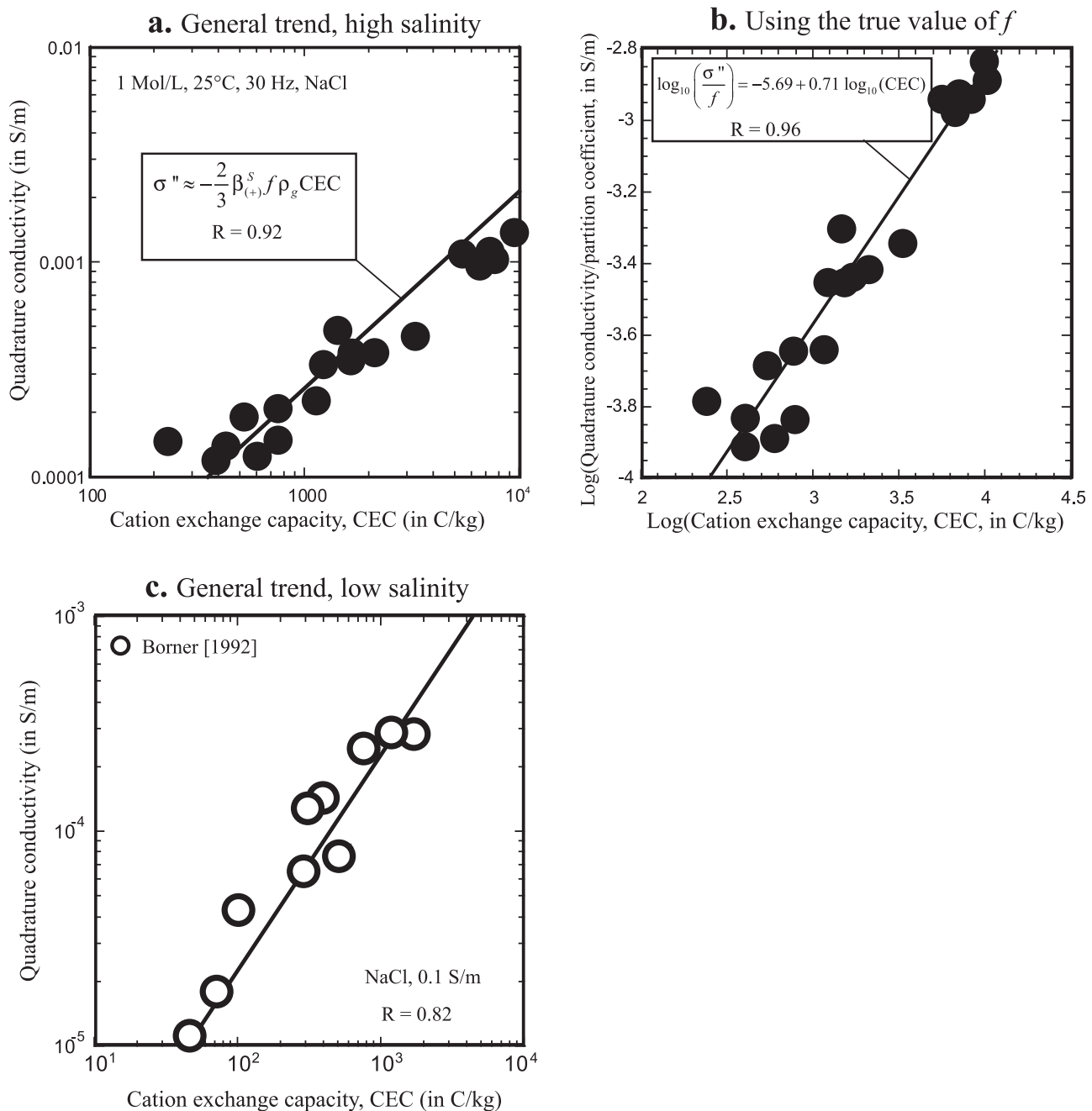


Figure 10. Relationship between the quadrature conductivity and the CEC (expressed in C kg^{-1}). (a and b) Data from the *Vinegar and Waxman* [1984] database. The model developed in the main text predicts a linear dependence between the quadrature conductivity and the CEC (lines), in agreement with the data. (c) Data from *Börner* [1992]. The line corresponds to the predicted response. The CEC is computed from the measured specific surface areas using $\text{CEC} = Q_s S_{sp}$, with the mean charge density for clay minerals Q_s being 2 elementary charges per nm^2 (0.32 C m^{-2}); see section 3.3.

the analysis of the in-phase conductivity. This improves the fit of the relationship between the scaled quadrature conductivity with the CEC. In Figure 10c, I used the database of *Börner* [1992] to test further the relationship between the quadrature conductivity and the CEC. This database does not contain CEC measurements, but the specific surface areas of

the core samples were measured. I use the relationship between the surface area and the CEC discussed below in section 3.3, equation (44), to convert the specific surface areas into CEC (see Table 2). Figure 10c is showing a strong correlation between the quadrature conductivity and the CEC. The slope of the trend is $(2.1 \pm 0.2) \times 10^{-7} \text{ S kg m}^{-1} \text{ C}^{-1}$, very

close to the slope given above ($2.4 \times 10^{-7} \text{ S kg C}^{-1} \text{ m}^{-1}$). Therefore, the model developed in this paper can clearly explain two published databases. The proportionality of the quadrature conductivity with the CEC is also supported by the data of *Garrouch and Sharma* [1994] (see their low-frequency dielectric dependence with the CEC, their Figures 3 and 4, for which one decade in the values of the CEC corresponds to one decade in the values of the apparent dielectric constant).

3.3. Influence of the Surface Area

[40] The charge density per unit surface area (in C m^{-2}) is defined as the ratio of the CEC by the specific surface area of the material S_{sp} (in $\text{m}^2 \text{ kg}^{-1}$)

$$Q_s = \frac{\text{CEC}}{S_{sp}}. \quad (44)$$

From Figure 11, the mean charge density for clay minerals is 2 elementary charges per nm^2 (0.32 C m^{-2} [see *Woodruff and Revil*, 2011]). Therefore the quadrature conductivity is also a measure of the specific surface area (in kg m^{-2}) of the material:

$$\sigma'' \approx -\frac{2}{3} \beta_{(+)}^s f \rho_g Q_s S_{sp}. \quad (45)$$

Taking $\beta_{(+)}^s (\text{Na}^+) = 1.5 \times 10^{-10} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$ at 25°C , $f = 0.90$, $Q_s = 0.32 \text{ C m}^{-2}$, $\rho_g = 2650 \text{ kg m}^{-3}$, we can write

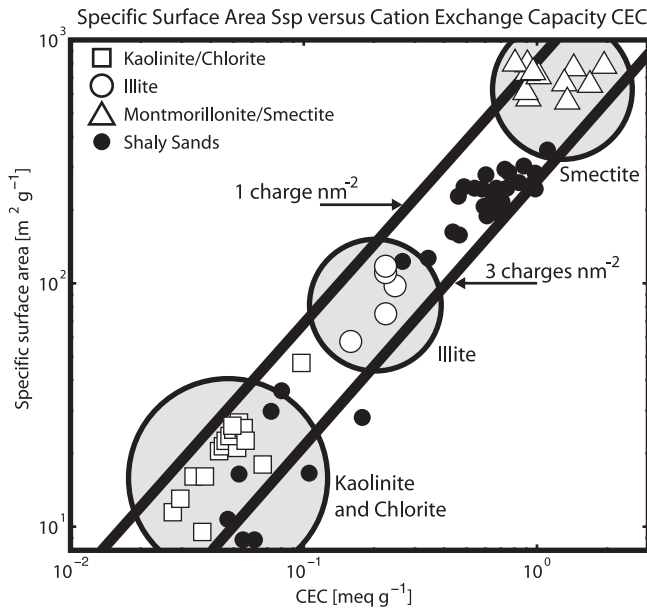


Figure 11. Specific surface area of clay minerals (in $\text{m}^2 \text{ g}^{-1}$) as a function of the CEC (in meq g^{-1} , where $1 \text{ meq g}^{-1} = 96,320 \text{ C kg}^{-1}$ in SI units) for various clay minerals. The ratio between the CEC and the specific surface area gives the equivalent total surface charge density of the mineral surface. Generalized regions corresponding to kaolinite, illite, and smectite are represented by shaded circles. Figure adapted from *Revil and Leroy* [2004]. The two lines correspond to 1–3 elementary charges per nm^2 . The average charge density is therefore 2 ± 1 elementary charges per nm^2 .

$\sigma'' \approx -a S_{sp}$ with $a = 7.6 \times 10^{-8} \text{ S kg m}^{-3}$. In Figure 12, I show the correlation between the quadrature conductivity and the specific surface area for the sandstones core samples investigated by *Börner* [1992]. There is a high correlation between the two parameters while the porosity of the core samples is inside a broad range (see Table 2). I find $\sigma'' \approx -a S_{sp}$ with $a = (8.0 \pm 0.9) \times 10^{-8} \text{ S kg m}^{-3}$. In other words, there is again a good agreement between the theory and the experimental data using this specific distinct data set.

[41] *Weller et al.* [2010] observed a linear relationship between the quadrature conductivity and the specific surface per unit pore volume at 1 Hz for 114 samples of sandstones and mix of sands and clays. The specific surface per unit pore volume can be related to the charge density Q_V and the surface charge density Q_s by

$$S_{por} = \frac{S}{V_p} = \frac{Q_V}{Q_s} \quad (46)$$

$$S_{por} = \rho_g \left(\frac{1 - \phi}{\phi} \right) S_{sp}. \quad (47)$$

Therefore the quadrature conductivity is given by

$$\sigma'' \approx -\frac{2}{3} \left(\frac{\phi}{1 - \phi} \right) \beta_{(+)}^s f Q_s S_{por}. \quad (48)$$

Clearly this relationship depends also on the porosity and the partition coefficient f , which explains the scatter in the data reported by *Weller et al.* [2010, Figure 2]. Taking average values of $\beta_{(+)}^s (\text{Na}^+) = 1.5 \times 10^{-10} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$, $\phi = 0.30$, $Q_s = 0.32 \text{ C m}^{-2}$, and $f = 0.90$, I obtain $\sigma'' \approx -c_p S_{por}$ with $c_p = 12.3 \times 10^{-12} \text{ S}$ with S_{por} expressed in m^{-1} and σ'' expressed in S m^{-1} . *Weller et al.* [2010, Table 1 and Figure 2] obtained $c_p = 9.9 \times 10^{-12} \text{ S}$. To my knowledge, this is the first time that a model can explain the dependence of the quadrature conductivity of shaly sands with the specific surface per unit pore volume. That said, the relationship between the quadrature conductivity and the specific surface area should have less scatter because it does not depend on the porosity (compare equations (45) and (48)). Indeed, from equation (47), it is clear that S_{por} play a role similar to Q_V in equation (15) while S_{sp} plays a role similar to the CEC in equation (15). Figure 12, using again the data set of *Börner* [1992] comprising core samples with a broad porosity range (see Table 2), shows that there is a higher correlation between σ'' and S_{sp} than between σ'' and S_{por} . A similar observation was reported by *Slater and Glaser* [2003].

3.4. Critical Phase Angle at Very Low Salinities

[42] The model of *Vinegar and Waxman* [1984] predicts the existence of a limiting value of the phase lag at very low salinities. According to my model, at very low salinities ($Du \gg 1$), the in-phase conductivity is given by $\sigma' \approx \sigma'_s$ and therefore the phase lag reaches indeed a critical value given by

$$\lim_{Du \gg 1} \varphi = -\frac{\beta_{(+)}^s f}{\beta_{(+)}(1 - f)}. \quad (49)$$

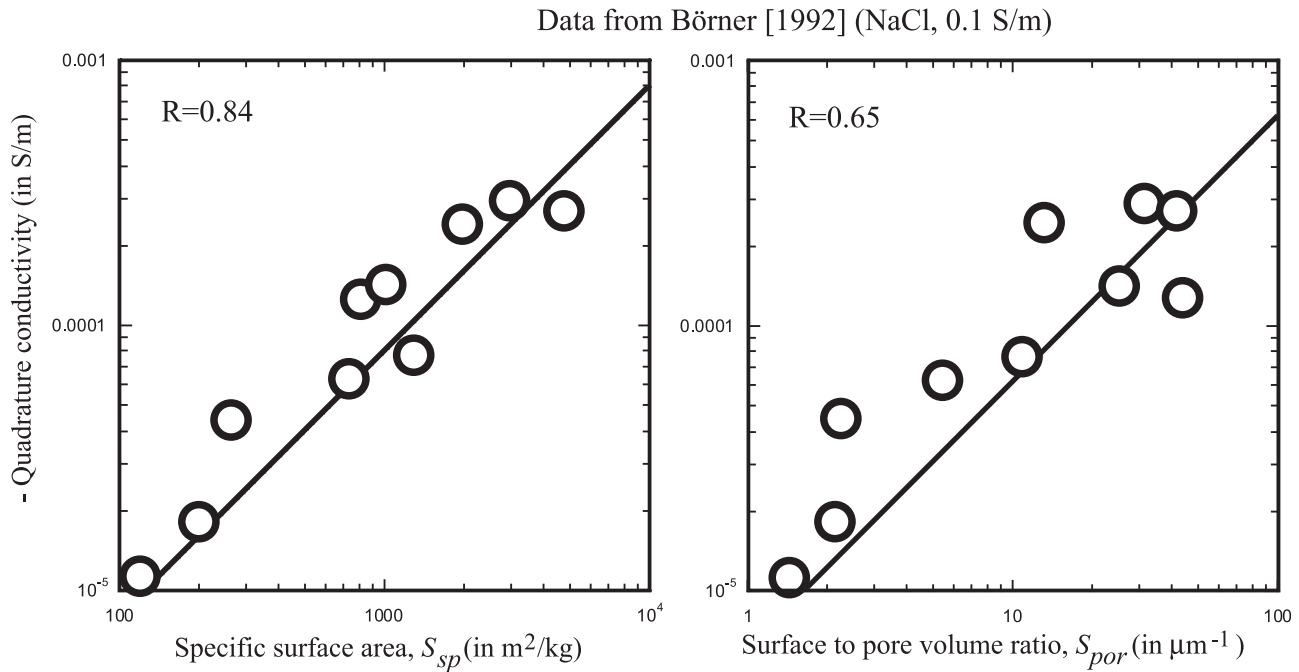


Figure 12. Quadrature conductivity (left) versus specific surface area and (right) versus the surface to pore volume ratio. The solid lines correspond to the linear relationships $\sigma'' \approx -a S_{sp}$ and $\sigma'' \approx -a S_{por}$, respectively, as discussed in the main text. Note that the correlation between the quadrature conductivity is much better than with the surface to pore volume ratio, in agreement with the prediction of the model.

This critical value is independent of the microstructure of the porous material. *Vinegar and Waxman* [1984] analyzed the experimental values obtained by *Fraser* [1964] at 10^{-4} N NaCl to show that this critical value is on the order of -30 mrad. Taking $\beta_{(+)}^S(\text{Na}^+) = 1.5 \times 10^{-10} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$ and $\beta_{(+)}(\text{Na}^+) = 5.2 \times 10^{-8} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$, this yields $f = 0.91$, a value consistent with the estimates given above. The existence of a maximum in the value of the phase lag on the order of -30 mrad is also shown in the values of the phase lag obtained by *Börner* [1992] (see Figure 13). For kaolinite samples, taking $f = 0.98$, the phase lag can reach higher values close to 140 mrad.

3.5. Influence of Temperature

[43] The temperature dependence of the mobility of sodium in the pore water and the apparent surface mobility entering in the surface conductivity term of the in-phase conductivity are given by [*Sen and Goode, 1992; Revil et al., 1998*]

$$\beta_{(+)}(T) = \beta_{(+)}(T_0)[1 + \alpha_f(T - T_0)], \quad (50)$$

$$\beta_S(T) = \beta_S(T_0)[1 + \alpha_S(T - T_0)], \quad (51)$$

with $T_0 = 25^\circ\text{C}$ and $\alpha_f = 0.019/^\circ\text{C}$ [*Sen and Goode, 1992; Binley et al., 2010*] and α_S represent the temperature sensitivity coefficient (in $^\circ\text{C}$) for the mobility in the bulk pore water and in the diffuse layer [*Revil et al., 1998*], respectively. As f_M is temperature independent (because it is associated only with two crystalline parameters), it is expected that $\alpha_f \approx \alpha_S$ because the viscosity in the bulk pore water

and in the diffuse layer are the same. At very low salinity, f could be temperature dependent and therefore the two temperature sensitivity coefficients could be different.

[44] The mobility of the counterions in the stern layer is written as

$$\beta_S^{(+)}(T) = \beta_S^{(+)}(T_0)[1 + \alpha_S^{(+)}(T - T_0)], \quad (52)$$

where $\alpha_S^{(+)}$ is unknown. At high salinity, the in-phase conductivity is dominated by the brine conductivity and therefore its temperature dependence is controlled by α_f . Therefore, the temperature dependence of the phase lag at high salinity is given by

$$\varphi(T) = \varphi(T_0) \left[\frac{1 + \alpha_f(T - T_0)}{1 + \alpha_S^{(+)}(T - T_0)} \right]. \quad (53)$$

At low salinity, the in-phase conductivity is dominated by the surface conductivity associated with the diffuse layer and therefore its temperature dependence is controlled by α_S . Therefore, the temperature dependence of the phase lag at high salinity is given by

$$\varphi(T) = \varphi(T_0) \left[\frac{1 + \alpha_S(T - T_0)}{1 + \alpha_S^{(+)}(T - T_0)} \right]. \quad (54)$$

A simple assumption is to consider that $\alpha_f = \alpha_S = \alpha_S^{(+)} = 0.023/^\circ\text{C}$ at high salinities to reduce the number of unknowns (this value is also slightly higher than provided above $\alpha_f = 0.019/^\circ\text{C}$). *Zisser et al.* [2010] found that the

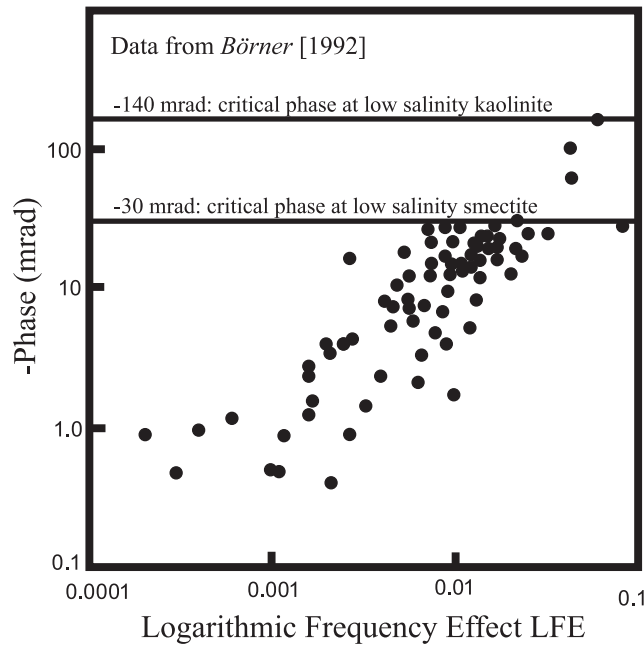


Figure 13. Measured phase lag for the sandstones investigated by Börner [1992]. Note that most of the measurements are below the critical phase lag value predicted by the model developed in this paper (approximately -30 mrad at 25°C for smectite and -140 mrad for kaolinite). The logarithmic frequency effect (LFE) is defined as $\text{LFE}_{12} = [\log |\sigma^*(\omega_2)| - \log |\sigma^*(\omega_1)|] / [\log \omega_2 - \log \omega_1]$.

temperature dependence of the quadrature conductivity of the Bentheimer sandstone (silica: 96%, feldspar: 3%, clay: 1%) is equal to $\alpha_S^{(+)} = 0.0254/^\circ\text{C}$ (in the temperature range 0 – 80°C , at 1 Hz, close to the relaxation peak), therefore close to the temperature dependence of the counterions in the pore water. Binley *et al.* [2010] found for 7 sandstones (Berea, Cottaer, and Sherwood) that the temperature dependence of the in-phase conductivity was $0.022 \pm 0.002/^\circ\text{C}$ and the temperature dependence of the quadrature conductivity was $\alpha_S^{(+)} = 0.023 \pm 0.002/^\circ\text{C}$. For the low-temperature range 0°C to 25°C , Hayley *et al.* [2007] found that $\alpha_f = \alpha_S = 0.020 \pm 0.002/^\circ\text{C}$ for the in-phase conductivity.

[45] With the assumption that $\alpha_f = \alpha_S = \alpha_S^{(+)}$, the phase lag is expected to be nearly temperature independent. This seems to be a good approximation at least for small temperature changes (see Figure 14, which is based on the data reported by Vinegar and Waxman [1984]).

3.6. Prediction of Permeability

[46] The determination of permeability from the complex conductivity is a topic of active research in geophysics [see Börner, 1992; Slater and Lesmes, 2002; Kemna *et al.*, 2005; Binley *et al.*, 2005; Börner, 2006; Hördt *et al.*, 2007; Slater, 2007]. I am only looking here for a simple scaling law in the present paper that is consistent with dimensional analysis (most empirical relationships in the literature do not satisfy dimensional analysis). The permeability scales as the function of the hydraulic radius square (R^2) divided by the formation factor F (Kozeny-Carman equation [e.g., Bernabé and Revil, 1995]) Using the scaling

of the hydraulic radius with the charge density Q_V ($R \propto 1/Q_V$), the scaling of the quadrature conductivity with the charge density Q_V , and the scaling of the formation factor with the porosity (using a high-porosity asymptotic behavior), I propose the following scaling law between the permeability, the formation factor, and the quadrature conductivity:

$$k = \frac{1}{a} \frac{1}{F^3 (\sigma'')^2}, \quad (55)$$

where a is a universal constant. If I consider σ'' expressed in S m^{-1} , the permeability expressed in mD (approximately 10^{-15} m^2) then using $a = 120 \text{ S}^2 \text{ m}^{-2} \text{ mD}^{-1}$ allows to fit the permeability data of Vinegar and Waxman [1984] reported in Table 1 and those of Börner [1992] reported in Table 2 (see Figure 15). The predictive power of equation (55) is inside an order of magnitude and no more accuracy can be probably expected. Equation (55) is different from the equation proposed by Slater and Lesmes [2002] (the permeability appears as a power law function of only the quadrature conductivity) but is consistent with equation (79) of Revil and Florsch [2010].

3.7. Influence of Saturation

[47] I first assume that the presence of a second nonwetting phase does not change the cation exchange capacity or the surface area able to influence the quadrature conductivity. With this additional assumption, the influence of saturation is easy to take into account using the following transform in the expressions of the in-phase and quadrature conductivity [Vinegar and Waxman, 1984; Revil, 1999]:

$$\frac{1}{F} \Rightarrow \frac{1}{F} S_w^n, \quad (56)$$

$$Q_V \Rightarrow Q_V S_w^{-1}, \quad (57)$$

where S_w is the relative water saturation ($S_w = 1$ at full water saturation, $0 \leq S_w \leq 1$). Equation (57) is called the second Archie's law and n is called the saturation or second Archie exponent (in absence of information it is customary to assume that $n \approx m \approx 2$ [e.g., Waxman and Smits, 1968]). Because the term $[\phi/(1-\phi)]$, at high porosity is proportional to $1/F$, I apply the transform given by equation (56) to this term. Using equations (27), (56), and (57) yields

$$\sigma''(S_w) \approx -\frac{2}{3} \left(\frac{\phi}{1-\phi} \right) S_w^{n-1} \lambda Q_V, \quad (58)$$

$$\sigma''(S_w) \approx S_w^p \sigma''_{\text{Sat}}, \quad (59)$$

with $p = n - 1$, in agreement with the results of Vinegar and Waxman [1984]. We test in Table 3 the prediction of equation (55) with $p = n - 1$ for some sands and sandstones. The prediction of this simple model is shown in Figure 16 using the data reported in Table 3. Equation (59) seems in agreement with not only the results of Vinegar and Waxman [1984] but also those of Slater and Glaser [2003] and Schmutz *et al.* [2010]. That said, the previous

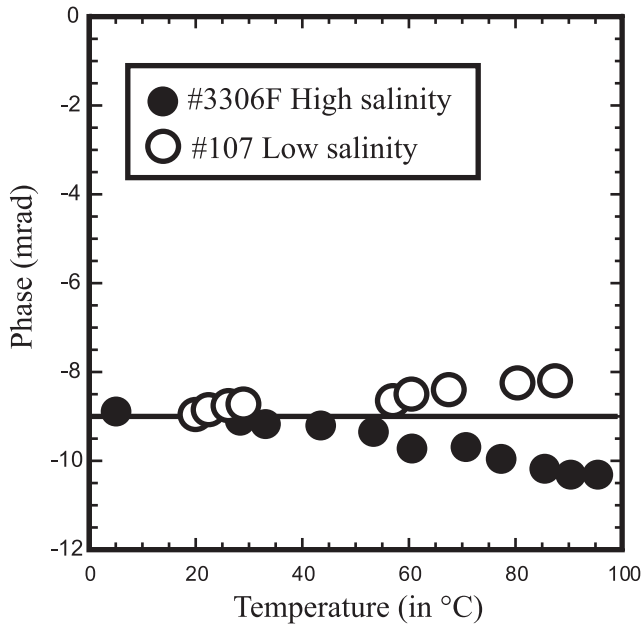


Figure 14. Temperature dependence of the phase lag for two shaly sandstones investigated by *Vinegar and Waxman* [1984]. At first approximation and for small temperature changes, the phase lag is temperature independent (line). This means in turn that the temperature dependence of the quadrature conductivity is roughly the same as the temperature dependence of the in-phase conductivity. The size of the symbols is on the order of the magnitude of the uncertainty.

model equation does not include the effect of the CEC attached to the interface between the wetting and the non-wetting phases and a more complete model needs to be developed to account for this contribution.

3.8. Tests of POLARIS

[48] For isothermal conditions, I test now the ability of POLARIS to predict the quadrature conductivity measurements reported by *Vinegar and Waxman* [1984]. The input parameters are the excess charge density Q_V , the salinity C_f , and the porosity ϕ . Nothing else is required. Indeed, combining equations (28) and (39) yields

$$\sigma'' \approx -\frac{2}{3} \lambda_M \left(\frac{\phi}{1-\phi} \right) \left(\frac{K_{Na} C_f}{10^{-\text{pH}}(1-f_M) + K_H + K_{Na} C_f} \right) Q_V. \quad (60)$$

Equation (60) provides an explicit dependence of the quadrature conductivity with the porosity, the salinity, and excess charge density for pH values close to neutral (5 to 8). At 25°C, to perform my comparison I use the model parameters determined in section 3.1, $\lambda_M = (1.32 \pm 0.03) \times 10^{-10} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$, $K_{Na} = (3.0 \pm 0.4) \times 10^{-4}$, $K_H = 8 \times 10^{-6}$, and pH = 5.6 (pH of water in equilibrium with the atmosphere, slightly acidic because of the solubility of the CO_2 in the pore water). The comparison between the predicted quadrature conductivity and the measured quadrature conductivity is shown in Figure 17. The model offers

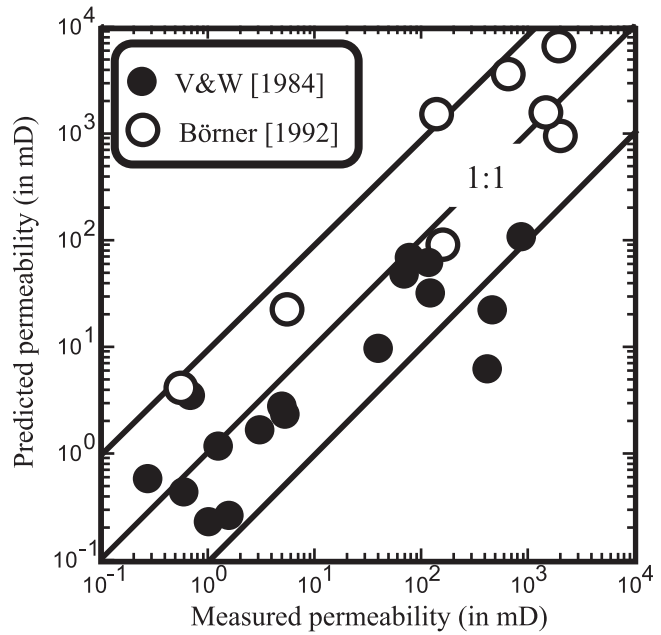


Figure 15. Predicted permeability (using the formation factor and the quadrature conductivity) versus the measured brine permeability (in mD, $1 \text{ mD} \approx 10^{-12} \text{ m}^2$) using the databases of *Vinegar and Waxman* [1984] and *Börner* [1992]. The data covers 5 orders of magnitude in permeability. The confidence interval corresponds to 1 order of magnitude ($R = 0.69$).

an excellent prediction of the data with very simple input parameters.

[49] Equation (60) is also tested with the data shown in Figure 4 to predict the salinity dependence of the quadrature conductivity. I used the measured values of the porosity and excess charge density Q_V and $\lambda_M = (1.32 \pm 0.03) \times 10^{-10} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$, $K_{Na} = (3.0 \pm 0.4) \times 10^{-4}$, $K_H = 8 \times 10^{-6}$, and pH = 5.6 for samples 3847A and B2. The model prediction works quite well for both samples.

[50] Equation (60) can be also written directly as a function of the CEC without using the porosity as an input parameter. This means that the CEC and the salinity are truly the two fundamental parameters needed to compute the quadrature conductivity, all the other parameters (pH, λ_M , K_{Na} , K_H) are known as discussed above. The quadrature conductivity can be also computed directly from the specific surface area. Using equations (46) and (47) in equation (60) yields

$$\sigma'' \approx -\frac{2}{3} \lambda_M \left(\frac{K_{Na} C_f}{10^{-\text{pH}} + K_{Na} C_f} \right) Q_S \rho_g S_{sp}. \quad (61)$$

In Figure 17 (right), I use $\lambda_M = (1.32 \pm 0.03) \times 10^{-10} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$, $K_{Na} = (3.0 \pm 0.4) \times 10^{-4}$, $K_H = 8 \times 10^{-6}$, pH = 5.6 (exactly like in the case above), $\rho_g = 2650 \text{ kg m}^{-3}$, and $Q_S = 0.8 \text{ C m}^2$ (5 charges nm^{-2}) to fit the data of *Börner* [1992]. The only input parameters are the salinity and the specific surface area, both independently measured by *Börner* [1992]. Again, there is a good agreement between the predictions of POLARIS and the experimental data.

Table 3. Prediction of the Exponent $p = n - 1$ for the Quadrature Conductivity as a Function of the Saturation Exponent n in the In-Phase Conductivity and Comparison With the Measured Value Using Quadrature Conductivity Data at Different Saturations of the Pore Water

Sample	Type ^a	Nonwetting Fluid	Measured n	Predicted p	Measured p
N2 55–60	sand (1)	air	1.3 ± 0.1	0.3 ± 0.1	0.6 ± 0.1
A and B	sand (2)	oil	2.1 ± 0.1	1.1 ± 0.1	1.3 ± 0.1
V104	clayey sand (3)	crude oil	2.17 ± 0.05	1.17 ± 0.05	1.18 ± 0.05
V106	clayey sand (3)	crude oil	2.43 ± 0.05	1.43 ± 0.05	1.28 ± 0.05
V110	clayey sand (3)	Soltrol	2.11 ± 0.05	1.11 ± 0.05	1.27 ± 0.05
V112	clayey sand (3)	Soltrol	1.63 ± 0.05	0.63 ± 0.05	0.78 ± 0.05
1883	clayey sand (3)	Soltrol	1.81 ± 0.05	0.81 ± 0.05	1.20 ± 0.05
3371	clayey sand (3)	Soltrol	2.40 ± 0.05	1.40 ± 0.05	1.17 ± 0.05
3372C	clayey sand (3)	Soltrol	2.35 ± 0.05	1.35 ± 0.05	1.16 ± 0.05
3374	clayey sand (3)	Soltrol	1.79 ± 0.05	0.79 ± 0.05	0.93 ± 0.05
3307C	clayey sand (3)	Soltrol	1.89 ± 0.05	0.89 ± 0.05	1.03 ± 0.05
5P-3A	sand (4)	heavy oil	1.67 ± 0.07	0.67 ± 0.07	0.59 ± 0.05
P-I-A	sand (4)	heavy oil	1.67 ± 0.07	0.67 ± 0.07	0.59 ± 0.05
Sample 1	sand (5)	air	1.38 ± 0.02	0.38 ± 0.02	0.61 ± 0.01

^aThe numbers in parentheses indicate the following: 1, unconsolidated sediments from the Olathe study site [from Slater and Glaser, 2003]; 2, unconsolidated sand [from Schmutz et al., 2010]; (3) clayey sands [from Vinegar and Waxman, 1984]; 4, tar sands (from Martinez, F. J., M. L. Batzle, and A. Revil, Influence of temperature on seismic velocities and complex conductivity of heavy oil bearing sands, submitted to *Geophysics*, 2012); 5, sand [Abdel-Aal et al., 2006].

4. Discussion

[51] A simple induced polarization model named POLARIS has been introduced to explain the dependence of the complex, frequency-independent, conductivity with the microstructure (porosity, specific surface area), interfacial properties (charge density or CEC) as well as with environmental variables (temperature and salinity). This is the first time that such a model is achieved keeping both scientific rigor and simplicity and it is therefore expected to be broadly applied, as discussed further below, to various problems.

[52] I discuss briefly two potential applications of POLARIS. This model can be used for mapping soil properties in agriculture (like their CEC because of the direct relationship between the quadrature conductivity and the CEC). Indeed, because of the broad range of polarization length scales for clayey soils, it is expected that the phase and the

quadrature conductivity are rather frequency independent. Mapping the phase lag can be done therefore at 10 Hz for instance. It is however possible that, in some circumstances, the Maxwell Wagner polarization [see Chen and Or, 2006] will have to be accounted for to interpret the phase lag measurements at this frequency. Additional information regarding the water content can be obtained from GPR for instance. In the case of clay-free soil, the mobility of the counterions in the Stern layer is expected to be the same that the mobility of the same ions in the pore water (like for silica). This would explain the high polarization observed in this type of soils (L. Slater, personal communication, 2011).

[53] The model described above can be applied to monitor salt tracer tests. Recently, Karaoulis et al. [2011] have developed a time-lapse spectral induced polarization inversion algorithm to monitor and image changes in the induced polarization parameters over time. This inversion algorithm can be combined with the model developed above to follow salt tracer tests at a given frequency (e.g., 1 Hz). This approach would be much more powerful than the traditional approach which is to use DC resistance tomography usually neglecting surface conductivity. A very innovative approach would be to use time-lapse inversion of both induced polarization and self-potential data because such geophysical measurements can be performed with the same apparatus and the physics of self-potential signals to follow salt tracer tests is now well understood [see Revil and Jardani, 2010].

5. Concluding Statements

[54] I have developed an analytical model called POLARIS for the quadrature conductivity on the basis of the solution of the differential effective medium theory. This model accounts for the two components of the electrical double layer coating the pore water mineral interface. The following conclusions have been reached.

[55] 1. The mobility of the counterions in the Stern layer is much lower (by a factor 350 for sodium) than the

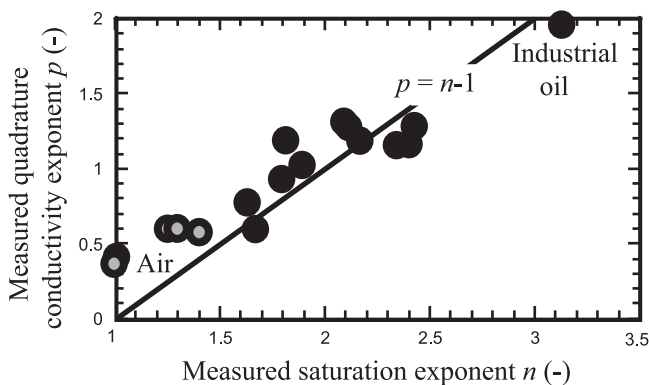


Figure 16. Predicted versus measured saturation exponent p for the quadrature conductivity. The predicted value is determined using the relationship $p = n - 1$, where n is the saturation exponent determined from the in-phase conductivity dependence on saturation. The uncertainty is roughly equal to the size of the symbols, and $R = 0.81$. References can be found in Table 3.

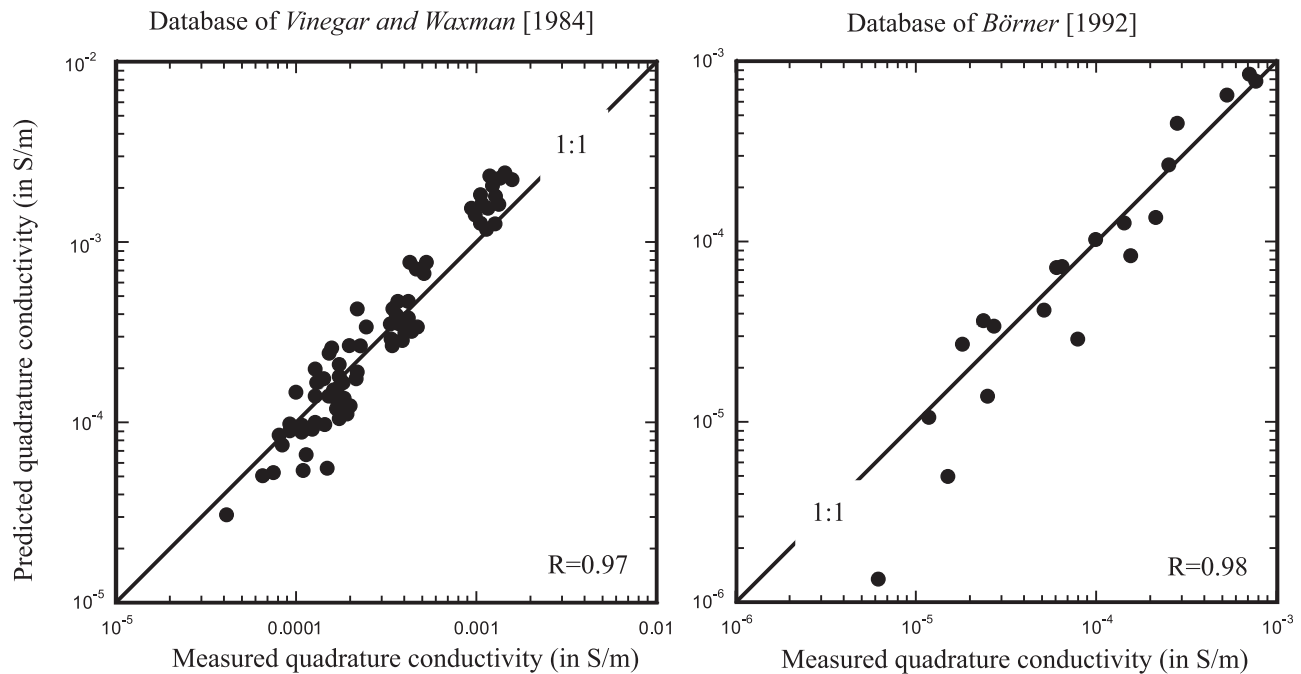


Figure 17. Predicted versus measured quadrature conductivity using, as input variables, the porosity, the excess charge density, and the salinity of the pore water for the quadrature conductivity data from Vinegar and Waxman [1984] and the specific surface area and the salinity for the data from Börner [1992] (samples E10, E12, and B2).

mobility of the counterions in the pore water in a clay material. This is in sharp contrast with the case of silica for which the mobility of the counterions in the Stern layer is the same than in the bulk pore water. This observations is explained by the fraction of counterions in the Stern layer, which is much more important in clays by comparison with silica grains.

[56] 2. The surface conductivity entering into the in-phase component of the electrical conductivity is controlled by the diffuse layer (the contribution of the Stern layer can be safely neglected because of the very small mobility of the counterions in the Stern layer).

[57] 3. The polarization of the Stern layer represents the dominating polarization mechanism in shaly sands except maybe at very high salinities for which membrane polarization is expected to dominate the overall polarization response. The quadrature conductivity seems to be controlled by (1) the mobility of the counterions in the Stern layer, (2) the excess of charge per unit volume determined by the CEC, (3) the porosity, and (4) by the fraction of counterions present in the Stern layer (partition coefficient).

[58] 4. The partition coefficient can be determined from a simple sorption isotherm, which predicts its salinity dependence and therefore the salinity dependence of the quadrature conductivity. From this model, it is possible to predict the effect of the composition of the pore water upon the quadrature conductivity using a sorption isotherm that would take into account the composition of the pore water. This step will be investigated in a future contribution.

[59] 5. The model explains the small dependence of the phase lag with the temperature and the observed dependence of the quadrature conductivity with the specific surface area of the material.

[60] 6. Permeability can be predicted inside 1 order of magnitude using the formation factor (determined from the in phase conductivity) and the quadrature conductivity.

[61] The model appears very robust in being able to predict the dependency of the complex conductivity with both textural, electrochemical, and environmental parameters. The next steps will be (1) to develop a model for the membrane polarization and to combine this model with the Stern layer polarization model developed above to get a unified model of low-frequency induced polarization in shaly sands, clean sands, and clay-sand mixtures and (2) to apply POLARIS to monitor bacteria growth in porous media [see Atekwana and Slater, 2009].

Notation

$\mathbf{E}$	electrical field.
$\mathbf{E}_0$	constant electrical field.
$\mathbf{J}$	conduction current density.
$\mathbf{J}_d$	displacement current density.
$\mathbf{H}$	magnetic field.
$\beta_{(+)}^S$	mobility of the counterions in the Stern layer.
$\beta_{(+)}$	mobility of the cation in the pore water.
β_S	apparent surface mobility for the double layer.
ϵ^*	complex permittivity.
ϵ'	real part of the permittivity.
ϵ''	imaginary part of the permittivity.
ϵ_0	permittivity of vacuum.
ϵ_{eff}	apparent permittivity.
ϕ	connected porosity.
λ	apparent mobility for the counterions.
λ_M	maximum value reached by the apparent mobility.
σ^*	complex electrical conductivity.

σ'	real part of the conductivity.
σ''	imaginary part of the conductivity.
$ \sigma $	amplitude of the complex conductivity.
σ_{eff}	apparent (real) conductivity.
σ_f	pore water (real) conductivity.
σ_s^*	surface (complex) conductivity.
σ'_S	real (in phase) surface conductivity.
σ''_S	imaginary (quadrature) surface conductivity.
ω	angular frequency.
φ	phase lag.
ρ_g	the grain density.
Σ_S	specific surface conductivity of the Stern layer.
Γ_S^0	total surface site density of the mineral surface.
Γ_{SNa}^0	surface charge density of the sites $>S^-Na^+$.
Γ_{S-}^0	surface charge density of the sites $>S^-Na^+$.
Γ_{SH}^0	surface charge density of the sites $>SH^0$.
Γ_X^0	equivalent surface charge density for the isomorphic substitutions.
α_S	temperature sensitivity coefficient for the mobility of the counterions.
α_f	temperature sensitivity coefficient for the ionic mobilities in the pore water.
α_S	temperature sensitivity coefficient for the mobility of the counterions in the Stern layer.
Du	Dukhin number.
d_0	single grain diameter.
D	grain diameter.
C_f	salinity.
E_h	expectation of $f(D)/D$.
F	formation factor.
f	partition coefficient of the electrical double layer.
f_M	maximum value reached by the partition coefficient.
$f(D)$	grain diameter probability distribution.
i	pure imaginary number.
k	permeability.
K_{Na}	equilibrium sorption coefficient for Na^+ .
K_H	equilibrium sorption coefficient for H^+ .
m	cementation exponent.
n	saturation exponent.
p	quadrature conductivity saturation exponent.
pH	Log of the activity of the proton concentration.
Q_V	total volumetric (Stern and diffuse layers) charge density.
$\bar{Q}_V$	volumetric diffuse layer charge density.
Q_S	charge density per unit surface area.
R	hydraulic radius.
S_w	relative water saturation.
S_{Sp}	specific surface area of the material.
S_{por}	specific surface per unit pore volume.
t	time.
T	temperature.
T_0	reference temperature.

[62] **Acknowledgments.** I thank the Office of Science (BER), US Department of Energy (awards DE-FG02-08ER646559 and DE-SC0007118), and Chevron Energy Technology Company for financial support. I thank the Associate Editor, Fred Day-Lewis, and the referees (Lee Slater, Andrew Binley, and an anonymous referee) for their very constructive reviews.

References

- Abdel Aal, G. Z., L. D. Slater, and E. A. Atekwana (2006), Induced-polarization measurements on unconsolidated sediments from a site of active hydrocarbon biodegradation, *Geophysics*, *71*, H13–H24, doi:10.1190/1.2187760.
- Archie, G. E. (1942), The electrical resistivity log as an aid in determining some reservoir characteristics, *Trans. Am. Inst. Min. Metall. Eng.*, *146*, 54–62.
- Atekwana, A., and L. Slater (2009), Biogeophysics: A new frontier in earth science research, *Rev. Geophys.*, *47*, RG4004, doi:10.1029/2009RG000285.
- Bernabé, Y., and A. Revil (1995), Pore-scale heterogeneity, energy dissipation and the transport properties of rocks, *Geophys. Res. Lett.*, *22*(12), 1529–1552.
- Binley, A., L. D. Slater, M. Fukes, and G. Cassiani (2005), Relationship between spectral induced polarization and hydraulic properties of saturated and unsaturated sandstone, *Water Resour. Res.*, *41*, W12417, doi:10.1029/2005WR004202.
- Binley, A., S. Kruschwitz, D. Lesmes, and N. Kettridge (2010), Exploiting the temperature effects on low frequency electrical spectra of sandstone: A comparison of effective diffusion path lengths, *Geophysics*, *75*, A43–A46, doi:10.1190/1.3483815.
- Bolève, A., A. Crespy, A. Revil, F. Janod, and J. L. Mattiuzzo (2007), Streaming potentials of granular media: Influence of the Dukhin and Reynolds numbers, *J. Geophys. Res.*, *112*, B08204, doi:10.1029/2006JB004673.
- Börner, F. D. (1992), Complex conductivity measurements of reservoir properties, in *Proceedings of the Third European Core Analysis Symposium*, pp. 359–386, Advances in Core Evaluation v. 3: Reservoir Management—Reviewed Proceedings of the Society for Core Analysis Third European Core Analysis Symposium—Paris, edited by P. F. Worthington and Catherine Chardaire-Riviere, Harwood Academic, Neward, New Jersey.
- Börner, F. D. (2006), Complex conductivity measurements, in: *Groundwater Geophysics—A Tool for Hydrogeology*, edited by R. Kirsch, 119–153, Springer, Berlin.
- Bourg, I. C., and G. Sposito (2011), Molecular dynamics simulations of the electrical double layer on smectite surfaces contacting concentrated mixed electrolyte ($NaCl-CaCl_2$) solutions, *J. Colloid Interface Sci.*, *360*, 701–715.
- Brovelli, A., and G. Cassiani (2011), Combined estimation of effective electrical conductivity and permittivity for soil monitoring, *Water Resour. Res.*, *47*, W08510, doi:10.1029/2011WR010487.
- Bruggeman, D. A. G. (1935), Berechnung verschiedener physikalischer konstanten von heterogenen substansen, *Ann. Phys. (Leipzig)*, *24*, 636–679.
- Bussian, A. E. (1983), Electrical conductance in a porous media, *Geophysics*, *48*, 1258–1268.
- Chen, Y., and D. Or (2006), Effects of Maxwell-Wagner polarization on soil complex dielectric permittivity under variable temperature and electrical conductivity, *Water Resour. Res.*, *42*, W06424, doi:10.1029/2005WR004590.
- de Lima, O. A. L., and M. M. Sharma (1990), A grain conductivity approach to shaly sandstone, *Geophysics*, *55*, 1347–1356.
- de Lima, O. A. L., and M. M. Sharma (1992), A generalized Maxwell-Wagner theory for membrane polarization in shaly sands, *Geophysics*, *57*, 431–440.
- Drolon, H., B. Hoyez, F. Druaux, and A. Faure (2003), Multiscale roughness analysis of particles: Application to the classification of detrital sediments, *Math. Geol.*, *35*(7), 805–817.
- Dukhin, S. S., and V. N. Shilov (2002), Nonequilibrium electric surface phenomena and extended electrokinetic characterization of particles, in *Interfacial Electrokinetics and Electrophoresis*, Surfactant Sci. Ser., vol. 106, edited by A. V. Delgado, pp. 55–85, Marcel Dekker, New York.
- Fraser, D. C. (1964), Electrical properties of clay-containing sandstone: B. Electrical polarization, *Rep. MT-64-4*, Inst. of Eng. Res., Univ. of Calif., Berkeley.
- Garrouch, A. A., and M. M. Sharma (1994), The influence of clay content, salinity, stress and wettability on the dielectric properties of brine-saturated rocks: 10 Hz to 10 MHz, *Geophysics*, *59*, 909–917.
- Glover, P. W. J., M. J. Hole, and J. Pous (2000), A modified Archie's law for two conducting phases, *Earth Planet. Sci. Lett.*, *180*, 369–384.
- Grosse, C. (2011), Extension of the classic thin double layer polarization theory of colloidal suspensions to include the stagnant layer conductivity, *J. Phys. Chem. B.*, *115*, 8996–9004.

- Hanai, T. (1968), Electrical properties of emulsions, in *Emulsion Science*, edited by P. Sherman, pp. 354–478, Academic, London.
- Hayley, K., L. R. Bentley, M. Gharihi, and M. Nightingale (2007), Low temperature dependence of electrical resistivity: Implications for near surface geophysical monitoring, *Geophys. Res. Lett.*, **34**, L18402, doi:10.1029/2007GL031124.
- Hördt, A., R. Blaschek, A. Kemna, and N. Zisser (2007), Hydraulic conductivity estimation from induced polarisation data at the field scale—The Krauthausen case history, *J. Appl. Geophys.*, **62**, 33–46.
- Hyslip, J. P., and L. E. Vallejo (1997), Fractal analysis of the roughness and size distribution of granular materials, *Eng. Geol.*, **48**(3–4), 231–244.
- Jougnot, D., A. Ghorbani, A. Revil, P. Leroy, and P. Cosenza (2010), Spectral induced polarization of partially saturated clay-rocks: A mechanistic approach, *Geophys. J. Int.*, **180**, 210–224, doi:10.1111/j.1365-246X.2009.04426.x.
- Kan, R., and P. N. Sen (1987), Electrolytic conduction in periodic arrays of insulators with charges, *J. Chem. Phys.*, **86**, 5748–5756.
- Karaoulis, M., A. Revil, D. D. Werkema, B. Minsley, W. F. Woodruff, and A. Kemna (2011), Time-lapse 3D inversion of complex conductivity data using an active time constrained (ATC) approach, *Geophys. J. Int.*, **187**, 237–251, doi:10.1111/j.1365-246X.2011.05156.x.
- Kemna, A. (2000), Tomographic inversion of complex resistivity: theory and application, PhD thesis, 196 pp., Ruhr-Univ. Bochum, Bochum, Germany.
- Kemna, A., H.-M. Münch, K. Titov, E. Zimmermann, and H. Vereecken (2005), Relation of SIP relaxation time of sands to salinity, grain size and hydraulic conductivity, in *Proceedings of the 11th European Meeting of Environmental and Engineering Geophysics, Extended Abstracts: Near Surface 2005: P054*, 11th European Meeting of Environmental and Engineering Geophysics, Palermo, Italy.
- Le Méhauté, A., and G. Crépy (1983), Introduction to transfer and motion in fractal media: The geometry of kinetics, *Solid State Ionics*, **9**–10, 17–30.
- Leroy, P., and A. Revil (2004), A triple layer model of the surface electrochemical properties of clay minerals, *J. Colloid Interface Sci.*, **270**(2), 371–380.
- Leroy, P., and A. Revil (2009), Spectral induced polarization of clays and clay-rocks, *J. Geophys. Res.*, **114**, B10202, doi:10.1029/2008JB006114.
- Leroy, P., A. Revil, S. Altmann, and C. Tournassat (2007), Modeling the composition of the pore water in a clay-rock geological formation (Callovo-Oxfordian, France), *Geochim. Cosmochim. Acta*, **71**(5), 1087–1097, doi:10.1016/j.gca.2006.11.009.
- Leroy, P., A. Revil, A. Kemna, P. Cosenza, and A. Gorbani (2008), Spectral induced polarization of water-saturated packs of glass beads, *J. Colloid Interface Sci.*, **321**(1), 103–117.
- Lesmes, D. P., and F. D. Morgan (2001), Dielectric spectroscopy of sedimentary rocks, *J. Geophys. Res.*, **106**(B7), 13,329–13,346.
- Marshall, D. J., and T. R. Madden (1959), Induced polarization, a study of its causes, *Geophysics*, **24**, 790–816.
- Meier, L. P., and G. Kahr (1999), Determination of the cation exchange capacity (CEC) of clay minerals using the complexes of copper(II) ion with triethylenetetramine and tetraethylenepentamine, *Clays Clay Miner.*, **47**(3), 386–388.
- Morgan, F. D., F. Scira-Scappuzzo, W. Shi, W. Rodi, J. Sogade, Y. Vichabian, and D. Lesmes (1999), Induced polarization imaging of a jet fuel plume, Symposium on the Application of Geophysics to Engineering and Environmental Problems “SAGEEP,” Environmental and Engineering Geophysical Society, Proceedings, 541–548, Oakland, California.
- Niwas, N., P. K. Gupta, and O. A. L. de Lima (2007), Nonlinear electrical conductivity response of shaly-sand reservoir, *Curr. Sci.*, **92**(5), 612–617.
- Olhoeft, G. R. (1979), Electrical properties, in *Initial Report of the Petrophysics Laboratory*, U.S. Geol. Surv. Circ., 789, 1–26.
- Olhoeft, G. R. (1986), Direct detection of hydrocarbon and organic chemicals with ground-penetrating radar and complex resistivity: Petroleum, hydrocarbons and organic chemicals, in *Proceedings of the NWWA/API Conference on Petroleum Hydrocarbons and Organic Chemicals in Ground Water—Prevention, Detection, and Restoration*, pp. 284–305, Natl. Well Water Assoc., Dublin, Ohio.
- Pelton, W. H., S. H. Ward, P. G. Hallöf, W. R. Sill, and P. H. Nelson (1978), Mineral discrimination and removal of inductive coupling with multifrequency IP, *Geophysics*, **43**, 588–609.
- Revil, A. (1999), Ionic diffusivity, electrical conductivity, membrane and thermoelectric potentials in colloids and granular porous media: A unified model, *J. Colloid Interface Sci.*, **212**, 503–522.
- Revil, A. (2001), Pervasive pressure solution transfer in a quartz sand, *J. Geophys. Res.*, **106**(B5), 8665–8690.
- Revil, A., and L. M. Cathles (1999), Permeability of shaly sands, *Water Resour. Res.*, **35**(3), 651–662.
- Revil, A., D. Hermitte, E. Spangenberg, and J. J. Cochémé (2002), Electrical properties of zeolitized volcanoclastic materials, *J. Geophys. Res.*, **107**(B8), 2168, doi:10.1029/2001JB000599.
- Revil, A., and P. Leroy (2004), Governing equations for ionic transport in porous shales, *J. Geophys. Res.*, **109**, B03208, doi:10.1029/2003JB002755.
- Revil, A., and N. Florsch (2010), Determination of permeability from spectral induced polarization in granular media, *Geophys. J. Int.*, **181**, 1480–1498, doi:10.1111/j.1365-246X.2010.04573.x.
- Revil, A., and A. Jardani (2010), Stochastic inversion of permeability and dispersivities from time lapse self-potential measurements: A controlled sandbox study, *Geophys. Res. Lett.*, **37**, L11404, doi:10.1029/2010GL043257.
- Revil, A., and M. Skold (2011), Salinity dependence of spectral induced polarization in sands and sandstones, *Geophys. J. Int.*, **187**, 813–824, doi:10.1111/j.1365-246X.2011.05181.x.
- Revil, A., L. M. Cathles, S. Losh, and J. A. Nunn (1998), Electrical conductivity in shaly sands with geophysical applications, *J. Geophys. Res.*, **103**(B10), 23,925–23,936.
- Revil, A., W. F. Woodruff, and N. Lu (2011), Constitutive equations for coupled flows in clay materials, *Water Resour. Res.*, **47**, W05548, doi:10.1029/2010WR010002.
- Schmutz, M., A. Revil, P. Vaudelet, M. Batzle, P. Femenía Viñao, and D. D. Werkema (2010), Influence of oil saturation upon spectral induced polarization of oil bearing sands, *Geophys. J. Int.*, **183**, 211–224, doi:10.1111/j.1365-246X.2010.04751.x.
- Schwarz, G. (1962), A theory of the low-frequency dielectric dispersion of colloidal particles in electrolyte solution, *J. Phys. Chem.*, **66**, 2636–2642.
- Scott, J., and R. Barker (2003), Determining pore-throat size in Permian-Triassic sandstones from low-frequency electrical spectroscopy, *Geophys. Res. Lett.*, **30**(9), 1450, doi:10.1029/2003GL016951.
- Sen, P. N., and P. A. Goode (1992), Influence of temperature on electrical conductivity on shaly sands, *Geophysics*, **57**, 89–96.
- Sen, P. N., C. Scala, and M. H. Cohen (1981), A self-similar model for sedimentary rocks with application to the dielectric constant of fused glass beads, *Geophysics*, **46**, 781–795.
- Sistat, P., A. Kozmaib, N. Pismenskayab, C. Larchet, G. Pourcellya, and V. Nikonenkob (2008), Low-frequency impedance of an ion-exchange membrane system, *Electrochim. Acta*, **53**, 6380–6390.
- Skold, M., and A. Revil (2011), The pH dependence of spectral induced polarization of silica sands: Experiment and modeling, *Geophys. Res. Lett.*, **38**, L12304, doi:10.1029/2011GL047748.
- Slater, L. (2007), Near surface electrical characterization of hydraulic conductivity: From petrophysical properties to aquifer geometries: A review, *Surv. Geophys.*, **28**(2–3), 169–197.
- Slater, L., and D. Glaser (2003), Controls on induced polarization in sandy unconsolidated sediments and application to aquifer characterization, *Geophysics*, **68**, 1547–1558.
- Slater, L., and D. P. Lesmes (2002), Electrical-hydraulic relationships observed for unconsolidated sediments, *Water Resour. Res.*, **38**(10), 1213, doi:10.1029/2001WR001075.
- Tarasov, A., and K. Titov (2007), Relaxation time distribution from time domain induced polarization measurements, *Geophys. J. Int.*, **170**, 31–43.
- Tertre, E., E. Ferrage, I. Bihannic, L. J. Michot, and D. Prêt (2011), Influence of the ionic strength and solid/solution ratio on Ca(II)-for-Na⁺ exchange on montmorillonite. Part 2: Understanding the effect of the m/V ratio. Implications for pore water composition and element transport in natural media, *J. Colloid Interface Sci.*, **363**, 334–347.
- Thomas, E. C. (1976), Determination of Qv from membrane potential measurements on shaly sands, *Trans. Am. Inst. Min. Metall. Pet. Eng.*, **261**, 1087–1096.
- Titov, K., V. Komarov, V. Tarasov, and A. Levitski (2002), Theoretical and experimental study of time domain induced polarization in water-saturated sands, *J. Appl. Geophys.*, **50**, 417–433.
- Tournassat, C., Y. Chapron, P. Leroy, M. Bizi, and F. Boulahya (2009), Comparison of molecular dynamics simulations with triple layer and modified Gouy-Chapman models in 0.1 M NaCl-montmorillonite system, *J. Colloid Interface Sci.*, **339**(2), 533–541.
- Urtinov, M. A.-K., E. V. Kirillova, N. M. Seidova, and V. V. Nikonenko (2007), Decoupling of the Nernst-Planck and Poisson equations. Application to a membrane system at overlimiting currents, *J. Phys. Chem. B*, **111**, 14,208–14,222.

- Vanhala, H. (1997), Mapping oil-contaminated sand and till with the spectral induced polarization (SIP) method, *Geophys. Prospect.*, *45*, 303–326.
- Vaudelet, P., A. Revil, M. Schmutz, M. Franceschi, and P. Bégassat (2011a), Induced polarization signature of the presence of copper in saturated sands, *Water Resour. Res.*, *47*, W02526, doi:10.1029/2010WR009310.
- Vaudelet, P., A. Revil, M. Schmutz, M. Franceschi, and P. Bégassat (2011b), Changes in induced polarization associated with the sorption of sodium, lead, and zinc on silica sands, *J. Colloid Interface Sci.*, *360*, 739–752.
- Vinegar, H. J., and M. H. Waxman (1984), Induced polarization of shaly sands, *Geophysics*, *49*, 1267–1287.
- Wang, M., J. Liu, and S. Chen (2007), Electrical potential distribution in nanoscale electroosmosis: from molecules to continuum, *Molecular Simulation*, *33*(15), 1273–1277.
- Waxman, M. H., and L. J. M. Smits (1968), Electrical conductivities in oil-bearing shaly sands, *J. Soc. Pet. Eng.*, *8*, 107–122.
- Weller, A., L. Slater, S. Nordsiek, and D. Ntarlagiannis (2010), On the estimation of specific surface per unit pore volume from induced polarization: A robust empirical relation fits multiple data sets, *Geophysics*, *75*, WA105–WA112.
- Wong, P. Z. (1987), Fractal surface in porous media, in *Physics and Chemistry of Porous Media*, vol. 2, edited by J. R. Banavar, J. Koplik, and K. W. Winkler, *AIP Conf. Proc.*, *154*, 304–318.
- Woodruff, W. F., and A. Revil (2011), CEC-normalized clay-water sorption isotherm, *Water Resour. Res.*, *47*, W11502, doi:10.1029/2011WR010919.
- Zisser, N., A. Kemna, and G. Nover (2010), Dependence of spectral-induced polarization response of sandstone on temperature and its relevance to permeability estimation, *J. Geophys. Res.*, *115*, B09214, doi:10.1029/2010JB007526.

A. Revil, Department of Geophysics, Colorado School of Mines, Green Center, 1500 Illinois St., Golden, CO 80401, USA. (arevil@mines.edu)

Salinity dependence of complex conductivity of unconsolidated and consolidated materials: Comparisons with electrical double layer models

Andreas Weller¹ and Lee Slater²

ABSTRACT

We examined the dependence of imaginary conductivity (σ'') on pore fluid conductivity (σ_w) for an extensive database of 67 samples acquired from twelve independent studies. We compared fitting of functions describing the salinity dependence of σ'' for two models of the electrical double layer (EDL) polarization, both of which predict asymptotic behavior of σ'' at high σ_w . We define these models as the diffuse layer polarization (DLP) and Stern layer polarization (SLP) models based on the physical description of the salinity dependence of the surface polarization. We also examined the database for evidence of a high salinity decrease in σ'' not predicted by either model. The dependence of σ'' on σ_w prior to the polarization plateau predicted by both models approximates a simple empirical power law with an average exponent of 0.34. The salinity dependence predicted by the DLP model adequately describes

most data sets. A fitting parameter representing the high salinity σ'' asymptote is strongly correlated ($R^2 = 0.822$) with pore normalized specific surface (S_{por}). The SLP model describes well the observations when a recently proposed additive polarization term representing the contribution of the protons is included. In this case, the SLP model provides an excellent fit to the data sets, including a low salinity asymptote (in log-log conductivity space) seen in some samples. Predicted values of the fitting parameters of the SLP model generally are consistent with the values expected based on the theory; the fitting parameter describing the high salinity asymptote of the SLP model is also strongly correlated ($R^2 = 0.890$) with S_{por} . The SLP and DLP models neglect a high salinity decrease in the polarization that is observed in numerous data sets from independent studies. New data acquired on a sandstone sample demonstrate that this high salinity decrease is likely not attributable to the limited phase accuracy of earlier measurements.

INTRODUCTION

The induced polarization (IP) method has emerged as a potentially powerful tool for subsurface imaging due to the dependence of the measurements on rock or soil internal surface area. For example, Weller et al. (2010b) showed how 114 samples from 10 independent data sets could be described well by a linear relationship between imaginary conductivity (σ'') and the surface area to pore volume (S_{por}). Earlier evidence of such a relationship (Börner and Schön, 1991; Börner, 1992; Börner et al., 1996; Slater and Lesmes, 2002) inspired efforts to use IP measurements to predict permeability (Börner et al., 1996; Slater and Lesmes, 2002; Revil and Florsch, 2010; Weller et al., 2010a).

As a result of such work, understanding of the dependence of IP measurements on the physical characteristics of rocks has evolved to the point that Weller et al. (2010b) cautiously proposed that the relationship between σ'' and S_{por} essentially represents the IP equivalent of Archie's Law (Archie, 1942), an empirical formulation long used to relate resistivity to porosity and water saturation. However, the contribution of the pore fluid chemical composition to the electrical double layer (EDL) polarization recorded with IP is less well defined, although several researchers have previously studied the controls of pore fluid salinity (Klein and Sill, 1982; Vinegar and Waxman, 1984; Lesmes and Frye, 2001; Revil and Skold, 2011; Weller et al., 2011), in addition to pH (Lesmes and Frye, 2001; Skold et al., 2011), and valence (Börner, 1992; Revil and Skold,

Manuscript received by the Editor 26 January 2012; revised manuscript received 21 May 2012; published online 23 August 2012.

¹Technische Universität Clausthal, Institut für Geophysik, Clausthal-Zellerfeld, Germany. E-mail: andreas.weller@tu-clausthal.de.

²Rutgers-Newark, Department of Earth & Environmental Sciences, New Jersey, USA. E-mail: lslater@rutgers.edu.

© 2012 Society of Exploration Geophysicists. All rights reserved.

2011; Weller et al., 2011). A notable advance in the mechanistic understanding of fluid chemistry effects is the work of Vaudelet et al. (2011), who combined a mechanistic model (discussed later) with a surface complexation model to predict the effect of differing sorption properties for the mineral surface for different cations.

From a theoretical standpoint, models have been proposed that include a prediction for the expected salinity dependence of IP measurements. One such early model is based on the polarization of the EDL surrounding the pores making up the porous medium, which is conceptualized as a bunch of capillary tubes (Börner, 1991, 1992). This model, which we will refer to here as the “diffuse layer polarization (DLP) model” because the salinity dependence relates to a varying capacity of the diffuse layer, predicts an approximate dependence of σ'' on the square root of the fluid electrical conductivity (σ_w) at low salinity. More recently, mechanistic models for the polarization of porous media have been based around the polarization of the EDL surrounding the mineral grains making up a porous medium (Leroy et al., 2008; Revil and Florsch, 2010). This model, which we will refer to here as the “Stern layer polarization (SLP) model” assumes that, for low frequencies (typically less than 100 Hz), the main polarization of a granular porous material is due to the Stern layer. The SLP model uses a triple layer model (TLM) of the specific surface conductivity of the Stern layer to define the expected salinity dependence associated with the polarization of absorbed counterions.

The DLP model and SLP models predict an asymptotic high-salinity value for σ'' , having been observed in numerous data sets (Vinegar and Waxman, 1984; Lesmes and Frye, 2001; Slater and Glaser, 2003; Weller et al., 2011). Examples of this behavior will be presented later. However, some previous works have observed an apparent decrease in the polarization at high salinity that is not predicted by either model (Vinegar and Waxman, 1984; Flath, 1989; Lesmes and Frye, 2001; Slater and Glaser, 2003). Some workers have suggested that this polarization decrease may result from a decrease in the mobility of the counterions in the EDL at high-ionic concentrations (Lesmes and Frye, 2001). However, Revil and Skold (2011) note that the high-salinity polarization peak is not consistent between data sets, suggesting that the apparent decrease in polarization at high salinity could be an artifact of the measurements resulting from the limited accuracy in the measurement of very low-phase angles at high salinity (discussed later).

In this study, we seek to resolve some of the uncertainty in the dependence of IP measurements on salinity. Whereas Weller et al. (2011) examined the salinity dependence for three sandstone samples, this paper compiles and examines a wide database comprised of 67 samples that have multiple salinity measurements available. Similar to Weller et al. (2010b), we seek to identify consistent empirical relationships across a broad range of consolidated and unconsolidated sedimentary materials recorded by multiple investigators. Furthermore, we extend the work of Weller et al. (2011) by examining the predictive capabilities of the DLP and SLP models with respect to the salinity dependence of the polarizability, and we also present new data that addresses the question regarding a decrease in polarization at high salinity.

ELECTRICAL PROPERTIES

The complex conductivity (σ^*) of a sample is determined from measurements of the conductivity magnitude ($|\sigma|$) and phase shift (φ) of a sample relative to a reference resistor. The real (σ') and

imaginary (σ'') components of the complex conductivity represent the ohmic conduction and polarization charge transport mechanisms respectively. These electrical properties directly are determined from the measurements. The phase shift is related to the real and imaginary parts of the complex conductivity according to

$$\varphi = \arctan\left(\frac{\sigma''}{\sigma'}\right) \approx \frac{\sigma''}{\sigma'} (\varphi < 100 \text{ mrad}). \quad (1)$$

By convention, the phase is defined as positive in conductivity space.

Most models for the complex electrical conductivity of a porous material at low frequencies (e.g., less than 100 Hz) are based on a parallel addition of two conduction terms representing, (1) an electrolytic contribution via conduction through the interconnected pore space (σ_{el}), and (2) a mineral surface contribution via conduction and polarization within the electrical double layer (EDL) of the interconnected pore surface (σ^*_{surf}) (e.g., Vinegar and Waxman, 1984),

$$\sigma^* = \sigma_{el} + \sigma^*_{surf}. \quad (2)$$

Polarization only is associated with the surface conductivity at low frequencies. In terms of the real and imaginary components,

$$\sigma' = \frac{1}{F} \sigma_w + \sigma'_{surf}, \quad (3)$$

and

$$\sigma'' = \sigma''_{surf}. \quad (4)$$

This paper focuses on how σ'' measurements, which directly represent the surface polarization (equation 4), depend on salinity.

Weller et al. (2010b, 2011) defined the fitting parameter of their linear relationship between σ'' and S_{por} as the specific polarizability (c_p),

$$c_p = \frac{\sigma''}{S_{por}}. \quad (5)$$

This parameter therefore represents the polarizability of the mineral-fluid interface per unit S_{por} . Based on a NaCl saturating solution with σ_w approximately equal to 100 mS/m, Weller et al. (2010b) found $c_p \approx 10^{-11} \text{ S} = 10 \text{ pS}$. In this work, we seek to investigate the dependence of c_p on salinity, i.e., $c_p(\sigma_w)$. The dependence of c_p on σ_w is expected, assuming that changes in surface charge density and/or surface ionic mobility (among other factors) modify the surface polarization. This property bears similarities to the specific surface conductance that has been defined in the development of mechanistic models to describe the surface conductivity of a porous medium (Revil and Florsch, 2010). Although c_p as presented by Weller et al. (2010b) is an empirically derived coefficient directly defining the relation between σ'' and S_{por} , it was recently defined in a mechanistic model for shaly sands (Revil, 2012).

The DLP model

Physics-based descriptions of the surface polarization can offer valuable insights into the expected dependence of σ'' on σ_w . We first consider an approach that uses as a starting point a model describing the modulus of the surface conductivity ($|\sigma_{surf}|$) proposed

by [Rink and Schopper \(1974\)](#). The geometrical and physical basis for this model briefly is stated here. Assuming a capillary bundle model, the measurable surface conductivity of a sample can be described by the ratio of cross section of the EDL (A'') to cross section of the sample (A) multiplied by the intrinsic conductivity within the double layer (σ_x),

$$\left| \sigma_{\text{surf}} \right| = \frac{A''}{A} \sigma_x. \quad (6)$$

Parameter A'' can be determined from the circumference of the pore multiplied by the thickness (δ) of the EDL. Considering a cylindrical cross section of the capillaries with a uniform radius r , we get the area of the cross section of a single pore $A' = \pi r^2$. Recognizing that (1) the ratio A'/A is equal to the porosity (Φ), and (2) $A''/A' = 2\delta/r$ is represented by the specific pore space surface area (S_{por}) in capillary bundle models multiplied by the EDL thickness (δ), equation 6 takes the form:

$$\left| \sigma_{\text{surf}} \right| = \frac{A''}{A'} \frac{A'}{A} \sigma_x = \frac{2}{r} \delta \Phi \sigma_x = S_{\text{por}} \delta \Phi \sigma_x. \quad (7)$$

Regarding the tortuous conduction path through the sample, the tortuosity (T) causes a reduction in surface conductivity. Using the ratio T/Φ as an equivalent for the formation factor F , the modulus of surface conductivity can be formulated as

$$\left| \sigma_{\text{surf}} \right| = \frac{S_{\text{por}}}{F} \delta \sigma_x. \quad (8)$$

[Börner \(1991, 1992\)](#) showed that the real (σ'_{surf}) and imaginary (σ''_{surf}) parts of σ^*_{surf} linearly are related and introduced parameter l to represent the ratio of σ''_{surf} to σ'_{surf} . Therefore, assuming the validity of equation 4,

$$\sigma'' \propto \frac{S_{\text{por}}}{F}. \quad (9)$$

The DLP model conceptualizes the fixed (Stern) and diffuse layers as two capacitors and assumes that the polarization of the EDL can be described by the capacitive contributions from both layers. We stress that the model assumes polarization of the fixed (Stern) and diffuse part of the double layers in response to an electrical field. We refer to this model as the DLP model only because it is the polarization of the diffuse layer that is assumed to be dependent on the salinity, and our work specifically is focused on this salinity dependence. This model implicitly assumes that the secondary electric field associated with the polarization is normal to the pore (or grain) surface. The model predicts

$$\sigma'' = l \frac{S_{\text{por}}}{F} f(\sigma_w) \propto \frac{S_{\text{por}}}{F} f(\sigma_w), \quad (10)$$

where l is considered salinity-independent. This factor is justified by the fact that the surface conductivity and polarizability share the same geometry ([Kulenkampff and Schopper, 1988](#)); the same salinity dependence is therefore assumed for the real and imaginary parts of surface conductivity through the dependence on variation in the thickness of the EDL. [Börner \(1992\)](#) presents supporting experimental evidence, finding that l (his equation 15) is nearly

independent of salinity but slightly different in various types of sandstones. Furthermore, [Börner \(1991, 1992\)](#) experimentally confirms the proportionality of the real and imaginary parts of surface conductivity.

The diffuse layer has a salinity-dependent thickness due to the decrease in thickness of the diffuse layer (δ) with increasing ionic concentration (C_f) as predicted, e.g., by traditional Gouy-Chapman double layer theory,

$$\delta = \left(\frac{\epsilon k T}{8 \pi N Z^2 e^2 C_f} \right)^{0.5}, \quad (11)$$

where ϵ is the mean relative permittivity of the diffuse layer, Z is electrolyte valence, N is Avogadro's number, k is Boltzman's constant, T is absolute temperature (in K) and e is the electronic charge ([Stumm and Morgan, 1996](#)).

The thickness is therefore approximately proportional to $C_f^{-0.5}$, implying an increase in capacity that is approximately proportional to $C_f^{0.5}$. The Stern layer is assumed to have a fixed capacity, which implies an asymptotic value of σ'' for high σ_w . The salinity dependence of σ'' predicted by this model can be described as ([Börner, 1991, 1992](#)),

$$\sigma''(\sigma_w) = a_d \frac{\sqrt{\sigma_w}}{b_d + \sqrt{\sigma_w}}, \quad (12)$$

where a and b are fitting parameters, and subscript d is to signify that these are the fitting parameters for the DLP model. Parameter a_d should be linearly dependent on the ionic mobility of the counterions in the diffuse layer (β_d), which for simplicity is considered to be independent of salinity.

The SLP model

The SLP model has been extensively described in recent literature ([Leroy et al., 2008](#); [Revil and Florsch, 2010](#); [Revil and Skold, 2011](#)) and only briefly is described here, with specific focus on the implications for the dependence of σ'' on salinity, for brevity. This model combines a macroscopic conductivity model for the polarization of a porous medium with a triple layer model for the specific surface conductivity of the Stern layer. Such triple layer models are more sophisticated than the traditional double layer model because they incorporate an inner and outer Stern layer and account for specific adsorption of ions that are tightly bound to the surface in the inner layer.

In the grain model, the surface polarization is specifically associated with the Stern layer surrounding the mineral grains, although surface conduction is assumed to occur in the diffuse and Stern layers. The polarization of the Stern layer around a mineral grain is based on a modification of the model of [Schwarz \(1962\)](#) for colloidal suspensions. Ion migration is tangential and results in accumulation of counterions on the side of the grain in the direction of the electric field. This model predicts,

$$\sigma''(\omega = 1/\tau) = \frac{F - 1}{F} \frac{\sum^s(\sigma_w)}{2d_0}, \quad (13)$$

where d_0 is the grain diameter (when the material is composed of uniform grains) and a decisive parameter describing the

polarization, Σ^s is the specific surface conductivity (in S), and τ (in s) is the main relaxation time. Note that this formulation can be extended for a distribution of grain sizes as described in [Revil and Florsch \(2010\)](#).

Assuming spherical particles, we can write,

$$d_0 = \frac{1 - \Phi}{\Phi} \times \frac{6}{S_{\text{por}}}, \quad (14)$$

with Φ being the porosity.

Therefore, by combining equation 14 with equation 13 and using the approximation

$$\Phi \approx \frac{1}{F}, \quad (15)$$

we get

$$\sigma''(\omega = 1/\tau) \propto \frac{S_{\text{por}}}{F} \Sigma^s(\sigma_w). \quad (16)$$

The specific surface conductivity represents the excess conductivity due to the presence of the Stern layer. The salinity dependence of the imaginary conductivity is attributed to the salinity dependence Σ^s determined by the sorption of counterions. The model assumes that the imaginary conductivity exhibits the same relative dependence on salinity of the pore water as Σ^s , i.e., $\partial \sigma'' / \partial C_f = \partial \Sigma^s / \partial C_f$ ([Revil and Skold, 2011](#)).

[Revil and Skold \(2011\)](#) derive the salinity dependence of Σ^s for a binary symmetric 1:1 electrolyte from first principles based on a triple layer model of the mineral-fluid interface for silica. In the case of Na counterions,

$$\Sigma^s(C_f) = \beta_{\text{Na}} e \Gamma_s^0 \frac{K_{\text{Na}} C_f}{10^{-\text{pH}} + K_{\text{Na}} C_f}, \quad (17)$$

where C_f is in mol l^{-1} , K_{Na} is the sorption coefficient for sodium (dimensionless), Γ_s^0 is total site density at the silica surface, and e is the electronic charge. A generalization to a multicomponent electrolyte is presented in [Revil and Florsch \(2010\)](#).

Assuming a constant pH this model predicts the following salinity dependence,

$$\sigma''(\sigma_w) \propto \Sigma^s(C_f) = a_s \frac{\sigma_w}{b_s + \sigma_w}, \quad (18)$$

where a_s and b_s are fitting parameters, and subscript s is to signify that these are the fitting parameters for the SLP model. Parameter a_s is considered linearly dependent on the ionic mobility of the counterions in the Stern layer (β_s), which for simplicity is considered to be independent of salinity. Equation 18 can be directly compared with equation 12.

[Skold et al. \(2011\)](#) recently presented an extension of this model that accounts for an additional polarization mechanism beyond the Stern layer polarization that these authors attribute to hopping of the protons on the mineral surface. This extension to the SLP model is used to explain the pH dependence of the imaginary conductivity observed for silica sands.

To account for this effect, [Skold et al. \(2011\)](#) include an additional contribution to the specific surface conductivity,

$$\Sigma_s = \Sigma_s(C_f) + \Sigma_s(H^+), \quad (19)$$

where the second term represents the possibility of mobile protons. [Skold et al. \(2011\)](#) propose the following form for the proton term,

$$\Sigma_s(H^+) = e \beta(H^+) \Gamma_{>\text{SiOH}_2^+}, \quad (20)$$

where $\Gamma_{>\text{SiOH}_2^+}$ is the concentration of surface sites for which protons are mobile and $\beta(H^+)$ represents an effective mobility of the proton along the mineral surface. The addition of this proton contribution implies a modification to equation 18 of the form,

$$\sigma''(\sigma_w, H^+) \propto \Sigma^s(C_f, H^+) = c_s + \left(a_s \frac{\sigma_w}{b_s + \sigma_w} \right), \quad (21)$$

where c_s is a third model parameter linearly related to $\Gamma_{>\text{SiOH}_2^+}$ and $\beta(H^+)$, both of which are considered independent of salinity. [Skold et al. \(2011\)](#) state that this proton hopping mechanism can dominate at very low pH, but may not be negligible at high pH.

METHODS AND DATA SETS

Our database is made up of 67 samples with σ'' recorded as a function of salinity from multiple sources ([Vinegar and Waxman, 1984](#); [Flath, 1989](#); [Lesmes and Frye, 2001](#); [Slater and Lesmes, 2002](#); [Slater and Glaser, 2003](#); [Comas and Slater, 2004](#); [Breede, 2006](#); [Kruschwitz, 2008](#); [Schröder, 2008](#); [Revil and Skold, 2011](#)). Table 1 summarizes the database, and includes the σ_w range investigated, the selected measurement frequency, the source of the data set and a sample description. The database includes fifteen previously unreported unconsolidated samples (# 43–57). Samples were measured at a pH close to neutral.

All measurements reported in Table 1 were obtained by measuring the impedance magnitude and phase shift of the voltage waveform recorded across the sample relative to the current waveform recorded on a reference resistor, the source typically being a sine signal (e.g., [Slater and Lesmes, 2002](#)). All measurements reported here were recorded using a four-electrode device, whereby separate electrode pairs were used to inject the current into the sample and record the potential waveform across the sample. Measured magnitude and phase were converted into a measured complex conductivity (σ^*) using the geometric factor defining the measurement geometry and the resulting current flow path in the test device.

Complex conductivity measurements can exhibit significant frequency dependence. It may then be necessary to model this frequency dependence to extract parameters that quantify the integrated polarization response over the measured frequency range. For example, a normalized chargeability can be defined from fitting either Cole-Cole ([Slater et al., 2005](#)), generalized Cole-Cole ([Scott and Barker, 2005](#)), or Debye decomposition ([Nordsieck and Weller, 2008](#)) models of measured spectra. Many of the data sets we use, which go back up to 28 years (Table 1), only report the dependence of the polarization (σ'') on salinity for a single frequency. To examine the largest database possible, we therefore focus on analyzing just a single frequency measurement. Assuming that the frequency dependence is not very strong, we can expect normalized chargeability and σ_w to show a similar dependence as σ'' and σ_w ([Weller et al., 2011](#)). However, the relationship between σ'' and σ_w is likely to show some degree of frequency dependence. This

Table 1. Summary of the 67 samples used in this study, showing source for each sample, description of sample, key measurement parameters, and findings (b is power law exponent of fit of σ'' versus σ_w below the polarization maximum).

	Sample	Salinity range (mS/m)	Frequency (Hz)	b	Asymptote reached?	Polarization peak? (σ_w mS/m)	Description
	1. Sandstones						
	Breede (2006)						
1	BK	7–2200	0.05	0.2	Y	Y(1200)	Helsby formation
2	BS4	7–2200	0.05	0.34	Y	Y(1100)	Kiddminster formation
3	BR5	7–2200	0.05	0.27	N	N	Helsby formation
	Schröder (2008)						
4	B49H	25–630	0.05	0.25	N	N	Bahariya formation
5	B49V	25–630	0.05	0.25	N	N	Bahariya formation
6	B4H	25–630	0.05	0.34	N	N	Bahariya formation
7	B4V	25–630	0.05	0.3	N	N	Bahariya formation
8	ES14	20–3000	0.09	0.4	N	N	Elbe sandstone
9	OK4	20–2000	0.05	0.32	Y	Y(320)	Oberkirchen sandstone
10	BU1	20–3000	0.05	0.3	N	N	Bunter sandstone
	Breede (2006); also shown in Weller et al. (2011)						
11	GR	24–2100	0.05	0.45	Y	Y(1200)	Glauconitic sandstone
12	BU12	96–2200	2.93	0.6	N	N	Bunter sandstone
13	BU3	24–2300	0.05	0.38	N	N	silty Bunter sandstone
	Kruschwitz (2008)						
14	B15	30–1000	0.15	0.59	N	N	Baumberg sandstone
15	C15	30–1000	0.1	0.46	N	N	Cotta sandstone
	Lesmes and Frye (2001)						
16	B_LF	20–8000	5	0.26	Y	Y(1000)	Berea sandstone
	Börner (1992)						
17	E10	3–5100	1	0.38	Y	Y(900)	Elbe sandstone
18	E12	3–5100	1	0.4	N	N	Elbe sandstone
19	B2	3–5300	1	0.25	Y	Y(900)	Bentheim sandstone
20	F3	2–1600	1	0.19	Y	Y(1600)	Fontainebleau sandstone
	Flath (1989)						
21	H18H	20–10000	1	0.34	Y	Y(1600)	Cheshire Basin
22	HCF120H	20–10000	1	0.38	Y	Y (2100)	Cheshire Basin
23	HCF236V	20–10000	1	0.31	Y	Y(2900)	Cheshire Basin
	Vinegar and Waxman (1984)						
24	3411	100–15000	30	0.21 ³	N	N	shaly sandstones
25	3336A	100–15000	30	0.1 ³	N	N	shaly sandstones
26	3478	100–15000	30	0.17 ³	N	N	shaly sandstones
27	101	100–1500	30	0.23	Y	N	shaly sandstones
28	102	100–8500	30	0.35	Y	Y(1100)	shaly sandstones
29	CZ10	100–15000	30	0.16 ³	N	N	shaly sandstones
30	3833A	100–15000	30	0.2 ³	N	N	shaly sandstones
31	3126B	100–15000	30	0.28 ³	N	N	shaly sandstones
32	3847A	100–8300	30	0.2	Y	Y (2500)	shaly sandstones
33	3823A	100–15000	30	0.32 ³	Y	Y(4600)	shaly sandstones
34	3885B	100–15000	30	0.34 ³	Y	Y(4600)	shaly sandstones
35	3972E	100–15000	30	0.31 ³	N	N	shaly sandstones

Table 1. Summary of the 67 samples used in this study, showing source for each sample, description of sample, key measurement parameters, and findings (*b* is power law exponent of fit of σ'' versus σ_w below the polarization maximum). (continued)

	Sample	Salinity range (mS/m)	Frequency (Hz)	<i>b</i>	Asymptote reached?	Polarization peak? (σ_w mS/m)	Description
36	3258A	100–15000	30	0.36 ³	Y	Y(4600)	shaly sandstones
37	3891A	100–15000	30	0.35 ³	Y	Y(4600)	shaly sandstones
38	3308A	100–15000	30	0.37 ³	Y	Y(4600)	shaly sandstones
39	3323F	1000–15000	30	n.d.	Y	Y(4600)	shaly sandstones
40	3324A	1000–15000	30	n.d.	Y	Y(4600)	shaly sandstones
41	3323E	1000–15000	30	n.d.	Y	Y(4600)	shaly sandstones
42	3324B	1000–15000	30	n.d.	Y	Y(4600)	shaly sandstones
2. Unconsolidated sediments							
43	B3_2	3–300	1	0.3	N	N	sands and gravels
44	B2_4	3–300	1	0.45	N	N	sands and gravels
45	B6_7	3–300	1	0.26	N	N	sands and gravels
46	C5_5	3–300	1	0.33	N	N	sands and gravels
47	B1	3–300	1	0.28	N	N	sands and gravels
48	C1_5	3–300	1	0.28	N	N	sands and gravels
49	C2_7	3–300	1	0.33	N	N	sands and gravels
50	C5	3–300	1	0.23	N	N	sands and gravels
51	C3_5	3–300	1	0.23	N	N	sands and gravels
52	B4_7	3–300	1	0.34	N	N	sands and gravels
53	C3_3	3–300	1	0.25	N	N	sands and gravels
54	C4_2	3–300	1	0.28	N	N	sands and gravels
55	C6197	80–400	1	0.43	N	N	sand and gravel flood deposits
56	C6189	10–120	1	0.4	N	N	sand and gravel flood deposits
57	C6208	10–120	1	0.3	N	N	sand and gravel flood deposits
Slater and Lesmes (2002)							
58	G1	2–200	1	0.36	Y	Y(30)	Maine sand and gravel
Comas and Slater (2004)							
59	S1	2–900	1	0.55	N	N	Peat soils (organics)
60	C1	2–1600	1	0.51	N	N	Peat soils (organics)
61	C9	2–1600	1	0.51	N	N	Peat soils (organics)
Slater and Glaser (2003)							
62	N2: 27–35	50–6600	1	0.42	Y	Y(850)	Sand/silt floodplain sediments
63	N2: 19–25	57–3500	1	0.64	Y	Y(140)	Sand/silt floodplain sediments
64	N1: 26.5–35	57–3800	1	0.35	Y	Y(260)	Sand/silt floodplain sediments
65	N1: 51–53	57–3300	1	0.37	Y	Y(3300)	Sand/silt floodplain sediments
Revil and Skold (2011)							
66	SB	3–200	0.1	0.58	N	N	Artificial sand
67	U#30	10–1000	0.1	0.21	N	N	Artificial sand

³Regression is only two points below 1000 ms/m.

should be recognized when comparing the data sets in Table 1, because the measurement frequency varies from a minimum of 0.05 to a maximum of 30 Hz.

The analysis of each data set first involved fitting a simple power law function to the observed general increase in σ'' with σ_w at the lower salinities,

$$\sigma'' = a(\sigma_w)^b, \quad (22)$$

where *a* and *b* are fitting parameters. Although equation 22 has no physical basis, it does provide an assessment of the first-order variations in the strength of the salinity dependence of σ'' over the entire sample database. Many of the data sets are characterized by a high salinity asymptote reported in previous measurements and predicted by the DLP and SLP models. Therefore, equation 22 was only fit to all available data below 1000 mS/m (representing potable groundwater) unless the asymptote occurred at lower

salinities, as observed for some unconsolidated samples. In the latter case, equation 22 was only fit to the σ_w measurements at salinities lower than the asymptote. In all except 12 samples, a minimum of three data points were available to define b . The exceptions occur for 12 of the 19 data sets used here that were reported by Vinegar and Waxman (1984) and highlighted (superscript a) in Table 1. Vinegar and Waxman (1984) were primarily interested in the characterization of briny oil reservoirs, and therefore focused on a higher salinity range than the other data sets reported in Table 1. Consequently, most of their measurements capture the σ'' dependence close to/on the asymptote, with limited coverage at the lower salinities that are more representative of potable groundwater. Values of the power law exponent b are presented in Table 1 and discussed later.

The remaining data analysis involved evaluating the fits of the DLP and SLP salinity dependence described in equations 12 and 21 to the σ'' data recorded across the σ_w range prior to the asymptote for each sample. Each data set also was evaluated for evidence of a σ'' peak and decrease in polarization at highest salinity.

Finally, to investigate evidence of the polarization decrease at high σ_w further, we took one sandstone sample (#11, GR from Breede, 2006) with measurements up to 2200 mS/m and extended these measurements to $\sim 10,000$ mS/m by acquiring data with three additional saturating electrolytes ($\sigma_w = 2800, 4500$, and 9000 mS/m). Complex conductivity measurements were acquired for each saturating electrolyte over a frequency range from 0.003 to 100 Hz using two commercially available SIP instruments, the SIP Fuchs (Radic Research) and the SIP Jülich described in Zimmermann et al. (2008). This comparison of instruments was performed to investigate whether the polarization decrease could be an artifact of measurement errors at high σ_w . Equation 1 shows that the phase measurement used to quantify the polarization response is inversely proportional to σ' . Therefore, φ decreases with increasing σ_w and

could approach the accuracy of the instrumentation. Whereas modern commercially produced SIP instruments have a high φ accuracy (e.g., ~ 0.1 mrad, Zimmermann et al., 2008), the accuracy of the instruments used to acquire the historic data sets reported in this paper is somewhat uncertain.

RESULTS

We organize the results by first presenting the fit of the data sets to salinity dependence predicted by the DLP and SLP models, as well as the fit to the simple empirical function described by equation 22. We then proceed to consider the significance of the parameters with respect to their dependence on available petrophysical properties and their predicted values in the case of the SLP model.

Fitting quality of the models

Table 1 summarizes the findings of the analysis performed on all 67 samples. The power law exponent (b) obtained from fitting equation 22 is reported for all samples excluding #39–42 from Vinegar and Waxman (1984). These four samples do not contain at least two data points for the fitting below the asymptote, but are retained in this analysis to support our investigation of the significance of a distinct polarization peak. Data sets exemplifying the power law behavior below 1000 mS/m are shown in Figure 1. Representative sandstone samples from four previous independent studies are plotted in Figure 1a. Representative unconsolidated samples from two previous independent studies and two new databases of our own are plotted in Figure 1b. The sandstone data sets are plotted to higher σ_w and show evidence of the plateau or maximum being reached above 1000 mS/m.

Considering the entire database ($n = 63$), the exponents vary from a minimum of 0.1 to a maximum of 0.64, with a mean (μ)

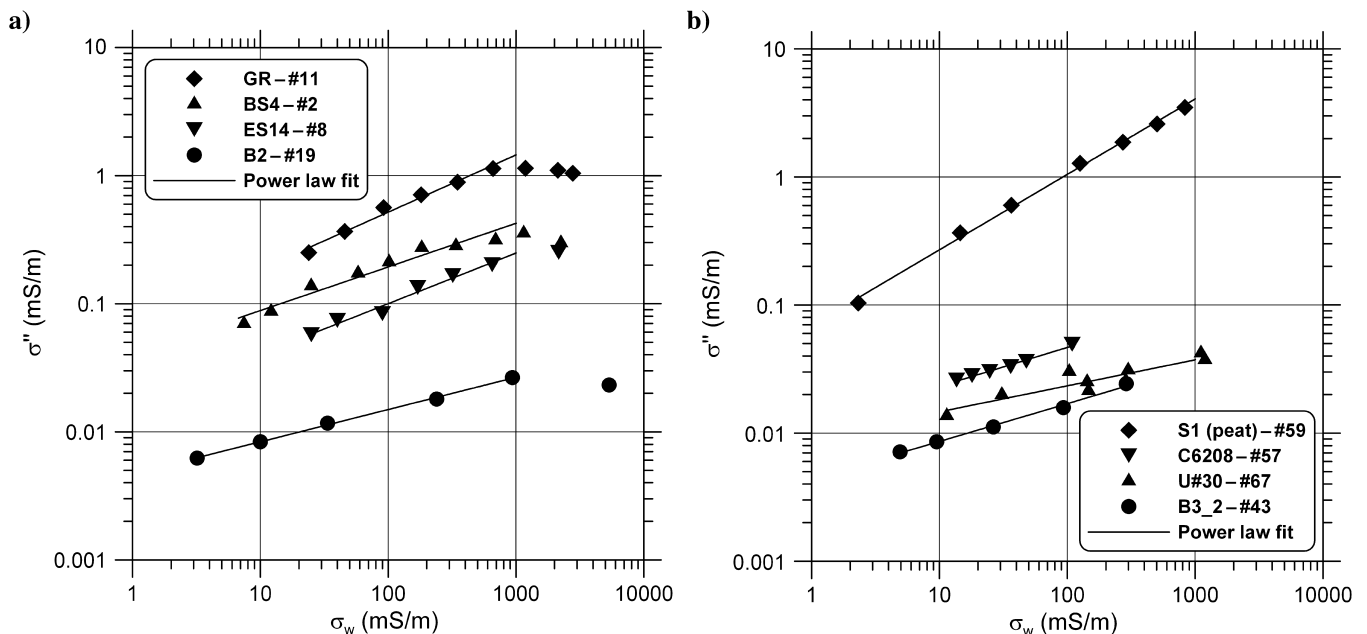


Figure 1. Example data sets showing the fit of the power law dependence of σ'' on σ_w below the asymptote (where present): (a) sandstone samples; (b) unconsolidated samples. Sample descriptions and power law exponents (b) are given in Table 1.

of 0.34 and a standard deviation of (σ_d) of 0.11. Excluding the high-salinity (Vinegar and Waxman, 1984) data set that contains more data close to the polarization asymptote ($n = 48$), the minimum increases to 0.19 and μ increases to 0.36, whereas the maximum and σ_d are unchanged. Considering the sandstone samples alone, the minimum is 0.1, the maximum is 0.59, $\mu = 0.31$, and $\sigma_d = 0.11$ ($n = 38$). Excluding the Vinegar and Waxman (1984) data set, this minimum increases to 0.19, μ increases to 0.35, and $\sigma_d = 0.11$ ($n = 23$). Considering the unconsolidated samples alone, the minimum is 0.21, the maximum is 0.64, and μ increases to 0.37 and $\sigma_d = 0.12$ ($n = 25$).

Twenty-three sandstone samples and five unconsolidated samples spanning nine independent data sets were found to display strong evidence of a high-salinity asymptote in the $\sigma''(\sigma_w)$ data. Figure 1a illustrates examples of this asymptotic behavior at $\sigma_w > 1000$ mS/m for sandstone samples from four independent data sets. This asymptote is predicted by the DLP (equation 12) and SLP (equation 21) models. However, twenty seven of these samples (again spanning nine independent data sets) exhibit some evidence of a polarization peak, with a decrease in polarization at high salinities that is not predicted by either model. Examples of this polarization peak are shown in Figure 2 for sandstone samples from three independent data sets (Figure 2a) and unconsolidated samples from one data set (Figure 2b). Table 1 indicates the salinity at which this peak occurs ($\sigma_{w[p]}$). For the 27 sandstone samples, the minimum $\sigma_{w[p]}$ is 320 mS/m, the maximum is 4600 mS/m, $\mu = 2719$ mS/m and $\sigma_d = 1690$ mS/m. These statistics are influenced strongly by the Vinegar and Waxman (1984) data set, containing nine samples where the peak is at 4600 mS/m, being the maximum within the identified range. For the five unconsolidated samples, the minimum $\sigma_{w[p]}$ is 30 mS/m, the maximum is 3200 mS/m, $\mu = 916$ mS/m and $\sigma_d = 1370$ mS/m. This analysis demonstrates that this polarization peak occurs at significantly lower salinity for the unconsolidated sediments.

Most of the data sets could be well-fit by the salinity dependence of the DLP model below the asymptote as described by equation 12. Figure 3 shows selected examples of the fitting quality to (a) sandstone samples, and (b) unconsolidated samples. The samples were selected to represent results obtained from independent studies and over different salinity ranges. The data shown above the polarization peak for sample # 21 are excluded from the fitting as the high σ_w polarization decrease is not predicted by the model. In total, the salinity dependence of the DLP model was found to describe 56 of the 67 samples well.

The fit of the SLP model to the available data sets was found to be critically dependent on the inclusion of the additive constant c_s recently proposed by Skold et al. (2011) to describe the additional contribution of the protons to the polarization. We examined the fit of the data sets to two forms of the model (a) with $c_s = 0$ as proposed by Revil and Skold (2011) in their studies of the salinity dependence of the induced polarization response of sandstones, and (b) with $c_s > 0$ as proposed by Skold et al. (2011) to account for the protons. In the case of $c_s = 0$ (equation 18), the dependence of σ'' on σ_w predicted by the SLP model was found only to provide a better fit than the DLP model (equation 12) for seven samples in our database (this included only one unconsolidated sample). Under these conditions, the deviation of the SLP model from the experimental data results from the fact that the slope of the σ'' dependence on σ_w approaches one at low salinity. However, the inclusion of the additional polarization contribution represented by c_s (equation 21) resulted in an excellent fitting to most data sets (excluding any data points suggesting a decrease in σ'' at high salinity). Figure 4 shows example fits to selected sandstone samples (Figure 4a) and unconsolidated samples (Figure 4b). Samples shown in Figure 4 were selected to represent results obtained from independent studies and to capture evidence of the asymptote (in log-log conductivity space) predicted when $c_s > 0$. The fits for $c_s = 0$ are shown by the dashed lines and the fits for $c_s > 0$ are

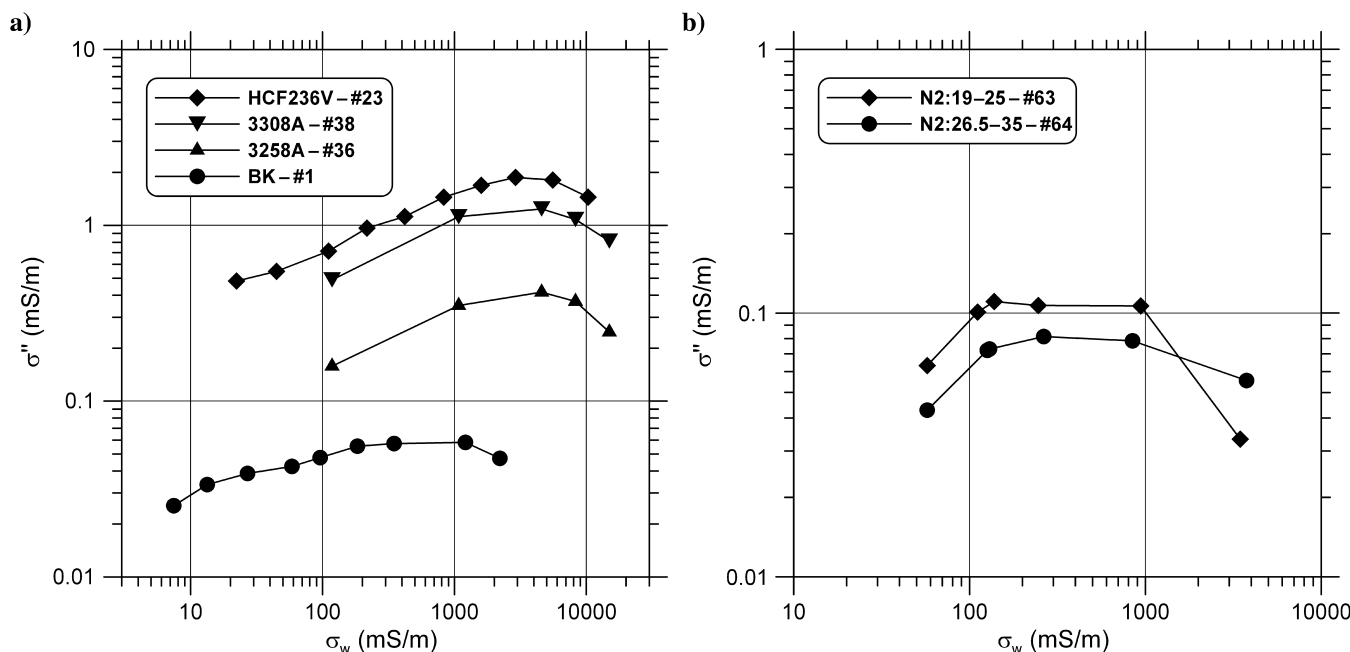


Figure 2. Example data sets showing evidence for a decrease in σ'' at high salinity; (a) sandstone samples, (b) unconsolidated samples.

shown by the solid lines. Two unconsolidated samples capture evidence for the low salinity asymptote predicted by the SLP model with $c_s > 0$ (samples # 44 and # 61). Sample # 61, an organic peat sample with measurements over a wide salinity range, provides strong support for an additive contribution not predicted by either equation 12 or 18. We note, however, that these peat samples are not representative of the clean sands for which the SLP is derived.

Four of the samples from the Vinegar and Waxman (1984) database could not be fit to either the SLP or DLP model because data points prior to the polarization peak are missing. These are samples that show a pronounced peak in the polarization and contain multiple data points indicating a drop in σ'' at high salinity. As noted previously, this high-salinity decrease in polarization is not predicted by either the DLP or SLP models. Results of our extended

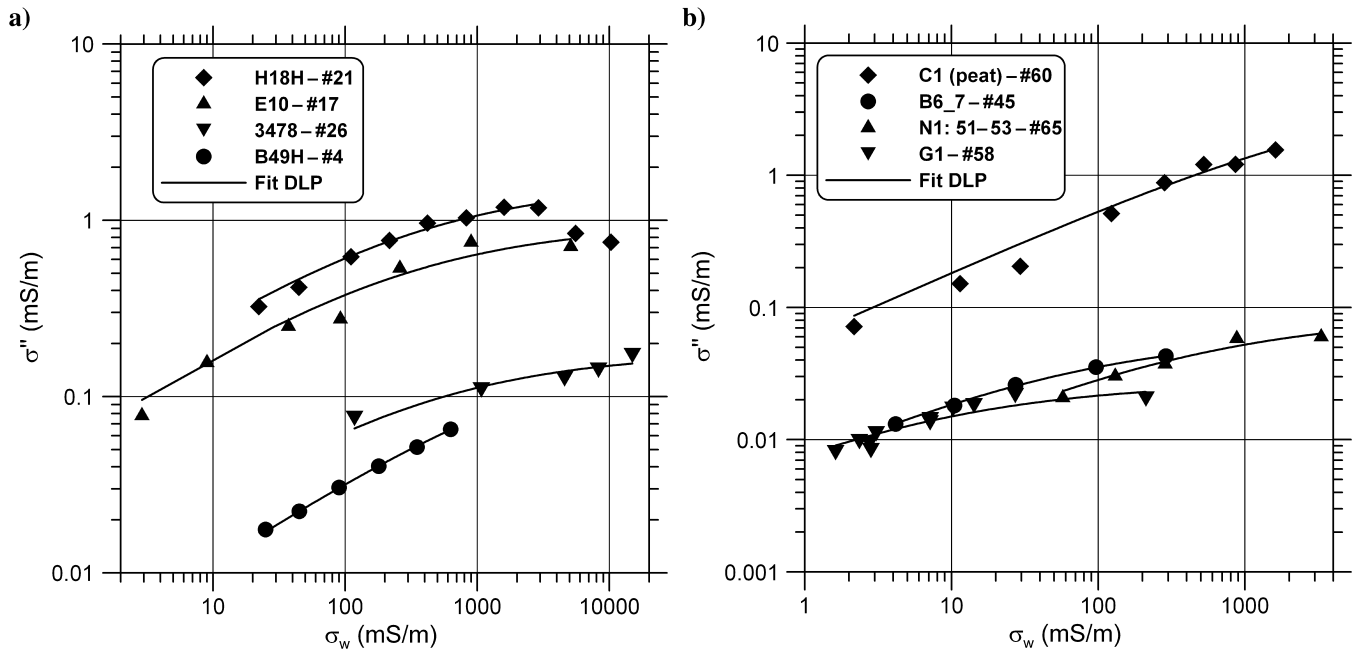


Figure 3. Example data sets showing the fit of the DLP model to experimental data sets; (a) selected sandstone samples, (b) selected unconsolidated sediments.

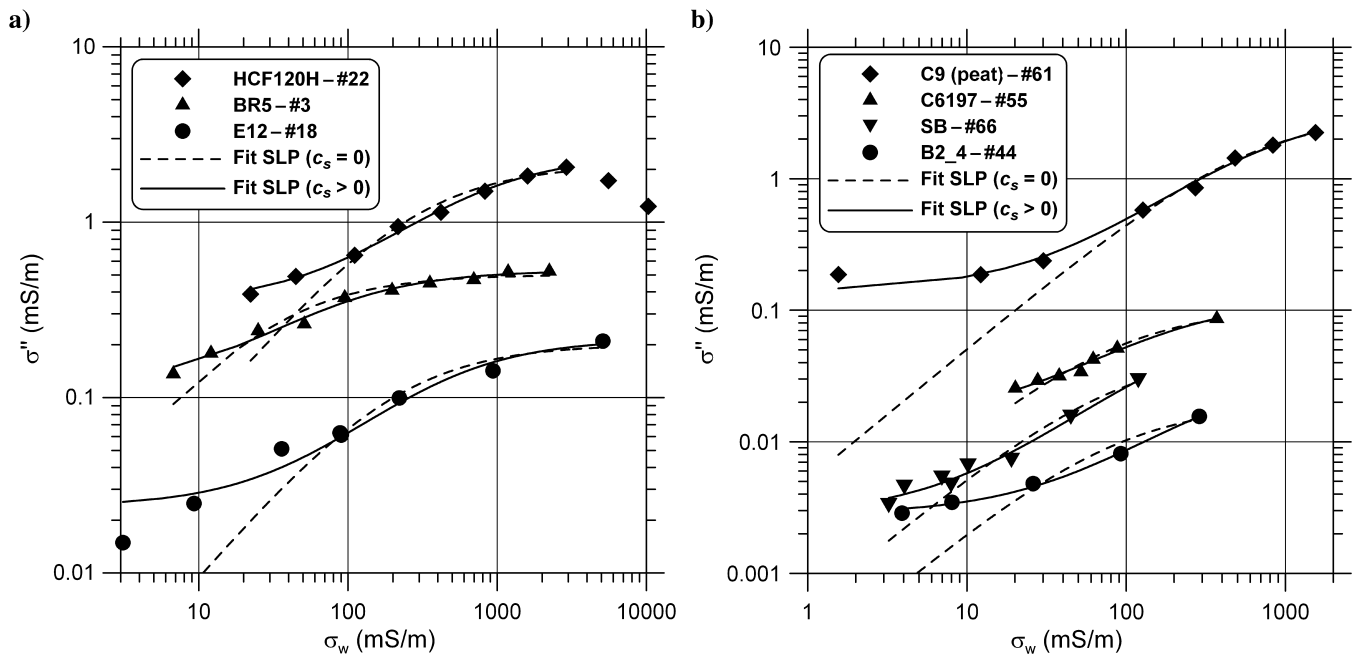


Figure 4. Example data sets showing the fit of the SLP model to experimental data sets; (a) selected sandstone samples, (b) selected unconsolidated sediments. The fits when the contribution of the protons is neglected ($c_s = 0$, e.g., Revil and Skold, 2011) and when the additional contribution is considered ($c_s > 0$, e.g., Skold et al., 2011) are shown for comparison.

high-salinity measurements using two high-accuracy, commercially produced complex resistivity instruments are shown in Figure 5. Figure 5a compares the full phase spectra for the three high-salinity measurements acquired with the two instruments. The response from the two instruments is near identical over the entire frequency range of the measurements. Figure 5b shows the extension of the measured σ'' dependence on σ_w to high salinities beyond the asymptote originally recorded in this data set. These measurements support the existence of a polarization peak and a high-salinity decrease in σ'' as has been previously recorded for 27 samples from nine independent data sets (Table 1).

Physical significance of the fitting parameters

We next consider the physical significance of the fitting parameters derived from the DLP and SLP models. We use 20 sandstone samples that have estimates of S_{por} and F , along with complex conductivity measurements at more than five salinities. The petrophysical properties and fitting parameters for the DLP and SLP models are shown in Table 2. We first consider the DLP model and the significance of the model parameters shown in equation 12. Data for 19 of the 20 samples reported in Table 2 were described well by the DLP model. Because no reasonable fitting parameters of sample B15 were achieved, this sample was excluded from the interpretation.

Although the DLP model is not parameterized sufficiently to assign expected values to the fitting parameters, equation 12 does provide information on the expected dependence of the model parameters on physical properties. Parameter a_d , which varies over almost three orders of magnitude (4.2×10^{-3} to 2.2 mS/m), describes the asymptotic value of σ'' at high salinity. This parameter should be strongly related to S_{por}/F (equation 10). Indeed, the variation of the normalized parameter $a_d F/S_{\text{por}}$ is much reduced to a little over one order of magnitude (202–3961 pS). However, the normalized parameter a_d/S_{por} , according to the concept of specific

polarizability (Weller et al., 2011), results in a further reduced range from 13–132 pS (with an average value 42 pS), thereby indicating a near-linear relation between a_d and S_{por} . The power law fit describing this relation is given by

$$a_d = 0.0304 S_{\text{por}}^{1.04}, \quad (23)$$

(a_d in mS/m and S_{por} in $1/\mu\text{m}$) with a coefficient of determination $R^2 = 0.822$ (Figure 6a).

The parameter b_d is related to the salinity dependence of imaginary conductivity. Lower values of b_d indicate asymptotic behavior of $\sigma''(\sigma_w)$ being reached at lower salinity. The numerical value of b_d varies between 4 and 134 with a mean value of 25. A slight increase in b_d with increasing S_{por} is observed. A power law fit of b_d to S_{por} results in an exponent of 0.3, but the coefficient of determination is only 0.197.

We next turn to the significance of the fitting parameters (equation 21) with respect to the SLP model. Unlike the DLP model, the expected values of these parameters are defined by the theory (Revil and Skold, 2011; Skold et al., 2011). We found that all 20 samples reported in Table 2 could be fit by equation 21. However, two samples (samples # 9 and # 16 in Table 2) gave a negative value of c_s , which is physically unreasonable but can be explained by the fact that c_s is only well-resolved when the low salinity asymptote is visible in the fitted curves (not the case in these samples). These two samples are therefore excluded from this analysis.

We first note that a_s should be an analog of a_d in the DLP model. Parameter a_s , which varies by almost three orders of magnitude (0.00272 to 1.21 mS/m), describes the asymptotic value of σ'' at high salinity. This parameter should be strongly related to S_{por}/F (equation 16). Considering equations 13, 14, 15, and 17, the normalized parameter $a_s F/S_{\text{por}}$ can be expressed by

$$\frac{a_s F}{S_{\text{por}}} = \frac{1}{12} \beta_{\text{Na}} e \Gamma_s^0. \quad (24)$$

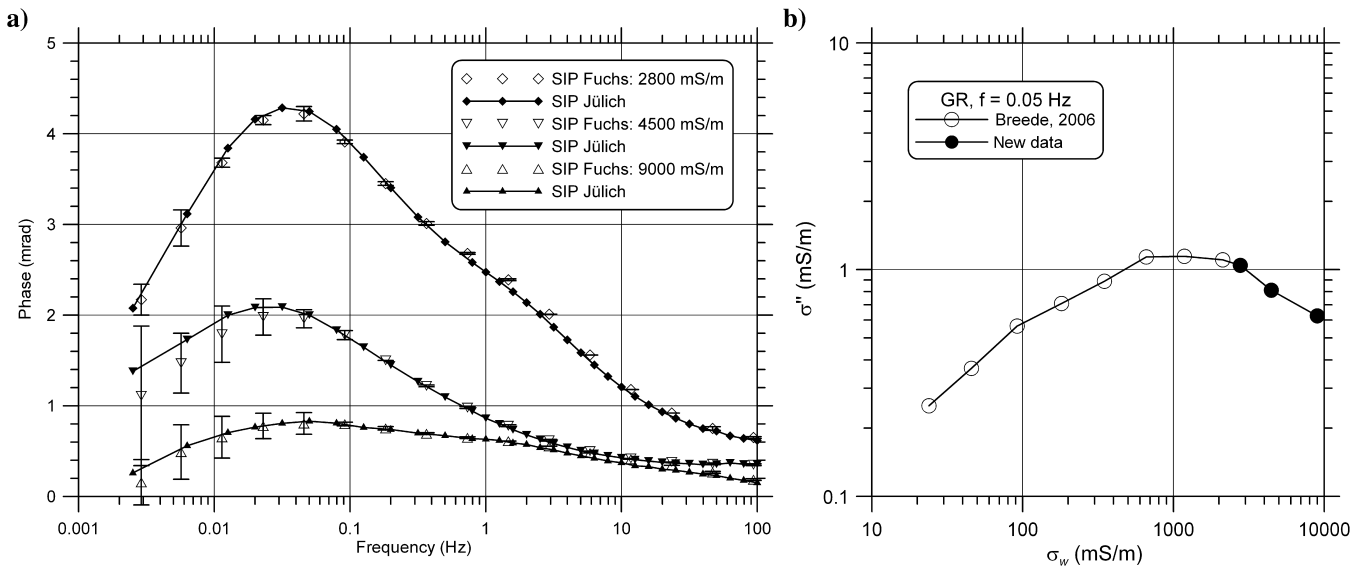


Figure 5. Results for sample GR #11. (a) Phase spectra recorded at high salinity with two different instruments (phase is shown positive for conductivity space and individual error bars are plotted for SIP Fuchs instrument); (b) σ'' at 0.05 Hz versus σ_w showing the extension of the original data to high salinity and the clear evidence for the high-salinity polarization decrease.

Assuming an average site density of 5 sites nm⁻² and a mobility $\beta_{\text{Na}} = 5.19 \times 10^{-8} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$ (adopted from [Revil and Skold, 2011](#)), this normalized parameter should be on the order of 3500 pS. Our fitting predicts $a_s F / S_{\text{por}}$ between 128 and 2544 pS with an average value of 539 pS. This difference might indicate a lower site density and ionic mobility in our samples than what is assumed by the model. The normalization of a_s to S_{por} according to the concept of specific polarizability (c_p) ([Weller et al., 2011](#)) results in a further reduced range from 7.7 to 48 pS (average value 20 pS) and confirms a nearly linear relation between a_s and S_{por} . The power law fit describing this relation is given by

$$a_s = 0.0158 S_{\text{por}}^{1.04} \quad (25)$$

(a_s in mS/m and S_{por} in 1/ μm) with $R^2 = 0.890$ (see Figure 6b). [Revil \(2012, equation 48\)](#) gives an explicit relation for a_s developed for shaly sands. Taking average values for all parameters and ignoring the contribution of protons, he calculates a value of 12.3 pS for the specific polarizability (Equation 5), being close to our average of 20 pS.

Similar to b_d , from the DLP model, parameter b_s is related to the salinity dependence of imaginary conductivity. Lower values of b_s indicate asymptotic behavior of $\sigma''(\sigma_w)$ being reached at lower salinity. The numerical value of b_s estimated from the fitting varies between 37 and 747 with a geometric mean value of 190. Considering the approximation,

$$\sigma_w [\text{mS/m}] \cong 10000 \times C_f [\text{mol/l}], \quad (26)$$

the dimensionless parameter b_s can be expressed by

$$b_s = \frac{10000 \times 10^{-\text{pH}}}{K_{\text{Na}}}, \quad (27)$$

and depends on the pH and the sorption coefficient of sodium. Assuming pH = 6 and $K_{\text{Na}} = 10^{-3.25}$ (adopted from [Revil and Skold \[2011\] Figure 8](#)), we get $b_s = 18$. This value is approximately a factor of 10 lower than the resulting geometric mean; this could indicate a lower average pH or K_{Na} for our sample database than the assumed values.

Table 2. Petrophysical parameters for 20 sandstone samples from the compiled database and fitting parameters for the DLP and SLP models: All samples listed in this table had a minimum of five data points describing the dependence of σ'' on σ_w .

Sample		Porosity	F	$S_{\text{por}} (\mu\text{m}^{-1})$	DLP model		SLP model		
					a_d (mS m $^{-1}$)	b_d	a_s (mS m $^{-1}$)	b_s	c_s (mS m $^{-1}$)
Breede (2006)									
1	BK	0.160	16.6	5.12	0.678	4.02	0.039	37.26	0.021
2	BS4	0.166	17.6	28	0.500	13.82	0.308	86.25	0.052
3	BR5	0.243	8.2	15	0.637	8.47	0.415	72.83	0.114
Schröder (2008)									
4	B49H	0.160	26.8	3.29	0.058	10.99	0.032	125.20	0.013
5	B49V	0.160	31.0	3.54	0.061	12.31	0.035	224.70	0.016
6	B4H	0.120	44.9	8.93	0.224	30.68	0.128	484.40	0.030
7	B4V	0.130	59.6	8.29	0.129	19.40	0.071	251.20	0.022
8	ES14	0.180	18.7	5.2	0.424	26.98	0.251	218.70	0.031
9	OK4	0.180	22.1	8.23	0.236	12.62			
10	BU1	0.140	33.8	25.4	0.852	12.31	0.552	100.00	0.113
Breede (2006); also shown in Weller et al. (2011)									
11	GR	0.246	9.1	31.3	1.775	20.45	1.171	130.80	0.066
12	BU12	0.181	18.0	10.5	1.106	134.71	0.350	581.30	0.002
13	BU3	0.089	66.7	46.2	1.540	60.00	0.738	746.80	0.121
Kruschwitz (2008)									
14	B15	0.191	12.6	69.17			1.211	582.20	0.066
15	C15	0.217	11.6	50.8	2.219	45.60	0.926	142.90	0.029
Lesmes and Frye (2001)									
16	B_LF	0.180	15.9	8.3	0.126	7.41			
Börner (1992)									
17	E10	0.224	17.0	41.6	0.949	15.19	0.694	156.10	0.088
18	E12	0.259	11.0	5.5	0.311	34.97	0.191	389.70	0.024
19	B2	0.194	15.6	2.15	0.028	7.08	0.019	97.10	0.006
20	F3	0.068	118.0	0.126	0.004	6.38	0.003	192.50	0.001

We finally consider parameter c_s introduced by Skold et al. (2011) to describe a contribution of the protons to the polarization. In our fitting, this additive constant varies between 1.34 and 121 $\mu\text{S}/\text{m}$. Similar to the parameter a_s , a normalized parameter $c_s F/S_{\text{por}}$ can be explicitly formulated:

$$\frac{c_s F}{S_{\text{por}}} = \frac{1}{12} \beta(H^+) e \Gamma_{>\text{SiOH}_2^+} = \frac{1}{12} \Sigma^S(H^+). \quad (28)$$

Assuming $\Sigma^S(H^+) \approx 2$ nS as shown in Skold et al. (2011) (see their Figure 2a), the value of the normalized parameter $c_s F/S_{\text{por}}$ is expected to be on the order of 160 pS, which is very close to the average value of 143 pS that is determined by the fitting procedure. However, this normalized parameter varies over a wide range from 3.2 to 1250 pS; this is likely attributable to the poor resolution of parameter c_s for many samples due to limited data points defining the low salinity asymptote (in double logarithmic space) predicted by equation 21.

DISCUSSION

This analysis of an extensive database of multisalinity, complex conductivity data sets demonstrates that σ'' universally follows a simple positive power law dependence on σ_w at low salinities representative of natural groundwater. Such an increase in polarization with increasing σ_w is predicted by the DLP and SLP models to result from an increase in surface charge density as σ_w increases. In most of the samples reported, this power law dependence is valid below $\sigma_w = 1000$ mS/m. A minority of unconsolidated samples display a polarization asymptote at σ_w significantly less than 1000 mS/m. In this case, the power law dependence only is observed for salinities lower than the asymptote. The mean value of the exponent is 0.34 for the entire database, 0.31 for the sandstone samples, and 0.37 for the unconsolidated samples. The observed dependence of σ'' on σ_w is weak when compared to the

expected linear dependence of σ' on σ_w (equation 3). Such a relatively weak dependence on σ_w , coupled with the strong linear dependence of σ'' on S_{por} (Weller et al., 2010b), highlights the value of complex conductivity measurements in lithologic discrimination and permeability estimation.

Although the empirical fitting and the fitting constants of equation 22 lack physical meaning, this relation does provide a first-order expected dependence of σ'' on σ_w that could be used to compensate for changes in σ'' driven by σ_w variation. Empirical laws frequently are used to constrain field geophysical measurements in the absence of sufficient information on physicochemical properties to populate the model space of physics-based models. Examples of such laws are Archie's law (Archie, 1942) discussed previously and the widely used Topp equation (Topp et al., 1980), a third-order polynomial for estimating moisture content in soils from dielectric permittivity measurements. We have shown here that the empirical power law relation fits the widest available range of sandstone and unconsolidated sediment materials with a relatively narrow range in the power law exponent. This empirical relation could be used to improve complex conductivity derived estimates of S_{por} and permeability for cases where the salinity varies between samples or between field locations. For example, the specific polarizability defined by Weller et al. (2010b) could be defined as a function of salinity.

The salinity dependence predicted by the DLP model mostly satisfies the observation at low salinities quite well for sandstone and unconsolidated sediment samples. The DLP model is based on a relatively simple electrical double layer where the polarization of the Stern layer is considered salinity independent. The DLP model assumes that the salinity dependence arises from an increase in capacity of the diffuse part of the EDL resulting from a decrease in the diffuse layer thickness accompanying an increasing ionic concentration. The mean of the power law exponents is 0.34, and therefore lower than the approximate value of 0.5 (Kulenkampff and

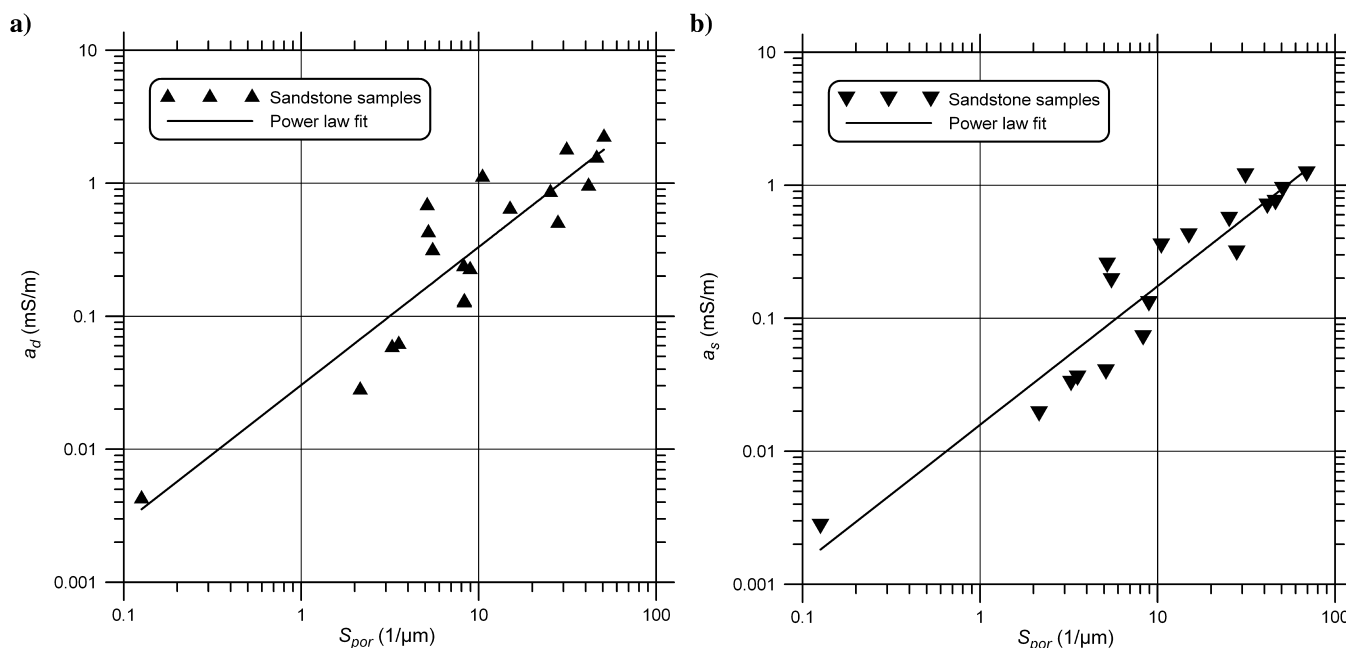


Figure 6. The correlation between model parameters and S_{por} : (a) a_d versus S_{por} ($R^2 = 0.8822$); (b) a_s versus S_{por} ($R^2 = 0.890$).

Schopper, 1988), predicted by the DLP model and previously observed based on the three sandstone samples described in Weller et al. (2011). The value of 0.5 represents the maximum of the slope at sufficiently low σ_w to be well away from the asymptote. The lower average value found here can be interpreted to result from the asymptotic character of the curves and the fact that this curvature is in part captured in many of the power law fits reported here (Table 1). Clearly, fitting data that in part capture the curvature as the asymptote is approached will lower the b estimate relative to that for data well away from the asymptote. For example, the organic sediments (samples # 59–# 61) were all recorded over a large salinity range that captured low salinities well below the asymptote (an example in Figure 3). These samples are fit with a value of $b = 0.5$ – 0.55 , being very close to the expected value from the DLP model.

The fit of the salinity dependence predicted by the SLP model to the observed data sets was found to be critically dependent on whether the recently proposed additive term representing the polarization of the protons (Skold et al., 2011) was included in the model. The form of the SLP model without this additional contribution (e.g., Revil and Skold, 2011) generally did not provide an adequate fit to most of the available experimental data (e.g., Figure 4). However, the inclusion of the term representing the protons generally resulted in an excellent fitting quality to all available data sets (e.g., Figure 4). The addition of this additive term results in a predicted low-salinity asymptote (in double logarithmic space) and evidence of this behavior is captured in a few of the available data sets that extend to low salinity. The strongest evidence for this behavior predicted by the SLP model was found for sample # 61 (shown in Figure 4b), although other samples shown in Figure 4 (# 22 in Figure 4a and # 44 in Figure 4b) also exhibit evidence of this low-salinity behavior predicted by the SLP model when the protons are considered, but not predicted by the DLP model. One clear finding from our work is that the SLP model must include the addition of the recently proposed proton contribution to adequately represent the available data sets describing the salinity dependence of σ_w . The importance of this additional term may therefore be greater than envisaged by Skold et al. (2011), who suggested it mostly would be significant in low-pH environments. It seems reasonable to assume that the role of the protons also would become more dominant toward low salinity due to the reduction in surface charge density within, and hence polarization of, the Stern layer.

Our analysis of the physical significance of the DLP and SLP models provides support for both model frameworks. Although the DLP model is not parameterized sufficiently to derive expected parameter values, we find that the predicted dependence of parameter a_d , representing the polarization at the high-salinity asymptote, is very well supported by the available petrophysical data. The same was observed for parameter a_s , representing the polarization at the high-salinity asymptote for the SLP model. Parameters a_d and a_s are predicted to be controlled strongly by the ratio S_{por}/F . Interestingly, in both cases, we found that a_d and a_s were much more strongly correlated with S_{por} than S_{por}/F . Although the explanation for this finding is unclear and contrary to both models, the result is consistent with the concept of specific polarizability as defined by Weller et al. (2010b). Further work is needed to investigate why the normalization of S_{por} by F results in a weaker correlation with imaginary conductivity.

The SLP model is a sophisticated model in that it combines a macroscopic conductivity model for the polarization of a porous medium with a triple layer model for the specific surface conductivity of the Stern layer. Model parameter values describing the salinity dependence found by fitting the experimental data sets were largely consistent with the expected values based on the available petrophysical properties and SLP model parameter estimates given by others. The relatively small deviations of our fitting parameter values from those predicted from the theory may reflect different pH and salinity conditions in our experiment relative to assumed pH and salinity conditions for cited values of some of the SLP model parameters not available in our database. The good correspondence between observed and predicted fitting parameter values lends additional support to the SLP model framework.

The asymptotic behavior of σ'' at high σ_w predicted by the DLP and SLP model formulations is observed clearly in the samples making up this extensive database. However, there also is compelling evidence of a polarization peak and a subsequent reduction in σ'' at higher salinities. This decrease is not predicted by either the DLP or SLP models. Careful high-salinity measurements using two modern CR instruments indicate that this high-salinity decrease in σ'' is not an artifact resulting from limited measurement precision of small phase angles at high salinity. Lesmes and Frye (2001) suggested that such a decrease in σ'' at high σ_w would result from a reduction in the mobility of the counterions due to the accumulation of charge in the EDL and greater ion-ion interactions. In contrast, the DLP and SLP formulations assume that the surface ionic mobility of the polarized counterions is constant. Revil and Skold (2011) offer an alternative suggestion, being that such a decrease might result from an additional membrane polarization effect not included in their mechanistic model. Irrespective of the physical explanation, it is clear that $\sigma''(\sigma_w)$ may deviate from both models at high salinities. Although this deviation often may occur at salinities greater than the normal range of interest in groundwater investigations, caution must be exercised in the interpretation of low-surface-area unconsolidated sediments where the peak may occur well below 1000 mS/m (e.g., Figure 2b and samples # 58, # 63, and # 64 in Table 1).

CONCLUSIONS

This analysis of a broad database suggests a well-defined salinity dependence of σ'' that is consistent for sandstone and unconsolidated sediments. The existence of an empirical relation, the IP equivalent of Archie's law, previously has been proposed. The results of this new study can improve this relationship by constraining the likely dependence of the specific polarizability, a parameter defining the linear correlation between σ'' and S_{por} , on fluid conductivity. Based on an extensive data set, we find that the power law dependence of σ'' on σ_w at low salinities (up to the polarization maximum representing an asymptote or peak) is on average 0.34. This value is lower than the maximum of 0.5 that is expected from the DLP model and has been reported in a previous study using a limited data set of only three samples.

The data sets generally can be well fit by the salinity dependence predicted by two existing physical models describing induced polarization. Both models result in fitting model parameters that are closely related to pore volume normalized internal surface (S_{por}) as predicted by both theories. However, our findings clearly show that the popular SLP model must include a recently proposed additional

contribution from proton hopping to reproduce the salinity dependence of σ'' toward low salinity. With this addition, the SLP model faithfully reproduces a wide range of experimental data sets, with fitting parameter values generally consistent with values expected from the SLP model theory. However, numerous samples from multiple databases support the existence of a polarization peak at high salinity, with further salinity increases result in a decrease in σ'' . Although the physical explanation for this decrease is not entirely certain, existing models for describing the complex conductivity response of soils and rocks would require modification to capture this high-salinity behavior.

ACKNOWLEDGMENTS

This material is based partly upon work supported by the National Science Foundation under grant EAR-0711053, and by the German Research Foundation (DFG) under grant WE1557/10. We thank Katrin Breede, Sabine Kruschwitz, Jeanette Montrey, and Henning Schröder for providing data sets used in this work. We also thank André Revil (Colorado School of Mines) for numerous insightful discussions regarding induced polarization source mechanisms and his constructive review comments. Financial support for acquiring data on samples # 55–# 57 was provided by the Hanford Integrated Field Research Challenge (IFRC) project.

REFERENCES

- Archie, G. E., 1942, The electrical resistivity log as an aid in determining some reservoir characteristics: *Transactions of American Institute of Mineralogists Metallurgists and Petroleum Engineers*, **146**, 54–62.
- Börner, F. D., 1991, Untersuchungen zur komplexen elektrischen Leitfähigkeit von Gesteinen im Frequenzbereich von 1 Millihertz bis 10 Kiloherz: Unpublished Dissertation, Bergakademie Freiberg, Germany.
- Börner, F. D., 1992, Complex conductivity measurements of reservoir properties: *Proceedings of the Third European Core Analysis Symposium*, 359–386.
- Börner, F. D., and J. H. Schön, 1991, A relation between the quadrature component of electrical conductivity and the specific surface area of sedimentary rocks: *Log Analyst*, **32**, 612–613.
- Börner, F. D., J. R. Schopper, and A. Weller, 1996, Evaluation of transport and storage properties in the soil and groundwater zone from induced polarization measurements: *Geophysical Prospecting*, **44**, 583–601, doi: [10.1111/gpr.1996.44.issue-4](https://doi.org/10.1111/gpr.1996.44.issue-4).
- Breede, K., 2006, SIP-Messungen an Sandsteinen: Diploma thesis, Technische Universität.
- Comas, X., and L. Slater, 2004, Low-frequency electrical properties of peat: *Water Resources Research*, **40**, W12414, doi: [10.1029/2004WR003534](https://doi.org/10.1029/2004WR003534).
- Flath, D., 1989, The low-frequency complex electrical response of brine-saturated shaly sandstones: Ph.D. thesis, University of Birmingham.
- Klein, J. D., and W. R. Sill, 1982, Electrical properties of artificial clay-bearing sandstone: *Geophysics*, **47**, 1593–1605, doi: [10.1190/1.1441310](https://doi.org/10.1190/1.1441310).
- Kruschwitz, S., 2008, Assessment of the complex resistivity behavior of salt affected building materials: BAM-Dissertationsreihe, Berlin, **30**.
- Kulenkampff, J., and J. R. Schopper, 1988, Low frequency complex conductivity — A means for separating volume and interlayer conductivity: *Transactions of the 12th European Formation Evaluation Symposium*.
- Leroy, P., A. Revil, A. Kemna, P. Cosenza, and A. Ghorbani, 2008, Complex conductivity of water-saturated packs of glass beads: *Journal of Colloid and Interface Science*, **321**, 1, 103–117, doi: [10.1016/j.jcis.2007.12.031](https://doi.org/10.1016/j.jcis.2007.12.031).
- Lesmes, D. P., and K. M. Frye, 2001, Influence of pore fluid chemistry on the complex conductivity and induced polarization responses of Berea sandstone: *Journal of Geophysical Research*, **106**, 4079–4090, doi: [10.1029/2000JB900392](https://doi.org/10.1029/2000JB900392).
- Nordsiek, S., and A. Weller, 2008, A new approach to fitting induced-polarization spectra: *Geophysics*, **73**, no. 6, F235–F245, doi: [10.1190/1.2987412](https://doi.org/10.1190/1.2987412).
- Revil, A., 2012, Spectral induced polarization of shaly sands: Influence of the electrical double layer: *Water Resources Research*, **48**, W02517, doi: [10.1029/2011WR011260](https://doi.org/10.1029/2011WR011260).
- Revil, A., and N. Florsch, 2010, Determination of permeability from spectral induced polarization in granular media: *Geophysical Journal International*, **181**, no. 3, 1480–1498.
- Revil, A., and M. Skold, 2011, Salinity dependence of spectral induced polarization in sands and sandstones: *Geophysical Journal International*, **187**, no. 2, 813–824, doi: [10.1111/gji.2011.187.issue-2](https://doi.org/10.1111/gji.2011.187.issue-2).
- Rink, M., and J. R. Schopper, 1974, Interface conductivity and its implication to electric logging: *Transactions of the SPWLA 15th Annual Logging Symposium*, Paper J.
- Schröder, H., 2008, SIP-Messungen an mit unterschiedlichen Salzlösungen gesättigten Sandsteinen: Diploma thesis, Technische Universität.
- Schwarz, G., 1962, A theory of the low-frequency dielectric dispersion of colloidal particles in electrolyte solution: *The Journal of Physical Chemistry*, **66**, 2636–2642.
- Scott, J. B. T., and R. D. Barker, 2005, Characterization of sandstone by electrical spectroscopy for stratigraphical and hydrogeological investigations: *Quarterly Journal of Engineering Geology and Hydrogeology*, **38**, no. 2, 143–154.
- Skold, M., A. Revil, and P. Vaudelet, 2011, The pH dependence of spectral induced polarization of silica sands: Experiment and modeling: *Geophysical Research Letters*, **38**, L12304, doi: [10.1029/2011GL047748](https://doi.org/10.1029/2011GL047748).
- Slater, L., and D. P. Lesmes, 2002, Electrical-hydraulic relationships observed for unconsolidated sediments: *Water Resour.*, **38**, 1213, doi: [10.1029/2001WR001075](https://doi.org/10.1029/2001WR001075).
- Slater, L. D., J. Choi, and Y. Wu, 2005, Electrical properties of iron-sand columns: Implications for induced polarization investigation and performance monitoring of iron-wall barriers: *Geophysics*, **70**, no. 4, G87–G94, doi: [10.1190/1.1990218](https://doi.org/10.1190/1.1990218).
- Slater, L. D., and D. R. Glaser, 2003, Controls on induced polarization in sandy unconsolidated sediments and application to aquifer characterization: *Geophysics*, **68**, 1547–1558, doi: [10.1190/1.1620628](https://doi.org/10.1190/1.1620628).
- Stumm, W., and J. J. Morgan, 1996, Aquatic chemistry: Chemical equilibria and rates in natural waters: *Environmental Science and Technology*: John Wiley & Sons.
- Topp, G. C., J. L. Davis, and A. P. Annan, 1980, Electromagnetic determination of soil water content: measurements in coaxial transmission lines: *Water Resources Research*, **16**, no. 3, 574–582, doi: [10.1029/WR016i003p00574](https://doi.org/10.1029/WR016i003p00574).
- Vaudelet, P., A. Revil, M. Schmutz, M. Franceschi, and P. Bégassat, 2011, Induced polarization signatures of cations exhibiting differential sorption behaviors in saturated sands: *Water Resources Research*, **47**, W02526, doi: [10.1029/2010WR009310](https://doi.org/10.1029/2010WR009310).
- Vinegar, H. J., and M. H. Waxman, 1984, Induced polarization of shaly sands: *Geophysics*, **49**, 1267–1287, doi: [10.1190/1.1441755](https://doi.org/10.1190/1.1441755).
- Weller, A., K. Breede, L. Slater, and S. Nordsiek, 2011, Effect of changing water salinity on complex conductivity spectra of sandstones: *Geophysics*, **76**, no. 5, F315–F327, doi: [10.1190/geo2011-0072.1](https://doi.org/10.1190/geo2011-0072.1).
- Weller, A., S. Nordsiek, and W. Debschütz, 2010a, Estimating permeability of sandstone samples by nuclear magnetic resonance and spectral-induced polarization: *Geophysics*, **75**, no. 6, E215–E226, doi: [10.1190/1.3507304](https://doi.org/10.1190/1.3507304).
- Weller, A., L. Slater, S. Nordsiek, and D. Ntarlagiannis, 2010b, On the estimation of specific surface per unit pore volume from induced polarization: A robust empirical relation fits multiple data sets: *Geophysics*, **75**, no. 4, WA105–WA112, doi: [10.1190/1.3471577](https://doi.org/10.1190/1.3471577).
- Zimmermann, E., A. Kemna, J. Berwix, W. Glaas, and H. Vereecken, 2008, EIT measurement system with high phase accuracy for the imaging of spectral induced polarization properties of soils and sediments: *Measurement Science and Technology*, **19**, 094010, doi: [10.1088/0957-0233/19/9/094010](https://doi.org/10.1088/0957-0233/19/9/094010).

Analytical modelling of membrane polarization with explicit parametrization of pore radii and the electrical double layer

Matthias Bucker^{1,2} and Andreas Hördt¹

¹*Institute of geophysics and extraterrestrial physics, TU Braunschweig, Mendelssohnstr. 3, 38106 Braunschweig, Germany.*

E-mail: a.hoerd@tu-braunschweig.de

²*Department of Geodynamics and Geophysics, University of Bonn, Bonn, Germany*

Accepted 2013 April 4. Received 2013 April 4; in original form 2012 November 27

SUMMARY

Processes related to the membrane-like behaviour of pore throats are generally considered relevant for the induced polarization of granular media. A model to describe the impedance of membrane polarization analytically was suggested previously for a 1-D system, consisting of a sequence of narrow and wide pores of different lengths. In that model, the membrane effect is caused by different mobilities of the ion species, resulting in different ion transference numbers in the two pore types. We extend the model by explicitly including the properties of the electrical double layer and the pore radii into the impedance calculation. Our basic assumption is that in a cylindrical pore system, the ion concentrations in the direction perpendicular and parallel to the pore are independent of each other. We then obtain concentrations averaged over the pore section from known theories of the charge distribution in a cylindrical pore. We also include the effect of the Stern layer in form of a partition coefficient. Our extension enables us to use the analytical expression of the impedance, but with an extended parametrization of the transference numbers. We demonstrate the validity of our approach by comparing the results of our calculations with previously published numerical simulations for a particular model. The usefulness of our method is illustrated with parameter studies that illustrate the possible dependence of relaxation times on pore radii.

Key words: Numerical solutions; Electrical properties; Hydrogeophysics; Permeability and porosity.

INTRODUCTION

The potential of the spectral induced polarization (SIP) method to estimate hydraulic conductivity of sediments has been widely recognized. However, there is no general agreement yet which parameters derived from SIP measurements should be used. Some authors suggest to base hydraulic conductivity estimation on a characteristic timescale derived from the spectral shape of the complex conductivity (Kemna *et al.* 2005; Revil & Florsch 2010; Weller *et al.* 2010a; Revil *et al.* 2012). The alternative is to use the imaginary conductivity obtained at a single-frequency to derive the specific inner surface area, from which hydraulic conductivity may be estimated (e.g. Weller *et al.* 2010b). This approach, because of its small required bandwidth, was already applied at the field scale (e.g. Kemna *et al.* 2004; Hördt *et al.* 2009). Both methods have their theoretical justifications and were demonstrated to have some predictive abilities with laboratory data, yet the underlying concept seems to be quite different.

The uncertainty which parameters are suitable and in which form is partly because the dominating processes at the pore scale are not yet fully understood. Two main processes are being discussed: The electrochemical polarization, which is based on charge motion

within the electrical double layer (EDL) at the surface of the mineral grains, and the so-called membrane polarization, caused by narrow throats within the pore space.

The theory of electrochemical polarization, first suggested by Schwarz (1962) to describe the complex conductivity of a spherical grain, was recently extended to derive equations for more complex grain size distributions and relaxation times (Leroy *et al.* 2008; Leroy & Revil 2009; Revil & Florsch 2010). The theory is well developed and allows a number of geometrical and electrochemical parameters, such as grain size distribution, pH, ion valence and temperature, to be included in the model. The theory of membrane polarization, suggested by Marshall & Madden (1959), is based on the model of a sequence of narrow and wide pores, in which a variation of ion mobilities causes an ion concentration gradient that leads to a frequency-dependent conductivity.

The two processes (electrochemical polarization and membrane polarization) do not exclude each other, and may both be relevant. Some authors have suggested to merge them in a single model (e.g. Leroy & Revil 2009; Jougnot *et al.* 2010), but the implementation is not trivial and no breakthrough was obtained yet. Lacking such a comprehensive model, there is an ongoing debate whether one of the processes dominates the spectra observed in real measurements

(e.g. Revil *et al.* 2012). This question seems to be difficult to answer, partly because the theories generate similar falsifiable hypotheses, such as the relationship between spatial scales and relaxation times (e.g. Blaschek & Hördt 2009). Few experiments have been designed explicitly to decide between the two theories, but even then the interpretation of the results is complicated and it is difficult to arrive at definitive conclusions (e.g. Scott 2006; Hördt & Milde 2012). However, it is of course important to identify the dominating processes and to advance the corresponding theories, because even minor differences in the equations may lead to significant differences in the prediction of hydraulic parameters (e.g. Koch *et al.* 2012; Revil *et al.* 2012).

The present paper deals with the membrane polarization theory, which may be considered less developed than the electrochemical polarization, in the sense that it does not yet include the properties of the EDL. It is based on the assumption that anion and cation mobilities are different in narrow pores, and the geometry is described in terms of the lengths of pores, and not their widths (e.g. Marshall & Madden 1959). As a result, discussions of membrane polarization are usually based on the pore length as the key parameter, and existing ideas to estimate hydraulic conductivity from pore radii distributions (e.g. Juang & Holtz 1986) could not yet be fully exploited. Sometimes, the assumption is made that pore widths have the same spatial scale as the pore length (e.g. Titov *et al.* 2010). On the contrary, for electrochemical polarization, a complete theory to determine hydraulic conductivity from SIP measurements for granular media has been suggested, which is based on the convolution of a single grain size response with the grain size probability distribution (Revil & Florsch 2010).

Some extensions of the original model by Marshall & Madden (1959) have been suggested to include the pore section and the EDL properties as additional parameters into membrane polarization theory. Buchheim & Irmer (1979) included the pore section explicitly into the boundary conditions. They also suggested to use averaged ion concentrations to calculate ion transference numbers, without, however, carrying out the corresponding derivations. Sidhorova & Fridrikhsberg (1978) consider the pore section in form of an efficiency parameter. Titov *et al.* (2002) also use ion concentrations averaged over the pore section to calculate ion transference numbers, without, however, accounting for the influence of the EDL on the ion distribution within the pore. A finite pore size was considered explicitly by Volkmann & Klitzsch (2010), who carried out numerical simulations to study the dependence of the SIP spectra on geometrical parameters and water salinity. We use their results below for comparison with our analytical solutions.

Here, we include the properties of the EDL by taking into account the ion concentration as function of distance from the pore surface when averaging over the radius. We explicitly consider the Stern layer (SL) in form of a partition coefficient as suggested by Leroy *et al.* (2008). Our approach allows us to use the analytical solution originally developed by Marshall & Madden (1959) for a 1-D pore system, with pore radii as additional parameters. This opens up a whole new field of possibilities to simulate membrane polarization and possibly find better prediction equations of hydraulic parameters that are based not only on the lengths of pores, but also on their widths.

THEORY

We start with the equations for the concentrations of positive and negative univalent ions c_p and c_n (in mol m^{-3}), and the correspond-

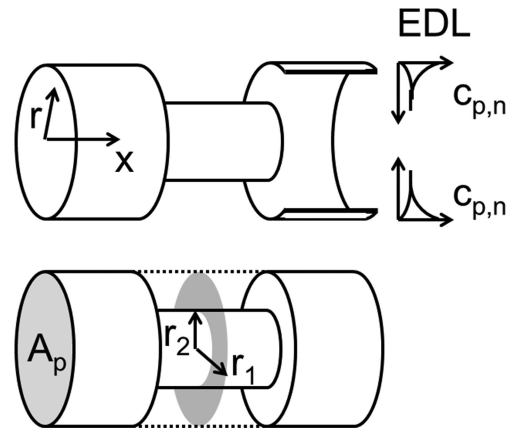


Figure 1. Schematic sketch of our model consisting of a sequence of wide and narrow pores. Top panel: definition of the coordinate system, and schematic behaviour of anomalous cation and anion concentration within the EDL. Bottom panel: definition of the pore section A_p to normalize the current, and definition of the radii r_1 and r_2 of wide and narrow pore, respectively.

ing current densities j_p and j_n (in A m^{-2}), in their 1-D form, as derived by Marshall & Madden (1959):

$$j_p = -F D_p \frac{\partial c_p}{\partial x} + F \mu_p c_p E, \quad (1)$$

$$j_n = -F D_n \frac{\partial c_n}{\partial x} - F \mu_n c_n E, \quad (2)$$

where F is Faraday's constant ($F = 96\,500 \text{ As mol}^{-1}$), D_p and D_n (in $\text{m}^2 \text{ s}^{-1}$) are diffusion coefficients of positive and negative ions, μ_p and μ_n (in $\text{m}^2 \text{ Vs}^{-1}$) denote the corresponding mobilities, respectively, and E is the electric field (in V m^{-1}). The x -direction is the direction along the pore (Fig. 1). The first term on the right-hand side describes the concentration diffusion, i.e. the tendency of ions to migrate from high to low concentration in order to maintain an equilibrium state. The second term on the right describes the transport of ions driven by the electric field. Note the difference in sign in the anion and cation equations: Both ion species tend to diffuse in the same direction (in opposite direction of the concentration gradients), but, due to the different signs of the ionic charge, in different directions when an electric field is applied.

Eqs (1) and (2) are coupled through Poisson's equation in the following form:

$$\frac{\partial E}{\partial x} = \frac{F}{\epsilon_0 \epsilon_r} (c_p - c_n), \quad (3)$$

where $\epsilon_0 = 8.85 \times 10^{-12} \text{ As Vm}^{-1}$ is the vacuum electrical permittivity and ϵ_r the relative permittivity. We used a typical value for water ($\epsilon_r = 80$) throughout this work.

In the model suggested by Marshall & Madden (1959), the induced polarization (IP) effect is generated by a variation of the mobilities along the pore. They assumed a cation selective zone where the current is mainly carried by the cations. The solution was then expressed in terms of cation and anion transference numbers t_p and t_n , representing the contribution of cations and anions to the total current, and defined as:

$$t_p = \frac{\mu_p}{\mu_p + \mu_n}, \quad (4)$$

$$t_n = \frac{\mu_n}{\mu_p + \mu_n}. \quad (5)$$

This definition results from a more general expression of transference numbers normally used in electrochemistry, which also depends on ion concentration and ion valence (e.g. Butt *et al.* 2003), by assuming univalent ions and the same concentration for both ion types. At the end of such a cation selective zone (in the direction of cation flow), cations will accumulate, and at the opposite side, a deficit will arise. Anions and cations are coupled through eq. (3). Even small concentration differences between the two ion species cause large electric field gradients, which will equilibrate anion and cation concentration. Essentially, $c_p = c_n$ can be assumed for low frequencies, which was also used by Marshall & Madden (1959) in their analytical solution and illustrated by Volkmann & Klitzsch (2010) based on numerical calculations. The ion selective zones cause an electrically neutral concentration gradient. The concentration gradients drive diffusion currents superposing the currents driven by the electric field, the net effect of which is an increase of resistance. All these processes are associated with a characteristic timescale. At high frequencies, the concentration gradients will have no time to build-up, at low frequencies they are fully expressed. The resistance of the pore will be frequency-dependent, constituting the IP-effect.

Marshall & Madden (1959) obtained equations of motion for the two ion species by taking the derivative with respect to the direction along the pore and applying the continuity equations:

$$F \frac{\partial c_p}{\partial t} + \frac{\partial j_p}{\partial x} = 0, \quad (6a)$$

$$F \frac{\partial c_n}{\partial t} + \frac{\partial j_n}{\partial x} = 0. \quad (6b)$$

They applied appropriate boundary conditions and linearized the equations system (eqs 1–3 and 6a,b) to obtain an analytical solution for the frequency-dependent impedance, hereafter called Marshall-Madden impedance, and discussed further below.

Extension to include pore radii

Marshall & Madden (1959) did not justify in detail why the transference numbers should vary along a pore, but it is natural to assume that the ion selectivity is related to the EDL. The (usually negative) charge at the surface of a mineral grain causes a fixed layer of (positive) ions attached to the mineral surface, called the SL (e.g. Schwarz 1962; Leroy *et al.* 2008). In the diffuse layer outside the SL, both ion species can move freely. Here, a cation surplus and an anion deficit develop which decrease with the distance to the pore surface (Fig. 1). The characteristic spatial scale is given by the Debye length λ_D (in m), which defines the distance from the surface of the SL, at which the charge deficit has decayed to the inverse of Euler's number of its original value:

$$\lambda_D = \sqrt{\frac{\epsilon_r \epsilon_0 k_B T}{2 c_0 e F}}. \quad (7)$$

Here, k_B is Boltzmann's constant ($1.38 \times 10^{-23} \text{ J K}^{-1}$), T is the temperature (in K), e is the electron charge ($1.6 \times 10^{-19} \text{ As}$) and c_0 (in mol m^{-3}) is the equilibrium concentration in the free electrolyte.

In narrow pores, with a radius in the range of the Debye length, the cation surplus will be significant, causing the difference in transference numbers as postulated by Marshall & Madden (1959). Therefore, it seems self-evident to formulate a model where the cation selectivity is generated by a variation of pore radii instead of mobilities. Buchheim & Irmer (1979) suggested a similar approach

to account for the EDL by including concentrations averaged over the pore radii into the definition of transference numbers, without, however, explicitly formulating the theory.

In order to replace the parametrization in terms of varying mobility by one in terms of pore radii, we describe the pore space as a sequence of cylindrical units with different radii (Fig. 1). We define an average current density $\bar{j}_p$ by integrating over the cross-section of a pore A_p . From the cation eq. (1) we obtain

$$\bar{j}_p = \frac{1}{A_p} \int_{A_p} \left(-F D_p \frac{\partial c_p}{\partial x} + F \mu_p c_p E \right) dA \quad (8)$$

and analogously for the anions. Note that $\bar{j}_p$ is normalized to the section A_p of the larger of the two pores, to account for the fact that the total current will be limited by the pore section. In the smaller pore, the integrand is zero in the space between inner and outer cylinder.

The ion concentration depends on both x and r . We now assume that c_p and c_n can be factorized as:

$$c_{p,n}(x, r) = c'_{p,n}(x) \times b_{p,n}(r), \quad (9)$$

where the functions $b_{p,n}(r)$ (dimensionless) describe the variation of $c_{p,n}$ with distance from the pore surface, which are known from theories of the electric double layer used explicitly further below. The functions $c'_{p,n}(x)$ (in mol m^{-3}) describe the variation of the concentration along the pore. Eq. (9) means that the variation along the pore and perpendicular to it do not depend on each other. This is a necessary step in order to be able to use the analytical solution with explicit parametrization of pore radii. To support the assumption, we argue that in a 2-D cylindrical pore the concentration is known to vary with distance from the pore surface, and the variation can be described with a functional form. The concentration variation in radial direction is significantly stronger (orders of magnitude) than that in the x -direction (just a small fraction of one, e.g. Blaschek & Hördt 2009). The influence, for example, of a $c'_p(x)$ variation on the $b_p(r)$ shape will be a second-order effect, probably most serious in the vicinity of the contact zone between two pores. The alternative, a full solution of the 3-D equations for particular pore systems, is far more complicated and only possible numerically. In the results section, we will discuss a comparison of our results with such a numerical solution.

In the following, we use the Nernst-Einstein relationship between diffusion constant and mobility:

$$D_p = \frac{\mu_p k_B T}{e}. \quad (10)$$

We write out the pore radii r_1 and r_2 explicitly, via

$$A_p = \pi r_1^2. \quad (11)$$

Making use of the radial symmetry and inserting eq. (9) into eq. (8), we obtain

$$\begin{aligned} \bar{j}_p = \frac{1}{\pi r_1^2} \int_0^{r_1} \left(-F \frac{\mu_p k_B T}{e} \frac{\partial (c'_p(x) b_p(r))}{\partial x} + F \mu_p c'_p(x) b_p(r) E \right) \\ \times 2\pi r dr \end{aligned} \quad (12)$$

in the larger pore, and analogously in the smaller pore, where the integration limit is r_2 , and the normalization is still carried out with r_1 . Rearrangement yields

$$\bar{j}_p = \left(-F \frac{\mu_p k_B T}{e} \frac{\partial c'_p(x)}{\partial x} + F \mu_p c'_p(x) E \right) \frac{1}{\pi r_1^2} \int_0^{r_1} b_p(r) 2\pi r dr, \quad (13)$$

which may also be written as

$$\bar{j}_p = -F \frac{\mu_p \bar{b}_p k_B T}{e} \frac{\partial c'_p(x)}{\partial x} + F \mu_p \bar{b}_p c'_p(x) E, \quad (14)$$

where $\bar{b}_p$ was defined as

$$\bar{b}_p = \frac{1}{\pi r_1^2} \int_0^{r_1} b_p(r) 2\pi r dr. \quad (15)$$

The derivation of the corresponding anion equation is completely analogous.

In eq. (14), the current density is expressed as function of the concentration gradient and the product of concentration and electric field in the same form as in eq. (1). Therefore, we may use the same solution for the impedance, but with $\bar{b}_p$ (and $\bar{b}_n$) as additional parameters. The dimensionless numbers $\bar{b}_p$ and $\bar{b}_n$ represent average ion concentrations, normalized to the concentration c_0 of the free electrolyte. They include the geometrical information, and thus the cation-selective properties of the narrow pores. It is no longer necessary to postulate the cation-selective properties and parametrize them in terms of mobility, since they come out naturally as a result of the difference in pore radius in relationship with the EDL properties.

From eq. (14), we see that the transference numbers can easily be extended to include geometry, because the parameters $\bar{b}_p$ (and $\bar{b}_n$) and the mobility occur as a product in both terms. Therefore, we define averaged transference numbers as:

$$\bar{t}_p = \frac{\mu_p \bar{b}_p}{\mu_p \bar{b}_p + \mu_n \bar{b}_n}, \quad (16)$$

$$\bar{t}_n = \frac{\mu_n \bar{b}_n}{\mu_p \bar{b}_p + \mu_n \bar{b}_n}. \quad (17)$$

These averaged transference numbers are now consistent with the conventional definition, because they include ion concentrations, the calculation of which will be explained in the next section.

Calculation of ion concentration

The calculation of the parameters $\bar{b}_{p,n}$ is equivalent to the calculation of ion concentration with distance from the mineral surface, which is based on the electrical potential. For a cylindrical pore with radius r_1 , the potential φ_{cyl} (in V) is given by (e.g. Hunter 1981)

$$\varphi_{cyl}(r) = \zeta \frac{J_0(j\kappa r)}{J_0(j\kappa r_1)}, \quad (18)$$

where $j^2 = -1$, J_0 is the zero-order Bessel function, and κ , the Debye parameter, is defined as the inverse of the Debye length as given in eq. (7). The Zeta potential ζ is the electrical potential at the surface of shear, an 'imaginary surface which is considered to lie close to the solid surface and within which the fluid is stationary' (Hunter 1981). In the model by Leroy & Revil (2004), it corresponds to the interface between the SL and the diffusive layer. Eq. (18) describes the potential from inside the pore to the surface of shear. The zone between the surface of shear and the surface of the mineral grain (the SL) will be considered in form of a correction factor.

The ion concentrations, normalized by the concentration in the free electrolyte c_0 , are then calculated from (e.g. Butt *et al.* 2003).

$$b_{n,p}(r) = e^{\pm \frac{e\varphi_{cyl}(r)}{k_B T}}, \quad (19)$$

Note that there is an asymmetry in b_p and b_n caused by the different sign of the exponent in eq. (19), which reflects, depending on the Zeta potential, the increase of the concentration of one ion species and the decrease of the concentration of the other ion species within the diffuse layer.

The approach defined by eqs (18) and (19) describes the ion concentration in the diffuse layer. The SL also contributes to the cation concentration in the EDL, but is not included in the above formalism because the cations are directly attached to the mineral surface and do not follow the differential equation describing the distribution of free ions. The contribution of the ions in the SL to the overall conductivity is a subject of current debate. For example Gimmi & Kosakowski (2011) stressed the unresolved contribution of SL cations to surface mobility values when presenting their model for surface diffusion of sorbed cations. In the absence of a generally accepted theoretical model to include the SL contribution, we follow the suggestion of Leroy & Revil (2004) and Leroy *et al.* (2008), who introduce the surplus charge per unit area σ (in As m⁻²) parallel to the pore surface. These authors define σ^d and σ^s as the charge densities in the diffuse and in the SL, respectively, and the partition coefficient f_Q as the contribution of the SL to the total surplus charge:

$$f_Q = \frac{\sigma^s}{\sigma^s + \sigma^d}. \quad (20)$$

The total surplus charge density, i.e. the sum of σ^d and σ^s , must be equal to the surface charge density of the mineral, and depends on electrochemical processes at the mineral surface, the pH and ion species of the electrolyte. More detailed treatments including tables with typical values for partition coefficients can be found in Leroy *et al.* (2008) and Leroy & Revil (2009).

Using the partition coefficient as defined in eq. (20), it can be shown (see Appendix A) that the contribution of the SL may be considered in the form of a correction to the average cation concentration

$$\tilde{b}_p = \frac{\bar{b}_p - f_Q}{1 - f_Q}. \quad (21)$$

Since $\bar{b}_p$ is larger than 1, $\tilde{b}_p$ increases monotonously with f_Q . The average cation concentration is increased by the correction, because the cations in the SL, which were so far ignored, are now added to the overall concentration. For small f_Q , corresponding to a small SL, the correction becomes negligible. The corrected form (eq. 21) will be used in the averaged transference numbers (eqs (16) and (17)), which completes our model. The full impedance equation according to Marshall & Madden (1959) with our extension and notation is given in Appendix B. The theory has been worked out here for two univalent ions, but could easily be extended to the case of several different ion types.

The pore radii will have the most notable effect on the parameters $\bar{b}_{p,n}$, because they enter with the power of two into eq. (15). However, all other parameters determining the structure of the EDL, such as temperature, equilibrium concentration of the free electrolyte and the potential at the mineral surface, enter as well through the Debye-length (eqs 7 and 18).

COMPARISON WITH NUMERICAL SIMULATION

Since our concept involves some assumptions, we evaluate its validity by comparison with numerical simulation results. The work

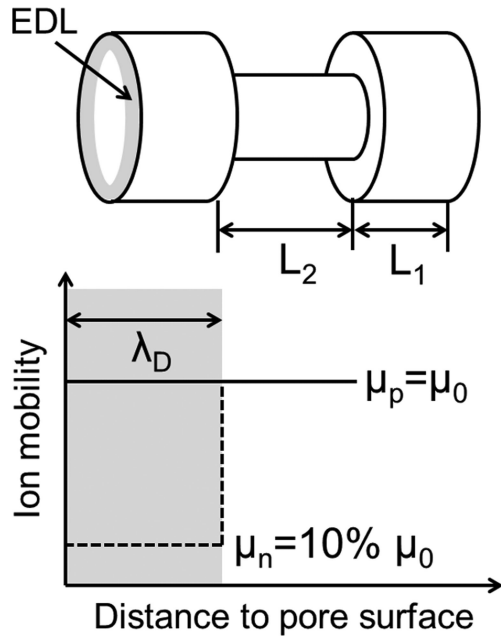


Figure 2. Sketch of the model defined by Volkmann & Klitzsch (2010) to simulate membrane polarization. Top panel: cylindrical model with pore lengths $L_1 = 5 \mu\text{m}$, $L_2 = 0.05 \mu\text{m}$. Bottom panel: definition of the outer shell with anion mobility within the EDL reduced to 10 percent compared to the free electrolyte. The pore radii are $r_1 = 0.5 \mu\text{m}$, $r_2 = 0.05 \mu\text{m}$, respectively, and the mobility in the free electrolyte is $5 \times 10^{-8} \text{ m}^2 \text{ Vs}^{-1}$.

most suitable for this purpose seems to be Volkmann & Klitzsch (2010), who modelled membrane polarization by solving the 3-D equivalent of the underlying equation system (eqs 1–3) with a commercial finite element package. Fig. 2 shows a sketch of their model we chose for comparison. It consists of two cylindrical pore types with different radii. Each pore consists of two zones, where the outer shell describes the EDL, the inner cylinder the free electrolyte. Due to restrictions in the spatial resolution, it was not possible to account for the EDL by assuming a fixed potential at the mineral surface. Instead, the EDL was simulated by assuming a layer of finite thickness, equal to the Debye length, where the anion and cation mobilities differ.

In order to find an equivalent model with our concept that should yield comparable results, we replace the change of mobilities by a change of ion concentrations (Fig. 3). Close to the pore surface, there is an anion deficit, whereas equilibrium is assumed in the inner cylinder. The parameters $\bar{b}_{p,n}$ defined above are calculated such that

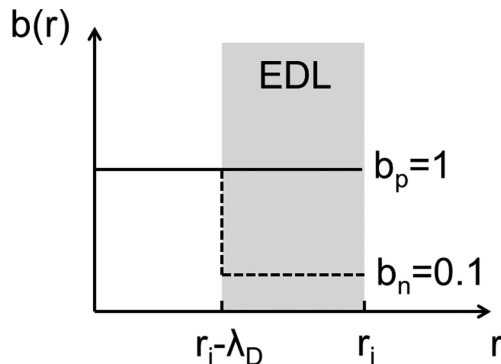


Figure 3. Sketch of the model to simulate a layered EDL, equivalent to the one described in Fig. 2. The EDL is simulated by a reduced anion concentration compared to the concentration of the free electrolyte c_0 .

the transport numbers in the two pores are the same as those in the model by Volkmann and Klitzsch. The evaluation of eq. (15) for the geometry shown in Fig. 3 yields for $\bar{b}_{p,n}$

$$\bar{b}_{pi} = 1, \quad (22)$$

$$\bar{b}_{ni} = \frac{1}{r_i^2} (r_i^2 - 0.9 (2\lambda_D r_i - \lambda_D^2)), \quad (23)$$

where the index i denotes the two pore types of different size, with $i = 1$ being the wider pore. Note that the normalization with the cross-section of the larger pore is considered in eq. (23) by using r_1 in the denominator. The corresponding averaged transference numbers are then simply obtained from eqs (16) and (17); the correction for the SL is obsolete in this case.

Several model parameters may be varied, and different parameters extracted from the resulting spectra may be used to compare the numerical results with our analytical approximation. Here, we chose the phase shifts with the maximum absolute value as a measure of the polarization effect, displayed as function of the pore radii, which is actually the new feature that can be modelled with our approach. The numerical solution and our approach agree qualitatively in the sense that the polarization first increases and then decreases again with increasing pore radius, for both ion concentrations (Fig. 4, top panel). The minimum phase values also agree

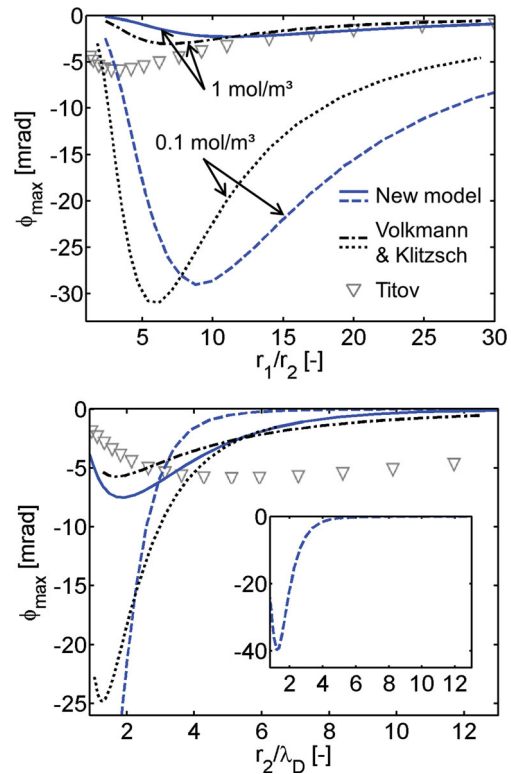


Figure 4. Phase shift with maximum absolute value of the impedance calculated for the model shown in Fig. 2 with the numerical approach (black dash-dot and dotted line) by Volkmann & Klitzsch (2010), and by our analytical approach for the model defined in Fig. 3 (blue solid and dashed lines). Top panel: maximum phase shift versus ratio of pore radii (r_1 is varied), for two different concentrations of the free electrolyte. Bottom panel: maximum phase shift versus radius of the smaller pore, normalized by the EDL thickness. The inset shows the dashed line with a different scaling. The triangles denote an attempt by Volkmann & Klitzsch (2010) to compare their results with a model by Titov *et al.* (2002).

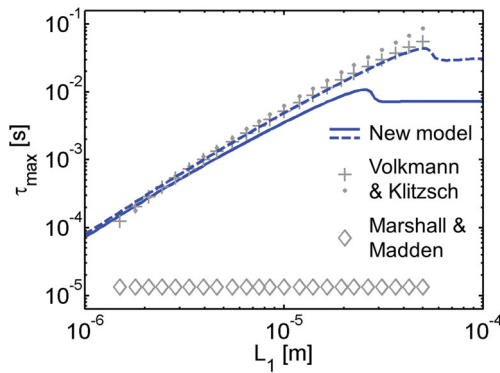


Figure 5. Relaxation time (inverse of angular frequency of maximum phase shift), versus length of the wider pore L_1 . Radii and mobilities are as given in Fig. 2. Dots and crosses: Numerical results from Volkman & Klitzsch (2010) for $c_0 = 0.1 \text{ mol m}^{-3}$ and $c_0 = 1 \text{ mol m}^{-3}$, respectively. Diamonds: Marshall–Madden impedance without expansion, as calculated by Volkman & Klitzsch (2010). Lines: expanded Marshall–Madden impedance for $c_0 = 0.1 \text{ mol m}^{-3}$ (dashed) and 1 mol m^{-3} (solid).

very well (e.g. approx. -30 mrad for 0.1 mol m^{-3}), whereas there is some difference in the radius at which the minimum is observed. On the other hand, the agreement between the results of Volkman & Klitzsch (2010) and the results obtained with the so-called ‘short narrow pore (SNP)’ model by Titov *et al.* (2002) is poor, indicating that assumptions of the SNP-model are poorly fulfilled.

The qualitative agreement between the numerical solution and our model also holds as function of the smaller pore radius (Fig. 4, bottom panel). The polarization adopts a maximum if the smaller pore radius is in the range of the Debye length (10^{-8} m for the chosen parameters), and then quickly decreases. For the smaller ion concentration, there is a difference of approx. 50 per cent in the minimum phase shifts between the numerical and our solution. Again, the comparison with the SNP-model is considerably more difficult.

The relaxation times obtained from the numerical simulation also agree very well with those obtained from our expanded Marshall–Madden impedance (Fig. 5). The relaxation times are defined by

$$\tau_{\max} = \frac{1}{\omega_{\max}}, \quad (24)$$

where $\omega_{\max}$ is the angular frequency at which the maximum phase shift occurs. The length of the wide pore is varied, because this parameter yields the largest dynamics of $\tau_{\max}$ for this particular set of parameters. The comparison of the relaxation times by Volkman & Klitzsch (2010) with those obtained from the conventional Marshall–Madden impedance is apparently difficult because pore radii cannot be included. We also conclude from Fig. 5 that the relaxation times are only weakly dependent on salinity: the curves for the two concentrations differ only slightly, consistent with experimental observations (e.g. Weller *et al.* 2011).

The causes of the differences between our approach and the numerical solution are not exactly known. A plausible candidate may be our basic assumption that concentrations in radial and axial directions are independent (eq. 9), which basically ignores 2-D and 3-D effects. Secondly, the numerical solution will neither be perfect. We conclude from this study that our concept is feasible at least for studies of the qualitative behaviour of the impedances. It may be used for quantitative estimates as well; in particular the relaxation times seem to be consistent.

The model used for comparison with numerical simulations (Fig. 3) may be oversimplified, not only because the EDL is de-

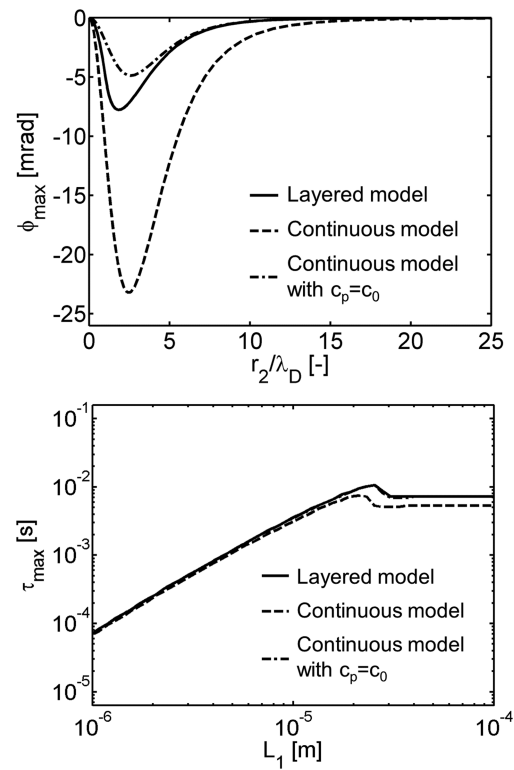


Figure 6. Maximum phase shift and relaxation time calculated with our new model. In both panels, the solid line was calculated to approximate the layered model shown in Fig. 3 and is identical to the solid line in Figs 4 (bottom panel) and 5, respectively, for $c_0 = 1 \text{ mol m}^{-3}$. The dashed and dash-dotted lines were calculated for more realistic continuous double layer models, further explained in the main section. Top panel: phase shift with maximum absolute value versus normalized radius of the smaller pore. Bottom panel: relaxation time versus length of the wider pore.

scribed as a discrete layer, but also because only the anion transference number is varied. We carry out the transition from the layered model to a model with a realistic continuous EDL as described by eqs (18) and (19) in two steps (Fig. 6). First, instead of forcing the EDL thickness and the concentrations to certain values, as in Fig. 3, we use the Zeta potential as a free parameter. The Zeta potential is determined such that the anion concentration averaged over the pore section is the same as that defined by Fig. 3 with the corresponding parameters for c_0 (1 mol m^{-3}) and μ ($5 \times 10^{-8} \text{ m}^2 \text{ Vs}^{-1}$). Assuming that the radius of the pore is large compared to the Debye length, we can determine a single value for the Zeta potential, independent of the pore radius ($\zeta = -0.0293 \text{ V}$ in this case). If the assumption of a large pore radius is violated, the concentration will not relax to c_0 in the middle of the pore and the equivalent Zeta potential will depend on the pore radius. The cation concentration is still kept constant at $c_p = c_0$ in this first step. The difference between this model and the layered model is small (dash dotted lines in Fig. 6), except for small pore radii (Fig. 6, top panel), which is due to the violation of the assumption of a large pore radius.

In a second step, we use the Zeta potential calculated above, but now allow the cation concentration to vary as well. The variation of cation concentration has a large impact on the maximum phase shifts (Fig. 6, top panel). The influence on the relaxation times (bottom panel) is smaller, and limited to relatively long narrow pores. We conclude that a layered EDL model may constitute a useful approximation that does not produce dramatic errors compared to a

continuous EDL model. For quantitative estimates and small pore radii, however, the continuous model should be used.

PARAMETER STUDIES

The inclusion of pore radii as new parameters in the analytical impedance formulation renders systematic parameter studies possible, to investigate under which conditions membrane polarization might make a significant contribution to the overall polarization of natural media. With numerical simulations, only a limited parameter regime can be exploited for practical or technical reasons, such as computing time and discretization issues (e.g. Blaschek & Hördt 2009; Volkmann & Klitzsch 2010). From previous studies it is evident that the parameter space of the Marshall–Madden impedance has different regimes in which different relationships between, for example, relaxation times and geometrical parameters exist. Thus, important relationships might be overlooked when investigating limited regions.

Here, we have further increased the number of parameters by the properties of the EDL and the radii of the two pores, and it is not trivial to derive general relationships or to fully exploit the parameter space. Therefore, we only discuss two examples here to demonstrate the new possibilities of our expansion, and leave a systematic analysis for future work.

First, we display the dependence of the IP parameters (maximum phase shift and relaxation time) on the radii of the two pores (Fig. 7). This is one of the new possibilities compared to previous analytical studies, where the geometrical parameters are usually the lengths of pores, and not their widths. All other parameters are fixed and given in the figure caption. This time, we apply the model with the continuous EDL (eq. 19), and thus there is no exact correspondence with any model by Volkmann & Klitzsch (2010). However, the parameters were chosen such that the transference numbers are comparable, i.e. the parameters for their figs 15 and 16 would each form a horizontal line in our figure, crossing in the centre.

The phase shift adopts a maximum in a region where the ratio between the two pore radii is approximately 10:1. On this diagonal, there is a maximum if the radius of the smaller pore is approx. equal to the Debye length (10^{-8} m for the chosen parameters), which is a measure of the thickness of the EDL. This means that maximum phase shifts should be observed for extremely small pore radii. The phase values are relatively large, with a maximum in the order of 100 mrad, and even for larger pore radii considerable phase shifts in the mrad range are obtained.

The relaxation time (bottom panel) also depends on the pore radii. This fact is important in itself, because in the original Marshall–Madden impedance, there is only a weak dependence of relaxation time on the mobility contrast (e.g. Blaschek & Hördt 2009), which might imply that the dependence on pore radii is also weak. In fact, the largest variation of relaxation time occurs where the radii of the two pores are approximately equal, which is the range where the two pores change their role as narrow and wide pore. In this region, the IP-effect itself (measured by the maximum phase shift) is weak (Fig. 7, top panel), and thus the dependence of relaxation times might be considered somewhat artificial. However, the relationships are apparently not simple, as also indicated by the broken symmetry in the lower left corner of Fig. 7 (bottom panel) which is caused by the asymmetry in pore lengths.

Our second parameter study shows that the ratio of pore radii that provides the maximum phase shift depends on the length ratio (Fig. 8, top panel). The line of maximum phase shift indicated in

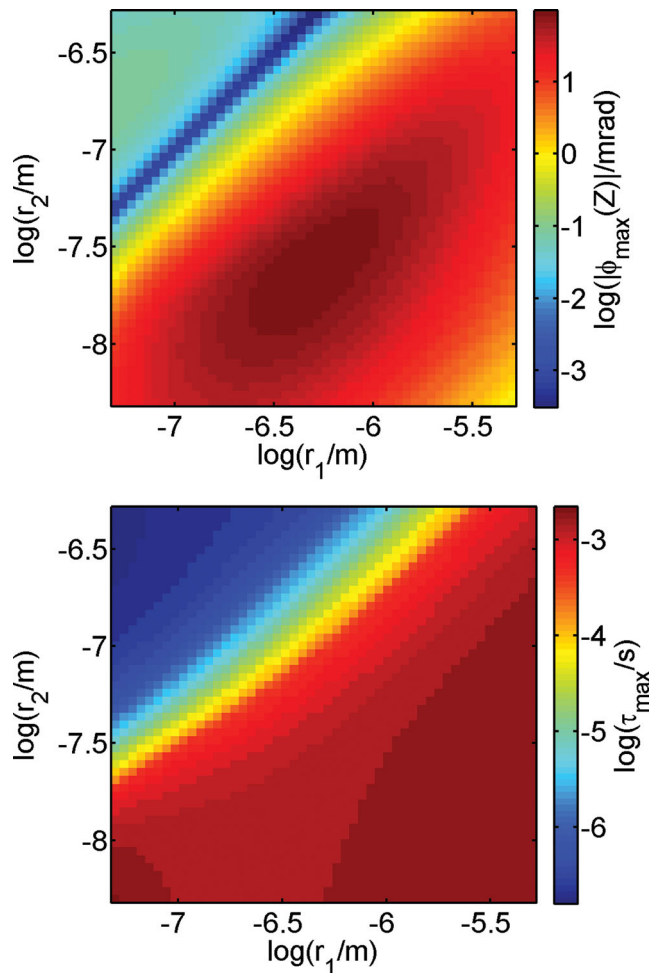


Figure 7. Top panel: maximum phase shift, colour coded with logarithmic colour scale as function of the two pore radii, for the model sketched in Fig. 1. Bottom panel: relaxation time, calculated as the inverse of the angular frequency at which the maximum phase shift occurs. The pore lengths are $L_1 = 5 \mu\text{m}$, $L_2 = 0.05 \mu\text{m}$, the mobility and the ion concentration in the free electrolyte are $5 \times 10^{-8} \text{ m}^2 \text{ Vs}^{-1}$ and 1 mol m^{-3} , respectively. The partition coefficient is $f_Q = 0.6$ and the Zeta potential is -75 mV .

the figure is defined by

$$\frac{r_1}{r_2} = \sqrt{\frac{L_1}{L_2}}. \quad (25)$$

Obviously, the 10:1 ratio of the pore radii found in Fig. 7 is a specific case that was derived for the length ratio 100:1. The relationship given by eq. (25) may not be generally valid, because the radius and length of the second pore are fixed, and the figure may depend on them. A full generalization is difficult, because the Debye length introduces an absolute length scale.

Most of the phase shift values in Fig. 8 are out of the currently measurable range of approx. 0.1 mrad, and only the dark red section in the lower left corner would be considered relevant from a practical point of view. We also observe that point symmetry is lacking in the figure. When crossing the horizontal line $r_1 = r_2$, the two pores apparently change their role, and one might expect point symmetry with respect to the point (0,0) in this double logarithmic scale. The symmetry is broken because we had to choose a fixed value for L_2 , and the geometrical parameters are all associated with an absolute value (the Debye length).

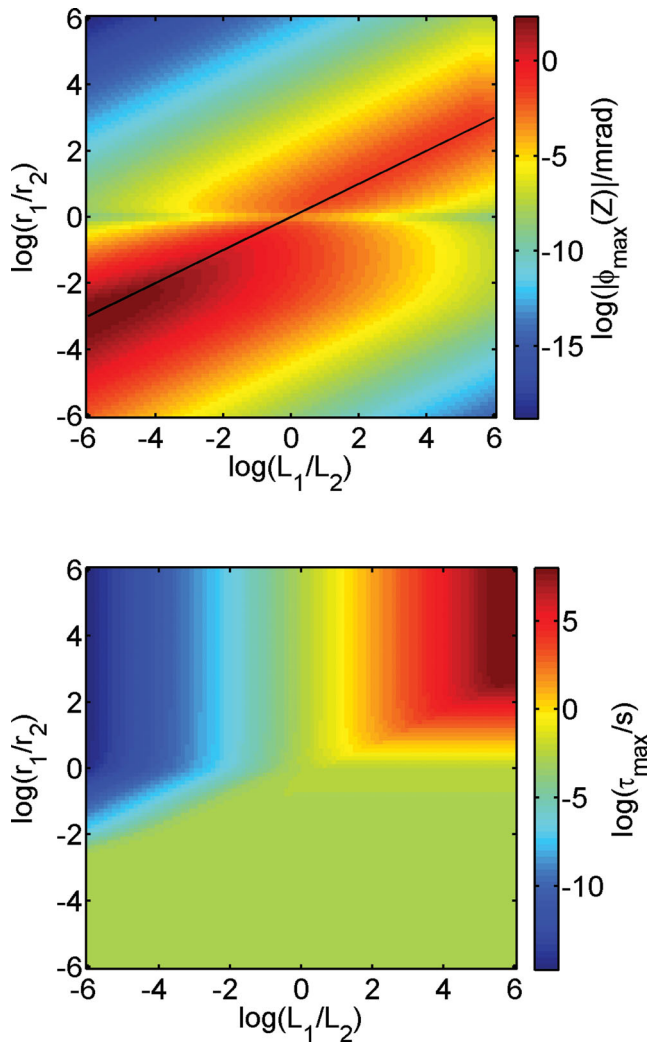


Figure 8. Top panel: maximum phase shift, colour coded with logarithmic colour scale as function of pore radii ratio and the pore length ratio, for the model sketched in Fig. 1. The line indicates the maximum phase shift. Bottom panel: relaxation time, calculated as the inverse of the angular frequency at which the maximum phase shift occurs. The pore radius r_2 is fixed to $r_2 = 5 \times 10^{-6}$ m, whereas r_1 is varied, and the pore length L_2 is fixed to $L_2 = 5 \times 10^{-6}$ m. The mobility and the ion concentration in the free electrolyte are 5×10^{-8} ($\text{m}^2 \text{Vs}^{-1}$) and 1 mol m^{-3} , respectively. The partition coefficient is $f_Q = 0$ and the Zeta potential is -75 mV .

The relaxation times exhibit different regimes characterized by different dependencies on the geometrical parameters (Fig. 8, bottom panel). For narrow pores (small r_1), relaxation times are independent of pore radii and pore length L_1 . The relaxation times depend on the length of the wide pore (L_2), which does not cause any observable variation because L_2 was fixed. For large radii r_1 (top section of Fig. 8, bottom panel), relaxation times depend on L_1 because the pores changed role, and pore 1 is the wide pore. The relaxation times depend on pore radii preferably in the transition zone around the horizontal line $r_1 = r_2$, consistent with the finding from the previous discussion of Fig. 7. The role of the transition zone can be understood from the behaviour of the Marshall–Madden impedance. It can be shown (Bücker 2011) that each pore is associated with its own relaxation time, the shorter of which dominates the impedance behaviour. In the transition zone, where the pores change role, the dominating relaxation time will also change. The phase shifts in the transition region are small, and the range where

this dependence is relevant in practice might also be small. We can conclude that while the relaxation times are independent of pore widths in certain parameter regimes, the independence does not hold as a general rule.

CONCLUSIONS

We have suggested an extension of the 1-D Marshall–Madden impedance to include the properties of the EDL and the pore radii. The inevitable, but well justified assumption is that the ion concentrations in the directions parallel and perpendicular to the axis of the cylindrical pore system are independent of each other. This ignores 2-D effects, which will be significant in particular in the vicinity of contact zones between the pores. Our comparison with numerical simulation results indicates that this does not cause dramatic errors in the impedance.

Our approach will be useful to verify the feasibility of numerical simulations at least qualitatively. Since the required spatial scales may span several orders of magnitude, the discretization of the equation system can be an issue, and the possibility to compare the results for simple models with an analytical solution may be helpful.

The extension opens up a wide range of new simulation possibilities. A much wider parameter range can be explored, which should compensate for the minor drawback that 2-D effects have to be ignored. The parameter space has become quite large: in addition to the two pore lengths and two pore radii, the equilibrium concentration and the ion mobilities, the parameters characterizing the EDL, i.e. Zeta potential (which depends on further parameters, such as pH), temperature, and partition coefficient, can also be considered in the model. In spite of the large number of parameters that can be varied, it should be possible through systematic studies, analytical derivations and scaling laws to increase the general understanding of membrane polarization and to derive general conclusions for the behaviour of the impedance.

The parameter studies indicate that the relaxation time derived from the impedance spectra may depend on the pore radii. This might be important for the prediction of hydraulic conductivity. The importance of using pore throat size in hydraulic conductivity prediction was also recognized by Revil *et al.* (2012) who suggested a new relationship that includes the dynamic pore radius rather than grain diameter, without a related mechanistic model, however. Kruschwitz *et al.* (2010) hypothesized that pore throat size controls different regimes of relaxation times. Our second preliminary conclusion is that the maximum polarization occurs when the narrower pore has the size of the Debye length and the ratio of the pore widths equals the square root of the pore length ratio.

Finally, our theory might constitute a step towards merging grain-based and pore space-based theories. The explicit consideration of the EDL properties in a membrane polarization model provides a connection between the two types of theories that might be exploited in future work.

ACKNOWLEDGEMENTS

We would like to thank Norbert Klitzsch and Jan Volkmann for fruitful discussion of this work, and for providing their data displayed in Figs 4 and 5. Christian Fulda gave some useful hints that improved the original version of the manuscript. We thank Damien Jougnot and an anonymous reviewer for their deep and helpful comments. The work was carried out within the gebo research association, sponsored by Baker Hughes and the Ministry of Science and Culture of Lower Saxony, Germany.

REFERENCES

- Blaschek, R. & Hördt, A., 2009. Numerical modelling of the IP-effect at the pore scale, *Near Surf. Geophys.*, **7**, 579–588.
- Buchheim, W. & Irmer, G., 1979. Zur Theorie der induzierten galvanischen Polarisation in Festkörpern mit elektrolytischer Porenfüllung, *Gerlands Beitr. Geophysik*, **88**, 53–73.
- Bucker, M., 2011. Die Membranpolarisation als Ursache des IP-Effekts in porösen Medien, *MSc thesis*, TU Braunschweig.
- Butt, H.-J., Graf, K. & Kappel, M., 2003, *Physics and Chemistry of Interfaces*, Wiley-VCH, Weinheim.
- Gimmi, T. & Kosakowski, G., 2011. How mobile are sorbed cations in clays and clay rocks? *Environ. Sci. Technol.*, **45**, 1443–1449.
- Hördt, A., Blaschek, R., Binot, F., Druventak, A., Kemna, A., Kreye, P. & Zisser, N., 2009. Case histories of hydraulic conductivity estimation with induced polarization at the field scale, *Near Surface Geophys.*, **7**, 529–545.
- Hördt, A. & Milde, S., 2012. Studies on the origin of induced polarisation with gel-filled sandstone samples, *Near Surface Geophys.*, **6**, 469–478.
- Hunter, R.J., 1981. *Zeta Potential in Colloid Science: Principles and Applications*, Academic Press, London.
- Jougnot, D., Ghorbani, A., Revil, A., Leroy, P. & Cosenza, P., 2010. Spectral induced polarization of partially saturated clay-rocks: a mechanistic approach, *Geophys. J. Int.*, **180**, 210–224.
- Juang, C.H. & Holtz, R.D., 1986. A probabilistic permeability model and the pore size density function, *Int. J. Numer. Anal. Methods Geomech.*, **10**, 543–553.
- Kemna, A., Binley, A. & Slater, L., 2004. Crosshole IP imaging for engineering and environmental applications, *Geophysics*, **69**, 97–107.
- Kemna, A., Münch, H.M., Titov, K., Zimmermann, E. & Vereecken, H., 2005. Relation of SIP relaxation time of sands to salinity, grain size and hydraulic conductivity, in *Proceedings of the 11th Eur. Meeting Environ. Eng. Geophys., Exp. Abst.*, European Association of Geoscientists & Engineers (EAGE), Palermo, Italy.
- Koch, K., Revil, A. & Holliger, K., 2012. Relating the permeability of quartz sands to their grain size and spectral induced polarization characteristics, *Geophys. J. Int.*, **190**, 230–242.
- Kruschwitz, S., Binley, A., Lesmes, D. & Elshenawy, A., 2010. Textural controls on low-frequency electrical spectra of porous media, *Geophysics*, **75**, WA113–WA123.
- Leroy, P. & Revil, A., 2004. A triple-layer model of the surface electrochemical properties of clay minerals, *J. Colloid Interface Sci.*, **270**, 371–380.
- Leroy, P. & Revil, A., 2009. A mechanistic model for the spectral induced polarization of clay materials, *J. geophys. Res.*, **114**, B10202, doi:10.1029/2008JB006114.
- Leroy, P., Revil, A., Kemna, A., Cosenza, A. & Ghorbani, A., 2008. Complex conductivity of water-saturated packs of glass beads, *J. Colloid Interface Sci.*, **321**, 103–117.
- Marshall, D.J. & Madden, T.R., 1959. Induced polarization: a study of its causes, *Geophysics*, **24**, 780–816.
- Revil, A. & Florsch, N., 2010. Determination of permeability from spectral induced polarization in granular media, *Geophys. J. Int.*, **181**, 1480–1498.
- Revil, A., Koch, K. & Holliger, K., 2012. Is it the grain size or the characteristic pore size that controls the induced polarization relaxation time of clean sands and sandstones? *Water Res. Res.*, **48**, W005602, doi:10.1029/211WR011561.
- Schwarz, G., 1962. A theory of the low-frequency dielectric dispersion of colloidal particles in electrolyte solution, *J. Phys. Chem.*, **66**, 2636–2642.
- Scott, J.B., 2006. The origin of the observed low-frequency electrical polarization in sandstones, *Geophysics*, **71**, 235–238.
- Sidhorova, M. & Fridrikhsberg, D., 1978. Study of induced polarization of systems simulating soils during freezing, in *Proceedings of the Permafrost: The USSR Contribution to the Second International Conference*, National Academy of Sciences, pp. 347–349.
- Titov, K., Komarov, V., Tarasov, V. & Levitski, A., 2002. Theoretical and experimental study of time domain-induced polarization in water-saturated sands, *J. Appl. Geophys.*, **50**, 417–433.
- Titov, K., Tarasov, A., Ilyin, Y., Seleznev, N. & Boyd, A., 2010. Relationships between induced polarization relaxation time and hydraulic properties of sandstone, *Geophys. J. Int.*, **180**, 1095–1106.
- Volkman, J. & Klitzsch, N., 2010. Frequency-dependent electric properties of microscale rock models for frequencies from one millihertz to ten kilohertz, *Vadoze Zone J.*, **9**, 858–870.
- Weller, A., Nordsiek, S. & Debschütz, W., 2010a. Estimating permeability of sandstone samples by nuclear magnetic resonance and spectral-induced polarization, *Geophysics*, **75**, E215–E226.
- Weller, A., Slater, L. & Nordsiek, S., 2010b. On the estimation of specific surface per unit volume from induced polarization: a robust empirical relation fits multiple data sets, *Geophysics*, **75**, WA105–WA112.
- Weller, A., Breede, K., Slater, L. & Nordsiek, S., 2011. Effect of changing water salinity on complex conductivity spectra of sandstones, *Geophysics*, **76**, F315–F327.

APPENDIX A: STERN LAYER CORRECTION

In order to derive the correction for the contribution to the Stern layer (SL), we first relate the surplus charge per unit area in the diffuse layer σ^d to the normalized average cation concentration $\tilde{b}_p$

$$2\pi r \sigma^d = F c_0 A_p (\tilde{b}_p - 1). \quad (A1)$$

The left-hand side is a surplus charge per unit length, where the length is in the direction of the pore. The right-hand side is the normalized surplus ion density ($\tilde{b}_p - 1$) multiplied with F and c_0 to convert it into a charge density, and with the pore section to convert it into a charge per unit length density. In the corrected form, the contribution of the SL is considered as well, and thus we can write the equation for the corrected normalized cation concentration $\tilde{b}_p$

$$2\pi r (\sigma^d + \sigma^s) = F c_0 A_p (\tilde{b}_p - 1). \quad (A2)$$

Division of eq. (A2) by (A1) yields:

$$\frac{\tilde{b}_p - 1}{\tilde{b}_p - 1} = \frac{\sigma^d + \sigma^s}{\sigma^d}. \quad (A3)$$

The final form (eq. (21) of the main section) is obtained after some algebra by solving for $\tilde{b}_p$ and using the definition of the partition coefficient (eq. 20).

APPENDIX B: EXTENDED MARSHALL-MADDEN IMPEDANCE

The impedance derived by Marshall & Madden (1959) for a 1-D pore system, extended for the 2-D system defined in Figs 1 and 2, writes

$$Z(\omega) = \frac{L_1}{\mu_{p1} \tilde{b}_{p1} c_0 F} \left[\tilde{t}_{p1} + \frac{\tilde{B}}{A} \tilde{t}_{p2} + \frac{(\tilde{S}_2 - \tilde{S}_1)^2}{\frac{\tilde{X}_1(\omega) \tilde{S}_1}{\tilde{t}_{p2}^2 \tilde{b}_{p1} \tanh(\tilde{X}_1(\omega))} + \frac{A}{B} \frac{\tilde{X}_2(\omega) \tilde{S}_2}{\tilde{t}_{p1}^2 \tilde{b}_{p2} \tanh(\tilde{X}_2(\omega))}} \right], \quad (B1)$$

where

$$\tilde{X}_i(\omega) = \left(\frac{j\omega}{2D_{pi} \tilde{t}_{pi} \tilde{b}_{pi}} \right)^{\frac{1}{2}} \frac{L_i}{2}. \quad (B2)$$

This is the same notation as that used in the original work, extended by the tilde to indicate values averaged over the pore section, and the

correction for the SL. Parameter A is the length ratio, and remains unchanged compared to the original definition:

$$A = \frac{L_1}{L_2}. \quad (\text{B3})$$

Parameter $\tilde{B}$ is the ratio of diffusivities, multiplied with the ratio of averaged, normalized and corrected concentrations

$$\tilde{B} = \frac{D_{p1}}{D_{p2}} \frac{\tilde{b}_{p1}}{\tilde{b}_{p2}}. \quad (\text{B4})$$

$\tilde{S}$ is the ratio of anion and cation transference numbers in each of the two zones

$$\tilde{S}_i = \frac{\tilde{t}_{ni}}{\tilde{t}_{pi}}. \quad (\text{B5})$$

The transference numbers are given by

$$\tilde{t}_{p,ni} = \frac{\mu_{p,ni} \tilde{b}_{p,ni}}{\mu_{pi} \tilde{b}_{pi} + \mu_{ni} \tilde{b}_{ni}} \quad (\text{B6})$$

with

$$\tilde{b}_{pi} = \frac{\bar{b}_{pi} - f_Q}{1 - f_Q}, \quad (\text{B7})$$

whereas

$$\tilde{b}_{ni} = \bar{b}_{ni} \quad (\text{B8})$$

because the anion concentration is unaffected by the SL.

In the examples shown in this work, we assume that diffusivities (and mobilities) are equal in the two zones and for the two ion species, i.e.

$$D_{pi} = D_{ni} = D_{0i} \quad (\text{B9})$$

$$D_{01} = D_{02} = D_0 \quad (\text{B10})$$

and analogously for the mobilities, which simplifies some of the equations. We leave them here in the general form to maintain the option to consider varying diffusivities.

Long and short narrow pore models for membrane polarization

Matthias Buecker¹ and Andreas Hoerdt²

ABSTRACT

Many recent approaches to derive hydraulic soil and rock parameters from spectral induced polarization data are based on the determination of the characteristic time of measured impedance spectra. The characteristic time, sometimes also termed *relaxation time*, depends on the characteristic length scale of the pore geometry. One commonly used theory describing the complex electrical impedance of porous media is the membrane polarization. The pore space is modeled as a sequence of wide and narrow pores with different anion and cation mobilities. A question of particular interest is the dependence of the characteristic time on the pore lengths. The dominance of diffusion processes leads to a quadratic dependence of the characteristic time on the length scale of the pore system. However, analytical and numerical studies revealed an inconsistency concerning the pore length that determines the characteristic time.

Whereas the analytical short narrow pore (SNP) model predicts a dependence on the length of the narrow pore, the length of the wide pore dominates in various numerical studies. We propose a new SNP model, which involves fewer assumptions concerning the ionic mobilities. Our new SNP model is a special case of the Cole-Cole impedance. Its characteristic time is controlled by the length of the narrow pore. A second model is derived for the case of long narrow pores and termed the *long narrow pore* model. The resulting impedance compares, but is not equal to a special case of the Cole-Davidson impedance and its characteristic time varies with the length of the wide pore. Both new models are special cases of the Marshall-Madden impedance for membrane polarization. The limitation of their respective validity with respect to certain combinations of mobilities and pore lengths, allows for the coexistence of both behaviors and resolves the apparent contradiction between theoretical and numerical studies.

INTRODUCTION

Induced polarization (IP) is a geophysical method based on the frequency-dependent complex resistivity of the near subsurface. Depending on the composition of the subsurface, different polarization mechanisms give rise to a low-frequency dispersion (typically between 1 mHz and 1 kHz) of the complex resistivity (or IP effect). In rocks with metallic constituents, electrode polarization causes a pronounced IP effect (e.g., Bleil, 1953; Pelton et al., 1978; Wong, 1979). With the appearance of more precise data acquisition systems, new applications beyond ore exploration emerged: Today, even small phases (below 1 mrad) caused by clayey sediments, sands, and microfractured rocks can be measured. In these cases, IP signatures are primarily caused by polarization mechanisms related to the electrical double layer (EDL) at the mineral-fluid interface (e.g., Schwarz, 1962; Vinegar and Waxman, 1984), giving rise to new

applications in environmental and hydrogeophysics. For instance, the specific surface area shows a strong correlation with the imaginary part of the complex conductivity (e.g., Börner and Schön, 1991; Slater et al., 2006; Kruschwitz et al., 2010; Weller et al., 2010b) offering a geophysical access to this soil parameter. Because hydraulic permeability also depends on the specific surface area, it may be related to the imaginary part of the complex conductivity by means of empirical equations (e.g., Pape et al., 1987; Börner et al., 1996; Slater and Lesmes, 2002). A similar approach uses the characteristic time scale of the measured resistivity spectrum to derive the hydraulic permeability (e.g., Binley et al., 2005; Kemna et al., 2005; Revil and Florsch, 2010). Therefore, spectral induced polarization (SIP) seems to be a promising method for noninvasive mapping and sounding of hydraulic soil and rock parameters also at the field scale (e.g., Hoerdt et al., 2009). Because of its sensibility with respect to variations in pore fluid chemistry (e.g., Angoran and Madden, 1977; Weller et al.,

Manuscript received by the Editor 30 December 2012; revised manuscript received 29 May 2013; published online 7 October 2013.

¹University of Bonn, Department of Geodynamics and Geophysics, Bonn, Germany and TU Braunschweig, Institute of Geophysics and Extraterrestrial Physics, Braunschweig, Germany. E-mail: buecker@geo.uni-bonn.de.

²TU Braunschweig, Institute of Geophysics and Extraterrestrial Physics, Braunschweig, Germany. E-mail: a.hoerdt@tu-braunschweig.de.

© 2013 Society of Exploration Geophysicists. All rights reserved.

2011) and to geometric alterations of the pore space caused by different contaminants, SIP has also been used for the delineation and characterization of contaminant plumes (e.g., Vanhala, 1997; Schmutz et al., 2010; Flores Orozco et al., 2012).

To reduce the information in the full spectra to a few meaningful parameters, empirical models have been used in data processing and interpretation. The most universal of these models — expressed in terms of a frequency-dependent impedance — is the generalized Cole-Cole model (Pelton et al., 1978), also known as *Havriliak-Negami relaxation* (Havriliak and Negami, 1967)

$$Z_{GCC}(\omega) = Z_{DC} \left[1 - \eta_0 \left(1 - \frac{1}{(1 + (i\omega\tau_{GCC})^c)^a} \right) \right]. \quad (1)$$

Here, ω is the angular frequency, Z_{DC} the low-frequency limit of the impedance, η_0 the polarizability (also termed *chargeability*), and τ is the time constant. The generalized Cole-Cole model, equation 1, is equivalent to the Cole-Cole model (Cole and Cole, 1941), if the exponent a equals unity, and to the Cole-Davidson model, if the exponent c equals unity. Generalized Cole-Cole models or superpositions of these models are apt to fit a wide range of experimental data, but provide little information about the underlying physical processes.

To overcome the shortcoming of the empirical models, and to understand the physical basics of the IP-effect in absence of metallic constituents, mainly two pore-scale theories have been discussed in literature. The grain-based model developed by O'Konski (1960), Schwarz (1962), and Schurr (1964) for dispersed spherical mineral particles describes the polarization of the Stern layer, a thin layer of ions directly adsorbed to the mineral surface. This model has recently been extended by Leroy et al. (2006, 2008) and Leroy and Revil (2009) who incorporated a detailed electrochemical description of the EDL and successfully predicted the impedance of granular media. The second model attributes the IP effect to the membrane-like behavior of narrow passages or clay particles distributed along the pore channel. Membrane polarization was first studied by Marshall and Madden (1959) for a pore space model consisting of a sequence of alternating zones. The zones differ in their length, and to account for the membrane-like behavior, reduced transport numbers (product of ion mobility and concentration) are assigned to the zones representing narrow passages. Subsequently, the pore space based model was modified to include the influence of the pore cross section (Fridrikhsberg and Sidorova, 1961) and to allow for a quantitative estimation of the influence of the EDL on membrane polarization (Buchheim and Irmer, 1979).

Motivated by the strong link between soil hydraulic parameters and SIP measurements, more recent studies have focused on the relation between the characteristic time and pore dimensions (i.e., length and diameter). Starting from the original membrane polarization model, Titov et al. (2002) developed an impedance model for the special case of very short narrow pores (SNP). One of their findings is that the characteristic time shows a quadratic dependence on the length of the narrow passages. On the other hand, based on numerical simulations Blaschek and Hördt (2009) and Volkmann and Klitzsch (2010) report a quadratic dependence on the length of the wide pore.

The small number of existing theoretical studies on membrane polarization is in contrast to a high demand for a better understanding of the underlying physical processes, necessary to profoundly

interpret field and laboratory data. The objective of this study is to clarify the apparently contradicting results of the aforementioned studies concerning the relation between characteristic time and pore length and at the same time provide a deeper insight into the physics of membrane polarization.

In the first section, we briefly review the basics of membrane polarization theory as described by Marshall and Madden (1959) for a 1D pore system. We express the impedance formula of these authors in terms of DC-limit, polarizability, and two time constants, one for each zone. A parameter study varying the lengths of the two zones reveals two clearly distinguishable regimes with respect to the characteristic time-pore length relation.

In the section that follows, we analytically derive two new approximate impedance models, which allow for the analytic determination of their characteristic times. The assumption of long narrow pores leads to an impedance model with a characteristic time that is determined solely by the length of the wide pore. This finding is in accordance with the results of Blaschek and Hördt (2009) and Volkmann and Klitzsch (2010). The assumption of SNP, on the other hand, results in a model with a quadratic dependence of the characteristic time on the length of the narrow pore as reported earlier by Titov et al. (2002).

In the last section, we provide a quantitative criterion for the distinction between the two characteristic time regimes and provide a equivalent circuit representation of the Marshall-Madden impedance involving finite-length Warburg impedances. We also discuss the practical significance of the two new impedance models.

MEMBRANE POLARIZATION THEORY

In the following, we briefly review the mathematical formulation of membrane polarization by Marshall and Madden (1959). An external electrical field accelerates the ions of the electrolytic pore fluid corresponding to their respective charges. In presence of a concentration gradient, this electrically excited migration current superimposes a diffusion current. As no ions are generated or removed, continuity equations for both ion species complete the system of partial differential equations, which describes the electrical current densities (for univalent cations and anions)

$$\mathbf{j}_p = -FD_p \nabla c_p + F\mu_p c_p \mathbf{E}, \quad (2)$$

$$\mathbf{j}_n = -FD_n \nabla c_n - F\mu_n c_n \mathbf{E}, \quad (3)$$

$$F \frac{\partial c_p}{\partial t} = -\nabla \mathbf{j}_p, \quad (4)$$

$$F \frac{\partial c_n}{\partial t} = -\nabla \mathbf{j}_n. \quad (5)$$

Here, $\mathbf{j}_i$ are the electrical current densities in C/(m²s), D_i the diffusion coefficients in m²/s, μ_i the ion mobilities in m²/(Vs), c_i the ion concentrations in mol/m³, F is the Faraday constant (96,485 C/mol), and $\mathbf{E}$ the electrical field in V/m. The subscript p denotes the corresponding quantities of the cations and the subscript n those of the anions. Diffusion coefficients and mobilities are related by the Nernst-Einstein equation

$$D = \frac{\mu k_B T}{e}, \quad (6)$$

where k_B is the Boltzmann constant ($1.38 \cdot 10^{-23}$ J/K), e the unit charge ($1.60 \cdot 10^{-19}$ C), and T the absolute temperature in K. Equations 2–5 are coupled via the Poisson equation, as the buildup of a concentration difference between ion species of opposite charge causes a nonvanishing charge density

$$\Delta\psi = -\nabla\mathbf{E} = -\frac{F}{\varepsilon}(c_p - c_n). \quad (7)$$

Here, ψ denotes the electrical potential in V, $\varepsilon = \varepsilon_0 \varepsilon_r$ the permittivity of the pore fluid, where ε_0 is the vacuum permittivity ($8.854 \cdot 10^{-12}$ C/(Vm)) and ε_r the relative permittivity (≈ 80 for water).

To describe the processes within a complex pore system, the pore system is reduced to an alternating sequence of two interconnected pore types (see Figure 1). Wide pores and narrow pores (pore throats) differ with respect to their lengths ΔL_i and cross sections. At the interface between mineral matrix and pore fluid, (usually) negative charges at the mineral surface cause an EDL to build up. The smaller the pore radius, the bigger the portion of the pore cross section, which is covered by the EDL. The asymmetric ion concentrations in the EDL cause an increase of the cationic and a reduction of the anionic contribution to the total electrolytic current through the pore, which is accounted for by different transport numbers for the two ion species. The transport numbers of a univalent symmetric electrolyte filling the pore space depend on average ion mobilities μ_i and concentrations c_i (e.g., Revil and Glover, 1997)

$$t_p = \frac{\mu_p c_p}{\mu_p c_p + \mu_n c_n}, \quad t_n = \frac{\mu_n c_n}{\mu_p c_p + \mu_n c_n}. \quad (8)$$

Obviously, the effect of different concentrations in the EDL on the total current through the pore is more pronounced in the case of small pore radii, which results in a reduced transport number of the anions particularly in the narrow pore ($t_n < t_p$).

If mobilities are assumed to be the same for both ion species ($\mu_p = \mu_n$, e.g., for KCl), the transport numbers depend on the average concentration only (e.g., Sidorova and Fridrikhsberg, 1978; Buchheim and Irmner, 1979; Titov et al., 2002). Marshall and Madden (1959) considered a 1D system, in which the ion selective behavior of the narrow pore is accounted for by assigning different ion mobilities to wide and narrow pores. Thus, narrow pores become “active” zones, wide pores “passive” zones. Here, differences in ion concentrations are neglected, $c_p = c_n$, and equation 8 reduces to

$$t_p = \frac{\mu_p}{\mu_p + \mu_n}, \quad t_n = \frac{\mu_n}{\mu_p + \mu_n}. \quad (9)$$

Throughout this study, this definition of the transport numbers will be used. However, Bückner and Hördt (2013) recently expanded the originally 1D Marshall-Madden impedance to the case of a 2D system of cylindrical pores. They introduced average transport numbers, which depend on the ion concentration averaged over the cross section of the pore, and thus on pore radii and the properties of

the EDL. Therefore, we keep in mind that the mobility contrasts considered in the present work can be caused not only by true mobility variations, but also by pore radii contrasts and EDL properties.

For small excess ion concentrations (i.e., small electrical fields and currents), Marshall and Madden (1959) solved the 1D correspondent of the above system of differential equations in the frequency domain. The solution is expressed in terms of a complex frequency-dependent impedance (subsequently termed *Marshall-Madden impedance*):

$$Z(\omega) = \frac{\Delta L_1}{\mu_{p1} c_0 F} \times \left[t_{p1} + \frac{B}{A} t_{p2} + \frac{(S_2 - S_1)^2}{\frac{X_1(\omega) S_1}{(t_{p2})^2 t_{p1} \tanh X_1(\omega)} + \frac{A}{B} \frac{X_2(\omega) S_2}{t_{p2} (t_{p1})^2 \tanh X_2(\omega)}} \right]. \quad (10)$$

Here, c_0 is the bulk concentration of the electrolyte in mol/m³ and $\omega = 2\pi f$ the angular frequency in rad/s. The subscript 1 denotes the corresponding quantities of the passive zone and subscript 2 those of the active zone. In addition, the abbreviations

$$X_i(\omega) = \left(\frac{i\omega}{2D_{pi} t_{ni}} \right)^{1/2} \frac{\Delta L_i}{2}, \quad (11)$$

$$A = \frac{\Delta L_1}{\Delta L_2}, \quad (12)$$

$$B = \frac{D_{p1}}{D_{p2}}, \quad (13)$$

and

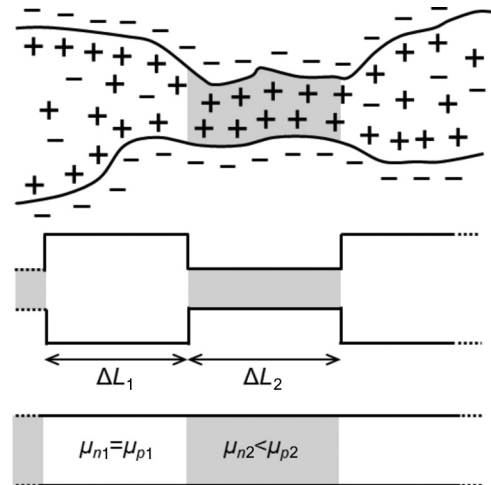


Figure 1. Top: Schematic illustration of the complex pore structure of a natural material with negative surface charge at the mineral-fluid interface and increased cation concentration in the pore fluid. Center: Simplified model consisting in a sequence of two pores with different lengths ΔL_i and cross sections. Bottom: 1D pore space model after Marshall and Madden (1959) consisting in a sequence of two zones of different lengths ΔL_i and ion mobilities μ_{pi} and μ_{ni} .

$$S_i = \frac{t_{ni}}{t_{pi}} \quad (14)$$

have been introduced. The last quantity S_i relates the transport numbers, equation 9, of the respective zones. Note that due to the one-dimensionality of the underlying model, the Marshall-Madden impedance, equation 10, is given as an impedance times unit area (in Ωm^2).

If there was no stronger influence of the EDL on the ion transport numbers in one of the zones (i.e., $S_1 = S_2$), the Marshall-Madden impedance, equation 10, would loose its dependence on frequency and reduce to the simple impedance of two electrolytic conductors in series

$$Z_\infty = \frac{1}{c_0 F} \left[\frac{\Delta L_1}{\mu_{p1} + \mu_{n1}} + \frac{\Delta L_2}{\mu_{p2} + \mu_{n2}} \right]. \quad (15)$$

The same expression can be found in the limiting case of very high frequencies ($\omega \rightarrow \infty$). The case of very low frequencies, i.e., direct current (DC), is treated in the next section.

The Marshall-Madden impedance

Usually, impedance models are formulated in terms of their corresponding DC-limits, polarizabilities (also termed *chargeability*), and time constants. The Marshall-Madden impedance may also be expressed in this form (for a detailed derivation see Appendix A)

$$Z(\omega) = Z_{\text{DC}} \left[1 - \eta_0 \times \left(1 - \frac{\frac{\Delta L_1}{\tau_1} + \frac{\Delta L_2}{\tau_2}}{\frac{\Delta L_1}{\tau_1} \sqrt{i\omega\tau_1} \coth \sqrt{i\omega\tau_1} + \frac{\Delta L_2}{\tau_2} \sqrt{i\omega\tau_2} \coth \sqrt{i\omega\tau_2}} \right) \right], \quad (16)$$

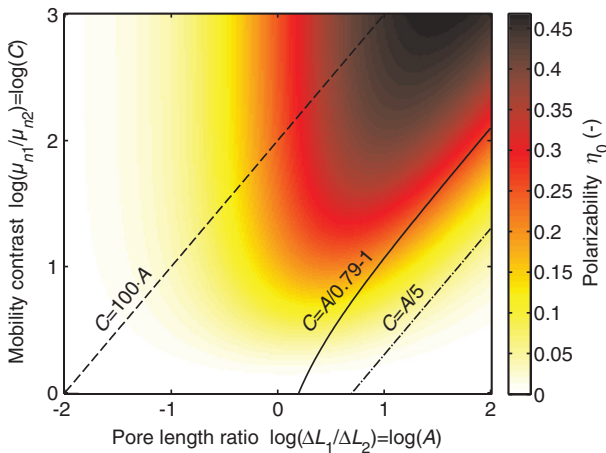


Figure 2. Variation of the polarizability η_0 with the length ratio $A = \Delta L_1/\Delta L_2$ and the mobility contrast $C = \mu_{n1}/\mu_{n2}$. The length of the active zone $\Delta L_1 = 16 \mu\text{m}$ and the mobilities μ_{p1} , μ_{p2} , and μ_{n1} are kept constant, while ΔL_2 and μ_{n2} are varying. Subsequently, the lines will be related to the limits of the LNP regime (dashed line) and the new SNP regime (dash-dot line) as well as the position of the transition between the two regimes (solid line). The basic model parameters are listed in Table 1.

where Z_{DC} is the low frequency limit ($\omega \rightarrow 0$) of the Marshall-Madden impedance, equation 10

$$Z_{\text{DC}} = \frac{k_B T}{e c_0 F} \left[\frac{\Delta L_1}{D_{p1} + D_{n1}} + \frac{\Delta L_2}{D_{p2} + D_{n2}} + \frac{8(t_{n2}t_{p1} - t_{n1}t_{p2})^2}{\frac{\Delta L_1}{\tau_1} + \frac{\Delta L_2}{\tau_2}} \right], \quad (17)$$

and the polarizability is given by

$$\eta_0 = \frac{k_B T}{e c_0 F} \frac{8(t_{n2}t_{p1} - t_{n1}t_{p2})^2}{\frac{\Delta L_1}{\tau_1} + \frac{\Delta L_2}{\tau_2}} \frac{1}{Z_{\text{DC}}}. \quad (18)$$

Here, all mobilities have been expressed in terms of the corresponding diffusion coefficients using the Nernst-Einstein relation (equation 6). The dependence on frequency, formerly represented by X_i , is now expressed in terms of time constants

$$\tau_i = \frac{\Delta L_i^2}{8D_{pi}t_{ni}} = \frac{\Delta L_i^2(D_{pi} + D_{ni})}{8D_{pi}D_{ni}}. \quad (19)$$

In this way, two new time constants, one for each zone, are defined. Note that, throughout this study, the term *time constant* refers to the physical quantities τ_i that directly appear in the impedance models and have the dimension of time. The term *characteristic time* describes the dominant time scale of the relaxation process and is calculated as the inverse of the spectral position of the phase maximum $\tau_c = 1/\omega_c$. Some authors use the term *relaxation time* for this quantity. In accordance with previous studies (e.g., Titov et al., 2002; Blaschek and Hördt, 2009; Volkman and Klitzsch, 2010), both time constants (equation 19) show a quadratic dependence on the respective pore length and are inversely proportional to a diffusion coefficient. The impedance is invariant with regard to a permutation of the pore indices (1, 2). This symmetry is broken, as soon as limiting cases are considered. Examples are the SNP model by Titov et al. (2002) (only valid for $\Delta L_1 \gg \Delta L_2$) and the new models proposed further ahead in this study.

The polarizability reflects the magnitude of the polarization effect. It is worthwhile to review some aspects of the polarizability's dependence on the model parameters by studying its variation with the pore length ratio $A = \Delta L_1/\Delta L_2$ and the ratio of the anion mobilities in the two pores $C = \mu_{n1}/\mu_{n2}$. Note that C is the anionic analog of B (see equation 13) and will subsequently be termed *mobility contrast*. Figure 2 shows that high polarizabilities are caused by reduced mobilities in the shorter pore, i.e., by short active zones (narrow passages, $A \gg 1$). However, high pore length ratios A only lead to high polarizabilities, if the mobility contrast is increased too. These findings agree with those derived from numerical simulations (cf., Blaschek and Hördt, 2009). The significance of the lines and the corresponding equations specified in the figure will be explained further below.

Instead of the polarizability, the full frequency spectrum of the Marshall-Madden impedance can be analyzed. Figure 3 shows huge phase values for mobility contrasts $C > 1$, where the short pore is the active zone. Maximum phase values increase with increasing mobility contrast and saturate at about $C \approx 10^3$. In the range $C < 1$, phases are much smaller and phase maxima are shifted by approximately one order of magnitude to higher frequencies.

To enhance the phases and facilitate analytic derivations in this study, we will use high mobility contrasts C between 100 and 500 (see Table 1). Note that maximum phases of up to -200 mrad as reached for the chosen mobility contrasts are much higher than those found experimentally for samples without metallic constituents. As mentioned above, we understand the low anion mobility in the active zone as a parameterization of the average anion transport number (see Bucker and Hördt, 2013). Therefore, high mobility contrasts may be caused by high contrasts in pore radius and ion concentration in addition to true mobility contrasts.

The variation of the phase spectra with the pore lengths, as illustrated in Figure 4, reveals a basic property of the Marshall-Madden impedance. The spectral position of the phase maxima varies with the lengths of both pores. However, the dependence is restricted to pore lengths smaller than a limit value. In Figure 4a, the phase maxima vary with the lengths of the wide pore as long as $\Delta L_1 < 10^{-3}$ m, and in Figure 4b with the lengths of the narrow pore as long as $\Delta L_2 < 10^{-7}$ m.

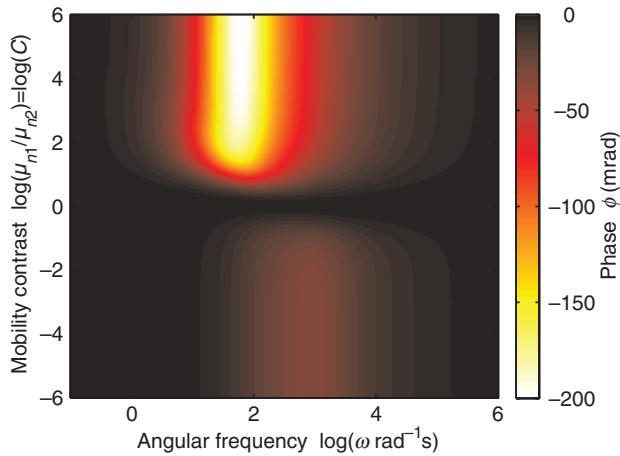


Figure 3. Variation of the phase ϕ of the Marshall-Madden impedance with the frequency ω and the mobility contrast $C = \mu_{n1}/\mu_{n2}$. The mobilities μ_{p1} , μ_{p2} , and μ_{n1} are kept constant, while μ_{n2} is varying. Here, $\Delta L_1 = 16 \mu\text{m}$ and $\Delta L_2 = 4 \mu\text{m}$.

Table 1. Basic model properties for Figures 2–11. Any property that differs from its value indicated in this table is specified in the caption to the corresponding figure. Ionic mobilities are close to the one of Na^+ in dilute solution at 25°C ($\mu_{\text{Na}^+} = 5.19 \cdot 10^{-8} \text{ m}^2/(\text{Vs})$). The reduction of the anion mobility in the active zone by 1/100 was selected to yield a significant mobility contrast.

Property	Symbol	Value
Absolute temperature	T	291 K
Concentration bulk electrolyte	c_0	1 mol/m
Anion mobility, passive zone	μ_{n1}	$5 \cdot 10^{-8} \text{ m}^2/(\text{Vs})$
Anion mobility, active zone	μ_{n2}	$5 \cdot 10^{-10} \text{ m}^2/(\text{Vs})$
Cation mobility, passive zone	μ_{p1}	$5 \cdot 10^{-8} \text{ m}^2/(\text{Vs})$
Cation mobility, active zone	μ_{p2}	$5 \cdot 10^{-8} \text{ m}^2/(\text{Vs})$

For lengths lesser and greater than the limit value, the positions of the phase maxima form straight lines. The slopes of lines g and h, both ≈ -0.5 , can be used to derive preliminary relations between pore lengths and characteristic times. Via $\tau_c = 1/\omega_c$, the spectral position of the phase maximum ω_c is related to the characteristic time τ_c . For line g, we find a dependence on the length of the active zone $\tau_c \sim \Delta L_2^2$. This relation corresponds to the one reported by Blaschek and Hördt (2009) and Volkman and Klitzsch (2010). On the other hand, line h leads to a quadratic variation of the characteristic time with the length of the narrow pore $\tau_c \sim \Delta L_1^2$ as found earlier by Titov et al. (2002). These preliminary results suggest that both behaviors are included in the Marshall-Madden impedance and can be derived as limiting cases.

For a more detailed analysis of the relation between the characteristic time and the two pore lengths, the characteristic time is computed based on the spectra of every pair of pore lengths. Although in this representation all information on magnitude and shape of the phase spectra is lost, Figure 5 allows a clear distinction

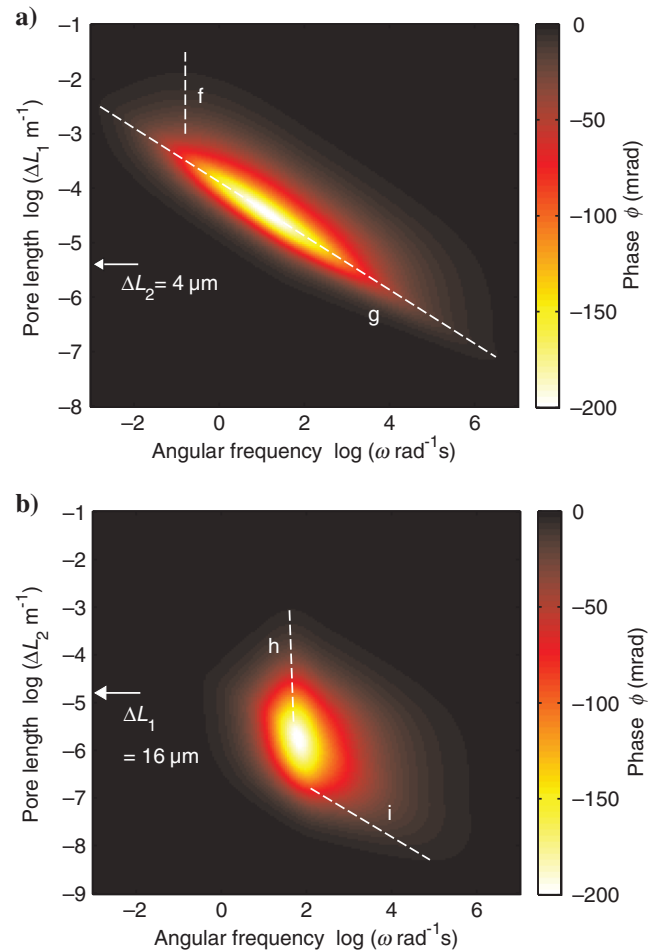


Figure 4. (a) Variation of the phase ϕ of the Marshall-Madden impedance with the frequency and the length ΔL_1 of the passive zone. The length of the active zone $\Delta L_2 = 4 \mu\text{m}$ is kept constant. (b) Variation of the phase ϕ with the frequency and the length ΔL_2 of the active zone. Here, the length of the passive zone $\Delta L_1 = 16 \mu\text{m}$ is kept constant. Lines “g” and “i” indicate a trend of the spectral position of the phase peak varying as a function of the respective pore length. For greater lengths, this trend disappears as indicated by lines “f” and “h”.

between two regimes: For $A \leq C/100$ (long active zone and/or large mobility contrast), the characteristic times only vary with the length ΔL_1 of the passive zone. Again, in this regime the relation $\tau_c \sim \Delta L_1^2$ holds. For $A \geq 5 \cdot C$ (long passive zone), the characteristic times solely depend on the length ΔL_2 of the active zone. As expected, this dependence can be expressed as $\tau_c \sim \Delta L_2^2$. In the range between these two regimes, the characteristic time varies with both lengths simultaneously. The formulation of the criteria in terms of the mobility contrast C ($A \leq C/100$, $A \geq 5 \cdot C$) is illustrated here for $C = 100$, but was verified for different values of C between 1 and 1000.

TWO NEW IMPEDANCE MODELS

Although the preceding analysis showed that the characteristic time versus pore length relation of the Marshall-Madden impedance is composed of two very simple functional relationships, no sufficiently simple analytical expression could be derived for the whole range of pore lengths. However, simple analytical expressions are made possible by the derivation of two new approximate impedance models for the two limiting cases $A \leq C/100$ and $A \gg 5 \cdot C$.

A long narrow pore model

Based on the reformulated Marshall-Madden impedance given in equation 16, an approximate impedance model for the regime $A \leq C/100$ and approximately equal mobilities $\mu_{p1} \approx \mu_{p2} \approx \mu_{n1}$ is derived (see Appendix A). Because this regime is characterized by relatively long narrow pores, the corresponding model shall be termed the *long narrow pore (LNP) model*:

$$Z_{\text{LNP}}(\omega) = Z_{\text{DC}} \left[1 - \eta_0 \left(1 - \frac{\tanh \sqrt{i\omega\tau_1}}{\sqrt{i\omega\tau_1}} \right) \right]. \quad (20)$$

The DC-limit, polarizability, and time constant are the same as for the Marshall-Madden impedance and given by equations 17, 18, and 19, respectively.

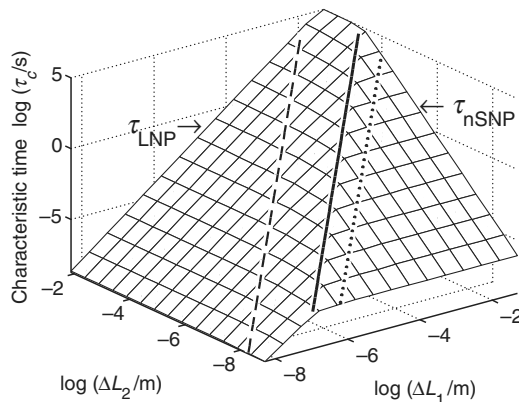


Figure 5. Variation of the characteristic time τ_c with simultaneously varying pore lengths ΔL_1 and ΔL_2 . The mobilities are chosen to provide a high-mobility contrast of $C = \mu_{n1}/\mu_{n2} = 100$ and high polarizabilities. Note the clear separation of the two regimes $A < C/100$ (delimited by the black dashed line) and $A > C/5$ (black dotted line). The black solid line ($A = 0.79 \cdot (C + 1)$) illustrates the approximate position of the transition between the two regimes.

Figure 6 illustrates the good agreement of the new LNP model with the Marshall-Madden impedance. Under the given condition $A \leq C/100$, a quantitative analysis of the deviations confirmed a small relative error $< 3\%$ for the magnitude of the impedance (not shown here for brevity). Note that, in this limiting case, the phase has a clearly asymmetrical shape. The low-frequency branch of the phase dispersion increases proportionally to ω while the high-frequency branch decreases at $-1/\sqrt{\omega}$. In terms of empirical models, asymmetric dispersions are often fitted by Cole-Davidson models. Effectively, the impedance spectrum matches a generalized Cole-Cole model, equation 1, if the time constant is $\tau_{\text{GCC}} = \tau_1$ and the exponents are $a = 0.5$ and $c = 1$ (Cole-Davidson model).

Figure 7 shows that the time constant of the impedance model does not necessarily coincide with its characteristic time. The variation of the characteristic times with the pore length ΔL_1 in Figure 7 was computed on the basis of spectra for each pore length. For each of the respective phase spectra, the characteristic angular frequency was determined and the characteristic time calculated as $\tau_c = 1/\omega_c$.

Due to the good agreement between the corresponding phase spectra, the characteristic times of the Marshall-Madden impedance

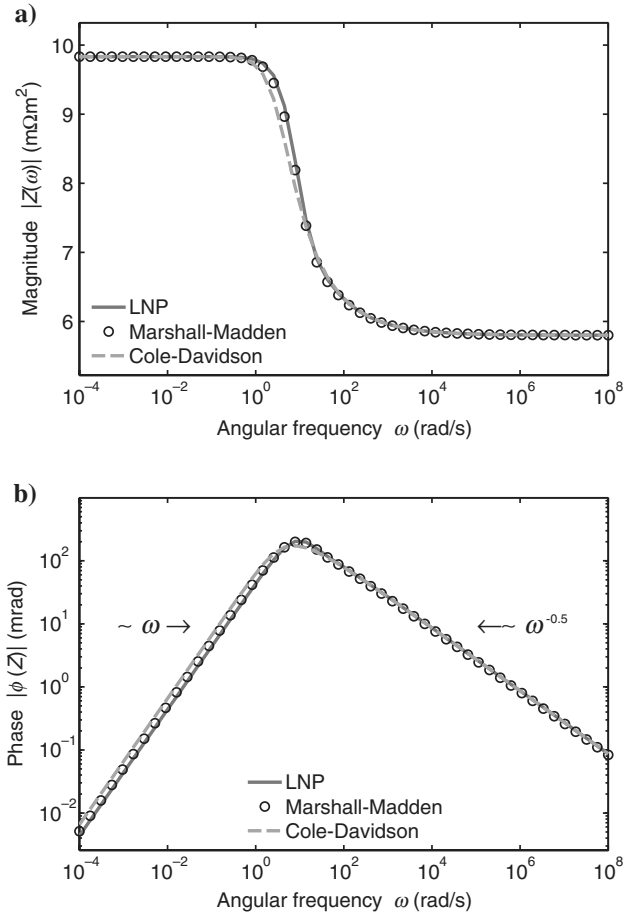


Figure 6. (a) Magnitude and (b) phase spectra of the Marshall-Madden impedance (circles), the LNP model (black solid line), and the Cole-Davidson model (gray dashed line) according to equation 1, where $c = 1$, $a = 0.5$, and $\tau_{\text{GCC}} = \tau_1$. The pore lengths $\Delta L_1 = 40 \mu\text{m}$ and $\Delta L_2 = 8 \mu\text{m}$ as well as the anion mobility in the active zone $\mu_{n2} = 1 \cdot 10^{-10} \text{ m}^2/(\text{Vs})$ are chosen to match the conditions $A \leq C/100$ and $C \gg 1$.

(see Figure 7) are recovered quite well by those of the LNP model. For pore lengths $\Delta L_1 > 10^{-3}$ m, the assumption $A \leq C/100$ is violated and the transition to the SNP model (no variation with ΔL_1 , see next section) explains the increasing differences. We also see in Figure 7 that the actual characteristic time of the LNP model, equation 20, is smaller than the time constant τ_1 . As shown in Appendix A, this can be explained theoretically. The corresponding factor between the actual characteristic time of the model τ_{LNP} and the time constant τ_1 is (<1)

$$\tau_{\text{LNP}} \approx [0.0222 + (0.1374 - 0.1245\eta_0)^{1/2}] \tau_1. \quad (21)$$

This analytically derived characteristic time is proportional to $\tau_1 = \Delta L_1^2 / (8D_{p1}t_{n1})$ for a given polarizability, which explains the observed quadratic dependence of the characteristic time in the LNP regime on the length of the wide (passive) pore.

The characteristic time of the LNP model, equation 21, may be compared to the empirical relationship reported by Blaschek and Hördt (2009), whose corresponding equation may be expressed as

$$\tau_c = 0.0998 \frac{\Delta L_1^2}{D}, \quad (22)$$

based on various numerical simulations. Here, D denotes the diffusion coefficient in the active zone ($D = D_{p1} = D_{n1}$) and τ_c is the characteristic time. A similar expression may be derived for our analytic LNP model. To this end, we consider the limiting case of very small polarizabilities $\eta_0 \ll 1$ in which the characteristic time, equation 21, reduces to $\tau_{\text{LNP}} \approx 0.4 \cdot \tau_1$. We get

$$\tau_{\text{LNP}} \approx 0.4 \cdot \tau_1 = 0.4 \cdot \frac{\Delta L_1^2}{8D_{p1}t_{n1}} \approx 0.1 \frac{\Delta L_1^2}{D}, \quad (23)$$

where in the last step $D = D_{p1} = D_{n1}$ was used to yield $t_{n1} = 0.5$. Both relations, equations 22 and 23, match very well and thus the characteristic time of the LNP model, equation 21, represents a theoretical explanation of the numerical findings by Blaschek and Hördt (2009).

A new short narrow pore model

The regime $A \geq 5 \cdot C$ is characterized by relatively SNP. The existing SNP model by Titov et al. (2002) offers a good approximation of the impedance for very pronounced length ratios $A \gg 1$. However, its applicability is limited to small mobility contrasts $C \approx 1$. In this section, a new short narrow pore (nSNP) model is presented, which overcomes this limitation. Instead, the condition for the length ratio is formulated in terms of the mobility contrast, as our previous discussion suggests $A \gg 5 \cdot C$.

An approximate SNP impedance model, which is able to predict the relation between characteristic time and pore lengths, must hold at least in the vicinity of the phase maximum. For frequencies ω close to the phase maximum at ω_c the condition $A \gg 5 \cdot C$ with the additional assumptions of approximately equal mobilities $\mu_{p1} \approx \mu_{p2} \approx \mu_{n1}$ and reduced anion mobilities in the active zone $\mu_{n1} > \mu_{n2}$ imply $\omega\tau_1 \gg 1$ and $\omega\tau_2 < 1$ (see Appendix A). Under these five assumptions, based on the reformulated Marshall-Madden impedance, equation 16, an nSNP can be derived (Appendix A) given by

$$Z_{\text{nSNP}}(\omega) = Z_{\text{DC}} \left[1 - \eta_0 \left(1 - \frac{1}{1 + \sqrt{i\omega\tau'}} \right) \right]. \quad (24)$$

Here again, the DC-limit and the polarizability are given by equations 17 and 18. The modified time constant

$$\tau' = \tau_2 \frac{D_{p1}t_{n1}}{D_{p2}t_{n2}} = \frac{\Delta L_2^2}{8D_{p2}t_{n2}} \frac{D_{p1}t_{n1}}{D_{p2}t_{n2}} \quad (25)$$

depends on the length of the active zone and on the ion mobilities in both zones. Note that the nSNP model is equivalent to a Cole-Cole model with the frequency exponent $c = 0.5$.

Figure 8a shows a good agreement between the magnitude spectra of our nSNP model and the Marshall-Madden impedance. A more extensive analysis revealed a relative error $<1\%$, as long as the condition $A \gg 5C$ is fulfilled. Due to the unfavorable combination of pore length ratio and mobility contrast ($A \geq 5 \cdot C$, lower right corner of Figure 2), spectral changes in the magnitude (i.e., polarizabilities) are much smaller than in the LNP regime. Note that the chosen scaling of the ordinate exaggerates existing deviations between the models.

The phase dispersion of the nSNP model in Figure 8b shows a symmetric behavior, which is typical of the Cole-Cole model. The low-frequency branch increases proportionally to $\sqrt{\omega}$, whereas the high-frequency branch decreases at a rate of $-\sqrt{\omega}$. The phases of the nSNP model and the Marshall-Madden impedance match very well, except for the low-frequency range, where the model assumption $\omega\tau_1 \gg 1$ is violated and the phase of the Marshall-Madden impedance shows an ω -like behavior. The poor agreement with the original SNP model by Titov et al. (2002) is due to the violation of the additional condition $C \approx 1$.

The deviations in the low-frequency range, the phase maximum, and therefore the characteristic frequency, are reproduced very well. As long as the model assumptions are fulfilled, this results in the good agreement between the characteristic times of both models in Figure 9. In the upper range of pore lengths ($\Delta L_1 > 2 \cdot 10^{-6}$ m),

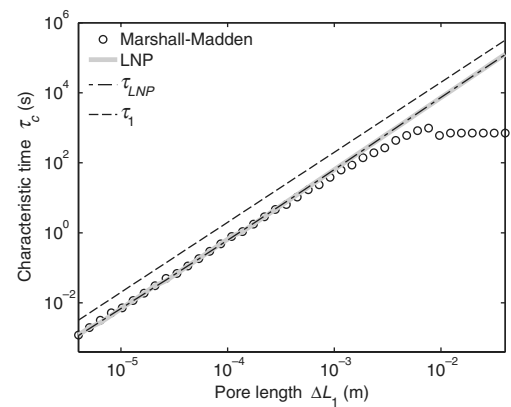


Figure 7. Variation of the characteristic times τ_c with the length ΔL_1 of the active zone. The characteristic times of the Marshall-Madden impedance (circles) and the LNP model (gray solid line) were computed based on the characteristic frequencies ω_c of spectra for each pore length ΔL_1 . The characteristic time τ_{LNP} (black dashed-dot line) was calculated according to equation 21. The time constant τ_1 (black dashed line) is given by equation 19. The pore length $\Delta L_2 = 8 \mu\text{m}$ and the mobility $\mu_{n2} = 1 \cdot 10^{-10} \text{ m}^2/(\text{Vs})$ are chosen to match the conditions $A \leq C/100$ and $C \gg 1$.

the characteristic times of the Marshall-Madden impedance tend toward a constant value (transition to LNP regime). This trend is not followed by the characteristic time of the nSNP model. As shown in Appendix A, the characteristic time can be expressed in terms of the modified time constant τ' and the polarizability

$$\tau_{nSNP} = (1 - \eta_0)\tau'. \quad (26)$$

For small polarizabilities $\eta_0 \ll 1$, the characteristic time τ_{nSNP} can be approximated by the modified time constant τ' .

Titov's short narrow pore model

Although Titov et al. (2002) developed the SNP model in the time domain, their approach is based on the membrane polarization theory by Marshall and Madden (1959). Therefore, it is possible to derive the SNP model starting from the frequency-domain Marshall-Madden impedance, equation 16. Adding the assumption $\omega\tau_1 \gg 1$, which is partly implied by their conditions $A \gg 1$ and $C \approx 1$, we obtain (Appendix A)

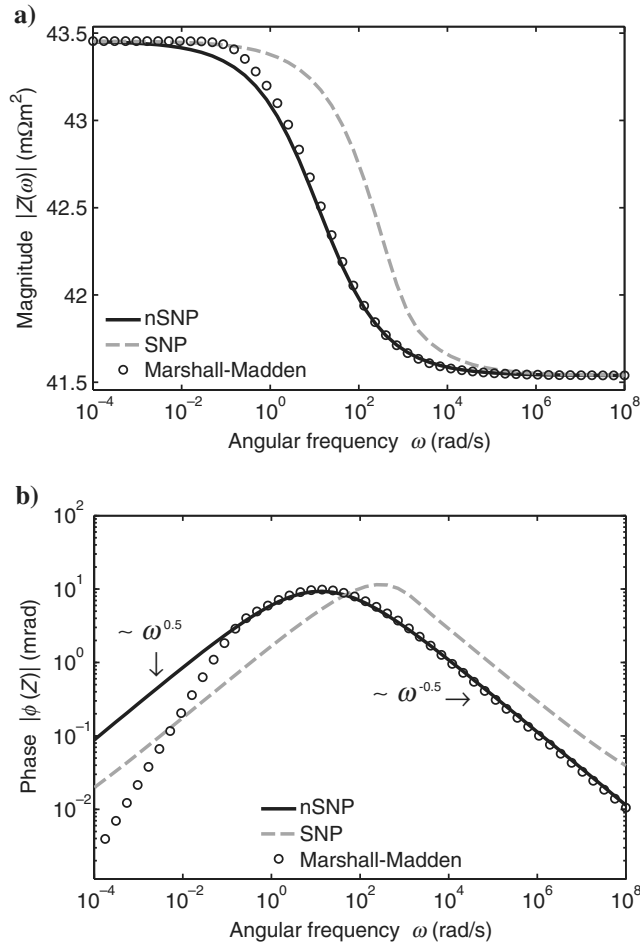


Figure 8. (a) Magnitude and (b) phase spectra of the Marshall-Madden impedance (circles), Titov's SNP model (gray dashed line), and the nSNP model (black solid line). The pore lengths $\Delta L_1 = 400 \mu\text{m}$ and $\Delta L_2 = 0.4 \mu\text{m}$ as well as the anion mobility in the active zone $\mu_{n2} = 1 \cdot 10^{-10} \text{ m}^2/(\text{Vs})$ are chosen to match the conditions $A \gg 5C$ and $C \gg 1$.

$$Z_{\text{SNP}}(\omega) = Z_{\text{DC}} \left[1 - \eta_0 \left(1 - \frac{1 - \exp(-2(i\omega\tau_2)^{1/2})}{2(i\omega\tau_2)^{1/2}} \right) \right], \quad (27)$$

where the time constant is given by

$$\tau_2 \approx \frac{\Delta L_2^2}{4D_{n2}}. \quad (28)$$

This is essentially the same impedance model as the one obtained by Titov et al. (2002) after the transformation to the frequency domain. The main difference between the two models consists of the expressions for the DC-limits of the impedances and the respective polarizabilities. In our approach, both quantities are given by the 1D Marshall-Madden impedance and expressed in terms of pore lengths and the transport numbers of both pores (see equations 17 and 18). The corresponding quantities in the original model by Titov et al. (2002) are given in terms of the same parameters plus the pore cross sections and an efficiency, which accounts for the different anion and cation concentrations in the EDL. That is why for $C \approx 1$ Titov et al. (2002) obtained reasonable polarizabilities (0.5%–2%) while the Marshall-Madden impedance predicts values close to zero (see Figure 2).

An additional minor difference is the choice of the diffusion coefficient for the calculation of the time constant. Although in the original model it is expressed in terms of the diffusion coefficient of the bulk electrolyte, our approach yields a dependence on the anionic diffusion coefficient in the narrow pore. For the case of small mobility contrasts (where the SNP model matches the corresponding limiting case of the Marshall-Madden impedance), this detail does not make a numeric difference, but is essential for the understanding of the underlying process.

The poor agreement between the respective magnitude and phase spectra of the SNP and the Marshall-Madden model in Figure 8 improves when the assumption $C \approx 1$ of the SNP model is fulfilled (not shown here).

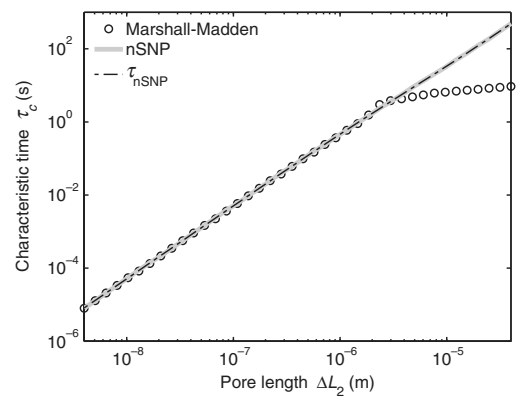


Figure 9. Variation of the characteristic times τ_c with the length ΔL_2 of the active zone. The characteristic times of the Marshall-Madden impedance (circles) and the nSNP model (gray solid line) were computed based on the characteristic frequencies ω_c of spectra for each pore length ΔL_2 . The characteristic time τ_{nSNP} (black dashed line) was calculated according to equation 21. The pore length $\Delta L_1 = 400 \mu\text{m}$ and the anion mobility in the active zone $\mu_{n2} = 1 \cdot 10^{-10} \text{ m}^2/(\text{Vs})$ are chosen to match the conditions $A \gg 5C$ and $C \gg 1$.

For small polarizabilities ($\eta_0 \ll 1$) — which are expected for the low mobility contrasts required to fulfill the SNP condition — the dependence of the characteristic time of the SNP impedance on the time constant τ_2 can be derived analytically (Appendix A)

$$\tau_{\text{SNP}} \approx 1.12 \cdot \tau_2. \quad (29)$$

The misfit between τ_{SNP} and the characteristic time of the SNP impedance in Figure 10 can only be reduced by decreasing the mobility contrast even more. The characteristic time of the nSNP model τ_{nSNP} , on the other hand, shows a much better agreement with the Marshall-Madden impedance.

In a mathematically strict sense, in particular with respect to the predicted characteristic time, the validity of the original SNP model is restricted to a narrow range of small mobility contrasts. Our nSNP model shares the main features of the SNP model at low mobilities, but overcomes its limitations and describes the impedances for high and low mobility contrasts at the same time.

IMPLICATIONS OF THE TWO NEW IMPEDANCE MODELS

Two characteristic time regimes

In other contexts, quantities of the type $\sim x^2/4D$, where x is a length and D a diffusion coefficient, are known as diffusion times. The diffusion time represents the time scale, on which a concentration distortion propagates along a profile of the length x . In the case of membrane polarization, the diffusion time — our characteristic time — controls the buildup and relaxation of excess and defect concentrations at the zone boundaries. For both SNP models, the dependence of the characteristic time was found to be basically $\sim \Delta L_2^2/D_{n2}$. In this case, the relaxation process seems to be dominated by diffusion via the narrow pore. For the LNP model, we

arrived at the relation $\sim \Delta L_1^2/D_{n1}$, which indicates the predominance of diffusion relaxation via the wide pore.

To understand the existence of these two characteristic time regimes, we revisit the variation of the characteristic time with both pore lengths. The 3D representation of the characteristic time in Figure 5 is dominated by two planar surfaces. Within the parameter range of each regime, these surfaces are given by the two characteristic times, respectively. The geometrical shape of the entire function can be reproduced by intersecting the two planes $\tau_{\text{LNP}}(\Delta L_1, \Delta L_2)$ and $\tau_{\text{nSNP}}(\Delta L_1, \Delta L_2)$ and removing the half-planes representing characteristic times that are bigger than the ones on the other plane. Physically speaking, in each case, the shorter characteristic time dominates the relaxation process and excess (defect) concentrations at the zone boundaries are mainly built up and balanced via the diffusion path with the shorter diffusion time.

In a more quantitative manner, this results in a criterion for the discrimination of the two regimes. The transition occurs approximately where the two planes intersect. Equating the two characteristic times (equations 21 and 26) gives

$$A = \frac{\Delta L_1}{\Delta L_2} = \frac{1}{\sqrt{0.4}} \frac{D_{p1} t_{n1}}{D_{p2} t_{n2}} \approx 0.79 \cdot (C + 1). \quad (30)$$

Here, both characteristic times have been taken in their limits for small polarizabilities $\tau_{\text{LNP}} \approx 0.4\tau_1$ and $\tau_{\text{nSNP}} \approx \tau'$ and the ratio of the diffusion coefficients has been approximated by $D_{p1}/D_{p2} \approx 1$. The ratio t_{n1}/t_{n2} can be approximated by $(C + 1)/2$, following from $\mu_{p1} \approx \mu_{p2} \approx \mu_{n1}$, which yields $t_{n1} \approx 0.5$ and $t_{n2} \approx 1/(C + 1)$. Figure 5 shows a good agreement between the corresponding line $A = 0.79 \cdot (C + 1)$ and the real position of the transition between the two regimes.

The existence of this criterion affirms the dependence of the model assumptions ($A \leq C/100$ and $A \geq 5 \cdot C$) on the mobility contrast C , which had only been suggested from empirical studies so far.

Transition between the two relaxation time regimes

The impedance spectra of the LNP model and the nSNP model do not only differ with respect to their characteristic times, but also show clearly distinct characteristic shapes. Because it represents a special case of the Cole-Cole model, the nSNP model inherits the typical symmetrical behavior. The LNP model, in contrast, has clearly asymmetrical spectra similar to those of a Cole-Davidson model. In addition, the LNP model shows a narrower phase peak than the nSNP model, which goes along with a steeper decrease in the magnitude spectrum at the characteristic frequency. Beside the existing differences, the high-frequency behavior of the phase spectra of both models is controlled by the same $1/\sqrt{\omega}$ -decrease. Most of the behavior of the Marshall-Madden impedance can be described and understood based on these two fundamental constituents.

The following comparison with the phase spectra of the Marshall-Madden impedance explains some of the mentioned differences and agreements between the two models. In Figure 11, phase spectra for different lengths of the wide pore are plotted simultaneously, illustrating the evolution of the Marshall-Madden impedance from the LNP regime (small ΔL_1) to the SNP regime (great ΔL_1). Beginning with the spectra corresponding to long wide pores (great ΔL_1), the symmetrical shape of the SNP phases can be

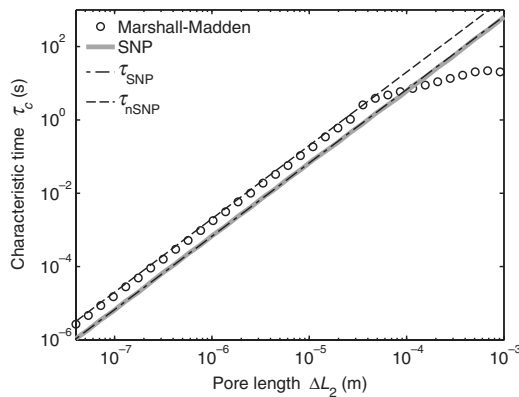


Figure 10. Variation of the characteristic times τ_c with the length ΔL_2 of the active zone. The characteristic times of the Marshall-Madden impedance (circles) and the SNP model (gray solid line) were computed based on the characteristic frequencies ω_c of spectra for each pore length ΔL_2 . The characteristic time τ_{SNP} (black dashed-dot line) was calculated according to equation 29. The characteristic time of the nSNP model τ_{nSNP} (black dashed line) is given by equation 26. Even though the pore length $\Delta L_1 = 400 \mu\text{m}$ and the anion mobility in the active zone $\mu_{n2} = 1 \cdot 10^{-8} \text{ m}^2/(V_s)$ are chosen to match the conditions $A \gg 1$ and $C \approx 1$, τ_{nSNP} matches the characteristic time of the Marshall-Madden impedance slightly better than τ_{SNP} .

recognized. Here, the SNP conditions are fulfilled, and while the phase maximum increases with decreasing lengths of the wide pore, it remains at its spectral position. The sharp bend at low frequencies, which is not included in the SNP impedance and was attributed to the violation of the assumption of relatively high frequencies ($\omega\tau_{n\text{SNP}} \ll 1$) earlier, approaches the phase maximum. It is obvious that, at low frequencies, the whole set of possible phase spectra is confined by the characteristic ω -dependence of the LNP phase. The transition between the two regimes coincides with the length of the wide pore, for which the bend reaches the phase maximum and the phase peak begins to wander toward higher frequencies ($\Delta L_1 = 0.1$ mm). The resulting envelope for the whole set of possible phase spectra is confined by $\sim\omega$ at low frequencies and $\sim 1/\sqrt{\omega}$ at high frequencies.

Strictly speaking, the two new impedance models are limited to their respective regimes and to defined frequencies that extend from a proximity of the phase maximum to high frequencies.

Practical significance

A summary of the membrane polarization impedances based on the model by Marshall and Madden (1959), including their respective characteristic times, is given in Table 2. Beside providing a basic understanding of the fundamental processes, they may also serve as generic models to calculate the impedances of more realistic pore geometries. The answer to the question whether any of the

impedances is more relevant for practical applications is beyond the scope of this paper. Intuitively, the two SNP models might seem more suitable, as the narrow passages are generally imagined to be relatively short. However, realistic pore geometries can be very complex, and the simulation based on fundamental models is not trivial. Also note that, up to now, only one mobility has been changed during the derivation of the relaxation models. More complication might arise if an extension of the model family might be necessary after having dropped the assumption $D_{p1} = D_{p2} = D_{n1}$. At this point, both models are relevant.

With respect to the interpretation of spectra obtained from real materials, the two new impedance models in the first line offer a strong physical link between the empirical models of the Cole-Cole family, and in this sense legitimize their application. In addition, they facilitate the discussion of the characteristic shape of experimentally obtained spectra. The widely used quadratic dependence of the characteristic time on the length scale of the pore space has been confirmed.

In real materials, simple relaxation or characteristic time relations are not always able to explain the experimental observations. Kruschwitz et al. (2010) and Titov et al. (2010), for instance, recently reported the presence of long characteristic times in samples with relatively small pore throat diameter (<7 μm), which do not obey any of the theoretically predicted relationships. Basically, two mechanisms were speculated to cause these unexpected observations. Ionic mobility in the EDL is reduced in comparison to the bulk electrolyte. In materials with small pore sizes and a high specific surface area, the relatively slow surface diffusion might predominate and increase the observed characteristic time considerably. The second mechanism is based on the idea that narrow pores could form a connected pathway, which acts as one long membrane zone. Both mechanisms would result in an increased characteristic time.

Especially for the second mechanism, the consequences of the present study are relevant. One of our main results is that the relaxation process is dominated by the shorter characteristic time. The observation of long characteristic times can therefore only be explained by assuming that even the shortest characteristic time is relatively long. The idea of interconnected narrow passages, which form one long active zone, is reminiscent of the LNP model. However, in the LNP model, characteristic times are determined by the length of the wide pore. To generate long characteristic times, as they were found experimentally, the wide pore would have to be long as well. Therefore, assuming the LNP model, one would arrive at a pore space with approximately equal lengths of the wide and the narrow pore. An interpretation based on the SNP model would result in

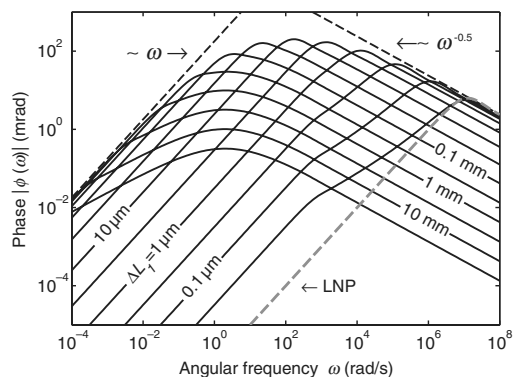


Figure 11. Variation of the phase spectra with the length ΔL_1 of the wide pore. The length of the narrow pore $\Delta L_2 = 1$ μm is kept constant. The gray dashed line represents the phase response of the LNP model for the smallest value of ΔL_1 . The two black dashed lines mark the low- and high-frequency envelopes of the whole set of phase spectra.

Table 2. Summary of the assumptions made for the derivation of the impedance models and the corresponding characteristic times. With respect to the length ratio, the mobility contrast, and the remaining mobilities, no assumptions are made for the Marshall-Madden impedance, which is the most general model.

Model	Length ratio	Mobility contrast	Remaining mobilities	Characteristic time
Marshall-Madden	—	—	—	—
LNP (this work)	$A < C/100$	—	$\mu_{p1} \approx \mu_{p2} \approx \mu_{n1}$	$\tau_{\text{LNP}} \approx 0.4 \cdot \frac{\Delta L_1^2}{8D_{p1}t_{n1}}$
SNP (Titov)	$A \gg 1$	$C \approx 1$	$\mu_{p1} \approx \mu_{p2} \approx \mu_{n1}$	$\tau_{\text{SNP}} \approx 1.12 \cdot \frac{\Delta L_2^2}{4D_{p2}}$
nSNP (this work)	$A \gg 5C$	—	$\mu_{p1} \approx \mu_{p2} \approx \mu_{n1}$	$\tau_{\text{nSNP}} \approx \frac{\Delta L_2^2}{8D_{p2}t_{n2}} \frac{D_{p1}t_{n1}}{D_{p2}t_{n2}}$

even longer wide pores or in interconnected wide pores, which are separated by short narrow passages. Any of the two models would require interconnected wide pores as well as interconnected narrow passages.

Another interesting experimental result was recently reported by Weller et al. (2010a), who observed a negative correlation between measured permeability and characteristic time (their mean relaxation time) for a set of sandstone samples. They explained the increase of the characteristic time by longer pathways in narrow pores. However, it seems unlikely that this explanation is consistent with this membrane polarization model. The present study shows that, in this model, the characteristic time is usually dominated by the shorter time constant, which in presence of LNP — as assumed by Weller et al. (2010a) — would be the one of the wider pores. An explanation might have to consider the expanded Marshall-Madden impedance developed in the work of Bucker and Hördt (2013). Their studies are focused on the effect of pore radii, and they did not find a negative correlation between characteristic time and pore radii, which might explain the observation of Weller et al. (2010a). However, true mobility variations caused by concentration dependence in the wide and narrow pores might also be important. A decrease of mobilities in narrower pores due to an increase of concentration might cause increasing time constants in both types of pores. This effect was not implemented in the model by Bucker and Hördt (2013), and thus it remains unclear at this stage whether, and under which conditions this effect is significant.

EQUIVALENT CIRCUIT REPRESENTATION OF THE MARSHALL-MADDEN IMPEDANCE

Equivalent circuits are often used for the illustration and interpretation of impedance models (e.g., Pelton et al., 1978; Wong, 1979). Writing the Marshall-Madden impedance, equation 16, as

$$Z(\omega) = Z_{\infty} + [1/Z_{W1}(\omega) + 1/Z_{W2}(\omega)]^{-1}, \quad (31)$$

reveals that it can be represented by an equivalent circuit with the high-frequency limit Z_{∞} (equation 15) in series with two impedances

$$Z_{Wi}(\omega) = Z_{Di} \frac{\tanh \sqrt{i\omega\tau_i}}{\sqrt{i\omega\tau_i}} \quad (32)$$

in parallel. These impedances have the form of finite-length Warburg impedances (e.g., Franceschetti et al., 1991) that are characteristic for diffusion processes in cases where the diffusion length $L_d \propto \sqrt{D/\omega}$ (i.e., distance concentrations propagate due to diffusion in time $\tau = 1/\omega$) is equal to or greater than the length of the region available for diffusion. The magnitudes of the finite-length Warburg impedances,

$$Z_{Di} = \frac{k_B T}{ec_0 F} 8(t_{n2} t_{p1} - t_{n1} t_{p2})^2 \frac{\tau_i}{\Delta L_i}, \quad (33)$$

are the maximum diffusion-related resistance of the corresponding pores. Note that this diffusion resistance does not only depend on the length and the diffusion coefficients of the pore i (see definition of time constant, equation 19) but also on the transport numbers of both pores.

CONCLUSIONS

The Marshall-Madden impedance as a model for membrane polarization bears a whole family of impedance models. We cast it into the classical form involving polarizability, and find that two time constants are required to describe the spectral behavior. To identify the dominating time scales and corresponding length scales, we propose approximate models, which allow for analytical expressions describing the relations between characteristic time and model parameters. We not only present for the first time the derivation of the SNP model in the frequency domain, but also propose two new models — an LNP model and an nSNP model. In contrast to the original SNP model, which is limited to the case of small mobility contrasts, our models are valid for a wide range of mobility contrasts.

Our results confirm the quadratic dependence of the characteristic time on pore lengths. The apparent contradiction between earlier studies regarding the influence of the two pore types on the characteristic time is resolved. The characteristic time results to be dominated by the length of the narrow pore in the limiting case of SNP and by the length of the wide pore in the case of LNP. A physical explanation of the observed coexistence of two different characteristic time regimes is also suggested, as the diffusion relaxation seems to be dominated by the shorter of both possible characteristic times.

Beside the quadratic dependence of the characteristic time on the pore length, which was confirmed by all three models, our results suggest that pore throat diameter and pore fluid chemistry, which are parameterized in terms of ionic mobilities, play an important role in the determination of the characteristic time.

Our nSNP model is equal to a Cole-Cole model, if the frequency exponent of the latter equals 0.5. The LNP model derived here matches a Cole-Davidson model, again if the exponent is set to 0.5. Both analogies help to physically legitimize the application of generalized Cole-Cole models, which are widely used to fit spectral electrical impedance data. In addition, we were able to give a simple equivalent circuit representation of the Marshall-Madden impedance involving finite-length Warburg impedances that are typical for diffusion-dominated conduction mechanisms.

ACKNOWLEDGMENTS

We thank Andreas Weller and two anonymous reviewers for their helpful comments that improved the original version of the manuscript. This work was funded by the gebo research association. The authors wish to acknowledge the donors Baker Hughes and the Ministry of Science and Culture of Lower Saxony, Germany, for the financial support of this study.

APPENDIX A

DERIVATION OF TWO NEW IMPEDANCE MODELS

At first, the impedance formula after Marshall and Madden (1959) is rearranged to yield an appropriate form for the subsequent mathematical derivation. Therefore, the frequency-dependent terms X_i in of the Marshall-Madden impedance, equation 10, are expressed in terms of the newly defined time constants $\tau_i = \frac{\Delta L_i^2}{8D_{pi}t_{ni}}$

$$X_i = \left(\frac{i\omega}{2D_{pi}t_{ni}} \right)^{1/2} \frac{\Delta L_i}{2} = \sqrt{i\omega\tau_i}. \quad (\text{A-1})$$

Replacing the abbreviations A , B , and S_i by the corresponding full expressions given in equations 12, 13, and 14 and using the relation $1/\tanh(x) = \coth(x)$ leads to

$$Z(\omega) = \frac{\Delta L_1}{\mu_{p1}c_0F} \left[t_{p1} + \frac{D_{p1}\Delta L_2}{D_{p2}\Delta L_1} t_{p2} + \frac{(t_{n2}t_{p1} - t_{n1}t_{p2})^2}{t_{n1}\sqrt{i\omega\tau_1} \coth \sqrt{i\omega\tau_1} + \frac{D_{p2}\Delta L_1}{D_{p1}\Delta L_2} t_{n2}\sqrt{i\omega\tau_2} \coth \sqrt{i\omega\tau_2}} \right]. \quad (\text{A-2})$$

For the next conversion, we have applied the Nernst-Einstein relation (equation 6) and multiplied the whole term in brackets by $\Delta L_1/D_{p1}$

$$Z(\omega) = \frac{k_B T}{ec_0 F} \left[\frac{\Delta L_1 t_{p1}}{D_{p1}} + \frac{\Delta L_2 t_{p2}}{D_{p2}} + \frac{(t_{n2}t_{p1} - t_{n1}t_{p2})^2}{\frac{D_{p1}t_{n1}}{\Delta L_1} \sqrt{i\omega\tau_1} \coth \sqrt{i\omega\tau_1} + \frac{D_{p2}t_{n2}}{\Delta L_2} \sqrt{i\omega\tau_2} \coth \sqrt{i\omega\tau_2}} \right]. \quad (\text{A-3})$$

Using the definition of the transport numbers (equation 9) and the Nernst-Einstein relation (equation 6) for the first two terms and the definition of the time constant $\tau_i = \frac{\Delta L_i^2}{8D_{pi}t_{ni}}$ for the last term gives the following expression

$$Z(\omega) = \frac{k_B T}{ec_0 F} \left[\frac{\Delta L_1}{D_{p1} + D_{n1}} + \frac{\Delta L_2}{D_{p2} + D_{n2}} + \frac{8(t_{n2}t_{p1} - t_{n1}t_{p2})^2}{\frac{\Delta L_1}{\tau_1} \sqrt{i\omega\tau_1} \coth \sqrt{i\omega\tau_1} + \frac{\Delta L_2}{\tau_2} \sqrt{i\omega\tau_2} \coth \sqrt{i\omega\tau_2}} \right]. \quad (\text{A-4})$$

Obviously, the frequency dependence of this complex impedance is solely determined by the terms $\sqrt{i\omega\tau_i} \coth(\sqrt{i\omega\tau_i})$. In the limiting case of very small arguments $\omega\tau_i \ll 1$, these terms become 1. Thus, we can finally rewrite the Marshall-Madden impedance in terms of its DC limit Z_{DC} and its polarizability η_0

$$Z(\omega) = Z_{DC} \left[1 - \eta_0 \times \left(1 - \frac{\frac{\Delta L_1}{\tau_1} + \frac{\Delta L_2}{\tau_2}}{\frac{\Delta L_1}{\tau_1} \sqrt{i\omega\tau_1} \coth \sqrt{i\omega\tau_1} + \frac{\Delta L_2}{\tau_2} \sqrt{i\omega\tau_2} \coth \sqrt{i\omega\tau_2}} \right) \right], \quad (\text{A-5})$$

where the DC limit is given by

$$Z_{DC} = Z(\omega \rightarrow 0) = \frac{k_B T}{ec_0 F} \left[\frac{\Delta L_1}{D_{p1} + D_{n1}} + \frac{\Delta L_2}{D_{p2} + D_{n2}} + \frac{8(t_{n2}t_{p1} - t_{n1}t_{p2})^2}{\frac{\Delta L_1}{\tau_1} + \frac{\Delta L_2}{\tau_2}} \right], \quad (\text{A-6})$$

and the polarizability by

$$\eta_0 = \frac{k_B T}{ec_0 F} \frac{8(t_{n2}t_{p1} - t_{n1}t_{p2})^2}{\frac{\Delta L_1}{\tau_1} + \frac{\Delta L_2}{\tau_2}} \frac{1}{Z_{DC}}. \quad (\text{A-7})$$

Long narrow pore model

In this section, an approximate impedance model for the case of (1) relatively LNP $A < C/100$ and (2) approximately equal mobilities $\mu_{p1} \approx \mu_{p2} \approx \mu_{n1}$ is derived. Expanding the last term of equation A-5 by $\tau_1/\Delta L_1$ gives

$$Z(\omega) = Z_{DC} \left[1 - \eta_0 \times \left(1 - \frac{1 + \frac{\Delta L_2 \tau_1}{\Delta L_1 \tau_2}}{\sqrt{i\omega\tau_1} \coth \sqrt{i\omega\tau_1} + \frac{\Delta L_2 \tau_1}{\Delta L_1 \tau_2} \sqrt{i\omega\tau_2} \coth \sqrt{i\omega\tau_2}} \right) \right]. \quad (\text{A-8})$$

The ratio of the time constants

$$\frac{\tau_1}{\tau_2} = \frac{\Delta L_1^2}{8D_{p1}t_{n1}} \frac{8D_{p2}t_{n2}}{\Delta L_2^2} \quad (\text{A-9})$$

may be expressed in terms of the length ratio $A = \Delta L_1/\Delta L_2$ and the mobility contrast $C = \mu_{n1}/\mu_{n2}$. Condition (2) $\mu_{p1} \approx \mu_{p2} \approx \mu_{n1}$ yields $t_{n1} \approx 0.5$ and $D_{p1}/D_{p2} \approx 1$. For the anionic transport number in the active zone, we get

$$t_{n2} = \frac{\mu_{n2}}{\mu_{p2} + \mu_{n2}} = \frac{1}{\mu_{p2}/\mu_{n2} + 1} \approx \frac{1}{\mu_{n1}/\mu_{n2} + 1} = \frac{1}{C + 1}. \quad (\text{A-10})$$

Considering condition (1) $A < C/100$, the ratio of the time constants (equation A-9) may then be approximated as follows

$$\frac{\tau_1}{\tau_2} \approx \frac{2A^2}{C + 1} < \frac{2C}{100^2(1 + 1/C)}. \quad (\text{A-11})$$

In the case of moderate mobility contrasts (approximately $C \leq 100$), this ratio is very small ($\tau_1/\tau_2 \ll 1$ or $\tau_1 \ll \tau_2$). Moreover, the last term of the numerator in equation A-8 gives

$$\frac{\Delta L_2 \tau_1}{\Delta L_1 \tau_2} \approx \frac{2A}{C + 1} < \frac{2}{100(1 + 1/C)} \leq \frac{2}{100} \quad (\text{A-12})$$

and may therefore be neglected. In the denominator, the approximate ratio of the of the time constants (equation A-11) can be used for the substitution $\tau_2 \approx \tau_1(C + 1)/(2A^2)$

$$Z(\omega) \approx Z_{\text{DC}} \left[1 - \eta_0 \times \left(1 - \frac{1}{\sqrt{i\omega\tau_1} (\coth \sqrt{i\omega\tau_1} + (\frac{2}{C+1})^{1/2} \coth \sqrt{i\omega\tau_2})} \right) \right]. \quad (\text{A-13})$$

Furthermore, the magnitude of the imaginary part of $\coth(\sqrt{i\omega\tau_i})$ decreases at a sufficient rate with growing arguments $\omega\tau_i$, so that in conjunction with $\tau_1 \ll \tau_2$ we have $|\text{Im}(\coth \sqrt{i\omega\tau_1})| > |\text{Im}(\coth \sqrt{i\omega\tau_2})|$. For the real part, this relation does not apply in a similarly rigorous manner, but still holds $|\text{Re}(\coth \sqrt{i\omega\tau_1})| \geq |\text{Re}(\coth \sqrt{i\omega\tau_2})|$. As the mobility contrast is assumed to be $C \geq 1$, the factor $(2/(C+1))^{1/2}$ is smaller than 1. Neglecting the last term in the denominator of equation A-13, this leads to our new LNP model

$$Z_{\text{LNP}}(\omega) = Z_{\text{DC}} \left[1 - \eta_0 \left(1 - \frac{\tanh \sqrt{i\omega\tau_1}}{\sqrt{i\omega\tau_1}} \right) \right]. \quad (\text{A-14})$$

Although the derivation of this expression requires the additional assumption of moderate mobility contrasts ($C \leq 100$), it can be shown in a similar fashion that it holds for very pronounced mobility contrasts ($C \gg 100$) as well.

The characteristic time of the impedance model (equation A-14) is not equal to the time constant τ_1 , but can be calculated as the inverse of the characteristic frequency $\omega_c = 1/\tau_{\text{LNP}}$, where the phase maximum is observed. For $\omega = \omega_c$, the derivative of the phase (inverse tangent of the ratio of imaginary and real part of the impedance) with respect to the frequency is set to zero. This leads to a relation between ω_c , τ_1 , and η_0 , which cannot be solved explicitly for ω_c . Here, it is solved for η_0

$$\eta_0(\omega_c \tau_1) = 1 + 2 \times \frac{-\cot \sqrt{2\omega_c \tau_1} + \coth \sqrt{2\omega_c \tau_1}}{2\sqrt{2\omega_c \tau_1} - \frac{\cosh \sqrt{2\omega_c \tau_1}}{\sinh \sqrt{2\omega_c \tau_1}} + \frac{\cos \sqrt{2\omega_c \tau_1}}{\sinh \sqrt{2\omega_c \tau_1}} + \cot \sqrt{2\omega_c \tau_1} - \coth \sqrt{2\omega_c \tau_1}}. \quad (\text{A-15})$$

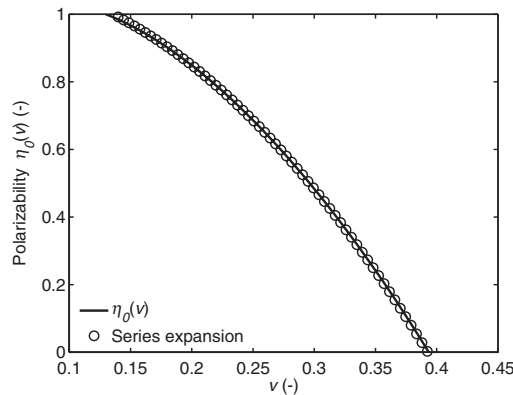


Figure A-1. The polarizability η_0 as a function of $v = 1/(\omega_c \tau_1)$ (black solid line) according to equation A-15 and the second-order approximation about $v = 0.25$ according to equation A-16 (gray dashed line) match very well within the range $0 \leq \eta_0 \leq 1$.

We are aiming at a relation $\tau_c = v(\eta_0) \cdot \tau_1$, where v is a dimensionless factor. For this purpose, $\sqrt{2\omega_c \tau_1}$ is expressed as $\sqrt{2/v}$. Thereafter, the function $\eta_0(v)$ is expanded in a Taylor series about $v = 0.25$, which gives a good fit (see Figure A-1). The numerical expansion yields

$$\eta_0(v) \approx \alpha + \beta(v - 0.25) + \gamma(v - 0.25)^2 \quad (\text{A-16})$$

with the coefficients

$$\alpha = 0.686272, \quad \beta = -3.65885 \quad \text{and} \quad \gamma = -8.03007. \quad (\text{A-17})$$

To finally yield the desired relation, the quadratic equation A-16 is solved for v .

$$v(\eta_0) = -\frac{\beta - 0.5\gamma}{2\gamma} \pm \left[\left(\frac{\beta - 0.5\gamma}{2\gamma} \right)^2 - \frac{\alpha - 0.25\beta + 0.0625\gamma - \eta_0}{\gamma} \right]^{1/2}. \quad (\text{A-18})$$

As only values $\eta_0 \geq 0$ are meaningful from a physical point of view, we are left with the positive root. The sought-after factor is given by

$$v(\eta_0) \approx 0.022178 + (0.137366 - 0.124532\eta_0)^{1/2}. \quad (\text{A-19})$$

In summary, the characteristic time of the LNP model, equation A-14, can be expressed in terms of the factor v as follows

$$\tau_{\text{LNP}} \approx [0.022178 + (0.137366 - 0.124532\eta_0)^{1/2}] \tau_1. \quad (\text{A-20})$$

For small polarizabilities $\eta_0 \ll 1$, this characteristic time reduces to $\tau_{\text{LNP}} \approx 0.4\tau_1$.

New short narrow pore model

In a similar fashion, an approximate impedance model can be found for the case of (1) relatively SNP $A \gg 5C$, (2) approximately equal mobilities $\mu_{p1} \approx \mu_{p2} \approx \mu_{n1}$, and (3) reduced anion mobilities in the active zone $\mu_{n1} > \mu_{n2}$. Further below, we will see that in the targeted frequency range ($\omega \approx \omega_c$) these three conditions imply $\omega\tau_1 \gg 1$ and $\omega\tau_2 < 1$. Using the latter relations, $\sqrt{i\omega\tau_1} \coth \sqrt{i\omega\tau_1}$ becomes $\sqrt{i\omega\tau_1}$ and $\sqrt{i\omega\tau_2} \coth \sqrt{i\omega\tau_2}$ may be approximated by 1. As a result, the Marshall-Madden impedance, equation A-5, becomes

$$Z(\omega) \approx Z_{\text{DC}} \left[1 - \eta_0 \left(1 - \frac{\frac{\Delta L_1}{\tau_1} + \frac{\Delta L_2}{\tau_2}}{\sqrt{i\omega\tau_1} + \frac{\Delta L_2}{\tau_2}} \right) \right]. \quad (\text{A-21})$$

Expansion of the last term of this equation by $\tau_2/\Delta L_2$ gives

$$Z(\omega) \approx Z_{\text{DC}} \left[1 - \eta_0 \left(1 - \frac{\frac{\Delta L_1 \tau_2}{\Delta L_2 \tau_1} + 1}{\frac{\Delta L_1 \tau_2}{\Delta L_2 \tau_1} \sqrt{i\omega\tau_1} + 1} \right) \right]. \quad (\text{A-22})$$

As before, condition (2) yields $t_{n1} \approx 0.5$, $D_{p1}/D_{p2} \approx 1$, and $t_{n2} \approx 1/(C+1)$ (see equation A-10). In conjunction with condition (1) $A \gg 5C$, the enumerator term becomes

$$\frac{\Delta L_1 \tau_2}{\Delta L_2 \tau_1} = \frac{\Delta L_1}{\Delta L_2} \frac{\Delta L_2^2}{8D_{p2} t_{n2}} \frac{8D_{p1} t_{n1}}{\Delta L_1^2} \approx \frac{0.5(C+1)}{A} \ll 1, \quad (\text{A-23})$$

and may therefore be neglected. After the introduction of a modified time constant

$$\tau' = \tau_2 \frac{D_{p1} t_{n1}}{D_{p2} t_{n2}}, \quad (\text{A-24})$$

equation A-22 reduces to our nSNP model given by

$$Z_{\text{nSNP}}(\omega) = Z_{\text{DC}} \left[1 - \eta_0 \left(1 - \frac{1}{1 + \sqrt{i\omega\tau'}} \right) \right]. \quad (\text{A-25})$$

To calculate the characteristic time τ_{nSNP} , the phase of the impedance, equation A-25, is differentiated with respect to the frequency and set to zero. The resulting characteristic frequency ω_c , where the phase maximum is observed is

$$\omega_c = \frac{1}{\tau'(1 - \eta_0)}. \quad (\text{A-26})$$

The characteristic time of the nSNP model may be expressed in terms of the polarizability η_0 and the modified time constant τ' .

$$\tau_{\text{nSNP}} = \frac{1}{\omega_c} = \tau'(1 - \eta_0). \quad (\text{A-27})$$

For small polarizabilities $\eta_0 \ll 0$, the characteristic time reduces to $\tau_{\text{nSNP}} \approx \tau'$.

Finally, the initially claimed implication of $\omega\tau_1 \gg 1$ and $\omega\tau_2 < 1$ from conditions (1), (2), and (3) can be demonstrated in the proximity of ω_c , where it should hold. For this purpose, we set $\omega = \omega_c$ and approximate ω_c by $1/\tau'$. With the modified time constant, equation A-24, this gives

$$\omega_c \tau_1 \approx \frac{\tau_1 D_{p2} t_{n2}}{\tau_2 D_{p1} t_{n1}}. \quad (\text{A-28})$$

As we already know, condition (2) $\mu_{p1} \approx \mu_{p2} \approx \mu_{n1}$ yields $t_{n1} \approx 0.5$ and $D_{p2}/D_{p1} \approx 1$. In addition, we have $t_{n2} \approx 1/(C+1)$ (see equation A-10). Substituting the time constant ratio, equation A-11, into equation A-28 and considering condition (1) $A \gg 5C$ gives

$$\omega_c \tau_1 \approx \left(\frac{2A}{C+1} \right)^2 \gg 1. \quad (\text{A-29})$$

With condition (3) $C = \mu_{n1}/\mu_{n2} > 1$ and the approximation $\omega_c \approx 1/\tau'$, the second implication may be easily shown

$$\omega_c \tau_2 \approx \frac{\tau_2}{\tau'} = \frac{D_{p2} t_{n2}}{D_{p1} t_{n1}} \approx \frac{2}{C+1} < 1. \quad (\text{A-30})$$

Titov's short narrow pore model

The original SNP model by Titov et al. (2002) can be derived in a similar manner. Here, the narrow pore is assumed to be absolutely shorter than the wide pore $A \gg 1$, unlike the nSNP model where the assumption was expressed relative to the mobility contrast. In addition, all mobilities are assumed to be approximately the same $\mu_{p1} \approx \mu_{p2} \approx \mu_{n1} \approx \mu_{n2}$ (i.e., $C \approx 1$). Expansion of the last term of equation A-5 by $\Delta L_2 \tau_1 / \Delta L_1^2$ gives

$$Z(\omega) \approx Z_{\text{DC}} \left[1 - \eta_0 \times \left(1 - \frac{\frac{1}{A} + 1}{\frac{1}{A} \sqrt{i\omega\tau_1} \coth \sqrt{i\omega\tau_1} + \sqrt{i\omega\tau_2} \coth \sqrt{i\omega\tau_2}} \right) \right]. \quad (\text{A-31})$$

Here, we have already considered the condition $\mu_{p1} \approx \mu_{p2} \approx \mu_{n1} \approx \mu_{n2}$ and the definition of the time constants $\tau_i = \Delta L_i^2 / (8D_{pi} t_{ni})$, leading to

$$\frac{\tau_1 \Delta L_2^2}{\tau_2 \Delta L_1^2} = \frac{D_{p2} t_{n2}}{D_{p1} t_{n1}} \approx 1. \quad (\text{A-32})$$

Similarly, one can show that $1/A\sqrt{i\omega\tau_1}$ can be approximated by $\sqrt{i\omega\tau_2}$. As $A \gg 1$, the term $1/A$ in the enumerator may be neglected and we get

$$Z(\omega) \approx Z_{\text{DC}} \left[1 - \eta_0 \times \left(1 - \frac{1}{\sqrt{i\omega\tau_2} (\coth \sqrt{i\omega\tau_1} + \coth \sqrt{i\omega\tau_2})} \right) \right]. \quad (\text{A-33})$$

This impedance model is supposed to hold in the proximity of $\omega_c \approx 1/\tau_2$. Therefore, the expression $\omega\tau_1$ can be approximated by $\omega_c \tau_1 \approx \frac{\tau_1}{\tau_2} = A^2 \gg 1$, so that $\coth \sqrt{i\omega_c \tau_1} \approx 1$. Altogether, we get the SNP impedance model

$$Z_{\text{SNP}}(\omega) = Z_{\text{DC}} \left[1 - \eta_0 \left(1 - \frac{1}{\sqrt{i\omega\tau_2} (1 + \coth \sqrt{i\omega\tau_2})} \right) \right]. \quad (\text{A-34})$$

When the cotangent function is expressed in terms of exponentials, a last rearrangement finally leads to the same expression as derived by Titov et al. (2002)

$$Z_{\text{SNP}}(\omega) = Z_{\text{DC}} \left[1 - \eta_0 \left(1 - \frac{1 - e^{-2\sqrt{i\omega\tau_2}}}{2\sqrt{i\omega\tau_2}} \right) \right]. \quad (\text{A-35})$$

Under the condition $\mu_{p2} \approx \mu_{n2}$, the time constant becomes

$$\tau_2 = \frac{\Delta L_2^2}{8D_{p2} t_{n2}} \approx \frac{\Delta L_2^2}{4D_{p2}}. \quad (\text{A-36})$$

To derive the corresponding characteristic time τ_{SNP} , the phase of the impedance, equation A-35, is differentiated with respect to the frequency. By setting this derivative to zero, we arrive at the

following equation, which relates the characteristic frequency ω_c with the polarizability η_0 and the time constant τ_2

$$\begin{aligned} & -\cos \sqrt{2\omega_c \tau_2} \left(\frac{1}{\eta_0} - 2 \right) 2e^{\sqrt{2\omega_c \tau_2}} - 2 \\ & - \sin \sqrt{2\omega_c \tau_2} \frac{1}{\eta_0} 2e^{\sqrt{2\omega_c \tau_2}} \\ & - 2 \sin \sqrt{2\omega_c \tau_2} \left(\frac{1}{\eta_0} - 1 \right) 2\sqrt{2\omega_c \tau_2} e^{\sqrt{2\omega_c \tau_2}} \\ & + 2e^{2\sqrt{2\omega_c \tau_2}} \left(\frac{1}{\eta_0} - 1 \right) = 0. \end{aligned} \quad (\text{A-37})$$

Because small mobility contrasts $C \approx 1$ are considered, only small polarizabilities are expected. Therefore, it is sufficient to analyze equation A-37 in the limit $\eta_0 \rightarrow 0$, where it becomes

$$\begin{aligned} & \cos \sqrt{2\omega_c \tau_2} + \sin \sqrt{2\omega_c \tau_2} + 2\sqrt{2\omega_c \tau_2} \sin \sqrt{2\omega_c \tau_2} \\ & - e^{\sqrt{2\omega_c \tau_2}} = 0. \end{aligned} \quad (\text{A-38})$$

Solving this equation numerically yields only one real-valued root $\omega_c \tau_2 \approx 0.892$. The approximate characteristic time τ_{SNP} is related to the time constant τ_2 as follows:

$$\tau_{\text{SNP}} = \frac{1}{\omega_c} \approx 1.12 \cdot \tau_2. \quad (\text{A-39})$$

REFERENCES

- Angoran, Y., and T. R. Madden, 1977, Induced polarization: A preliminary study of its chemical basis: *Geophysics*, **42**, 788–803, doi: [10.1190/1.1440747](#).
- Binley, A., L. Slater, M. Fukes, and G. Cassiani, 2005, Relationship between spectral induced polarization and hydraulic properties of saturated and unsaturated sandstone: *Water Resources Research*, **41**, W12417, doi: [10.1029/2005WR004202](#).
- Blaschek, R., and A. Hördt, 2009, Numerical modelling of the IP effect at the pore scale: *Near Surface Geophysics*, **7**, 579–588.
- Bleil, D., 1953, Induced polarisation: A method of geophysical prospecting: *Geophysics*, **18**, 636–661, doi: [10.1190/1.1437917](#).
- Börner, F., and J. Schön, 1991, A relation between the quadrature component of electrical conductivity and the specific surface area of sedimentary rocks: *The Log Analyst*, **32**, 612–613.
- Börner, F., J. Schopper, and A. Weller, 1996, Evaluation of transport and storage properties in the soil and groundwater zone from induced polarization measurements: *Geophysical Prospecting*, **44**, 583–601, doi: [10.1111/j.1365-2478.1996.tb00167.x](#).
- Buchheim, W., and G. Irmer, 1979, Zur Theorie der induzierten galvanischen Polarisation in Festkörpern mit elektrolytischer Porenfüllung: *Gerlands Beiträge zur Geophysik*, **88**, 53–72.
- Bücker, M., and A. Hördt, 2013, Analytical modeling of membrane polarization with explicit parameterization of pore radii and the electrical double layer: *Geophysical Journal International*, **194**, 804–813, doi: [10.1093/gji/ggt136](#).
- Cole, K. S., and R. H. Cole, 1941, Dispersion and absorption in dielectrics — I. Alternating current characteristics: *Journal of Chemical Physics*, **9**, 341–351, doi: [10.1063/1.1759096](#).
- Flores Orozco, A., A. Kemna, C. Oberdörster, L. Zschornack, C. Leven, P. Dietrich, and H. Weiss, 2012, Delineation of subsurface hydrocarbon contamination at a former hydrogenation plant using spectral induced polarization imaging: *Journal of Contaminant Hydrology*, **136–137**, 131–144, doi: [10.1016/j.jconhyd.2012.06.001](#).
- Franceschetti, D. R., J. R. Macdonald, and R. P. Buck, 1991, Interpretation of finite-length-Warburg-type impedances in supported and unsupported electrochemical cells with kinetically reversible electrodes: *Journal of the Electrochemical Society*, **138**, 1368–1371, doi: [10.1149/1.2085788](#).
- Fridriksberg, D., and M. Sidorova, 1961, Issledovanie svyazi yavleniya vyzvannoi polarizatsii s elektrokineticheskimi svoystvami kapillyarnykh sistem (A study of relationship between the induced polarization phenomenon and the electro-kinetic properties of capillary systems): *Vestnik Leningradskogo Universiteta, Seria Khimiya*, **4**, 222–226.
- Havriliak, S., and S. Negami, 1967, A complex plane representation of dielectric and mechanical relaxation processes in some polymers: *Polymer*, **8**, 161–210, doi: [10.1016/0032-3861\(67\)90021-3](#).
- Hördt, A., R. Blaschek, F. Binot, A. Druiventak, A. Kemna, P. Kreye, and N. Zisser, 2009, Case histories of hydraulic conductivity estimation with induced polarization at the field scale: *Near Surface Geophysics*, **7**, 529–545.
- Kemna, A., H. Münch, K. Titov, E. Zimmermann, and H. Vereecken, 2005, Relation of SIP relaxation time of sands to salinity, grain size and hydraulic conductivity: 11th European Meeting of Environmental and Engineering Geophysics, EAGE, Expanded Abstracts, P054.
- Kruschwitz, S., A. Binley, D. Lesmes, and A. Elshenawy, 2010, Textural controls on low-frequency electrical spectra of porous media: *Geophysics*, **75**, no. 4, WA113–WA123, doi: [10.1190/1.3479835](#).
- Leroy, P., and A. Revil, 2009, A mechanistic model for the spectral induced polarization of clay minerals: *Journal of Geophysical Research*, **114**, doi: [10.1029/2008JB006114](#).
- Leroy, P., A. Revil, and D. Coelho, 2006, Diffusion of ionic species in bentonite: *Journal of Colloid and Interface Science*, **296**, 248–255, doi: [10.1016/j.jcis.2005.08.034](#).
- Leroy, P., A. Revil, A. Kemna, P. Cosenza, and A. Ghorbani, 2008, Complex conductivity of water-saturated packs of glass beads: *Journal of Colloid and Interface Science*, **321**, 103–117, doi: [10.1016/j.jcis.2007.12.031](#).
- Marshall, D. J., and T. R. Madden, 1959, Induced polarization: A study of its causes: *Geophysics*, **24**, 790–816, doi: [10.1190/1.1438659](#).
- O’Konski, C. T., 1960, Electric properties of macromolecules: V. Theory of theory of ionic polarization in polyelectrolytes: *Journal of Physical Chemistry*, **64**, 605–619, doi: [10.1021/j100834a023](#).
- Pape, H., L. Riepe, and J. R. Schopper, 1987, Theory of self-similar network structures in sedimentary and igneous rocks and their investigation with microscopical and physical methods: *Journal of Microscopy*, **148**, 121–147, doi: [10.1111/j.1365-2818.1987.tb02861.x](#).
- Pelton, W. H., S. H. Ward, P. G. Hallof, W. R. Sill, and P. H. Nelson, 1978, Mineral discrimination and removal of inductive coupling with multifrequency IP: *Geophysics*, **43**, 588–609, doi: [10.1190/1.1440839](#).
- Revil, A., and N. Florsch, 2010, Determination of permeability from spectral induced polarization in granular media: *Geophysical Journal International*, **181**, 1480–1498.
- Revil, A., and P. W. J. Glover, 1997, Theory of ionic surface electrical conduction in porous media: *Physical Review B: Condensed Matter and Materials Physics*, **55**, 1757–1773, doi: [10.1103/PhysRevB.55.1757](#).
- Schmutz, M., A. Revil, P. Vaudelet, M. Batzle, P. Femenía Viñao, and D. D. Werkema, 2010, Influence of oil saturation upon spectral induced polarization of oil-bearing sands: *Geophysical Journal International*, **183**, 211–224, doi: [10.1111/j.1365-246X.2010.04751.x](#).
- Schurr, J. M., 1964, On the theory of the dielectric dispersion of spherical colloidal particles in electrolyte solution: *Journal of Physical Chemistry*, **68**, 2407–2413, doi: [10.1021/j100791a004](#).
- Schwarz, G., 1962, A theory of the low-frequency dielectric dispersion of colloidal particles in electrolyte solution: *Journal of Physical Chemistry*, **66**, 2636–2642, doi: [10.1021/j100818a067](#).
- Sidorova, M., and D. Fridriksberg, 1978, Study of induced polarization of systems simulating soils during freezing: *Permafrost: The USSR Contribution to the Second International Conference*, National Academy of Sciences, **34**, 7–349.
- Slater, L., D. Ntarlagiannis, and D. Wishart, 2006, On the relationship between induced polarization and surface area in metal-sand and clay-sand mixtures: *Geophysics*, **71**, no. 2, A1–A5, doi: [10.1190/1.2187707](#).
- Slater, L. D., and D. P. Lesmes, 2002, Electric hydraulic relationships observed for unconsolidated sediments: *Water Resources Research*, **38**, 31–31–31–13, doi: [10.1029/2001WR001075](#).
- Titov, K., V. Komarov, V. Tarasov, and A. Levitski, 2002, Theoretical and experimental study of time domain-induced polarization in water-saturated sands: *Journal of Applied Geophysics*, **50**, 417–433, doi: [10.1016/S0926-9851\(02\)00168-4](#).
- Titov, K., A. Tarasov, Y. Ilyin, N. Seleznev, and A. Boyd, 2010, Relationships between induced polarization relaxation time and hydraulic properties of sandstone: *Geophysical Journal International*, **180**, 1095–1106, doi: [10.1111/j.1365-246X.2009.04465.x](#).
- Vanhala, H., 1997, Mapping oil-contaminated sand and till with the spectral induced polarization (SIP) method: *Geophysical Prospecting*, **45**, 303–326, doi: [10.1046/j.1365-2478.1997.00338.x](#).
- Vinegar, H., and M. Waxman, 1984, Induced polarization of shaly sands: *Geophysics*, **49**, 1267–1287, doi: [10.1190/1.1441755](#).
- Volkman, J., and N. Klitzsch, 2010, Frequency-dependent electric properties of microscale rock models for frequencies from one millihertz to ten kilohertz: *Vadose Zone Journal*, **9**, 858–870.

- Weller, A., K. Breede, L. Slater, and S. Nordsiek, 2011, Effect of changing water salinity on complex conductivity spectra of sandstones: *Geophysics*, **76**, no. 5, F315–F327, doi: [10.1190/geo2011-0072.1](https://doi.org/10.1190/geo2011-0072.1).
- Weller, A., S. Nordsiek, and W. Debschütz, 2010a, Estimating permeability of sandstone samples by nuclear magnetic resonance and spectral-induced polarization: *Geophysics*, **75**, no. 6, E215–E226, doi: [10.1190/1.3507304](https://doi.org/10.1190/1.3507304).
- Weller, A., L. Slater, S. Nordsiek, and D. Ntarlagiannis, 2010b, On the estimation of specific surface per unit pore volume from induced polarization: A robust empirical relation fits multiple data sets: *Geophysics*, **75**, no. 4, WA105–WA112, doi: [10.1190/1.3471577](https://doi.org/10.1190/1.3471577).
- Wong, J., 1979, An electrochemical model of the induced-polarization phenomenon in disseminated sulfide ores: *Geophysics*, **44**, 1245–1265, doi: [10.1190/1.1441005](https://doi.org/10.1190/1.1441005).

Temperature dependence of spectral induced polarization data: experimental results and membrane polarization theory

Katharina Bairlein,¹ Matthias Bucker,^{2,3} Andreas Hördt¹ and Björn Hinze⁴

¹Institute of Geophysics and Extraterrestrial Physics, TU Braunschweig, Mendelssohnstr. 3, D-38106 Braunschweig, Germany. E-mail: k.bairlein@tu-bs.de

²Department of Geophysics, Steinmann Institute, University of Bonn, Bonn, Germany

³Research Group Geophysics, Department of Geodesy and Geoinformation, Vienna University of Technology, Vienna, Austria

⁴Institute for Materials, TU Braunschweig, Braunschweig, Germany

Accepted 2016 January 18. Received 2016 January 18; in original form 2015 February 13

SUMMARY

Spectral induced polarization measurements are affected by temperature variations due to a variety of temperature-dependent parameters that control the complex electrical conductivity. Most important is the influence of the ion mobility, which increases with increasing temperature. It is responsible for the increase of the conductivity of the fluid in the pores with temperature and influences the electrical double layer on the mineral surface. This work is based on laboratory measurements of 13 sandstone samples from different sources with different geological and petrophysical characteristics. We measured the complex impedance in a frequency range from 0.01 to 100 Hz and a temperature range from 0 to 40 °C. The main observation is a decrease of the characteristic time (defined by the inverse of the frequency, at which the phase shift is maximum) with increasing temperature. The strength of this decrease differs from one sample to another. The temperature dependence of the phase shift magnitude cannot easily be generalized, as it depends on the particular sample. The experimental findings suggest that neglecting the influence of temperature on complex conductivity may lead to significant errors when estimating hydraulic conductivity from relaxation time. We also simulate the temperature dependence with a theoretical model of membrane polarization and review some of the model properties, with an emphasis on the temperature dependence of the parameters. The model reproduces several features characterizing the measured data, including the temperature dependence of the characteristic times. Computed tomography and microscope images of the pore structure of three samples also allow us to associate differences in the geometrical parameters used in the modelling with pore scale parameters of the actual samples.

Key words: Electrical properties; Hydrogeophysics; Hydrothermal systems; Permeability and porosity.

1 INTRODUCTION

The complex electrical conductivity of sediments is influenced by different physical properties, including the temperature. Consequently, variations of the temperature of the subsurface may change the results of induced polarization (IP) measurements in the field. Seasonal and weather-related differences in temperature may lead to notable variations particularly for the shallow subsurface, and measurement results might not be comparable if taken at different temperatures. Relations between IP parameters and petrophysical or hydrological parameters are often based on laboratory measurements, which are usually performed at room temperature or in climate chambers. The typical temperature is around 25 °C, which does not necessarily coincide with the ambient temperature of the

subsurface under *in situ* conditions. The differences in temperature near the surface are usually small, but it is also conceivable that induced polarization measurements may be used for exploration of geothermal areas, where a large temperature range is covered, as IP is particularly suitable for estimating the hydraulic conductivity of the subsurface (e.g. Börner *et al.* 1996; Hördt *et al.* 2007).

The temperature dependence of DC conductivity was studied by several authors, using various physical and empirical approaches (e.g. Llera *et al.* 1990; Sen & Goode 1992; Hayley *et al.* 2007) and is well known. However, there is a number of temperature-dependent parameters, which influence the complex frequency-dependent electrical conductivity additionally to the DC conductivity of porous media. The complex electrical conductivity increases with increasing temperature, both in the real and imaginary part, as reported,

for example, in Binley *et al.* (2010) for temperatures between 5 and 30 °C. The time constant decreases with increasing temperature as a consequence of increasing ion mobility. This is described for timescales determined from the inverse of the spectral position of the phase maximum (Binley *et al.* 2010) and for mean relaxation times from a Debye decomposition (Zisser *et al.* 2010a) and also by Tong & Tao (2007) for IP measurements in the time domain. Tong & Tao (2007) also proposed a correction of temperature effects on the relaxation time of shaly sands by shifting the relaxation time spectra along the relaxation time axis.

Observations of measured parameters describing the polarization strength differ depending on the investigated material. Collett (1959) reported an increase of the polarization strength of andesite particles with increasing temperature for clean material, but a decrease if a small amount of pyrite is added. A similar relation was found by Vinegar & Waxman (1984) for a sandstone sample with low clay content compared to a sandstone sample with high clay content for temperatures between −5 and 95 °C. An influence of the clay content on the temperature dependence was observed also by Treichel *et al.* (2014). However, the phase shift of the sandstone samples investigated by Binley *et al.* (2010) remained constant. Worthington & Collar (1984) refer to measurements of other authors, who observed a change in the polarization with temperature that was not monotonous.

Although there are several investigations of the temperature dependence of spectral induced polarization (SIP), the existing data comprise only a few different sample materials and some of the observations differ. Especially, the behaviour of the phase shift is poorly constrained. Moreover, it is unclear if conclusions made on observations of individual samples are valid also for other materials.

The theoretical background of the temperature dependence of the complex electrical conductivity is based mainly on empirical or semi-empirical approaches, which do not cover all characteristics of the measured data. An approach to explain the temperature dependence of the phase shift can be found in Revil (2012), who describes the influence of temperature on SIP with different temperature dependencies of the ion mobilities in the free electrolyte, the diffuse layer and the Stern layer of the electrical double layer (EDL). However, a general conclusion about the temperature behaviour is not suggested.

Temperature-dependent SIP data may also be useful to test existing theories about the origin of the IP effect, including mechanistic models. Mechanistic models in turn provide a good opportunity to improve our understanding of the temperature dependence of the polarization effects in porous media. Furthermore, models can help to generalize the observations of measurements and predict the temperature dependence of SIP data.

Currently, there are principally two types of models, which describe polarization mechanisms at the pore scale; the grain-based electrochemical polarization model (e.g. Schwarz 1962; Leroy *et al.* 2008; Revil & Florsch 2010) and the pore-based membrane polarization model (e.g. Marshall & Madden 1959; Buchheim & Irmer 1979; Titov *et al.* 2002). The membrane polarization model, suggested by Marshall & Madden (1959), solves the equations describing the motion of ions for a 1-D geometry. Changing ion mobilities along the pore system leads to a concentration gradient, when an external electric field is applied. Bückner & Hördt (2013a) extended this model by explicitly parametrizing the EDL and the pore radii of cylindrical pores.

In order to gain further insight into the temperature dependence of SIP, we add additional data acquired with different materials and compare them with previous observations and empirical

relationships. We discuss the extended membrane polarization model by Bückner & Hördt (2013a) with an emphasis on the temperature dependence of the model parameters and compare measured with calculated phase spectra.

2 TEMPERATURE DEPENDENCE OF THE COMPLEX ELECTRICAL CONDUCTIVITY

There are two main factors causing the temperature dependence of the polarization response: (1) the electrical conductivity of the pore fluid, the temperature dependence of which basically stems from changes in ion mobility and (2) the EDL, coating the surface of the rock matrix or grains. The surface of the solid phase usually acquires a negative charge, when brought into contact with the pore fluid. Close to the surface within the electrolyte, the net surface charge results in an increased concentration of ions of opposite charge (counter ions). This region is known as the EDL, which consists of two parts; an inner Stern layer of strongly bound counter ions and an outer diffuse layer. Within the diffuse layer, ions can move freely but are affected by diffusion and electrostatic forces. The properties of the EDL are mainly controlled by the temperature-dependent parameters ion mobility, Zeta potential and Debye length.

For a small temperature range of 0–30 °C, the temperature dependence of the DC conductivity $\sigma(T)$ of the bulk electrolyte can be approximated by a simple linear relation (e.g. Sorensen & Glass 1987; Hayley *et al.* 2007; Binley *et al.* 2010):

$$\sigma(T) = \sigma(T_0)(1 + a(T - T_0)). \quad (1)$$

Here, $\sigma(T_0)$ is the DC conductivity (in S m^{-1}) at the reference temperature T_0 and a is a temperature compensation factor, which assumes values between 0.02 and 0.025 K^{-1} for different sample types and salinities. For a wider temperature range, the temperature dependencies of dynamic viscosity η , hydrodynamic radius and Coulomb interaction result in a non-linear relation. In this case, and for materials where surface conductivity can be neglected, the semi-empirical ‘modified Walden product’ appropriately describes the variation of the DC conductivity with temperature (e.g. Sorensen & Glass 1987).

$$\sigma(T) = \sigma(T_0) \left(\frac{\eta(T_0)}{\eta(T)} \right)^{\alpha_\sigma}. \quad (2)$$

The empirical constant α_σ is a measure of the strength of the temperature dependence. Eq. (2) is based on the assumption, that the temperature dependency of the ion mobility in the pore fluid is the basic cause of the temperature dependency of DC conductivity. The underlying relation between ion mobility and dynamic viscosity is:

$$\mu(T) = \mu(T_0) \left(\frac{\eta(T_0)}{\eta(T)} \right)^{\alpha_\mu}. \quad (3)$$

At temperatures between −20 and 110 °C and standard pressure, the dynamic viscosity of water can be calculated from the empirical equation by Pátek *et al.* (2009):

$$\eta(T) = 10^{-6} \cdot \sum_{i=1}^4 a_i \left(\frac{T}{300} \right)^{b_i}, \quad (4)$$

with parameters $a_i = [280.68, 511.45, 61.131, 0.45903]$ and $b_i = [-1.9, -7.7, -19.6, -40]$. η is given in Pa·s and T in K.

So far we have only considered the temperature dependence of the ion mobility and the DC conductivity of the bulk electrolyte. However, polarization effects taking place at the solid–fluid interface of a

porous medium give rise to a frequency dependence of the complex electrical conductivity. One important parameter that describes the spectral behaviour is the characteristic time τ of the polarization process, which is proportional to the square of a geometrical scale of the pores/grains as well as reciprocal to the diffusion coefficient D (e.g. Titov *et al.* 2002; Leroy *et al.* 2008; Bucker & Hördt 2013b). The relation between the diffusion coefficient and ion mobility μ is described by the Nernst–Einstein relation:

$$D = \frac{\mu k_B T}{e}. \quad (5)$$

Based on the changes in diffusion coefficient (eq. 5) and mobility (eq. 3), a first approximate relation between the characteristic time and the temperature can be derived. Following Zisser *et al.* (2010a), the characteristic time decreases with temperature as follows:

$$\tau(T) = \tau(T_0) \frac{T_0}{T} \left(\frac{\eta(T)}{\eta(T_0)} \right)^{\alpha_\tau}. \quad (6)$$

We use the symbols α_τ and α_μ for the Walden exponent if applied to the calculations of the characteristic times and mobilities, respectively. These two exponents are identical ($\alpha = \alpha_\mu = \alpha_\tau$) if the temperature dependence of the characteristic time only depends on the dynamic viscosity described by the modified Walden product. Depending on ion species and mobility, the Walden exponents range from 0.8 to 0.97 (Zisser *et al.* 2010a, and references therein).

However, eq. (6) does not account for the significant temperature variations of the Zeta potential ζ and Debye length λ_D , which control the properties of the EDL and consequently also affect the polarization process. The Zeta potential ζ is the electrical potential at the surface of shear, which is often used as the electrical potential at the interface between Stern and diffuse layers (e.g. Leroy & Revil 2004). Although the temperature dependence of the Zeta potential stems from changes in a number of parameters (e.g. pH, adsorption equilibria), the relation between Zeta potential and temperature is nearly linear (Revil *et al.* 1999):

$$\zeta(T) = \zeta(T_0)(1 + g(T - T_0)) \quad (7)$$

In the case of silica surfaces ζ increases approximately 1.7 per cent per K ($g \approx 0.017 \text{ K}^{-1}$).

The Debye length defines the characteristic spatial scale of the diffuse layer

$$\lambda_D = \sqrt{\frac{\epsilon_0 \epsilon_r k_B T}{2e c_0 z^2 F}}. \quad (8)$$

Here, $\epsilon_0 = 8.85 \times 10^{-12} \text{ As Vm}^{-1}$ is the vacuum permittivity and z the ion valence (we assume $z = 1$ throughout this work). The Debye length (eq. 8) depends on temperature both directly and indirectly, because the relative electrical permittivity ϵ_r of the pore fluid decreases with temperature (Pátek *et al.* 2009):

$$\epsilon_r(T) = \sum_{i=1}^4 e_i \left(\frac{T}{300} \right)^{f_i}. \quad (9)$$

Where $e_i = [-43.7527, 299.504, -399.364, 221.327]$ and $f_i = [-0.05, -1.47, -2.11, -2.31]$.

The effect of the temperature dependence of Debye length and Zeta potential on the complex electrical conductivity will be discussed in the context of a membrane polarization model.

3 EXPERIMENTAL DATA

Measurements were carried out on 13 different sandstone samples (see Table 1). Samples TU-B1, TU-B4 and TU-B5 stem from different drilling cores obtained at the Ketzin pilot site, Germany, (e.g. Fleury *et al.* 2013; Martens *et al.* 2013, 2014) and sample P8 is a Bunter sandstone from an outcrop in the eastern Thuringian Basin. Samples BeRo1, Biu1, FrUK1, GiUK1, koQ1, koVe1, OK1 and smVG1 are sandstones from different outcrops at the North German Basin (see Reyer & Philipp (2014) for more details). Sample S3 is an Elb Sandstone, which was previously investigated by Hördt & Milde (2012) in a different context. All samples were cut into cylinders with a diameter of 25 mm and a length of 50 mm. Available geological and petrophysical parameters of the samples are listed in Table 1. The porosities were determined from the dry and saturated weight and the volume of the samples. Dominant pore-throat diameters and the surface-area/pore-volume ratio were measured by mercury injection and nitrogen adsorption, respectively.

For three investigated rock samples (TU-B1, P8 and S3) X-ray computed tomography (CT) scans could be obtained using a GE phoenix nanotom[®] s system. The CT scans of these sandstone samples were acquired on small pieces of a few millimetres in diameter. The resolution of the scans was 1.25 μm (sample TU-B1), 1.5 μm (sample P8) and 2.0 μm (sample S3), respectively. 2-D images of the scans of the sandstone samples are shown in Fig. 1.

The main differences between the three sandstone samples are: (1) the size of the grains is smallest for sample TU-B1 and largest for S3. The grains of P8 and S3 are more rounded than the grains of TU-B1. (2) Samples P8 and S3 are more consolidated and compacted than TU-B1. The visually higher compaction is consistent with the lower porosity of sample P8 and S3 (0.11 and 0.19, for comparison: TU-B1 0.27), and the fact that P8 and S3 are robust while TU-B1 is brittle. Due to the different degrees of consolidation we assume that the pores of TU-B1 are wider than those of P8. The CT image shows that sample S3 has large pores compared to P8. This assumption is supported by mercury injection measurements, which reveal dominant pore-throat diameters of 18.3 μm for TU-B1, 2.9 μm for sample P8 and 44.4 μm for sample S3 (Table 1). However, sandstone samples TU-B1 and P8 are inhomogeneous and show regions with high cementation, where much smaller pores may exist. For estimating the pore length, we refer to the mean grain size, which is approximately 180 and 330 μm for P8 and S3, respectively (see Table 1). Because of the weak consolidation of sample TU-B1 we assume that the pores are longer than the mean grain size and similar to the length of sample P8.

3.1 SIP measurements

SIP measurements were performed with a VMP3 impedance analyser by Princeton Applied Research in the frequency range of 0.01–100 Hz. We used four-point measuring cells with non-polarizing potential electrodes as described in detail by Kruschwitz (2008), Hördt & Milde (2012) and Bairlein *et al.* (2014). The filled measuring cell was stored within a climate chamber with an accuracy of 0.1 °C to keep the temperature of the sample at the desired value. In order to assess the accuracy of the current measuring system, the measuring cell was filled with a sodium chloride solution ($\sigma_F(20^\circ\text{C}) = 50.0 \text{ mS m}^{-1}$) and measurements at temperatures between 0 and 90 °C were carried out. Fig. 2 shows that the measured electrical conductivity coincides with the increase of water conductivity with increasing temperature as predicted by the modified Walden product

Table 1. Geological and petrophysical parameters of the sandstone samples. Parameters listed are: porosity Φ , pore-fluid conductivity σ_f at 20 °C, dominant pore-throat radius r_{dom} , mean grain size d , specific surface area S_{por} , clay content, mineralogy, cementation c , sorting s and rounding r .

Sample	Local name	Φ [per cent]	σ_f [mS m ⁻¹]	r_{dom} [μm]	d [μm]	S_{por} [1 μm^{-1}]	Clay [per cent]	Mineralogy [per cent]	c [per cent]	s^c	r^c
TU-B1	Stuttgart Fm.	27.5	47.9	9.1	90	116.6	16.0 $\pm$ 10.4 ^a	36.6 $\pm$ 6.1 Q, 21.8 $\pm$ 4.6 F ^a	—	—	—
TU-B4	Exter Fm.	21.3	53.8	0.7	—	147.1	—	—	—	—	—
TU-B5	Exter Fm.	17.0	55.4	2.8	210	213.6	—	—	—	—	—
P8	Bunter Sst.	11.2	34.2	1.4	180	47.2	—	—	—	—	—
BeRo1	Rotliegend Sst.	6.9	64.6	6.0	400	64.9	3.3 ^b	59.1 Q, 4.2 F, 36.7 L ^b	17.6 ^b	4	2.5
BiSu1	Bernburg-F.	19.8	54.4	6.4	210	20.3	0.9 ^b	72.2 Q, 17.1 F, 10.7 L ^b	12.3 ^b	2	3
FrUK1	Bentheimer Sst.	14.6	57.4	5.9	100	46.6	4.3 ^b	96.1 Q, 2.1 F, 1.8 L ^b	17.0 ^b	3.5	3.5
GiUK1	Gildehaus Sst.	22.7	54.9	16.8	250	1.8	1.3 ^b	94.9 Q, 2.4 F, 2.7 L	16.8	2	2.5
koQ1	Rhät Sst.	17.1	56.9	9.7	200	20.6	0.0 ^b	78.6 Q, 4.7 F, 16.7 L ^b	31.0 ^b	1	3
koVe1	Rhät Sst.	13.0	60.6	0.9	100	96.8	0.0 ^b	83.2 Q, 3.8 F, 13.0 L ^b	40.0 ^b	1	3
OK1	Wealden Sst.	18.9	58.4	1.5	200	14.6	2.8 ^b	94.2 Q, 4.0 F, 1.8 L ^b	17.4 ^b	2	3
smVG1	Volpriehausen-F.	20.1	46.6	5.3	210	11.3	0.8 ^b	80.1 Q, 15.8 F, 4.1 L ^b	23.8 ^b	3.5	3
S3	Elb Sst.	19.3	54.8	22.2	330	9.7	—	—	—	—	—

Notes: Q: quartz, F: feldspar, L: lithoclasts.

^aFrom Förster *et al.* (2009), determined for the formation.

^bFrom Reyer (2014), measured on similar samples.

^c1: very good, 2: good, 3: intermediate, 4: poor, 5: very poor.

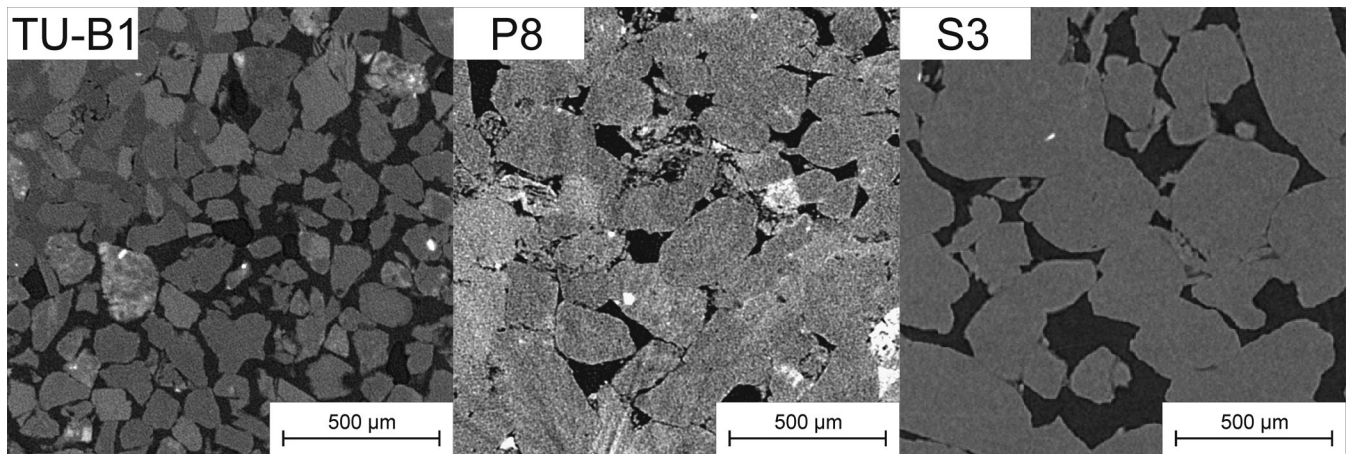


Figure 1. 2-D images of computed tomography (CT) scans of the pore space of three investigated samples. Left: sample TU-B1. Middle: sample P8. Right: sample S3.

(eq. 2) for $\alpha_\mu = 0.91$ and $T_0 = 25^\circ\text{C}$. This observation is independent of frequency. The measured phase shift is zero with a standard deviation of 0.2 mrad (not shown here for brevity) for frequencies between 1 mHz and 10 Hz. During the measurements, particularly at high temperatures, a small amount of water (a few millilitres) evaporated through narrow chinks between the individual parts of the sample holder, which led to a slight increase of fluid conductivity to $\sigma_f(20^\circ\text{C}) = 54.9 \text{ mS m}^{-1}$ by the end of this test. Due to this change in fluid conductivity α_μ might be slightly overestimated, but it is still in the range between 0.8 and 0.97 given by Zisser *et al.* (2010a).

Before the actual measurements, the rock samples were dried in a vacuum chamber at 30°C for at least 48 hr to remove residual water from the pores. Afterwards, the samples were saturated with a sodium chloride solution. The saturation process was performed slowly and under vacuum conditions in order to avoid air being trapped inside the pores. After saturation, the sample and the fluid had to equilibrate, which took several days, differing from one sample to another. When water conductivity had remained constant for at least two days, the sample was inserted into the sample holder. At the time of the measurements, the ion concentration had increased,

most probably due to the dissolution of salt that precipitated during the drying procedure and/or the dissolution of ions from the mineral matrix of the samples. Pore fluid conductivities of the samples taken before the SIP measurements are listed in Table 1.

Temperature was increased from 0 to 40°C in steps of 5°C . The time between increasing the temperature and starting the measurement was at least 24 hr to ensure that the sample was at the required temperature and chemical equilibrium during the measurement. Tests showed that the major change in conductivity occurred within the first two hours after a temperature change of 5°C . Before each of these measurement series from 0 to 40°C , one reference measurement at 20°C was performed to assess any change of the measured spectra with time. A comparison of the reference spectra with those taken at 20°C during the measurement series shows a slight change of electrical conductivity with time, which is supposed to arise from ions dissolved during the measurement series. The real part of the conductivity at 20°C increases stronger during the measurements series than the imaginary part, resulting in a slight decrease of the phase shift with time.

Both real and imaginary part of the electrical conductivity increase with temperature, shown exemplarily for sample TU-B1 in

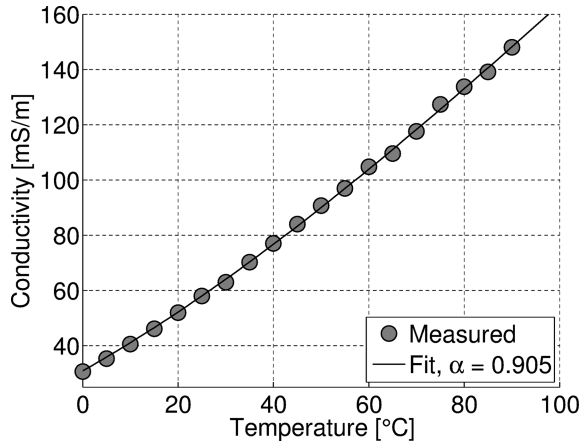


Figure 2. Magnitude of the electrical conductivity at 1 Hz versus temperature for a water-filled sample holder (circles) fitted with eq. (2) and $\alpha_\sigma = 0.905$ (solid line). The fluid conductivity at 20 °C increased during the measurement series from 50.0 to 54.9 mS m⁻¹.

Fig. 3. In agreement with eq. (1), the real part increases linearly by about 2 per cent per °C regardless of the frequency. The temperature increase of the imaginary part though depends on frequency. Its increase is largest (2.4 per cent per K for sample TU-B1) at the peak frequency f_{peak} , where the maximum of the imaginary conductivity can be observed, and smallest (1.9 per cent per K) at 0.01 Hz, where the phase shift is smallest. The different temperature dependencies of real and imaginary conductivity result in slight changes in the phase shift with temperature.

The temperature dependence of the real part of conductivity can be described quantitatively by fitting eq. (2) to the data. The Walden exponents α_σ presented in Table 2 are determined from the real conductivity at 1 Hz, which is typically used for evaluating SIP data at a single frequency (e.g. Weller *et al.* 2010). As stated before, for the real part of conductivity, the temperature dependence is independent of frequency. Most of the values are in the range previously reported in literature. Some of the samples exhibit stronger temperature dependencies ($\alpha_\sigma > 1$). Exponents > 1 mean that temperature-dependent parameters other than ion mobility become important. One possible explanation is that surface conductivity, which also depends on Zeta potential and the EDL thickness, becomes important.

The phase spectra of six of the investigated samples are displayed in Fig. 4. A common feature of the phase spectra is the increase of the frequency of the maximum phase shift with temperature, that is, the spectra shift towards higher frequencies. This is particularly obvious for all samples except GiUK1. Also the samples not presented in Fig. 3 show this behaviour. The phase spectra of sample GiUK1 appear a bit more noisy, because the phase shifts are much smaller (max. 4.5 mrad compared to up to 30 mrad for the others). Sample GiUK1 is also the only sample, which exhibits two phase maxima. While the low-frequency maximum shifts towards higher frequencies with temperature, which is in agreement with the other four samples. The high-frequency maximum appears to be independent of frequency, which might be caused by the overlay of electromagnetic coupling effects.

In order to quantify the shift towards higher frequencies and compare it with theoretical predictions, we calculated a characteristic time as the inverse of the frequency where the maximum phase shift occurs ($\tau_{\text{peak}} = (2\pi f_{\text{peak}})^{-1}$) (Fig. 5). The peak frequencies f_{peak} were determined with the help of a spline interpolation between the measured frequencies. The characteristic times decrease with

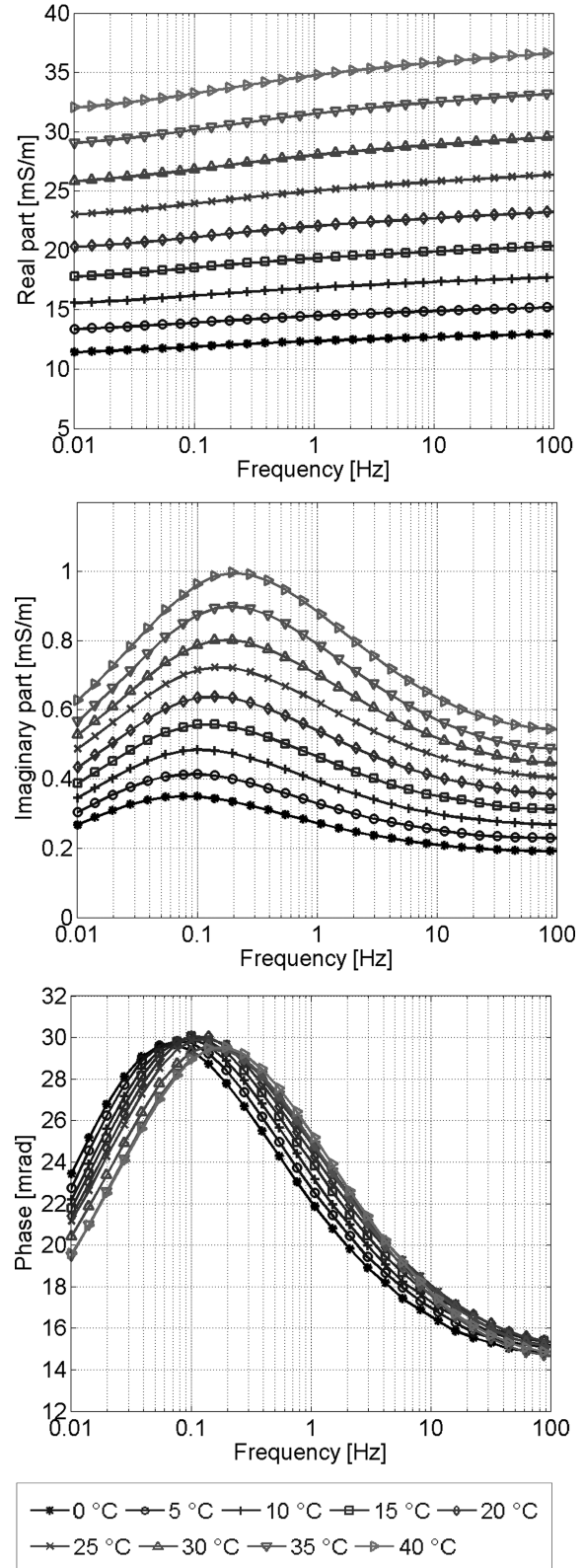


Figure 3. Real part (top), imaginary part (middle) and phase shift (bottom) of the electrical conductivity of sandstone sample TU-B1 for temperatures from 0 to 40 °C.

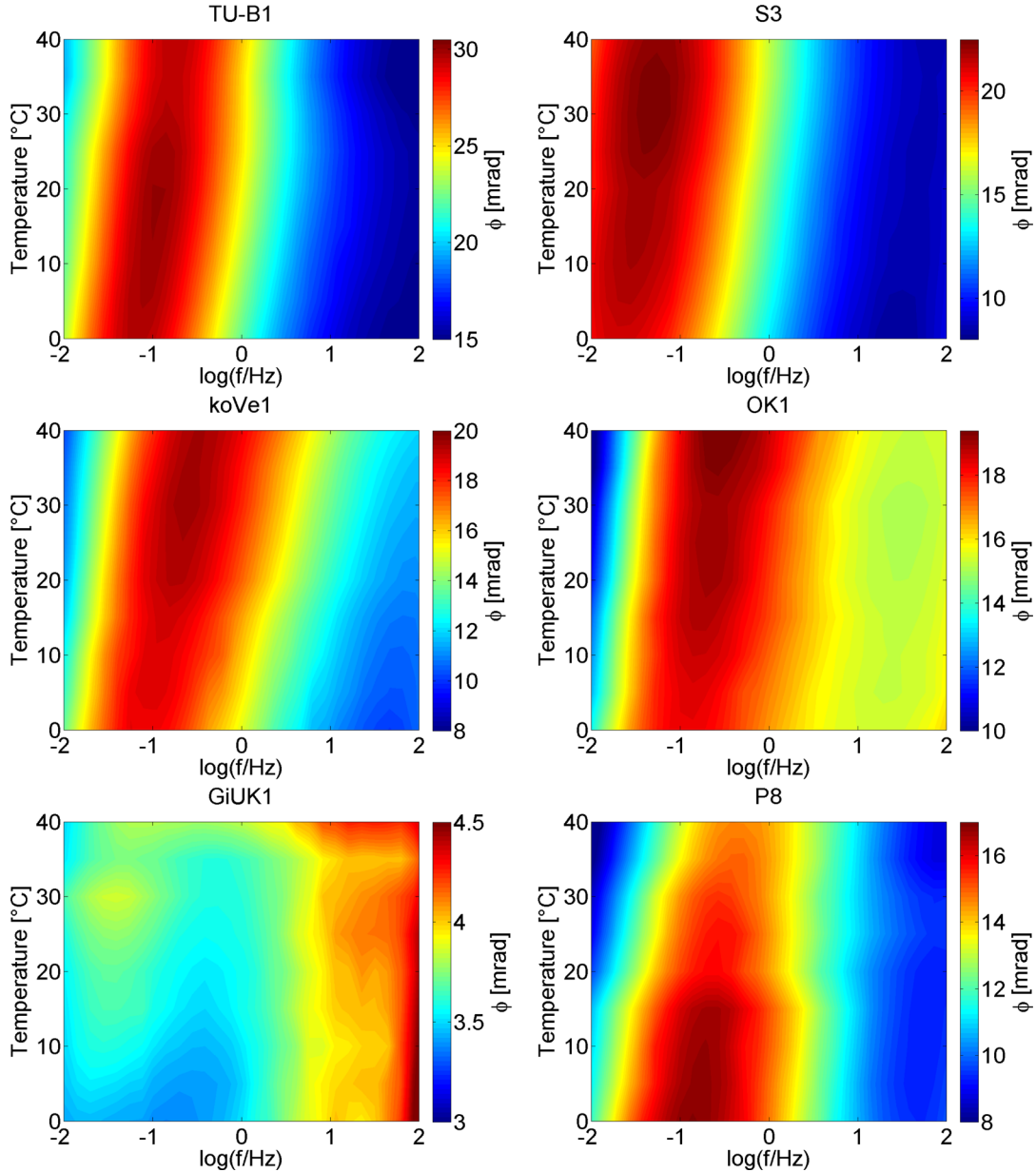


Figure 4. Phase shift of the electrical conductivity colour coded versus temperature and frequency of samples TU-B1, S3, koVe1, OK1 and GiUK and P8. Note that the colour scales are different.

increasing temperature and can be fitted with eq. (6) and Walden exponents listed in Table 2. To reduce the effect due to the error of the characteristic time at the reference temperature $\tau(T_0)$, we used every measured temperature as reference temperature to realize a separate fit. The final Walden exponents are the averages of these separate realizations along with the corresponding standard deviations $\delta\alpha_{\text{tau}}$ (Table 2). The exponents α_{tau} range from 0.52 to 1.42, which is a wider range than the values reported in literature (0.8–0.97). Compared to the real part of conductivity, the characteristic times show a wider variation in temperature dependence and also values α_{τ} that are significantly smaller than the temperature dependence of fluid conductivity ($\alpha_{\mu} = 0.905$). For some of the samples the standard deviations of the fits $\delta\alpha_{\text{tau}}$ are large, resulting from the scatter in the characteristic times, that may arise from the precision of the phase shift of 0.2 mrad.

The temperature dependencies of the amplitude of the maximum phase shifts (Fig. 4) are less uniform (i.e. the maximum phase shift can either increase or decrease with temperature) than the characteristic times. While the phase spectra of samples TU-B1, S3 and GiUK1 show an absolute maximum at 20, 35 and 30 °C, respectively, the maximum phase shift of samples koVe1 and OK1 slightly increases over the entire temperature range, indicating an absolute maximum at higher temperatures. The maximum phase shift of sample P8 decreases with increasing temperature, indicating an absolute maximum that is shifted to smaller temperatures compared to samples TU-B1, S3 and GiUK1. The temperatures of the absolute maximum in the phase shift T_{max} of all samples are listed in Table 2.

In order to analyse the cause of the differences in the temperature dependence of different samples, we examined if there is any

Table 2. Characteristic times τ_{peak} at 25 °C and Walden exponents obtained from fitting eqs (2) and (6) to the temperature dependence of the real conductivity α_σ and the characteristic times α_τ , respectively, along with standard deviations $\delta\alpha$ of the fits. T_{max} is the temperature, where the phase peak is largest within the temperature range of 0–40 °C. Sample FrUK1 does not show a maximum phase shift in the investigated frequency range.

Sample	α_σ	$\delta\alpha_\sigma$	$\tau_{\text{peak}}(25^\circ\text{C})$ [s]	α_τ	$\delta\alpha_\tau$	T_{max} [°C]
TU-B1	0.98	0.00	1.24	0.81	0.04	20
TU-B4	1.05	0.01	0.45	0.75	0.14	40
TU-B5	1.02	0.02	0.34	0.52	0.06	0
P8	1.13	0.01	0.52	0.92	0.16	0
BeRo1	1.07	0.00	1.99	1.30	0.29	15
BiSu1	0.95	0.01	0.23	0.91	0.07	10
FrUK1	0.85	0.03	—	—	—	—
GiUK1	0.87	0.01	4.12	0.92	0.15	30
koQ1	0.85	0.01	0.35	0.72	0.15	40
koVe1	0.93	0.01	0.84	1.42	0.13	40
OK1	0.97	0.00	0.71	0.83	0.08	40
smVG1	0.86	0.00	0.18	0.68	0.08	40
S3	0.88	0.02	3.83	0.89	0.06	35

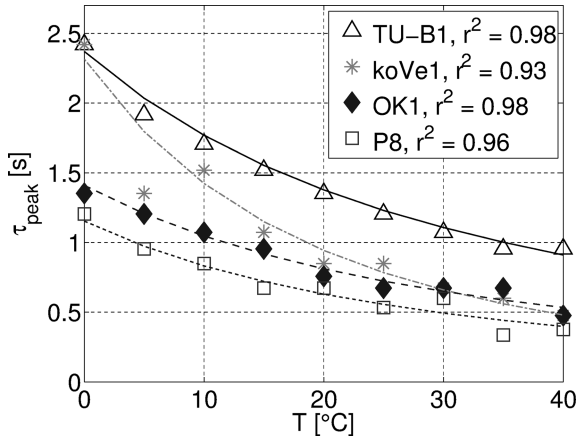


Figure 5. Characteristic times of four sandstone samples as a function of temperature. The lines show the characteristic times fitted with the modified Walden product (eq. 6), with a reference temperature of 25 °C, fitting parameters listed in Table 2 and coefficients of determination r^2 .

correlation to the sample characteristics. It seems natural that the influence of the EDL gets stronger with decreasing pore diameter, thus the Walden exponents of samples with small dominant pore-throat radius r_{dom} should deviate stronger from the Walden exponent of the electrolyte ($\alpha = 0.91$) than the exponents of samples with large pores. The temperature dependence of the real part of conductivity versus r_{dom} (Fig. 6) suggests a weak correlation of both parameters r_{dom} and α_σ , but the poor coefficient of determination ($r^2 = 0.29$) indicates that also other parameters have a strong influence. For the Walden exponents of the characteristic times, no correlation with r_{dom} could be observed. Possibly, r_{dom} is too simplified to characterize the pore space sufficiently and the entire pore- (or grain) size distribution needs to be taken into account. For none of the remaining parameters in Table 1, we could find a direct correlation to the temperature dependence of α_σ or α_τ . However, we cannot exclude that a combination of surface characteristics and the geometry of the pore space determines the temperature dependence.

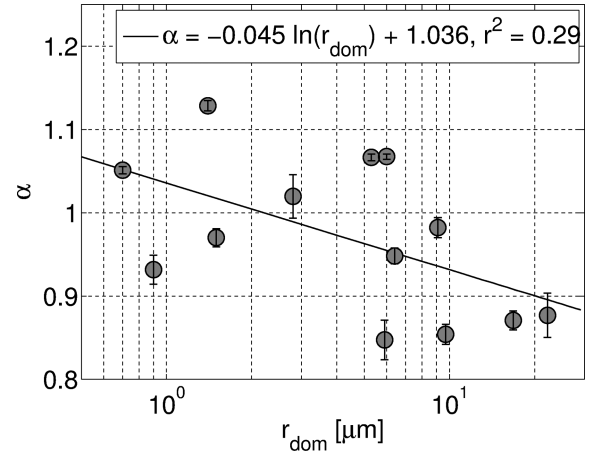


Figure 6. Walden exponent of the real conductivity versus dominant pore-throat diameter. The solid line corresponds to the logarithmic fit.

4 MODELLING THE TEMPERATURE DEPENDENCE

Our experimental results suggest that the temperature dependence of SIP data cannot be explained solely by the temperature dependence of ion mobilities. In order to obtain a deeper theoretical understanding, we carry out simulations with an existing membrane polarization model that allows to simulate temperature dependence, including the EDL properties. While we do not expect to obtain a quantitative fit to the data, we hope to be able to explain at least some of the observations with processes at the pore scale.

4.1 Membrane polarization model

The theory of membrane polarization, first established by Marshall & Madden (1959), is based on a model of alternating active and passive zones. It considers electromigration and diffusion processes within a symmetric electrolyte. Varying ion mobilities are used to describe the ion-selective (or membrane-like) behaviour of the active zones. When subject to an external electric field, an ion concentration gradient builds up resulting in a frequency-dependent impedance of the 1-D pore model. Using the approach of Bückner & Hördt (2013b), the Marshall–Madden impedance Z_{MM} can be written in terms of its DC limit R_0 (in Ωm^2), a dimensionless polarizability η_0 and two time constants τ_i (in s).

$$Z_{\text{MM}}(\omega) = R_0 \left[1 - \eta_0 \left(1 - \frac{\frac{L_1}{\tau_1} + \frac{L_2}{\tau_2}}{\frac{L_1}{\tau_1} \sqrt{j\omega\tau_1} \coth \sqrt{j\omega\tau_1} + \frac{L_2}{\tau_2} \sqrt{j\omega\tau_2} \coth \sqrt{j\omega\tau_2}} \right) \right] \quad (10)$$

$$R_0 = \frac{k_B T}{ec_0 F} \left[\frac{L_1}{D_{p1} + D_{n1}} + \frac{L_2}{D_{p2} + D_{n2}} + \frac{8(t_{n2}t_{p1} - t_{n1}t_{p2})^2}{\frac{L_1}{\tau_1} + \frac{L_2}{\tau_2}} \right], \quad (11)$$

$$\eta_0 = \frac{k_B T}{ec_0 F} \frac{8(t_{n2}t_{p1} - t_{n1}t_{p2})^2}{\frac{L_1}{\tau_1} + \frac{L_2}{\tau_2}} \frac{1}{R_0}. \quad (12)$$

$$\tau_i = \frac{L_i^2}{8D_{pi}t_{ni}} \quad (13)$$

Here, ω is the angular frequency of the external electric field (in rad s⁻¹), $k_B = 1.38 \times 10^{-23}$ J K⁻¹ the Boltzmann constant, T the absolute temperature (in K), $e = 1.6 \times 10^{-19}$ C the elementary charge, c_0 the bulk concentration in the free electrolyte (in mol m⁻³), $F = 96,500$ C mol⁻¹ the Faraday constant and $j^2 = -1$. L_1 and L_2 are the lengths (in m) of the two zones (the subscripts 1 and 2 denote the properties of the passive and the active zone, respectively). The impedance is calculated in units of Ωm^2 , because it is related to a 1-D pore system and scaled to the cross-sectional area of the pores. Due to the varying ion mobilities, the dimensionless transference numbers t_{pi} and t_{ni} , which describe the contribution of the cations and the anions to the total current, also differ between active and passive zone.

The original model by Marshall & Madden (1959) was designed to roughly estimate the magnitude of possible membrane polarization processes. The authors postulated the required mobility variations along the pores without suggesting a mechanism or quantitative estimates. Therefore, Bückner & Hördt (2013a) explicitly attributed the ion-selective behaviour of the active zone to the EDL, which coats the solid–fluid interface of porous media. In the following, we briefly review their model with an emphasis on the temperature dependence. The model describes the pore space as a sequence of wide and narrow cylindrical pores and allows to include pore radii and the properties of the EDL explicitly into the original membrane polarization model.

The electrical potential φ (in V) within the diffuse layer coating the inner surface of a cylindrical pore with radius r_1 (in m) is given by (Hunter 1981)

$$\varphi(r) = \zeta \frac{J_0(j\kappa r)}{J_0(j\kappa r_1)}, \quad (14)$$

where J_0 is the zero-order Bessel function. The decay constant κ is the inverse of the Debye length (eq. 8). The variations of the ion concentrations over the pore cross-section are then calculated from (e.g. Butt *et al.* 2003).

$$c_{n,p}(r) = c_0 \exp\left(\frac{\pm e\varphi(r)}{k_B T}\right) \quad (15)$$

In narrow pores with a radius in the order of some tens of Debye lengths or smaller, the excess of cations (in the case of a negative surface charge) and the lack of anions within the diffuse layer will result in a significant difference between average cation and anion concentrations. For a cylindrical pore with radius r_i , the average ion concentrations, normalized by the bulk concentration c_0 and the cross-sectional area of the wide pore A_1 (in m²), write

$$\bar{b}_{n,p} = \frac{1}{c_0 A_1} \int_0^{r_i} c_{p,n}(r) 2\pi r dr. \quad (16)$$

Although cations within the Stern layer are strongly bound, they can move tangentially to the surface and therefore contribute to the total cation-carried current along the pore. In order to consider this contribution, we use the partition coefficient f_Q defined by Leroy *et al.* (2008) as the ratio of charge density in the Stern layer and total charge density in the EDL (Stern and diffuse layers). The normalized average cation concentration then becomes

$$\tilde{b}_p = \frac{\bar{b}_p - f_Q}{1 - f_Q}. \quad (17)$$

As Bückner & Hördt (2013a) show, the normalized average concentrations can be used to define average transference numbers

$$t_p = \frac{\mu_p \tilde{b}_p}{\mu_p \tilde{b}_p + \mu_n \bar{b}_n} \quad (18)$$

$$t_n = \frac{\mu_n \bar{b}_n}{\mu_p \tilde{b}_p + \mu_n \bar{b}_n} \quad (19)$$

and average diffusion coefficients

$$D_p = \frac{\mu_p \tilde{b}_p k_B T}{e} \quad (20)$$

$$D_n = \frac{\mu_n \bar{b}_n k_B T}{e} \quad (21)$$

for each of the two cylindrical pores. Bückner & Hördt (2013a) showed that using the above defined average transference numbers and diffusion coefficients the analytical solution (eqs 10–13) can be used to describe the impedance of a sequence of two cylindrical pores with different pore radii.

So far, the impedance magnitude Z_{MM} was given in Ωm , because it is based on a 1-D pore system. Here, we suggest to convert it into an effective conductivity that enables us to compare the conductivity magnitudes to measured data. The conversion requires an assumption about the porosity Φ . If L_1 and L_2 are the pore lengths and A_1 and A_2 the cross-sectional areas we show in Appendix A that the effective conductivity may be defined by:

$$\sigma_{\text{eff}} = \frac{A_1}{Z_{MM}} \cdot \frac{L^2 \Phi}{A_1 L_1 + A_2 L_2}. \quad (22)$$

The temperature dependence of the model parameters was included by implementing the temperature dependence of mobility, Zeta potential, Debye length and permittivity as described in chapter 2.

4.2 Modelling results

In order to explore how the overall polarization process responds to the temperature dependencies of the different model parameters described above, we carry out a modelling study. If no other value is specified, the following model parameters will be used: pore lengths are chosen to fit the typical pore size of a fine-grained sandstone (e.g. Howard *et al.* 1993; Lindquist *et al.* 2000; Peng *et al.* 2012) with the length of the wide pore $L_1 = 50$ μm and the length of the narrow pore $L_2 = 10$ μm . Pore radii are chosen to cause a significant phase response of the model impedance ($r_1 = 0.2$ μm and $r_2 = 0.02$ μm for the wide pore and the narrow pore, respectively). The ion concentration of the pore fluid was set to 1 mol m⁻³. An ion mobility of $5 \times 10^{-8} \frac{\text{m}^2}{\text{Vs}}$ was used, which is approximately the mobility of a Na⁺ ion (e.g. Atkins & De Paula 2013). Following Leroy *et al.* (2008), a Zeta potential of -75 mV and a (constant) partition coefficient of 0.6 were used. For calculating the ion mobility (eq. 3), we used a Walden exponent of $\alpha_\mu = 0.91$, which was the best matching value for the data of Zisser *et al.* (2010a). The reference temperature T_0 for the calculation of the temperature dependence of the mobilities and the Zeta potential (eq. 7) was 25 °C. The porosity for calculating the effective conductivity was $\Phi = 0.2$, which is a typical value for sandstones.

For all frequencies in the range from 10 mHz to 100 Hz, the real part of the effective conductivity increases monotonously with temperature (Fig. 7). The modeled temperature dependence of the real part of the effective conductivity can be fitted by a Walden product (eq. 2) with an exponent $\alpha_\sigma \approx 2.19$, which is large compared to the value that was chosen for the ion mobility in the free electrolyte (0.91). This large value implies, that the temperature dependence of the effective conductivity is strongly influenced by surface conductivity and cannot be described solely by the ion mobility in the

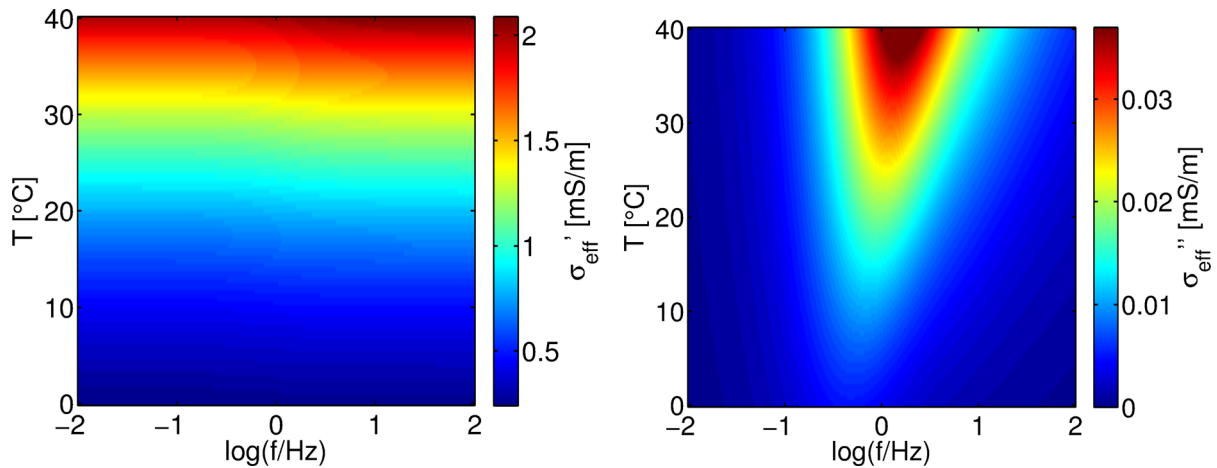


Figure 7. Effective conductivity of the model colour coded versus temperature and frequency calculated with parameters $L_1 = 50$, $L_2 = 10$, $r_1 = 0.2$ and $r_2 = 0.02 \mu\text{m}$. Left: real conductivity. Right: imaginary conductivity.

free electrolyte. The temperature dependence of the surface conductivity is stronger than the dependence of the fluid conductivity on temperature. We conclude that besides the mobility other factors, most likely the Zeta potential and Debye length, also control the admittance. The conclusion is supported by the observation that the temperature dependence of the modeled real conductivity adopts the value of the mobility (0.91) for large pore radii ($r_1, r_2 > 100 \mu\text{m}$, not shown here for brevity) where the double layer becomes unimportant. The imaginary part of the effective conductivity also increases with temperature, depending on the frequency.

Fig. 8 shows the variation of the phase shift with frequency and temperature for three different pore lengths. The ratio of the two pore lengths L_1 and L_2 is kept constant, because it is more likely that in natural materials with long pores both narrow and wide pores are long compared to materials with short pores. The maximum phase shift attains highest values at medium temperatures (approximately 30°C), which remain unchanged when the pore length is increased. Moreover, for all studied pore lengths, the peak frequency f_{peak} , where the maximum phase shift occurs, increases with increasing temperature. This translates into a decrease of the characteristic time (Fig. 9) with increasing temperature (calculated the same way as for the measured data), resulting mainly from the increase of ion mobility and diffusion coefficient as described by eq. (6). As reported before by B cker & H rdt (2013a,b), the critical frequency of the membrane polarization model shifts towards lower frequencies, when increasing the pore length. Fitting the characteristic times with eq. (6) results in a slightly stronger temperature dependence of $\alpha_\tau = 1.07$ than $\alpha_\mu = 0.91$ used for the ion mobility. As for the admittance, mobility is not the only factor controlling the characteristic times.

Fig. 9 also shows that it is possible to fit the behaviour of the characteristic time of the data (using P8 as an example) almost exactly by choosing appropriate pore lengths. Measured and simulated curves of the characteristic times can be fitted with the modified Walden product with exponents of 0.92 (sample P8) and 1.07 (model).

Fig. 10 shows the variation of the phase shift with frequency and temperature for three different pore radii. Note that the ratio between the pore radii of the two pores is kept constant. Again, the peak frequency increases with temperature. Changing the pore radii in the range presented here has no influence on the characteristic timescale and its temperature dependence (not shown here). The absolute phase maximum though moves to higher temperatures when increasing pore radii, and changes its magnitude.

5 DISCUSSION

Our measurement results exhibit a clear dependence of SIP parameters on temperature. The real part of the complex electrical conductivity of our measured IP spectra shows the expected increase of approximately 2 per cent per $^\circ\text{C}$, which is in agreement with previous observations of other authors (e.g. Hayley *et al.* 2007; Binley *et al.* 2010). Similarly, the imaginary part increases about 2 per cent in average (consistent with Binley *et al.* 2010), but the strength of the increase depends on frequency. The small deviations of the temperature dependencies of the imaginary part from the real part leads to a slight change in the magnitude of the phase shift with temperature.

For the changes in chargeability with temperature, Worthington & Collar (1984) reported both an increase and a non-monotonic behaviour for different studies. As the phase shift is comparable to chargeability, obtained from time-domain measurements, these results are consistent with the increase of the maximum phase shift and the absolute maximum found here for some of the samples (Fig. 4). The temperature dependence of the magnitude of the phase shift cannot be generalized. Our data show that both increasing and decreasing phase shifts with increasing temperature can occur, which was also observed by Vinegar & Waxman (1984) on two samples with different clay content. However, for the samples investigated here, we could not find a correlation of the temperature behaviour of the phase shift and the clay content. Binley *et al.* (2010) reported a temperature-independent phase shift (magnitude) for a small temperature range. This is not a contradiction to our results, as the change in the phase shift in our data is weak and could be interpreted as irrelevant for small temperature changes of a few degree centigrade. Nevertheless, for large temperature differences, a significant change in the phase shift of several milliradians can occur, which may lead to misinterpretations of measurement data if neglected. Additionally, a detailed observation of the temperature dependence of the phase shift helps to understand the temperature dependence of the underlying polarization process.

5.1 Characteristic time of measured spectra

From the CT images of sandstone samples TU-B1, P8 and S3, we can make some qualitative statements about the average pore size of the samples relative to each other. The characteristic times are associated with a characteristic geometrical scale of the pores or

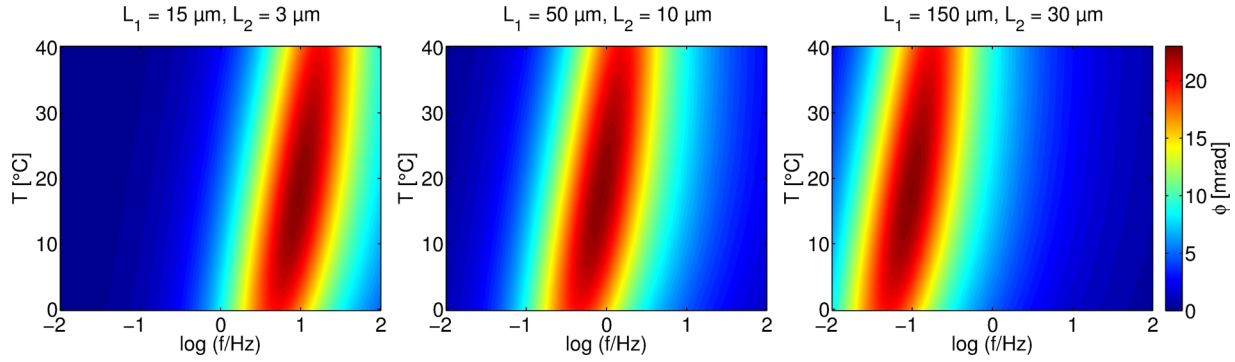


Figure 8. Phase shift of the modeled admittance colour coded versus temperature and frequency for different pore lengths L_1 and L_2 . Pore lengths increase from left to right. Pore radii are $r_1 = 0.2$ and $r_2 = 0.02$ μm .

grains. For the membrane polarization model, the dominating factor is the length scale of the pores. Pores of sandstone samples TU-B1 and P8 are estimated to have a similar average length, while S3 has longer pores (Fig. 1). Consequently, the maxima of the phase shift of samples TU-B1 and P8 should occur at similar frequencies and for sample S3 at smaller frequencies. At 25 °C, the peak frequency of sample P8 is found at 0.31 Hz and the one of TU-B1 at 0.13 Hz, which translates into characteristic times of 0.51 and 1.20 s (Table 2), respectively. Even though the mean grain size of sample TU-B1 is smaller than the one of sample P8, the latter is much more compacted and cemented, which seems to result in shorter pores and explain the shorter characteristic timescale. The characteristic time at 25 °C of sample S3 is larger than τ_{peak} of the two other samples and thus consistent with the geometrical considerations.

Previous publications (e.g. Scott & Barker 2005; Kruschwitz *et al.* 2010; Revil 2013) related the time constant to the dominant pore-throat diameter $D_{\text{dom}} = 2r_{\text{dom}}$. Fig. 11 shows that our data are consistent with their results, as we also see an increase of the characteristic time τ_{peak} with increasing D_{dom} . A fit of the same form as that of Scott & Barker (2005) can be obtained. This observation does not contradict the membrane polarization model, where the characteristic time is dominated by the pore length. Pore lengths are experimentally not accessible, except indirectly, because pore throats and pore lengths in real rocks usually are not independent from each other. Therefore the correlations between τ_{peak} and D_{dom} do not allow the conclusion that D_{dom} , and not the pore length, is in fact the controlling parameter at the pore scale. For sandstones with similar shaped grains and cementation, smaller pore radii are associated with shorter pores. Moreover, the fitting parameters differ from Scott & Barker (2005) and the scattering of the data are indications for the influence of other or additional parameters than D_{dom} on the time constant.

For increasing temperature, a decrease of the characteristic times was found for all samples that showed a peak in the measured phase spectra, agreeing with the results of Binley *et al.* (2010) and Zisser *et al.* (2010a). However, the strength of the temperature dependence varies for different samples. In order to quantify the temperature dependencies of the different characteristic times, they were fitted using eq. (6). The exponents α_τ range from 0.52 to 1.42, which clearly exceeds the range from 0.8 to 0.97 reported in Zisser *et al.* (2010a). The equation that was used for fitting the temperature dependence of the characteristic times (eq. 6) is based on the assumption that the only temperature-dependent parameter is the ion mobility. Zisser *et al.* (2010a) assumed, that the ion mobility in the Stern and diffuse layers are similar to the mobility in the free electrolyte, valid for clay-free sandstones. Our results with Walden

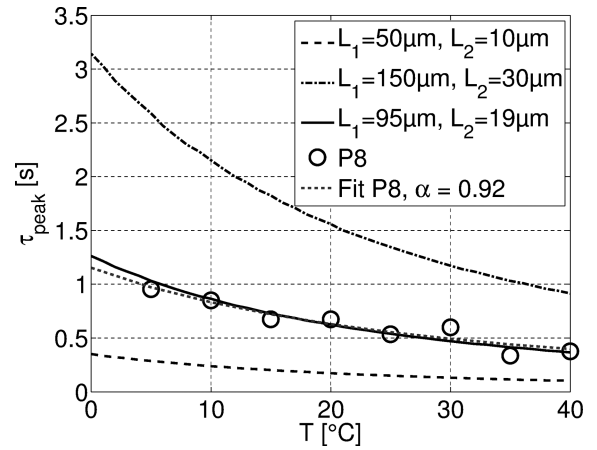


Figure 9. Modeled characteristic times of three different combinations of pore lengths L_1 and L_2 compared to the characteristic times of sample P8, fitted with $\alpha_\tau = 0.92$. The pore radii are $r_1 = 0.2$ and $r_2 = 0.02$ μm .

exponents different from that of the free electrolyte ($\alpha_\mu = 0.905$), suggest that this assumption is not valid for most of the samples of this study. However, we could not find a direct relation to the clay content or other petrophysical parameters.

5.2 Comparison of theoretical and experimental results

Although we are able to fit the temperature dependence of the characteristic time of the model to that of a sandstone sample with realistic pore lengths (Fig. 9), the following considerations are not intended to demonstrate that the model exactly describes all mechanisms taking place in the pores of a sandstone. The model represents a strong simplification of the pore geometry of a sandstone and might therefore underestimate or neglect the influence of several contributing factors. However, a qualitative comparison of the temperature dependence of the IP spectra with the model can help to assess whether the theory describes relevant processes, and possibly help to understand the processes at the pore scale underlying the observed temperature dependencies.

The effective conductivity of the model (Fig. 7) is about one order of magnitude smaller than the measured conductivity (Fig. 3). This discrepancy is small considering the simple model geometry compared to the complex network of different-sized pores in series and parallel of a natural rock. Comparing the temperature dependence of the real part of the measured conductivity and the effective model conductivity, respectively, gives a qualitative agreement in

the increasing behaviour for both quantities, but a strong difference of the Walden exponent with $\alpha_\sigma(\text{measurements}) = 0.85\text{--}1.13$ and $\alpha_\sigma(\text{model}) = 2.19$. Although the Walden exponent of the model is only valid for one pore size and can change when the pore size is changed, exponents similar to the ones measured only occur when both pore radii are larger than $1\text{ }\mu\text{m}$. At large pore radii ($r_i \gg \lambda_D$), the EDL covers a negligible part of the pore cross section. Therefore, bulk conductivity controls the absolute value of the admittance and the temperature dependence only stems from ion mobilities and diffusion coefficients (Walden exponent $\alpha_\mu = 0.91$). At much smaller pore radii ($\ll 0.1\text{ }\mu\text{m}$), as used in this study to produce sufficiently large phase shifts, the EDL covers a considerable part of the pore cross-section and the temperature dependence of the total current through the pore system also depends on the properties of the EDL. In particular, the Zeta potential increases rapidly with temperature and leads to a much stronger temperature dependence ($\alpha_\sigma = 2.19$) of the absolute value of the admittance than predicted by the modified Walden product. The difference in the Walden exponents between theoretical and measured values might imply that the model in its present form tends to overestimate the influence of narrow pores.

In the following we focus on the phase shift. The most prominent similarities of the measurements and the model are (1) the shift of the phase maximum to higher frequencies (which is equivalent to a decrease of the characteristic time) with increasing temperature and (2) different positions of the absolute maximum on the frequency and temperature axis for different samples (measurements) or pore geometries (model). These two aspects are discussed in more detail below.

5.2.1 Characteristic time

A detailed discussion of the increase of the modeled characteristic time (decrease of the peak frequency) with the pore length can be found in Bückner & Hördt (2013b). As the polarization process is mainly controlled by diffusion, the characteristic time is proportional to the square of the characteristic length of the system; $\tau \propto L^2$, with L being the total pore length.

Figs 5 and 9 show that both the characteristic times of the model and those of the sandstone samples decrease with temperature. The fitted exponent (eq. 6) of $\alpha_\tau = 1.07$ for the characteristic times of the model lies within the range of $0.52\text{--}1.42$ of the exponents fitted to the characteristic times of the sandstone samples. It is slightly higher than the Walden exponent used for the calculation of the temperature dependence of the ion mobility ($\alpha_\mu = 0.91$). This means, that also for the characteristic time, the temperature dependence of the model is not only controlled by ion mobility and diffusion coefficient, but also by the properties of the EDL.

5.2.2 Influence of pore geometry

The membrane polarization model predicts a strong temperature dependence of the magnitude of the maximum phase shift, which is related to polarizability. From eq. (12), it is evident that the polarizability (and thus the maximum phase shift) depends on a complex combination of the geometrical model parameters (pore lengths and radii) and the properties of the EDL. Earlier studies (Bückner & Hördt 2013b) revealed that the polarization magnitude adopts a maximum for certain combinations of pore radii r_1 and r_2 depending on the properties of the EDL. Our modelling results show that the temperature, at which the phase shift reaches a maximum, strongly depends on pore radii (Fig. 10). Apparently, at the temperature of

maximum phase shift, the combination of Zeta potential and Debye length maximizes the polarizability for a certain geometry (predominantly pore radii). When the pore radii increase, the fraction of the EDL in the cross-section of the pore becomes smaller and the influence of the EDL decreases. Presumably, this decrease is compensated by a higher magnitude of the Zeta potential, that is reached at a higher temperature, shifting the absolute maximum to a higher temperature.

The measurements of samples TU-B1, P8 and S3 are also consistent with our modelling results, when taking into account the pore structure as observed from the CT images. The measured phase shifts of samples TU-B1 and S3 show an absolute maximum at approximately 20 and 35°C , respectively. The phase shift of sample P8 has its maximum at a lower temperature compared to TU-B1 and S3 (Fig. 4). Actually, pores and pore throats of sample TU-B1 are narrower than those of sample S3 and wider than those of sample P8, which is in agreement with the prediction of the model. However, this observation does not hold for some of the samples, when only the dominant pore-throat radius is considered, for example, sample TU-B4 with small r_{dom} but a maximum at 40°C .

Another observation when comparing simulated and measured spectra is the different width of the phase peak, which arises from the pore-size distribution. The model considers only two pore lengths (and two radii), which result in a superposition of two characteristic times and corresponding phase peaks. However, Bückner & Hördt (2013b) point out, that generally only one of the pores dominates the relaxation process, which gives rise to a very narrow phase peak with steep slopes. In natural rock samples, on the other hand, a wide range of pore lengths may exist resulting in a distribution of characteristic times and a broadened phase peak. Similarly, the wide range of different pore radii in natural rock samples is supposed to cause a broad maximum of the phase shift in the temperature domain, leading to a weaker temperature dependence of the measured phase shift compared to the modeled one.

5.2.3 Properties of the model

The parameter sets used here to produce results comparable to the measured data include ratios between pore lengths and pore radii of up to 500. This seems relatively large, considering that typical pore-throat sizes of sandstone determined by mercury injection measurements are only about $1/10$ of the mean grain size (e.g. Binley et al. 2005; Nelson 2009). The mean grain size, in turn, is a suitable approximation for the mean pore length. The ratio $1/10$ is also in agreement with the grain and pore sizes observed in the CT images of our sandstone samples.

The CT images give an idea about characteristic length scales, but they do not include information on the possible existence of submicron pores due to the resolution of approximately $1\text{ }\mu\text{m}$. The pore radii required for calculating a phase shift similar to measured phase shifts are smaller than mean pore radii of rock samples. This discrepancy could be attributed to an underestimation of polarization effects in large pores by the model, but it is also possible that only very small pores are responsible for the polarization effect. The existence of submicron pores in our samples has been observed by the mercury injection measurements, although in most samples the dominant pore-throat diameter is larger. Also other authors have reported submicron pores for sandstone samples (e.g. Bloomfield et al. 2001; Kruschwitz et al. 2010). Anyway, at this stage we cannot say whether these submicron pores are relevant to the overall polarization response of our samples or not.

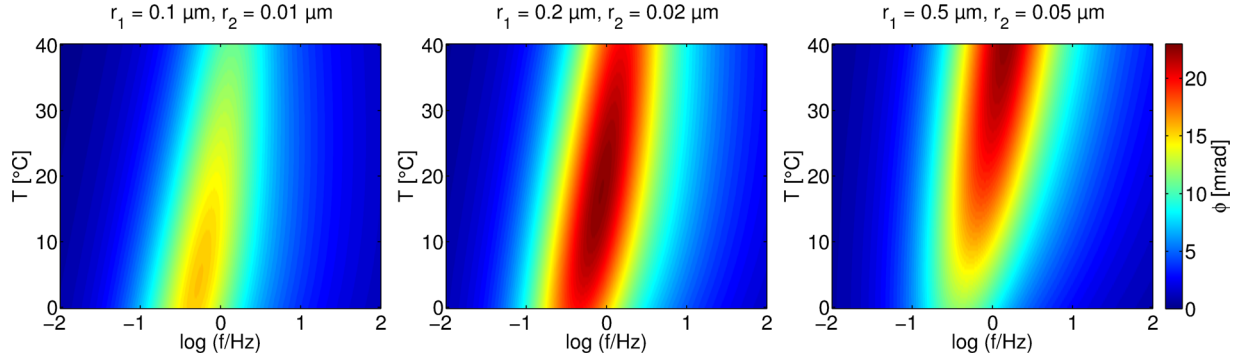


Figure 10. Phase shift of the modeled admittance colour coded versus temperature and frequency for different pore radii r_1 and r_2 . Pore radii increase from left to right. Pore lengths are $L_1 = 50$ and $L_2 = 10$ μm .

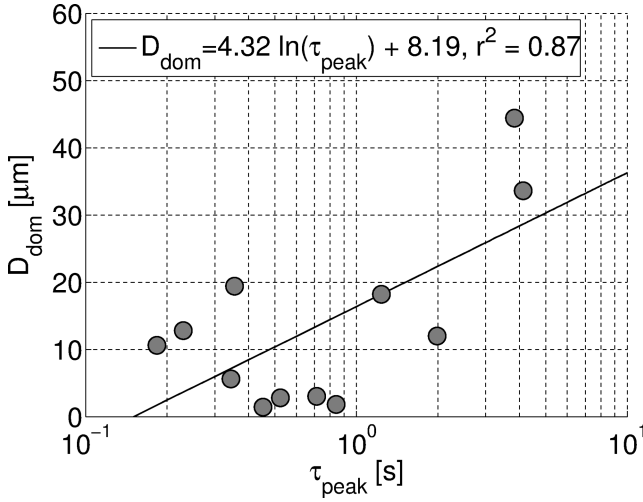


Figure 11. Dominant pore-throat diameters from mercury injection versus characteristic times of the IP spectra with logarithmic fit.

The model response is calculated with the assumption that the mobility in the Stern layer is equal to the mobility of the ions in the free electrolyte. This assumption is used for clean sands in several publications (e.g. Leroy *et al.* 2008; Revil & Florsch 2010), but it might be different in the presence of clays. For example, Revil (2013) calculate a considerably smaller ion mobility of $\mu_s = 1.5 \times 10^{-10} \frac{\text{m}^2}{\text{Vs}}$ for the Stern layer of clayey sandstones. In the membrane polarization model, an ion mobility in the Stern layer much lower than in the diffuse layer leads to $\tilde{b}_p \approx \bar{b}_p$ (see Appendix B). This means that the Stern layer does not contribute significantly to the overall conduction, if the mobility in the Stern layer is much lower than in the bulk electrolyte and the diffuse layer, which contradicts our previous idea that surface conductivity might play an important role for the temperature dependence for samples with clay content.

We conclude that the interplay between Zeta potential, partition coefficient and Stern layer mobility may not be fully understood and that further work might lead to future modifications of the model. For example, the consideration of surface conductance through the partition coefficient might not consider relevant conduction mechanisms, such as the proton hopping or Grotthuss mechanism suggested by Skold *et al.* (2011). Although the pore geometry might not include all characteristics of the material, that are relevant for the temperature dependence, as, for example, the material composition, the comparison of modelling results and experimental findings with regard to the pore size provides a good qualitative agreement

in various respects. However, the model includes many parameters and we have not yet fully exploited the entire parameter space.

5.3 Consequences for practical applications

Seasonal temperature variations mainly affect the first 10 m of the subsurface (e.g. Hayley *et al.* 2007). In the temperate zone, the differences in subsurface temperature are usually in the range of a few Kelvin to 10 K. Considerably larger differences can occur in geothermal areas. As shown by the experiments in the previous sections, temperature variations of a few Kelvin already influence the IP measurement parameters, confirming that temperature dependence might have to be included in the interpretation of IP data. Especially quantitative evaluations based on empirical relationships, which were developed under laboratory conditions, have to be adjusted to different temperatures. A parameter often correlated with different IP parameters and important for geothermal investigations is the hydraulic permeability. For a brief estimation of the influence of temperature we focus on an approach which relates hydraulic permeability k to the time constant. Both quantities depend on a characteristic geometric scale (e.g. Slater 2007), leading to empirical relationships reported by different authors in the form (e.g. Binley *et al.* 2005; Zisser *et al.* 2010b)

$$k = a\tau^b \quad (23)$$

with empirical parameters a and b , which depend on the material. The temperature dependence of the time constant (eq. 6) can easily be inserted into this equation:

$$k(T) = k(T_0) \left[\frac{T_0}{T} \left(\frac{\eta(T)}{\eta(T_0)} \right)^{\alpha_\tau} \right]^b. \quad (24)$$

For Walden exponents $\alpha_{\text{tau}} = 0.5$ –1.4 observed in this work and using $b = 3.85$ as reported by Binley *et al.* (2005) for sandstones, a temperature difference of 10 °C would lead to an error of 43–74 percent in permeability. Therefore, investigations of geothermal fields where the reservoir temperature is considerably higher than the laboratory temperature, ignoring the temperature may lead to a strong underestimation of permeability in the high temperature region.

6 CONCLUSIONS

The investigated samples show a clear temperature dependence of the SIP response, which is apparent both in the real and imaginary part of the conductivity. The absolute maximum of the phase shift appears at different temperatures, depending on the samples. The

characteristic time scales decrease with increasing temperature, displaying different strengths of the temperature dependence (Walden exponents) of different samples. This observation is in good agreement with the decrease with increasing temperature, that is predicted by previous publications. The comparison with CT images indicates that differences in the Walden exponents are related to the pore space geometry. However, we did not quantify robust quantitative relationships with measurable parameters, such as dominant pore-throat diameter or clay content. If we assume that at least 'part of the polarization is caused by an interaction of different pore radii, a correlation with a single parameter characterizing the pore space cannot necessarily be expected.

Furthermore, we simulated the temperature dependence with a membrane polarization model that is based on a sequence of wide and narrow pores. Considering the simplicity of the model geometry, the simulated spectra reproduce several features of the measured data remarkably well. Characteristic times decrease with increasing temperature and the position of the phase maximum is affected by the pore geometry. While the mean pore length mainly controls the spectral position of the maximum phase, the pore radius (or width) determines the temperature, at which the absolute phase maximum occurs. The decrease of the modeled amplitude of the admittance and the characteristic time is stronger than the predicted decrease resulting from ion mobility only. This indicates that, additionally to ion mobility, also other temperature-dependent parameters of the EDL, such as Debye length and Zeta potential, are important.

ACKNOWLEDGEMENTS

This work was funded by Baker Hughes and the Ministry of Science and Culture of Lower Saxony, Germany, within the Geothermal Energy and High-Performance Drilling (gebo) research association. We give our thanks to the Helmholtz Centre Potsdam GFZ, namely Inga Moeck, Nicole Pastrik, Sonja Martens and Ben Norden, and Dorothea Reyer for supplying the samples. We also thank Daniel Albrecht and Manfred Stövesand for cutting the samples and Annett Zimathies for the mercury intrusion measurements.

REFERENCES

- Atkins, P. & De Paula, J., 2013. *Elements of Physical Chemistry*, Oxford Univ. Press.
- Bairlein, K., Hördt, A. & Nordsiek, S., 2014. The influence on sample preparation on spectral induced polarization of unconsolidated sediments, *Near Surf. Geophys.*, **1**(1), 1–12.
- Binley, A., Slater, L., Fukes, M. & Cassiani, G., 2005. Relationship between spectral induced polarization and hydraulic properties of saturated and unsaturated sandstone, *Water Resour. Res.*, **41**(12), W12417, doi:10.1029/2005WR004202.
- Binley, A., Kruschwitz, S., Lesmes, D. & Kettridge, N., 2010. Exploiting the temperature effects on low frequency electrical spectra of sandstone: a comparison of effective diffusion path lengths, *Geophysics*, **75**(6), A43–A46.
- Bloomfield, J., Goody, D., Bright, M. & Williams, P., 2001. Pore-throat size distributions in Permo-Triassic sandstones from the United Kingdom and some implications for contaminant hydrogeology, *Hydrogeol. J.*, **9**(3), 219–230.
- Börner, F., Schopper, J. & Weller, A., 1996. Evaluation of transport and storage properties in the soil and groundwater zone from induced polarization measurements, *Geophys. Prospect.*, **44**(4), 583–601.
- Buchheim, W. & Irmer, G., 1979. Zur Theorie der induzierten galvanischen Polarisation in Festkörpern mit elektrolytischer Porenfüllung, *Gerlands Beitr. Geophys.*, **88**, 53–72.
- Bücker, M. & Hördt, A., 2013a. Analytical modelling of membrane polarization with explicit parametrization of pore radii and the electrical double layer, *Geophys. J. Int.*, **194**(2), 804–813.
- Bücker, M. & Hördt, A., 2013b. Long and short narrow pore models for membrane polarization, *Geophysics*, **78**(6), E299–E314.
- Butt, H., Graf, K. & Kappl, M., 2003. *Physics and Chemistry of Interfaces*, Wiley Online Library.
- Collett, L., 1959. Laboratory investigation of overvoltage, in *Overvoltage Research and Geophysical Applications*, pp. 50–70, Pergamon, New York, NY.
- Fleury, M., Gautier, S., Gland, N., Boulin, P., Norden, B. & Schmidt-Hattenberger, C., 2013. Advanced and integrated petrophysical characterization for CO₂ storage: application to the Ketzin site, *Oil Gas Sci. Technol. Rev. IFP Energ. Nouvelles*, **68**(3), 557–576.
- Förster, A., Giese, R., Juhlin, C., Norden, B., Springer, N. & Group, C., 2009. The geology of the CO₂SINK site: from regional scale to laboratory scale, *Energy Procedia*, **1**(1), 2911–2918.
- Hayley, K., Bentley, L., Gharibi, M. & Nightingale, M., 2007. Low temperature dependence of electrical resistivity: implications for near surface geophysical monitoring, *Geophys. Res. Lett.*, **34**(18), doi:10.1029/2007GL031124.
- Hördt, A. & Milde, S., 2012. Studies with gel-filled sandstone samples with implications for the origin of induced polarization, *Near Surf. Geophys.*, **10**(6), 469–478.
- Hördt, A., Blaschek, R., Kemna, A. & Zisser, N., 2007. Hydraulic conductivity estimation from induced polarisation data at the field scale—the Krauthausen case history, *J. appl. Geophys.*, **62**(1), 33–46.
- Howard, J.J., Kenyon, W.E. & Straley, C., 1993. Proton magnetic resonance and pore size variations in reservoir sandstones, *SPE Formation Evaluation*, **8**(03), 194–200.
- Hunter, R.J., 1981. *Zeta Potential in Colloid Science: Principles and Applications*, Academic Press.
- Kruschwitz, S., 2008. Assessment of the complex resistivity behavior of salt affected building materials, *PhD thesis*, Federal Institute for Materials Research and Testing, Berlin, Germany.
- Kruschwitz, S., Binley, A., Lesmes, D. & Elshenawy, A., 2010. Textural controls on low-frequency electrical spectra of porous media, *Geophysics*, **75**(4), WA113–WA123.
- Leroy, P. & Revil, A., 2004. A triple-layer model of the surface electrochemical properties of clay minerals, *J. Colloid Interface Sci.*, **270**(2), 371–380.
- Leroy, P., Revil, A., Kemna, A., Cosenza, P. & Ghorbani, A., 2008. Complex conductivity of water-saturated packs of glass beads, *J. Colloid Interface Sci.*, **321**(1), 103–117.
- Lindquist, W.B., Venkatarangan, A., Dunsmuir, J. & Wong, T.-f., 2000. Pore and throat size distributions measured from synchrotron X-ray tomographic images of fontainebleau sandstones, *J. geophys. Res.*, **105**(B9), 21 509–21 527.
- Llera, F.J., Sato, M., Nakatsuka, K. & Yokoyama, H., 1990. Temperature dependence of the electrical resistivity of water-saturated rocks, *Geophysics*, **55**(5), 576–585.
- Marshall, D. & Madden, T., 1959. Induced polarization, a study of its causes, *Geophysics*, **24**(4), 790–816.
- Martens, S. et al., 2013. CO₂ storage at the Ketzin pilot site, Germany: fourth year of injection, monitoring, modelling and verification, *Energy Procedia*, **37**, 6434–6443.
- Martens, S., Möller, F., Streibel, M., Liebscher, A. & Group, T.K., 2014. Completion of five years of safe CO₂ injection and transition to the post-closure phase at the Ketzin pilot site, *Energy Procedia*, **59**, 190–197.
- Nelson, P.H., 2009. Pore-throat sizes in sandstones, tight sandstones, and shales, *AAPG Bull.*, **93**(3), 329–340.
- Pátek, J., Hrub, J., Klomfar, J., Součková, M. & Harvey, A.H., 2009. Reference correlations for thermophysical properties of liquid water at 0.1 Mpa, *J. Phys. Chem. Ref. Data*, **38**(1), 21–29.
- Peng, S., Hu, Q., Dultz, S. & Zhang, M., 2012. Using X-ray computed tomography in pore structure characterization for a Berea sandstone: resolution effect, *J. Hydrol.*, **472**, 254–261.

- Revil, A., 2012. Spectral induced polarization of shaly sands: influence of the electrical double layer, *Water Resour. Res.*, **48**(2), doi:10.1029/2011WR011260.
- Revil, A., 2013. Effective conductivity and permittivity of unsaturated porous materials in the frequency range 1 mHz–1 GHz, *Water Resour. Res.*, **49**(1), 306–327.
- Revil, A. & Florsch, N., 2010. Determination of permeability from spectral induced polarization in granular media, *Geophys. J. Int.*, **181**(3), 1480–1498.
- Revil, A., Pezard, P. & Glover, P., 1999. Streaming potential in porous media: 1. Theory of the zeta potential, *J. geophys. Res.*, **104**(B9), 20 021–20 031.
- Reyer, D., 2014. Outcrop analogue studies of rocks from the north-west German Basin for geothermal exploration and exploitation: fault zone structure, heterogeneous rock properties, and application to reservoir conditions, *PhD thesis*, Georg-August-Universität Göttingen.
- Reyer, D. & Philipp, S., 2014. Empirical relations of rock properties of outcrop and core samples from the Northwest German Basin for geothermal drilling, *Geotherm. Energy Sci.*, **2**, 21–37.
- Schwarz, G., 1962. A theory of the low-frequency dielectric dispersion of colloidal particles in electrolyte solution, *J. Phys. Chem.*, **66**(12), 2636–2642.
- Scott, J. & Barker, R., 2005. Characterization of sandstone by electrical spectroscopy for stratigraphical and hydrogeological investigations, *Q. J. Eng. Geol. Hydrogeol.*, **38**(2), 143–154.
- Sen, P.N. & Goode, P.A., 1992. Influence of temperature on electrical conductivity on shaly sands, *Geophysics*, **57**(1), 89–96.
- Skold, M., Revil, A. & Vaudelet, P., 2011. The pH dependence of spectral induced polarization of silica sands: experiment and modeling, *Geophys. Res. Lett.*, **38**(12), doi:10.1029/2011GL047748.
- Slater, L., 2007. Near surface electrical characterization of hydraulic conductivity: from petrophysical properties to aquifer geometries—a review, *Surv. Geophys.*, **28**(2–3), 169–197.
- Sorensen, J.A. & Glass, G.E., 1987. Ion and temperature dependence of electrical conductance for natural waters, *Anal. Chem.*, **59**(13), 1594–1597.
- Titov, K., Komarov, V., Tarasov, V. & Levitski, A., 2002. Theoretical and experimental study of time domain-induced polarization in water-saturated sands, *J. appl. Geophys.*, **50**(4), 417–433.
- Tong, M. & Tao, H., 2007. Experimental study of induced polarization relaxation time spectra of shaly sands, *J. Petrol. Sci. Eng.*, **59**(3), 239–249.
- Treichel, A., Binley, A., Kemna, A., Esser, O., Zimmermann, E., Vereecken, H. & Huismann, J., 2014. Temperature dependence of broadband complex electrical conductivity in unconsolidated porous media with variable clay content, in *3rd International Workshop on Induced Polarization*, pp. 38–39, Ile d'Oleron, France.
- Vinegar, H. & Waxman, M., 1984. Induced polarization of shaly sands, *Geophysics*, **49**(8), 1267–1287.
- Weller, A., Slater, L., Nordsiek, S. & Ntarlagiannis, D., 2010. On the estimation of specific surface per unit pore volume from induced polarization: a robust empirical relation fits multiple data sets, *Geophysics*, **75**(4), WA105–WA112.
- Worthington, P.F. & Collar, F.A., 1984. Relevance of induced polarization to quantitative formation evaluation, *Mar. Petrol. Geol.*, **1**(1), 14–26.
- Zisser, N., Kemna, A. & Nover, G., 2010a. Dependence of spectral-induced polarization response of sandstone on temperature and its relevance to permeability estimation, *J. geophys. Res.*, **115**(B9), doi:10.1029/2010JB007526.
- Zisser, N., Kemna, A. & Nover, G., 2010b. Relationship between low-frequency electrical properties and hydraulic permeability of low-permeability sandstones, *Geophysics*, **75**(3), E131–E141.

APPENDIX A: CONVERSION OF THE 1-D IMPEDANCE TO AN EFFECTIVE CONDUCTIVITY

To convert the 1-D Marshall–Madden impedance (in Ωm^2) to an effective resistivity, we consider a unit cell consisting of a pore sequence (one wide and one narrow cylindrical pore) within a cuboid of length $L = L_1 + L_2$ and a cross-sectional area $A_S = a_S^2$ with $a_S \geq 2 \cdot r_1$. The volume surrounding the pores is assumed to be non-conducting. The porosity of this unit cell is given by

$$\Phi = \frac{A_1 L_1 + A_2 L_2}{A_S L}. \quad (\text{A1})$$

By dividing the Marshall–Madden impedance by the cross-sectional area of the larger pore $Z = Z_{\text{MM}}/A_1$, it is converted to the resistance of the pore sequence. From these considerations, we get the effective resistivity of the unit cell

$$\rho_{\text{eff}} = Z \cdot \frac{A_S}{L} \quad (\text{A2})$$

$$= \frac{Z_{\text{MM}}}{A_1} \cdot \frac{A_1 L_1 + A_2 L_2}{L^2 \Phi}, \quad (\text{A3})$$

and the effective conductivity $\sigma_{\text{eff}} = 1/\rho_{\text{eff}}$.

APPENDIX B: CONSIDERATION OF A REDUCED ION MOBILITY IN THE STERN LAYER

A reduced ion mobility in the Stern layer μ_s can be included to the membrane polarization model by modifying the equations of the Stern layer correction in Bückner & Hördt (2013a):

$$2\pi r \left(\sigma_d + \frac{\mu_s}{\mu} \sigma_s \right) = F c_0 A_p (\tilde{b}_p - 1) \quad (\text{B1})$$

with the surface charge densities associated with the Stern and the diffuse layer σ_d and σ_s and the pore section A_p . Following Bückner & Hördt (2013a), we get for the average cation concentration

$$\tilde{b}_p = \frac{\bar{b}_p \left[1 + f_Q \left(\frac{\mu_s}{\mu} - 1 \right) \right] - f_Q \frac{\mu_s}{\mu}}{1 - f_Q}, \quad (\text{B2})$$

which in the case $\mu_s = \mu$ becomes eq. (16). In the case $\mu_s \ll \mu$ eq. (B2) reduces to $\tilde{b}_p \approx \bar{b}_p$.

Measuring surface charge: why experimental characterization and molecular modeling should be coupled

Remco Hartkamp^a, Anne-Laure Biance^b, Li Fu^b, Jean-François Dufrêche^c, Oriane Bonhomme^b, Laurent Joly^b

©2018. This manuscript version is made available under the CC-BY-NC-ND 4.0 license <http://creativecommons.org/licenses/by-nc-nd/4.0/>
DOI: <https://doi.org/10.1016/j.cocis.2018.08.001>

^aProcess & Energy Department, Delft University of Technology, Leeghwaterstraat 39, 2628 CB Delft, The Netherlands

^bUniv Lyon, Université Claude Bernard Lyon 1, CNRS, Institut Lumière Matière, F-69622 Villeurbanne, France

^cInstitut de Chimie Séparative de Marcoule ICSM, UMR 5257 CEA-CNRS-ENSCM-Université Montpellier, Bâtiment 426, F-30207 Bagnols-sur-Cèze, France

Abstract

Surface charge controls many static and dynamic properties of soft matter and micro/nanofluidic systems, but its unambiguous measurement forms a challenge. Standard characterization methods typically probe an effective surface charge, which provides limited insight into the distribution and dynamics of charge across the interface, and which cannot predict consistently all surface-charge-governed properties. New experimental approaches provide local information on both structure and transport, but models are typically required to interpret raw data. Conversely, molecular dynamics simulations have helped showing the limits of standard models and developing more accurate ones, but their reliability is limited by the empirical interaction potentials they are usually based on. This review highlights recent developments and limitations in both experimental and computational research focusing on the liquid-solid interface. Based on recent studies, we make the case that coupling of experiments and simulations is pivotal to mitigate methodological shortcomings and address open problems pertaining to charged interfaces.

Keywords: surface charge, electrical double layer, zeta potential, electrokinetics, scanning probe microscopy, spectroscopy, molecular dynamics, ab initio methods

1. Introduction

When a solid surface meets an aqueous electrolyte, physical or chemical mechanisms can generate an electric surface charge [1–4]. Ions in the liquid reorganize to form a nanometric layer to balance the surface charge, the electrical double layer (EDL). Surface charge governs the stability and dynamics of soft matter systems, and as such it is a key property to characterize. Surface charge also drives the response of nanofluidic systems to thermodynamic gradients [5]. The development of new membranes to harvest e.g. blue energy (the osmotic energy of sea water) [6–8] has led to a renewed interest for finding new functional interfaces with optimal surface charge.

In that context, however, it is not clear that all interfacial properties governed by surface charge can be described with a single, well-defined quantity. For instance, equilibrium interactions between colloids depend on the static distribution of ions in the EDL, but their electrophoretic motion also depends on interfacial hydrodynamics [9, 10]. Similarly, different types of osmotic flows [5] and surface conductivity [11] might be controlled by a differently defined surface charge. Consequently, the results of standard characterization methods might not be easily used to simultaneously predict all properties governed by surface charge.

In this short review, we will first discuss the difficulty in defining surface charge. We will then give a brief overview of standard models and experimental tools (detailed descriptions can be found in textbooks or reviews, e.g. [1–4]), and discuss their limits. Next, we will give an overview of recent developments in experimental characterization, and show how progress in molecular modeling has transformed our understanding of the EDL. Finally, we will discuss benefits of coupling state-of-the-art experimental tools with molecular modeling to obtain a comprehensive picture of the interfacial structure and dynamics, necessary to accurately predict all surface-charge-related properties of liquid-solid interfaces. A complementary point of view can be found in a recent review focused on water at interfaces [12].

2. Surface charge: an ill-defined concept

The concept of surface charge cannot be defined without ambiguity because liquid-solid interfaces are globally uncharged, with any surface charge being compensated by an oppositely charged EDL. It is therefore a question of separating charges between the surface and the liquid. However, both the charged species at the surface and those in the EDL can have complex structure and dynamics. For instance, interfacial charged species can be strongly bound to the solid, or free to diffuse along the surface [13]. Furthermore, ions in the EDL can have a reduced mobility, or belong to a hydrodynamic stagnant layer.

Email addresses: oriane.bonhomme@univ-lyon1.fr (Oriane Bonhomme), laurent.joly@univ-lyon1.fr (Laurent Joly)

From this complex atomistic picture, different effective surface charges can be defined and measured, which quantify different physical phenomena at a larger scale. First, interactions between solid surfaces in solution result from the long-range distribution of ions between the surfaces. This can be used to define a static surface charge, which will for instance control the stability of colloidal systems. One can also define dynamic surface charges. In particular, the electroosmotic mobility can be used to define an electrokinetic charge [9, 10]. However, it is not clear that the same electrokinetic charge can also describe other osmotic flows, e.g., diffusioosmosis and thermoosmosis. Finally, surface conductivity [11] can be used to define yet another surface charge.

Therefore, a suitable characterization method has to measure the relevant effective surface charge corresponding to a given phenomenon, or to provide a detailed enough description of the interface, which can be used to evaluate the adequate effective surface charge. In the second case, an accurate model of the interface is also needed. The discrepancy between surface charge measurements, and the use of macroscopic theories – which are not *a priori* justified at the nanometer scale – to interpret them, underline the needs of coupling experimental measurements to molecular modeling in order to take proper account of the concept of surface charge. In the following section, we will review the standard models of the EDL, the standard characterization tools based on these models, and highlight their limits.

3. Standard approaches

3.1. Standard models

A comprehensive model of a charged interface in an aqueous electrolyte (but even pure water is one) needs to describe the distribution of charge, the dynamics of charged species, and interfacial hydrodynamics (see Fig. 1).

For the charge distribution, models generally separate the EDL into two regions. Beyond a few molecular sizes of the solid surface, ions can be considered as point charges and one commonly uses the mean-field Poisson-Boltzmann (PB) equation to describe their distribution, assuming that the dielectric permittivity of the solvent is local, isotropic and homogeneous [1]. This equation predicts that the electric potential and the local charge density decrease exponentially with the distance from the charged interface. The decay range is given by the Debye length λ_D , which scales as the inverse square root of the salt concentration (Fig. 1). The PB equation also predicts that the charge of the EDL can concentrate in a region thinner than the Debye length with a non-exponential decay. Specifically, this happens when the so-called Gouy-Chapman length ℓ_{GC} , which scales as the inverse of the surface charge, becomes smaller than λ_D . The charge of the EDL then concentrates in a region of thickness ℓ_{GC} (Fig. 1).

Very close to the surface (a few molecular sizes), the hypotheses underlying the PB equation are especially poorly justified. One generally introduces the so-called Stern layer to describe this region [14, 15]. The Stern layer is usually assumed to consist of adsorbed ions – specifically or not, which may be

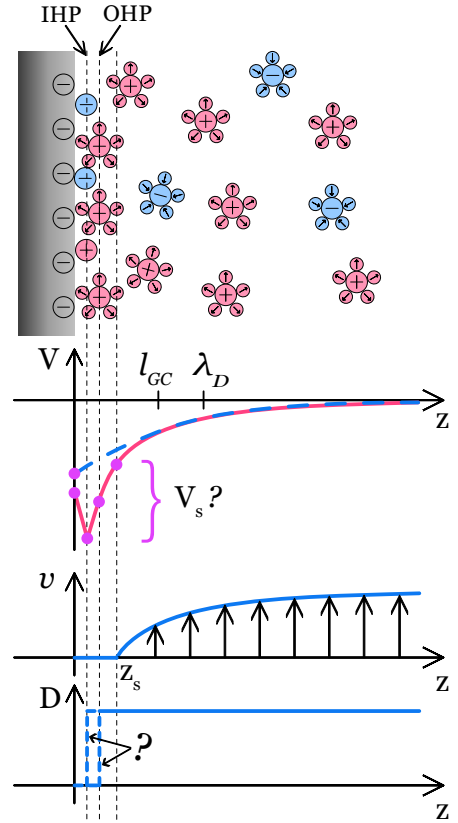


Figure 1: Standard model of the electrical double layer. From top to bottom: ion distribution; electric potential profile V (full red line: true potential; dashed blue line: apparent exponential potential as seen far from the interface); velocity profile v ; ion diffusion coefficient profile D . Even from this traditional description, a number of surface potentials V_s (represented by magenta points) and corresponding surface charges can be defined, see text for details.

partially or fully dehydrated. A number of planes and layers are defined accordingly, e.g., the inner Helmholtz plane (IHP), below which ions are specifically adsorbed and at least partially dehydrated, and the outer Helmholtz plane (OHP) separating the adsorbed hydrated ions and the diffuse layer obeying the PB equation (Fig. 1). It has also been recognized that the dielectric permittivity of the solvent below the IHP can deviate from that beyond the IHP, due to a preferred orientation of solvent molecules in response to surface charge [16]. Finally, theoretical models beyond the standard PB framework have been developed to account explicitly for, e.g., correlations, image charges, finite-ion-size effects, or specific adsorption in the Stern layer [1, 2].

For hydrodynamics, a local and homogeneous viscosity is usually assumed, together with a no-slip boundary condition. A stagnant layer is often introduced, defining the shear plane where the hydrodynamic velocity vanishes (Fig. 1). While the stagnant layer does not participate to the flow, whether its diffusion dynamics is bulk-like, hindered or even frozen remains unclear and may depend on the specifics of the interface. The hydrodynamic shear plane position z_s is not simply related to the “static” IHP and OHP. However, z_s is commonly confused with the IHP or OHP, i.e., the stagnant layer and the Stern layer are assumed to share the same boundaries.

Finally, regarding ion mobility, standard models assume bulk-like diffusion for ions in the diffuse layer [11, 17], and immobile ions in the Stern layer, although a dynamic Stern layer is sometimes introduced to explain anomalies in surface conductivity [18–21].

Standard models provide an effective description of the interface, introducing a limited number of adjustable parameters with a simple physical interpretation. These models can relate macroscopic experimental measurements to microscopic parameters such as surface potential or shear plane position. They can also be adjusted or extended to consistently describe different effective surface charges, for instance the electrokinetic charge or the surface conductivity. However, being fundamentally effective models, which are not based on an atomistic description of the interface, they cannot be expected to provide a comprehensive and consistent prediction of all the different effective surface charges.

3.2. Standard experimental tools

On standard basis, different methods are used to characterize surface charge. Some are based on transport properties (Fig. 2a-b), and in particular electrokinetic characterization of colloids, porous materials or surfaces. Others rely on probing static properties of the interface, such as electrostatic potentials (Fig. 2d).

3.2.1. Transport measurements

Electrokinetic methods have been widely used to investigate the mobile part of electrolyte solutions at the interface with suspended colloids or in micro/nanochannels. In these methods, for example the electrophoretic mobility is measured,

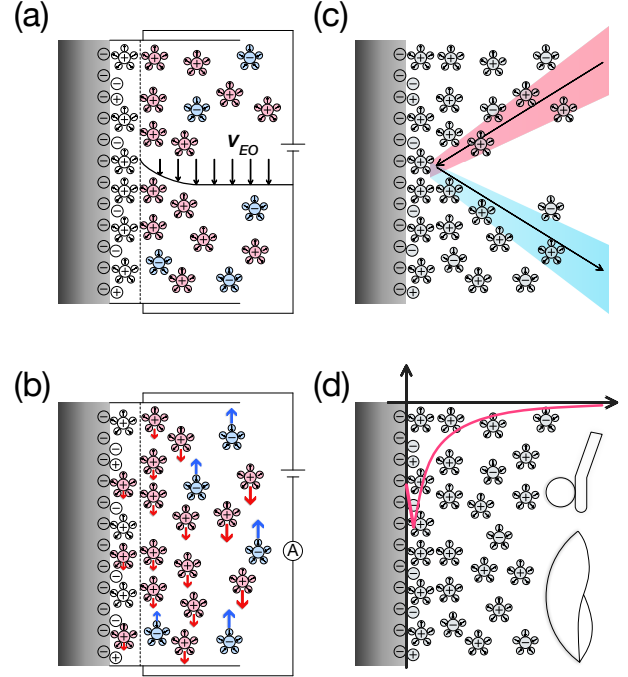


Figure 2: Different experimental tools: (a) electrokinetic transport; (b) surface conductivity; (c) X-ray reflectivity or nonlinear optics; (d) AFM/SFA-like tools. Methods illustrated in (a) and (b) only provide information on ions highlighted in color, while methods illustrated in (c) and (d) are non-specific.

from which an electrokinetic potential, or ζ -potential, is inferred [9]. Alternatively, an electroosmotic plug flow in micro/nanochannels can be determined, using for example the all-electric current monitoring method [22] or some fluorescent probe [23]. According to Onsager reciprocal relations, the ζ -potential can also be determined from the streaming current [24]. Within the standard description of the EDL, the ζ -potential can be identified with the value of the electric potential at the shear plane, separating the stagnant layer from the flowing liquid [9], see Fig. 1. This electrostatic potential can in turn be related to the electrokinetic surface charge, i.e., the total charge contained behind the shear plane, thanks to the Grahame equation [15]. This equation is based on the mean-field PB description of the diffuse EDL, so that it relies on the assumptions that the shear plane is located outside the Stern layer and that the dielectric permittivity does not vary beyond the shear plane. Another method is to probe the conductivity in porous materials or nanopores at low salt concentration [3, 25, 26]. In this regime, a so-called surface conductivity is measured, which is a signature of the electrostatic environment in the vicinity of the surfaces. However, surface conductivity also depends on ion mobility close to the surface, and can include contributions from the solid or from quantum charge transport at the interface, so that its relation to the bare surface charge is complex.

Generally, electrokinetic experiments provide no direct insight into the bare surface properties of a colloid or channel wall, to which ions are adsorbed. Moreover, despite the good match between results obtained from streaming current or electroosmosis experiments, a discrepancy with the surface charge obtained from surface conductivity remains [20, 27–29], under-

lining the complexity of the systems when transport properties are considered.

3.2.2. Static measurements

Static measurements typically either measure the electrostatic surface potential, or the number of ionized groups at the surface, to determine the surface charge.

Surface potential measurements have been expended in the 1990s by measuring the forces involved between a sphere and a plane using AFM/SFA-like tools [30–34], see Fig. 2d.

The static surface charge can be measured in various ways. For example, potentiometric titration is commonly used to probe the bare charge of a solid surface [35]. Other seminal techniques involve specific adsorption of organic conjugated compounds onto the charged surface, which is titrated using UV-Vis spectroscopy [36]. Similarly, neutron reflectivity is a powerful tool to determine non-selectively species adsorption [37] or water structure [38] at interfaces.

Attempts to probe both electrokinetic transport properties and static properties for the same sample remain scarce due to the technical constraints of the different measurements [39]. For colloids, the relation between static charge (obtained by titration) and electrokinetic charge depends on the nature of the surface charge [40]: while electrokinetic and static potentials are consistent for AgI, for which the charge results from an excess of the ionically bonded constituents, the electrokinetic potential is lower than the static potential for metal oxides, for which the charge arises from the dissociation of hydroxyl groups.

4. Latest innovations in experimental characterization

Recent innovations in determining the structure of liquid-solid interfaces focused on probing the chemical nature and position of ions in the vicinity of the interface [41], see Fig. 2c. Among them, specific set-ups based on X-ray reflectivity represent a powerful tool to investigate the interfacial ion distribution, for example in the case of electrolytes in the vicinity of electrodes [42].

Thermodynamic adsorption energies have been measured using resonant anomalous X-ray reflectivity [43, 44], and specific adsorption of chloride versus iodide near hydrophilic surfaces has been observed with X-ray standing waves [45]. A more sophisticated method based on X-ray photoelectron spectroscopy applied to a microjet containing silica nanoparticles was used to determine precisely the influence of ion specificity on the surface potential and on the Stern layer composition [46]. This method has also been compared to other, more standard, surface potential determination [47]. However, most of the experimental results in the above-mentioned studies require support of modeling (density functional theory [43, 44], molecular dynamics [45], analytical [46]) to get a full picture of the processes taking place at the interface. Moreover, X-ray based techniques are limited because they can only identify the chemical nature of the ion and not its ionization state and electrostatic

environment. For instance, these techniques are blind to hydroxide and hydronium ions, which have a strong impact on the structure and dynamics of charged interfaces.

The electrostatic potential of interfaces has also been probed using nonlinear optics, sum frequency generation (SFG) [48–50] – evidencing a cationic specific Stern layer structure – and second harmonic generation (SHG) [51]. Furthermore, surface potential measurements in recent ion-sensitive field-effect transistor (ISFET) studies have provided new insight into the effect of pH and salinity of various electrolytes on the EDL structure and found that standard complexation models cannot explain the observed behavior [52–54]. Here also, molecular simulations have been used to interpret the results [55, 56].

More direct probing techniques have also been used to gain insight into the interfacial fluid structure. Following huge improvement of AFM resolution, it became possible to probe the structure of the interface [57–59], and even the residence time of single ions in the Stern layer [60, 61].

Radically different approaches also focused on the determination of ion repartition at interfaces using transport property characterization. Beyond standard electrokinetic characterization, new electrokinetic properties have been investigated, and in particular the diffusiophoretic [62, 63] and diffusioosmotic [6, 64] response to ion concentration gradients in solution. Whereas the ζ -potential measured from diffusioosmotic and electroosmotic velocities coincide for KI, LiI and NaI salts in the vicinity of a silica interface [64], discrepancies are observed when considering the diffusioosmotic current of a KCl solution flowing through a membrane formed by boron nitride nanotubes [6]. In the latter, the surface potential is in good agreement with the one measured by surface conductivity, rather than the one calculated from electrokinetic mobility. A similar discrepancy between the ζ -potential and the thermophoretic mobility was recently highlighted for functionalized polystyrene particles [65].

To disentangle the couplings between static and transport properties of interfacial ions, a few attempts to measure both the electrokinetic response of the interface together with its structure have been documented [66]. For example, Jalil and Pyell [67] combined their own standard electrokinetic measurements with X-ray spectrometry data of Brown et al. [46] to get a more refined picture of the EDL for monovalent electrolyte solutions in contact with silica nanoparticles. In a more direct approach, in situ SFG experiments at liquid-solid interfaces under flow have shown a signature of the flow on the surface potential [68]. Conversely, SHG experiments performed at liquid-gas interfaces evidenced no modification of the signal when electroosmotic flow was generated [69]. Recent experiments coupling SHG and streaming potential at hydrophilic and hydrophobic solid-liquid interfaces also did not observe any effect of the flow [70]. These contradicting observations underline the complexity of the mechanisms involved and indicate that the relation between the EDL structure and dynamics depends on the chemical nature of the interface.

5. Latest developments in molecular modeling

Whereas state-of-the-art experiments revealed new information on the structure and dynamics of the EDL, molecular simulations have highlighted the limits of standard models [71–78], and can help refining models using a bottom-up approach. Molecular dynamics (MD) simulation is a particularly powerful tool to explore the structure and dynamics of the EDL. MD provides an explicit description of the atomic structure of the liquid-solid interface, with its time evolution computed based on empirical interaction potentials between atoms. MD simulations provide accurate control over environmental conditions and full access to microscopic information that is inaccessible in experiments. As such, simulations can be used to explain experimental observations, or to improve models and assumptions for interpreting experimental measurements [79]. Indeed, the suitability of standard models becomes questionable when screening lengths compare with the molecular size, as well as typical values for surface roughness. In fact, both the Debye length and the Gouy-Chapman length can easily reach one nanometer for realistic salt concentrations ($> 10^{-2}$ M) or surface charges (> 40 mC/m²). In that situation, standard models can fail to describe the EDL in many ways and a more sophisticated description is needed.

First, it is possible that the ion distribution does not follow the mean-field Boltzmann law, especially when only electrostatic interactions are taken into account. Here, MD can provide information on specific interactions that need to be included in the Boltzmann factor [80–89]. Furthermore, ion-ion correlations, which are particularly important with multivalent species and concentrated solutions, can strongly affect the ion distribution, and even reverse the apparent surface charge as seen far from the interface [90]. Also here, MD can help to refine the existing models [91–94].

More importantly, the dielectric permittivity of the solvent can become inhomogeneous and anisotropic near the interface, or the local permittivity approximation can break down [89, 95–100]. Extended theories need to be used in those cases, although it has been shown that simple effective models, with just a step in permittivity at a given distance from the wall, can reproduce static and electrokinetic charges obtained by molecular simulations (which make no assumption on the dielectric permittivity) [97]. Whether such simple models can also predict the surface response to other thermodynamic gradients (e.g. osmotic or thermal) remains to be explored.

The standard picture for hydrodynamics, with a stagnant layer followed by a liquid with constant viscosity, has also been questioned by MD simulations at the nanoscale [101, 102]. First, the no-slip boundary condition fails on some surfaces [103]. The electrokinetic mobility is then amplified by hydrodynamic slip for a given surface charge [104–106]. In turn, the amplitude of slip depends on surface charge [83, 107–109]. Corrected continuum descriptions [110, 111] describe MD results well [105, 107], and the amplification effect of slip was confirmed by two independent experiments [39, 112]. Hydrodynamic slip is for instance key to understanding anomalous electrokinetic charge in foam films [113, 114]. Other osmotic

flows can be strongly affected by slip [115–117], and more generally by nanoscale dynamics [118].

Second, the stagnant layer concept has been questioned by MD results. For instance, no stagnant layer was observed on amorphous silica [94], beyond a monolayer of strongly adsorbed water molecules, whose thickness was comparable to the roughness of the disordered surface. Zhang et al. [119] also did not observe a stagnant layer on silica, but instead they showed that viscosity increased smoothly near the surface. Smoothly increasing viscosity profiles were also observed by others near hydrophilic surfaces [87, 97, 120], but not near hydrophobic surfaces [97]. More generally, various simulation studies have indicated that viscosity can be inhomogeneous [87, 120–123] and even nonlocal [124]. Continuum theory taking the viscosity profile into account can predict electroosmotic flow rates [119]. The complex viscosity profiles can also be described through effective sharp boundary models, for instance by a constant viscosity combined with a few angstroms thick stagnant layer on silica [119], by a constant viscosity on slipping surfaces, or by the inclusion of a step in viscosity on non-slipping surfaces [97]. Although such simple descriptions can be convenient, their parametrization hinges on detailed insight into the interfacial region. Moreover, because these effective descriptions do not correspond to the real microscopic picture, their transferability to describe all surface-charge-governed properties is not guaranteed.

MD simulations have also given new insight into the diffusion dynamics of ions in the different subsections of the EDL [125]. In their pioneering work, Lyklema et al. [126] used MD to confirm the emerging picture of a stagnant layer behaving like a gel, in which the ions can diffuse almost freely, but which does not flow globally. This picture explains the large possible differences between the surface charge one can extract from electrokinetic or surface conductivity measurements [20, 27–29]. Recent MD simulations on amorphous silica [94] found that the standard decomposition of the EDL into a Stern and a diffuse layer was inadequate. The EDL was instead decomposed into a mobile and an immobile ion population, of which the distribution overlaps. The diffusion coefficient of free ions continuously decreases close to the wall, an effect that can be described by continuum hydrodynamics [127], and can dramatically affect surface conductivity. Finally, on hydrophobic surfaces where the surface charge is carried by specifically adsorbed ions, the surface mobility of these ions can affect the electrokinetic response of the interface [13].

Even though simulations have proven valuable in the study of EDL properties, classical MD simulations are reaching their limits, because of two strong weaknesses. First, interactions between atoms are based on empirical force fields. By definition, these force fields are built to reproduce a given set of data, and their transferability to different situations is questionable. Specifically, most standard force fields are designed to reproduce equilibrium structural properties of bulk systems, and there is no guarantee that they can correctly describe the dynamical and transport properties of interfaces. A striking example concerns the effect that ions have on water diffusion and viscosity [128]. While some salts enhance water diffusion, most

empirical force fields predict a systematic decrease in diffusion with increasing ion concentration [129, 130]. However, recently developed force fields have been able to qualitatively reproduce the effect of large ions [131], and to quantitatively describe the effect of small ions [132–134] on water diffusion.

The second major weakness of force field-based simulations is related to the systems described. Indeed, the atomic wall structure and charge distribution need to be constructed before the simulation can be run, often from limited information. Notably, the surface charge is usually imposed by assigning partial charges to the atoms in the substrate, conform the force field employed. These partial charges are kept constant throughout a classical simulation – rendering the bare surface charge unaffected by the rearrangement of interfacial ions or solvent molecules. Alternatively, surface reactivity is considered by using reactive force fields [135–138] and *ab initio* methods [139–141]. Such reactive simulations are important to investigate, for example, charge regulation [142, 143], which occurs in narrow channels due to overlapping EDLs.

Ab initio molecular dynamics (AIMD) has been used to explore the structure and dynamics of water-oxide interfaces [144–159]. In particular, AIMD can be used to compute vibrational spectra and non-linear optical response [147, 160–164], and as such is a key tool to help interpret experimental observations. Recently, liquid-solid friction has been characterized with AIMD [165, 166], opening perspectives for the investigation of other transport properties with these methods. At present, however, *ab initio* methods are typically based on density functional theory, and take electronic exchange and correlations into account through an approximate functional. This limits the accuracy of *ab initio* methods to describe the structure and dynamics of water-solid interfaces [139]. Most AIMD works also do not take quantum nuclear effects into account. Finally, the computational cost of reactive and *ab initio* simulations is very large – effectively limiting the accessible simulation time and the number of atoms. These restrictions are currently prohibitive for studying rare events, dilute electrolytes, or for simulating a sufficiently large system to accurately represent the structure and heterogeneous charge distribution of amorphous surfaces.

Size restrictions can be mitigated drastically with primitive-model simulations, in which the solvent is included implicitly [167]. For instance, such an approach was used to investigate the origin of surface charge on graphene and boron nitride [168]. However, Lee et al. [169] found that various physical quantities that depend on the orientation of solvent molecules were not accurately predicted by the primitive model approach. Additionally, Vangara et al. [170] recently showed, by comparing explicit and implicit solvent DFT models, that both the solvent and the ions contribute to the chemical balance between surface groups and the solution.

Apart from the importance of solvent molecules and dissolved salts, various experiments have revealed intricate effects of pH on for example specific ion adsorption [171, 172]. Furthermore, local enhancement of proton mobility affects the Stern conductivity [173], but may also have important consequences for surface reactivity. Simulations are, in princi-

ple, suitable for elucidating the molecular-level mechanisms responsible for such pH dependence, but current computing power does not permit explicitly accounting for near-neutral pH in molecular simulation because of the immense system size required. Interestingly, this numerical difficulty echoes the limits of experimental methods based on X-rays, which are blind to hydronium and hydroxide ions and hence unable to provide insight on the role of pH.

Explicit-pH simulations and large-scale quantum-based simulations are currently beyond the feasible. However, with the continuous improvement of computing power, *ab initio* simulations should progressively be able to tackle an increasingly large panel of problems. Meanwhile, the information obtained from small *ab initio* simulations is already transferred to classical simulations using *ab initio*-based force fields [174–178].

6. Why coupling experiments and modeling is needed, and recent attempts

Despite recent developments in experimental characterization and molecular modeling, the study of local structure and dynamics at the solid-electrolyte interface remains restricted by the inherent limitations of the respective methods. The ambiguity in what quantity is measured by each method is a considerable limitation, making it difficult to form a unified understanding of the EDL by combining data from multiple experimental techniques. Ambiguity or uncertainty of measurements can even propagate when combining techniques. For example, the charge held within the Stern layer can be estimated by combining titration experiments to determine the bare surface charge density and electrokinetic experiments to infer the charge contained in the mobile region [10]. However, the latter relies on the assumption that the electrokinetic charge equals the charge held in the diffuse layer, i.e., the shear plane is assumed to coincide with the OHP. Similarly, the difference between the static and electrokinetic potentials has been used to estimate the Stern layer thickness, assuming a constant permittivity across the Stern layer [179]. The calculated thickness depends linearly on the assumed Stern-layer permittivity and on the approximated potential drop between the surface potential and the electrokinetic potential, which were obtained using two different techniques and material samples. Furthermore, this approach again assumes that the shear plane marks the outside of the Stern layer. Evidently, inferred quantities, such as the Stern layer thickness and its charge, can be highly sensitive to assumptions and models.

On the other hand, simulations can help to interpret and complement experiments, without the need for assumptions or theoretical models. However, direct and quantitative comparison between experiments and simulations is often difficult for two major reasons: first, due to empirical force fields in classical MD, or approximations in *ab initio* methods, and second, because atomistically-detailed computations are limited to short simulation times and small systems, whereas many experiments are limited by their spatial and temporal resolution. Direct coupling between simulations and experiments thus presents a major challenge. Yet, combining these disciplines can help to

leverage their complementary strengths. Specifically, experiments are essential to validate simulation results and to improve simulation force fields, while accurate simulations are helpful to interpret and explain experimental measurements.

Thus far, few studies have combined experiments and simulations to gain deep understanding of interfacial fluid properties [38, 42, 44, 54–57, 60, 70, 85, 164, 180–192]. For example, Předota et al. [186] performed MD simulations and titration experiments for different electrolytes near a hydroxylated (110) rutile surface. The simulations suggested that different slopes in the titration curves were caused by different adsorption mechanisms. In a recent study, Bourg et al. combined X-ray reflectivity, MD simulation, and complexation theory to provide detailed insight into the interfacial structure of 0.1 M alkali chloride solutions on a muscovite mica surface [190]. The authors showed that the structure of the first molecular layer of water was determined predominantly by interaction with the surface, whereas the structure of the second layer depended also on interactions with adsorbed interfacial ions. Water beyond the first two monolayers exhibited density layering, but showed no evidence of sensitivity to short-range interactions with either the surface or adsorbed ions. With the exception of Li^+ , the experimentally and computationally measured ion exchange energies were in close agreement. The trend in the exchange energy of different ions suggested that not only the hydration free energy was important, but also the match between the surface structure and the hydration structure of the ions.

Combined experiments and simulations were also used to understand the mechanisms underlying charge inversion [90]. Labbez and coworkers combined surface titration measurements, electrophoretic experiments, and implicit-solvent grand canonical Monte Carlo simulations to study the charging behavior of calcium silicate hydrate [181, 183]. The authors found that the apparent charge inversion observed for concentrated divalent solutions decreased, or even disappeared, upon the addition of monovalent electrolytes to the electrolyte mixture. This contrasts the idea that charge inversion increases with ion concentration. Using MD simulations and electrophoresis experiments, Calero et al. observed charge inversion for large organic monovalent ions near a hydrophobic colloid, but not near a hydrophilic surface [85]. This demonstrated that charge inversion can occur also when ion-ion correlations are negligible. Semenov et al. [185] calculated the electrokinetic mobility of a single latex colloid in a trivalent electrolyte solution from implicit-solvent MD simulations combined with hydrodynamic theory. Electrostatic and specific adsorption were both essential to predict the mobility reversal observed in their optical tweezer experiments.

The insights obtained in the studies described above could not be obtained solely using experiments, due to the need for non-invasive measurements with a sub-nanometer resolution, or detailed insight into the interactions between individual atoms. On the other hand, only modeling the problem would also be unsatisfactory, as the validity of the results is not guaranteed.

7. Conclusions and outlook

A complete understanding of the dynamics and structure of ions repartition near surfaces requires first a full experimental investigation. Most of the techniques nowadays rely on determining charge transport properties (probing electrokinetics mainly) *or* static properties such as surface potential or ion repartition structure. However, most recent advances show that both are intimately coupled; transport can indeed change the ion repartition and *vice versa*. Only a full characterization of the liquid structure near the interface while electrokinetic transport takes place, under a range of conditions, would allow giving realistic inputs in this subject. Another direction that needs to be tackled is the identification of specific solid systems. Indeed, most oxide surfaces, especially silica, are complex: their properties depend a lot on the preparation and despite huge efforts in developing methodology in the experimental community, results are poorly reproducible from one group to the other. Identifying robust model materials would be a real asset.

Molecular simulations can be a valuable tool in the quest for identifying suitable model materials and the coupling between static and transport properties. Leveraging the strengths of AIMD, and combining this with computationally cheaper tools, may hold much promise for the future of computational molecular research. In fact, AIMD has adopted an increasingly important role in the study of molecular fluids and fluid-solid interfaces in recent years. The main drawback of this technique has thus far been its large computational cost, which strongly limits the accessible simulation time and system sizes. Although the accessible time scale in AIMD is insufficient to directly probe transport properties, AIMD could instead be used to explore free energy profiles [193, 194]. Alternatively, the limitation of accessible time scales can be mitigated using machine learning, e.g., to predict infrared spectra [195]. Machine learning was also key to mitigate high computational costs to calculate free energy differences in a classical MD system [196]. Machine learning, combined either with classical or quantum simulation, can be a powerful tool in the development of more versatile and transferable simulation force fields, which require optimization against a large set of conditions and variables. Particularly, fingerprint algorithms, an aspect of machine learning, were recently suggested as a ‘*useful building block for constructing data-driven next generation force fields*’ [197]. In addition to the potential speedup to be achieved with machine learning, the arrival of quantum computing may hold promise for drastically accelerating molecular simulations. Finally, advanced techniques capable of taking quantum nuclear effects into account are increasingly accessible [184, 198–200] and could help improving the description of water-based systems.

In conclusion, both experimental characterization and numerical modeling of charged interfaces have vastly progressed over the past years, and current developments give hope for a bright future. MD does not only help to understand the microscopic phenomena; it also helps to extend the validity of the experimental measurements. Indeed, numerous macroscopic parameters that are necessary to interpret the measurements (ad-

sorption constant, slip length, *etc.*) can be calculated from MD, so that the treatment of experimental data is facilitated. Furthermore, very often, simulations and experiments can be directly compared without requiring the use of ill-defined concepts such as ζ -potential or effective surface charges. This does not mean that such macroscopic parameters are not useful anymore. Any macroscopic theory needs such average quantities. It means that thanks to this direct comparison between experiments and simulations, one can decide which effective quantity is relevant for a given system. New, more efficient models have been developed in recent years, but there is still room for improvement. For instance, future models could go beyond traditional assumptions of distinct layers separated by sharp interfaces. In particular, the complex 3D structure and dynamics of the EDL due to surface roughness and chemical heterogeneities could be taken explicitly into account. Such 3D models could indeed provide more insight on the discrepancy between static and dynamic surface charge, and identify parameters controlling this discrepancy. Coupling experiments and atomistic modeling is the best method to assess the value of recent and future models.

We hope we have convinced the reader that major advances in the understanding and detailed characterization of surface charge(s) will come from the coupling between state-of-the-art experiments and molecular simulations.

Acknowledgement

This work is supported by the ANR, project ANR-16-CE06-0004-01 NECtAR. LJ is supported by the Institut Universitaire de France.

Recommended reading

- of special interest
- of outstanding interest

[9] • Review on electrokinetic transport measurements, discussing the models used for their interpretation.

[10] •• This article presents a critical discussion of the difference between bare surface properties and electrokinetic quantities among other things.

[12] • Review on latest advances in understanding the structure and dynamics of water at various interfaces.

[24] •• This work compared the streaming current using various channels and ion concentrations.

[39] • Comparison between AFM and electrokinetic measurements, showing the amplification of electrokinetic mobility by liquid-solid slip.

[44] • X-ray reflectivity and *ab initio* simulations are used to explore cation adsorption at the water-quartz interface.

[45] • X-ray standing waves are used to explore ion distribution at the water-silica interface.

[46] •• This work introduces X-ray photoelectron spectroscopy from a liquid microjet, which provides insight into the potential drop across the Stern layer.

[57] • The authors combine AFM experiments with DFT simulations to measure and explain how different electrolytes and concentrations affect the effective surface charge.

[60] • This study succeeded in probing adsorbed ions with high-resolution AFM experiments, supported by MD simulation.

[68] •• SFG experiments showed that surface charge can be reversibly affected by a flow, suggesting a strong coupling between chemistry and transport.

[82] •• MD work showing anomalous electroosmosis due to ionic specificity, resulting for instance in a nonzero zeta potential for a neutral surface.

[85] • This study combines MD simulation and electrophoresis experiments to demonstrate that solvation free energy can cause or prevent charge inversion, depending on the colloid surface hydrophobicity.

[94] • This computational study reveals interesting insights by deviating from the traditional way of looking at the electrical double layer.

[97] •• MD work showing that both viscosity and dielectric permittivity are affected by solvent structure close to interfaces.

[101] • Very complete review on both experimental and theoretical aspects on coupled electrokinetic transport near solid interfaces.

[105] • MD work showing that the zeta potential depends critically on hydrodynamic boundary condition, and can be amplified by liquid-solid slip.

[120] •• This study uses MD simulation to provide deep understanding of the origin of the macroscopically measured zeta potential.

[122] • MD work discussing the limits of traditional models of electroosmosis in the context of lipid membranes.

[130] •• An article illustrating the power of *ab initio* methods, which can describe the effects of salts on water diffusion, in contrast with force field-based simulations.

[139] • A review discussing the interest and shortcomings of *ab initio* methods to describe aqueous systems.

[164] • A recent work showing how coupling SFG and *ab initio* simulations can help to understand water-mineral interfaces at the molecular level.

[165] • AIMD was used to show that liquid-solid friction can depend crucially on specific electronic structure effects.

[168] • DFT calculations were used to explore the origin of surface charge on graphene and boron nitride.

[180] • A combined MD and X-ray study of ion adsorption at the water-rutile interface

[181] • This study combined simulations with potentiometric titration and electrophoresis experiments, as well as theory, to study the effect of pH and ion concentration of surface charging.

[184] • A combination of SFG spectroscopy and path integral MD simulations showed that the molecular structure of water-vapor interfaces is affected by nuclear quantum effects.

[190] •• The consistent views provided by the experiments and simulations in this study contribute to a better understanding of the Stern layer.

References

References

- [1] D. Andelman, Electrostatic Properties of Membranes: The Poisson-Boltzmann Theory, in: *Handbook of Biological Physics*, volume 1B, 1995, pp. 603–642.
- [2] J. N. Israelachvili, *Intermolecular and surface forces*, Academic Press, 2011.
- [3] R. J. Hunter, *Foundations of colloid science*, Oxford University Press, 2001.
- [4] J. Lyklema, *Fundamentals of Interface and Colloid Science*, volume 2: Solid-Liquid Interfaces, 1995.
- [5] J. Anderson, Colloid Transport By Interfacial Forces, *Annual Review of Fluid Mechanics* 21 (1989) 61–99.
- [6] A. Siria, P. Poncharal, A.-L. Biance, R. Fulcrand, X. Blase, S. T. Purcell, L. Bocquet, Giant osmotic energy conversion measured in a single transmembrane boron nitride nanotube, *Nature* 494 (2013) 455–458.
- [7] J. Feng, M. Graf, K. Liu, D. Ovchinnikov, D. Dumcenco, M. Heiranian, V. Nandigana, N. R. Aluru, A. Kis, A. Radenovic, Single-layer MoS₂ nanopores as nanopower generators, *Nature* 536 (2016) 197–200.
- [8] A. Siria, M.-L. Bocquet, L. Bocquet, New avenues for the large-scale harvesting of blue energy, *Nature Reviews Chemistry* 1 (2017) 0091.
- [9] A. Delgado, F. González-Caballero, R. Hunter, L. Koopal, J. Lyklema, Measurement and interpretation of electrokinetic phenomena, *Journal of Colloid and Interface Science* 309 (2007) 194–224.
- [10] J. Lyklema, Surface charges and electrokinetic charges: Distinctions and juxtapositionings, *Colloids and Surfaces A: Physicochemical and Engineering Aspects* 376 (2011) 2–8.
- [11] J. Lyklema, Surface conduction, *Journal of Physics: Condensed Matter* 13 (2001) 5027–5034.
- [12] O. Björneholm, M. H. Hansen, A. Hodgson, L.-M. Liu, D. T. Limmer, A. Michaelides, P. Pedevilla, J. Rossmeisl, H. Shen, G. Tocci, E. Tyrode, M.-M. Walz, J. Werner, H. Bluhm, Water at Interfaces, *Chemical Reviews* 116 (2016) 7698–7726.
- [13] S. R. Maduar, A. V. Belyaev, V. Lobaskin, O. I. Vinogradova, Electrohydrodynamics near hydrophobic surfaces, *Physical Review Letters* 114 (2015) 118301.
- [14] O. Stern, Zur Theorie der Elektrolytischen Doppelschicht, *Zeitschrift für Elektrochemie* 30 (1924) 508–516.
- [15] D. C. Grahame, The Electrical Double Layer and the Theory of Electrocapillarity, *Chemical Reviews* 41 (1947) 441–501.
- [16] J. O. Bockris, M. A. V. Devanathan, K. Muller, On the Structure of Charged Interfaces, *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* 274 (1963) 55–79.
- [17] J. J. Bikerman, Iontentheorie der Elektrosmose, der Strömungsströme und der Oberflächenleitfähigkeit, *Zeitschrift für Physikalische Chemie* 163A (1933) 378–394.
- [18] C. Zukoski IV, D. Saville, The interpretation of electrokinetic measurements using a dynamic model of the stern layer: I. The dynamic model, *Journal of Colloid and Interface Science* 114 (1986) 32–44.
- [19] C. Zukoski IV, D. Saville, The interpretation of electrokinetic measurements using a dynamic model of the stern layer: II. Comparisons between theory and experiment, *Journal of Colloid and Interface Science* 114 (1986) 45–53.
- [20] L. A. Rosen, J. C. Baygents, D. A. Saville, The interpretation of dielectric response measurements on colloidal dispersions using the dynamic Stern layer model, *J. Chem. Phys.* 98 (1993) 4183–4194.
- [21] R. R. Netz, Electrofriction and Dynamic Stern Layers at Planar Charged Surfaces, *Physical Review Letters* 91 (2003) 138101.
- [22] X. Huang, M. J. Gordon, R. N. Zare, Current-monitoring method for measuring the electroosmotic flow rate in capillary zone electrophoresis, *Analytical Chemistry* 60 (1988) 1837–1838.
- [23] J. G. Santiago, S. T. Wereley, C. D. Meinhart, D. J. Beebe, R. J. Adrian, A particle image velocimetry system for microfluidics, *Experiments in Fluids* 25 (1998) 316–319.
- [24] F. H. J. van der Heyden, D. Stein, C. Dekker, Streaming currents in a single nanofluidic channel, *Phys. Rev. Lett.* 95 (2005) 116104.
- [25] D. Stein, M. Kruithof, C. Dekker, Surface-Charge-Governed Ion Transport in Nanofluidic Channels, *Physical Review Letters* 93 (2004) 035901.
- [26] S. Li, P. Leroy, F. Heberling, N. Devau, D. Jougnot, C. Chiaberge, Influence of surface conductivity on the apparent zeta potential of calcite, *J. Colloid Interface Sci.* 468 (2016) 262–275.
- [27] J. Kijlstra, H. P. van Leeuwen, J. Lyklema, Low-Frequency Dielectric Relaxation of Hematite and Silica Sols, *Langmuir* 9 (1993) 1625–1633.
- [28] W. N. Rowlands, R. W. O'Brien, The Dynamic Mobility and Dielectric Response of Kaolinite Particles, *J. Colloid Interface Sci.* 175 (1995) 190–200.
- [29] M. Rasmusson, W. Rowlands, R. W. O'Brien, R. J. Hunter, The dynamic mobility and dielectric response of sodium bentonite, *J. Colloid Interface Sci.* 189 (1997) 92–100.
- [30] W. A. Ducker, T. J. Senden, R. M. Pashley, Direct measurement of colloidal forces using an atomic force microscope, *Nature* 353 (1991) 239–241.
- [31] D. T. Atkins, R. M. Pashley, Surface forces between zinc sulfide and mica in aqueous electrolytes, *Langmuir* 9 (1993) 2232–2236.
- [32] I. Larson, C. J. Drummond, D. Y. C. Chan, F. Grieser, Direct force measurements between titanium dioxide surfaces, *Journal of the American Chemical Society* 115 (1993) 11885–11890.
- [33] Y. Q. Li, N. J. Tao, J. Pan, A. A. Garcia, S. M. Lindsay, Direct measurement of interaction forces between colloidal particles using the scanning force microscope, *Langmuir* 9 (1993) 637–641.
- [34] S. Biggs, P. Mulvaney, C. F. Zukoski, F. Grieser, Study of Anion Adsorption at the Gold-Aqueous Solution Interface by Atomic Force Microscopy, *Journal of the American Chemical Society* 116 (1994) 9150–9157.
- [35] J. Lützenkirchen, T. Preočanin, D. Kovačević, V. Tomišić, L. Lövgren, N. Kallay, Potentiometric Titrations as a Tool for Surface Charge Determination, *Croat. Chem. Acta* 85 (2012) 391–417.
- [36] P. W. Atkins, P. W. Atkins, *The elements of physical chemistry*, volume 496, Oxford University Press Oxford United Kingdom, 1992.
- [37] J. R. Lu, R. K. Thomas, Neutron reflection from wet interfaces, *J. Chem. Soc., Faraday Trans.* 94 (1998) 995–1018.
- [38] H.-W. Wang, M. J. DelloStritto, N. Kumar, A. I. Kolesnikov, P. R. C. Kent, J. D. Kubicki, D. J. Wesolowski, J. O. Sofo, Vibrational Density of States of Strongly H-Bonded Interfacial Water: Insights from Inelastic Neutron Scattering and Theory, *The Journal of Physical Chemistry C* 118 (2014) 10805–10813.
- [39] M.-C. Audry, A. Piednoir, P. Joseph, E. Charlaix, Amplification of electro-osmotic flows by wall slippage: direct measurements on OTS-surfaces, *Faraday Discussions* 146 (2010) 113.
- [40] Ph. Attard D. Antelmi I. Larson, Comparison of the zeta potential with the diffuse layer potential from charge titration, *Langmuir* 16 (2000) 1542.
- [41] G. S. Perera, C. B. Nettles, Y. Zhou, S. Zou, T. K. Hollis, D. Zhang, Direct Observation of Ion Pairing at the Liquid/Solid Interfaces by Surface Enhanced Raman Spectroscopy, *Langmuir* 31 (2015) 8998–9005.
- [42] H.-G. Steinrück, C. Cao, Y. Tsao, C. J. Takacs, O. Konovalov, J. Vatamanu, O. Borodin, M. F. Toney, The nanoscale structure of the electrolyte-metal oxide interface, *Energy Environ. Sci.* 4 (2018) 1166–1169.
- [43] S. S. Lee, P. Fenter, K. L. Nagy, N. C. Sturchio, Changes in adsorption free energy and speciation during competitive adsorption between monovalent cations at the muscovite (001)-water interface, *Geochim. Cosmochim. Acta* 123 (2013) 416–426.
- [44] F. Bellucci, S. S. Lee, J. D. Kubicki, A. Bandura, Z. Zhang, D. J. Wesolowski, P. Fenter, Rb + Adsorption at the Quartz(101)Aqueous Interface: Comparison of Resonant Anomalous X-ray Reflectivity with ab Initio Calculations, *The Journal of Physical Chemistry C* 119 (2015) 4778–4788.
- [45] S. ben Jabrallah, F. Malloggi, L. Belloni, L. Girard, D. Novikov, C. Mocuta, D. Thiaudière, J. Daillant, Electrolytes at interfaces: accessing the first nanometers using X-ray standing waves, *Physical Chemistry Chemical Physics* 19 (2017) 167–174.
- [46] M. A. Brown, Z. Abbas, A. Kleibert, R. G. Green, A. Goel, S. May, T. M. Squires, Determination of Surface Potential and Electrical Double-Layer Structure at the Aqueous Electrolyte-Nanoparticle Interface, *Phys. Rev. X* 6 (2016) 011007.
- [47] T. A. Gmür, A. Goel, M. A. Brown, Quantifying Specific Ion Effects on the Surface Potential and Charge Density at Silica Nanoparticle-Aqueous Electrolyte Interfaces, *J. Phys. Chem. C* 120 (2016) 16617–

- 16625.
- [48] A. M. Jubb, H. C. Allen, Sulfate Adsorption at the Buried Fluorite-Solution Interface Revealed by Vibrational Sum Frequency Generation Spectroscopy, *The Journal of Physical Chemistry C* 116 (2012) 9085–9091.
 - [49] K. A. Lovering, A. K. Bertram, K. C. Chou, New Information on the Ion-Identity-Dependent Structure of Stern Layer Revealed by Sum Frequency Generation Vibrational Spectroscopy, *J. Phys. Chem. C* 120 (2016) 18099–18104.
 - [50] J. Schaefer, G. Gonella, M. Bonn, E. H. G. Backus, Surface-specific vibrational spectroscopy of the water/silica interface: screening and interference, *Physical Chemistry Chemical Physics* 19 (2017) 16875–16880.
 - [51] C. Lütgebaucks, G. Gonella, S. Roke, Optical label-free and model-free probe of the surface potential of nanoscale and microscopic objects in aqueous solution, *Phys. Rev. B* 94 (2016) 195410.
 - [52] A. Tarasov, M. Wipf, R. L. Stoop, K. Bedner, W. Fu, V. A. Guzenko, O. Knopfmacher, M. Calame, C. Schönenberger, Understanding the electrolyte background for biochemical sensing with ion-sensitive field-effect transistors, *ACS Nano* 6 (2012) 9291–9298.
 - [53] K. B. Parizi, X. Xu, A. Pal, X. Hu, H. S. P. Wong, ISFET pH Sensitivity: Counter-Ions Play a Key Role, *Sci. Rep.* 7 (2017) 41305.
 - [54] R. Sivakumarasamy, R. Hartkamp, B. Siboulet, J.-F. Dufrêche, K. Nishiguchi, A. Fujiwara, N. Clément, Selective layer-free blood serum ionogram based on ion-specific interactions with a nanotransistor, *Nature Materials* 17 (2018) 464–470.
 - [55] J. P. Fitts, M. L. Machesky, D. J. Wesolowski, X. Shang, J. D. Kubicki, G. W. Flynn, T. F. Heinz, K. B. Eiseenthal, Second-harmonic generation and theoretical studies of protonation at the water/ α -TiO₂ (110) interface, *Chemical Physics Letters* 411 (2005) 399–403.
 - [56] S. Hosseinpour, F. Tang, F. Wang, R. A. Livingstone, S. J. Schlegel, T. Ohto, M. Bonn, Y. Nagata, E. H. Backus, Chemisorbed and Physisorbed Water at the TiO₂/Water Interface, *Journal of Physical Chemistry Letters* 8 (2017) 2195–2199.
 - [57] I. Siretanu, D. Ebeling, M. P. Andersson, S. L. S. Stipp, A. Philipse, M. C. Stuart, D. van den Ende, F. Mugele, Direct observation of ionic structure at solid-liquid interfaces: a deep look into the Stern Layer, *Scientific reports* 4 (2014) 4956.
 - [58] C. Zhao, D. Ebeling, I. Siretanu, D. van den Ende, F. Mugele, Extracting local surface charges and charge regulation behavior from atomic force microscopy measurements at heterogeneous solid-electrolyte interfaces, *Nanoscale* 7 (2015) 16298–16311.
 - [59] F. Liu, A. Klaassen, C. Zhao, F. Mugele, D. van den Ende, Electroviscous Dissipation in Aqueous Electrolyte Films with Overlapping Electric Double Layers, *The Journal of Physical Chemistry B* 122 (2018) 933–946.
 - [60] M. Ricci, P. Spijker, F. Stellacci, J.-F. Molinari, K. Voitchovsky, Direct Visualization of Single Ions in the Stern Layer of Calcite, *Langmuir* 29 (2013) 2207–2216.
 - [61] M. Ricci, W. Trewby, C. Cafolla, K. Voitchovsky, Direct observation of the dynamics of single metal ions at the interface with solids in aqueous solutions, *Scientific Reports* 7 (2017) 43234.
 - [62] B. Abécassis, C. Cottin-Bizonne, C. Ybert, A. Ajdari, L. Bocquet, Boosting migration of large particles by solute contrasts, *Nature materials* 7 (2008) 785.
 - [63] N. Shi, R. Nery-Azevedo, A. I. Abdel-Fattah, T. M. Squires, Diffusiophoretic focusing of suspended colloids, *Phys. Rev. Lett.* 117 (2016) 258001.
 - [64] C. Lee, C. Cottin-Bizonne, A.-L. Biance, P. Joseph, L. Bocquet, C. Ybert, Osmotic flow through fully permeable nanochannels, *Physical review letters* 112 (2014) 244501.
 - [65] J. Burelbach, M. Zupkauskas, R. Lamboll, Y. Lan, E. Eiser, Colloidal motion under the action of a thermophoretic force, *The Journal of Chemical Physics* 147 (2017) 094906.
 - [66] R. Scheu, Y. Chen, M. Subinya, S. Roke, Stern Layer Formation Induced by Hydrophobic Interactions: A Molecular Level Study, *Journal of the American Chemical Society* 135 (2013) 19330–19335.
 - [67] A. H. Jalil, U. Pyell, Quantification of Zeta-Potential and Electrokinetic Surface Charge Density for Colloidal Silica Nanoparticles Dependent on Type and Concentration of the Counterion: Probing the Outer Helmholtz Plane, *The Journal of Physical Chemistry C* 122 (2018) 4437–4453.
 - [68] D. Lis, E. H. G. Backus, J. Hunger, S. H. Parekh, M. Bonn, Liquid flow along a solid surface reversibly alters interfacial chemistry, *Science* 344 (2014) 1138–1142.
 - [69] B. Blanc, O. Bonhomme, P.-F. Brevet, E. Benichou, C. Ybert, A.-L. Biance, Electroosmosis near surfactant laden liquid-air interfaces, *Soft Matter* 14 (2018) 2604–2609.
 - [70] J. Lützenkirchen, T. Scharnweber, T. Ho, A. Striolo, M. Sulpizi, A. Abdelmonem, A set-up for simultaneous measurement of second harmonic generation and streaming potential and some test applications, *J. Colloid Interface Sci.* 529 (2018) 294–305.
 - [71] V. Knecht, H. J. Risselada, A. E. Mark, S. J. Marrink, Electrophoretic mobility does not always reflect the charge on an oil droplet, *Journal of Colloid and Interface Science* 318 (2008) 477–486.
 - [72] I. Pagonabarraga, B. Rotenberg, D. Frenkel, Recent advances in the modelling and simulation of electrokinetic effects: bridging the gap between atomistic and macroscopic descriptions, *Physical Chemistry Chemical Physics* 12 (2010) 9566.
 - [73] J. Lyklema, Molecular interpretation of electrokinetic potentials, *Current Opinion in Colloid & Interface Science* 15 (2010) 125–130.
 - [74] B. Rotenberg, I. Pagonabarraga, Electrokinetics: insights from simulation on the microscopic scale, *Molecular Physics* 111 (2013) 827–842.
 - [75] H. Yoshida, H. Mizuno, T. Kinjo, H. Washizu, J.-L. Barrat, Molecular dynamics simulation of electrokinetic flow of an aqueous electrolyte solution in nanochannels, *The Journal of Chemical Physics* 140 (2014) 214701.
 - [76] S. Dewan, V. Carnevale, A. Bankura, A. Eftekhari-Bafrooei, G. Fiorin, M. L. Klein, E. Borguet, Structure of Water at Charged Interfaces: A Molecular Dynamics Study, *Langmuir* 30 (2014) 8056–8065.
 - [77] B. Lowe, C.-K. Skylaris, N. Green, Y. Shibuta, T. Sakata, Calculation of surface potentials at the silica-water interface using molecular dynamics: Challenges and opportunities, *Japanese Journal of Applied Physics* 57 (2018) 04FM02.
 - [78] Y. Uematsu, R. R. Netz, D. J. Bonhuis, Analytical interfacial layer model for the capacitance and electrokinetics of charged aqueous interfaces, *Langmuir* (2018) acs.langmuir.7b04171.
 - [79] Y. Nagata, T. Ohto, E. H. G. Backus, M. Bonn, Molecular Modeling of Water Interfaces: From Molecular Spectroscopy to Thermodynamics, *The Journal of Physical Chemistry B* 120 (2016) 3785–3796.
 - [80] J.-F. Dufrêche, V. Marry, O. Bernard, P. Turq, Models for electrokinetic phenomena in montmorillonite, *Coll. Surf. A* 195 (2001) 171–180.
 - [81] D. Horinek, R. R. Netz, Specific Ion Adsorption at Hydrophobic Solid Surfaces, *Physical Review Letters* 99 (2007) 226104.
 - [82] D. Huang, C. Cottin-Bizonne, C. Ybert, L. Bocquet, Ion-Specific Anomalous Electrokinetic Effects in Hydrophobic Nanochannels, *Physical Review Letters* 98 (2007) 177801.
 - [83] D. M. Huang, C. Cottin-Bizonne, C. Ybert, L. Bocquet, Aqueous Electrolytes near Hydrophobic Surfaces: Dynamic Effects of Ion Specificity and Hydrodynamic Slip, *Langmuir* 24 (2008) 1442–1450.
 - [84] D. Ben-Yaakov, D. Andelman, R. Podgornik, D. Harries, Ion-specific hydration effects: Extending the Poisson-Boltzmann theory, *Current Opinion in Colloid & Interface Science* 16 (2011) 542–550.
 - [85] C. Calero, J. Faraudo, D. Bastos-González, Interaction of Monovalent Ions with Hydrophobic and Hydrophilic Colloids: Charge Inversion and Ionic Specificity, *Journal of the American Chemical Society* 133 (2011) 15025–15035.
 - [86] P.-A. Cazade, R. Hartkamp, B. Coasne, Structure and Dynamics of an Electrolyte Confined in Charged Nanopores, *The Journal of Physical Chemistry C* 118 (2014) 5061–5072.
 - [87] R. Hartkamp, B. Siboulet, J.-F. Dufrêche, B. Coasne, Ion-specific adsorption and electroosmosis in charged amorphous porous silica, *Phys. Chem. Chem. Phys.* 17 (2015) 24683–24695.
 - [88] S. Hocine, R. Hartkamp, B. Siboulet, M. Duvail, B. Coasne, P. Turq, J.-F. Dufrêche, How Ion Condensation Occurs at a Charged Surface: A Molecular Dynamics Investigation of the Stern Layer for Water/Silica Interfaces, *J. Phys. Chem. C* 120 (2016) 963–973.
 - [89] Y. Uematsu, R. R. Netz, D. J. Bonhuis, The effects of ion adsorption on the potential of zero charge and the differential capacitance of charged aqueous interfaces, *Journal of Physics: Condensed Matter* 30 (2018) 064002.
 - [90] A. Grosberg, T. Nguyen, B. Shklovskii, Colloquium: the physics of charge inversion in chemical and biological systems, *Reviews of modern physics* 74 (2002) 329.

- [91] C. D. Lorenz, A. Traveset, Charge inversion of divalent ionic solutions in silica channels, *Physical Review E* 75 (2007) 061202.
- [92] C. D. Lorenz, P. S. Crozier, J. A. Anderson, A. Traveset, Molecular Dynamics of Ionic Transport and Electrokinetic Effects in Realistic Silica Channels, *The Journal of Physical Chemistry C* 112 (2008) 10222–10232.
- [93] M. Jardat J.-F. Dufr che V. Marry B. rotenberg P. Turq, Salt exclusion in charged porous media: a coarse-graining strategy in the case of montmorillonite clays, *Phys. Chem. Chem. Phys.* 11 (2009) 2023.
- [94] B. Siboulet, S. Hocine, R. Hartkamp, J.-F. Dufr che, Scrutinizing Electro-Osmosis and Surface Conductivity with Molecular Dynamics, *The Journal of Physical Chemistry C* 121 (2017) 6756–6769.
- [95] D. Bonthuis, S. Gekle, R. Netz, Dielectric Profile of Interfacial Water and its Effect on Double-Layer Capacitance, *Physical Review Letters* 107 (2011) 1–5.
- [96] H. Zhu, A. Ghoufi, A. Szymczyk, B. Balanec, D. Morineau, Anomalous Dielectric Behavior of Nanoconfined Electrolytic Solutions, *Physical Review Letters* 109 (2012) 107801.
- [97] D. J. Bonthuis, R. R. Netz, Beyond the Continuum: How Molecular Solvent Structure Affects Electrostatics and Hydrodynamics at Solid-Electrolyte Interfaces, *The Journal of Physical Chemistry B* 117 (2013) 11397–11413.
- [98] C. Zhang, F. Gygi, G. Galli, Strongly Anisotropic Dielectric Relaxation of Water at the Nanoscale, *The Journal of Physical Chemistry Letters* 4 (2013) 2477–2481.
- [99] S. Parez, M. P redota, M. Machesky, Dielectric Properties of Water at Rutile and Graphite Surfaces: Effect of Molecular Structure, *The Journal of Physical Chemistry C* 118 (2014) 4818–4834.
- [100] S. De Luca, S. K. Kannam, B. D. Todd, F. Frascoli, J. S. Hansen, P. J. Davis, Effects of Confinement on the Dielectric Response of Water Extends up to Mesoscale Dimensions, *Langmuir* 32 (2016) 4765–4773.
- [101] L. Bocquet, E. Charlaix, Nanofluidics, from bulk to interfaces, *Chem. Soc. Rev.* 39 (2010) 1073–1095.
- [102] J. S. Hansen, J. C. Dyre, P. Davis, B. D. Todd, H. Bruus, Continuum Nanofluidics, *Langmuir* 31 (2015) 13275–13289.
- [103] L. Bocquet, J.-L. Barrat, Flow boundary conditions from nano- to micro-scales, *Soft Matter* 3 (2007) 685–693.
- [104] V. Marry J.-F. Dufr che M. Jardat P. Turq, Equilibrium and electrokinetic phenomena in charged porous media from microscopic and mesoscopic models: electro-osmosis in montmorillonite, *Mol. Phys.* 101 (2003) 3111.
- [105] L. Joly, C. Ybert, E. Trizac, L. Bocquet, Hydrodynamics within the electric double layer on slipping surfaces, *Phys. Rev. Lett.* 93 (2004) 257805.
- [106] J.-F. Dufr che V. Marry N. Malikova P. Turq, Molecular hydrodynamics for electro-osmosis in clays: from kubo to smoluchowski, *J. Mol. Liq* 118 (2005) 145.
- [107] L. Joly, C. Ybert, E. Trizac, L. Bocquet, Liquid friction on charged surfaces: from hydrodynamic slippage to electrokinetics., *The Journal of Chemical Physics* 125 (2006) 204716.
- [108] A. Botan, V. Marry, B. Rotenberg, P. Turq, B. Noetinger, How Electrostatics Influences Hydrodynamic Boundary Conditions: Poiseuille and Electro-osmotic Flows in Clay Nanopores., *The Journal of Physical Chemistry C* 117 (2013) 978–985.
- [109] D. Jing, B. Bhushan, The coupling of surface charge and boundary slip at the solidliquid interface and their combined effect on fluid drag: A review, *Journal of Colloid and Interface Science* 454 (2015) 152–179.
- [110] V. M. Muller, I. P. Sergeeva, V. D. Sobolev, N. V. Churaev, Boundary effects in the theory of electrokinetic phenomena, *Colloid Journal of the USSR* 48 (1986) 606–614.
- [111] H. A. Stone, A. D. Stroock, A. Ajdari, Engineering Flows in Small Devices: Microfluidics Toward a Lab-on-a-Chip, *Annual Review of Fluid Mechanics* 36 (2004) 381–411.
- [112] C. I. Bouzigues, P. Tabeling, L. Bocquet, Nanofluidics in the Debye layer at hydrophilic and hydrophobic surfaces, *Phys. Rev. Lett.* 101 (2008) 114503.
- [113] L. Joly, F. Detcheverry, A.-L. Biance, Anomalous ζ potential in foam films, *Physical Review Letters* 113 (2014) 088301.
- [114] A. Barbosa De Lima, L. Joly, Electro-osmosis at surfactant-laden liquidgas interfaces: beyond standard models, *Soft Matter* 13 (2017) 3341–3351.
- [115] A. Ajdari, L. Bocquet, Giant Amplification of Interfacially Driven Transport by Hydrodynamic Slip: Diffusio-Osmosis and Beyond, *Physical Review Letters* 96 (2006) 186102.
- [116] J. Morthomas, A. W rger, Thermophoresis at a charged surface: the role of hydrodynamic slip, *J. Phys.: Condens. Matter* 21 (2009) 035103.
- [117] L. Fu, S. Merabia, L. Joly, What Controls Thermo-osmosis? Molecular Simulations Show the Critical Role of Interfacial Hydrodynamics, *Physical Review Letters* 119 (2017) 214501.
- [118] C. Lee, C. Cottin-Bizonne, R. Fulcrand, L. Joly, C. Ybert, Nanoscale Dynamics versus Surface Interactions: What Dictates Osmotic Transport?, *The Journal of Physical Chemistry Letters* 8 (2017) 478–483.
- [119] H. Zhang, A. a. Hassanali, Y. K. Shin, C. Knight, S. J. Singer, The water-amorphous silica interface: analysis of the Stern layer and surface conduction., *The Journal of Chemical Physics* 134 (2011) 024705.
- [120] M. P redota, M. L. Machesky, D. J. Wesolowski, Molecular Origins of the Zeta Potential, *Langmuir* 32 (2016) 10189–10198.
- [121] M. P redota, P. T. Cummings, D. J. Wesolowski, Electric Double Layer at the Rutile (110) Surface. 3. Inhomogeneous Viscosity and Diffusivity Measurement by Computer Simulations, *The Journal of Physical Chemistry C* 111 (2007) 3071–3079.
- [122] V. Knecht, B. Klasczyk, R. Dimova, Macro- versus Microscopic View on the Electrokinetics of a Water-Membrane Interface, *Langmuir* 29 (2013) 7939–7948.
- [123] Y. Uematsu, R. R. Netz, D. J. Bonthuis, Power-law electrokinetic behavior as a direct probe of effective surface viscosity, *Chemical Physics Letters* 670 (2017) 11–15.
- [124] B. D. Todd, J. S. Hansen, P. J. Davis, Nonlocal Shear Stress for Homogeneous Fluids, *Physical Review Letters* 100 (2008) 195901.
- [125] M. Bouhadja, A. A. Skelton, Dynamical Properties of Water and Ions at the Quartz (101)-Water Interface at a Range of Solution Conditions: A Classical Molecular Dynamics Study, *J. Phys. Chem. C* 122 (2018) 1535–1546.
- [126] J. Lyklema, S. Rovillard, J. De Coninck, Electrokinetics: The Properties of the Stagnant Layer Unraveled, *Langmuir* 14 (1998) 5659–5663.
- [127] J. Happel, H. Brenner, Low Reynolds number hydrodynamics: with special applications to particulate media, volume 1, Springer Science & Business Media, 2012.
- [128] Y. Marcus, Effect of Ions on the Structure of Water : Structure Making and Breaking Effect of Ions on the Structure of Water : Structure Making and Breaking, *Chemical Reviews* 109 (2009) 1346–1370.
- [129] J. S. Kim, Z. Wu, A. R. Morrow, A. Yethiraj, Self-Diffusion and Viscosity in Electrolyte Solutions, *The Journal of Physical Chemistry B* 116 (2012) 12007–12013.
- [130] Y. Ding, A. a. Hassanali, M. Parrinello, Anomalous water diffusion in salt solutions., *Proceedings of the National Academy of Sciences of the United States of America* 111 (2014) 3310–5.
- [131] Z. R. Kann, J. L. Skinner, A scaled-ionic-charge simulation model that reproduces enhanced and suppressed water diffusion in aqueous salt solutions, *The Journal of Chemical Physics* 141 (2014) 104507.
- [132] Y. Yao, Y. Kanai, M. L. Berkowitz, Role of Charge Transfer in Water Diffusivity in Aqueous Ionic Solutions, *The Journal of Physical Chemistry Letters* 5 (2014) 2711–2716.
- [133] Y. Yao, M. L. Berkowitz, Y. Kanai, Communication: Modeling of concentration dependent water diffusivity in ionic solutions: Role of intermolecular charge transfer, *The Journal of Chemical Physics* 143 (2015) 241101.
- [134] J. Li, F. Wang, Pairwise-additive force fields for selected aqueous monovalent ions from adaptive force matching, *The Journal of Chemical Physics* 143 (2015) 194505.
- [135] A. C. T. van Duin, S. Dasgupta, F. Lorant, W. A. Goddard, ReaxFF: A Reactive Force Field for Hydrocarbons, *J. Phys. Chem. A* 105 (2001) 9396–9409.
- [136] D. Raymand, A. C. van Duin, W. A. Goddard, K. Hermansson, D. Sp ngberg, Hydroxylation Structure and Proton Transfer Reactivity at the Zinc Oxide/Water Interface, *The Journal of Physical Chemistry C* 115 (2011) 8573–8579.
- [137] S.-Y. Kim, N. Kumar, P. Persson, J. Sofo, A. C. T. van Duin, J. D. Kubicki, Development of a ReaxFF Reactive Force Field for Titanium Dioxide/Water Systems, *Langmuir* 29 (2013) 7838–7846.
- [138] T. P. Senftle, S. Hong, M. M. Islam, S. B. Kylasa, Y. Zheng, Y. K. Shin, C. Junkermeier, R. Engel-Herbert, M. J. Janik, H. M. Aktulga, T. Ver-

- straalen, A. Grama, A. C. T. van Duin, The ReaxFF reactive force-field: development, applications and future directions, *npj Computational Materials* 2 (2016) 15011.
- [139] M. J. Gillan, D. Alfè, A. Michaelides, Perspective: How good is DFT for water?, *Journal of Chemical Physics* 144 (2016) 130901.
- [140] Y. Nagata, T. Ohto, M. Bonn, T. D. Kühne, Surface tension of ab initio liquid water at the water-air interface, *The Journal of Chemical Physics* 144 (2016) 204705.
- [141] M. Chen, H.-Y. Ko, R. C. Remsing, M. F. Calegari Andrade, B. Santra, Z. Sun, A. Selloni, R. Car, M. L. Klein, J. P. Perdew, X. Wu, Ab initio theory and modeling of water, *Proceedings of the National Academy of Sciences* 114 (2017) 10846–10851.
- [142] T. Markovich, D. Andelman, R. Podgornik, Charge regulation: a generalized boundary condition?, *Epl* 113 (2016) 1–6.
- [143] G. Trefalt, S. H. Behrens, M. Borkovec, Charge Regulation in the Electrical Double Layer: Ion Adsorption and Surface Interactions, *Langmuir* 32 (2016) 380–400.
- [144] P. Masini, M. Bernasconi, Ab initio simulations of hydroxylation and dehydroxylation reactions at surfaces: amorphous silica and brucite, *Journal of Physics: Condensed Matter* 14 (2002) 4133–4144.
- [145] L. J. Criscenti, J. D. Kubicki, S. L. Brantley, Silicate glass and mineral dissolution: Calculated reaction paths and activation energies for hydrolysis of a Q 3 Si by H₃O⁺ using ab initio methods, *Journal of Physical Chemistry A* 110 (2006) 198–206.
- [146] F. Tielens, C. Gervais, J. F. Lambert, F. Mauri, D. Costa, Ab Initio Study of the Hydroxylated Surface of Amorphous Silica: A Representative Model, *Chemistry of Materials* 20 (2008) 3336–3344.
- [147] N. Kumar, S. Neogi, P. R. C. Kent, A. V. Bandura, J. D. Kubicki, D. J. Wesolowski, D. Cole, J. O. Sofo, Hydrogen Bonds and Vibrations of Water on (110) Rutile, *The Journal of Physical Chemistry C* 113 (2009) 13732–13740.
- [148] C. Liu, Z. Li, Molecular Dynamics Simulation of Composite Nanochannels as Nanopumps Driven by Symmetric Temperature Gradients, *Physical Review Letters* 105 (2010) 1–4.
- [149] A. Cimas, F. Tielens, M. Sulpizi, M.-P. Gaigeot, D. Costa, The amorphous silicaliquid water interface studied by ab initio molecular dynamics (AIMD): local organization in global disorder, *Journal of Physics: Condensed Matter* 26 (2014) 244106.
- [150] M.-P. Gaigeot, M. Sprik, M. Sulpizi, Oxide/water interfaces: how the surface chemistry modifies interfacial water properties, *Journal of Physics: Condensed Matter* 24 (2012) 124106.
- [151] M. Sulpizi, M.-P. Gaigeot, M. Sprik, The Silica/Water Interface: How the Silanols Determine the Surface Acidity and Modulate the Water Properties, *Journal of Chemical Theory and Computation* 8 (2012) 1037–1047.
- [152] S. Tazi, B. Rotenberg, M. Salanne, M. Sprik, M. Sulpizi, Absolute acidity of clay edge sites from ab-initio simulations, *Geochimica et Cosmochimica Acta* 94 (2012) 1–11.
- [153] D. J. Wesolowski, J. O. Sofo, A. V. Bandura, Z. Zhang, E. Mamontov, M. Předota, N. Kumar, J. D. Kubicki, P. R. C. Kent, L. Vlcek, M. L. Machesky, P. A. Fenter, P. T. Cummings, L. M. Anovitz, A. A. Skelton, J. Rosenqvist, Comment on "Structure and dynamics of liquid water on rutile TiO₂ (110)", *Physical Review B* 85 (2012) 167401.
- [154] G. Tocci, A. Michaelides, Solvent-Induced Proton Hopping at a Water/Oxide Interface, *The Journal of Physical Chemistry Letters* 5 (2014) 474–480.
- [155] S. Laporte, F. Finocchi, L. Paulatto, M. Blanchard, E. Balan, F. Guyot, A. M. Saitta, Strong electric fields at a prototypical oxide/water interface probed by ab initio molecular dynamics: MgO(001), *Phys. Chem. Chem. Phys.* 17 (2015) 20382–20390.
- [156] M. Pfeiffer-Laplaud, M.-P. Gaigeot, Adsorption of Singly Charged Ions at the Hydroxylated (0001) α -Quartz/Water Interface, *The Journal of Physical Chemistry C* 120 (2016) 4866–4880.
- [157] M. Pfeiffer-Laplaud, M.-P. Gaigeot, Electrolytes at the Hydroxylated (0001) α -Quartz/Water Interface: Location and Structural Effects on Interfacial Silanols by DFT-Based MD, *The Journal of Physical Chemistry C* 120 (2016) 14034–14047.
- [158] M. J. DelloStritto, J. D. Kubicki, J. O. Sofo, Effect of Ions on H-Bond Structure and Dynamics at the Quartz(101)-Water Interface, *Langmuir* 32 (2016) 11353–11365.
- [159] R. Réocreux, T. Jiang, M. Iannuzzi, C. Michel, P. Sautet, Structuration and Dynamics of Interfacial Liquid Water at Hydrated γ -Alumina Determined by ab Initio Molecular Simulations: Implications for Nanoparticle Stability, *ACS Applied Nano Materials* 1 (2018) 191–199.
- [160] C. Zhang, J. Wu, G. Galli, F. Gygi, Structural and vibrational properties of liquid water from van der Waals density functionals, *Journal of Chemical Theory and Computation* 7 (2011) 3054–3061.
- [161] Q. Wan, L. Spanu, G. A. Galli, F. Gygi, Raman Spectra of Liquid Water from Ab Initio Molecular Dynamics: Vibrational Signatures of Charge Fluctuations in the Hydrogen Bond Network, *Journal of Chemical Theory and Computation* 9 (2013) 4124–4130.
- [162] M. Thomas, M. Brehm, R. Fligg, P. Vöhringer, B. Kirchner, Computing vibrational spectra from ab initio molecular dynamics, *Physical Chemistry Chemical Physics* 15 (2013) 6608.
- [163] M. Sulpizi, M. Salanne, M. Sprik, M.-P. Gaigeot, Vibrational Sum Frequency Generation Spectroscopy of the Water Liquid/Vapor Interface from Density Functional Theory-Based Molecular Dynamics Simulations, *The Journal of Physical Chemistry Letters* 4 (2013) 83–87.
- [164] R. Khatib, E. H. G. Backus, M. Bonn, M.-J. Perez-Haro, M.-P. Gaigeot, M. Sulpizi, Water orientation and hydrogen-bond structure at the fluoride/water interface, *Scientific Reports* 6 (2016) 24287.
- [165] G. Tocci, L. Joly, A. Michaelides, Friction of water on graphene and hexagonal boron nitride from ab initio methods: very different slippage despite very similar interface structures, *Nano Lett.* 14 (2014) 6872–6877.
- [166] L. Joly, G. Tocci, S. Merabia, A. Michaelides, Strong Coupling between Nanofluidic Transport and Interfacial Chemistry: How Defect Reactivity Controls Liquid/Solid Friction through Hydrogen Bonding, *The Journal of Physical Chemistry Letters* 7 (2016) 1381–1386.
- [167] D. M. Heyes, Molecular dynamics simulations of restricted primitive model 1:1 electrolytes, *Chem. Phys.* 69 (1982) 155–163.
- [168] B. Grosjean, C. Pean, A. Siria, L. Bocquet, R. Vuilleumier, M.-L. Bocquet, Chemisorption of Hydroxide on 2D Materials from DFT Calculations: Graphene versus Hexagonal Boron Nitride, *The Journal of Physical Chemistry Letters* 7 (2016) 4695–4700.
- [169] J. W. Lee, J. A. Templeton, K. K. Mandadapu, J. A. Zimmerman, Comparison of molecular and primitive solvent models for electrical double layers in nanochannels, *J. Chem. Theory Comp.* 9 (2013) 3051–3061.
- [170] R. Vangara, D. Brown, F. van Swol, D. Petsev, Electrolyte solution structure and its effect on the properties of electric double layers with surface charge regulation, *J. Colloid Interface Sci.* 488 (2017) 180–189.
- [171] A. M. Darlington, T. A. Jarisz, E. L. DeWalt-Kerian, S. Roy, S. Kim, M. S. Azam, D. K. Hore, J. M. Gibbs, Separating the pH-Dependent Behavior of Water in the Stern and Diffuse Layers with Varying Salt Concentration, *J. Phys. Chem. C* 121 (2017) 20229–20241.
- [172] E. L. DeWalt-Kerian, S. Kim, M. S. Azam, H. Zeng, Q. Liu, J. M. Gibbs, pH-Dependent Inversion of Hofmeister Trends in the Water Structure of the Electrical Double Layer, *J. Phys. Chem. Lett.* 8 (2017) 2855–2861.
- [173] H. Chinen, K. Mawatari, Y. Pihosh, K. Morikawa, Y. Kazoe, T. Tsukahara, T. Kitamori, Enhancement of Proton Mobility in Extended-Nanospace Channels, *Angew. Chemie Int. Ed.* 51 (2012) 3573–3577.
- [174] M. Předota, A. V. Bandura, P. T. Cummings, J. D. Kubicki, D. J. Wesolowski, A. A. Chialvo, M. L. Machesky, Electric Double Layer at the Rutile (110) Surface. 1. Structure of Surfaces and Interfacial Water from Molecular Dynamics by Use of ab Initio Potentials, *The Journal of Physical Chemistry B* 108 (2004) 12049–12060.
- [175] M. Masia, Ab initio based polarizable force field parametrization, *J. Chem. Phys.* 128 (2008) 184107.
- [176] M. L. Machesky, M. Předota, D. J. Wesolowski, L. Vlcek, P. T. Cummings, J. Rosenqvist, M. K. Ridley, J. D. Kubicki, A. V. Bandura, N. Kumar, J. O. Sofo, Surface Protonation at the Rutile (110) Interface: Explicit Incorporation of Solvation Structure within the Refined MUSIC Model Framework, *Langmuir* 24 (2008) 12331–12339.
- [177] S. Tazi, J. J. Molina, B. Rotenberg, P. Turq, R. Vuilleumier, M. Salanne, A transferable ab initio based force field for aqueous ions., *The Journal of Chemical Physics* 136 (2012) 114507.
- [178] A. Butenuth, G. Moras, J. Schneider, M. Koleini, S. Köppen, R. Meißner, L. B. Wright, T. R. Walsh, L. C. Ciacchi, Ab initio derived force-field parameters for molecular dynamics simulations of deprotonated amorphous-SiO₂/water interfaces, *Phys. Status Solidi Basic Res.* 249 (2012) 292–305.
- [179] M. A. Brown, A. Goel, Z. Abbas, Effect of Electrolyte Concentration on the Stern Layer Thickness at a Charged Interface, *Angew. Chemie - Int.*

- Ed. 55 (2016) 3790–3794.
- [180] M. Předota, Z. Zhang, P. Fenter, D. J. Wesolowski, P. T. Cummings, Electric Double Layer at the Rutile (110) Surface. 2. Adsorption of Ions from Molecular Dynamics and X-ray Experiments, *The Journal of Physical Chemistry B* 108 (2004) 12061–12072.
 - [181] C. Labbez, B. Jonsson, I. Pochard, A. Nonat, B. Cabane, Surface charge density and electrokinetic potential of highly charged minerals: Experiments and monte carlo simulations on calcium silicate hydrate, *J. Phys. Chem. B* 110 (2006) 9219–9230.
 - [182] C. Labbez, A. Nonat, I. Pochard, B. Jönsson, Experimental and theoretical evidence of overcharging of calcium silicate hydrate, *J. Colloid Interface Sci.* 309 (2007) 303–307.
 - [183] C. Labbez, I. Pochard, B. Jönsson, A. Nonat, C-S-H/solution interface: Experimental and Monte Carlo studies, *Cem. Concr. Res.* 41 (2011) 161–168.
 - [184] Y. Nagata, R. E. Pool, E. H. Backus, M. Bonn, Nuclear quantum effects affect bond orientation of water at the water-vapor interface, *Physical Review Letters* 109 (2012) 1–5.
 - [185] I. Semenov, S. Raafatnia, M. Sega, V. Lobaskin, C. Holm, F. Kremer, Electrophoretic mobility and charge inversion of a colloidal particle studied by single-colloid electrophoresis and molecular dynamics simulations, *Phys. Rev. E - Stat. Nonlinear, Soft Matter Phys.* 87 (2013) 1–7.
 - [186] M. Predota, M. L. Machesky, D. J. Wesolowski, P. T. Cummings, Electric double layer at the rutile (110) surface. 4. effect of temperature and pH on the adsorption and dynamics of ions, *J. Phys. Chem. C* 117 (2013) 22852–22866.
 - [187] A. J. Page, A. Elbourne, R. Stefanovic, M. a. Addicoat, G. G. Warr, K. Voitchovsky, R. Atkin, 3-Dimensional atomic scale structure of the ionic liquidgraphite interface elucidated by AM-AFM and quantum chemical simulations, *Nanoscale* 6 (2014) 8100–8106.
 - [188] Z. Brkljača, M. Klimczak, Z. Miličević, M. Weisser, N. Taccardi, P. Wasserscheid, D. M. Smith, A. Magerl, A. S. Smith, Complementary molecular dynamics and x-ray reflectivity study of an imidazolium-based ionic liquid at a neutral sapphire interface, *J. Phys. Chem. Lett.* 6 (2015) 549–555.
 - [189] H. Hussain, G. Tocci, T. Woolcot, X. Torrelles, C. L. Pang, D. S. Humphrey, C. M. Yim, D. C. Grinter, G. Cabailh, O. Bikondoa, R. Lindsay, J. Zegenhagen, A. Michaelides, G. Thornton, Structure of a model TiO₂ photocatalytic interface, *Nature Materials* 16 (2016) 461–466.
 - [190] I. C. Bourg, S. S. Lee, P. Fenter, C. Tournassat, Stern Layer Structure and Energetics at Mica-Water Interfaces, *J. Phys. Chem. C* 121 (2017) 9402–9412.
 - [191] K. A. Perrine, K. M. Parry, A. C. Stern, M. H. C. Van Spyk, M. J. Makowski, J. A. Freites, B. Winter, D. J. Tobias, J. C. Hemminger, Specific cation effects at aqueous solutionvapor interfaces: Surfactant-like behavior of Li⁺ revealed by experiments and simulations, *Proceedings of the National Academy of Sciences* 114 (2017) 13363–13368.
 - [192] Z. Ge, Y. Wang, Estimation of Nanodiamond Surface Charge Density from Zeta Potential and Molecular Dynamics Simulations, *J. Phys. Chem. B* 121 (2017) 3394–3402.
 - [193] M. D. Baer, C. J. Mundy, Toward an Understanding of the Specific Ion Effect Using Density Functional Theory, *The Journal of Physical Chemistry Letters* 2 (2011) 1088–1093.
 - [194] Y. Yao, Y. Kanai, Free Energy Profile of NaCl in Water: First-Principles Molecular Dynamics with SCAN and ω B97X-V ExchangeCorrelation Functionals, *Journal of Chemical Theory and Computation* 14 (2018) 884–893.
 - [195] M. Gastegger, J. Behler, P. Marquetand, Machine learning molecular dynamics for the simulation of infrared spectra, *Chem. Sci.* 8 (2017) 6924–6935.
 - [196] S. Riniker, Molecular Dynamics Fingerprints (MDFP): Machine Learning from MD Data To Predict Free-Energy Differences, *J. Chem. Inf. Model.* 57 (2017) 726–741.
 - [197] Y.-H. Tang, D. Zhang, G. E. Karniadakis, An atomistic fingerprint algorithm for learning ab initio molecular force fields, *J. Chem. Phys.* 148 (2018) 034101.
 - [198] G. A. Cisneros, K. T. Wikfeldt, L. Ojamäe, J. Lu, Y. Xu, H. Torabifard, A. P. Bartók, G. Csányi, V. Molinero, F. Paesani, Modeling Molecular Interactions in Water: From Pairwise to Many-Body Potential Energy Functions, *Chemical Reviews* 116 (2016) 7501–7528.
 - [199] T. E. Markland, M. Ceriotti, Nuclear quantum effects enter the mainstream, *Nature Reviews Chemistry* 2 (2018) 0109.
 - [200] Y. Litman, D. Donadio, M. Ceriotti, M. Rossi, Decisive role of nuclear quantum effects on surface mediated water dissociation at finite temperature, *The Journal of Chemical Physics* 148 (2018) 102320.

JGR Solid Earth

RESEARCH ARTICLE

10.1029/2019JB017679

Key Points:

- We investigate Maxwell-Wagner, Stern-layer and diffuse-layer, and membrane polarization in porous media in a generalized mathematical framework
- Numerical and analytical solutions are mutually verified, and existing analytical models of the four mechanisms are improved
- Low-frequency Stern-layer polarization is stronger than diffuse-layer polarization if the electrical double layer is discontinuous, only

Correspondence to:

M. Bucker,
 m.buecker@tu-bs.de

Citation:

Bücker, M., Flores Orozco, A., Undorf, S., & Kemna, A. (2019). On the role of Stern- and diffuse-layer polarization mechanisms in porous media. *Journal of Geophysical Research: Solid Earth*, 124, 5656–5677. <https://doi.org/10.1029/2019JB017679>

Received 12 MAR 2019

Accepted 24 MAY 2019

Accepted article online 31 MAY 2019

Published online 26 JUN 2019

On the Role of Stern- and Diffuse-Layer Polarization Mechanisms in Porous Media

Matthias Bucker^{1,2,3} , Adrián Flores Orozco² , Sabine Undorf^{3,4} , and Andreas Kemna³ 

¹Institute for Geophysics and extraterrestrial Physics, TU Braunschweig, Braunschweig, Germany, ²Department of Geodesy and Geoinformation, Research Group Geophysics, TU Wien, Vienna, Austria, ³Institute of Geosciences, Geophysics Section, Rheinische Friedrich-Wilhelms-Universität Bonn, Bonn, Germany, ⁴School of GeoSciences, University of Edinburgh, Edinburgh, UK

Abstract Water-saturated porous media exhibit a low-frequency (<1 MHz) dispersion of the electrical conductivity caused by the polarization of the electrical double layer (EDL) coating the charged solid-liquid interface. We develop a mathematical framework describing the polarization caused by field-induced perturbations of the ion densities in the Stern and the diffuse layer of the EDL for two different geometrical configurations of solid and liquid phase. For spherical grains immersed in an electrolyte we derive an improved analytical description by combining suitable models for diffuse- and Stern-layer polarization. The selected models differ from those usually used in geophysical literature and improve the agreement with the corresponding finite-element solution significantly. We then employ the validated finite-element model to examine the EDL in a pore-constriction geometry, which is often used to study membrane polarization. Here, a suitable analytical model can only be set up for a pure diffuse-layer polarization. The results for the coupled Stern- and diffuse-layer polarization in both geometries indicate that (1) the polarization of the Stern layer is much stronger than the polarization of the diffuse layer as long as the EDL is not connected at the system scale; (2) this dominance of the Stern-layer polarization can be observed in both geometries, but (3) the contribution of the diffuse layer increases with increasing compaction as represented by the pore-constriction geometry; and (4) the contributions of both parts of the EDL reach similar levels, when the EDLs on different surfaces are interconnected at the system scale.

1. Introduction

Complex-conductivity measurements assess the frequency-dependent electrical conduction and polarization properties of soil and subsurface materials. Because macroscopic measurements are correlated to pore and/or grain geometry, electrochemical characteristics of the solid-liquid interface, and of the pore-filling electrolyte, the method has a huge potential for novel hydrogeophysical (e.g., Börner et al., 1996; Hördt et al., 2009) and biogeophysical (e.g., Atekwana & Slater, 2009; Flores Orozco et al., 2011; Wainwright et al., 2015) applications. Although empirical relations based on these correlations are increasingly used in near-surface studies, there is no widely accepted model linking macroscopic polarization effects to microscopic properties (e.g., Kemna et al., 2012). In absence of conductive minerals, four mechanisms contribute to the low-frequency (<1 MHz) dispersion of complex conductivity, all of which are sensitive to the polarization of different parts of the electrical double layer (EDL) at the solid-liquid interface (e.g., Lesmes & Morgan, 2001): (1) Maxwell-Wagner polarization, (2) polarization of the Stern layer, (3) polarization of the diffuse layer, and (4) membrane polarization.

Maxwell (1892) and Wagner (1914) studied interfacial polarization in heterogeneous media consisting of two or more phases of different electrical conductivity and/or dielectric constant: unequal conduction and displacement current densities in the different phases are balanced by an accumulation of charge along the geometrical boundaries. O'Konski (1960) extended the Maxwell-Wagner theory to include the polarization of charges bound to the surface of charged particles. Later, Garcia et al. (1985) treated the case of uncharged dielectric particles in electrolyte solution. In contrast to earlier theories, which assume homogeneous conductivities in both phases resulting in true surface charge distributions, their treatment accounts for local field-induced concentration variations in the electrolyte and thus volume charge of finite extension. The characteristic time scales of the different types of Maxwell-Wagner polarization are short, such that this relaxation usually occurs at the high-frequency limit (kHz) of the complex-conductivity response (e.g., Leroy et al., 2008; Lesmes & Morgan, 2001).

A first theory describing the polarization of the Stern layer was presented by Schwarz (1962) and improved by Schurr (1964). Schwarz' theory accounts for the diffusion-controlled polarization of the Stern layer of bound counterions. Schurr combined this model with the one by O'Konski to include the effect of surface conductivity due to the diffuse layer. Because the contribution of the diffuse layer is frequency independent in this model, the dispersion of the complex conductivity is attributed to Stern-layer polarization, and the diffuse layer only results in a uniform increase of the real part of the conductivity. Later, Schurr's model provided the basis for the treatment by Leroy et al. (2008), which includes a detailed electrochemical model to quantify the partition of charges into Stern and diffuse layer (see, e.g., Revil & Glover, 1997, 1998) and also accounts for a grain-size distribution. This model successfully predicted the response of glass beads (Leroy et al., 2008) and was extended to sand-oil-water mixtures later (Schmutz et al., 2010).

The polarization of the diffuse layer has been studied by Dukhin and Shilov (1974), who developed a theory for the ion fluxes through thin diffuse layers, which induce concentration variations in the diffuse layer and the adjacent electrolyte. This concentration-polarization mechanism has also been treated analytically by Fixman (1980), Chew and Sen (1982a), and Hinch et al. (1984) among others. A numerical solution removing many of the limitations of the analytical approaches—for example, the assumption of a thin diffuse layer compared to the particle radius, a sufficiently small surface potential, and a monovalent symmetric electrolyte—was presented by DeLacey and White (1981). Later, Shilov et al. (2001) included the effect of Maxwell-Wagner polarization into the classical Dukhin-Shilov theory yielding a good agreement with the numerical model by DeLacey and White (1981).

Different attempts have been made to determine the relative importance of Stern- and diffuse-layer polarization and to develop combined models. Lyklema et al. (1983) generalized the Schwarz-Schurr model by including the coupling of charges in the Stern layer to those in the diffuse layer, which mainly results in a decrease of the relaxation time scale. Leroy et al. (2017) applied this approach to model the polarization response of calcite precipitations on glass beads. de Lima and Sharma (1992) analyzed the models by Schwarz, Schurr, and Fixman separately in order to assess their relative contribution to the overall polarization response. By superposing the individual responses, Lesmes and Morgan (2001) developed a combined model considering all three polarization mechanisms, that is, Stern-/diffuse-layer and Maxwell-Wagner polarization. The authors also included a volume-averaging approach to study water-particle mixtures characterized by a grain-size distribution. Based on the work of Kijlstra et al. (1992), Shilov et al. (2001) modified the Dukhin-Shilov theory to account for the contribution of the Stern layer to surface conductivity.

Most theories describing the first three polarization mechanisms rest on the analytical solution of the underlying system of partial differential equations (PDEs) around one isolated spherical particle. The induced dipole moment of the polarized particle obtained from such treatment can then be used to derive the effective conductivity (or the effective dielectric constant) of ensembles of more than one particle. The effective medium theory provides mixing laws for dilute suspensions (e.g., Maxwell, 1892; Wagner, 1914) or mixtures with higher particle concentrations (e.g., Bruggeman, 1935; Hanai, 1960).

Membrane polarization is generally studied on pore networks with different levels of complexity. Marshall and Madden (1959) developed the first model for a sequence of two types of one-dimensional pores or zones. The membrane effect is introduced by assuming different mobilities for cations and anions in the "active" zone. While these authors do not further specify the origin of mobility variations, later developments related ion selectivity to the unequal contributions of cations and anions to the surface conductivity at the pore wall (Buchheim & Irmer, 1979; Fridrikhsberg & Sidorova, 1961; Titov et al., 2002, 2004). Blaschek and Hördt (2009) carried out numerical simulations on one- and two-dimensional pore networks, where the ion-selective behavior of narrow pores is still parameterized in terms of ion mobilities, which are constant over the pore cross section. Volkmann and Klitzsch (2010) improved this approach and limited the ion selectivity—expressed in terms of unequal effective ion mobilities—to a thin layer covering the pore walls.

Bücker and Hördt (2013a) proposed an analytical model, which allows to explicitly include pore radii and surface conductivity due to Stern and diffuse layer into the one-dimensional impedance model by Marshall and Madden. This model has later been extended to model the effect of temperature, fluid salinity, pH, and immiscible hydrocarbon contaminants on the polarization response (Bairlein et al., 2016; Bücker et al., 2017; Hördt et al., 2016). Based on the same model, Stebner et al. (2017) used impedance networks to model the membrane polarization of porous media. Because the model by Bücker and Hördt typically requires large

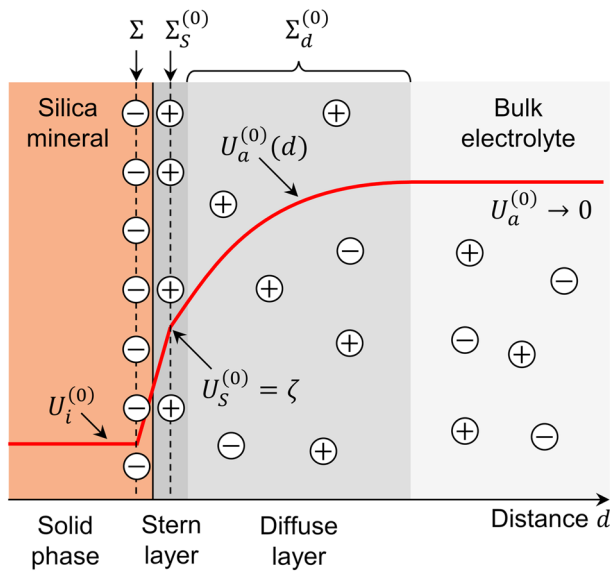


Figure 1. Simplified model of the equilibrium electrical double layer (EDL) at the charged silica surface. Due to the deprotonation of silanol surface sites, the mineral surface acquires the surface charge Σ . In the electrolyte, this usually negative charge is screened by an equal number of the positive charges distributed over the Stern layer ($\Sigma_S^{(0)}$) and the diffuse layer ($\Sigma_d^{(0)}$). The red line sketches the spatial variation of the electrical potential $U^{(0)}$ across the different parts of the electrical double layer.

aspect ratios, length of a pore divided by its diameter, to produce measurable polarization magnitudes, Hördt et al. (2017) further investigated into the geometrical constraints of this membrane-polarization model.

Besides geometrical constraints, the simplified consideration of the Stern layer is a major limitation of the existing theoretical treatment of membrane polarization. While combined treatments of Stern-layer, diffuse-layer, and Maxwell-Wagner polarization mechanisms exist for grain-based models, the coupling between these three mechanisms is often largely simplified in pore-constriction models. Irrespective from the model geometry, the complexity of the equations for the fully coupled system describing all three mechanisms puts a challenge on analytical solutions, which therefore all suffer from strong simplifications or only treat limiting cases.

The present paper addresses the repeatedly stated need for “a mechanistic approach and [...] general framework in which all these mechanisms are explained and quantified in their relative importance” (Kemna et al., 2012). In the theory section, we compile the mathematical descriptions of all relevant physical processes and provide a mathematical framework, which allows modeling the fully coupled interplay of Stern- and diffuse-layer polarization on arbitrary geometries. Subsequently, we derive an improved analytical approximation for the polarization of a single grain, which matches the results obtained from numerically solving the equations of the generalized mathematical framework. The validated numerical model is then applied to the pore-constriction geometry. Based on the comparison of numerical and analytical results, we improve the membrane-polarization model by Bückner and Hördt (2013a) and for the first time study the fully coupled Stern- and diffuse-layer polarization in a pore-constriction geometry.

2. Theory

2.1. EDL Model

Most solid surfaces in contact with aqueous solutions are charged. We will consider silica surfaces in contact with a monovalent electrolyte, such as NaCl, where the deprotonation of silanol surface sites produces a negative surface-charge density Σ over a wide pH range (e.g., Leroy et al., 2008; Somasundaran, 2006). In the electrolyte next to the silica surface, the electric field of Σ attracts counterions (cations, if $\Sigma < 0$) and repels coions (here anions) giving rise to the development of two layers: The inner layer—also known as Stern or Helmholtz layer—consists of counterions adsorbed to the silica surface. The outer layer—also known as diffuse or Gouy-Chapman layer—is mainly populated by counterions and a minor fraction of coions both obeying Poisson-Boltzmann statistics.

We adopt the simplified model displayed in Figure 1. The Stern layer is treated as an infinitely thin layer of counterions (e.g., Leroy et al., 2008; Schwarz, 1962; Schurr, 1964) and is characterized by the uniform surface charge density $|\Sigma_S^{(0)}| < |\Sigma|$ (the superscript (0) indicates quantities in the equilibrium state, that is, without external excitation), which partly shields the electric field of Σ . Furthermore, we assume that charges in the Stern layer can only move along the surface, which reflects a strong binding of counter-ions to the surface.

The movement of the ions in the diffuse layer is not restricted to the surface; that is, these ions can move in all directions. As a result of an equilibrium between simultaneously acting electrostatic forces and thermal fluctuations, counterion excess and coion deficit concentrations smoothly decay with the distance d from the surface. At a far distance from the surface, both ion concentrations approach their values in the bulk electrolyte. Together with $\Sigma_S^{(0)}$, the total positive charge density $\rho(d)$ in the diffuse layer screens the negative surface charge Σ . By integrating $\rho(d)$ across the diffuse layer, we obtain the equivalent surface charge density $\Sigma_d^{(0)}$ of the diffuse layer. In equilibrium, the EDL consisting of charged surface sites, Stern layer, and diffuse layer is electroneutral, that is, $\Sigma_S^{(0)} + \Sigma_d^{(0)} = -\Sigma$.

The electric potential at the inner limit of the diffuse layer is usually identified with the ζ -potential at the plane of shear (e.g., Leroy et al., 2008; Bückner & Hördt, 2013a). Due to the space charge ρ , the electric potential $U_a^{(0)}$ in the electrolyte decays from ζ at the solid-liquid interface to zero in the bulk electrolyte. Note that in the simplified model sketched in Figure 1, the (equilibrium) potential in the Stern layer $U_s^{(0)}$ is equal to ζ .

2.2. Basic Equations

2.2.1. Bulk Electrolyte and Diffuse Layer

The bulk electrolyte is characterized by the ion valences $z_{\pm}$, mobilities $\mu_{\pm}$, and bulk concentrations $C_{\pm}^{\infty}$, as well as the relative permittivity ϵ_a . For the sake of simplicity, we assume that the mobilities of cations (subindex +) and anions (subindex −) are equal, that is, $\mu_+ = \mu_- = \mu$, and limit our treatment to monovalent electrolytes, that is, $z_{\pm} = \pm 1$. In the case of a dilute solution, the electrical conductivity of the bulk electrolyte is then given by $\sigma_a = 2e\mu C_{\pm}^{\infty}$.

Spatial and temporal variations of the ion concentrations $C_{\pm}(\mathbf{r}, t)$ and the electrical potential $U_a(\mathbf{r}, t)$ in the solution, that is, in the diffuse layer and the bulk electrolyte, are controlled by the steady state Nernst-Planck, mass continuity, and Poisson equations (e.g., Garcia et al., 1985):

$$\mathbf{J}_{\pm}(\mathbf{r}, t) = -D\nabla C_{\pm}(\mathbf{r}, t) - z_{\pm}\mu C_{\pm}(\mathbf{r}, t)\nabla U_a(\mathbf{r}, t), \quad (1)$$

$$\nabla \mathbf{J}_{\pm}(\mathbf{r}, t) = -\partial_t C_{\pm}(\mathbf{r}, t), \quad (2)$$

$$\nabla^2 U_a(\mathbf{r}, t) = -\frac{e}{\epsilon_0 \epsilon_a} [C_+(\mathbf{r}, t) - C_-(\mathbf{r}, t)]. \quad (3)$$

Here, D denotes the diffusion coefficient, $e = 1.602 \times 10^{-19}$ C is the elementary charge, and $\epsilon_0 = 8.85 \times 10^{-12}$ F/m is the vacuum permittivity. In absence of specific interaction between the different ion species, the Einstein relation $D = \mu k_B T / e$ can be used to connect diffusion coefficient and mobility of the ions. Here, $k_B = 1.381 \times 10^{-23}$ J/K is Boltzmann's constant, and T is the absolute temperature. The current densities $\mathbf{J}_{\pm}$ defined by the steady state Nernst-Planck equation (1) consider diffusion $-D\nabla C_{\pm}$ and electromigration $-z_{\pm}\mu C_{\pm}\nabla U_a$ ion fluxes.

If the system is excited by a time-harmonic electric field $E_0 e^{i\omega t}$, where ω and t denote angular frequency and time, respectively, the ion concentrations $C_{\pm}(\mathbf{r}, t)$ can be expressed by the sums of the static equilibrium concentrations $C_{\pm}^{(0)}$ and perturbation concentrations $\delta C_{\pm}$ (e.g., Garcia et al., 1985):

$$C_{\pm}(\mathbf{r}, t) = C_{\pm}^{(0)}(\mathbf{r}) + \delta C_{\pm}(\mathbf{r}, \omega) \cdot e^{i\omega t}. \quad (4)$$

The electrical potential $U_a(\mathbf{r}, t)$ can also be decomposed into the static equilibrium potential $U_a^{(0)}$ and the perturbation $|\delta U_a| \ll U_a^{(0)}$:

$$U_a(\mathbf{r}, t) = U_a^{(0)}(\mathbf{r}) + \delta U_a(\mathbf{r}, \omega) \cdot e^{i\omega t}. \quad (5)$$

By inserting equations (4) and (5) into equations (1) through (3) and Fourier-transforming the resulting system, the problem can be decomposed into a static part and a frequency-dependent part (e.g., Chew & Sen, 1982b, 1982a). While the frequency-dependent part couples to the static solution, the static part can be solved independently.

To obtain the static part of the system, we set $\delta C_{\pm}, \delta U_a, \partial_t = 0$. After a few additional manipulations, this yields the Boltzmann-distributed equilibrium ion concentrations (Chew & Sen, 1982b)

$$C_{\pm}^{(0)}(\mathbf{r}) = C_{\pm}^{\infty} \exp \left[-\frac{z_{\pm} e}{k_B T} U_a^{(0)}(\mathbf{r}) \right] \quad (6)$$

and the Poisson-Boltzmann equation (e.g., Chew & Sen, 1982b)

$$\nabla^2 U_a^{(0)}(\mathbf{r}) = -\kappa^2 \frac{k_B T}{e} \sinh \left[\frac{e}{k_B T} U_a^{(0)}(\mathbf{r}) \right], \quad (7)$$

where $\kappa = [2e^2 C_{\pm}^{\infty} / (\epsilon_0 \epsilon_r k_B T)]^{1/2}$ is the inverse Debye screening length. Together with the boundary conditions discussed below, equations (6) and (7) describe the equilibrium ion concentrations $C_{\pm}^{\infty}$ and the static electrical potential $U_a^{(0)}$ in the diffuse layer and the bulk electrolyte.

In order to obtain the solution of the frequency-dependent system under the influence of a weak external field $E_0 \exp(i\omega t)$, the steady state Nernst-Planck equation (1) is inserted into the continuity equation (2), Fourier-transformed and linearized, which gives (e.g., Chew & Sen, 1982a)

$$i\omega \delta C_{\pm}(\mathbf{r}, \omega) = \nabla \cdot \left\{ D \nabla \delta C_{\pm}(\mathbf{r}, \omega) + \mu z_{\pm} \left[C_{\pm}^{(0)}(\mathbf{r}) \nabla \delta U_a(\mathbf{r}, \omega) + \delta C_{\pm}(\mathbf{r}, \omega) \nabla U_a^{(0)}(\mathbf{r}) \right] \right\} + \mathcal{O}(E_0^2). \quad (8)$$

The frequency-dependent perturbation concentrations of the two ion species and the potentials are coupled to each other by the Fourier-transformed Poisson equation (e.g., Chew & Sen, 1982a)

$$\nabla^2 \delta U_a(\mathbf{r}, \omega) = -\frac{e}{\epsilon_0 \epsilon_r} [\delta C_+(\mathbf{r}, \omega) - \delta C_-(\mathbf{r}, \omega)]. \quad (9)$$

Equations (8) and (9) constitute three coupled PDEs that describe the spatial variations of the perturbation quantities. As the boundary conditions at the solid surface link the solution in the electrolyte to the corresponding solutions in the Stern layer and the interior of the solid, we will discuss the boundary conditions further below.

2.2.2. Stern Layer

The Stern layer is modeled as a thin layer situated at the solid-liquid interface. In absence of an external excitation, the surface-charge density $\Sigma_S^{(0)}$ in this layer is uniform. Under the influence of the electrical field $E_0 \exp(i\omega t)$, the counterions move along the surface, but no charge exchange with the electrolyte nor the solid is considered. We describe the surface-charge density in the Stern layer in terms of the constant equilibrium value $\Sigma_S^{(0)}$ and the perturbation $|\delta \Sigma_S| \ll \Sigma_S^{(0)}$, such that

$$\Sigma_S(\mathbf{r}_S, t) = \Sigma_S^{(0)} + \delta \Sigma_S(\mathbf{r}_S, \omega) \exp(i\omega t), \quad (10)$$

where $\mathbf{r}_S$ denotes the position vector in local coordinates along the solid-liquid interface. Adopting the treatment of the bound surface-charge densities on spherical particles proposed by Schwarz (1962) and Schurr (1964), the perturbation of the surface-charge density in the Stern layer is controlled by

$$i\omega \delta \Sigma_S(\mathbf{r}_S, \omega) = \nabla_S \cdot \left[D_S \nabla_S \delta \Sigma_S(\mathbf{r}_S, \omega) + \mu_S \Sigma_S^{(0)} \nabla_S \delta U_S(\mathbf{r}_S, \omega) \right] + \mathcal{O}(E_0^2), \quad (11)$$

where ∇_S is the surface Laplacian operating on functions defined on the geometrical boundary (also known as Laplace-Beltrami operator), μ_S is the mobility, D_S is the diffusion coefficient, and δU_S is the perturbation potential in the Stern layer. Note that equation (11) is the surface equivalent of equation (8) describing diffusion and electromigration surface-flux densities within the Stern layer. Only the third term on the right-hand side of equation (8) has no equivalent because the constant potential results in a vanishing tangential electrical field $-\nabla_S U_S^{(0)}$.

For spherical particles of radius a centered at the origin of the spherical coordinate system (r, θ, ϕ) and an excitation parallel to the polar axis $\theta = 0$, equation (11) takes the form (e.g., Schurr, 1964; Schwarz, 1962)

$$i\omega \delta \Sigma_S(\theta, \omega) = \frac{1}{a^2 \sin \theta} \frac{\partial}{\partial \theta} \left[D_S \sin \theta \frac{\partial}{\partial \theta} \delta \Sigma_S(\theta, \omega) + \mu_S \Sigma_S^{(0)} \sin \theta \frac{\partial}{\partial \theta} \delta U_S(\theta, \omega) \right]. \quad (12)$$

There is no particular surface equivalent of Poisson's equation (9). Instead, the continuity of the electrical potential at the solid surface (in conjunction with the vanishing thickness of the Stern layer) directly couples δU_S and the surface-charge density $\delta \Sigma_S$ to the adjacent perturbation potentials in electrolyte and solid.

2.2.3. Solid Dielectric

The solid dielectric has a relative permittivity ϵ_i and zero electrical conductivity. The spatial variation of the potential within the solid is governed by the Laplace equation $\nabla^2 U_i(\mathbf{r}, t) = 0$. In equilibrium, the static electrical potential $U_i^{(0)}$ must be equal to the (constant) ζ -potential on the surface, and $U_i^{(0)}$ must be constant throughout the solid. The Fourier-transformed frequency-dependent Laplace equation writes

$$\nabla^2 \delta U_i(\mathbf{r}, \omega) = 0, \quad (13)$$

which in conjunction with the spatially varying surface potential $\delta U_S(\mathbf{r}_S, \omega)$ determines the perturbation potential $\delta U_i(\mathbf{r}, \omega)$ within the solid.

2.3. Boundary Conditions

At far distances d from the surface, the static background potential in the electrolyte should approach zero, that is,

$$U_a^{(0)} \xrightarrow{d \rightarrow \infty} 0. \quad (14)$$

At the surface, we take advantage of our knowledge of the constant potential $U_i^{(0)}$ inside of the solid (see above) and use the continuity of the displacement current to define the Neumann boundary condition

$$-\epsilon_0 \epsilon_a \nabla U_a^{(0)} \Big|_{\text{surface}} \cdot \mathbf{n} = \Sigma + \Sigma_S^{(0)}, \quad (15)$$

where $\mathbf{n}$ denotes the unit normal vector to the solid surface (pointing out of the solid into the electrolyte) and $\Sigma + \Sigma_S^{(0)} = -\Sigma_d^{(0)}$ is the net surface-charge density. The boundary conditions on the equilibrium ion concentrations $C_{\pm}^{(0)}$ are implicit to equation (6).

The perturbation potential at far distances d from the surface must be equal to values corresponding to the external electrical field, that is,

$$\delta U_a(\mathbf{r}, \omega) \xrightarrow{d \rightarrow \infty} -\mathbf{E}_{\text{ext}}(\mathbf{r}) \cdot \mathbf{r}, \quad (16)$$

while the ion concentrations should approach their bulk values, that is,

$$\delta C_{\pm}(\mathbf{r}, \omega) \xrightarrow{d \rightarrow \infty} 0. \quad (17)$$

At the surface of the solid, the solutions for the three model domains—electrolyte, Stern layer, and solid—are pieced together. The continuity of the electrical potential demands the three perturbation potentials to be equal at any point $\mathbf{r}_S$ on the surface, that is,

$$\delta U_i(\mathbf{r}_S, \omega) = \delta U_S(\mathbf{r}_S, \omega) = \delta U_a(\mathbf{r}_S, \omega). \quad (18)$$

We assume that on the time scales of interest, ions of the solution do not engage in surface reactions; thus, they are neither produced nor consumed, and the normal fluxes through the surface are zero, that is,

$$\left\{ -D \nabla \delta C_{\pm}(\mathbf{r}, \omega) - \mu z_{\pm} \left[C_{\pm}^{(0)}(\mathbf{r}, \omega) \nabla \delta U_a(\mathbf{r}, \omega) + \delta C_{\pm}(\mathbf{r}, \omega) \nabla U_a^{(0)}(\mathbf{r}, \omega) \right] \right\} \Big|_{\mathbf{r}=\mathbf{r}_S} \cdot \mathbf{n} = 0. \quad (19)$$

The continuity of the displacement current implies that (e.g., Schwarz, 1962)

$$\left[-\epsilon_0 \epsilon_a \nabla \delta U_a(\mathbf{r}, \omega) + \epsilon_0 \epsilon_i \nabla \delta U_i(\mathbf{r}, \omega) \right] \Big|_{\mathbf{r}=\mathbf{r}_S} \cdot \mathbf{n} = \delta \Sigma_S(\mathbf{r}_S), \quad (20)$$

which completes the set of boundary conditions.

3. Polarization of Spherical Grains

The model setup in the previous section describes the charge polarization of the Stern layer and the concentration polarization produced by unequal contributions of anions and cations to electrical conduction through the diffuse layer. In this section, we will study the relative importance of both polarization mechanisms for the case of dilute suspensions of dielectric spheres. Because even for spherical particles no analytical solution of the fully coupled problem is known, we first obtain a suitable analytical approximation of the coupled polarization process, which combines (i) the Stern-layer polarization model by Schwarz (1962), including the correction of the corresponding relaxation time proposed by Lyklema et al. (1983), and (ii) the diffuse-layer polarization model developed by Dukhin and Shilov (1974). In order to assess the quality of our analytical model and open the possibility to model the polarization response of more complex geometrical configurations, we also present a numerical finite-element solution.

3.1. Analytical Model

At distances far from a spherical particle of radius a centered at the origin of the spherical coordinate system $\mathbf{r} = (r, \theta, \phi)$, all approximate analytical solutions take the form (e.g., Dukhin & Shilov, 1974; Maxwell, 1892; Schurr, 1964; Wagner, 1914)

$$\delta U_a(\mathbf{r}, \omega) = E_0 \left[-r + \frac{f(\omega)a^3}{r^2} \right] \cos \theta, \quad (21)$$

if the external excitation is parallel to the polar axis, that is, $\theta = 0$. The first term of this expression accounts for the potential due to the external field, and the second term describes the effective long-range dipole moment of the polarized particle. The reflection coefficient $f(\omega)$ contains the complete information on the macroscopic polarization response of the particle. For a pure Maxwell-Wagner polarization, it writes

$$f(\omega) = \frac{\sigma_i^*(\omega) - \sigma_a^*(\omega)}{2\sigma_a^*(\omega) + \sigma_i^*(\omega)}, \quad (22)$$

where $\sigma_a^*(\omega) = \sigma_a + i\omega\epsilon_0\epsilon_a$ and $\sigma_i^*(\omega) = i\omega\epsilon_0\epsilon_i$ are the complex conductivity of the bulk electrolyte and the nonconducting particle, respectively.

With $f(\omega)$ for one spherical particle at hand, the effective complex conductivity $\sigma^*(\omega)$ of a dilute suspension of a number of equal particles can be obtained using a generalized form of the theory by Wagner (1914)

$$\frac{\sigma^*(\omega)}{\sigma_a^*} = \frac{1 + 2\nu f(\omega)}{1 - \nu f(\omega)}, \quad (23)$$

where ν denotes the volume fraction of suspended particles. Together with the complex conductivities σ_a^* and σ_i^* , equations (22) and (23) describe the Maxwell-Wagner polarization of suspensions of dielectric particles in a medium with homogeneous complex conductivity $\sigma_a^*(\omega)$.

O'Konski (1960) included the effect of a uniform surface charge density Σ_x into this model. He found that the effect of charge carriers that move freely along the particle surface can be taken into account by adding the effective conductivity increment $\sigma_x = 2\mu_x\Sigma_x/a$ to the complex conductivity σ_i^* of the particle. According to this idea, the effective conductivity of the diffuse layer can be expressed as

$$\sigma_d = |\sigma_{d+} + \sigma_{d-}| = \frac{2\mu|\Sigma_{d+}^{(0)} - \Sigma_{d-}^{(0)}|}{a}, \quad (24)$$

where the contributions of the two types of ions to the effective conductivity of the diffuse layer are defined as

$$\sigma_{d\pm} = \frac{\pm 2\mu\Sigma_{d\pm}^{(0)}}{a}. \quad (25)$$

For sufficiently thin diffuse layers, that is, $\kappa a \ll 1$, the equivalent surface charge densities in the diffuse layer can be related to the surface charge using a variation of Bikerman's equation for the surface conductivity near a (highly) charged plane surface (e.g., Shilov et al., 2001)

$$\Sigma_{d\pm}^{(0)} = \pm \frac{2eC_{\pm}^{\infty}}{\kappa} \left[\exp\left(\mp \frac{e\zeta}{2k_B T}\right) - 1 \right]. \quad (26)$$

Note that while the total charge stored in the diffuse layer is represented by the sum of both contributions, that is, $\Sigma_d^{(0)} = \Sigma_{d+}^{(0)} + \Sigma_{d-}^{(0)}$, the total surface conductivity of the diffuse layer σ_d is proportional to their difference. For a given surface charge density $\Sigma_D^{(0)}$, the ζ -potential can be obtained from

$$\zeta(\Sigma_d^{(0)}) = -\frac{2k_B T}{e} \sinh^{-1} \left(\Sigma_d^{(0)} \frac{\kappa}{4eC_{\pm}^{\infty}} \right). \quad (27)$$

This relation can readily be obtained from equation (26). Resolved for $\Sigma_d^{(0)}(\zeta)$, it is also known as Grahame equation (Grahame, 1947).

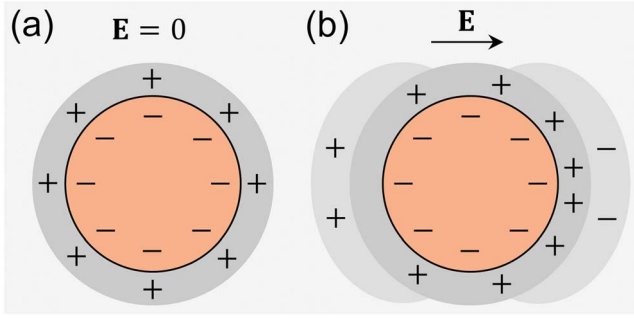


Figure 2. Sketch of the polarization of the Stern layer: (a) negatively charged particle and Stern layer in equilibrium; (b) under the influence of the external field $\mathbf{E}$, cations in the Stern layer move along the surface, deplete on the left side and accumulate on the right side of the particle. The resulting net surface charge $\delta\Sigma_S$ causes field-induced diffuse layers of opposite charge to build up in the electrolyte next to the charged surfaces (e.g., Lyklema et al., 1983).

The surface-charge density of the Stern layer can be expressed in terms of the effective conductivity

$$\sigma_S = \frac{2\mu_S |\Sigma_S^{(0)}|}{a}. \quad (28)$$

In the present study, the partition of counterions into diffuse layer and Stern layer will be expressed in terms of the ratio $p = -\Sigma_S^{(0)}/\Sigma$. Note that this definition is slightly different from the partition coefficient $f_Q = \Sigma_S^{(0)}/(\Sigma_{d+}^{(0)} + \Sigma_S^{(0)})$ introduced by Leroy and Revil (2004).

3.1.1. Stern-Layer Polarization

If the entire counter charge is located in the Stern layer (i.e., $p = 1$), and no diffuse layer is present (i.e., $\Sigma_d^{(0)}, \zeta = 0$), the low-frequency response of the particle is caused by the polarization of the Stern layer only (see Figure 2). In the thin double layer limit, that is, $\kappa a \ll 1$, the solution of the polarization problem can be approximated as done by Schwarz (1962). In terms of the reflection coefficient f defined in equation (21), the final result obtained by Schwarz (1962) can be expressed as (from his equation 13)

$$f_S(\omega) = \frac{\sigma_S^*(\omega) - \sigma_a^*(\omega)}{2\sigma_a^*(\omega) + \sigma_S^*(\omega)}, \quad (29)$$

where σ_S^* is the effective complex conductivity of the spherical particle, which writes (from Schwarz, 1962, equations 14 and 16)

$$\sigma_S^*(\omega) = \sigma_S \frac{i\omega\tau_S}{1 + i\omega\tau_S} + i\omega\epsilon_0\epsilon_i. \quad (30)$$

Note that the particle is assumed to be nonconducting, that is, $\sigma_i = 0$. The relaxation time of the Stern-layer polarization τ_S can be expressed as (Lyklema et al., 1983, equation 36)

$$\tau_S = \frac{a^2}{2D_S M}, \quad (31)$$

where the coefficient M defined as (Lyklema et al., 1983, equation 34)

$$M = 1 + \frac{\kappa \Sigma_S^{(0)}}{2eC_\pm^\infty \cosh[e\zeta/(2k_B T)]} \quad (32)$$

accounts for the coupling of the charges in the Stern layer to the electrolyte, which had not been considered in the original model by Schwarz (1962).

Because $\zeta = 0$ if $p = 1$, there is no equilibrium diffuse layer, and the polarization response is only due to the polarization of the Stern layer, first term in equation (30), and the Maxwell-Wagner polarization caused by the conductivity and permittivity contrasts between particle and electrolyte.

3.1.2. Diffuse-Layer Polarization

In the opposite case, in which no Stern layer exists (i.e., $p = 0$) and the entire counter charge is located in the diffuse layer, only the diffuse layer polarizes (see Figure 3). This scenario has been treated by Dukhin and Shilov (1974) for the limit of thin diffuse layers. If electro-osmotic effects are neglected; that is, in the limit of an infinitely large fluid viscosity, the result of the classical Dukhin-Shilov theory can be expressed in terms of the reflection coefficient (e.g., Grosse & Shilov, 1996; Shilov et al., 2001)

$$f_d(\omega) = \frac{2Du(\zeta) - 1}{2Du(\zeta) + 2} - \frac{3S}{2} \frac{(\sigma_{d+} - \sigma_{d-})^2}{\sigma_a^2 [2Du(\zeta) + 2]^2} \left[1 - \frac{i\omega\tau_a}{1 + \sqrt{i\omega 2\tau_a/S + i\omega\tau_a}} \right], \quad (33)$$

where $Du(\zeta) = \sigma_d/(2\sigma_a)$ is the Dukhin number. The coefficient S , which appears in the expression for f_d , writes

$$S = \frac{[Du(\zeta) + 1] \sigma_a^2}{(\sigma_{d+} + \sigma_a)(\sigma_{d-} + \sigma_a)} \quad (34)$$

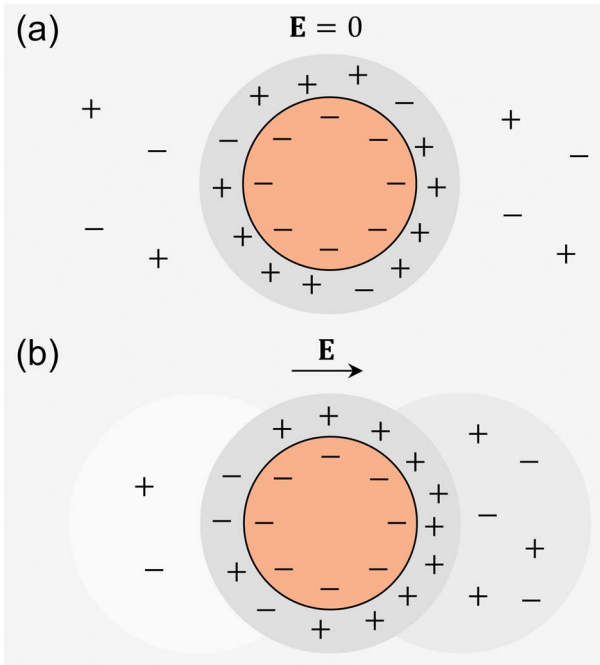


Figure 3. Sketch of the polarization of the diffuse layer: (a) negatively charged particle, diffuse layer, and bulk electrolyte in equilibrium; (b) under the influence of the external field $\mathbf{E}$, the high effective conductivity of the diffuse layer leads to an accumulation of positive charge on the right side and negative charge on the left side of the particle. In addition, unequal migration flux densities of cations and anions through the diffuse layer are counterbalanced by an electroneutral salinity gradient that builds up next to the particle.

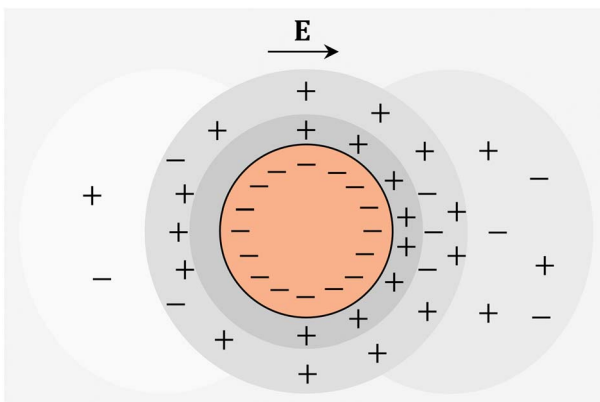


Figure 4. Sketch of the coupled polarization of Stern and diffuse layer: Due to the presence of a diffuse layer, the net surface density in the polarized Stern layer $\delta\Sigma_S$ is larger than in the case without diffuse layer (Lyklema et al., 1983). In the inner part of the diffuse layer, $\delta\Sigma_S$ attracts opposite charges (here positive on the right and negative on the left side of the particle). In the outer part, the charge in the diffuse layer changes its sign, which compensates for the high effective conductivity of this layer (as in Figure 3). The salinity gradient in the electrolyte next to the particle is a result of the unequal migration flux densities of cations and anions through the diffuse layer.

and the time constant is defined as $\tau_a = a^2 S / (2D)$. The effective conductivity of the suspension can be obtained by substituting f by f_d in equation (23). Note that this formulation only accounts for the polarization of the diffuse layer; an extension including Maxwell-Wagner polarization can be found in Shilov et al. (2001).

3.1.3. Coupled Polarization

For any value $0 < p < 1$, the counter charges are distributed over Stern and diffuse layer, which leads to a simultaneous polarization of both parts of the EDL (see Figure 4). As argued by Lesmes and Morgan (2001), the coupled polarization can be approximated by a superposition of the individual responses, that is, by adding the effective complex dielectric constants of the particle (or the corresponding effective complex-conductivity increments). In order to obtain an effective conductivity increment describing the diffuse-layer polarization, we rearrange equation (22) as follows:

$$\sigma_d^*(\omega) = \sigma_a^*(\omega) \frac{1 + 2f_d(\omega)}{1 - f_d(\omega)}. \quad (35)$$

The effective conductivity σ_s^* , which accounts for Stern-layer and Maxwell-Wagner polarization, is given by equations (30) through (32), and, thus, the total effective conductivity of the particle including all three mechanisms writes

$$\sigma_c^*(\omega) = \sigma_d^*(\omega) + \sigma_s^*(\omega) = \sigma_d^*(\omega) + \sigma_s \frac{i\omega\tau_s}{1 + i\omega\tau_s} + i\omega\epsilon_0\epsilon_i. \quad (36)$$

Note that this expression is almost the same as the one obtained by Schurr (1964) except for the decrease of τ_s by the factor M and the substitution of σ_d by $\sigma_d^*(\omega)$, that is, including the frequency-dependent contribution of the diffuse layer defined by equations (33) through (35).

The new coupled model described here is also similar to the one proposed by Lesmes and Morgan (2001), from which it differs with regard to (i) the relaxation time of the Stern layer τ_s , where we use the correction by Lyklema et al. (1983); (ii) the low-frequency contribution σ_s of the Stern layer (their equations 5 and 6), which we do not consider because it contradicts the assumption of a strongly bound Stern-layer; and (iii) the selection of the model describing the diffuse-layer polarization, where we use the Dukhin-Shilov theory instead of the model by Fixman (1980). Fixman's assumption that the coion contribution to the surface conductivity can be ignored for sufficiently high ζ -potentials does not hold for the range of ζ -potentials studied here. We found that this simplification significantly affects the real part of the complex conductivity (not shown here for brevity).

3.2. Numerical Model

Figure 5 shows the geometrical setup used for the numerical modeling. The dielectric particle of radius a is centered at the origin of coordinates, and the external electric field is imposed in x direction, that is, $\mathbf{E}_{\text{ext}} = E_0 \mathbf{e}_x$, where $\mathbf{e}_x$ denotes the unit vector in x direction. Due to the cylindrical symmetry of the problem, the numerical simulation can be carried out on the two-dimensional model domain in Cartesian coordinates $\mathbf{r} = (x, y)$ marked in red in Figure 5.

We use the finite-element software COMSOL Multiphysics to successfully obtain the static and the frequency-dependent solution. We first

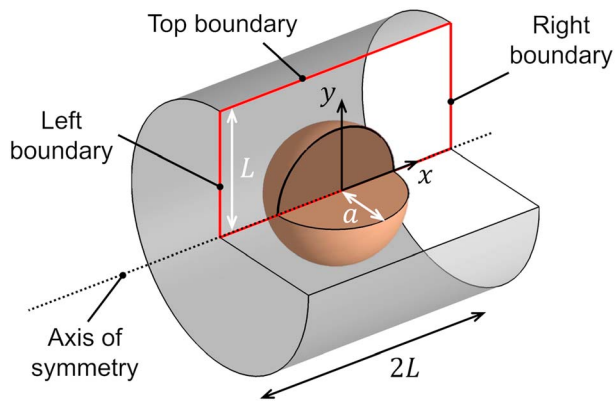


Figure 5. Three-dimensional sketch of the geometrical setup used for the numerical simulation of the polarization around spherical particles. The particle of radius a is enclosed by a cylinder of radius L and height $2L$, which represents the electrolyte. The red rectangle marks the actual two-dimensional model domain.

solve the Poisson-Boltzmann equation (7), which contains $U^{(0)}(\mathbf{r})$ as only unknown variable. For the numerical implementation, boundary conditions (14) and (15) are translated as follows: $U^{(0)} = 0$ on the left, right, and top boundary; $[\epsilon_0 \epsilon_i \nabla U_i^{(0)} - \epsilon_0 \epsilon_a \nabla U_a^{(0)}] \mathbf{n} = -\Sigma_d^{(0)}$ on the particle surface; and $\partial U(0)/\partial y = 0$ on the axis of symmetry. The static background ion concentrations are computed by inserting $U_a^{(0)}$ into equation (6).

Subsequently, we solve the frequency-dependent part described by equations (8), (9), (11), and (13) and the boundary conditions (16) through (20). For the numerical solution, the latter are adjusted as follows: $\delta n_{\pm} = 0$ and $\delta U_a = \pm E_0 L$ on the left and right boundary, respectively; $\mathbf{J}_{\pm} = 0$ and $\partial \delta U / \partial y = 0$ on top boundary and on the axis of symmetry; unchanged on the particle surface. While the static solution only needs to be computed once for each set of model parameters (i.e., a , Σ , and p), the frequency-dependent problem has to be solved for each value of the angular frequency ω separately.

Because bulk values are implicitly assumed on the left, right, and top boundary, these boundaries must be placed sufficiently far away from the particle surface, where the polarization is expected to cause nonzero per-

turbations. As a trade-off between precision and computational cost, we define a standard domain size of $L = 10a$. For more technical details on the implementation, see the appendix.

The effective conductivity of the model can be obtained from the numerical integration of the total ion flux densities through the left (or likewise the right) boundary, that is,

$$\sigma_{\text{mod}}^* = \frac{2}{E_0 L^2} \int_0^L [\mathbf{J}_+(y) + \mathbf{J}_-(y)] \cdot \mathbf{e}_x y dy, \quad (37)$$

where the term $y dy$ accounts for the area element of the boundary and the factor $2/L^2$ stems from the normalization with the total area.

The standard domain size is $L = 10a$ and corresponds to a rather small volume fraction of dielectric particles. Therefore, we scale the modeled effective conductivities $\sigma_{\text{mod}}^*(\omega)$ to a more realistic volumetric content of $\nu = 0.4$ using the mixing rule defined in equation (23) with

$$f(\omega) = \frac{1}{\nu_{\text{mod}}} \frac{\sigma_{\text{mod}}^*(\omega) - \sigma_a^*(\omega)}{\sigma_{\text{mod}}^*(\omega) + 2\sigma_a^*(\omega)}. \quad (38)$$

3.3. Comparison of Analytical and Numerical Solution

Unless otherwise stated, the following parameter values are used to obtain both numerical and analytical results: The relative permittivities are $\epsilon_a = 80$ for the aqueous electrolyte and $\epsilon_i = 4.5$ for the solid, which is a typical value for quartz sand (e.g., Robinson & Friedman, 2003). The uniform ion mobility is $\mu = 5 \cdot 10^{-8} \text{ m}^2/(\text{Vs})$, which is approximately equal to the mobility of the sodium cation (e.g., Atkins & De Paula, 2013) and will be used for both ion species in the electrolyte. The mobility of the counterions in the Stern layer is only 10% of the ion mobility in the bulk electrolyte, that is, $\mu_s = \mu/10$. This value corresponds to the reduction of the cation mobility in the Stern layer inferred for K^+ on latex surfaces by Zukoski and Saville (1986) and for Na^+ on clay surfaces by Revil and Glover (1998) and Revil et al. (1998). For near-neutral pH of the solution and ion concentrations in the bulk electrolyte of $C_{\pm}^{\infty} = 1 \text{ mol/m}^3$, it is adequate to assume a surface charge density of $\Sigma = -0.01 \text{ C/m}^2$ (e.g., Kosmulski, 2006). The absolute temperature of $T = 293 \text{ K}$ (room temperature) and the above-mentioned volumetric content of $\nu = 0.4$ complete the set of standard model parameters.

Figure 6 displays the conductivity spectra of a suspension of spherical particles of radius $a = 5 \mu\text{m}$ for six values of p between 0 and 1. The effective conductivity of the mixture σ^* is expressed in terms of the corresponding real (σ') and imaginary (σ'') parts. In Figure 6a, we observe a continuous decrease of σ' with increasing p . This means that the surface conductivity due to counterions in the diffuse layer is much larger than the one of the Stern layer. This is due, on the one hand, to the significantly reduced mobility of the counterions in the Stern layer, which affects all frequencies equally. On the other hand, at low frequencies,

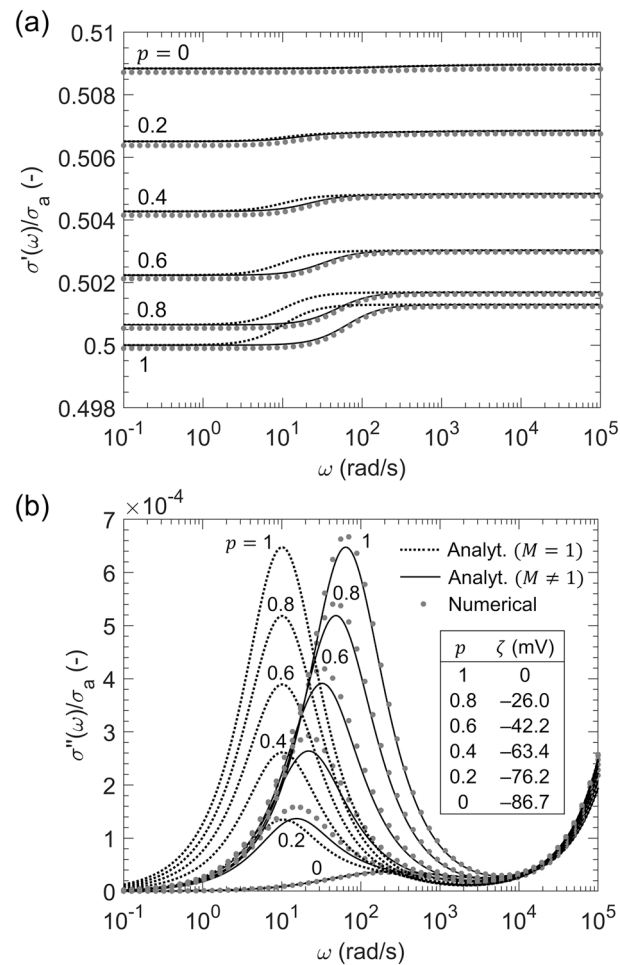


Figure 6. Complex-conductivity spectra of a suspension of dielectric particles of radius $a = 5 \mu\text{m}$ at different ratios $p = -\Sigma_S^{(0)}/\Sigma$, that is, the surface charge in the Stern layer increases with p . Complex-valued conductivities in terms of real (a) and imaginary (b) parts normalized to the bulk conductivity σ_a . Numerical results (gray circles) are displayed along with the corresponding analytical models according to equation (36) using the simple relaxation time after Schwarz, (1962; $M = 1$ in equation (31), dotted line) and using the corrected relaxation time after Lyklema et al. (1983; $M \neq 1$, solid line). The values of ζ shown in (b) are computed from equation (27).

the Stern layer is completely polarized and does not contribute to the DC conductivity because we assume that it cannot exchange ions with the bulk electrolyte. However, the Stern layer does contribute to the high-frequency limit of σ' , that is, at frequencies larger than the characteristic frequency of the Stern-layer polarization. The higher p and thus the surface conductivity σ_S , the larger becomes the difference between high- and low-frequency limits of σ' , which is largest in the case of a pure Stern-layer polarization ($p = 1$). In contrast, in the case of a sole diffuse-layer polarization ($p = 0$), we hardly recognize any variation of σ' with the angular frequency.

The peak imaginary conductivity σ'' (Figure 6b) increases almost linearly with p , that is, the charge density in the Stern layer. For the same surface-charge densities, the Stern-layer polarization ($p = 1$) results in a maximum of σ'' , which is approximately an order of magnitude larger than the peak produced by the corresponding diffuse-layer polarization ($p = 0$). From Figure 7, which shows a close-up of the small imaginary conductivities generated in the case of a depopulated Stern layer, it is evident that even a small fraction of 20% of positive charge located in the Stern layer produces a stronger response than the other 80% located in the diffuse layer. It is worth mentioning that the polarization magnitude (here in terms of the maximum of σ'') largely varies with the mobility of the counterions in the Stern layer. In a separate analysis, we observed a difference between the magnitudes of the two polarization processes of 2 orders of magnitude when a larger mobility of $\mu_S = \mu/2$ was assumed (not shown). This observation is in agreement with the

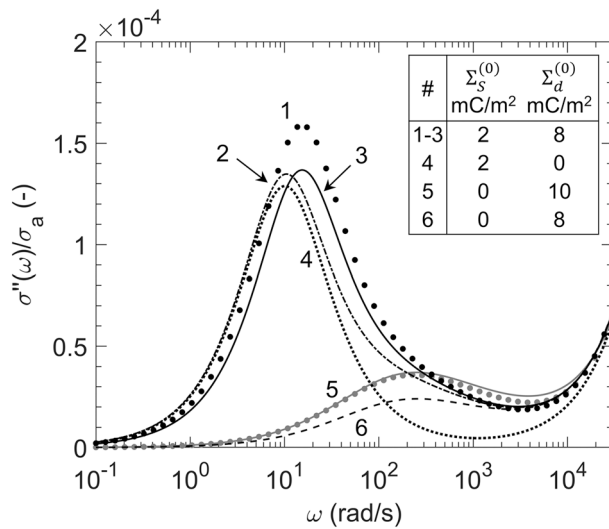


Figure 7. Numerical (filled circles) and analytical (lines) imaginary conductivity spectra for small surface charge densities $\Sigma_s^{(0)}$ in the Stern layer. Analytical curves are calculated from equation (36), using the values of $\Sigma_s^{(0)}$ and $\Sigma_d^{(0)}$ given in this Figure. Curve 1, numerical solution for the coupled Stern and diffuse-layer polarization; curves 2 and 3, analytical solution for the coupled Stern and diffuse-layer polarization using $M = 1$ and $M \neq 1$ in equation (31), respectively; curve 4, analytical solution for a pure Stern-layer polarization using $M = 1$; curves 5 and 6, numerical and analytical solutions for a pure diffuse-layer polarization and two different surface charge densities $\Sigma_d^{(0)}$ in the diffuse layer.

results obtained earlier by Lesmes and Morgan (2001) and confirms their conclusion that the Stern-layer polarization produces a much stronger frequency dispersion than the diffuse-layer polarization.

For sufficiently small values of p , Figure 7 also shows that the contribution of the diffuse-layer polarization results in a slight increase of the polarization magnitude and the characteristic angular frequency (curves 1 and 3) as compared to the pure Stern-layer polarization (curve 4). The latter is related to the higher characteristic frequency of the diffuse-layer relaxation as a consequence of the shorter relaxation time, here $\tau_a \approx \tau_s/2$ because $S \approx 1$.

The comparison of the corresponding curves in Figures 6 and 7 also serves as a mutual validation of our finite-difference implementation and the new analytical approximation for the coupled polarization response. For angular frequencies between 0.1 and 10^4 mrad/s, the relative deviation between analytical and numerical results is $<0.3\%$ in the real part and $<20\%$ in the imaginary part of the effective conductivity. Here, the analytical approximation clearly underestimates the polarization magnitude. A good agreement, however, is observed between the characteristic frequency, that is, the angular frequency at which the σ'' peaks are observed. Apart from the underestimated polarization magnitude, the good agreement between numerical and analytical curves confirms the approach to model the coupled polarization process by a simple superposition of the individual responses of Stern and diffuse layer.

4. Membrane Polarization

Besides the polarization of Stern and diffuse layer around grains, the polarization of ion-selective pore constrictions can also cause a low-frequency dispersion of the complex conductivity. In the present section, we study this membrane-polarization mechanism based on an alternating series of wide and narrow cylinders. The cylinder walls are negatively charged and covered by an EDL consisting of a Stern and a diffuse layer, both contributing to the macroscopic polarization of the system. In order to assess their relative contributions, effective conductivity spectra are computed for varying charge densities in the two layers. Because no analytical model is available for the coupled response in this geometry, we resort to the numerical finite-element implementation validated in the previous section. We also show that after a slight modification presented here, the analytical membrane-polarization model proposed by Bückner and Hördt (2013a) reproduces the numerical results for the sole diffuse-layer polarization fairly well.

4.1. Analytical Model for Diffuse-Layer Polarization

Bückner and Hördt (2013a) considered the sequence of wide and narrow pores sketched in Figure 8. The pores are characterized by their radii R_i and lengths L_i ; the indices 1 and 2 denote properties of the wide and the narrow pore, respectively. A nonzero ζ -potential at the cylinder walls causes diffuse layers to build up. According to Bückner and Hördt (2013a), upon excitation by an external electrical field, the electrical current parallel to the symmetry axis will be controlled by the mean ion concentrations

$$b_{\pm,i} = \frac{2\pi}{C_{\pm}^{\infty} A_1} \int_0^{R_i} r C_{\pm,i}^{(0)}(r) dr. \quad (39)$$

These mean ion concentrations are averaged over the pore cross-section and normalized with the bulk ion concentration $C_{\pm}^{\infty}$. An additional normalization with the area of the wide pore $A_1 = \pi R_1^2$ accounts for the reduction of the total current through the narrow pore due to the reduced cross-section (see Bückner & Hördt, 2013a).

For sufficiently small ζ -potentials, that is, $\zeta \ll k_B T/e$, the radial variation of the electrical potential in the pore can be approximated by solving the linearized Poisson-Boltzmann equation in cylindrical coordinates,

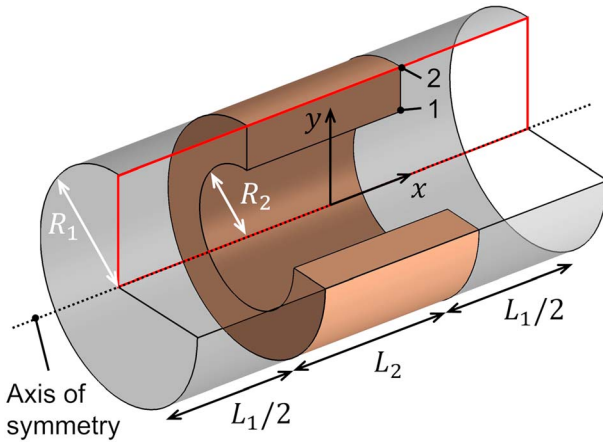


Figure 8. Three-dimensional sketch of the geometrical setup used for the numerical simulation of membrane polarization in a pore constriction. The two wide cylinders of length $L_1/2$ and radius R_1 and the narrow cylinder of length L_2 and radius R_2 are saturated with electrolyte solution. The volume around the narrow cylinder belongs to the nonconducting matrix with dielectric constant ϵ_i . The red rectangle marks the actual two-dimensional model domain.

which gives (Hunter, 1981; Bückner & Hördt, 2013a)

$$U_a^{(0)}(r) = \zeta \frac{J_0(ikr)}{J_0(ikR)}, \quad (40)$$

where J_0 is the Bessel function of the first kind and order zero. The radial variation of the ion concentrations $C_{\pm,i}^{(0)}(r)$ are obtained by inserting $U_a^{(0)}(r)$ into equation (6).

For the typically much higher ζ -potential values of up to -100 mV on silica surfaces (here, $\Sigma = -0.01$ C/m²), the solution of the linearized Poisson-Boltzmann equation becomes imprecise. If we instead limit our treatment to sufficiently wide pore radii, that is, $\kappa R_i \gg 1$, we can make use of Bikerman's equation 26, which is more adequate for highly charged surfaces. The dimensionless mean ion concentrations can then be approximated by

$$b_{\pm,i} \approx \frac{A_i}{A_1} \left(\frac{\pm 2\Sigma_{d\pm}^{(0)}}{eC_{\pm}^{\infty}R_i} + 1 \right). \quad (41)$$

Either of these definitions of $b_{\pm,i}$, that is, equation 39 proposed by Bückner and Hördt (2013a) or equation (41) proposed here, can be used to express the effective transference numbers

$$t_{\pm,1} = \frac{b_{\pm,i}}{b_{+,i} + b_{-,i}}. \quad (42)$$

By means of this approximation, the three-dimensional cylindrical pore system is collapsed to a sequence of one-dimensional pores, the frequency-dependent impedance of which was derived by Marshall and Madden (1959). Following Bückner and Hördt (2013b) and Bückner et al. (2017), the Marshall-Madden impedance can be written as

$$Z(\omega) = \frac{2}{\sigma_a} \left[\frac{L_1}{b_{+,1} + b_{-,1}} + \frac{L_2}{b_{+,2} + b_{-,2}} + \frac{8D(t_{+,1} - t_{+,2})^2}{\frac{L_1}{\tau_1} \sqrt{i\omega\tau_1} \coth \sqrt{i\omega\tau_1} + \frac{L_2}{\tau_2} \sqrt{i\omega\tau_2} \coth \sqrt{i\omega\tau_2}} \right], \quad (43)$$

where the frequency dependence is controlled by the two time constants

$$\tau_i = \frac{L_i^2}{2D} S_i \quad \text{with} \quad S_i = \frac{1}{4b_{+,i}t_{-,i}}. \quad (44)$$

In order to stress the structural similarities between these two relaxation times and the relaxation time of the diffuse layer around spherical grains, we can rewrite the dimensionless mean ion concentrations $b_{\pm,i}$ defined in equation (41) in terms of a Dukhin number for cylindrical geometries, which we define as

$$Du(\zeta, R_i) = \frac{\sigma_d(R_i)}{2\sigma_a} = \frac{|\sigma_{d+}(R_i) + \sigma_{d-}(R_i)|}{2\sigma_a}. \quad (45)$$

In analogy to equation (25), the surface conductivities write $\sigma_{d\pm}(R_i) = \pm 2\mu\Sigma_{d\pm}^{(0)}/R_i$. The coefficients S_i take the form

$$S_i = \frac{A_1}{2A_i} \frac{[2Du(\zeta, R_i) + 1]\sigma_a^2}{[2\sigma_{d+}(R_i) + 1][2\sigma_{d-}(R_i) + 1]}. \quad (46)$$

Besides the factor $A_1/(2A_i)$ and the factors 2, with which the Dukhin numbers and the individual conductivities $\sigma_{d\pm}$ are multiplied, this definition of S_i is equivalent to the definition of S in equation (34), which controls the relaxation time of the diffuse layer around spherical particles.

4.2. Numerical Model for the Coupled Polarization

The numerical modeling is carried out using COMSOL Multiphysics. Again, the cylindrical symmetry of the problem permits limiting the computation to a two-dimensional model domain (red rectangle in Figure 8).

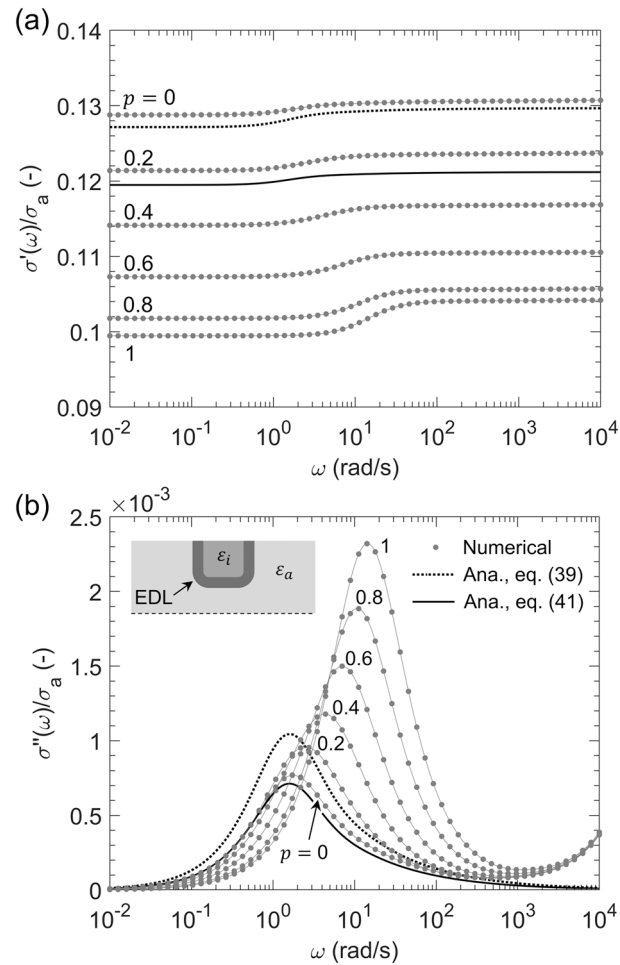


Figure 9. Complex-conductivity spectra of the membrane-polarization model for different ratios $p = -\Sigma_s^{(0)}/\Sigma$. Complex-valued conductivity in terms of real (a) and imaginary parts (b) normalized to the bulk conductivity σ_a . Numerical results (gray circles, lines are to guide the eye only) for all values of p are displayed along with the analytical models for $p = 0$ after Bückner and Hördt (2013a) using the average ion concentrations from equation (39) (dotted line) and from equation (41) (solid line). The sketch in (b) shows that the electrical double layer (EDL) is discontinuous in the wide pore. Pore lengths and radii are $L_1 = 90 \mu\text{m}$, $L_2 = 10 \mu\text{m}$, $R_1 = 2 \mu\text{m}$, and $R_2 = 0.2 \mu\text{m}$; all other parameter values are the same as in Figure 6.

The systems of PDEs describing the static and the frequency-dependent parts of the polarization problem remain unchanged, and only boundary conditions and finite-element mesh need to be adapted to the different geometry.

The boundary conditions for the static solution, that is, equations (14) and (15), merge into $U^{(0)} = 0$ on the left and right boundary; $[\epsilon_0 \epsilon_i \nabla U_i^{(0)} - \epsilon_0 \epsilon_a \nabla U_a^{(0)}] \mathbf{n} = -\Sigma_d^{(0)}$ on the solid-liquid interface; and $\partial U(0)/\partial y = 0$ on the axis of symmetry and the entire top boundary. This setup represents a discontinuous EDL, which only covers the surfaces of the volume shown in brown in Figure 8. Additionally, we compute the response of a setup, where the EDL is continuous in the wide pore and the boundary condition on the corresponding parts of the top boundary writes $[-\epsilon_0 \epsilon_a \nabla U_a^{(0)}] \mathbf{n} = -\Sigma_d^{(0)}$.

For the frequency-dependent part of the problem, the boundary conditions described in equations (16) through (20) are adjusted as follows: $\delta n_{\pm} = 0$ and $\delta U_a = \pm E_0 L$ on the left and right boundary, respectively; $\mathbf{J}_{\pm} = 0$ and $\partial \delta U / \partial y = 0$ on the axis of symmetry and the top boundary; equations (18) through (20) remain unchanged on the solid-liquid interface. In the setup with the continuous EDL, the boundary conditions on those parts of the top boundary, which delimit the wide pore, are given by the unchanged zero-flux condition (19) and the continuity of the displacement current, which here writes $-\epsilon_0 \epsilon_a \nabla \delta U_a(\mathbf{r}_S) \cdot \mathbf{n} = \delta \Sigma_s(\mathbf{r}_S)$.

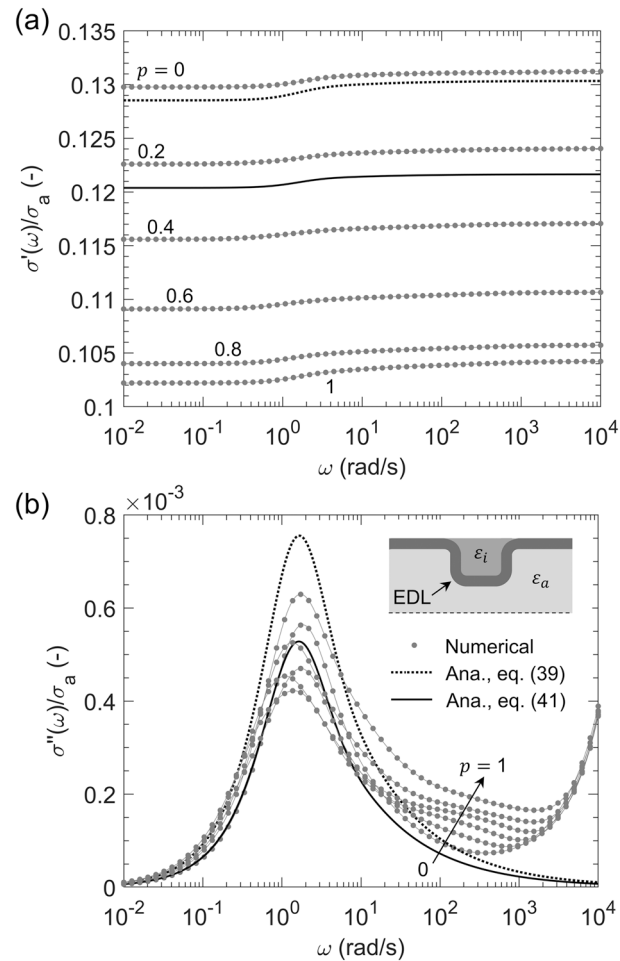


Figure 10. Complex-conductivity spectra of the membrane-polarization model with a continuous electrical double layer (EDL) in terms of the normalized real (a) and imaginary conductivity (b). Numerical results (gray circles lines are to guide the eye only) for different values of p are displayed along with the analytical models for $p = 0$ after Bückner and Hördt (2013a) using the average ion concentrations from equation (39) (dotted line) and from equation (41) (solid line). The sketch in (b) shows the continuous EDL in the wide pore; all other parameter values are the same as in Figure 9.

In order to avoid artifacts related to sharp corners covered by an EDL, the vertex 1 in Figure 8 is rounded off for the model with a discontinuous EDL, whereas the vertices 1 and 2 are rounded off for the model with a continuous EDL. In both cases, the radius of curvature is $(R_1 - R_2)/2$. For further technical details on the implementation, see the appendix.

The effective conductivity of the model can be obtained from numerically integrating the total ion flux densities through the left (or right) boundary. The resulting expression is equal to equation (37), if L substituted by R_1 . In the model with the continuous EDL, the surface current through the Stern layer has to be considered, too. The corresponding conductivity increment $-2/(E_0 R_1)[D_S \partial \delta \Sigma_S / \partial x + \mu_S \Sigma_S \partial \delta U_S / \partial x]$ has to be added to σ_{mod}^* . The analytical expression for the effective conductivity writes $\sigma_{\text{eff}}(\omega) = (L_1 + L_2)/Z(\omega)$, where $Z(\omega)$ is defined in equation (43).

4.3. Comparison of Analytical and Numerical Solution

The effective-conductivity spectra obtained from the model with the discontinuous EDL are shown in Figure 9. In the analytical models (for the case $p = 0$), the absence of the EDL in the wide pore has to be taken into account by setting $b_{\pm,1} = 1$ instead of using equation (39) or (41). The length of the narrow pore $L_2 = 10 \mu\text{m}$ was selected to match the diameter of the spherical particle treated above; the length of the wide pore $L_1 = 90 \mu\text{m}$ as well as the two pore radii $R_1 = 2 \mu\text{m}$ and $R_2 = 0.2 \mu\text{m}$ were adjusted to achieve a significant polarization response. As discussed in more detail in Hördt et al. (2017), the polarization magnitude of

the model by Bücker and Hördt (2013a) largely depends on the relation between the two ratios R_1/R_2 and L_1/L_2 .

In Figure 9a, we observe a similar variation of the magnitude of σ' with the ratio p as for the polarization around a spherical particle. In both cases, the surface conductivity increases σ' if the countercharges are mainly located in the diffuse layer. At low frequencies, the Stern layer polarizes and does not contribute to σ' ; at high frequencies, its contribution is limited by the low ion mobility in the Stern layer (remember that $\mu_S = \mu/10$). Moreover, the variation of the imaginary part σ'' with p (Figure 9b) is similar to the one observed for the spherical particle. Again, the σ'' peak increases with the amount of countercharges located in the Stern layer, that is, with increasing p , and shifts toward higher angular frequencies.

Figure 9 also shows that computing the mean ion concentrations from equation (41) instead of equation (39) improves the agreement with the numerical solution: for $p = 0$, the relative deviations between the analytical and the numerical σ'' curves are $<20\%$ for the modified and $<50\%$ for the original model at angular frequencies <100 rad/s. Because the analytical model does not consider Maxwell-Wagner polarization, which becomes dominant at high frequencies, the misfit increases at angular frequencies >100 rad/s. The original membrane polarization model, however, matches the real part (σ') better. Here, the relative deviation between the analytical models and the numerical solution are $\approx 7\%$ for the modified and $<1\%$ for the original model.

The fair overall agreement of analytical and numerical curves for $p = 0$ confirms the validity of the model developed by Bücker and Hördt (2013a) for a sole diffuse-layer polarization. The same authors also proposed a simple modification of the mean counterion concentration to take into account the contribution of the Stern layer. However, their model rather predicts a continuous decrease of the polarization magnitude with increasing p and practically a constant characteristic frequency (not shown here for brevity), indicating that the Stern-layer polarization is more complex than assumed by Bücker and Hördt (2013a).

The polarization response changes significantly if the EDL is assumed to be continuous in the wide pore (Figure 10). Here, magnitude and characteristic frequency of the main σ'' peak located around 2 rad/s decrease only slightly from $p = 0$ to $p = 0.4$ and increase not much steeper for larger values of p . In comparison to the significant variation of the main peak observed in Figure 9 for the discontinuous EDL, here, the σ'' peak remains almost unchanged. Furthermore, at frequencies between 100 and 1,000 rad/s, a smaller secondary polarization peak appears for $p > 0$ and increases monotonically with p . The real part σ' shows the usual decrease of surface conductivity with increasing p . The relative deviations between the numerical results for $p = 0$ and the two analytical models are similar to those reported above for a discontinuous EDL.

5. Discussion

The new analytical model presented here describes Maxwell-Wagner, Stern-layer, and diffuse-layer polarization around spherical grains and permits analyzing the frequency response due to a superposition of all three mechanisms. Numerical and analytical results for typical model parameters consistently confirm the results of earlier studies: The contribution of the diffuse-layer polarization to the macroscopic response is significantly smaller than the one of the Stern-layer polarization (e.g., de Lima & Sharma, 1992; Lesmes & Morgan, 2001). In most practical cases, the polarization of the diffuse layer can therefore safely be neglected.

The comparison of numerical and analytical results also confirmed the strong coupling of the surface charge in the Stern layer with charges in the electrolyte as predicted by Lyklema et al. (1983). The model proposed by Lyklema and coworkers can easily be obtained from the model by Schwarz (1962) by adjusting the relaxation time of the Stern layer: For a charge density in the Stern layer of $\Sigma_S^{(0)} \approx 0.01$ C/m² and a bulk ion concentration of 1 mol/m³, equation (32) predicts a reduction of the relaxation time by a factor $M \approx 6.4$. This effect can also be interpreted in terms of an effective diffusion coefficient $D_S^{\text{eff}} = D_S M$, which in our case (i.e., $D_S = D/10$) is only a factor ≈ 2 smaller than the diffusion coefficient of the ions in the bulk electrolyte.

This observation is of particular interest, as it offers an explanation for the large diffusion coefficients D_S needed to adjust the model by Leroy et al. (2008) to measured complex-conductivity spectra. The model by Leroy and coworkers does not include the correction of the relaxation time proposed by Lyklema et al. (1983). Thus, it often requires the assumption of similar diffusion coefficients in Stern layer and bulk electrolyte (e.g., Leroy & Revil, 2009; Leroy et al., 2008; Revil & Florsch, 2010; Schmutz et al., 2010). Independent determinations of the diffusion coefficient in the Stern layer from experimental surface-conductivity and

particle-mobility data, however, normally result in a reduction of the counterion mobility by a factor 10 or larger (e.g., Zukoski & Saville, 1986; Revil & Glover, 1998). A recent extension of the model by Leroy et al. (2008) already considers the corrected relaxation time and successfully adjusts the polarization response of calcite precipitations on glass beads (Leroy et al., 2017). Together with our findings that support the theory by Lyklema et al. (1983), this indicates that the difference between the predictions of the two experiments could be reduced significantly, if the Stern-layer relaxation time used in the model by Leroy et al. (2008) (and subsequent studies) was replaced by the one defined in equations (31) and (32).

Although the correction of the relaxation time might be able to justify the (to date) unexplained assumption inherent to the model by Leroy et al. (2008), it also questions the often-used simple relation between relaxation time and grain size by introducing an additional dependence on chemical properties of pore fluid and solid surface: The coefficient M strongly depends on the electrolyte concentration and the surface-charge density in the Stern layer. Consequently, besides the grain diameter and the diffusion coefficient in the Stern layer, variations of experimentally determined relaxation times might partly be due to variations of these chemical parameters.

We have also applied our finite-element model to study membrane polarization, which had not been investigated in the same detail as the polarization around spherical grains before. In the limiting case of a sole diffuse-layer polarization, we found that an improved analytical formulation based on the work by Marshall and Madden (1959) and Bückner and Hördt (2013a) yields a good agreement with the numerical results. However, the analytical model does not reproduce the numerical results for the coupled polarization of diffuse and Stern layer. Consequently, the incorporation of the Stern-layer polarization in the model by Bückner and Hördt (2013a) is insufficient in its current form, and our numerical results can contribute substantially to our understanding of the role of the Stern layer in the context of membrane polarization.

Even in the pore-constriction geometry, the polarization of the Stern layer can control the frequency response, especially if the EDL is discontinuous. Certain similarities of the coupled responses of both geometries stand out—namely, the generally larger response of the Stern-layer polarization and the simultaneous increase of characteristic frequency and polarization magnitude with the surface charge in the Stern layer. However, we also observed that the particular geometrical configuration of the EDL in the pore-constriction model can largely reduce the difference between the respective magnitudes of diffuse-layer and Stern-layer polarization: If the EDL becomes continuous, the contributions of both mechanisms can become practically the same.

Although we have taken an important step toward this long-term goal, it is beyond the scope of the present study to provide an analytical model that integrates membrane polarization and the polarization around spherical grains. Nevertheless, we can put our results into a broader context and draw some preliminary conclusions regarding the relative contributions of the various polarization mechanisms.

Grain-based polarization models are best suited for the modeling of dilute suspensions of dielectric particles because they only account for grain-electrolyte interactions and largely ignore grain-grain interactions. For practical purposes, these models can even be applied to high particle concentrations, and the responses of unconsolidated granular media (e.g., Leroy et al., 2008) have been matched successfully with this type of models. For typical parameter combinations, the polarization of isolated grains is only controlled by the Stern layer, and contributions of a simultaneously occurring diffuse-layer polarization can be neglected.

With increasing degree of compaction and cementation, grain-grain interactions are expected to become more important (e.g., Lesmes & Morgan, 2001). These interactions include (i) the interaction of polarization dipoles of adjacent grains as well as (ii) percolating diffuse and Stern layers. Based on the good agreement between experimental data and the responses of grain-based polarization models, the interaction of polarization dipoles can safely be neglected. To date, the effect of percolating diffuse and Stern layers on the polarization response has only been touched in passing: For instance, Leroy et al. (2008) argue that a diffuse layer above the percolation threshold cannot polarize and only consider an increase of the low-frequency conductivity. In the same sense, they postulate that the Stern layers of adjacent particles are discontinuous because the model based on the Stern-layer polarization accurately describes experimental observations. While the result of this argumentation—the small contribution of the diffuse-layer polarization and the dominating role of the Stern layer—agrees with our findings, the physical picture should be reconsidered: Around isolated particles, the polarization of the diffuse layer is negligible, too, such that it is not necessary

to assume that the diffuse layers are percolating. In addition, we have shown that even in the case of continuous (or percolating) diffuse and Stern layers in the pore-constriction model, both layers can polarize. Consequently, the discontinuity of neither of the two layers is needed to generate a polarization response.

The similarity of the polarization response of grain-based and pore-constriction geometry indicates a gradual transition between the two models with increasing degree of compaction and cementation. As long as the EDLs are assumed to be discontinuous, the Stern layer dominates the polarization response of both models, which makes the distinction between effects related to individual grains and effects related to pore constrictions a mere question of the point of view. Our results suggest that responses caused by pore constrictions, that is, typical membrane-polarization responses, can—at least to a certain degree—be adjusted using grain-based models and vice versa. This is particularly plausible, if we consider that in granular media made of near-spherical particles, pore diameters and pore lengths are of the same order of magnitude as the typical grain sizes and thus all relaxation times are controlled by similar characteristic lengths.

6. Conclusions

We have investigated the low-frequency electrical conductivity of porous media by means of analytical and numerical models for single-grain and pore-constriction geometries. Our results allowed us to assess the relative contributions of polarization mechanisms originating from the diffuse part and the Stern layer of the EDL covering charged mineral surfaces. Because our models also include Maxwell-Wagner polarization, they are useful for a broad frequency range.

To match our numerical results, we assembled a new analytical model for the grain-based geometry by combining the Dukhin-Shilov model (Dukhin & Shilov, 1974), which accounts for the diffuse-layer polarization, with Schurr's Stern-layer polarization model (Schurr, 1964) including a correction of the Stern-layer relaxation time proposed by Lyklema et al. (1983). The corrected relaxation time, which accounts for the interaction of the charges in the Stern layer with the electrolyte solution, significantly improves the agreement of analytical and numerical results. Without the correction, untypically high diffusion coefficients of the counterions in the Stern layer have to be assumed to fit the spectral response. For typical model parameters, the relative contribution of the diffuse-layer polarization was seen to be insignificant in comparison to the large response of the Stern layer.

Our numerical results for the pore-constriction geometry agree with the analytical membrane-polarization model by Bückner and Hördt (2013a) if we relate the mean ion concentrations of the cylindrical pores to Bikerman's expression for the surface conductivity of highly charged surfaces (Bikerman, 1933) and as long as no Stern layer is considered. We also propose detailed model to describe Stern-layer polarization in a typical membrane-polarization geometry. As in the grain-based model, the diffuse layer makes a much smaller contribution to the total response than the Stern layer—at least as long as the charged surfaces are below the percolation threshold; that is, the individual EDLs are not interconnected at the system scale.

In conclusion, the responses of grain-based and pore-constriction geometries are more similar than usually assumed, particularly if the polarization of both parts of the EDL are taken into account. Below the percolation threshold, the Stern-layer dominates the macroscopic response, but as soon as the EDL becomes percolating, this dominance breaks down and both mechanisms contribute to a similar extend. More detailed studies on specific pore geometries are required to conclusively assess the relative importance of the different polarization mechanisms, but our study is a significant step toward this long-term goal and sets the basis for extensive numerical studies.

Appendix A: Numerical Implementation

The COMSOL PDE interface in coefficient form allows the definition of PDEs and systems of PDEs of the general type

$$e_a \frac{\partial^2 u}{\partial t^2} + d_a \frac{\partial u}{\partial t} + \nabla \cdot (-c \nabla u - a u + \gamma) + \beta \cdot \nabla u + a u = f \quad (\text{A1})$$

with the general boundary conditions

$$-\mathbf{n} \cdot (-c \nabla u - a u + \gamma) = g - q u \quad \text{and} \quad (\text{A2})$$

$$u = s, \quad (\text{A3})$$

where u denotes the dependent variable and $\mathbf{n}$ the inward-pointing unit normal vector.

Due to their cylindrical symmetry, the problems can be solved on two-dimensional domains (see Figures 5 and 8) with coordinates $\mathbf{r} = (x, y)^T$ and gradient operator $\nabla = (\partial/\partial x, \partial/\partial y)^T$. As explained in more detail in Bückner et al. (2018), multiplying all coefficients in equations (A1) and (A2) by y yields a transformation to cylindrical coordinates.

A1. Static Solution

The static problem set up by equations (6) and (7) is implemented by setting $u_1 = U^{(0)}(\mathbf{r})$, $c_1 = y\epsilon_0\epsilon_a$, $f_1 = -y2C_{\pm}^{\infty}e \sinh(u_1 e/(kT))$ in the electrolyte as well as $c_1 = y\epsilon_0\epsilon_a$ and $f_1 = 0$ in the solid phase. Note that in this appendix, coefficients are assumed to be zero if not specified differently.

The fixed surface-charge boundary condition (15) is realized by setting $g = -y\Sigma_d^{(0)}$ on all charged boundaries. The definition of the zero reference potential, equation (14), depends on the specific model: In the grain geometry, we set $s = 0$ on the left, right, and top boundary; in the membrane geometry, we define the point-wise constraint $u_1 = 0$ at two positions $(x = \pm(L_1 + L_2)/2, y = 0)$ located on the left and right boundary, respectively. In this case, standard no-flow boundary conditions are set on the remaining parts of left and right boundary, as well as on the uncharged parts of the top boundary. Because $y = 0$ is the symmetry axis (selecting rotational symmetry in the model setup), no specific boundary conditions are defined on this boundary.

A2. Frequency-Dependent Solution

For the frequency-dependent problem set up by the coupled PDEs (8) and (9), which describe the physics in the electrolyte, the Laplace equation (13), which controls the electrical field in the solid phase, and equation (11), which controls the surface charge in the Stern layer, need to be solved simultaneously. We use the same dependent variable u_2 in the electrolyte and in the solid phase and define suitable PDE coefficients on each of the two subdomains. The variable u_2 writes

$$\mathbf{u}_2 = \begin{bmatrix} u_{21} \\ u_{22} \\ u_{23} \end{bmatrix} = \begin{bmatrix} \delta C_{-}(\mathbf{r}, \omega) \\ \delta C_{+}(\mathbf{r}, \omega) \\ \delta U(\mathbf{r}, \omega) \end{bmatrix}. \quad (\text{A4})$$

Assuming isotropic properties, the coefficient c_2 is a 3-by-3 coefficient matrix. In the electrolyte it writes

$$\underline{c}_2 = \begin{bmatrix} yD & 0 & -y\mu C_{-}^{\infty} \exp(\frac{e}{kT}u_1) \\ 0 & yD & y\mu C_{+}^{\infty} \exp(-\frac{e}{kT}u_1) \\ 0 & 0 & y\epsilon_0\epsilon_a \end{bmatrix}, \quad (\text{A5})$$

and in the solid phase

$$\underline{c}_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & y\epsilon_0\epsilon_i \end{bmatrix}. \quad (\text{A6})$$

The variable u_1 is the static part of the electrical potential. The coefficient matrix $\underline{\alpha}$ writes

$$\underline{\alpha}_2 = \begin{bmatrix} -y\mu\nabla u_1 & 0 & 0 \\ 0 & y\mu\nabla u_1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (\text{A7})$$

in the electrolyte and is equal to the null matrix in the solid phase. Note that on our two-dimensional modeling domain, each element of the matrix $\underline{\alpha}_2$ is a two-element vector. The matrix $\underline{a}_2$ is

$$\underline{a}_2 = \begin{bmatrix} y i \omega & 0 & 0 \\ 0 & y i \omega & 0 \\ y F & -y F & 0 \end{bmatrix} \quad (\text{A8})$$

in the electrolyte. In the solid phase, $\underline{a}_2$ is the same except for the last line, which has to be filled with zeros instead.

Equation (11) describing the perturbation surface-charge density $\delta\Sigma_S$ is solved on the solid-liquid interface only. By means of a so-called lower-dimensional physics interface, the curvature of this interface is taken into account. Because the electrical potential in the Stern layer has to be equal to the potentials δU_i and δU_a on the corresponding boundaries, the only dependent variable on the interface is $u_3 = \delta\Sigma_S(\mathbf{r}_S, \omega)$. The coefficients are $c_3 = yD_S$, $a_3 = y i \omega$, and $\gamma_3 = -y \mu_S \Sigma_S^{(0)} \nabla u_{23}$, where u_{23} denotes the perturbation potentials on the adjacent two-dimensional subdomains.

The boundary condition (16) describing the external excitation and the condition on the perturbation concentrations (17) are realized by setting $\mathbf{s}_2 = (0, 0, \pm E_0 L)^T$. In the membrane-polarization model with the continuous EDL, this boundary is not located at a far distance from the charged surface. However, because of the symmetry of the problem with respect to $x = 0$, we expect the perturbation ion concentrations to vanish in this geometry, too (see, e.g., concentration profiles in Blaschek & Hördt, 2009). For the same reason, the perturbation surface-charge density $\delta\Sigma_S$ is also expected to vanish on the left and right boundary, and we set $s_3 = 0$ (only for a continuous EDL).

On the solid-liquid interface, we implement the zero-flux boundary conditions (19) and the condition on the displacement current (20) by defining $\mathbf{g} = (0, 0, y u_3)^T$.

On the top boundary, we implement zero-flux boundary condition on all three components of u_2 resulting in vanishing normal ion fluxes and a vanishing normal electrical field. In the membrane-polarization model with the continuous EDL, parts of the top boundary represent charged surfaces and are furnished with the corresponding boundary conditions described above. In the membrane-polarization model with the discontinuous EDL, the one-dimensional domain representing the Stern layer ends at the top boundary. In this case, we also define no-flux boundary conditions for u_3 .

A3. Model Discretization

Special care has to be taken with the model discretization: While particle and pore sizes are in the micrometer to millimeter range, the thickness of the diffuse layer is orders of magnitude smaller. We use a special quadrangle boundary-layer mesh at the solid-liquid interface, which in radial direction consists of eight elements with sizes increasing from $\lambda_D/2$ at the surface to $\approx 1.8\lambda_D$ at the outer edge. In tangential direction (i.e., along the boundaries), the elements have a size of $\pi a/400$ in the grain-based model and $R_2/5$ in the pore-constriction geometry. The remaining volume is filled with triangular elements, the maximum size of which increases from $\pi a/400$ (or $R_2/5$) at the edge of the boundary-layer meshes to $L/20$ (or $R_1/5$) at the remote boundaries. For a spherical particle of radius $a = 5\ \mu\text{m}$, this results in a total of $\approx 20,900$ elements, of which 6,400 elements correspond to the boundary-layer mesh. For pore lengths $L_1 = 90\ \mu\text{m}$ and $L_2 = 10\ \mu\text{m}$ and pore radii $R_1 = 2\ \mu\text{m}$ and $R_2 = 0.2\ \mu\text{m}$, the meshes consist of $\approx 14,700$ elements ($\approx 11,000$ for the continuous EDL), of which $\approx 5,200$ ($\approx 7,600$) elements make the boundary-layer mesh.

Acknowledgments

M. Bückner acknowledges the Austrian Federal Ministry of Science, Research and Economy (project Development of Geophysical Exploration Methods for the Characterization of Mine-Tailings Towards Exploitation) and the German Academic Exchange Service (DAAD) for scholarships received for his work on this study. All authors gratefully acknowledge P. Leroy and one anonymous reviewer for their insightful comments that helped to improve the manuscript. Scripts, numerical models, and data used for this manuscript are available at Zenodo (<https://doi.org/10.5281/zenodo.3067277>).

References

- Atekwana, E. A., & Slater, L. D. (2009). Biogeophysics: A new frontier in earth science research. *Reviews of Geophysics*, 47, RG4004. <https://doi.org/10.1029/2009RG000285>
- Atkins, P., & De Paula, J. (2013). *Elements of Physical Chemistry*. USA: Oxford University Press.
- Bairlein, K., Bückner, M., Hördt, A., & Hinze, B. (2016). Temperature dependence of spectral induced polarization data: experimental results and membrane polarization theory. *Geophysical Journal International*, 205, 440–453.
- Bikerman, J. (1933). Ionentheorie der Elektromose, der Strömungsströme und der Oberflächenleitfähigkeit. *Zeitschrift für Physikalische Chemie A*, 163, 378–394.
- Blaschek, R., & Hördt, A. (2009). Numerical modelling of the IP effect at the pore scale. *Near Surface Geophysics*, 7(5-6), 579–588.
- Börner, F., Schopper, J., & Weller, A. (1996). Evaluation of transport and storage properties in the soil and groundwater zone from induced polarization measurements. *Geophysical Prospecting*, 44(4), 583–601.
- Bruggeman, D. A. G. (1935). Berechnung verschiedener physikalischer Konstanten von heterogenen Substanzen. I. Dielektrizitätskonstanten und Leitfähigkeiten der Mischkörper aus isotropen Substanzen. *Annalen der Physik*, 416(7), 636–664.
- Buchheim, W., & Irmer, G. (1979). Zur Theorie der induzierten galvanischen Polarisation in Festkörpern mit elektrolitischer Porenfüllung. *Gerlands Beiträge zur Geophysik*, 88, 53–72.
- Bückner, M., Flores Orozco, A., Hördt, A., & Kemna, A. (2017). An analytical membrane-polarization model to predict the complex conductivity signature of immiscible liquid hydrocarbon contaminants. *Near Surface Geophysics*, 15(6), 547–562.
- Bückner, M., & Hördt, A. (2013a). Analytical modelling of membrane polarization with explicit parametrization of pore radii and the electrical double layer. *Geophysical Journal International*, 194, 804–813.
- Bückner, M., & Hördt, A. (2013b). Long and short narrow pore models for membrane polarization. *Geophysics*, 78(6), E299–E314.

- Bücker, M., Undorf, S., Flores Orozco, A., & Kemna, Andreas. (2018). Electro-chemical polarization around metallic particles-Part 2: The role of diffuse surface charge. *Geophysics*, 84(2), 1–64.
- Chew, W., & Sen, P. (1982a). Potential of a sphere in an ionic solution in thin double layer approximations. *The Journal of Chemical Physics*, 77(4), 2042–2044.
- Chew, W., & Sen, P. (1982b). Dielectric enhancement due to electrochemical double layer: Thin double layer approximation. *The Journal of Chemical Physics*, 77(9), 4683–4693.
- de Lima, O. A., & Sharma, M. M. (1992). A generalized Maxwell-Wagner theory for membrane polarization in shaly sands. *Geophysics*, 57(3), 431–440.
- DeLacey, E. H., & White, L. R. (1981). Dielectric response and conductivity of dilute suspensions of colloidal particles. *Journal of the Chemical Society, Faraday Transactions 2: Molecular and Chemical Physics*, 77(11), 2007–2039.
- Dukhin, S., & Shilov, V. (1974). *Dielectric phenomena and the double layer in disperse systems and polyelectrolytes*. New York: Wiley.
- Fixman, M. (1980). Charged macromolecules in external fields. I. The sphere. *The Journal of Chemical Physics*, 72(9), 5177–5186.
- Flores Orozco, A., Williams, K. H., Long, P. E., Hubbard, S. S., & Kemna, A. (2011). Using complex resistivity imaging to infer biogeochemical processes associated with bioremediation of an uranium-contaminated aquifer. *Journal of Geophysical Research*, 116, G03001. <https://doi.org/10.1029/2010JG001591>
- Fridrikhsberg, D. A., & Sidorova, M. P. (1961). A study of relationship between the induced polarization phenomenon and the electro-kinetic properties of capillary systems (in Russian). *Vestnik Leningradskogo Universiteta, Seria Khimii*, 4, 222–226.
- Garcia, A., Grosse, C., & Brito, P. (1985). On the effect of volume charge distribution on the Maxwell-Wagner relaxation. *Journal of Physics D: Applied Physics*, 18(4), 739.
- Grahame, D. C. (1947). The electrical double layer and the theory of electrocapillarity. *Chemical Reviews*, 41(3), 441–501.
- Grosse, C., & Shilov, V. N. (1996). Theory of the low-frequency electrorotation of polystyrene particles in electrolyte solution. *The Journal of Physical Chemistry*, 100(5), 1771–1778.
- Hördt, A., Bairlein, K., Bielefeld, A., Bücker, M., Kuhn, E., Nordsiek, S., & Stebner, H. (2016). The dependence of induced polarization on fluid salinity and pH, studied with an extended model of membrane polarization. *Journal of Applied Geophysics*, 135, 408–417.
- Hanaï, T. (1960). Theory of the dielectric dispersion due to the interfacial polarization and its application to emulsions. *Kolloid-Zeitschrift*, 171(1), 23–31.
- Hinch, E. J., Sherwood, J. D., Chew, W. C., & Sen, P. N. (1984). Dielectric response of a dilute suspension of spheres with thin double layers in an asymmetric electrolyte. *Journal of the Chemical Society, Faraday Transactions 2: Molecular and Chemical Physics*, 80(5), 535–551.
- Hördt, A., Druiventak, A., Blaschek, R., Binot, F., Kemna, A., Kreye, P., & Zisser, N. (2009). Case histories of hydraulic conductivity estimation with induced polarization at the field scale. *Near Surface Geophysics*, 7(5–6), 529–545.
- Hördt, A., Bairlein, K., Bücker, M., & Stebner, H. (2017). Geometrical constraints for membrane polarization. *Near Surface Geophysics*, 15, 579–592.
- Hunter, R. (1981). *Zeta potential in colloid science: Principles and applications*. London: Academic Press.
- Kemna, A., Binley, A., Cassiani, G., Niederleithinger, E., Revil, A., Slater, L., et al. (2012). An overview of the spectral induced polarization method for near-surface applications. *Near Surface Geophysics*, 10(6), 453–468.
- Kijlstra, J., van Leeuwen, H. P., & Lyklema, J. (1992). Effects of surface conduction on the electrokinetic properties of colloids. *Journal of the Chemical Society, Faraday Transactions*, 88(23), 3441–3449.
- Kosmulski, M. (2006). Electric charge density of silica, alumina, and related surfaces. In P. Somasundaran (Ed.), *Encyclopedia of surface and colloid science* (Vol. 3, pp. 1857–1867). New York, London: CRC press.
- Leroy, P., Li, S., Jougnot, D., Revil, A., & Wu, Y. (2017). Modelling the evolution of complex conductivity during calcite precipitation on glass beads. *Geophysical Journal International*, 209(1), 123–140.
- Leroy, P., & Revil, A. (2004). A triple-layer model of the surface electrochemical properties of clay minerals. *Journal of Colloid and Interface Science*, 270(2), 371–380.
- Leroy, P., & Revil, A. (2009). A mechanistic model for the spectral induced polarization of clay materials. *Journal of Geophysical Research*, 114, B10202. <https://doi.org/10.1029/2008JB006114>
- Leroy, P., Revil, A., Kemna, A., Cosenza, P., & Ghorbani, A. (2008). Complex conductivity of water-saturated packs of glass beads. *Journal of Colloid and Interface Science*, 321(1), 103–117.
- Lesmes, D. P., & Morgan, F. D. (2001). Dielectric spectroscopy of sedimentary rocks. *Journal of Geophysical Research*, 106(B7), 13,329–13,346.
- Lyklema, J., Dukhin, S., & Shilov, V. (1983). The relaxation of the double layer around colloidal particles and the low-frequency dielectric dispersion: Part I. Theoretical considerations. *Journal of Electroanalytical Chemistry and Interfacial Electrochemistry*, 143(1–2), 1–21.
- Marshall, D. J., & Madden, T. R. (1959). Induced polarization, a study of its causes. *Geophysics*, 24(4), 790–816.
- Maxwell, J. C. (1892). *A treatise on electricity and magnetism* (3rd ed.). Oxford, England: Clarendon.
- O’Konski, C. T. (1960). Electric properties of macromolecules. V. Theory of ionic polarization in polyelectrolytes. *The Journal of Physical Chemistry*, 64(5), 605–619.
- Revil, A., Cathles, L., Losh, S., & Nunn, J. (1998). Electrical conductivity in shaly sands with geophysical applications. *Journal of Geophysical Research*, 103(B10), 23,925–23,936.
- Revil, A., & Florsch, N. (2010). Determination of permeability from spectral induced polarization in granular media. *Geophysical Journal International*, 181(3), 1480–1498.
- Revil, A., & Glover, P. W. J. (1997). Theory of ionic-surface electrical conduction in porous media. *Physical Review B*, 55(3), 1757.
- Revil, A., & Glover, P. W. J. (1998). Nature of surface electrical conductivity in natural sands, sandstones, and clays. *Geophysical Research Letters*, 25(5), 691–694.
- Robinson, D. A., & Friedman, S. P. (2003). A method for measuring the solid particle permittivity or electrical conductivity of rocks, sediments, and granular materials. *Journal of Geophysical Research*, 108(B2), 2076. <https://doi.org/10.1029/2001JB000691>
- Schmutz, M., Revil, A., Vaudelet, P., Batzle, M., Viñao, P. F., & Werkema, D. (2010). Influence of oil saturation upon spectral induced polarization of oil-bearing sands. *Geophysical Journal International*, 183(1), 211–224.
- Schurr, J. (1964). On the theory of the dielectric dispersion of spherical colloidal particles in electrolyte solution. *The Journal of Physical Chemistry*, 68(9), 2407–2413.
- Schwarz, G. (1962). A theory of the low-frequency dielectric dispersion of colloidal particles in electrolyte solution. *The Journal of Physical Chemistry*, 66(12), 2636–2642.
- Shilov, V. N., Delgado, A. V., Gonzalez-Caballero, F., & Grosse, C. (2001). Thin double layer theory of the wide-frequency range dielectric dispersion of suspensions of non-conducting spherical particles including surface conductivity of the stagnant layer. *Colloids and Surfaces A: Physicochemical and Engineering Aspects*, 192(1), 253–265.

- Somasundaran, P. (2006). *Encyclopedia of surface and colloid science* (Vol. 2). New York, London: CRC press.
- Stebner, H., Halisch, M., & Hördt, A. (2017). Simulation of membrane polarization of porous media with impedance networks. *Near Surface Geophysics*, 15, 563–578.
- Titov, K., Kemna, A., Tarasov, A., & Vereecken, H. (2004). Induced polarization of unsaturated sands determined through time domain measurements. *Vadose Zone Journal*, 3(4), 1160–1168.
- Titov, K., Komarov, V., Tarasov, V., & Levitski, A. (2002). Theoretical and experimental study of time domain-induced polarization in water-saturated sands. *Journal of Applied Geophysics*, 50(4), 417–433.
- Volkman, J., & Klitzsch, N. (2010). Frequency-dependent electric properties of microscale rock models for frequencies from one millihertz to ten kilohertz. *Vadose Zone Journal*, 9(4), 858–870.
- Wagner, Karl W. (1914). Erklärung der Dielektrischen Nachwirkungsvorgänge auf Grund Maxwellscher Vorstellungen. *Electrical Engineering (Archiv für Elektrotechnik)*, 2(9), 371–387.
- Wainwright, H. M., Flores Orozco, A., Bucker, M., Dafflon, B., Chen, J., Hubbard, S. S., & Williams, K. H. (2015). Hierarchical Bayesian method for mapping biogeochemical hot spots using induced polarization imaging. *Water Resources Research*, 52, 533–551. <https://doi.org/10.1002/2015WR017763>
- Zukoski, C. F., & Saville, D. A. (1986). The interpretation of electrokinetic measurements using a dynamic model of the stern layer: II. Comparisons between theory and experiment. *Journal of colloid and interface science*, 114(1), 45–53.